

Alex's Anthology of Algorithms
Common Code for Contests in Concise C++

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Preface

Visit github.com/alxli/algorithm-anthology for the most up-to-date digital version of this book.

Introduction

Welcome to a comprehensive collection of common algorithms and data structures. The ultimate goal of this book is not to explain concepts from the ground up, but instead to explore the finer details behind their *implementations*. There are many potential ways you can use this, for instance:

- as a reference to help you better understand topics that you have only studied on a high level,
- as a printed codebook, which is a permitted resource for contests such as the ACM ICPC, or
- to cross-check existing code you have written for contest or coding interview questions.

Before diving into any section, it is strongly recommended that you have already studied the algorithms involved. Reading the code first is never an ideal approach to properly understand an algorithm. You should instead try proving (its correctness and time/space complexity) and implementing it from scratch.

Every topic to be explored is easily researchable online. Thus instead of including theoretical discussions, I document just enough to establish the problem being solved, notation being used, and any special trickery involved. I have also included small, non-rigorous examples to demonstrate usage of the code.

We mentioned that the implementation itself is the focus, but what makes an implementation good? The code is written with the following principles in mind:

- *Clarity*: A reader already familiar with an algorithm should have no problem understanding how its implementation works. Consistency in naming conventions should be emphasized, and any tricks or language-specific hacks should be documented.
- *Concision*: To minimize the amount of scrolling and searching during the frenzy of time contests, it is helpful for code to be compact. Shorter code is also generally easier to understand, as long as it is not overly cryptic. Finally, each implementation should fit in a single source file as required by nearly all online judging systems.
- *Efficiency*: The code here is designed to be performant on real contests, and should maintain a low constant overhead. This is often challenging in the face of clarity and tweakability, but we can hope for contest setters to be liberal with time and memory limits. If the code here times out, you can reasonably rule out insufficient constant optimization and assume that you are choosing an algorithm from a suboptimal complexity class.
- *Genericness*: Implementations should be easy to adapt to achieve slightly different goals. One may want to tweak some core logic, parameters, data types, etc. In timed contests, we would certainly prefer this process to be as painless as possible. C++ templates are often used to increase tweakability at a slight cost to simplicity.

- *Portability*: Different contest environments use different compiler builds. The code targets C++17, which is broadly available on modern contest systems, while still avoiding unnecessary dependencies and non-standard features where practical.

As these points and the title both suggest, there is a slight bias towards contests. Compiling a codebook for my personal reference during contests was indeed how this project got started. This work has become much more multipurpose now. Whatever your use case is, I hope you discover something enlightening.

Cheers.
— Alex

Portability Notes

All programs were tested with GCC using the switches below:

```
g++ -std=gnu++17 -pedantic -Wall -O2
```

This means the following are assumed about data types:

- `bool` and `char` are 8-bit.
- `int` and `float` are 32-bit.
- `double` and `long long` are 64-bit.
- `long double` is 96-bit.

Programs target ISO C++17, except in the following regards:

- Usage of variable sized arrays. While easily replaced by vectors, they are generally simpler and avoid dynamic memory (which some argue is a bad idea for contests).
- Usage of GCC's built-in functions like `__builtin_popcount()` and `__builtin_clz()`. These can be extremely convenient, but are straightforward to implement if unavailable. See here for a reference: <https://gcc.gnu.org/onlinedocs/gcc/Other-Builtins.html>
- Hacks that may depend on the platform (e.g. endianness), such as getting the signbit with type-punned pointers. Be weary of portability for all bitwise/lower level code.

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Chapter 1

Elementary Algorithms

1.1 Array Transformations

1.1.1 Sorting Algorithms

These functions are equivalent to `std::sort()`, taking random-access iterators as a range `[lo, hi)` to be sorted. Elements between `lo` and `hi` (including the element pointed to by `lo` but excluding the element pointed to by `hi`) will be sorted into ascending order after the function call. Optionally, a comparison function object specifying a strict weak ordering may be specified to replace the default `operator<`.

These functions are not meant to compete with standard library implementations in terms of speed. Instead, they are meant to demonstrate how common sorting algorithms can be concisely implemented in C++.

```
#include <algorithm>
#include <functional>
#include <iterator>
#include <vector>
```

Quicksort repeatedly selects a pivot and partitions the range so that elements comparing less than the pivot precede it, elements comparing equal stay in the middle, and elements comparing greater follow it. Divide and conquer is then applied to the two outer ranges until the original range is sorted. Despite having a worst case of $O(n^2)$, quicksort is often faster in practice than merge sort and heapsort, which both have a worst case time complexity of $O(n \log n)$.

The pivot chosen in this implementation is always a middle element of the range to be sorted. To reduce the likelihood of encountering the worst case, the pivot can be chosen in better ways (e.g. randomly, or using the “median of three” technique).

Time Complexity (Average): $O(n \log n)$.

Time Complexity (Worst): $O(n^2)$.

Space Complexity: $O(\log n)$ auxiliary stack space.

Stable?: No.

```
template<class It, class Compare>
void quicksort(It lo, It hi, Compare comp) {
    while (hi - lo >= 2) {
        auto pivot = *(lo + (hi - lo) / 2);
        It lt = lo, i = lo, gt = hi;
        while (i != gt) {
            if (comp(*i, pivot)) {
```

```

        std::iter_swap(lt++, i++);
    } else if (comp(pivot, *i)) {
        std::iter_swap(i, --gt);
    } else {
        ++i;
    }
}
if (lt - lo < hi - gt) {
    quicksort(lo, lt, comp);
    lo = gt;
} else {
    quicksort(gt, hi, comp);
    hi = lt;
}
}
}

template<class It>
void quicksort(It lo, It hi) {
    using T = typename std::iterator_traits<It>::value_type;
    quicksort(lo, hi, std::less<T>());
}

```

Merge sort first divides a list into n sublists of one element each, then recursively merges the sublists into sorted order until only a single sorted sublist remains. Merge sort is a stable sort, meaning that it preserves the relative order of elements which compare equal by `operator<` or the custom comparator given.

An analogous function in the C++ standard library is `std::stable_sort()`, except that the implementation here requires sufficient memory to be available. When $O(n)$ auxiliary memory is not available, `std::stable_sort()` falls back to a time complexity of $O(n \log^2 n)$ whereas the implementation here will simply fail.

Time Complexity (Average): $O(n \log n)$.

Time Complexity (Worst): $O(n \log n)$.

Space Complexity: $O(\log n)$ auxiliary stack space and $O(n)$ auxiliary heap space.

Stable?: Yes.

```

template<class It, class Compare>
void mergesort(It lo, It hi, Compare comp) {
    if (hi - lo < 2) {
        return;
    }
    It mid = lo + (hi - lo - 1) / 2, a = lo, c = mid + 1;
    mergesort(lo, mid + 1, comp);
    mergesort(mid + 1, hi, comp);
    using T = typename std::iterator_traits<It>::value_type;
    std::vector<T> merged;
    while (a <= mid && c < hi) {
        merged.push_back(comp(*c, *a) ? *c++ : *a++);
    }
    merged.insert(merged.end(), a, mid + 1);
    merged.insert(merged.end(), c, hi);
    std::copy(merged.begin(), merged.end(), lo);
}

template<class It>
void mergesort(It lo, It hi) {
    using T = typename std::iterator_traits<It>::value_type;
    mergesort(lo, hi, std::less<T>());
}

```

Heapsort first rearranges an array to satisfy the max-heap property. Then, it repeatedly pops the max element of the heap (the left, unsorted subrange), moving it to the beginning of the right, sorted subrange until the entire

range is sorted. Heapsort has a better worst case time complexity than quicksort and also a better space complexity than merge sort.

The C++ standard library equivalent is calling `std::make_heap(lo, hi)`, followed by `std::sort_heap(lo, hi)`.

The loop below has two modes controlled by whether `i > lo`. While `i > lo`, it is heapifying (Floyd's $O(n)$ bottom-up heap build: sift down each non-leaf from right to left). Once `i == lo`, the heap is built, and the loop switches to extraction: repeatedly swap the root to the back of the unsorted region (`*j`) and sift down the new root.

Time Complexity (Average): $O(n \log n)$.

Time Complexity (Worst): $O(n \log n)$.

Space Complexity: $O(1)$ auxiliary.

Stable?: No.

```
template<class It, class Compare>
void heapsort(It lo, It hi, Compare comp) {
    using T = typename std::iterator_traits<It>::value_type;
    T tmp;
    It i = lo + (hi - lo) / 2, j = hi, parent, child;
    for (;;) {
        if (i <= lo) {
            if (--j == lo) {
                return;
            }
            tmp = *j;
            *j = *lo;
        } else {
            tmp = *(--i);
        }
        parent = i;
        child = lo + 2 * (i - lo) + 1;
        while (child < j) {
            if (child + 1 < j && comp(*child, *(child + 1))) {
                child++;
            }
            if (!comp(tmp, *child)) {
                break;
            }
            *parent = *child;
            parent = child;
            child = lo + 2 * (parent - lo) + 1;
        }
        *(lo + (parent - lo)) = tmp;
    }
}

template<class It>
void heapsort(It lo, It hi) {
    using T = typename std::iterator_traits<It>::value_type;
    heapsort(lo, hi, std::less<T>());
}
```

Comb sort is an improved bubble sort. While bubble sort increments the gap between swapped elements for every inner loop iteration, comb sort fixes the gap size in the inner loop, decreasing it by a particular shrink factor in every iteration of the outer loop. The shrink factor of 1.3 is empirically determined to be the most effective.

Even though the worst case time complexity is $O(n^2)$, a well chosen shrink factor ensures that the gap sizes are co-prime, in turn requiring astronomically large n to make the algorithm exceed $O(n \log n)$ steps. On random arrays, comb sort is only 2-3 times slower than merge sort. Its short code length relative to its good performance makes it a worthwhile algorithm to remember.

Time Complexity (Worst): $O(n^2)$.

Space Complexity: $O(1)$ auxiliary.

Stable?: No.

```
template<class It, class Compare>
void combsort(It lo, It hi, Compare comp) {
    int gap = hi - lo;
    bool swapped = true;
    while (gap > 1 || swapped) {
        if (gap > 1) {
            gap = gap * 10 / 13;
        }
        swapped = false;
        for (It it = lo; it + gap < hi; ++it) {
            if (comp(*(it + gap), *it)) {
                std::swap(*it, *(it + gap));
                swapped = true;
            }
        }
    }
}

template<class It>
void combsort(It lo, It hi) {
    using T = typename std::iterator_traits<It>::value_type;
    combsort(lo, hi, std::less<T>());
}
```

Radix sort is used to sort integer elements with a constant number of bits in linear time. This implementation only works on ranges pointing to unsigned integer primitives. The elements in the input range do not strictly have to be unsigned types, as long as their values are nonnegative integers.

In this implementation, a power of two is chosen to be the base for the sort so that bitwise operations can be easily used to extract digits. This avoids the need to use modulo and exponentiation, which are much more expensive operations. In practice, it's been demonstrated that 2^8 is the best choice for sorting 32-bit integers (approximately 5 times faster than `std::sort()`, and typically 2-4 times faster than radix sort using any other power of two chosen as the base).

Time Complexity: $O(n \cdot w)$ for n integers of w bits each.

Space Complexity: $O(n + 2^b)$ auxiliary for a radix of b bits, i.e. $O(n)$ for constant b .

```
template<class It>
void radix_sort(It lo, It hi) {
    if (hi - lo < 2) {
        return;
    }
    const int radix_bits = 8;
    const int radix_base = 1 << radix_bits; // e.g. 2^8 = 256
    const int radix_mask = radix_base - 1; // e.g. 2^8 - 1 = 0xFF
    int num_bits = 8 * sizeof(*lo); // 8 bits per byte
    using T = typename std::iterator_traits<It>::value_type;
    std::vector<T> buf(hi - lo);
    for (int pos = 0; pos < num_bits; pos += radix_bits) {
        std::vector<int> count(radix_base);
        for (It it = lo; it != hi; ++it) {
            count[( *it >> pos) & radix_mask]++;
        }
        std::vector<T *> bucket(radix_base);
        T *curr = buf.data();
        for (int i = 0; i < radix_base; curr += count[i++]) {
            bucket[i] = curr;
        }
        for (It it = lo; it != hi; ++it) {
            *bucket[( *it >> pos) & radix_mask]++ = *it;
        }
    }
}
```

```

    }
    std::copy(buf.begin(), buf.end(), lo);
}
}

```

Example Usage

```

#include <cassert>
#include <cstdlib>
#include <ctime>
#include <iomanip>
#include <iostream>
#include <vector>
using namespace std;

template<class It>
void print_range(It lo, It hi) {
    while (lo != hi) {
        cout << *lo++ << " ";
    }
    cout << endl;
}

template<class It>
bool sorted(It lo, It hi) {
    while (++lo != hi) {
        if (*lo < *(lo - 1)) {
            return false;
        }
    }
    return true;
}

bool compare_as_ints(double i, double j) {
    return static_cast<int>(i) < static_cast<int>(j);
}

int main() {
    { // Can be used to sort arrays like std::sort().
        vector<int> a{32, 71, 12, 45, 26, 80, 53, 33};
        quicksort(a.begin(), a.end());
        assert(sorted(a.begin(), a.end()));
    }
    { // STL containers work too.
        vector<int> v{32, 71, 12, 45, 26, 80, 53, 33};
        quicksort(v.begin(), v.end());
        assert(sorted(v.begin(), v.end()));
    }
    { // Reverse iterators work as expected.
        vector<int> v{32, 71, 12, 45, 26, 80, 53, 33};
        heapsort(v.rbegin(), v.rend());
        assert(sorted(v.rbegin(), v.rend()));
    }
    { // We can sort doubles just as well.
        vector<double> v{1.1, -5.0, 6.23, 4.123, 155.2};
        combsort(v.begin(), v.end());
        assert(sorted(v.begin(), v.end()));
    }
    { // Must use radix_sort with unsigned values, but sorting in reverse works!
        vector<unsigned int> v{32, 71, 12, 45, 26, 80, 53, 33};
        radix_sort(v.rbegin(), v.rend());
        assert(sorted(v.rbegin(), v.rend()));
    }

    // Example from: http://www.cplusplus.com/reference/algorithm/stable\_sort
    const vector<double> a{3.14, 1.41, 2.72, 4.67, 1.73, 1.32, 1.62, 2.58};

```



```

{
    vector<double> v(a);
    cout << "mergesort() with default comparisons: ";
    mergesort(v.begin(), v.end());
    print_range(v.begin(), v.end());
}
{
    vector<double> v(a);
    cout << "mergesort() with 'compare_as_ints()': ";
    mergesort(v.begin(), v.end(), compare_as_ints);
    print_range(v.begin(), v.end());
}
cout << "-----" << endl;

vector<int> v, v2;
for (int i = 0; i < 5000000; i++) {
    v.push_back((rand() & 0x7fff) | ((rand() & 0x7fff) << 15));
}
v2 = v;
cout << "Sorting five million integers..." << endl;
cout.precision(3);

#define test(sort_function) \
{ \
    clock_t start = clock(); \
    sort_function(v.begin(), v.end()); \
    double t = static_cast<double>((clock() - start)) / CLOCKS_PER_SEC; \
    cout << setw(14) << left << #sort_function "(): "; \
    cout << fixed << t << "s" << endl; \
    assert(sorted(v.begin(), v.end())); \
    v = v2; \
}
test(std::sort);
test(quicksort);
test(mergesort);
test(heapsort);
test(combsort);
test(radix_sort);
return 0;
}

```

Example Output

```

mergesort() with default comparisons: 1.32 1.41 1.62 1.73 2.58 2.72 3.14 4.67
mergesort() with 'compare_as_ints()': 1.41 1.73 1.32 1.62 2.72 2.58 3.14 4.67
-----
Sorting five million integers...
std::sort(): 0.355s
quicksort(): 0.426s
mergesort(): 1.263s
heapsort(): 1.093s
combsort(): 0.827s
radix_sort(): 0.076s

```

1.1.2 Array Rotation

These functions are equivalent to `std::rotate()`, taking three iterators `lo`, `mid`, and `hi` ($lo \leq mid \leq hi$) to perform a left rotation on the range `[lo, hi)`. After the function call, `[lo, hi)` will consist of the concatenation of the elements originally in `[mid, hi)` + `[lo, mid)`. That is, the range `[lo, hi)` will be rearranged in such a way that the element at `mid` becomes the first element of the new range and the element at `mid - 1` becomes the last element, all while preserving the relative ordering of elements within the two rotated subarrays.

All three versions below achieve the same result using in-place algorithms. Version 1 uses a straightforward swapping algorithm requiring ForwardIterators. Version 2 requires BidirectionalIterators, employing a well-known

trick with three simple inversions. Version 3 requires random-access iterators, applying a juggling algorithm which first divides the range into `gcd(hi - lo, mid - lo)` sets and then rotates the corresponding elements in each set.

Time Complexity: $O(n)$ per call to both functions, where n is the distance between `lo` and `hi`.

Space Complexity: $O(1)$ auxiliary for all versions

```
#include <algorithm>

template<class It>
void rotate1(It lo, It mid, It hi) {
    if (lo == mid || mid == hi) {
        return;
    }
    It next = mid;
    while (lo != next) {
        std::iter_swap(lo++, next++);
        if (next == hi) {
            next = mid;
        } else if (lo == mid) {
            mid = next;
        }
    }
}

template<class It>
void rotate2(It lo, It mid, It hi) {
    if (lo == mid || mid == hi) {
        return;
    }
    std::reverse(lo, mid);
    std::reverse(mid, hi);
    std::reverse(lo, hi);
}

int gcd(int a, int b) {
    return (b == 0) ? a : gcd(b, a % b);
}

template<class It>
void rotate3(It lo, It mid, It hi) {
    if (lo == mid || mid == hi) {
        return;
    }
    int n = hi - lo, jump = mid - lo;
    int g = gcd(jump, n), cycle = n / g;
    for (int i = 0; i < g; i++) {
        int curr = i, next;
        for (int j = 0; j < cycle - 1; j++) {
            next = curr + jump;
            if (next >= n) {
                next -= n;
            }
            std::iter_swap(lo + curr, lo + next);
            curr = next;
        }
    }
}
```

Example Usage

```
#include <algorithm>
#include <cassert>
#include <iostream>
#include <vector>
using namespace std;
```

```

int main() {
    vector<int> v0, v1, v2, v3;
    for (int i = 0; i < 10000; i++) {
        v0.push_back(i);
    }
    v1 = v2 = v3 = v0;
    int mid = 5678;
    std::rotate(v0.begin(), v0.begin() + mid, v0.end());
    rotate1(v1.begin(), v1.begin() + mid, v1.end());
    rotate2(v2.begin(), v2.begin() + mid, v2.end());
    rotate3(v3.begin(), v3.begin() + mid, v3.end());
    assert(v0 == v1 && v0 == v2 && v0 == v3);

    // Example from: http://en.cppreference.com/w/cpp/algorithm/rotate
    vector<int> v{2, 4, 2, 0, 5, 10, 7, 3, 7, 1};
    cout << "before sort: ";
    for (const auto &x : v) {
        cout << x << " ";
    }
    cout << endl;

    // Insertion sort.
    for (auto i = v.begin(); i != v.end(); ++i) {
        rotate1(std::upper_bound(v.begin(), i, *i), i, i + 1);
    }
    cout << "after sort: ";
    for (const auto &x : v) {
        cout << x << " ";
    }
    cout << endl;

    // Simple rotation to the left.
    rotate2(v.begin(), v.begin() + 1, v.end());
    cout << "rotate left: ";
    for (const auto &x : v) {
        cout << x << " ";
    }
    cout << endl;

    // Simple rotation to the right.
    rotate3(v.rbegin(), v.rbegin() + 1, v.rend());
    cout << "rotate right: ";
    for (const auto &x : v) {
        cout << x << " ";
    }
    cout << endl;

    return 0;
}

```

Example Output

```

before sort:  2 4 2 0 5 10 7 3 7 1
after sort:   0 1 2 2 3 4 5 7 7 10
rotate left:  1 2 2 3 4 5 7 7 10 0
rotate right: 0 1 2 2 3 4 5 7 7 10

```

1.1.3 Counting Inversions

The number of inversions for a sequence a is the number of ordered pairs (i, j) such that $i < j$ and $a[i] > a[j]$. This is roughly how “close” an array is to being sorted, but is *not* the minimum number of swaps required to sort the array. If the array is sorted, then the inversion count is 0. If the array is sorted in decreasing order, then the inversion count is maximal. The following two functions are each techniques to efficiently count inversions.

- `inversions(lo, hi)` uses merge sort to return the number of inversions given two random-access iterators as a range `[lo, hi)`. The input range will be sorted after the function call. This requires `operator<` to be defined on the iterators' value type.
- `inversions(a)` uses coordinate compression and a Fenwick tree to return the number of inversions in an integer vector without modifying it.

Time Complexity:

- $O(n \log n)$ per call to `inversion(lo, hi)`, where n is the distance between `lo` and `hi`.
- $O(n \log n)$ per call to `inversions(a)`.

Space Complexity:

- $O(n)$ auxiliary space and $O(\log n)$ stack space for `inversions(lo, hi)`.
- $O(n)$ auxiliary heap space for `inversions(a)`.

```
#include <algorithm>
#include <iterator>
#include <vector>

template<class It>
long long inversions(It lo, It hi) {
    if (hi - lo < 2) {
        return 0;
    }
    It mid = lo + (hi - lo - 1) / 2, a = lo, c = mid + 1;
    long long res = 0;
    res += inversions(lo, mid + 1);
    res += inversions(mid + 1, hi);
    using T = typename std::iterator_traits<It>::value_type;
    std::vector<T> merged;
    while (a <= mid && c < hi) {
        if (*c < *a) {
            merged.push_back(*(c++));
            res += (mid - a) + 1;
        } else {
            merged.push_back(*(a++));
        }
    }
    if (a > mid) {
        for (It it = c; it != hi; ++it) {
            merged.push_back(*it);
        }
    } else {
        for (It it = a; it <= mid; ++it) {
            merged.push_back(*it);
        }
    }
    for (It it = lo; it != hi; ++it) {
        *it = merged[it - lo];
    }
    return res;
}

long long inversions(const std::vector<int> &a) {
    int n = a.size();
    std::vector<int> values(a.begin(), a.end());
    std::sort(values.begin(), values.end());
    values.erase(std::unique(values.begin(), values.end()), values.end());
    std::vector<int> bit(values.size() + 1, 0);
    long long res = 0;
    for (int i = n - 1; i >= 0; i--) {
        int id = std::lower_bound(values.begin(), values.end(), a[i]) - values.begin() + 1;
        for (int j = id - 1; j > 0; j -= j & -j) {
            res += bit[j];
        }
        for (int j = id; j < static_cast<int>(bit.size()); j += j & -j) {
```

```

        bit[j]++;
    }
}
return res;
}

```

Example Usage

```

#include <cassert>

int main() {
    {
        std::vector<int> a{6, 9, 1, 14, 8, 12, 3, 2};
        assert(inversions(a.begin(), a.end()) == 16);
    }
    {
        std::vector<int> a{6, 9, 1, 14, 8, 12, 3, 2};
        assert(inversions(a) == 16);
    }
    return 0;
}

```

1.1.4 Coordinate Compression

Given two ForwardIterators as a range `[lo, hi)` of n numerical elements, reassign each element in the range to an integer in $[0, k)$, where k is the number of distinct elements in the original range, while preserving the initial relative ordering of elements. That is, if a is the array of original values and b is the array of compressed values, then every pair of indices i, j in $[0, n)$ shall satisfy $a[i] < a[j]$ if and only if $b[i] < b[j]$.

Both implementations below require `operator<` to be defined on the iterator's value type. Version 1 performs the compression by sorting the array, removing duplicates, and binary searching for the position of each original value. Version 2 achieves the same result by inserting all values in a balanced binary search tree (`std::map`) which automatically removes duplicate values and supports efficient lookups of the compressed values.

Time Complexity: $O(n \log n)$ per call to either function, where n is the distance between `lo` and `hi`.

Space Complexity: $O(n)$ auxiliary heap space.

```

#include <algorithm>
#include <iterator>
#include <map>
#include <vector>

template<class It>
void compress1(It lo, It hi) {
    using T = typename std::iterator_traits<It>::value_type;
    std::vector<T> v(lo, hi);
    std::sort(v.begin(), v.end());
    v.resize(std::unique(v.begin(), v.end()) - v.begin());
    for (It it = lo; it != hi; ++it) {
        *it = static_cast<int>((std::lower_bound(v.begin(), v.end(), *it) - v.begin()));
    }
}

template<class It>
void compress2(It lo, It hi) {
    using T = typename std::iterator_traits<It>::value_type;
    std::map<T, int> m;
    for (It it = lo; it != hi; ++it) {
        m[*it] = 0;
    }
    int id = 0;
}

```

```

for (auto &[key, val] : m) {
    val = id++;
}
for (It it = lo; it != hi; ++it) {
    *it = m[*it];
}
}

```

Example Usage

```

#include <iostream>
using namespace std;

template<class It>
void print_range(It lo, It hi) {
    while (lo != hi) {
        cout << *lo++ << " ";
    }
    cout << endl;
}

int main() {
    {
        vector<int> a{1, 30, 30, 7, 9, 8, 99, 99};
        compress1(a.begin(), a.end());
        print_range(a.begin(), a.end());
    }
    {
        vector<int> a{1, 30, 30, 7, 9, 8, 99, 99};
        compress2(a.begin(), a.end());
        print_range(a.begin(), a.end());
    }
    { // Non-integral types work too, as long as ints can be assigned to them.
        vector<double> a{0.5, -1.0, 3, -1.0, 20, 0.5};
        compress1(a.begin(), a.end());
        print_range(a.begin(), a.end());
    }
    return 0;
}

```

Example Output

```

0 4 4 1 3 2 5 5
0 4 4 1 3 2 5 5
1 0 2 0 3 1

```

1.1.5 Selection (Quickselect)

`nth_element2()` is equivalent to `std::nth_element()`, taking random-access iterators `lo`, `nth`, and `hi` as the range `[lo, hi)` to be partially sorted. The values in `[lo, hi)` are rearranged such that the value pointed to by `nth` is the element that would be there if the range were sorted. Furthermore, the range is partitioned such that no value in `[lo, nth)` compares greater than the value pointed to by `nth` and no value in `(nth, hi)` compares less. This implementation requires `operator <` to be defined on the iterator's value type.

The pivot is chosen uniformly at random, and the partition step groups equal keys together so arrays with many duplicate values remain efficient.

Time Complexity: $O(n)$ on average per call to `nth_element2()`, where n is the distance between `lo` and `hi`.

Space Complexity: $O(1)$ auxiliary.

```

#include <algorithm>

```

```

#include <iterator>
#include <random>

template<class It>
void nth_element2(It lo, It nth, It hi) {
    static std::mt19937 rng(std::random_device{}());
    while (hi - lo > 1) {
        std::uniform_int_distribution<int> dist(0, hi - lo - 1);
        auto pivot = *(lo + dist(rng));
        It lt = lo, i = lo, gt = hi;
        while (i != gt) {
            if (*i < pivot) {
                std::iter_swap(lt++, i++);
            } else if (pivot < *i) {
                std::iter_swap(i, --gt);
            } else {
                ++i;
            }
        }
        if (nth < lt) {
            hi = lt;
        } else if (nth >= gt) {
            lo = gt;
        } else {
            return;
        }
    }
}

```

Example Usage

```

#include <cassert>
#include <vector>
using namespace std;

int main() {
    vector<int> a{5, 6, 4, 3, 2, 6, 7, 9, 3};
    int n = a.size();
    nth_element2(a.begin(), a.begin() + n / 2, a.end());
    assert(a[n / 2] == 5);
    // Values left of the median are <= it and values right are >= it (the exact
    // order within each side is randomized, since the pivot is chosen at random).
    for (int i = 0; i < n / 2; i++) {
        assert(a[i] <= a[n / 2]);
    }
    for (int i = n / 2 + 1; i < n; i++) {
        assert(a[i] >= a[n / 2]);
    }
    return 0;
}

```

1.1.6 3-Way Partitioning (Dutch National Flag)

Partitions a range into three consecutive groups according to a pivot value: elements less than the pivot, elements equal to the pivot, and elements greater than the pivot. This is the Dutch National Flag algorithm, and is the core partitioning step used in three-way quicksort and arrays with many duplicate keys.

- `partition_three_way(lo, hi, pivot)` rearranges the range `[lo, hi)` in-place and returns iterators `(mid_lo, mid_hi)`, where `[lo, mid_lo)` is less than `pivot`, `[mid_lo, mid_hi)` is equal to `pivot`, and `[mid_hi, hi)` is greater than `pivot`.
- `sort_012(lo, hi)` sorts a range whose values are all 0, 1, or 2.

Time Complexity: $O(n)$ per call, where n is the distance between `lo` and `hi`.

Space Complexity: $O(1)$ auxiliary.

```
#include <algorithm>
#include <utility>

template<class It, class T>
std::pair<It, It> partition_three_way(It lo, It hi, const T &pivot) {
    It lt = lo, i = lo, gt = hi;
    while (i != gt) {
        if (*i < pivot) {
            std::iter_swap(lt++, i++);
        } else if (pivot < *i) {
            std::iter_swap(i, --gt);
        } else {
            ++i;
        }
    }
    return {lt, gt};
}

template<class It>
void sort_012(It lo, It hi) {
    partition_three_way(lo, hi, 1);
}
```

Example Usage

```
#include <cassert>
#include <vector>
using namespace std;

int main() {
    vector<int> a{2, 0, 1, 2, 1, 0, 1};
    sort_012(a.begin(), a.end());
    for (int i = 0; i < 7; i++) {
        assert(a[i] == (i < 2 ? 0 : i < 5 ? 1 : 2));
    }

    vector<int> b{4, 2, 5, 3, 3, 1};
    auto [mid_lo, mid_hi] = partition_three_way(b.begin(), b.end(), 3);
    for (auto it = b.begin(); it != mid_lo; ++it) {
        assert(*it < 3);
    }
    for (auto it = mid_lo; it != mid_hi; ++it) {
        assert(*it == 3);
    }
    for (auto it = mid_hi; it != b.end(); ++it) {
        assert(*it > 3);
    }
    return 0;
}
```


1.2 Arrays and Dynamic Programming

1.2.1 Prefix Sums

Precomputes prefix sums so range-sum queries can be answered in constant time. This is one of the most common array preprocessing tools, and also appears as a building block for subarray sums, difference arrays, and two-dimensional grids.

- `prefix_sums(a)` returns array `pref` with `pref[0] = 0` and `pref[i + 1] = a[0] + ... + a[i]`.
- `range_sum(pref, lo, hi)` returns the sum of the half-open range `[lo, hi)`.
- `prefix_sums_2d(a)` returns a two-dimensional prefix sum table for matrix `a`.
- `rectangle_sum(pref, r1, c1, r2, c2)` returns the sum of rows `[r1, r2)` and columns `[c1, c2)`.

Time Complexity:

- $O(n)$ per call to `prefix_sums(a)`, where n is the array size.
- $O(r \cdot c)$ per call to `prefix_sums_2d(a)`, where r and c are matrix dimensions.
- $O(1)$ per range or rectangle query.

Space Complexity:

- $O(n)$ for one-dimensional prefix sums.
- $O(r \cdot c)$ for two-dimensional prefix sums.

```
#include <vector>

std::vector<long long> prefix_sums(const std::vector<int> &a) {
    std::vector<long long> pref(a.size() + 1, 0);
    for (int i = 0; i < static_cast<int>(a.size()); i++) {
        pref[i + 1] = pref[i] + a[i];
    }
    return pref;
}

long long range_sum(const std::vector<long long> &pref, int lo, int hi) {
    return pref[hi] - pref[lo];
}

std::vector<std::vector<long long>> prefix_sums_2d(const std::vector<std::vector<int>> &a) {
    int rows = a.size(), cols = rows == 0 ? 0 : a[0].size();
    std::vector<std::vector<long long>> pref(rows + 1, std::vector<long long>(cols + 1, 0));
    for (int r = 0; r < rows; r++) {
        for (int c = 0; c < cols; c++) {
            pref[r + 1][c + 1] = a[r][c] + pref[r][c + 1] + pref[r + 1][c] - pref[r][c];
        }
    }
    return pref;
}

long long rectangle_sum(
    const std::vector<std::vector<long long>> &pref, int r1, int c1, int r2, int c2
) {
    return pref[r2][c2] - pref[r1][c2] - pref[r2][c1] + pref[r1][c1];
}
```

Example Usage

```
#include <cassert>
using namespace std;

int main() {
    vector<int> a{3, -1, 4, 1, 5};
    vector<long long> pref = prefix_sums(a);
```

```

assert(range_sum(pref, 0, 5) == 12);
assert(range_sum(pref, 1, 4) == 4);

vector<vector<int>> grid{{1, 2, 3}, {4, 5, 6}};
vector<vector<long long>> ps = prefix_sums_2d(grid);
assert(rectangle_sum(ps, 0, 1, 2, 3) == 16);
return 0;
}

```

1.2.2 Difference Array

Applies many range-add updates to an array in linear total time using a difference array. Instead of immediately adding `delta` to every element of `[lo, hi)`, add `delta` at `diff[lo]` and subtract it at `diff[hi]`; a final prefix sum reconstructs the updated values.

- `DifferenceArray(n)` constructs an initially zero array of size `n`.
- `add(lo, hi, delta)` adds `delta` to every position in the half-open range `[lo, hi)`.
- `build()` returns the final array after all range updates.

Time Complexity:

- $O(1)$ per call to `add(lo, hi, delta)`.
- $O(n)$ per call to `build()`.

Space Complexity: $O(n)$ for the difference array.

```

#include <vector>

class DifferenceArray {
    std::vector<long long> diff;

public:
    explicit DifferenceArray(int n) : diff(n + 1, 0) {}

    void add(int lo, int hi, long long delta) {
        diff[lo] += delta;
        diff[hi] -= delta;
    }

    std::vector<long long> build() const {
        std::vector<long long> res(diff.size() - 1);
        long long cur = 0;
        for (int i = 0; i < static_cast<int>(res.size()); i++) {
            cur += diff[i];
            res[i] = cur;
        }
        return res;
    }
};

```

Example Usage

```

#include <cassert>
using namespace std;

int main() {
    DifferenceArray d(5);
    d.add(0, 3, 2);
    d.add(1, 5, 4);
    d.add(2, 4, -1);
    vector<long long> a = d.build();
    vector<long long> expected{2, 6, 5, 3, 4};
}

```

```

    for (int i = 0; i < static_cast<int>(expected.size()); i++) {
        assert(a[i] == expected[i]);
    }
    return 0;
}

```

1.2.3 Longest Increasing Subsequence

Given two random-access iterators `lo` and `hi` specifying a range `[lo, hi)`, determine a longest subsequence of the range such that all of its elements are in strictly ascending order. This implementation requires `operator<` to be defined on the iterator's value type. The subsequence is not necessarily contiguous or unique, so only one such answer will be found. The answer is computed using binary search and dynamic programming.

Time Complexity: $O(n \log n)$ per call to `longest_increasing_subsequence()`, where n is the distance between `lo` and `hi`.

Space Complexity: $O(n)$ auxiliary heap space for `longest_increasing_subsequence()`.

```

#include <iterator>
#include <vector>

template<class It>
auto longest_increasing_subsequence(It lo, It hi) {
    using T = typename std::iterator_traits<It>::value_type;
    int len = 0, n = hi - lo;
    if (n == 0) {
        return std::vector<T>();
    }
    // tail[i] = index (into [lo, hi)) of the last element of the best known LIS of length i+1.
    // prev[i] = index of the predecessor of element i in its LIS (or -1 if first).
    std::vector<int> prev(n), tail(n);
    for (int i = 0; i < n; i++) {
        int left = -1, right = len;
        while (right - left > 1) {
            int mid = left + (right - left) / 2;
            if (*(lo + tail[mid]) < *(lo + i)) {
                left = mid;
            } else {
                right = mid;
            }
        }
        if (len < right + 1) {
            len = right + 1;
        }
        prev[i] = right > 0 ? tail[right - 1] : -1;
        tail[right] = i;
    }
    std::vector<T> res(len);
    for (int i = tail[len - 1]; i != -1; i = prev[i]) {
        res[--len] = *(lo + i);
    }
    return res;
}

```

Example Usage

```

#include <iostream>
using namespace std;

template<class It>
void print_range(It lo, It hi) {
    while (lo != hi) {

```

```

        cout << *lo++ << " ";
    }
    cout << endl;
}

int main() {
    vector<int> a{-2, -5, 1, 9, 10, 8, 11, 10, 13, 11};
    vector<int> res = longest_increasing_subsequence(a.begin(), a.end());
    print_range(res.begin(), res.end());
    return 0;
}

```

Example Output

```
-5 1 9 10 11 13
```

1.2.4 Subset Sum (Meet-in-the-Middle)

Given random-access iterators `lo` and `hi` specifying a range `[lo, hi)` of integers, return the maximum sum of any subset of the range that is at most a given integer `v`. This is a common meet-in-the-middle version of subset sum: enumerate subset sums for each half, sort one side, and binary search for the best compatible partner. Note that 64-bit integers are used in intermediate calculations to avoid overflow.

Time Complexity: $O(2^{n/2})$ per call to `max_subset_sum_at_most()`, where n is the distance between `lo` and `hi`.

Space Complexity: $O(2^{n/2})$ auxiliary heap space.

```

#include <algorithm>
#include <climits>
#include <vector>

template<class It>
long long max_subset_sum_at_most(It lo, It hi, long long v) {
    int n = hi - lo;
    long long llen = 1LL << (n / 2), hlen = 1LL << (n - n / 2);
    std::vector<long long> lsum(llen), hsum(hlen);
    for (long long mask = 1; mask < llen; mask++) {
        int bit = __builtin_ctzll(mask);
        lsum[mask] = lsum[mask ^ (1LL << bit)] + *(lo + bit);
    }
    for (long long mask = 1; mask < hlen; mask++) {
        int bit = __builtin_ctzll(mask);
        hsum[mask] = hsum[mask ^ (1LL << bit)] + *(lo + n / 2 + bit);
    }
    std::sort(hsum.begin(), hsum.end());
    long long res = LLONG_MIN;
    for (long long x : lsum) {
        auto it = std::upper_bound(hsum.begin(), hsum.end(), v - x);
        if (it != hsum.begin()) {
            res = std::max(res, x + *--it);
        }
    }
    return res;
}

```

Example Usage

```

#include <cassert>

int main() {
    std::vector<int> a{9, 1, 5, 0, 1, 11, 5};
    assert(max_subset_sum_at_most(a.begin(), a.end(), 8) == 7);
    std::vector<int> b{-7, -3, -2, 5, 8};
}

```

```

    assert(max_subset_sum_at_most(b.begin(), b.end(), 0) == 0);
    return 0;
}

```

1.2.5 Knapsack (0-1)

Solves the 0/1 knapsack problem: given items with nonnegative integer weights and values, choose each item at most once to maximize total value without exceeding a capacity limit.

The one-dimensional dynamic programming array `dp[w]` stores the best value that can be achieved with total weight at most `w` after processing some prefix of the items. Iterating weights in decreasing order is what enforces the 0/1 constraint; it prevents the same item from being reused during the current item update.

- `knapsack_01(weight, value, capacity)` returns the maximum value achievable with total weight at most `capacity`.
- `weight[i]` and `value[i]` are the weight and value of item `i`.
- All weights must be nonnegative integers. Items with weight `0` are allowed and can each be taken once.

Time Complexity: $O(nW)$, where n is the number of items and W is `capacity`.

Space Complexity: $O(W)$ auxiliary heap space.

```

#include <algorithm>
#include <cassert>
#include <vector>

long long knapsack_01(
    const std::vector<int> &weight, const std::vector<long long> &value, int capacity
) {
    int n = weight.size();
    std::vector<long long> dp(capacity + 1, 0);
    for (int i = 0; i < n; i++) {
        for (int w = capacity; w >= weight[i]; w--) {
            dp[w] = std::max(dp[w], dp[w - weight[i]] + value[i]);
        }
    }
    return dp[capacity];
}

```

Example Usage

```

#include <cassert>
using namespace std;

int main() {
    vector<int> weight{3, 4, 5, 9};
    vector<long long> value{4, 5, 7, 10};

    assert(knapsack_01(weight, value, 8) == 11); // Items of weight 3 and 5.
    assert(knapsack_01(weight, value, 10) == 12); // Items of weight 4 and 5.
    return 0;
}

```

1.2.6 Knapsack (Unbounded and Coin Change)

Solves two common unbounded knapsack variants, where each item may be used any number of times. In the maximum-value version, each item has a weight and value, and the goal is to maximize total value subject to a

capacity limit. In the coin change version, each coin denomination may be used any number of times, and the goal is to count unordered ways to form a target sum.

For unbounded maximum-value knapsack, weights are iterated in increasing order so the current item may be reused. For unordered coin change, coins are the outer loop so each multiset of coins is counted once, independent of ordering.

- `unbounded_knapsack(weight, value, capacity)` returns the maximum value achievable with total weight at most `capacity`.
- `count_coin_change(coins, target)` returns the number of unordered ways to make sum `target` using the given coin denominations.
- All capacities must be nonnegative integers. Weights and coin denominations should be positive to avoid infinitely many uses of a zero-weight item.

Time Complexity:

- $O(n \cdot W)$ for `unbounded_knapsack(weight, value, capacity)`, where n is the number of items and W is `capacity`.
- $O(c \cdot t)$ for `count_coin_change(coins, target)`, where c is the number of coin denominations and t is `target`.

Space Complexity:

- $O(W)$ auxiliary heap space for `unbounded_knapsack(weight, value, capacity)`.
- $O(t)$ auxiliary heap space for `count_coin_change(coins, target)`.

```
#include <algorithm>
#include <vector>

long long unbounded_knapsack(
    const std::vector<int> &weight, const std::vector<long long> &value, int capacity
) {
    int n = weight.size();
    std::vector<long long> dp(capacity + 1, 0);
    for (int i = 0; i < n; i++) {
        for (int w = weight[i]; w <= capacity; w++) {
            dp[w] = std::max(dp[w], dp[w - weight[i]] + value[i]);
        }
    }
    return dp[capacity];
}

long long count_coin_change(const std::vector<int> &coins, int target) {
    std::vector<long long> dp(target + 1, 0);
    dp[0] = 1;
    for (int coin : coins) {
        for (int sum = coin; sum <= target; sum++) {
            dp[sum] += dp[sum - coin];
        }
    }
    return dp[target];
}
```

Example Usage

```
#include <cassert>
using namespace std;

int main() {
    vector<int> weight{3, 4, 5};
    vector<long long> value{4, 5, 7};
    assert(unbounded_knapsack(weight, value, 10) == 14); // Two weight-5 items.

    vector<int> coins{1, 2, 5};
    assert(count_coin_change(coins, 5) == 4); // 5, 2+2+1, 2+1+1+1, 1*5.
    return 0;
}
```

1.2.7 Largest Zero Submatrix

Given a rectangular matrix with n rows and m columns consisting of only 0's and 1's as a two-dimensional vector of bool, return the area of the largest rectangular submatrix consisting of only 0's. This solution uses a reduction to the problem of finding the maximum rectangular area under a histogram, which is efficiently solved using a stack algorithm.

Time Complexity: $O(n \cdot m)$ per call to `max_zero_submatrix()`, where n is the number of rows and m is the number of columns in the matrix.

Space Complexity: $O(m)$ auxiliary heap space, where m is the number of columns in the matrix.

```
#include <algorithm>
#include <stack>
#include <vector>

int max_zero_submatrix(const std::vector<std::vector<bool>> &matrix) {
    if (matrix.empty()) {
        return 0;
    }
    int n = matrix.size(), m = matrix[0].size(), res = 0;
    // last1[c] = row index of the most recent 1 in column c (-1 if none yet).
    // left_wall[c] = nearest column left of c whose last1 value >= last1[c] (exclusive left bound).
    // right_wall[c] = nearest column right of c whose last1 value >= last1[c] (exclusive right bound).
    std::vector<int> last1(m, -1), left_wall(m), right_wall(m);
    for (int r = 0; r < n; r++) {
        for (int c = 0; c < m; c++) {
            if (matrix[r][c]) {
                last1[c] = r;
            }
        }
        std::stack<int> s;
        for (int c = 0; c < m; c++) {
            while (!s.empty() && last1[s.top()] <= last1[c]) {
                s.pop();
            }
            left_wall[c] = s.empty() ? -1 : s.top();
            s.push(c);
        }
        while (!s.empty()) {
            s.pop();
        }
        for (int c = m - 1; c >= 0; c--) {
            while (!s.empty() && last1[s.top()] <= last1[c]) {
                s.pop();
            }
            right_wall[c] = s.empty() ? m : s.top();
            s.push(c);
        }
        for (int j = 0; j < m; j++) {
            res = std::max(res, (r - last1[j]) * (right_wall[j] - left_wall[j] - 1));
        }
    }
    return res;
}
```

Example Usage

```
#include <cassert>
using namespace std;

int main() {
    vector<vector<bool>> matrix{
        {1, 0, 1, 1, 0, 0},
        {1, 0, 0, 1, 0, 0},
```

```

    {0, 0, 0, 0, 0, 1},
    {1, 0, 0, 1, 0, 0},
    {1, 0, 1, 0, 0, 1}
};
assert(max_zero_submatrix(matrix) == 6);
return 0;
}

```

1.2.8 Minimum Excluded Value (MEX)

Computes the minimum excluded nonnegative integer, commonly called the MEX. The MEX of a set is the smallest integer $x \geq 0$ that does not appear in the set. It appears frequently in array problems, game theory with Grundy numbers, and dynamic programming states.

- `mex(lo, hi)` returns the MEX of the values in `[lo, hi)`.
- `DynamicMex(n)` maintains counts of values in `[0, n]`, enough to track the MEX of a multiset of at most `n` relevant nonnegative values.
- `add(x)` inserts one copy of value `x` if $0 \leq x \leq n$.
- `remove(x)` removes one copy of value `x` if present.
- `get()` returns the current MEX.

Time Complexity:

- $O(n)$ per call to `mex(lo, hi)`, where n is the number of values.
- $O(\log n)$ per call to `add(x)`, `remove(x)`, and `get()` for `DynamicMex`.

Space Complexity: $O(n)$ auxiliary heap space for both approaches.

```

#include <set>
#include <vector>

template<class It>
int mex(It lo, It hi) {
    int n = hi - lo;
    std::vector<bool> seen(n + 1, false);
    for (It it = lo; it != hi; ++it) {
        if (0 <= *it && *it <= n) {
            seen[*it] = true;
        }
    }
    for (int x = 0; x <= n; x++) {
        if (!seen[x]) {
            return x;
        }
    }
    return n + 1;
}

class DynamicMex {
    std::vector<int> count;
    std::set<int> missing;

public:
    explicit DynamicMex(int n) : count(n + 1, 0) {
        for (int x = 0; x <= n; x++) {
            missing.insert(x);
        }
    }

    void add(int x) {
        if (0 <= x && x < static_cast<int>(count.size()) && count[x]++ == 0) {
            missing.erase(x);
        }
    }
}

```



```

}

void remove(int x) {
    if (0 <= x && x < static_cast<int>(count.size()) && count[x] > 0 && --count[x] == 0) {
        missing.insert(x);
    }
}

int get() const { return *missing.begin(); }
};

```

Example Usage

```

#include <cassert>
using namespace std;

int main() {
    vector<int> a{0, 1, 4, 2, 1};
    assert(mex(a.begin(), a.end()) == 3);

    DynamicMex m(5);
    m.add(0);
    m.add(1);
    m.add(3);
    assert(m.get() == 2);
    m.add(2);
    assert(m.get() == 4);
    m.remove(1);
    assert(m.get() == 1);
    return 0;
}

```

1.2.9 Two Pointers and Sliding Window

Uses two moving indices to process subarrays in linear time. This technique is most useful when expanding the right endpoint of a window monotonically and shrinking the left endpoint monotonically preserves enough state to test a condition or update an answer.

The examples below cover two common patterns: finding the shortest subarray with sum at least a target when all values are nonnegative, and finding the longest subarray with at most k distinct values.

- `min_length_at_least(a, target)` returns the minimum length of a contiguous subarray with sum at least `target`, or -1 if none exists. Values in `a` must be nonnegative.
- `longest_at_most_k_distinct(a, k)` returns the maximum length of a contiguous subarray containing at most k distinct values.

Time Complexity: $O(n)$ expected per call, where n is the array size.

Space Complexity:

- $O(1)$ auxiliary for `min_length_at_least(a, target)`.
- $O(k)$ auxiliary heap space for `longest_at_most_k_distinct(a, k)`.

```

#include <algorithm>
#include <unordered_map>
#include <vector>

int min_length_at_least(const std::vector<int> &a, long long target) {
    int best = static_cast<int>(a.size()) + 1;
    long long sum = 0;
    for (int l = 0, r = 0; r < static_cast<int>(a.size()); r++) {
        sum += a[r];

```

```

    while (sum >= target) {
        best = std::min(best, r - l + 1);
        sum -= a[l++];
    }
}
return best == static_cast<int>(a.size()) + 1 ? -1 : best;
}

int longest_at_most_k_distinct(const std::vector<int> &a, int k) {
    if (k <= 0) {
        return 0;
    }
    std::unordered_map<int, int> count;
    int best = 0;
    for (int l = 0, r = 0; r < static_cast<int>(a.size()); r++) {
        count[a[r]]++;
        while (static_cast<int>(count.size()) > k) {
            if (--count[a[l]] == 0) {
                count.erase(a[l]);
            }
            l++;
        }
        best = std::max(best, r - l + 1);
    }
    return best;
}

```

Example Usage

```

#include <cassert>
using namespace std;

int main() {
    vector<int> a{2, 3, 1, 2, 4, 3};
    assert(min_length_at_least(a, 7) == 2); // [4, 3].

    vector<int> b{1, 2, 1, 3, 4, 3, 5};
    assert(longest_at_most_k_distinct(b, 2) == 3); // [1, 2, 1].
    return 0;
}

```

1.3 Greedy and Scheduling

1.3.1 Interval Scheduling

Selects the maximum number of non-overlapping intervals using the classic greedy algorithm that sorts by earliest finish time. This is also known as activity selection.

Intervals are represented as half-open ranges `[start, finish)`, so two intervals are compatible if the next interval's `start` is at least the previous interval's `finish`.

- `interval_scheduling(intervals)` returns one maximum-size compatible subset of intervals, in the order they are selected.
- `Interval` stores `start`, `finish`, and the original `id`.

Time Complexity: $O(n \log n)$ per call due to sorting, where n is the number of intervals.

Space Complexity: $O(n)$ auxiliary heap space for sorting and the answer.

```

#include <algorithm>

```

```

#include <limits>
#include <vector>

struct Interval {
    int start, finish, id;
};

std::vector<Interval> interval_scheduling(std::vector<Interval> intervals) {
    std::sort(intervals.begin(), intervals.end(), [](const Interval &a, const Interval &b) {
        return a.finish != b.finish ? a.finish < b.finish : a.start < b.start;
    });
    std::vector<Interval> res;
    int last_finish = std::numeric_limits<int>::min();
    for (const auto &iv : intervals) {
        if (iv.start >= last_finish) {
            res.push_back(iv);
            last_finish = iv.finish;
        }
    }
    return res;
}

```

Example Usage

```

#include <cassert>
using namespace std;

int main() {
    vector<Interval> intervals{{1, 4, 0}, {3, 5, 1}, {0, 6, 2}, {5, 7, 3}, {8, 9, 4}};
    vector<Interval> chosen = interval_scheduling(intervals);
    assert(chosen.size() == 3);
    assert(chosen[0].id == 0);
    assert(chosen[1].id == 3);
    assert(chosen[2].id == 4);
    return 0;
}

```

1.3.2 Weighted Interval Scheduling

Selects a maximum-weight subset of non-overlapping intervals using dynamic programming and binary search. Unlike unweighted interval scheduling, the earliest finish-time greedy choice is not sufficient when intervals have weights.

Intervals are represented as half-open ranges `[start, finish)`, so two intervals are compatible if the next interval's `start` is at least the previous interval's `finish`.

- `weighted_interval_scheduling(intervals)` returns the maximum total weight of a compatible subset.
- `WeightedInterval` stores `start`, `finish`, and `weight`.

Time Complexity: $O(n \log n)$ per call due to sorting and binary searching compatible intervals.

Space Complexity: $O(n)$ auxiliary heap space.

```

#include <algorithm>
#include <vector>

struct WeightedInterval {
    int start, finish;
    long long weight;
};

long long weighted_interval_scheduling(std::vector<WeightedInterval> intervals) {

```

```

std::sort(
    intervals.begin(), intervals.end(), [](const WeightedInterval &a, const WeightedInterval &b) {
        return a.finish != b.finish ? a.finish < b.finish : a.start < b.start;
    }
);
int n = intervals.size();
std::vector<int> finish;
finish.reserve(n);
for (const auto &iv : intervals) {
    finish.push_back(iv.finish);
}
std::vector<long long> dp(n + 1, 0);
for (int i = 1; i <= n; i++) {
    int j = std::upper_bound(finish.begin(), finish.end(), intervals[i - 1].start) - finish.begin();
    dp[i] = std::max(dp[i - 1], dp[j] + intervals[i - 1].weight);
}
return dp[n];
}

```

Example Usage

```

#include <cassert>
using namespace std;

int main() {
    vector<WeightedInterval> intervals{{1, 3, 5}, {2, 5, 6}, {4, 6, 5}, {6, 7, 4}, {5, 8, 11}};
    assert(weighted_interval_scheduling(intervals) == 17);
    return 0;
}

```

1.3.3 Interval Partitioning

Assigns intervals to the minimum number of groups so that no two intervals in the same group overlap. This is the classic interval partitioning problem, also known as finding the minimum number of rooms needed for meetings.

The greedy algorithm sorts intervals by start time and reuses the group whose current finish time is earliest whenever possible. Otherwise, it creates a new group.

- `interval_partitioning(intervals)` returns `room[id]`, the assigned room for each original interval id.
- The number of rooms used is `1 + max(room[id])`, or 0 if there are no intervals.

Time Complexity: $O(n \log n)$ per call due to sorting and priority queue operations.

Space Complexity: $O(n)$ auxiliary heap space.

```

#include <algorithm>
#include <functional>
#include <queue>
#include <utility>
#include <vector>

struct PartitionInterval {
    int start, finish, id;
};

std::vector<int> interval_partitioning(std::vector<PartitionInterval> intervals) {
    std::sort(
        intervals.begin(), intervals.end(),
        [](const PartitionInterval &a, const PartitionInterval &b) {
            return a.start != b.start ? a.start < b.start : a.finish < b.finish;
        }
    );
    std::vector<int> room(intervals.size());
}

```

```

std::priority_queue<
    std::pair<int, int>, std::vector<std::pair<int, int>>, std::greater<std::pair<int, int>>>
    > pq;
int rooms = 0;
for (const auto &iv : intervals) {
    int id = iv.id;
    if (!pq.empty() && pq.top().first <= iv.start) {
        int r = pq.top().second;
        pq.pop();
        room[id] = r;
    } else {
        room[id] = rooms++;
    }
    pq.emplace(iv.finish, room[id]);
}
return room;
}

```

Example Usage

```

#include <algorithm>
#include <cassert>
using namespace std;

int main() {
    vector<PartitionInterval> intervals{{0, 30, 0}, {5, 10, 1}, {15, 20, 2}};
    vector<int> room = interval_partitioning(intervals);
    assert(1 + *max_element(room.begin(), room.end()) == 2);
    assert(room[1] == room[2]);
    assert(room[0] != room[1]);
    return 0;
}

```

1.3.4 Deadline Scheduling

Schedules unit-time jobs with deadlines and profits to maximize total profit. Each job takes one time slot and must be scheduled at or before its deadline. The greedy algorithm processes jobs in decreasing profit order and places each chosen job in the latest available slot before its deadline.

This is a compact scheduling pattern that combines sorting with a disjoint-set forest over available time slots.

- `schedule_deadline_jobs(jobs)` returns the maximum total profit.
- `Job` stores `deadline` and `profit`. Deadlines are positive integer time slots.

Time Complexity: $O(n \log n)$ per call due to sorting, with near-constant disjoint-set operations.

Space Complexity: $O(n)$ auxiliary heap space.

```

#include <algorithm>
#include <numeric>
#include <vector>

struct Job {
    int deadline;
    long long profit;
};

class SlotDSU {
    std::vector<int> root;

public:
    explicit SlotDSU(int n) : root(n + 1) { std::iota(root.begin(), root.end(), 0); }
}

```

```

int find_root(int u) {
    if (root[u] != u) {
        root[u] = find_root(root[u]);
    }
    return root[u];
}

void occupy(int u) { root[u] = find_root(u - 1); }

};

long long schedule_deadline_jobs(std::vector<Job> jobs) {
    std::sort(jobs.begin(), jobs.end(), [](const Job &a, const Job &b) {
        return a.profit != b.profit ? a.profit > b.profit : a.deadline < b.deadline;
    });
    int max_deadline = 0;
    for (const auto &j : jobs) {
        max_deadline = std::max(max_deadline, j.deadline);
    }
    SlotDSU slots(max_deadline);
    long long res = 0;
    for (const auto &j : jobs) {
        int slot = slots.find_root(j.deadline);
        if (slot > 0) {
            res += j.profit;
            slots.occupy(slot);
        }
    }
    return res;
}

```

Example Usage

```

#include <cassert>
using namespace std;

int main() {
    vector<Job> jobs{{2, 100}, {1, 19}, {2, 27}, {1, 25}, {3, 15}};
    assert(schedule_deadline_jobs(jobs) == 142);
    return 0;
}

```

1.3.5 Minimizing Late Jobs (Moore-Hodgson)

Selects a maximum-size subset of jobs that can all be completed by their deadlines. Each job has a processing time and deadline, and all jobs are available at time 0. The Moore-Hodgson algorithm sorts by deadline and keeps the accepted jobs in a max-heap by processing time; whenever the schedule becomes late, it removes the longest accepted job.

- `maximize_on_time_jobs(jobs)` returns the maximum number of jobs that can finish by their deadlines.
- `TimedJob` stores `duration` and `deadline`.

Time Complexity: $O(n \log n)$ per call due to sorting and heap operations.

Space Complexity: $O(n)$ auxiliary heap space.

```

#include <algorithm>
#include <queue>
#include <vector>

struct TimedJob {
    int duration, deadline;
};

```

```
int maximize_on_time_jobs(std::vector<TimedJob> jobs) {
    std::sort(jobs.begin(), jobs.end(), [](const TimedJob &a, const TimedJob &b) {
        return a.deadline != b.deadline ? a.deadline < b.deadline : a.duration < b.duration;
    });
    std::priority_queue<int> durations;
    int time = 0;
    for (const auto &j : jobs) {
        time += j.duration;
        durations.push(j.duration);
        if (time > j.deadline) {
            time -= durations.top();
            durations.pop();
        }
    }
    return static_cast<int>(durations.size());
}
```

Example Usage

```
#include <cassert>
using namespace std;

int main() {
    vector<TimedJob> jobs{{3, 4}, {2, 3}, {1, 2}, {2, 7}};
    assert(maximize_on_time_jobs(jobs) == 3);
    return 0;
}
```

1.4 Streaming and Randomized Algorithms

1.4.1 Fisher-Yates Shuffle

Randomly permutes a range in-place using the Fisher-Yates shuffle. Each possible permutation is produced with equal probability, assuming the random number source is uniform.

The algorithm scans from right to left. At position `i`, it chooses a random position `j` from `[0, i]` and swaps the two elements. This fixes one uniformly random remaining element at a time.

- `fisher_yates_shuffle(lo, hi)` randomly shuffles the range `[lo, hi)`.

Time Complexity: $O(n)$ per call, where n is the distance between `lo` and `hi`.

Space Complexity: $O(1)$ auxiliary.

```
#include <algorithm>
#include <random>

template<class It>
void fisher_yates_shuffle(It lo, It hi) {
    static std::mt19937 rng(std::random_device{}());
    int n = hi - lo;
    for (int i = n - 1; i > 0; i--) {
        std::uniform_int_distribution<int> dist(0, i);
        int j = dist(rng);
        std::iter_swap(lo + i, lo + j);
    }
}
```

Example Usage

```
#include <algorithm>
#include <cassert>
#include <vector>
using namespace std;

int main() {
    vector<int> a{1, 2, 3, 4, 5}, original(a);
    fisher_yates_shuffle(a.begin(), a.end());
    sort(a.begin(), a.end());
    assert(a == original);
    return 0;
}
```

1.4.2 Reservoir Sampling

Selects uniformly random samples from a stream whose length may be unknown in advance. Reservoir sampling keeps only the chosen sample or samples in memory while processing each element once.

Each class maintains its reservoir incrementally; call `add(x)` once per stream element in any order, then `get()` to retrieve the result.

- `ReservoirSampleOne<T>()` constructs a single-element sampler.
- `ReservoirSampleK<T>(k)` constructs a k-element sampler.
- `add(x)` incorporates one more stream element.
- `get()` returns the current sample. For `ReservoirSampleOne`, `get()` requires at least one `add()` call; for `ReservoirSampleK`, returns the full reservoir (may be fewer than `k` elements if the stream was shorter).
- `count()` returns the number of stream elements seen so far.

Time Complexity:

- $O(1)$ per call to `add(x)`.
- $O(n)$ to process a range of `n` elements.

Space Complexity:

- $O(1)$ auxiliary for `ReservoirSampleOne`.
- $O(k)$ auxiliary heap space for `ReservoirSampleK`.

```
#include <cassert>
#include <random>
#include <vector>

int rand_int(int lo, int hi) {
    static std::mt19937 rng(std::random_device{}());
    return std::uniform_int_distribution<int>(lo, hi)(rng);
}

template<class T>
class ReservoirSampleOne {
    T sample{};
    int seen;

public:
    ReservoirSampleOne() : seen(0) {}

    void add(const T &x) {
        if (rand_int(0, seen++) == 0) {
            sample = x;
        }
    }
}
```



```

    const T &get() const {
        assert(seen > 0);
        return sample;
    }

    int count() const { return seen; }
};

template<class T>
class ReservoirSampleK {
    int k, seen;
    std::vector<T> reservoir;

public:
    explicit ReservoirSampleK(int k) : k(k), seen(0) {}

    void add(const T &x) {
        ++seen;
        if (static_cast<int>(reservoir.size()) < k) {
            reservoir.push_back(x);
        } else {
            int j = rand_int(0, seen - 1);
            if (j < k) {
                reservoir[j] = x;
            }
        }
    }

    const std::vector<T> &get() const { return reservoir; }
    int count() const { return seen; }
};

```

Example Usage

```

#include <cassert>
using namespace std;

int main() {
    vector<int> a{10, 20, 30, 40, 50};

    // Streaming interface: feed elements one at a time.
    ReservoirSampleOne<int> s1;
    for (int x : a) {
        s1.add(x);
    }
    int one = s1.get();
    assert(one == 10 || one == 20 || one == 30 || one == 40 || one == 50);

    ReservoirSampleK<int> sk(3);
    for (int x : a) {
        sk.add(x);
    }
    assert(static_cast<int>(sk.get().size()) == 3);
    return 0;
}

```

1.4.3 Maximum Subarray Sum (Kadane)

Given an array of numbers, determine the maximum possible sum of any contiguous subarray. Kadane's algorithm scans through the array, at each index computing the maximum nonnegative sum subarray ending there. Either this subarray is empty (in which case its sum is zero), or it consists of one more element than the maximum sequence ending at the previous position. This can be adapted to compute the maximal submatrix sum as well.

- `max_subarray_sum(lo, hi, &res_lo, &res_hi)` returns the maximal subarray sum for the range `[lo, hi)`, where `lo` and `hi` are random-access iterators to numeric types. This implementation requires operators `+` and `<` to be defined on the iterators' value type. Optionally, two `int` pointers may be passed to store the inclusive boundary indices `[res_lo, res_hi]` of the resulting subarray. By convention, the empty subarray is allowed, so an input range consisting of only negative values returns `0` with an empty result interval.
- `max_submatrix_sum(matrix, &r1, &c1, &r2, &c2)` returns the largest sum of any rectangular submatrix for a matrix of n rows by m columns. The matrix should be given as a 2-dimensional vector, where the outer vector must contain n vectors each of size m . This implementation requires operators `+` and `<` to be defined on the iterators' value type. Optionally, four `int` pointers may be passed to store the boundary indices of the resulting subarray, with `(r1, c1)` specifying the top-left index and `(r2, c2)` specifying the bottom-right index. By convention, the empty submatrix is allowed, so an input matrix consisting of only negative values returns `0` with an empty result interval.

Time Complexity:

- $O(n)$ per call to `max_subarray_sum()`, where n is the distance between `lo` and `hi`.
- $O(n \cdot m^2)$ per call to `max_submatrix_sum()`, where n is the number of rows and m is the number of columns in the matrix.

Space Complexity:

- $O(1)$ auxiliary for `max_subarray_sum()`.
- $O(n)$ auxiliary heap space for `max_submatrix_sum()`, where n is the number of rows in the matrix.

```
#include <algorithm>
#include <cstddef>
#include <iterator>
#include <vector>

template<class It>
auto max_subarray_sum(It lo, It hi, int *res_lo = nullptr, int *res_hi = nullptr) {
    using T = typename std::iterator_traits<It>::value_type;
    if (lo == hi) {
        return T();
    }
    int curr_begin = 0, begin = 0, end = -1;
    T sum = 0, max_sum = 0;
    for (It it = lo; it != hi; ++it) {
        sum += *it;
        if (sum < 0) {
            sum = 0;
            curr_begin = (it - lo) + 1;
        } else if (max_sum < sum) {
            max_sum = sum;
            begin = curr_begin;
            end = it - lo;
        }
    }
    if (res_lo != nullptr && res_hi != nullptr) {
        *res_lo = begin;
        *res_hi = end;
    }
    return max_sum;
}

template<class T>
T max_submatrix_sum(
    const std::vector<std::vector<T>> &matrix, int *r1 = nullptr, int *c1 = nullptr,
    int *r2 = nullptr, int *c2 = nullptr
) {
    if (matrix.empty() || matrix[0].empty()) {
        return T();
    }
    int n = matrix.size(), m = matrix[0].size();
    std::vector<T> sums(n);
```

```

T sum, max_sum = 0;
for (int clo = 0; clo < m; clo++) {
    std::fill(sums.begin(), sums.end(), 0);
    for (int chi = clo; chi < m; chi++) {
        for (int i = 0; i < n; i++) {
            sums[i] += matrix[i][chi];
        }
        int rlo, rhi;
        sum = max_subarray_sum(sums.begin(), sums.end(), &rlo, &rhi);
        if (max_sum < sum) {
            max_sum = sum;
            if (r1 != nullptr && c1 != nullptr && r2 != nullptr && c2 != nullptr) {
                *r1 = rlo;
                *c1 = clo;
                *r2 = rhi;
                *c2 = chi;
            }
        }
    }
}
return max_sum;
}

```

Example Usage

```

#include <cassert>
#include <iostream>
using namespace std;

int main() {
    {
        vector<int> a{-2, -1, -3, 4, -1, 2, 1, -5, 4};
        int lo = 0, hi = 0;
        assert(max_subarray_sum(a.begin(), a.begin() + 3) == 0);
        assert(max_subarray_sum(a.begin(), a.end(), &lo, &hi) == 6);
        cout << "Maximal sum subarray:" << endl;
        for (int i = lo; i <= hi; i++) {
            cout << a[i] << " ";
        }
        cout << endl;
    }
    {
        vector<vector<int>> matrix{{0, -2, -7, 0, 5},
                                   {9, 2, -6, 2, -4},
                                   {-4, 1, -4, 1, 0},
                                   {-1, 8, 0, -2, 3}};

        int r1 = 0, c1 = 0, r2 = 0, c2 = 0;
        assert(max_submatrix_sum(matrix, &r1, &c1, &r2, &c2) == 15);
        cout << "\nMaximal sum submatrix:" << endl;
        for (int i = r1; i <= r2; i++) {
            for (int j = c1; j <= c2; j++) {
                cout << matrix[i][j] << " ";
            }
            cout << endl;
        }
    }
    return 0;
}

```

Example Output

Maximal sum subarray:
4 -1 2 1

Maximal sum submatrix:

```
9 2
-4 1
-1 8
```

1.4.4 Majority Element (Boyer-Moore)

Given two random-access iterators `lo` and `hi` specifying a range `[lo, hi)` of n elements, return an iterator to the first occurrence of the majority element, or the iterator `hi` if there is no majority element. The majority element is defined as an element which occurs strictly more than $\lfloor n/2 \rfloor$ times in the range. This implementation requires `operator==` to be defined on the iterator's value type.

Time Complexity: $O(n)$ per call to `majority()`, where n is the size of the array.

Space Complexity: $O(1)$ auxiliary.

```
template<class It>
It majority(It lo, It hi) {
    int count = 0;
    It candidate = lo;
    for (It it = lo; it != hi; ++it) {
        if (count == 0) {
            candidate = it;
            count = 1;
        } else if (*it == *candidate) {
            count++;
        } else {
            count--;
        }
    }
    count = 0;
    for (It it = lo; it != hi; ++it) {
        if (*it == *candidate) {
            count++;
        }
    }
    if (count <= (hi - lo) / 2) {
        return hi;
    }
    return candidate;
}
```

Example Usage

```
#include <cassert>
#include <vector>

int main() {
    std::vector<int> a{3, 2, 3, 1, 3};
    assert(*majority(a.begin(), a.end()) == 3);
    std::vector<int> b{2, 3, 3, 3, 2, 1};
    assert(majority(b.begin(), b.end()) == b.end());
    return 0;
}
```

1.4.5 Online Statistics (Welford)

Maintains the mean and variance of a stream of numbers in one pass using Welford's algorithm. This is more numerically stable than maintaining the sum of values and the sum of squares separately.

- `OnlineStatistics()` constructs an empty summary.

- `add(x)` incorporates one more value `x` into the summary.
- `count()` returns the number of values seen so far.
- `mean()` returns the current arithmetic mean.
- `variance_population()` returns population variance, dividing by n .
- `variance_sample()` returns sample variance, dividing by $n - 1$.

Time Complexity: $O(1)$ per call to `add(x)` and each query.

Space Complexity: $O(1)$ auxiliary.

```
class OnlineStatistics {
    int n;
    double avg, m2;

public:
    OnlineStatistics() : n(0), avg(0), m2(0) {}

    void add(double x) {
        n++;
        double delta = x - avg;
        avg += delta / n;
        double delta2 = x - avg;
        m2 += delta * delta2;
    }

    int count() const { return n; }
    double mean() const { return avg; }
    double variance_population() const { return n == 0 ? 0 : m2 / n; }
    double variance_sample() const { return n <= 1 ? 0 : m2 / (n - 1); }
};
```

Example Usage

```
#include <cassert>
#include <cmath>

bool close(double a, double b) {
    return fabs(a - b) < 1e-9;
}

int main() {
    OnlineStatistics stats;
    for (int x = 1; x <= 5; x++) {
        stats.add(x);
    }
    assert(stats.count() == 5);
    assert(close(stats.mean(), 3.0));
    assert(close(stats.variance_population(), 2.0));
    assert(close(stats.variance_sample(), 2.5));
    return 0;
}
```

1.4.6 Frequent Elements (Misra-Gries)

Finds candidate frequent elements in a stream using the Misra-Gries algorithm. With parameter k , the algorithm keeps at most $k - 1$ counters and guarantees that every value occurring more than $\lfloor n/k \rfloor$ times appears among the returned candidates.

The candidates are not automatically verified, since the algorithm intentionally uses sublinear memory and does not retain the stream. If exact frequencies are needed, make a second pass over the input and count only the returned candidates.

- `misra_gries(lo, hi, k)` returns a hash table of candidate values to their residual counters. Use `k = 2` for the Boyer-Moore majority-candidate special case.

Time Complexity: $O(n)$ amortized per call on average: each element either hits an existing counter ($O(1)$) or triggers a full-table decrement ($O(k)$), but the table-full event can occur at most $n/(k-1)$ times total, so the amortized cost per element is $O(1 + k/(k-1)) = O(1)$. In the worst case a single call is $O(nk)$, but that cannot be sustained across a long stream.

Space Complexity: $O(k)$ auxiliary heap space.

```
#include <unordered_map>
#include <vector>

template<class T, class It>
std::unordered_map<T, int> misra_gries(It lo, It hi, int k) {
    std::unordered_map<T, int> count;
    for (It it = lo; it != hi; ++it) {
        if (auto found = count.find(*it); found != count.end()) {
            found->second++;
        } else if (static_cast<int>(count.size()) < k - 1) {
            count[*it] = 1;
        } else {
            std::vector<T> erase;
            for (auto &[key, val] : count) {
                if (--val == 0) {
                    erase.push_back(key);
                }
            }
            for (const auto &key : erase) {
                count.erase(key);
            }
        }
    }
    return count;
}
```

Example Usage

```
#include <cassert>
using namespace std;

int main() {
    vector<int> a{1, 2, 1, 3, 1, 2, 1, 4, 2, 2, 2};
    unordered_map<int, int> candidates = misra_gries<int>(a.begin(), a.end(), 3);
    assert(candidates.count(1));
    assert(candidates.count(2));
    assert(!candidates.count(3));
    return 0;
}
```

1.5 Cycle Detection

1.5.1 Cycle Detection (Floyd)

Given a function f mapping a set of integers to itself and an x-coordinate in the set, return a pair containing the (position, length) of a cycle in the sequence of numbers obtained from repeatedly composing f with itself starting with the initial x . Formally, since f maps a finite set S to itself, some value is guaranteed to eventually repeat in the sequence: $x_0, x_1 = f(x_0), x_2 = f(x_1), \dots, x_n = f(x_{n-1}), \dots$

There must exist a pair of indices i and j ($i < j$) such that $x_i = x_j$. When this happens, the rest of the sequence will consist of the subsequence from x_i to x_{j-1} repeating indefinitely. The cycle detection problem asks to find such an i , along with the length of the repeating subsequence. A well-known special case is the problem of cycle-detection in a degenerate linked list.

Floyd’s cycle-finding algorithm, a.k.a. the “tortoise and the hare algorithm”, is a space-efficient algorithm that moves two pointers through the sequence at different speeds. Each step in the algorithm moves the “tortoise” one step forward and the “hare” two steps forward in the sequence, comparing the sequence values at each step. Their first meeting point is somewhere in the cycle; a second phase then finds the cycle’s starting position.

Time Complexity: $O(m + n)$ per call to `find_cycle_floyd()`, where m is the smallest index of the sequence which is the beginning of a cycle, and n is the cycle’s length.

Space Complexity: $O(1)$ auxiliary.

```
struct Cycle {
    int start, length;
};

template<class Fn>
Cycle find_cycle_floyd(Fn f, int x0) {
    int tortoise = f(x0), hare = f(f(x0));
    while (tortoise != hare) {
        tortoise = f(tortoise);
        hare = f(f(hare));
    }
    int start = 0;
    tortoise = x0;
    while (tortoise != hare) {
        tortoise = f(tortoise);
        hare = f(hare);
        start++;
    }
    int length = 1;
    hare = f(tortoise);
    while (tortoise != hare) {
        hare = f(hare);
        length++;
    }
    return {start, length};
}
```

Example Usage

```
#include <cassert>
#include <set>
using namespace std;

int f(int x) {
    return (123 * x * x + 4567890) % 1337;
}

void verify(int x0, int start, int length) {
    set<int> s;
    int x = x0;
    for (int i = 0; i < start; i++) {
        assert(!s.count(x));
        s.insert(x);
        x = f(x);
    }
    int startx = x;
    s.clear();
    for (int i = 0; i < length; i++) {
        assert(!s.count(x));
        s.insert(x);
    }
}
```

```

    x = f(x);
}
assert(startx == x);
}

int main() {
    int x0 = 0;
    auto [start, length] = find_cycle_floyd(f, x0);
    assert(start == 4 && length == 2);
    verify(x0, start, length);
    return 0;
}

```

1.5.2 Cycle Detection (Brent)

Given a function f mapping a set of integers to itself and an x -coordinate in the set, return a pair containing the (position, length) of a cycle in the sequence of numbers obtained from repeatedly composing f with itself starting with the initial x . Formally, since f maps a finite set S to itself, some value is guaranteed to eventually repeat in the sequence: $x_0, x_1 = f(x_0), x_2 = f(x_1), \dots, x_n = f(x_{n-1}), \dots$

There must exist a pair of indices i and j ($i < j$) such that $x_i = x_j$. When this happens, the rest of the sequence will consist of the subsequence from x_i to x_{j-1} repeating indefinitely. The cycle detection problem asks to find such an i , along with the length of the repeating subsequence. A well-known special case is the problem of cycle-detection in a degenerate linked list.

While Floyd's cycle-finding algorithm finds cycles by simultaneously moving two pointers at different speeds, Brent's algorithm keeps the tortoise pointer stationary in every iteration, only teleporting it to the hare pointer at every power of two. Once the two pointers meet, the cycle length is known, and a second phase finds the cycle's starting position. This improves upon the constant factor of Floyd's algorithm by reducing the number of calls made to f .

Time Complexity: $O(m + n)$ per call to `find_cycle_brent()`, where m is the smallest index of the sequence which is the beginning of a cycle, and n is the cycle's length.

Space Complexity: $O(1)$ auxiliary.

```

struct Cycle {
    int start, length;
};

template<class Fn>
Cycle find_cycle_brent(Fn f, int x0) {
    int power = 1, length = 1, tortoise = x0, hare = f(x0);
    while (tortoise != hare) {
        if (power == length) {
            tortoise = hare;
            power *= 2;
            length = 0;
        }
        hare = f(hare);
        length++;
    }
    hare = x0;
    for (int i = 0; i < length; i++) {
        hare = f(hare);
    }
    int start = 0;
    tortoise = x0;
    while (tortoise != hare) {
        tortoise = f(tortoise);
        hare = f(hare);
        start++;
    }
}

```



```
    }  
    return {start, length};  
}
```

Example Usage

```
#include <cassert>  
#include <set>  
using namespace std;  
  
int f(int x) {  
    return (123 * x * x + 4567890) % 1337;  
}  
  
void verify(int x0, int start, int length) {  
    set<int> s;  
    int x = x0;  
    for (int i = 0; i < start; i++) {  
        assert(!s.count(x));  
        s.insert(x);  
        x = f(x);  
    }  
    int startx = x;  
    s.clear();  
    for (int i = 0; i < length; i++) {  
        assert(!s.count(x));  
        s.insert(x);  
        x = f(x);  
    }  
    assert(startx == x);  
}  
  
int main() {  
    int x0 = 0;  
    auto [start, length] = find_cycle_brent(f, x0);  
    assert(start == 4 && length == 2);  
    verify(x0, start, length);  
    return 0;  
}
```

Chapter 2

Data Structures

2.1 Linked Lists

2.1.1 Singly Linked List Operations

Implements common singly linked list operations using raw nodes. Linked lists are rarely the best contest choice compared with arrays, vectors, and `std::list`, but the pointer patterns are useful for interview-style problems and for understanding constant-time splicing.

For production C++, prefer containers such as `std::list`, `std::forward_list`, or `std::vector` when they fit the problem. Use the raw-node style below when the problem statement already gives nodes, or when manual pointer manipulation is the point of the exercise.

- `reverse_list(head)` reverses a singly linked list and returns the new head.
- `merge_sorted_lists(a, b)` merges two sorted lists using a dummy node and returns the merged head.
- `split_half(head, &second)` cuts a list into two halves, returning the first half and storing the second half in `second`.

For linked-list cycle detection, use Floyd or Brent cycle detection from section 1.5.

Time Complexity:

- $O(n)$ per call to `reverse_list(head)` and `split_half(head, &second)`.
- $O(n + m)$ per call to `merge_sorted_lists(a, b)`.

Space Complexity: $O(1)$ auxiliary for all operations.

```
#include <cstddef>

struct ListNode {
    int value;
    ListNode *next;

    explicit ListNode(int value = 0) : value(value), next(nullptr) {}
};

ListNode *reverse_list(ListNode *head) {
    ListNode *prev = nullptr;
    while (head != nullptr) {
        ListNode *next = head->next;
        head->next = prev;
        prev = head;
        head = next;
    }
    return prev;
}
```

```

}

ListNode *merge_sorted_lists(ListNode *a, ListNode *b) {
    ListNode dummy;
    ListNode *tail = &dummy;
    while (a != nullptr && b != nullptr) {
        if (b->value < a->value) {
            tail->next = b;
            b = b->next;
        } else {
            tail->next = a;
            a = a->next;
        }
        tail = tail->next;
    }
    tail->next = a != nullptr ? a : b;
    return dummy.next;
}

ListNode *split_half(ListNode *head, ListNode **second) {
    if (head == nullptr || head->next == nullptr) {
        *second = nullptr;
        return head;
    }
    ListNode *slow = head, *fast = head->next;
    while (fast != nullptr && fast->next != nullptr) {
        slow = slow->next;
        fast = fast->next->next;
    }
    *second = slow->next;
    slow->next = nullptr;
    return head;
}

```

Example Usage

```

#include <cassert>
#include <vector>
using namespace std;

vector<int> values(ListNode *head) {
    vector<int> res;
    for (; head != nullptr; head = head->next) {
        res.push_back(head->value);
    }
    return res;
}

int main() {
    vector<ListNode> a{ListNode(1), ListNode(3), ListNode(5)};
    vector<ListNode> b{ListNode(2), ListNode(4), ListNode(6)};
    for (int i = 0; i + 1 < static_cast<int>(a.size()); i++) {
        a[i].next = &a[i + 1];
        b[i].next = &b[i + 1];
    }
    ListNode *merged = merge_sorted_lists(&a[0], &b[0]);
    vector<int> merged_values = values(merged);
    for (int i = 0; i < 6; i++) {
        assert(merged_values[i] == i + 1);
    }

    ListNode *second = nullptr;
    ListNode *first = split_half(merged, &second);
    assert(values(first).size() == 3);
    assert(values(second).size() == 3);
    assert(values(reverse_list(first))[0] == 3);
}

```

```
    return 0;
}
```

2.1.2 Linked List Splicing

Moves nodes or ranges between singly linked lists in $O(1)$ pointer changes once the neighboring nodes are known. Splicing is the main operation that makes linked lists useful: it moves existing nodes without copying values or invalidating pointers to the moved nodes.

The functions below use a dummy head node for each list. This makes insertions and removals at the beginning of a list match all other positions. In the C++ standard library, `std::list::splice()` provides analogous operations for doubly linked lists, while `std::forward_list::splice_after()` provides analogous operations for singly linked lists.

- `splice_after(pos, before)` moves the node after `before` so it appears after `pos`.
- `splice_range_after(pos, before_first, last)` moves the half-open range `(before_first, last)` so it appears after `pos`.

Time Complexity:

- $O(1)$ per call to `splice_after(pos, before)`.
- $O(k)$ per call to `splice_range_after(pos, before_first, last)`, where k is the number of moved nodes. The actual pointer splicing is $O(1)$, but this compact version walks to the end of the moved range.

Space Complexity: $O(1)$ auxiliary.

```
#include <cstddef>

struct ListNode {
    int value;
    ListNode *next;

    explicit ListNode(int value = 0) : value(value), next(nullptr) {}
};

void splice_after(ListNode *pos, ListNode *before) {
    ListNode *node = before->next;
    if (node == nullptr || pos == before || pos == node) {
        return;
    }
    before->next = node->next;
    node->next = pos->next;
    pos->next = node;
}

void splice_range_after(ListNode *pos, ListNode *before_first, ListNode *last) {
    ListNode *first = before_first->next;
    if (first == last) {
        return;
    }
    ListNode *tail = first;
    while (tail->next != last) {
        tail = tail->next;
    }
    before_first->next = last;
    tail->next = pos->next;
    pos->next = first;
}
```

Example Usage

```
#include <cassert>
#include <vector>
using namespace std;

vector<int> values(ListNode *dummy) {
    vector<int> res;
    for (ListNode *p = dummy->next; p != nullptr; p = p->next) {
        res.push_back(p->value);
    }
    return res;
}

int main() {
    ListNode dummy(0), a(1), b(2), c(3), d(4), e(5);
    dummy.next = &a;
    a.next = &b;
    b.next = &c;
    c.next = &d;
    d.next = &e;

    splice_after(&dummy, &c); // Move 4 to the front: 4, 1, 2, 3, 5.
    vector<int> v = values(&dummy);
    vector<int> expected1{4, 1, 2, 3, 5};
    for (int i = 0; i < static_cast<int>(expected1.size()); i++) {
        assert(v[i] == expected1[i]);
    }

    splice_range_after(&d, &a, &e); // Move 2, 3 after 4: 4, 2, 3, 1, 5.
    v = values(&dummy);
    vector<int> expected2{4, 2, 3, 1, 5};
    for (int i = 0; i < static_cast<int>(expected2.size()); i++) {
        assert(v[i] == expected2[i]);
    }
    return 0;
}
```

2.1.3 LRU Cache

Maintains a least-recently-used cache with fixed capacity. The cache supports lookups and updates by key, evicting the least recently used entry whenever an insertion would exceed capacity.

This implementation combines a doubly linked list (`std::list`) with a map from keys to list iterators. The front of the list stores the most recently used item. The operation `items.splice(items.begin(), items, it)` is the standard-library way to move an existing list node to the front in $O(1)$ time.

- `LRUCache(capacity)` constructs an empty cache holding at most `capacity` keys.
- `get(key, &value)` returns whether `key` is present. If present, it stores the value in `value` and marks the key as most recently used.
- `put(key, value)` inserts or updates `key`, marking it as most recently used and evicting the least recently used key if needed.
- `size()` returns the number of keys currently stored.

Time Complexity: $O(1)$ amortized per call to `get(key, &value)` and `put(key, value)`.

Space Complexity: $O(n)$, where n is the cache capacity.

```
#include <list>
#include <unordered_map>
#include <utility>

template<class Key, class Value>
```

```

class LRUCache {
    using ListIter = typename std::list<std::pair<Key, Value>>::iterator;

    int cap;
    std::list<std::pair<Key, Value>> items;
    std::unordered_map<Key, ListIter> where;

public:
    explicit LRUCache(int capacity) : cap(capacity) {}

    int size() const { return items.size(); }

    bool get(const Key &key, Value *value) {
        auto it = where.find(key);
        if (it == where.end()) {
            return false;
        }
        items.splice(items.begin(), items, it->second);
        it->second = items.begin();
        *value = it->second->second;
        return true;
    }

    void put(const Key &key, const Value &value) {
        auto it = where.find(key);
        if (it != where.end()) {
            it->second->second = value;
            items.splice(items.begin(), items, it->second);
            it->second = items.begin();
            return;
        }
        if (cap == 0) {
            return;
        }
        if (static_cast<int>(items.size()) == cap) {
            where.erase(items.back().first);
            items.pop_back();
        }
        items.push_front({key, value});
        where[key] = items.begin();
    }
};

```

Example Usage

```

#include <cassert>
using namespace std;

int main() {
    LRUCache<int, int> cache(2);
    int value = 0;
    cache.put(1, 10);
    cache.put(2, 20);
    assert(cache.get(1, &value) && value == 10);
    cache.put(3, 30); // Evicts key 2.
    assert(!cache.get(2, &value));
    assert(cache.get(3, &value) && value == 30);
    cache.put(1, 11);
    assert(cache.get(1, &value) && value == 11);
    return 0;
}

```

2.2 Heaps and Priority Queues

2.2.1 Binary Heap

Maintain a min-priority queue, that is, a collection of elements with support for querying and extraction of the minimum. This implementation requires an ordering on the set of possible elements defined by `operator<`. A binary min-heap implements a priority queue by inserting and deleting nodes into a binary tree such that the parent of any node is always less than its children.

- `BinaryHeap()` constructs an empty priority queue.
- `BinaryHeap(lo, hi)` constructs a priority queue from two ForwardIterators, consisting of elements in the range `[lo, hi)`.
- `size()` returns the size of the priority queue.
- `empty()` returns whether the priority queue is empty.
- `push(v)` inserts the value `v` into the priority queue.
- `pop()` removes the minimum element from the priority queue.
- `top()` returns the minimum element in the priority queue.

Time Complexity:

- $O(1)$ per call to the first constructor, `size()`, `empty()`, and `top()`.
- $O(\log n)$ per call to `push()` and `pop()`, where n is the number of elements in the priority queue.
- $O(n)$ per call to the second constructor on the distance between `lo` and `hi` (bottom-up heapify).

Space Complexity:

- $O(n)$ for storage of the priority queue elements.
- $O(1)$ auxiliary for all operations.

```
#include <algorithm>
#include <stdexcept>
#include <vector>

template<class T>
class BinaryHeap {
    std::vector<T> heap;

    void sift_down(int i) {
        for (;;) {
            int child = 2 * i + 1;
            if (child >= static_cast<int>(heap.size())) {
                break;
            }
            if (child + 1 < static_cast<int>(heap.size()) && heap[child + 1] < heap[child]) {
                child++;
            }
            if (heap[child] < heap[i]) {
                std::swap(heap[i], heap[child]);
                i = child;
            } else {
                break;
            }
        }
    }

public:
    BinaryHeap() {}

    template<class It>
    BinaryHeap(It lo, It hi) : heap(lo, hi) {
        for (int i = static_cast<int>(heap.size()) / 2 - 1; i >= 0; i--) {
            sift_down(i);
        }
    }
};
```

```

}

int size() const { return heap.size(); }
bool empty() const { return heap.empty(); }

void push(const T &v) {
    heap.push_back(v);
    int i = heap.size() - 1;
    while (i > 0) {
        int parent = (i - 1) / 2;
        if (!heap[i] < heap[parent]) {
            break;
        }
        std::swap(heap[i], heap[parent]);
        i = parent;
    }
}

void pop() {
    if (heap.empty()) {
        throw std::runtime_error("Cannot pop from empty heap.");
    }
    heap[0] = heap.back();
    heap.pop_back();
    if (!heap.empty()) {
        sift_down(0);
    }
}

T top() const {
    if (heap.empty()) {
        throw std::runtime_error("Cannot get top of empty heap.");
    }
    return heap[0];
}
};

```

Example Usage

```

#include <iostream>
using namespace std;

int main() {
    vector<int> a{0, 5, -1, 12};
    BinaryHeap<int> h(a.begin(), a.end());
    h.push(10);
    while (!h.empty()) {
        cout << h.top() << endl;
        h.pop();
    }
    return 0;
}

```

Example Output

```

-1
0
5
10
12

```


2.2.2 Skew Heap

Maintain a mergeable min-priority queue, that is, a collection of elements with support for querying and extraction of the minimum as well as efficient merging with other instances. This implementation requires an ordering on the set of possible elements defined by `operator<`. A skew heap attempts to maintain balance by unconditionally swapping all nodes in the merge path when merging.

- `SkewHeap()` constructs an empty priority queue.
- `SkewHeap(lo, hi)` constructs a priority queue from two ForwardIterators, consisting of elements in the range `[lo, hi)`.
- `size()` returns the size of the priority queue.
- `empty()` returns whether the priority queue is empty.
- `push(v)` inserts the value `v` into the priority queue.
- `pop()` removes the minimum element from the priority queue.
- `top()` returns the minimum element in the priority queue.
- `absorb(h)` inserts every value from `h` and sets `h` to the empty priority queue.

Time Complexity:

- $O(1)$ per call to the first constructor, `size()`, `empty()`, and `top()`.
- $O(\log n)$ amortized auxiliary per call to `push()`, `pop()`, and `absorb()`, where n is the number of elements in the priority queue.
- $O(n)$ per call to the second constructor, where n is the distance between `lo` and `hi`.

Space Complexity:

- $O(n)$ for storage of the priority queue elements.
- $O(\log n)$ amortized auxiliary stack space for `push()`, `pop()`, and `absorb()`.
- $O(1)$ auxiliary for all other operations.

```
#include <algorithm>
#include <cstdint>
#include <stdexcept>

template<class T>
class SkewHeap {
    struct Node {
        T value;
        Node *left, *right;

        explicit Node(const T &v) : value(v), left(nullptr), right(nullptr) {}
    } *root;

    int num_nodes;

    static Node *merge(Node *a, Node *b) {
        if (a == nullptr) {
            return b;
        }
        if (b == nullptr) {
            return a;
        }
        if (b->value < a->value) {
            std::swap(a, b);
        }
        std::swap(a->left, a->right);
        a->left = merge(b, a->left);
        return a;
    }

    static void clean_up(Node *n) {
        if (n != nullptr) {
            clean_up(n->left);
            clean_up(n->right);
        }
    }
};
```

```

        delete n;
    }
}

public:
    SkewHeap() : root(nullptr), num_nodes(0) {}

    template<class It>
    SkewHeap(It lo, It hi) : root(nullptr), num_nodes(0) {
        while (lo != hi) {
            push(*(lo++));
        }
    }

    ~SkewHeap() { clean_up(root); }
    SkewHeap(const SkewHeap &) = delete;
    SkewHeap &operator=(const SkewHeap &) = delete;
    int size() const { return num_nodes; }
    bool empty() const { return root == nullptr; }

    void push(const T &v) {
        root = merge(root, new Node(v));
        num_nodes++;
    }

    void pop() {
        if (empty()) {
            throw std::runtime_error("Cannot pop from empty heap.");
        }
        Node *tmp = root;
        root = merge(root->left, root->right);
        delete tmp;
        num_nodes--;
    }

    T top() const {
        if (empty()) {
            throw std::runtime_error("Cannot get top of empty heap.");
        }
        return root->value;
    }

    void absorb(SkewHeap &h) {
        root = merge(root, h.root);
        num_nodes += h.num_nodes;
        h.root = nullptr;
        h.num_nodes = 0;
    }
};

```

Example Usage

```

#include <iostream>
using namespace std;

int main() {
    SkewHeap<int> h, h2;
    h.push(12);
    h.push(10);
    h2.push(5);
    h2.push(-1);
    h2.push(0);
    h.absorb(h2);
    while (!h.empty()) {
        cout << h.top() << endl;
        h.pop();
    }
}

```

```

    }
    return 0;
}

```

Example Output

```

-1
0
5
10
12

```

2.2.3 Pairing Heap

Maintain a mergeable min-priority queue, that is, a collection of elements with support for querying and extraction of the minimum as well as efficient merging with other instances. This implementation requires an ordering on the set of possible elements defined by `operator<`. A pairing heap is a heap-ordered multi-way tree, using a two-pass merge to self-adjust during each deletion.

- `PairingHeap()` constructs an empty priority queue.
- `PairingHeap(lo, hi)` constructs a priority queue from two ForwardIterators, consisting of elements in the range `[lo, hi)`.
- `size()` returns the size of the priority queue.
- `empty()` returns whether the priority queue is empty.
- `push(v)` inserts the value `v` into the priority queue.
- `pop()` removes the minimum element from the priority queue.
- `top()` returns the minimum element in the priority queue.
- `absorb(h)` inserts every value from `h` and sets `h` to the empty priority queue.

Time Complexity:

- $O(1)$ per call to the first constructor, `size()`, `empty()`, `top()`, `push()`, and `absorb()`.
- $O(\log n)$ amortized per call to `pop()`.
- $O(n)$ per call to the second constructor on the distance between `lo` and `hi`.

Space Complexity:

- $O(n)$ for storage of the priority queue elements.
- $O(\log n)$ amortized auxiliary stack space for `pop()`.
- $O(1)$ auxiliary for all other operations.

```

#include <cstddef>
#include <stdexcept>

template<class T>
class PairingHeap {
    struct Node {
        T value;
        Node *left, *next;

        explicit Node(const T &v) : value(v), left(nullptr), next(nullptr) {}

        void add_child(Node *n) {
            if (left == nullptr) {
                left = n;
            } else {
                n->next = left;
                left = n;
            }
        }
    } *root;
}

```

```

int num_nodes;

static Node *merge(Node *a, Node *b) {
    if (a == nullptr) {
        return b;
    }
    if (b == nullptr) {
        return a;
    }
    if (a->value < b->value) {
        a->add_child(b);
        return a;
    }
    b->add_child(a);
    return b;
}

static Node *merge_pairs(Node *n) {
    if (n == nullptr || n->next == nullptr) {
        return n;
    }
    Node *a = n, *b = n->next, *c = n->next->next;
    a->next = b->next = nullptr;
    return merge(merge(a, b), merge_pairs(c));
}

static void clean_up(Node *n) {
    if (n != nullptr) {
        clean_up(n->left);
        clean_up(n->next);
        delete n;
    }
}

public:
PairingHeap() : root(nullptr), num_nodes(0) {}

template<class It>
PairingHeap(It lo, It hi) : root(nullptr), num_nodes(0) {
    while (lo != hi) {
        push(*(lo++));
    }
}

~PairingHeap() { clean_up(root); }
PairingHeap(const PairingHeap &) = delete;
PairingHeap &operator=(const PairingHeap &) = delete;
int size() const { return num_nodes; }
bool empty() const { return root == nullptr; }

void push(const T &v) {
    root = merge(root, new Node(v));
    num_nodes++;
}

void pop() {
    if (empty()) {
        throw std::runtime_error("Cannot pop from empty heap.");
    }
    Node *tmp = root;
    root = merge_pairs(root->left);
    delete tmp;
    num_nodes--;
}

T top() const {
    if (empty()) {
        throw std::runtime_error("Cannot get top of empty heap.");
    }
}

```

```

    }
    return root->value;
}

void absorb(PairingHeap &h) {
    root = merge(root, h.root);
    num_nodes += h.num_nodes;
    h.root = nullptr;
    h.num_nodes = 0;
}
};

```

Example Usage

```

#include <iostream>
using namespace std;

int main() {
    PairingHeap<int> h, h2;
    h.push(12);
    h.push(10);
    h2.push(5);
    h2.push(-1);
    h2.push(0);
    h.absorb(h2);
    while (!h.empty()) {
        cout << h.top() << endl;
        h.pop();
    }
    return 0;
}

```

Example Output

```

-1
0
5
10
12

```

2.3 Dictionaries and Ordered Sets

2.3.1 Binary Search Tree

Maintain a map, that is, a collection of key-value pairs such that each possible key appears at most once in the collection. This implementation requires an ordering on the set of possible keys defined by `operator<` on the key type. A binary search tree implements this map by inserting and deleting keys into a binary tree such that every node's left child has a lesser key and every node's right child has a greater key.

- `BST()` constructs an empty map.
- `size()` returns the size of the map.
- `empty()` returns whether the map is empty.
- `insert(k, v)` adds an entry with key `k` and value `v` to the map, returning `true` if a new entry was added or `false` if the key already exists (in which case the map is unchanged and the old value associated with the key is preserved).
- `erase(k)` removes the entry with key `k` from the map, returning `true` if the removal was successful or `false` if the key to be removed was not found.
- `find(k)` returns a pointer to a const value associated with key `k`, or `nullptr` if the key was not found.
- `walk(f)` calls the function `f(k, v)` on each entry of the map, in ascending order of keys.

Time Complexity:

- $O(1)$ per call to the constructor, `size()`, and `empty()`.
- $O(n)$ per call to `insert()`, `erase()`, `find()`, and `walk()`, where n is the number of nodes currently in the map.

Space Complexity:

- $O(n)$ for storage of the map elements.
- $O(n)$ auxiliary stack space for `insert()`, `erase()`, and `walk()`.
- $O(1)$ auxiliary for all other operations.

```
#include <cstddef>

template<class K, class V>
class BST {
    struct Node {
        K key;
        V value;
        Node *left, *right;

        Node(const K &k, const V &v) : key(k), value(v), left(nullptr), right(nullptr) {}
    } *root;

    int num_nodes;

    static bool insert(Node *&n, const K &k, const V &v) {
        if (n == nullptr) {
            n = new Node(k, v);
            return true;
        }
        if (k < n->key) {
            return insert(n->left, k, v);
        } else if (n->key < k) {
            return insert(n->right, k, v);
        }
        return false;
    }

    static bool erase(Node *&n, const K &k) {
        if (n == nullptr) {
            return false;
        }
        if (k < n->key) {
            return erase(n->left, k);
        } else if (n->key < k) {
            return erase(n->right, k);
        }
        if (n->left != nullptr && n->right != nullptr) {
            Node *tmp = n->right, *parent = nullptr;
            while (tmp->left != nullptr) {
                parent = tmp;
                tmp = tmp->left;
            }
            n->key = tmp->key;
            n->value = tmp->value;
            if (parent != nullptr) {
                return erase(parent->left, parent->left->key);
            }
            return erase(n->right, n->right->key);
        }
        Node *tmp = (n->left != nullptr) ? n->left : n->right;
        delete n;
        n = tmp;
        return true;
    }
}
```

```

template<class Fn>
static void walk(Node *n, Fn f) {
    if (n != nullptr) {
        walk(n->left, f);
        f(n->key, n->value);
        walk(n->right, f);
    }
}

static void clean_up(Node *n) {
    if (n != nullptr) {
        clean_up(n->left);
        clean_up(n->right);
        delete n;
    }
}

public:
    BST() : root(nullptr), num_nodes(0) {}

    ~BST() { clean_up(root); }
    BST(const BST &) = delete;
    BST &operator=(const BST &) = delete;
    int size() const { return num_nodes; }
    bool empty() const { return root == nullptr; }

    bool insert(const K &k, const V &v) {
        if (insert(root, k, v)) {
            num_nodes++;
            return true;
        }
        return false;
    }

    bool erase(const K &k) {
        if (erase(root, k)) {
            num_nodes--;
            return true;
        }
        return false;
    }

    const V *find(const K &k) const {
        Node *n = root;
        while (n != nullptr) {
            if (k < n->key) {
                n = n->left;
            } else if (n->key < k) {
                n = n->right;
            } else {
                return &(n->value);
            }
        }
        return nullptr;
    }

    template<class Fn>
    void walk(Fn f) const {
        walk(root, f);
    }
};

```

Example Usage

```

#include <cassert>
#include <iostream>

```

```
using namespace std;

int main() {
    BST<int, char> t;
    t.insert(2, 'b');
    t.insert(1, 'a');
    t.insert(3, 'c');
    t.insert(5, 'e');
    assert(t.insert(4, 'd'));
    assert(*t.find(4) == 'd');
    assert(!t.insert(4, 'd'));
    auto printch = [](int k, char v) { cout << v; };
    t.walk(printch);
    cout << endl;
    assert(t.erase(1));
    assert(!t.erase(1));
    assert(t.find(1) == nullptr);
    t.walk(printch);
    cout << endl;
    return 0;
}
```

Example Output

```
abcde
bcde
```

2.3.2 Treap

Maintain a map, that is, a collection of key-value pairs such that each possible key appears at most once in the collection. This implementation requires an ordering on the set of possible keys defined by `operator<` on the key type. A Treap is a binary search tree where each node holds a randomly assigned priority. The tree satisfies the BST property on keys and the min-heap property on priorities (lower priority value is closer to root). Since priorities are random, this keeps the tree balanced with high probability, making insertions and deletions run in $O(\log n)$.

- `Treap()` constructs an empty map.
- `size()` returns the size of the map.
- `empty()` returns whether the map is empty.
- `insert(k, v)` adds an entry with key `k` and value `v` to the map, returning `true` if a new entry was added or `false` if the key already exists (in which case the map is unchanged and the old value associated with the key is preserved).
- `erase(k)` removes the entry with key `k` from the map, returning `true` if the removal was successful or `false` if the key to be removed was not found.
- `find(k)` returns a pointer to a const value associated with key `k`, or `nullptr` if the key was not found.
- `walk(f)` calls the function `f(k, v)` on each entry of the map, in ascending order of keys.

Time Complexity:

- $O(1)$ per call to the constructor, `size()`, and `empty()`.
- $O(\log n)$ on average per call to `insert()`, `erase()`, and `find()`, where n is the number of entries currently in the map.
- $O(n)$ per call to `walk()`.

Space Complexity:

- $O(n)$ for storage of the map elements.
- $O(\log n)$ auxiliary stack space on average for `insert()`, `erase()`, and `walk()`.
- $O(1)$ auxiliary for all other operations.

```
#include <cstdint>
```



```

template<class K, class V>
class Treap {
    struct Node {
        static uint32_t rand32() {
            static uint32_t x = 123456789;
            x ^= x << 13;
            x ^= x >> 17;
            x ^= x << 5;
            return x;
        }

        K key;
        V value;
        uint32_t priority;
        Node *left, *right;

        Node(const K &k, const V &v)
            : key(k), value(v), priority(rand32()), left(nullptr), right(nullptr) {}
    } *root;

    int num_nodes;

    static void rotate_left(Node *&n) {
        Node *tmp = n;
        n = n->right;
        tmp->right = n->left;
        n->left = tmp;
    }

    static void rotate_right(Node *&n) {
        Node *tmp = n;
        n = n->left;
        tmp->left = n->right;
        n->right = tmp;
    }

    static bool insert(Node *&n, const K &k, const V &v) {
        if (n == nullptr) {
            n = new Node(k, v);
            return true;
        }
        if (k < n->key && insert(n->left, k, v)) {
            if (n->left->priority < n->priority) {
                rotate_right(n);
            }
            return true;
        }
        if (n->key < k && insert(n->right, k, v)) {
            if (n->right->priority < n->priority) {
                rotate_left(n);
            }
            return true;
        }
        return false;
    }

    static bool erase(Node *&n, const K &k) {
        if (n == nullptr) {
            return false;
        }
        if (k < n->key) {
            return erase(n->left, k);
        } else if (n->key < k) {
            return erase(n->right, k);
        }
        if (n->left != nullptr && n->right != nullptr) {
            if (n->left->priority < n->right->priority) {

```

```

        rotate_right(n);
        return erase(n->right, k);
    }
    rotate_left(n);
    return erase(n->left, k);
}
Node *tmp = (n->left != nullptr) ? n->left : n->right;
delete n;
n = tmp;
return true;
}

template<class Fn>
static void walk(Node *n, Fn f) {
    if (n != nullptr) {
        walk(n->left, f);
        f(n->key, n->value);
        walk(n->right, f);
    }
}

static void clean_up(Node *n) {
    if (n != nullptr) {
        clean_up(n->left);
        clean_up(n->right);
        delete n;
    }
}

public:
    Treap() : root(nullptr), num_nodes(0) {}

    ~Treap() { clean_up(root); }
    Treap(const Treap &) = delete;
    Treap &operator=(const Treap &) = delete;
    int size() const { return num_nodes; }
    bool empty() const { return root == nullptr; }

    bool insert(const K &k, const V &v) {
        if (insert(root, k, v)) {
            num_nodes++;
            return true;
        }
        return false;
    }

    bool erase(const K &k) {
        if (erase(root, k)) {
            num_nodes--;
            return true;
        }
        return false;
    }

    const V *find(const K &k) const {
        Node *n = root;
        while (n != nullptr) {
            if (k < n->key) {
                n = n->left;
            } else if (n->key < k) {
                n = n->right;
            } else {
                return &(n->value);
            }
        }
        return nullptr;
    }
}

```

```

template<class Fn>
void walk(Fn f) const {
    walk(root, f);
}
};

```

Example Usage

```

#include <cassert>
#include <iostream>
using namespace std;

int main() {
    Treap<int, char> t;
    t.insert(2, 'b');
    t.insert(1, 'a');
    t.insert(3, 'c');
    t.insert(5, 'e');
    assert(t.insert(4, 'd'));
    assert(*t.find(4) == 'd');
    assert(!t.insert(4, 'd'));
    auto printch = [](int k, char v) { cout << v; };
    t.walk(printch);
    cout << endl;
    assert(t.erase(1));
    assert(!t.erase(1));
    assert(t.find(1) == nullptr);
    t.walk(printch);
    cout << endl;
    return 0;
}

```

Example Output

```

abcde
bcde

```

2.3.3 AVL Tree

Maintain a map, that is, a collection of key-value pairs such that each possible key appears at most once in the collection. This implementation requires an ordering on the set of possible keys defined by `operator<` on the key type. An AVL tree is a binary search tree balanced by height, guaranteeing $O(\log n)$ worst-case running time in insertions and deletions by making sure that the heights of the left and right subtrees at every node differ by at most 1.

- `AVLTree()` constructs an empty map.
- `size()` returns the size of the map.
- `empty()` returns whether the map is empty.
- `insert(k, v)` adds an entry with key `k` and value `v` to the map, returning `true` if a new entry was added or `false` if the key already exists (in which case the map is unchanged and the old value associated with the key is preserved).
- `erase(k)` removes the entry with key `k` from the map, returning `true` if the removal was successful or `false` if the key to be removed was not found.
- `find(k)` returns a pointer to a const value associated with key `k`, or `nullptr` if the key was not found.
- `walk(f)` calls the function `f(k, v)` on each entry of the map, in ascending order of keys.

Time Complexity:

- $O(1)$ per call to the constructor, `size()`, and `empty()`.
- $O(\log n)$ per call to `insert()`, `erase()`, and `find()`, where n is the number of entries currently in the map.

- $O(n)$ per call to `walk()`.

Space Complexity:

- $O(n)$ for storage of the map elements.
- $O(\log n)$ auxiliary stack space for `insert()`, `erase()`, and `walk()`.
- $O(1)$ auxiliary for all other operations.

```
#include <algorithm>
#include <cstdint>

template<class K, class V>
class AVLTree {
    struct Node {
        K key;
        V value;
        int height;
        Node *left, *right;

        Node(const K &k, const V &v) : key(k), value(v), height(1), left(nullptr), right(nullptr) {}
    } *root;

    int num_nodes;

    static int height(Node *n) { return (n != nullptr) ? n->height : 0; }

    static void update_height(Node *n) {
        if (n != nullptr) {
            n->height = 1 + std::max(height(n->left), height(n->right));
        }
    }

    static void rotate_left(Node *&n) {
        Node *tmp = n;
        n = n->right;
        tmp->right = n->left;
        n->left = tmp;
        update_height(tmp);
        update_height(n);
    }

    static void rotate_right(Node *&n) {
        Node *tmp = n;
        n = n->left;
        tmp->left = n->right;
        n->right = tmp;
        update_height(tmp);
        update_height(n);
    }

    static int balance_factor(Node *n) {
        return (n != nullptr) ? (height(n->left) - height(n->right)) : 0;
    }

    static void rebalance(Node *&n) {
        if (n == nullptr) {
            return;
        }
        update_height(n);
        int bf = balance_factor(n);
        if (bf > 1 && balance_factor(n->left) >= 0) {
            rotate_right(n);
        } else if (bf > 1 && balance_factor(n->left) < 0) {
            rotate_left(n->left);
            rotate_right(n);
        } else if (bf < -1 && balance_factor(n->right) <= 0) {
            rotate_left(n);
        } else if (bf < -1 && balance_factor(n->right) > 0) {
            rotate_right(n->right);
            rotate_left(n);
        }
    }
};
```

```

        rotate_right(n->right);
        rotate_left(n);
    }
}

static bool insert(Node *&n, const K &k, const V &v) {
    if (n == nullptr) {
        n = new Node(k, v);
        return true;
    }
    if ((k < n->key && insert(n->left, k, v)) || (n->key < k && insert(n->right, k, v))) {
        rebalance(n);
        return true;
    }
    return false;
}

static bool erase(Node *&n, const K &k) {
    if (n == nullptr) {
        return false;
    }
    if (!(k < n->key || n->key < k)) {
        if (n->left != nullptr && n->right != nullptr) {
            Node *tmp = n->right, *parent = nullptr;
            while (tmp->left != nullptr) {
                parent = tmp;
                tmp = tmp->left;
            }
            n->key = tmp->key;
            n->value = tmp->value;
            if (parent != nullptr) {
                if (!erase(parent->left, parent->left->key)) {
                    return false;
                }
            }
            else if (!erase(n->right, n->right->key)) {
                return false;
            }
        }
        else {
            Node *tmp = (n->left != nullptr) ? n->left : n->right;
            delete n;
            n = tmp;
        }
        rebalance(n);
        return true;
    }
    if ((k < n->key && erase(n->left, k)) || (n->key < k && erase(n->right, k))) {
        rebalance(n);
        return true;
    }
    return false;
}

template<class Fn>
static void walk(Node *n, Fn f) {
    if (n != nullptr) {
        walk(n->left, f);
        f(n->key, n->value);
        walk(n->right, f);
    }
}

static void clean_up(Node *n) {
    if (n != nullptr) {
        clean_up(n->left);
        clean_up(n->right);
        delete n;
    }
}

```

```

public:
    AVLTree() : root(nullptr), num_nodes(0) {}

    ~AVLTree() { clean_up(root); }
    AVLTree(const AVLTree &) = delete;
    AVLTree &operator=(const AVLTree &) = delete;
    int size() const { return num_nodes; }
    bool empty() const { return root == nullptr; }

    bool insert(const K &k, const V &v) {
        if (insert(root, k, v)) {
            num_nodes++;
            return true;
        }
        return false;
    }

    bool erase(const K &k) {
        if (erase(root, k)) {
            num_nodes--;
            return true;
        }
        return false;
    }

    const V *find(const K &k) const {
        Node *n = root;
        while (n != nullptr) {
            if (k < n->key) {
                n = n->left;
            } else if (n->key < k) {
                n = n->right;
            } else {
                return &(n->value);
            }
        }
        return nullptr;
    }

    template<class Fn>
    void walk(Fn f) const {
        walk(root, f);
    }
};

```

Example Usage

```

#include <cassert>
#include <iostream>
using namespace std;

int main() {
    AVLTree<int, char> t;
    t.insert(2, 'b');
    t.insert(1, 'a');
    t.insert(3, 'c');
    t.insert(5, 'e');
    assert(t.insert(4, 'd'));
    assert(*t.find(4) == 'd');
    assert(!t.insert(4, 'd'));
    auto printch = [](int k, char v) { cout << v; };
    t.walk(printch);
    cout << endl;
    assert(t.erase(1));
    assert(!t.erase(1));
}

```

```

assert(t.find(1) == nullptr);
t.walk(printch);
cout << endl;
return 0;
}

```

Example Output

```

abcde
bcde

```

2.3.4 Red-Black Tree

Maintain a map, that is, a collection of key-value pairs such that each possible key appears at most once in the collection. This implementation requires an ordering on the set of possible keys defined by `operator<` on the key type. A red black tree is a binary search tree balanced by coloring its nodes red or black, then constraining node colors on any simple path from the root to a leaf.

- `RedBlackTree()` constructs an empty map.
- `size()` returns the size of the map.
- `empty()` returns whether the map is empty.
- `insert(k, v)` adds an entry with key `k` and value `v` to the map, returning `true` if a new entry was added or `false` if the key already exists (in which case the map is unchanged and the old value associated with the key is preserved).
- `erase(k)` removes the entry with key `k` from the map, returning `true` if the removal was successful or `false` if the key to be removed was not found.
- `find(k)` returns a pointer to a const value associated with key `k`, or `nullptr` if the key was not found.
- `walk(f)` calls the function `f(k, v)` on each entry of the map, in ascending order of keys.

Time Complexity:

- $O(1)$ per call to the constructor, `size()`, and `empty()`.
- $O(\log n)$ per call to `insert()`, `erase()`, and `find()`, where n is the number of entries currently in the map.
- $O(n)$ per call to `walk()`.

Space Complexity:

- $O(n)$ for storage of the map elements.
- $O(\log n)$ auxiliary stack space for `walk()`.
- $O(1)$ auxiliary for all other operations.

```

#include <algorithm>
#include <cstdlib>

template<class K, class V>
class RedBlackTree {
    enum Color { RED, BLACK };
    struct Node {
        K key;
        V value;
        Color color;
        Node *left, *right, *parent;

        Node(const K &k, const V &v, Color c)
            : key(k), value(v), color(c), left(nullptr), right(nullptr), parent(nullptr) {}
    } *root, *LEAF_NIL;

    int num_nodes;

    void rotate_left(Node *n) {
        Node *tmp = n->right;
        if ((n->right = tmp->left) != LEAF_NIL) {

```

```

    n->right->parent = n;
}
if ((tmp->parent = n->parent) == LEAF_NIL) {
    root = tmp;
} else if (n->parent->left == n) {
    n->parent->left = tmp;
} else {
    n->parent->right = tmp;
}
tmp->left = n;
n->parent = tmp;
}

void rotate_right(Node *n) {
    Node *tmp = n->left;
    if ((n->left = tmp->right) != LEAF_NIL) {
        n->left->parent = n;
    }
    if ((tmp->parent = n->parent) == LEAF_NIL) {
        root = tmp;
    } else if (n->parent->right == n) {
        n->parent->right = tmp;
    } else {
        n->parent->left = tmp;
    }
    tmp->right = n;
    n->parent = tmp;
}

void insert_fix(Node *n) {
    while (n->parent->color == RED) {
        Node *parent = n->parent;
        Node *grandparent = n->parent->parent;
        if (parent == grandparent->left) {
            Node *uncle = grandparent->right;
            if (uncle->color == RED) {
                grandparent->color = RED;
                parent->color = BLACK;
                uncle->color = BLACK;
                n = grandparent;
            } else {
                if (n == parent->right) {
                    rotate_left(parent);
                    n = parent;
                    parent = n->parent;
                }
                rotate_right(grandparent);
                std::swap(parent->color, grandparent->color);
                n = parent;
            }
        } else if (parent == grandparent->right) {
            Node *uncle = grandparent->left;
            if (uncle->color == RED) {
                grandparent->color = RED;
                parent->color = BLACK;
                uncle->color = BLACK;
                n = grandparent;
            } else {
                if (n == parent->left) {
                    rotate_right(parent);
                    n = parent;
                    parent = n->parent;
                }
                rotate_left(grandparent);
                std::swap(parent->color, grandparent->color);
                n = parent;
            }
        }
    }
}

```



```

    }
    root->color = BLACK;
}

void replace(Node *n, Node *replacement) {
    if (n->parent == LEAF_NIL) {
        root = replacement;
    } else if (n == n->parent->left) {
        n->parent->left = replacement;
    } else {
        n->parent->right = replacement;
    }
    replacement->parent = n->parent;
}

void erase_fix(Node *n) {
    while (n != root && n->color == BLACK) {
        Node *parent = n->parent;
        if (n == parent->left) {
            Node *sibling = parent->right;
            if (sibling->color == RED) {
                sibling->color = BLACK;
                parent->color = RED;
                rotate_left(parent);
                sibling = parent->right;
            }
            if (sibling->left->color == BLACK && sibling->right->color == BLACK) {
                sibling->color = RED;
                n = parent;
            } else {
                if (sibling->right->color == BLACK) {
                    sibling->left->color = BLACK;
                    sibling->color = RED;
                    rotate_right(sibling);
                    sibling = parent->right;
                }
                sibling->color = parent->color;
                parent->color = BLACK;
                sibling->right->color = BLACK;
                rotate_left(parent);
                n = root;
            }
        } else {
            Node *sibling = parent->left;
            if (sibling->color == RED) {
                sibling->color = BLACK;
                parent->color = RED;
                rotate_right(parent);
                sibling = parent->left;
            }
            if (sibling->left->color == BLACK && sibling->right->color == BLACK) {
                sibling->color = RED;
                n = parent;
            } else {
                if (sibling->left->color == BLACK) {
                    sibling->right->color = BLACK;
                    sibling->color = RED;
                    rotate_left(sibling);
                    sibling = parent->left;
                }
                sibling->color = parent->color;
                parent->color = BLACK;
                sibling->left->color = BLACK;
                rotate_right(parent);
                n = root;
            }
        }
    }
}

```

```

    n->color = BLACK;
}

template<class Fn>
void walk(Node *n, Fn f) const {
    if (n != LEAF_NIL) {
        walk(n->left, f);
        f(n->key, n->value);
        walk(n->right, f);
    }
}

void clean_up(Node *n) {
    if (n != LEAF_NIL) {
        clean_up(n->left);
        clean_up(n->right);
        delete n;
    }
}

public:
RedBlackTree() : num_nodes(0) { root = LEAF_NIL = new Node(K(), V(), BLACK); }

~RedBlackTree() {
    clean_up(root);
    delete LEAF_NIL;
}

RedBlackTree(const RedBlackTree &) = delete;
RedBlackTree &operator=(const RedBlackTree &) = delete;
int size() const { return num_nodes; }
bool empty() const { return num_nodes == 0; }

bool insert(const K &k, const V &v) {
    Node *curr = root, *prev = LEAF_NIL;
    while (curr != LEAF_NIL) {
        prev = curr;
        if (k < curr->key) {
            curr = curr->left;
        } else if (curr->key < k) {
            curr = curr->right;
        } else {
            return false;
        }
    }
    Node *n = new Node(k, v, RED);
    n->parent = prev;
    if (prev == LEAF_NIL) {
        root = n;
    } else if (k < prev->key) {
        prev->left = n;
    } else {
        prev->right = n;
    }
    n->left = n->right = LEAF_NIL;
    insert_fix(n);
    num_nodes++;
    return true;
}

bool erase(const K &k) {
    Node *n = root;
    while (n != LEAF_NIL) {
        if (k < n->key) {
            n = n->left;
        } else if (n->key < k) {
            n = n->right;
        } else {

```

```

        break;
    }
}
if (n == LEAF_NIL) {
    return false;
}
Color color = n->color;
Node *replacement;
if (n->left == LEAF_NIL) {
    replacement = n->right;
    replace(n, n->right);
} else if (n->right == LEAF_NIL) {
    replacement = n->left;
    replace(n, n->left);
} else {
    Node *tmp = n->right;
    while (tmp->left != LEAF_NIL) {
        tmp = tmp->left;
    }
    color = tmp->color;
    replacement = tmp->right;
    if (tmp->parent == n) {
        replacement->parent = tmp;
    } else {
        replace(tmp, tmp->right);
        tmp->right = n->right;
        tmp->right->parent = tmp;
    }
    replace(n, tmp);
    tmp->left = n->left;
    tmp->left->parent = tmp;
    tmp->color = n->color;
}
delete n;
if (color == BLACK) {
    erase_fix(replacement);
}
return true;
}

const V *find(const K &k) const {
    Node *n = root;
    while (n != LEAF_NIL) {
        if (k < n->key) {
            n = n->left;
        } else if (n->key < k) {
            n = n->right;
        } else {
            return &(n->value);
        }
    }
    return nullptr;
}

template<class Fn>
void walk(Fn f) const {
    walk(root, f);
}
};

```

Example Usage

```

#include <cassert>
#include <iostream>
using namespace std;

```

```

int main() {
    RedBlackTree<int, char> t;
    t.insert(2, 'b');
    t.insert(1, 'a');
    t.insert(3, 'c');
    t.insert(5, 'e');
    assert(t.insert(4, 'd'));
    assert(*t.find(4) == 'd');
    assert(!t.insert(4, 'd'));
    auto printch = [](int k, char v) { cout << v; };
    t.walk(printch);
    cout << endl;
    assert(t.erase(1));
    assert(!t.erase(1));
    assert(t.find(1) == nullptr);
    t.walk(printch);
    cout << endl;
    return 0;
}

```

Example Output

```

abcde
bcde

```

2.3.5 Splay Tree

Maintain a map, that is, a collection of key-value pairs such that each possible key appears at most once in the collection. This implementation requires an ordering on the set of possible keys defined by `operator<` on the key type. A splay tree is a balanced binary search tree with the additional property that recently accessed elements are quick to access again.

- `SplayTree()` constructs an empty map.
- `size()` returns the size of the map.
- `empty()` returns whether the map is empty.
- `insert(k, v)` adds an entry with key `k` and value `v` to the map, returning `true` if a new entry was added or `false` if the key already exists (in which case the map is unchanged and the old value associated with the key is preserved).
- `erase(k)` removes the entry with key `k` from the map, returning `true` if the removal was successful or `false` if the key to be removed was not found.
- `find(k)` returns a pointer to a const value associated with key `k`, or `nullptr` if the key was not found.
- `walk(f)` calls the function `f(k, v)` on each entry of the map, in ascending order of keys.

Time Complexity:

- $O(1)$ per call to the constructor, `size()`, and `empty()`.
- $O(\log n)$ per call to `insert()`, `erase()`, and `find()`, where n is the number of entries currently in the map.
- $O(n)$ per call to `walk()`.

Space Complexity:

- $O(n)$ for storage of the map elements.
- $O(\log n)$ auxiliary stack space for `insert()`, `erase()`, and `walk()`.
- $O(1)$ auxiliary for all other operations.

```

#include <cstddef>

template<class K, class V>
class SplayTree {
    struct Node {
        K key;
        V value;

```

```

Node *left, *right;

Node(const K &k, const V &v) : key(k), value(v), left(nullptr), right(nullptr) {}
} *root;

int num_nodes;

static void rotate_left(Node *&n) {
    Node *tmp = n;
    n = n->right;
    tmp->right = n->left;
    n->left = tmp;
}

static void rotate_right(Node *&n) {
    Node *tmp = n;
    n = n->left;
    tmp->left = n->right;
    n->right = tmp;
}

static void splay(Node *&n, const K &k) {
    if (n == nullptr) {
        return;
    }
    if (k < n->key && n->left != nullptr) {
        if (k < n->left->key) {
            splay(n->left->left, k);
            rotate_right(n);
        } else if (n->left->key < k) {
            splay(n->left->right, k);
            if (n->left->right != nullptr) {
                rotate_left(n->left);
            }
        }
        if (n->left != nullptr) {
            rotate_right(n);
        }
    } else if (n->key < k && n->right != nullptr) {
        if (k < n->right->key) {
            splay(n->right->left, k);
            if (n->right->left != nullptr) {
                rotate_right(n->right);
            }
        } else if (n->right->key < k) {
            splay(n->right->right, k);
            rotate_left(n);
        }
        if (n->right != nullptr) {
            rotate_left(n);
        }
    }
}

static bool insert(Node *&n, const K &k, const V &v) {
    if (n == nullptr) {
        n = new Node(k, v);
        return true;
    }
    splay(n, k);
    if (k < n->key) {
        Node *tmp = new Node(k, v);
        tmp->left = n->left;
        tmp->right = n;
        n->left = nullptr;
        n = tmp;
    } else if (n->key < k) {
        Node *tmp = new Node(k, v);

```

```

    tmp->left = n;
    tmp->right = n->right;
    n->right = nullptr;
    n = tmp;
} else {
    return false;
}
return true;
}

static bool erase(Node *&n, const K &k) {
    if (n == nullptr) {
        return false;
    }
    splay(n, k);
    if (k < n->key || n->key < k) {
        return false;
    }
    Node *tmp = n;
    if (n->left == nullptr) {
        n = n->right;
    } else {
        splay(n->left, k);
        n = n->left;
        n->right = tmp->right;
    }
    delete tmp;
    return true;
}

template<class Fn>
static void walk(Node *n, Fn f) {
    if (n != nullptr) {
        walk(n->left, f);
        f(n->key, n->value);
        walk(n->right, f);
    }
}

static void clean_up(Node *n) {
    if (n != nullptr) {
        clean_up(n->left);
        clean_up(n->right);
        delete n;
    }
}

public:
    SplayTree() : root(nullptr), num_nodes(0) {}

    ~SplayTree() { clean_up(root); }
    SplayTree(const SplayTree &) = delete;
    SplayTree &operator=(const SplayTree &) = delete;
    int size() const { return num_nodes; }
    bool empty() const { return root == nullptr; }

    bool insert(const K &k, const V &v) {
        if (insert(root, k, v)) {
            num_nodes++;
            return true;
        }
        return false;
    }

    bool erase(const K &k) {
        if (erase(root, k)) {
            num_nodes--;
            return true;
        }
    }

```

```

    }
    return false;
}

const V *find(const K &k) {
    splay(root, k);
    return (k < root->key || root->key < k) ? nullptr : &(root->value);
}

template<class Fn>
void walk(Fn f) const {
    walk(root, f);
}
};

```

Example Usage

```

#include <cassert>
#include <iostream>
using namespace std;

int main() {
    SplayTree<int, char> t;
    t.insert(2, 'b');
    t.insert(1, 'a');
    t.insert(3, 'c');
    t.insert(5, 'e');
    assert(t.insert(4, 'd'));
    assert(*t.find(4) == 'd');
    assert(!t.insert(4, 'd'));
    auto printch = [](int k, char v) { cout << v; };
    t.walk(printch);
    cout << endl;
    assert(t.erase(1));
    assert(!t.erase(1));
    assert(t.find(1) == nullptr);
    t.walk(printch);
    cout << endl;
    return 0;
}

```

Example Output

```

abcde
bcde

```

2.3.6 Size Balanced Tree

Maintain a map, that is, a collection of key-value pairs such that each possible key appears at most once in the collection. In addition, support queries for keys given their ranks as well as queries for the ranks of given keys. This implementation requires an ordering on the set of possible keys defined by `operator<` on the key type. A size balanced tree augments each nodes with the size of its subtree, using it to maintain balance and compute order statistics.

- `SBTree()` constructs an empty map.
- `size()` returns the size of the map.
- `empty()` returns whether the map is empty.
- `insert(k, v)` adds an entry with key `k` and value `v` to the map, returning `true` if a new entry was added or `false` if the key already exists (in which case the map is unchanged and the old value associated with the key is preserved).

- `erase(k)` removes the entry with key `k` from the map, returning `true` if the removal was successful or `false` if the key to be removed was not found.
- `find(k)` returns a pointer to a const value associated with key `k`, or `nullptr` if the key was not found.
- `select(r)` returns a key-value pair of the node with a key of zero-based rank `r` in the map, throwing an exception if the rank is not between 0 and `size() - 1`.
- `rank(k)` returns the zero-based rank of key `k` in the map, throwing an exception if the key was not found in the map.
- `walk(f)` calls the function `f(k, v)` on each entry of the map, in ascending order of keys.

Time Complexity:

- $O(1)$ per call to the constructor, `size()`, and `empty()`.
- $O(\log n)$ per call to `insert()`, `erase()`, `find()`, `select()`, and `rank()`, where n is the number of entries currently in the map.
- $O(n)$ per call to `walk()`.

Space Complexity:

- $O(n)$ for storage of the map elements.
- $O(\log n)$ auxiliary stack space for `insert()`, `erase()`, and `walk()`.
- $O(1)$ auxiliary for all other operations.

```
#include <cstddef>
#include <stdexcept>
#include <utility>

template<class K, class V>
class SBTTree {
    struct Node {
        K key;
        V value;
        int size;
        Node *left, *right;

        Node(const K &k, const V &v) : key(k), value(v), size(1), left(nullptr), right(nullptr) {}

        inline Node *&child(int c) { return (c == 0) ? left : right; }

        void update() {
            size = 1;
            if (left != nullptr) {
                size += left->size;
            }
            if (right != nullptr) {
                size += right->size;
            }
        }
    } *root;

    static inline int size(Node *n) { return (n == nullptr) ? 0 : n->size; }

    static void rotate(Node *&n, int c) {
        Node *tmp = n->child(c);
        n->child(c) = tmp->child(!c);
        tmp->child(!c) = n;
        n->update();
        tmp->update();
        n = tmp;
    }

    static void maintain(Node *&n, int c) {
        if (n == nullptr || n->child(c) == nullptr) {
            return;
        }
        Node *&tmp = n->child(c);
```



```

    if (size(tmp->child(c)) > size(n->child(!c))) {
        rotate(n, c);
    } else if (size(tmp->child(!c)) > size(n->child(!c))) {
        rotate(tmp, !c);
        rotate(n, c);
    } else {
        return;
    }
    maintain(n->left, 0);
    maintain(n->right, 1);
    maintain(n, 0);
    maintain(n, 1);
}

```

```

static bool insert(Node *&n, const K &k, const V &v) {
    if (n == nullptr) {
        n = new Node(k, v);
        return true;
    }
    bool found;
    if (k < n->key) {
        found = insert(n->left, k, v);
        maintain(n, 0);
    } else if (n->key < k) {
        found = insert(n->right, k, v);
        maintain(n, 1);
    } else {
        return false;
    }
    n->update();
    return found;
}

```

```

static bool erase(Node *&n, const K &k) {
    if (n == nullptr) {
        return false;
    }
    bool found;
    int c = (k < n->key);
    if (k < n->key) {
        found = erase(n->left, k);
    } else if (n->key < k) {
        found = erase(n->right, k);
    } else {
        if (n->right == nullptr || n->left == nullptr) {
            Node *tmp = n;
            n = (n->right == nullptr) ? n->left : n->right;
            delete tmp;
            return true;
        }
        Node *p = n->right;
        while (p->left != nullptr) {
            p = p->left;
        }
        n->key = p->key;
        found = erase(n->right, p->key);
    }
    maintain(n, c);
    n->update();
    return found;
}

```

```

static std::pair<K, V> select(Node *n, int r) {
    int rank = size(n->left);
    if (r < rank) {
        return select(n->left, r);
    } else if (r > rank) {
        return select(n->right, r - rank - 1);
    }
}

```

```

    }
    return {n->key, n->value};
}

static int rank(Node *n, const K &k) {
    if (n == nullptr) {
        throw std::runtime_error("Cannot rank key that's not in tree.");
    }
    int r = size(n->left);
    if (k < n->key) {
        return rank(n->left, k);
    } else if (n->key < k) {
        return rank(n->right, k) + r + 1;
    }
    return r;
}

template<class Fn>
static void walk(Node *n, Fn f) {
    if (n != nullptr) {
        walk(n->left, f);
        f(n->key, n->value);
        walk(n->right, f);
    }
}

static void clean_up(Node *n) {
    if (n != nullptr) {
        clean_up(n->left);
        clean_up(n->right);
        delete n;
    }
}

public:
    SBTTree() : root(nullptr) {}

    ~SBTTree() { clean_up(root); }
    SBTTree(const SBTTree &) = delete;
    SBTTree &operator=(const SBTTree &) = delete;
    int size() const { return size(root); }
    bool empty() const { return root == nullptr; }
    bool insert(const K &k, const V &v) { return insert(root, k, v); }
    bool erase(const K &k) { return erase(root, k); }
    int rank(const K &k) const { return rank(root, k); }

    const V *find(const K &k) const {
        Node *n = root;
        while (n != nullptr) {
            if (k < n->key) {
                n = n->left;
            } else if (n->key < k) {
                n = n->right;
            } else {
                return &(n->value);
            }
        }
        return nullptr;
    }

    std::pair<K, V> select(int r) const {
        if (r < 0 || r >= size(root)) {
            throw std::runtime_error("Select rank must be between 0 and size() - 1.");
        }
        return select(root, r);
    }

    template<class Fn>

```

```

void walk(Fn f) const {
    walk(root, f);
}
};

```

Example Usage

```

#include <cassert>
#include <iostream>
using namespace std;

int main() {
    SBTTree<int, char> t;
    t.insert(2, 'b');
    t.insert(1, 'a');
    t.insert(3, 'c');
    t.insert(5, 'e');
    assert(t.insert(4, 'd'));
    assert(*t.find(4) == 'd');
    assert(!t.insert(4, 'd'));
    auto printch = [](int k, char v) { cout << v; };
    t.walk(printch);
    cout << endl;
    assert(t.erase(1));
    assert(!t.erase(1));
    assert(t.find(1) == nullptr);
    t.walk(printch);
    cout << endl;
    assert(t.rank(2) == 0);
    assert(t.rank(3) == 1);
    assert(t.rank(5) == 3);
    assert(t.select(0).first == 2);
    assert(t.select(1).first == 3);
    assert(t.select(2).first == 4);
    return 0;
}

```

Example Output

```

abcde
bcde

```

2.3.7 Interval Treap

Maintain a map from closed, one-dimensional intervals to values while supporting efficient reporting of any or all entries that intersect with a given query interval. This implementation uses `std::pair` to represent intervals, requiring operators `<` and `==` to be defined on the numeric key type. A treap is used to process the entries, where keys are compared lexicographically as pairs.

- `IntervalTreap()` constructs an empty map.
- `size()` returns the size of the map.
- `empty()` returns whether the map is empty.
- `insert(lo, hi, v)` adds an entry with key `[lo, hi]` and value `v` to the map, returning `true` if a new interval was added or `false` if the interval already exists (in which case the map is unchanged and the old value associated with the key is preserved).
- `erase(lo, hi)` removes the entry with key `[lo, hi]` from the map, returning `true` if the removal was successful or `false` if the interval was not found.
- `find_key(lo, hi)` returns a pointer to a const `std::pair` representing the key of some interval in the map which intersects with `[lo, hi]`, or `nullptr` if no such entry was found.

- `find_value(lo, hi)` returns a pointer to a const value of some entry in the map with a key that intersects with `[lo, hi]`, or `nullptr` if no such entry was found.
- `find_all(lo, hi, f)` calls the function `f(lo, hi, v)` on each entry in the map that overlaps with `[lo, hi]`, in lexicographically ascending order of intervals.
- `walk(f)` calls the function `f(lo, hi, v)` on each interval in the map, in lexicographically ascending order of intervals.

Time Complexity:

- $O(1)$ per call to the constructor, `size()`, and `empty()`.
- $O(\log n)$ on average per call to `insert()`, `erase()`, and `find_any()`, where n is the number of intervals currently in the set.
- $O(\log n + m)$ on average per call to `find_all()`, where m is the number of intersecting intervals that are reported.
- $O(n)$ per call to `walk()`.

Space Complexity:

- $O(n)$ for storage of the map elements.
- $O(1)$ auxiliary for `size()` and `empty()`.
- $O(\log n)$ auxiliary stack space on average for all other operations.

```
#include <cstdint>
#include <utility>

template<class K, class V>
class IntervalTreap {
    using Interval = std::pair<K, K>;

    struct Node {
        static uint32_t rand32() {
            static uint32_t x = 123456789;
            x ^= x << 13;
            x ^= x >> 17;
            x ^= x << 5;
            return x;
        }

        Interval interval;
        V value;
        K max;
        uint32_t priority;
        Node *left, *right;

        Node(const Interval &i, const V &v)
            : interval(i), value(v), max(i.second), priority(rand32()), left(nullptr), right(nullptr) {}

        void update() {
            max = interval.second;
            if (left != nullptr && left->max > max) {
                max = left->max;
            }
            if (right != nullptr && right->max > max) {
                max = right->max;
            }
        }
    } *root;

    int num_nodes;

    static void rotate_left(Node *&n) {
        Node *tmp = n;
        n = n->right;
        tmp->right = n->left;
        n->left = tmp;
    }
};
```

```

    tmp->update();
}

static void rotate_right(Node *&n) {
    Node *tmp = n;
    n = n->left;
    tmp->left = n->right;
    n->right = tmp;
    tmp->update();
}

static bool insert(Node *&n, const Interval &i, const V &v) {
    if (n == nullptr) {
        n = new Node(i, v);
        return true;
    }
    if (i < n->interval && insert(n->left, i, v)) {
        if (n->left->priority < n->priority) {
            rotate_right(n);
        }
        n->update();
        return true;
    }
    if (i > n->interval && insert(n->right, i, v)) {
        if (n->right->priority < n->priority) {
            rotate_left(n);
        }
        n->update();
        return true;
    }
    return false;
}

static bool erase(Node *&n, const Interval &i) {
    if (n == nullptr) {
        return false;
    }
    if (i < n->interval) {
        return erase(n->left, i);
    }
    if (i > n->interval) {
        return erase(n->right, i);
    }
    if (n->left != nullptr && n->right != nullptr) {
        bool res;
        if (n->left->priority < n->right->priority) {
            rotate_right(n);
            res = erase(n->right, i);
        } else {
            rotate_left(n);
            res = erase(n->left, i);
        }
        n->update();
        return res;
    }
    Node *tmp = (n->left != nullptr) ? n->left : n->right;
    delete n;
    n = tmp;
    return true;
}

static Node *find_any(Node *n, const Interval &i) {
    if (n == nullptr) {
        return nullptr;
    }
    if (n->interval.first <= i.second && i.first <= n->interval.second) {
        return n;
    }
}

```

```

    if (n->left != nullptr && i.first <= n->left->max) {
        return find_any(n->left, i);
    }
    return find_any(n->right, i);
}

template<class Fn>
static void find_all(Node *n, const Interval &i, Fn f) {
    if (n == nullptr || n->max < i.first) {
        return;
    }
    find_all(n->left, i, f);
    if (n->interval.first <= i.second && i.first <= n->interval.second) {
        f(n->interval.first, n->interval.second, n->value);
    }
    if (n->interval.first <= i.second) {
        find_all(n->right, i, f);
    }
}

template<class Fn>
static void walk(Node *n, Fn f) {
    if (n != nullptr) {
        walk(n->left, f);
        f(n->interval.first, n->interval.second, n->value);
        walk(n->right, f);
    }
}

static void clean_up(Node *n) {
    if (n != nullptr) {
        clean_up(n->left);
        clean_up(n->right);
        delete n;
    }
}

public:
IntervalTreap() : root(nullptr), num_nodes(0) {}

~IntervalTreap() { clean_up(root); }
IntervalTreap(const IntervalTreap &) = delete;
IntervalTreap &operator=(const IntervalTreap &) = delete;
int size() const { return num_nodes; }
bool empty() const { return root == nullptr; }

bool insert(const K &lo, const K &hi, const V &v) {
    if (insert(root, Interval{lo, hi}, v)) {
        num_nodes++;
        return true;
    }
    return false;
}

bool erase(const K &lo, const K &hi) {
    if (erase(root, Interval{lo, hi})) {
        num_nodes--;
        return true;
    }
    return false;
}

const Interval *find_key(const K &lo, const K &hi) const {
    Node *n = find_any(root, Interval{lo, hi});
    return (n == nullptr) ? nullptr : &(n->interval);
}

const V *find_value(const K &lo, const K &hi) const {

```

```

    Node *n = find_any(root, Interval{lo, hi});
    return (n == nullptr) ? nullptr : &(n->value);
}

template<class Fn>
void find_all(const K &lo, const K &hi, Fn f) const {
    find_all(root, Interval{lo, hi}, f);
}

template<class Fn>
void walk(Fn f) const {
    walk(root, f);
}
};

```

Example Usage

```

#include <cassert>
#include <iostream>
using namespace std;

int main() {
    IntervalTreap<int, char> t;
    t.insert(15, 20, 'a');
    t.insert(10, 30, 'b');
    t.insert(17, 19, 'c');
    t.insert(5, 20, 'd');
    t.insert(12, 15, 'e');
    t.insert(10, 40, 'f');
    assert(t.size() == 6);
    assert(!t.insert(5, 20, 'x'));
    t.erase(17, 19);
    assert(t.size() == 5);
    assert(*t.find_key(3, 9) == (pair<int, int>{5, 20}));
    assert(*t.find_value(3, 9) == 'd');
    cout << "Intervals intersecting [16, 20]:";
    auto print = [](int lo, int hi, char v) { cout << " [" << lo << ", " << hi << " ]"; };
    t.find_all(16, 20, print);
    cout << "\nAll intervals:";
    t.walk(print);
    cout << endl;
    return 0;
}

```

Example Output

```

Intervals intersecting [16, 20]: [5, 20] [10, 30] [10, 40] [15, 20]
All intervals: [5, 20] [10, 30] [10, 40] [12, 15] [15, 20]

```

2.3.8 Hash Map

Maintain a map, that is, a collection of key-value pairs such that each possible key appears at most once in the collection. This implementation requires `operator==` to be defined on the key type. A hash map implements a map by hashing keys into buckets using a hash function. This implementation resolves collisions by chaining entries hashed to the same bucket into a linked list.

- `HashMap()` constructs an empty map.
- `size()` returns the size of the map.
- `empty()` returns whether the map is empty.
- `insert(k, v)` adds an entry with key `k` and value `v` to the map, returning `true` if a new entry was added or `false` if the key already exists (in which case the map is unchanged and the old value associated with the key is preserved).

- `erase(k)` removes the entry with key `k` from the map, returning `true` if the removal was successful or `false` if the key to be removed was not found.
- `find(k)` returns a pointer to the value associated with key `k`, or `nullptr` if the key was not found.
- `operator[]` returns a reference to key `k`'s associated value (which may be modified), or if necessary, inserts and returns a new entry with the default constructed value if key `k` was not originally found.
- `walk(f)` calls the function `f(k, v)` on each entry of the map, in no guaranteed order.

Time Complexity:

- $O(1)$ per call to the constructor, `size()`, and `empty()`.
- $O(1)$ amortized per call to `insert()`, `erase()`, `find()`, and `operator[]`.
- $O(n)$ per call to `walk()`, where n is the number of entries in the map.

Space Complexity:

- $O(n)$ for storage of the map elements.
- $O(n)$ auxiliary heap space for `insert()`.
- $O(1)$ auxiliary for all other operations.

```
#include <cstddef>
#include <list>
#include <vector>

template<class K, class V, class Hash>
class HashMap {
    struct HashMapEntry {
        K key;
        V value;

        HashMapEntry(const K &k, const V &v) : key(k), value(v) {}
    };

    std::vector<std::list<HashMapEntry>> table;
    int table_size, num_entries;

    void double_capacity_and_rehash() {
        std::vector<std::list<HashMapEntry>> old = std::move(table);
        table_size = 2 * table_size;
        table.assign(table_size, std::list<HashMapEntry>());
        num_entries = 0;
        for (auto &bucket : old) {
            for (auto &entry : bucket) {
                insert(entry.key, entry.value);
            }
        }
    }

public:
    explicit HashMap(int size = 128) : table(size), table_size(size), num_entries(0) {}

    int size() const { return num_entries; }
    bool empty() const { return num_entries == 0; }

    bool insert(const K &k, const V &v) {
        if (find(k) != nullptr) {
            return false;
        }
        if (num_entries >= table_size) {
            double_capacity_and_rehash();
        }
        unsigned int i = Hash()(k) % table_size;
        table[i].emplace_back(k, v);
        num_entries++;
        return true;
    }
};
```



```

bool erase(const K &k) {
    unsigned int i = Hash()(k) % table_size;
    auto it = table[i].begin();
    while (it != table[i].end() && !(it->key == k)) {
        ++it;
    }
    if (it == table[i].end()) {
        return false;
    }
    table[i].erase(it);
    num_entries--;
    return true;
}

V *find(const K &k) {
    unsigned int i = Hash()(k) % table_size;
    auto it = table[i].begin();
    while (it != table[i].end() && !(it->key == k)) {
        ++it;
    }
    if (it == table[i].end()) {
        return nullptr;
    }
    return &(it->value);
}

V &operator[](const K &k) {
    if (V *ptr = find(k); ptr != nullptr) {
        return *ptr;
    }
    if (num_entries >= table_size) {
        double_capacity_and_rehash();
    }
    unsigned int i = Hash()(k) % table_size;
    table[i].emplace_back(k, V());
    num_entries++;
    return table[i].back().value;
}

template<class Fn>
void walk(Fn f) const {
    for (int i = 0; i < table_size; i++) {
        for (const auto &entry : table[i]) {
            f(entry.key, entry.value);
        }
    }
}

};

```

Example Usage

```

#include <cassert>
#include <iostream>
using namespace std;

struct ClassHash {
    unsigned int operator()(int k) { return ClassHash()(static_cast<unsigned int>(k)); }
    unsigned int operator()(long long k) { return ClassHash()(static_cast<unsigned long long>(k)); }

    // Knuth's one-to-one multiplicative method.
    unsigned int operator()(unsigned int k) {
        return k * 2654435761u; // Or just return k.
    }

    // Jenkins's 64-bit hash.
    unsigned int operator()(unsigned long long k) {

```

```

    k += ~(k << 32);
    k ^= (k >> 22);
    k += ~(k << 13);
    k ^= (k >> 8);
    k += (k << 3);
    k ^= (k >> 15);
    k += ~(k << 27);
    k ^= (k >> 31);
    return k;
}

// Jenkins's one-at-a-time hash.
unsigned int operator()(const std::string &k) {
    unsigned int hash = 0;
    for (char c : k) {
        hash += ((hash + static_cast<unsigned char>(c)) << 10);
        hash ^= (hash >> 6);
    }
    hash += (hash << 3);
    hash ^= (hash >> 11);
    return hash + (hash << 15);
}
};

int main() {
    HashMap<string, char, ClassHash> m;
    m["foo"] = 'a';
    m.insert("bar", 'b');
    assert(m["foo"] == 'a');
    assert(m["bar"] == 'b');
    assert(m["baz"] == '\0');
    m["baz"] = 'c';
    m.walk([](const string &k, char v) { cout << v; });
    cout << endl;
    assert(m.erase("foo"));
    assert(m.size() == 2);
    assert(m["foo"] == '\0');
    assert(m.size() == 3);
    return 0;
}

```

Example Output

```
cab
```

2.3.9 Skip List

Maintain a map, that is, a collection of key-value pairs such that each possible key appears at most once in the collection. This implementation requires operators `<` and `==` to be defined on the key type. A skip list maintains a linked hierarchy of sorted subsequences with each successive subsequence skipping over fewer elements than the previous one.

- `SkipList()` constructs an empty map.
- `size()` returns the size of the map.
- `empty()` returns whether the map is empty.
- `insert(k, v)` adds an entry with key `k` and value `v` to the map, returning `true` if a new entry was added or `false` if the key already exists (in which case the map is unchanged and the old value associated with the key is preserved).
- `erase(k)` removes the entry with key `k` from the map, returning `true` if the removal was successful or `false` if the key to be removed was not found.
- `find(k)` returns a pointer to the value associated with key `k`, or `nullptr` if the key was not found.

- `operator[]` returns a reference to key `k`'s associated value (which may be modified), or if necessary, inserts and returns a new entry with the default constructed value if key `k` was not originally found.
- `walk(f)` calls the function `f(k, v)` on each entry of the map, in ascending order of keys.

Time Complexity:

- $O(1)$ per call to the constructor, `size()`, and `empty()`.
- $O(\log n)$ on average per call to `insert()`, `erase()`, `find()`, and `operator[]`, where n is the number of entries currently in the map.
- $O(n)$ per call to `walk()`.

Space Complexity:

- $O(n)$ on average for storage of the map elements.
- $O(n)$ auxiliary heap space for `insert()` and `erase()`.
- $O(1)$ auxiliary for all other operations.

```
#include <random>
#include <vector>

template<class K, class V>
class SkipList {
    static const int MAX_LEVELS = 32; // log2(max possible keys)

    struct Node {
        K key;
        V value;
        std::vector<Node *> next;

        Node(const K &k, const V &v, int levels) : key(k), value(v), next(levels, (Node *)nullptr) {}
    } *head;

    int num_nodes;

    static int random_level() {
        static std::mt19937 rng(std::random_device{}());
        static std::uniform_int_distribution<int> coin(0, 1);
        int level = 1;
        while (coin(rng) && level < MAX_LEVELS) {
            level++;
        }
        return level;
    }

    static int node_level(const std::vector<Node *> &v) {
        int i = 0;
        while (i < static_cast<int>(v.size()) && v[i] != nullptr) {
            i++;
        }
        return i + 1;
    }

public:
    SkipList() : head(new Node(K(), V(), MAX_LEVELS)), num_nodes(0) {
        for (auto &ptr : head->next) {
            ptr = nullptr;
        }
    }

    ~SkipList() { delete head; }
    SkipList(const SkipList &) = delete;
    SkipList &operator=(const SkipList &) = delete;
    int size() const { return num_nodes; }
    bool empty() const { return num_nodes == 0; }

    bool insert(const K &k, const V &v) {
        std::vector<Node *> update(head->next);
```

```

    int curr_level = node_level(update);
    Node *n = head;
    for (int i = curr_level; i-- > 0;) {
        while (n->next[i] != nullptr && n->next[i]->key < k) {
            n = n->next[i];
        }
        update[i] = n;
    }
    n = n->next[0];
    if (n != nullptr && n->key == k) {
        return false;
    }
    int new_level = random_level();
    if (new_level > curr_level) {
        for (int i = curr_level; i < new_level; i++) {
            update[i] = head;
        }
    }
    n = new Node(k, v, new_level);
    for (int i = 0; i < new_level; i++) {
        n->next[i] = update[i]->next[i];
        update[i]->next[i] = n;
    }
    num_nodes++;
    return true;
}

bool erase(const K &k) {
    std::vector<Node *> update(head->next);
    Node *n = head;
    for (int i = node_level(update); i-- > 0;) {
        while (n->next[i] != nullptr && n->next[i]->key < k) {
            n = n->next[i];
        }
        update[i] = n;
    }
    n = n->next[0];
    if (n != nullptr && n->key == k) {
        for (int i = 0, levels = static_cast<int>(update.size()); i < levels; i++) {
            if (update[i]->next[i] != n) {
                break;
            }
            update[i]->next[i] = n->next[i];
        }
        delete n;
        num_nodes--;
        return true;
    }
    return false;
}

V *find(const K &k) const {
    Node *n = head;
    for (int i = node_level(n->next); i-- > 0;) {
        while (n->next[i] != nullptr && n->next[i]->key < k) {
            n = n->next[i];
        }
    }
    n = n->next[0];
    return (n != nullptr && n->key == k) ? &(n->value) : nullptr;
}

V &operator[](const K &k) {
    V *ptr = find(k);
    if (ptr != nullptr) {
        return *ptr;
    }
    insert(k, V());
}

```

```

    return *find(k);
}

template<class Fn>
void walk(Fn f) const {
    Node *n = head->next[0];
    while (n != nullptr) {
        f(n->key, n->value);
        n = n->next[0];
    }
}
};

```

Example Usage

```

#include <cassert>
#include <iostream>
using namespace std;

int main() {
    SkipList<int, char> l;
    l.insert(2, 'b');
    l.insert(1, 'a');
    l.insert(3, 'c');
    l.insert(5, 'e');
    assert(l.insert(4, 'd'));
    assert(*l.find(4) == 'd');
    assert(!l.insert(4, 'd'));
    auto printch = [](int k, char v) { cout << v; };
    l.walk(printch);
    cout << endl;
    assert(l.erase(1));
    assert(!l.erase(1));
    assert(l.find(1) == nullptr);
    l.walk(printch);
    cout << endl;
    return 0;
}

```

Example Output

```

abcde
bcde

```

2.4 Range Queries in One Dimension

2.4.1 Sparse Table (Range Minimum Query)

Given a static array with indices from 0 to $n - 1$, precompute a table that may later be used to perform range minimum queries on the array in constant time. This version is simplified to only work on integer arrays.

The dynamic programming state `dp[i][j]` holds the index of the minimum value in the sub-array starting at i and having length 2^j . Each `dp[i][j]` will always be equal to either `dp[i][j - 1]` or `dp[i + (1 << (j - 1))][j - 1]`, whichever of the indices corresponds to the smaller value in the array.

Time Complexity:

- $O(n \log n)$ per call to `build()`, where n is the size of the array.
- $O(1)$ per call to `query()`.

Space Complexity:

- $O(n \log n)$ for storage of the sparse table, where n is the size of the array.
- $O(1)$ auxiliary for `query()`.

```
#include <vector>

std::vector<int> table;
std::vector<std::vector<int>> dp;

void build(const std::vector<int> &a) {
    int n = a.size();
    table.resize(n + 1);
    dp.assign(n, std::vector<int>());
    for (int i = 2; i <= n; i++) {
        table[i] = table[i >> 1] + 1;
    }
    for (int i = 0; i < n; i++) {
        dp[i].resize(table[n] + 1);
        dp[i][0] = i;
    }
    for (int j = 1; (1 << j) <= n; j++) {
        for (int i = 0; i + (1 << j) <= n; i++) {
            int x = dp[i][j - 1];
            int y = dp[i + (1 << (j - 1))][j - 1];
            dp[i][j] = (a[x] < a[y]) ? x : y;
        }
    }
}

int query(const std::vector<int> &a, int lo, int hi) {
    int j = table[hi - lo + 1];
    int x = dp[lo][j];
    int y = dp[hi - (1 << j) + 1][j];
    return (a[x] < a[y]) ? a[x] : a[y];
}
```

Example Usage

```
#include <cassert>

int main() {
    std::vector<int> a{6, -2, 1, 8, 10};
    build(a);
    assert(query(a, 0, 3) == -2);
    return 0;
}
```

2.4.2 Square Root Decomposition

Maintain a fixed-size array (from 0 to `size() - 1`) while supporting dynamic queries of contiguous sub-arrays and dynamic updates of individual indices.

The query operation is defined by an associative aggregate function `combine(a, b)`. The default definition below assumes a numerical array type, supporting queries for the “min” of the target range. Another possible query operation is “sum”, in which case `combine(a, b)` should return $a + b$.

The point update operation is defined by `apply_delta(v, d)`, which returns the new value at a single updated index. The default definition below supports updates that “set” the chosen array index to a new value. Another possible update operation is “increment”, in which case `apply_delta(v, d)` should return $v + d$.

The operations supported by this data structure are identical to those of the point update segment tree found in this section.

Time Complexity:

- $O(n)$ per call to both constructors, where n is the size of the array.
- $O(1)$ per call to `size()`.
- $O(\sqrt{n})$ per call to `at()`, `update()`, and `query()`.

Space Complexity:

- $O(n)$ for storage of the array elements.
- $O(1)$ auxiliary for all operations.

```
#include <algorithm>
#include <cmath>
#include <vector>

template<class T>
class SqrtDecomposition {
    static T combine(const T &a, const T &b) { return std::min(a, b); }
    static T apply_delta(const T &v, const T &d) { return d; }

    int len, blocklen;
    std::vector<T> value, block;

    void init() {
        blocklen = std::max(1, static_cast<int>(sqrt(len)));
        int nblocks = (len + blocklen - 1) / blocklen;
        for (int i = 0; i < nblocks; i++) {
            T blockval = value[i * blocklen];
            int blockhi = std::min(len, (i + 1) * blocklen);
            for (int j = i * blocklen + 1; j < blockhi; j++) {
                blockval = combine(blockval, value[j]);
            }
            block.push_back(blockval);
        }
    }

public:
    explicit SqrtDecomposition(int n, const T &v = T()) : len(n), value(n, v) { init(); }

    template<class It>
    SqrtDecomposition(It lo, It hi) : len(hi - lo), value(lo, hi) {
        init();
    }

    int size() const { return len; }
    T at(int i) const { return query(i, i); }

    T query(int lo, int hi) const {
        T res;
        int blocklo = ceil(static_cast<double>(lo) / blocklen), blockhi = (hi + 1) / blocklen - 1;
        if (blocklo > blockhi) {
            res = value[lo];
            for (int i = lo + 1; i <= hi; i++) {
                res = combine(res, value[i]);
            }
        } else {
            res = block[blocklo];
            for (int i = blocklo + 1; i <= blockhi; i++) {
                res = combine(res, block[i]);
            }
            for (int i = lo; i < blocklo * blocklen; i++) {
                res = combine(res, value[i]);
            }
            for (int i = (blockhi + 1) * blocklen; i <= hi; i++) {
```

```

        res = combine(res, value[i]);
    }
}
return res;
}

void update(int i, const T &d) {
    value[i] = apply_delta(value[i], d);
    int b = i / blocklen;
    int blockhi = std::min(len, (b + 1) * blocklen);
    block[b] = value[b * blocklen];
    for (int i = b * blocklen + 1; i < blockhi; i++) {
        block[b] = combine(block[b], value[i]);
    }
}
};

```

Example Usage

```

#include <cassert>
#include <iostream>
using namespace std;

int main() {
    vector<int> arr{6, -2, 1, 8, 10};
    SqrtDecomposition<int> sd(arr.begin(), arr.end());
    sd.update(2, 4);
    cout << "Values: ";
    for (int i = 0; i < sd.size(); i++) {
        cout << " " << sd.at(i);
    }
    cout << endl;
    assert(sd.query(0, 3) == -2);
    return 0;
}

```

Example Output

Values: 6 -2 4 8 10

2.4.3 Segment Tree (Point Update)

Maintain a fixed-size array while supporting dynamic queries of contiguous sub-arrays and dynamic updates of individual indices.

The query operation is defined by an associative aggregate function `combine(a, b)`. The default code below assumes a numerical array type, defining queries for the “min” of the target range. Another possible query operation is “sum”, in which case `combine(a, b)` should return $a + b$.

The point update operation is defined by `apply_delta(v, d)`, which returns the new value at a single updated index. The default definition below supports updates that “set” the chosen array index to a new value. Another possible update operation is “increment”, in which case `apply_delta(v, d)` should return $v + d$.

- `SegTree(n, v)` constructs an array of size `n` with indices from 0 to `n - 1`, inclusive, and all values initialized to `v`.
- `SegTree(lo, hi)` constructs an array from two random-access iterators as a range `[lo, hi)`, initialized to the elements of the range in the same order.
- `size()` returns the size of the array.
- `at(i)` returns the value at index `i`.
- `query(lo, hi)` returns the result of `combine()` applied to all indices from `lo` to `hi`, inclusive. If `lo == hi`, then the single specified value is returned.

- `update(i, d)` assigns the value `v` at index `i` to `apply_delta(v, d)`.

Time Complexity:

- $O(n)$ per call to both constructors, where n is the size of the array.
- $O(1)$ per call to `size()`.
- $O(\log n)$ per call to `at()`, `update()`, and `query()`.

Space Complexity:

- $O(n)$ for storage of the array elements.
- $O(\log n)$ auxiliary stack space for `update()` and `query()`.
- $O(1)$ auxiliary for `size()`.

```
#include <algorithm>
#include <vector>

template<class T>
class SegTree {
    static T combine(const T &a, const T &b) { return std::min(a, b); }
    static T apply_delta(const T &v, const T &d) { return d; }

    int len;
    std::vector<T> value;

    void build(int i, int lo, int hi, const T &v) {
        if (lo == hi) {
            value[i] = v;
            return;
        }
        int mid = lo + (hi - lo) / 2;
        build(i * 2 + 1, lo, mid, v);
        build(i * 2 + 2, mid + 1, hi, v);
        value[i] = combine(value[i * 2 + 1], value[i * 2 + 2]);
    }

    template<class It>
    void build(int i, int lo, int hi, It arr) {
        if (lo == hi) {
            value[i] = *(arr + lo);
            return;
        }
        int mid = lo + (hi - lo) / 2;
        build(i * 2 + 1, lo, mid, arr);
        build(i * 2 + 2, mid + 1, hi, arr);
        value[i] = combine(value[i * 2 + 1], value[i * 2 + 2]);
    }

    T query(int i, int lo, int hi, int tgt_lo, int tgt_hi) const {
        if (lo == tgt_lo && hi == tgt_hi) {
            return value[i];
        }
        int mid = lo + (hi - lo) / 2;
        if (tgt_lo <= mid && mid < tgt_hi) {
            return combine(
                query(i * 2 + 1, lo, mid, tgt_lo, std::min(tgt_hi, mid)),
                query(i * 2 + 2, mid + 1, hi, std::max(tgt_lo, mid + 1), tgt_hi)
            );
        }
        if (tgt_lo <= mid) {
            return query(i * 2 + 1, lo, mid, tgt_lo, std::min(tgt_hi, mid));
        }
        return query(i * 2 + 2, mid + 1, hi, std::max(tgt_lo, mid + 1), tgt_hi);
    }

    void update(int i, int lo, int hi, int target, const T &d) {
        if (target < lo || target > hi) {
```

```

        return;
    }
    if (lo == hi) {
        value[i] = apply_delta(value[i], d);
        return;
    }
    int mid = lo + (hi - lo) / 2;
    update(i * 2 + 1, lo, mid, target, d);
    update(i * 2 + 2, mid + 1, hi, target, d);
    value[i] = combine(value[i * 2 + 1], value[i * 2 + 2]);
}

public:
    explicit SegTree(int n, const T &v = T()) : len(n), value(4 * len) {
        if (len > 0) {
            build(0, 0, len - 1, v);
        }
    }

    template<class It>
    SegTree(It lo, It hi) : len(hi - lo), value(4 * len) {
        if (len > 0) {
            build(0, 0, len - 1, lo);
        }
    }

    int size() const { return len; }
    T at(int i) const { return query(i, i); }
    T query(int lo, int hi) const { return query(0, 0, len - 1, lo, hi); }
    void update(int i, const T &d) { update(0, 0, len - 1, i, d); }
};

```

Example Usage

```

#include <cassert>
#include <iostream>
using namespace std;

int main() {
    vector<int> arr{6, -2, 1, 8, 10};
    SegTree<int> t(arr.begin(), arr.end());
    t.update(2, 4);
    cout << "Values:";
    for (int i = 0; i < t.size(); i++) {
        cout << " " << t.at(i);
    }
    cout << endl;
    cout << "The minimum value in the range [0, 3] is " << t.query(0, 3) << "." << endl;
    assert(t.query(0, 3) == -2);
    return 0;
}

```

Example Output

```

Values: 6 -2 4 8 10
The minimum value in the range [0, 3] is -2.

```

2.4.4 Segment Tree (Range Update)

Maintain a fixed-size array while supporting both dynamic queries and updates of contiguous subarrays via the lazy propagation technique.

The query operation is defined by an associative aggregate function `combine(a, b)`. The default code below assumes a numerical array type, defining queries for the “min” of the target range. Another possible query operation is “sum”, in which case `combine(a, b)` should return $a + b$.

Range updates are defined by `apply_delta(v, d, len)`, which applies an update delta `d` to an aggregate summary `v` representing `len` array values, and by `compose_deltas(old, d)`, which combines a pending older delta with a newer delta in that order. These functions do not support arbitrary combinations: applying a delta to a combined segment must be equivalent to applying it to each child segment and then combining the results, and composed deltas must be equivalent to performing their updates sequentially. The default code below defines range assignment. For range increment, `compose_deltas(old, d)` should return `old + d`; `apply_delta(v, d, len)` should return `v + d` for range-min/range-max queries, and `v + d * len` for range-sum queries.

- `SegTree(n, v)` constructs an array of size `n` with indices from 0 to `n - 1`, inclusive, and all values initialized to `v`.
- `SegTree(lo, hi)` constructs an array from two random-access iterators as a range `[lo, hi)`, initialized to the elements of the range in the same order.
- `size()` returns the size of the array.
- `at(i)` returns the value at index `i`, where `i` is between 0 and `size() - 1`.
- `query(lo, hi)` returns the result of `combine()` applied to all indices from `lo` to `hi`, inclusive. If `lo == hi`, then the single specified value is returned.
- `update(i, d)` assigns the value `v` at index `i` to `apply_delta(v, d)`.
- `update(lo, hi, d)` modifies the value at each array index from `lo` to `hi`, inclusive, by applying the delta `d` to each value.

Time Complexity:

- $O(n)$ per call to both constructors, where n is the size of the array.
- $O(1)$ per call to `size()`.
- $O(\log n)$ per call to `at()`, `update()`, and `query()`.

Space Complexity:

- $O(n)$ for storage of the array elements.
- $O(\log n)$ auxiliary stack space for `update()` and `query()`.
- $O(1)$ auxiliary for `size()`.

```
#include <algorithm>
#include <vector>

template<class T>
class SegTree {
    static T combine(const T &a, const T &b) { return std::min(a, b); }
    static T apply_delta(const T &v, const T &d, long long len) { return d; }
    static T compose_deltas(const T &d1, const T &d2) { return d2; }

    int len;
    std::vector<T> value, delta;
    std::vector<bool> pending;

    void build(int i, int lo, int hi, const T &v) {
        if (lo == hi) {
            value[i] = v;
            return;
        }
        int mid = lo + (hi - lo) / 2;
        build(i * 2 + 1, lo, mid, v);
        build(i * 2 + 2, mid + 1, hi, v);
        value[i] = combine(value[i * 2 + 1], value[i * 2 + 2]);
    }
};

template<class It>
void build(int i, int lo, int hi, It arr) {
    if (lo == hi) {
```

```

    value[i] = *(arr + lo);
    return;
}
int mid = lo + (hi - lo) / 2;
build(i * 2 + 1, lo, mid, arr);
build(i * 2 + 2, mid + 1, hi, arr);
value[i] = combine(value[i * 2 + 1], value[i * 2 + 2]);
}

void push_delta(int i, int lo, int hi) {
    if (pending[i]) {
        value[i] = apply_delta(value[i], delta[i], hi - lo + 1);
        if (lo != hi) {
            int l = 2 * i + 1, r = 2 * i + 2;
            delta[l] = pending[l] ? compose_deltas(delta[l], delta[i]) : delta[i];
            delta[r] = pending[r] ? compose_deltas(delta[r], delta[i]) : delta[i];
            pending[l] = pending[r] = true;
        }
        pending[i] = false;
    }
}

T query(int i, int lo, int hi, int tgt_lo, int tgt_hi) {
    push_delta(i, lo, hi);
    if (lo == tgt_lo && hi == tgt_hi) {
        return value[i];
    }
    int mid = lo + (hi - lo) / 2;
    if (tgt_lo <= mid && mid < tgt_hi) {
        return combine(
            query(i * 2 + 1, lo, mid, tgt_lo, std::min(tgt_hi, mid)),
            query(i * 2 + 2, mid + 1, hi, std::max(tgt_lo, mid + 1), tgt_hi)
        );
    }
    if (tgt_lo <= mid) {
        return query(i * 2 + 1, lo, mid, tgt_lo, std::min(tgt_hi, mid));
    }
    return query(i * 2 + 2, mid + 1, hi, std::max(tgt_lo, mid + 1), tgt_hi);
}

void update(int i, int lo, int hi, int tgt_lo, int tgt_hi, const T &d) {
    push_delta(i, lo, hi);
    if (hi < tgt_lo || lo > tgt_hi) {
        return;
    }
    if (tgt_lo <= lo && hi <= tgt_hi) {
        delta[i] = d;
        pending[i] = true;
        push_delta(i, lo, hi);
        return;
    }
    update(2 * i + 1, lo, lo + (hi - lo) / 2, tgt_lo, tgt_hi, d);
    update(2 * i + 2, lo + (hi - lo) / 2 + 1, hi, tgt_lo, tgt_hi, d);
    value[i] = combine(value[2 * i + 1], value[2 * i + 2]);
}

public:
explicit SegTree(int n, const T &v = T())
    : len(n), value(4 * len), delta(4 * len), pending(4 * len, false) {
    if (len > 0) {
        build(0, 0, len - 1, v);
    }
}

template<class It>
SegTree(It lo, It hi) : len(hi - lo), value(4 * len), delta(4 * len), pending(4 * len, false) {
    if (len > 0) {
        build(0, 0, len - 1, lo);
    }
}

```

```

    }
}

int size() const { return len; }
T at(int i) { return query(i, i); }
T query(int lo, int hi) { return query(0, 0, len - 1, lo, hi); }
void update(int i, const T &d) { update(0, 0, len - 1, i, i, d); }
void update(int lo, int hi, const T &d) { update(0, 0, len - 1, lo, hi, d); }
};

```

Example Usage

```

#include <cassert>
#include <iostream>
using namespace std;

int main() {
    vector<int> arr{6, -2, 1, 8, 10};
    SegTree<int> t(arr.begin(), arr.end());
    t.update(2, 4);
    cout << "Values:";
    for (int i = 0; i < t.size(); i++) {
        cout << " " << t.at(i);
    }
    cout << endl;
    assert(t.query(0, 3) == -2);
    t.update(0, 4, 5);
    t.update(3, 2);
    t.update(3, 1);
    cout << "Values:";
    for (int i = 0; i < t.size(); i++) {
        cout << " " << t.at(i);
    }
    cout << endl;
    assert(t.query(0, 3) == 1);
    return 0;
}

```

Example Output

```

Values: 6 -2 4 8 10
Values: 5 5 5 1 5

```

2.4.5 Segment Tree (Compressed)

Maintain an array over a large index range while supporting both dynamic queries and updates of contiguous subarrays via the lazy propagation technique. This implementation uses lazy initialization of nodes to conserve memory: only the nodes covering touched indices are ever allocated, so indices up to $\mathbb{N}$ are supported without preallocating the whole tree.

The query operation is defined by an associative aggregate function `combine(a, b)`. Since untouched nodes are implicit, `repeat_value(v, len)` must return the aggregate summary of `len` copies of the initial value `v`. The default code below assumes a numerical array type, defining queries for the “min” of the target range. For range-sum queries, `combine(a, b)` should return $a + b$ and `repeat_value(v, len)` should return $v \cdot len$.

Range updates are defined by `apply_delta(v, d, len)`, which applies an update delta `d` to an aggregate summary `v` representing `len` array values, and by `compose_deltas(old, d)`, which combines a pending older delta with a newer delta in that order. These functions do not support arbitrary combinations: applying a delta to a combined segment must be equivalent to applying it to each child segment and then combining the results, and composed deltas must be equivalent to performing their updates sequentially. The default code below defines range

assignment. For range increment, `compose_deltas(old, d)` should return `old + d`; `apply_delta(v, d, len)` should return `v + d` for range-min/range-max queries, and `v + d * len` for range-sum queries.

- `SegTree(v)` constructs an array over indices 0 to N , inclusive, with every value implicitly initialized to `v`. Nodes are allocated lazily as indices are touched.
- `at(i)` returns the value at index `i`, where `i` is between 0 and N .
- `query(lo, hi)` returns the result of `combine()` applied to all indices from `lo` to `hi`, inclusive. If `lo == hi`, then the single specified value is returned.
- `update(i, d)` assigns the value `v` at index `i` to `apply_delta(v, d)`.
- `update(lo, hi, d)` modifies the value at each array index from `lo` to `hi`, inclusive, by applying the delta `d` to each value.

Time Complexity:

- $O(1)$ per call to the constructor.
- $O(\log N)$ per call to `at()`, `update()`, and `query()`.

Space Complexity:

- $O(k \log(N))$ for storage after k index updates.
- $O(\log N)$ auxiliary stack space for `update()` and `query()`.

```
#include <algorithm>
#include <cstdlib>

template<class T>
class SegTree {
    static const int N = 1000000000;

    static T combine(const T &a, const T &b) { return std::min(a, b); }
    static T repeat_value(const T &v, long long len) { return v; }
    static T apply_delta(const T &v, const T &d, long long len) { return d; }
    static T compose_deltas(const T &d1, const T &d2) { return d2; }

    struct Node {
        T value, delta;
        bool pending;
        Node *left, *right;

        explicit Node(const T &v) : value(v), pending(false), left(nullptr), right(nullptr) {}
    } *root;

    T init;

    void update_delta(Node *&n, const T &d, long long len) {
        if (n == nullptr) {
            n = new Node(repeat_value(init, len));
        }
        n->delta = n->pending ? compose_deltas(n->delta, d) : d;
        n->pending = true;
    }

    void push_delta(Node *n, int lo, int hi) {
        if (n == nullptr) {
            return;
        }
        if (n->pending) {
            n->value = apply_delta(n->value, n->delta, hi - lo + 1);
            if (lo != hi) {
                int mid = lo + (hi - lo) / 2;
                update_delta(n->left, n->delta, mid - lo + 1);
                update_delta(n->right, n->delta, hi - mid);
            }
        }
        n->pending = false;
    }
};
```

```

T query(Node *n, int lo, int hi, int tgt_lo, int tgt_hi) {
    if (n == nullptr) {
        return repeat_value(init, hi - lo + 1);
    }
    push_delta(n, lo, hi);
    if (lo == tgt_lo && hi == tgt_hi) {
        return n->value;
    }
    int mid = lo + (hi - lo) / 2;
    if (tgt_lo <= mid && mid < tgt_hi) {
        return combine(
            query(n->left, lo, mid, tgt_lo, std::min(tgt_hi, mid)),
            query(n->right, mid + 1, hi, std::max(tgt_lo, mid + 1), tgt_hi)
        );
    }
    if (tgt_lo <= mid) {
        return query(n->left, lo, mid, tgt_lo, std::min(tgt_hi, mid));
    }
    return query(n->right, mid + 1, hi, std::max(tgt_lo, mid + 1), tgt_hi);
}

void update(Node *&n, int lo, int hi, int tgt_lo, int tgt_hi, const T &d) {
    if (n == nullptr) {
        if (hi < tgt_lo || lo > tgt_hi) {
            return;
        }
        n = new Node(repeat_value(init, hi - lo + 1));
    } else {
        push_delta(n, lo, hi);
    }
    if (hi < tgt_lo || lo > tgt_hi) {
        return;
    }
    if (tgt_lo <= lo && hi <= tgt_hi) {
        n->delta = d;
        n->pending = true;
        push_delta(n, lo, hi);
        return;
    }
    int mid = lo + (hi - lo) / 2;
    update(n->left, lo, mid, tgt_lo, tgt_hi, d);
    update(n->right, mid + 1, hi, tgt_lo, tgt_hi, d);
    T left_value = (n->left != nullptr) ? n->left->value : repeat_value(init, mid - lo + 1);
    T right_value = (n->right != nullptr) ? n->right->value : repeat_value(init, hi - mid);
    n->value = combine(left_value, right_value);
}

void clean_up(Node *n) {
    if (n != nullptr) {
        clean_up(n->left);
        clean_up(n->right);
        delete n;
    }
}

public:
    explicit SegTree(const T &v = T()) : root(nullptr), init(v) {}

    ~SegTree() { clean_up(root); }
    SegTree(const SegTree &) = delete;
    SegTree &operator=(const SegTree &) = delete;
    T at(int i) { return query(i, i); }
    T query(int lo, int hi) { return query(root, 0, N, lo, hi); }
    void update(int i, const T &d) { return update(i, i, d); }
    void update(int lo, int hi, const T &d) { return update(root, 0, N, lo, hi, d); }
};

```

Example Usage

```

#include <cassert>
#include <iostream>
using namespace std;

int main() {
    SegTree<int> t(0);
    t.update(0, 6);
    t.update(1, -2);
    t.update(2, 4);
    t.update(3, 8);
    t.update(4, 10);
    cout << "Values:";
    for (int i = 0; i < 5; i++) {
        cout << " " << t.at(i);
    }
    cout << endl;
    assert(t.query(0, 3) == -2);
    t.update(0, 4, 5);
    t.update(3, 2);
    t.update(3, 1);
    cout << "Values:";
    for (int i = 0; i < 5; i++) {
        cout << " " << t.at(i);
    }
    cout << endl;
    assert(t.query(0, 3) == 1);
    return 0;
}

```

Example Output

```

Values: 6 -2 4 8 10
Values: 5 5 5 1 5

```

2.4.6 Segment Tree (Persistent)

Maintains immutable versions of a segment tree under point updates. Each update creates a new root by copying only the nodes on the path from the root to the updated leaf, while all unchanged subtrees are shared with previous versions.

The implementation below stores sums over a fixed index range $[0, n - 1]$. Persistent segment trees are useful for offline order-statistics queries, rollback-like histories, and functional dynamic programming states.

- `PersistentSegTree(a)` constructs version 0 from array `a`.
- `update(version, index, value)` returns a new version where `a[index]` is set to `value`, leaving the old version unchanged.
- `query(version, lo, hi)` returns the sum of the inclusive range `[lo, hi]` in the specified version.
- `versions()` returns the number of stored roots.

Time Complexity:

- $O(n)$ for construction.
- $O(\log n)$ per call to `update(version, index, value)` and `query(version, lo, hi)`.

Space Complexity:

- $O(n + q \log n)$ for q updates.
- $O(\log n)$ auxiliary stack space per update and query.

```

#include <vector>

```



```

class PersistentSegTree {
    struct node {
        long long sum;
        int left, right;

        node(long long sum = 0, int left = -1, int right = -1) : sum(sum), left(left), right(right) {}
    };

    int n;
    std::vector<node> tree;
    std::vector<int> root;

    int build(const std::vector<int> &a, int lo, int hi) {
        if (lo == hi) {
            tree.emplace_back(a[lo]);
            return static_cast<int>(tree.size()) - 1;
        }
        int mid = lo + (hi - lo) / 2;
        int left = build(a, lo, mid);
        int right = build(a, mid + 1, hi);
        tree.emplace_back(tree[left].sum + tree[right].sum, left, right);
        return static_cast<int>(tree.size()) - 1;
    }

    int update(int cur, int lo, int hi, int index, int value) {
        if (lo == hi) {
            tree.emplace_back(value);
            return static_cast<int>(tree.size()) - 1;
        }
        int mid = lo + (hi - lo) / 2;
        int left = tree[cur].left, right = tree[cur].right;
        if (index <= mid) {
            left = update(left, lo, mid, index, value);
        } else {
            right = update(right, mid + 1, hi, index, value);
        }
        tree.emplace_back(tree[left].sum + tree[right].sum, left, right);
        return static_cast<int>(tree.size()) - 1;
    }

    long long query(int cur, int lo, int hi, int qlo, int qhi) const {
        if (qlo <= lo && hi <= qhi) {
            return tree[cur].sum;
        }
        int mid = lo + (hi - lo) / 2;
        long long res = 0;
        if (qlo <= mid) {
            res += query(tree[cur].left, lo, mid, qlo, qhi);
        }
        if (mid < qhi) {
            res += query(tree[cur].right, mid + 1, hi, qlo, qhi);
        }
        return res;
    }

public:
    explicit PersistentSegTree(const std::vector<int> &a) : n(a.size()) {
        if (n > 0) {
            root.push_back(build(a, 0, n - 1));
        }
    }

    int versions() const { return root.size(); }

    int update(int version, int index, int value) {
        root.push_back(update(root[version], 0, n - 1, index, value));
        return static_cast<int>(root.size()) - 1;
    }
}

```

```

long long query(int version, int lo, int hi) const {
    return query(root[version], 0, n - 1, lo, hi);
}
};

```

Example Usage

```

#include <cassert>
using namespace std;

int main() {
    vector<int> a{1, 2, 3, 4};
    PersistentSegTree t(a);
    int v1 = t.update(0, 1, 10);
    int v2 = t.update(v1, 3, -1);
    assert(t.query(0, 0, 3) == 10);
    assert(t.query(v1, 0, 3) == 18);
    assert(t.query(v2, 1, 3) == 12);
    assert(t.versions() == 3);
    return 0;
}

```

2.4.7 Cartesian Tree

Builds the min Cartesian tree of an array in linear time. A Cartesian tree is a binary tree whose inorder traversal is the original array order and whose heap property is determined by array values. For a min Cartesian tree, each parent has value less than or equal to its children.

Cartesian trees connect arrays, stacks, range minimum queries, treaps, and largest rectangle style problems. Equal values are broken by position, so earlier equal values become ancestors of later equal values.

- `CartesianTree(a)` constructs the tree for array `a`.
- `root` stores the root index.
- `parent[i]`, `left[i]`, and `right[i]` store neighboring node indices, or `-1` if absent.

Time Complexity: $O(n)$ construction time, where n is the array size.

Space Complexity: $O(n)$ for tree arrays and the monotone stack.

```

#include <vector>

struct CartesianTree {
    int root;
    std::vector<int> parent, left, right;

    explicit CartesianTree(const std::vector<int> &a)
        : root(-1), parent(a.size(), -1), left(a.size(), -1), right(a.size(), -1) {
        std::vector<int> st;
        for (int i = 0, n = static_cast<int>(a.size()); i < n; i++) {
            int last = -1;
            while (!st.empty() && a[i] < a[st.back()]) {
                last = st.back();
                st.pop_back();
            }
            if (!st.empty()) {
                right[st.back()] = i;
                parent[i] = st.back();
            }
            if (last != -1) {
                left[i] = last;
                parent[last] = i;
            }
        }
    }
};

```

```

    }
    st.push_back(i);
  }
  root = st.empty() ? -1 : st[0];
}
};

```

Example Usage

```

#include <cassert>
using namespace std;

void inorder(const CartesianTree &t, int u, vector<int> *order) {
    if (u == -1) {
        return;
    }
    inorder(t, t.left[u], order);
    order->push_back(u);
    inorder(t, t.right[u], order);
}

int main() {
    vector<int> a{3, 2, 6, 1, 9};
    CartesianTree t(a);
    assert(t.root == 3);
    assert(t.parent[1] == 3);
    assert(t.parent[4] == 3);

    vector<int> order;
    inorder(t, t.root, &order);
    for (int i = 0; i < 5; i++) {
        assert(order[i] == i);
    }
    return 0;
}

```

2.4.8 Implicit Treap

Maintain a dynamically-sized array using a balanced binary search tree while supporting both dynamic queries and updates of contiguous subarrays via the lazy propagation technique. A treap maintains a balanced binary tree structure by preserving the heap property on the randomly generated priority values of nodes, thereby making insertions and deletions run in $O(\log n)$ with high probability.

The query operation is defined by an associative aggregate function `combine(a, b)`. The default code below assumes a numerical array type, defining queries for the “min” of the target range. Another possible query operation is “sum”, in which case `combine(a, b)` should return $a + b$.

Range updates are defined by `apply_delta(v, d, len)`, which applies an update delta `d` to an aggregate summary `v` representing `len` array values, and by `compose_deltas(old, d)`, which combines a pending older delta with a newer delta in that order. These functions do not support arbitrary combinations: applying a delta to a combined segment must be equivalent to applying it to each child segment and then combining the results, and composed deltas must be equivalent to performing their updates sequentially. The default code below defines range assignment. For range increment, `compose_deltas(old, d)` should return `old + d`; `apply_delta(v, d, len)` should return `v + d` for range-min/range-max queries, and `v + d * len` for range-sum queries.

This data structure shares every operation of one-dimensional segment trees in this section, with the additional operations `empty()`, `insert()`, `erase()`, `push_back()`, and `pop_back()` analogous to those of `std::vector` (here, `insert()` and `erase()` both take an index instead of an iterator).

Time Complexity:

- $O(n)$ per call to both constructors, where n is the size of the array.
- $O(1)$ per call to `size()` and `empty()`.
- $O(\log n)$ on average per call to all other operations.

Space Complexity:

- $O(n)$ for storage of the array elements.
- $O(1)$ auxiliary for `size()` and `empty()`.
- $O(\log n)$ auxiliary stack space for all other operations.

```
#include <stdint>

template<class T>
class ImplicitTreap {
    static T combine(const T &a, const T &b) { return a < b ? a : b; }
    static T apply_delta(const T &v, const T &d, long long len) { return d; }
    static T compose_deltas(const T &d1, const T &d2) { return d2; }

    struct Node {
        static uint32_t rand32() {
            static uint32_t x = 123456789;
            x ^= x << 13;
            x ^= x >> 17;
            x ^= x << 5;
            return x;
        }

        T value, subtree_value, delta;
        bool pending;
        int size;
        uint32_t priority;
        Node *left, *right;

        explicit Node(const T &v)
            : value(v),
              subtree_value(v),
              pending(false),
              size(1),
              priority(rand32()),
              left(nullptr),
              right(nullptr) {}
    } *root;

    static int size(Node *n) { return (n == nullptr) ? 0 : n->size; }

    static void update_value(Node *n) {
        if (n == nullptr) {
            return;
        }
        n->subtree_value = n->value;
        if (n->left != nullptr) {
            n->subtree_value = combine(n->subtree_value, n->left->subtree_value);
        }
        if (n->right != nullptr) {
            n->subtree_value = combine(n->subtree_value, n->right->subtree_value);
        }
        n->size = 1 + size(n->left) + size(n->right);
    }

    static void update_delta(Node *n, const T &d) {
        if (n != nullptr) {
            n->value = apply_delta(n->value, d, 1);
            n->subtree_value = apply_delta(n->subtree_value, d, n->size);
            n->delta = n->pending ? compose_deltas(n->delta, d) : d;
            n->pending = true;
        }
    }
};
```

```

    }
}

static void push_delta(Node *n) {
    if (n == nullptr || !n->pending) {
        return;
    }
    if (n->size > 1) {
        update_delta(n->left, n->delta);
        update_delta(n->right, n->delta);
    }
    n->pending = false;
}

static void merge(Node *&n, Node *left, Node *right) {
    push_delta(left);
    push_delta(right);
    if (left == nullptr) {
        n = right;
    } else if (right == nullptr) {
        n = left;
    } else if (left->priority < right->priority) {
        merge(left->right, left->right, right);
        n = left;
    } else {
        merge(right->left, left, right->left);
        n = right;
    }
    update_value(n);
}

static void split(Node *n, Node *&left, Node *&right, int i) {
    push_delta(n);
    if (n == nullptr) {
        left = right = nullptr;
    } else if (i <= size(n->left)) {
        split(n->left, left, n->left, i);
        right = n;
    } else {
        split(n->right, n->right, right, i - size(n->left) - 1);
        left = n;
    }
    update_value(n);
}

static void insert(Node *&n, Node *new_node, int i) {
    push_delta(n);
    if (n == nullptr) {
        n = new_node;
    } else if (new_node->priority < n->priority) {
        split(n, new_node->left, new_node->right, i);
        n = new_node;
    } else if (i <= size(n->left)) {
        insert(n->left, new_node, i);
    } else {
        insert(n->right, new_node, i - size(n->left) - 1);
    }
    update_value(n);
}

static void erase(Node *&n, int i) {
    push_delta(n);
    if (i == size(n->left)) {
        Node *left = n->left, *right = n->right;
        delete n;
        merge(n, left, right);
    } else if (i < size(n->left)) {
        erase(n->left, i);
    }
}

```

```

    } else {
        erase(n->right, i - size(n->left) - 1);
    }
    update_value(n);
}

static Node *select(Node *n, int i) {
    push_delta(n);
    if (i < size(n->left)) {
        return select(n->left, i);
    }
    if (i > size(n->left)) {
        return select(n->right, i - size(n->left) - 1);
    }
    return n;
}

void clean_up(Node *&n) {
    if (n != nullptr) {
        clean_up(n->left);
        clean_up(n->right);
        delete n;
    }
}

public:
    explicit ImplicitTreap(int n = 0, const T &v = T()) : root(nullptr) {
        for (int i = 0; i < n; i++) {
            push_back(v);
        }
    }

    template<class It>
    ImplicitTreap(It lo, It hi) : root(nullptr) {
        for (; lo != hi; ++lo) {
            push_back(*lo);
        }
    }

    ~ImplicitTreap() { clean_up(root); }
    ImplicitTreap(const ImplicitTreap &) = delete;
    ImplicitTreap &operator=(const ImplicitTreap &) = delete;
    int size() const { return size(root); }
    bool empty() const { return root == nullptr; }
    void insert(int i, const T &v) { insert(root, new Node(v), i); }
    void erase(int i) { erase(root, i); }
    void push_back(const T &v) { insert(size(), v); }
    void pop_back() { erase(size() - 1); }
    T at(int i) const { return select(root, i)->value; }

    T query(int lo, int hi) {
        Node *l1, *r1, *l2, *r2, *t;
        split(root, l1, r1, hi + 1);
        split(l1, l2, r2, lo);
        T res = r2->subtree_value;
        merge(t, l2, r2);
        merge(root, t, r1);
        return res;
    }

    void update(int lo, int hi, const T &d) {
        Node *l1, *r1, *l2, *r2, *t;
        split(root, l1, r1, hi + 1);
        split(l1, l2, r2, lo);
        update_delta(r2, d);
        merge(t, l2, r2);
        merge(root, t, r1);
    }
}

```

```
void update(int i, const T &d) { update(i, i, d); }
};
```

Example Usage

```
#include <iostream>
#include <vector>
using namespace std;

void print(ImplicitTreap<int> &t) {
    cout << "Values:";
    for (int i = 0; i < t.size(); i++) {
        cout << " " << t.at(i);
    }
    cout << " (min: " << t.query(0, t.size() - 1) << ")" << endl;
}

int main() {
    vector<int> arr{99, -2, 1, 8, 10};
    ImplicitTreap<int> t(arr.begin(), arr.end());
    t.push_back(11);
    t.push_back(12);
    t.pop_back();
    print(t);
    t.insert(0, 90);
    t.erase(1);
    print(t);
    t.update(0, 1, 2);
    print(t);
    return 0;
}
```

Example Output

```
Values: 99 -2 1 8 10 11 (min: -2)
Values: 90 -2 1 8 10 11 (min: -2)
Values: 2 2 1 8 10 11 (min: 1)
```

2.5 Range Queries in Two Dimensions

2.5.1 Quadtree (Point Update)

Maintain a two-dimensional array while supporting dynamic queries of rectangular sub-arrays and dynamic updates of individual indices. This implementation uses lazy initialization of nodes to conserve memory while supporting large indices.

The query operation is defined by a commutative associative aggregate function `combine(a, b)`. Because untouched regions are implicit, `repeat_value(v, area)` must return the aggregate summary of `area` copies of the initial value `v`. The default code below assumes a numerical array type, defining queries for the “min” of the target range. For rectangle-sum queries, `combine(a, b)` should return $a + b$ and `repeat_value(v, area)` should return $v \cdot \text{area}$.

The point update operation is defined by `apply_delta(v, d)`, which returns the new value at one updated cell. The default code below defines updates that “set” the chosen cell to a new value. Another possible update operation is “increment”, in which case `apply_delta(v, d)` should return $v + d$.

- `Quadtree(v)` constructs a two-dimensional array with rows from 0 to `R` and columns from 0 to `C`, inclusive. All values are implicitly initialized to `v`.
- `at(r, c)` returns the value at row `r`, column `c`.

- `query(r1, c1, r2, c2)` returns the result of `combine()` applied to every value in the rectangular region consisting of rows from `r1` to `r2`, inclusive, and columns from `c1` to `c2`, inclusive.
- `update(r, c, d)` assigns the value `v` at `(r, c)` to `apply_delta(v, d)`.

Time Complexity:

- $O(1)$ per call to the constructor.
- $O(\max(R, C))$ per call to `at()`, `update()`, and `query()`.

Space Complexity:

- $O(n \log(\max(R, C)))$ for storage of the quadtree nodes after n point updates.
- $O(\log(\max(R, C)))$ auxiliary stack space for `update()`, `query()`, and `at()`.

```
#include <algorithm>
#include <cstdint>

template<class T>
class Quadtree {
    static const int R = 1000000000;
    static const int C = 1000000000;

    static T combine(const T &a, const T &b) { return std::min(a, b); }
    static T repeat_value(const T &v, long long area) { return v; }
    static T apply_delta(const T &v, const T &d) { return d; }

    struct Node {
        T value;
        Node *child[4];

        explicit Node(const T &v) : value(v) {
            for (auto &c : child) {
                c = nullptr;
            }
        }
    };

    Node *root;
    T init;

    // Helper variables for query().
    int tgt_r1, tgt_c1, tgt_r2, tgt_c2;
    T res;
    bool found;

    static long long area(int r1, int c1, int r2, int c2) {
        return static_cast<long long>(r2 - r1 + 1) * (c2 - c1 + 1);
    }

    void append_result(const T &v) {
        res = found ? combine(res, v) : v;
        found = true;
    }

    T child_value(Node *n, int r1, int c1, int r2, int c2) {
        return (n != nullptr) ? n->value : repeat_value(init, area(r1, c1, r2, c2));
    }

    void query(Node *n, int r1, int c1, int r2, int c2) {
        if (tgt_r2 < r1 || tgt_r1 > r2 || tgt_c2 < c1 || tgt_c1 > c2) {
            return;
        }
        if (n == nullptr) {
            int rlen = std::min(r2, tgt_r2) - std::max(r1, tgt_r1) + 1;
            int clen = std::min(c2, tgt_c2) - std::max(c1, tgt_c1) + 1;
            append_result(repeat_value(init, static_cast<long long>(rlen) * clen));
            return;
        }

```



```

    }
    if (tgt_r1 <= r1 && r2 <= tgt_r2 && tgt_c1 <= c1 && c2 <= tgt_c2) {
        append_result(n->value);
        return;
    }
    int rmid = r1 + (r2 - r1) / 2, cmid = c1 + (c2 - c1) / 2;
    query(n->child[0], r1, c1, rmid, cmid);
    query(n->child[1], rmid + 1, c1, r2, cmid);
    query(n->child[2], r1, cmid + 1, rmid, c2);
    query(n->child[3], rmid + 1, cmid + 1, r2, c2);
}

// Helper variables for update().
int tgt_r, tgt_c;
T delta;

void update(Node *&n, int r1, int c1, int r2, int c2) {
    if (tgt_r < r1 || tgt_r > r2 || tgt_c < c1 || tgt_c > c2) {
        return;
    }
    if (n == nullptr) {
        n = new Node(repeat_value(init, area(r1, c1, r2, c2)));
    }
    if (r1 == r2 && c1 == c2) {
        n->value = apply_delta(n->value, delta);
        return;
    }
    int rmid = r1 + (r2 - r1) / 2, cmid = c1 + (c2 - c1) / 2;
    update(n->child[0], r1, c1, rmid, cmid);
    update(n->child[1], rmid + 1, c1, r2, cmid);
    update(n->child[2], r1, cmid + 1, rmid, c2);
    update(n->child[3], rmid + 1, cmid + 1, r2, c2);
    n->value = combine(
        combine(
            child_value(n->child[0], r1, c1, rmid, cmid),
            child_value(n->child[1], rmid + 1, c1, r2, cmid)
        ),
        combine(
            child_value(n->child[2], r1, cmid + 1, rmid, c2),
            child_value(n->child[3], rmid + 1, cmid + 1, r2, c2)
        )
    );
}

static void clean_up(Node *n) {
    if (n != nullptr) {
        for (auto *c : n->child) {
            clean_up(c);
        }
        delete n;
    }
}

public:
    explicit Quadtree(const T &v = T()) : root(nullptr), init(v) {}

    ~Quadtree() { clean_up(root); }
    Quadtree(const Quadtree &) = delete;
    Quadtree &operator=(const Quadtree &) = delete;
    T at(int r, int c) { return query(r, c, r, c); }

    T query(int r1, int c1, int r2, int c2) {
        tgt_r1 = r1;
        tgt_c1 = c1;
        tgt_r2 = r2;
        tgt_c2 = c2;
        found = false;
        query(root, 0, 0, R, C);
    }

```

```

    return found ? res : repeat_value(init, area(r1, c1, r2, c2));
}

void update(int r, int c, const T &d) {
    tgt_r = r;
    tgt_c = c;
    delta = d;
    update(root, 0, 0, R, C);
}
};

```

Example Usage

```

#include <cassert>
#include <iostream>
using namespace std;

int main() {
    Quadtree<int> t(0);
    t.update(0, 0, 7);
    t.update(0, 1, 6);
    t.update(1, 0, 5);
    t.update(1, 1, 4);
    t.update(2, 1, 1);
    t.update(2, 2, 9);
    cout << "Values:" << endl;
    for (int i = 0; i < 3; i++) {
        for (int j = 0; j < 3; j++) {
            cout << t.at(i, j) << " ";
        }
        cout << endl;
    }
    assert(t.query(0, 0, 0, 1) == 6);
    assert(t.query(0, 0, 1, 0) == 5);
    assert(t.query(1, 1, 2, 2) == 0);
    assert(t.query(0, 0, 1000000000, 1000000000) == 0);
    t.update(500000000, 500000000, -100);
    assert(t.query(0, 0, 1000000000, 1000000000) == -100);
    return 0;
}

```

Example Output

```

Values:
7 6 0
5 4 0
0 1 9

```

2.5.2 Quadtree (Range Update)

Maintain a two-dimensional array while supporting both dynamic queries and updates of rectangular sub-arrays via the lazy propagation technique. This implementation uses lazy initialization of nodes to conserve memory while supporting large indices.

The query operation is defined by a commutative associative aggregate function `combine(a, b)`. Because untouched regions are implicit, `repeat_value(v, area)` must return the aggregate summary of `area` copies of the initial value `v`. The default code below assumes a numerical array type, defining queries for the “min” of the target range. For rectangle-sum queries, `combine(a, b)` should return $a + b$ and `repeat_value(v, area)` should return $v \cdot area$.

Range updates are defined by `apply_delta(v, d, area)`, which applies an update delta `d` to an aggregate summary `v` representing `area` cells, and by `compose_deltas(old, d)`, which combines a pending older delta with a newer delta in that order. These functions do not support arbitrary combinations: applying a delta to a combined region

must be equivalent to applying it to each child region and then combining the results, and composed deltas must be equivalent to performing their updates sequentially. The default code below defines rectangle assignment. For rectangle increment, `compose_deltas(old, d)` should return `old + d`; `apply_delta(v, d, area)` should return `v + d` for range-min/range-max queries, and `v + d * area` for range-sum queries.

- `Quadtree(v)` constructs a two-dimensional array with rows from 0 to `R` and columns from 0 to `C`, inclusive. All values are implicitly initialized to `v`.
- `at(r, c)` returns the value at row `r`, column `c`.
- `query(r1, c1, r2, c2)` returns the result of `combine()` applied to every value in the rectangular region consisting of rows from `r1` to `r2` and columns from `c1` to `c2`, inclusive.
- `update(r, c, d)` assigns the value `v` at `(r, c)` to `apply_delta(v, d)`.
- `update(r1, c1, r2, c2)` modifies the value at each index of the rectangular region consisting of rows from `r1` to `r2` and columns from `c1` to `c2`, inclusive, by applying the delta `d` to each value.

Time Complexity:

- $O(1)$ per call to the constructor.
- $O(\max(R, C))$ per call to `at()`, `update()`, and `query()`.

Space Complexity:

- $O(n \log(\max(R, C)))$ for storage of the quadtree nodes after n point or rectangle updates.
- $O(\log(\max(R, C)))$ auxiliary stack space for `update()`, `query()`, and `at()`.

```
#include <algorithm>
#include <cstdio>

template<class T>
class Quadtree {
    static const int R = 1000000000;
    static const int C = 1000000000;

    static T combine(const T &a, const T &b) { return std::min(a, b); }
    static T repeat_value(const T &v, long long area) { return v; }
    static T apply_delta(const T &v, const T &d, long long area) { return v + d; }
    static T compose_deltas(const T &d1, const T &d2) { return d2; }

    struct Node {
        T value, delta;
        bool pending;
        Node *child[4];

        explicit Node(const T &v) : value(v), pending(false) {
            for (auto &c : child) {
                c = nullptr;
            }
        }
    } *root;

    T init;

    static long long area(int r1, int c1, int r2, int c2) {
        return static_cast<long long>(r2 - r1 + 1) * (c2 - c1 + 1);
    }

    void update_delta(Node *&n, const T &d, long long area) {
        if (n == nullptr) {
            n = new Node(repeat_value(init, area));
        }
        n->delta = n->pending ? compose_deltas(n->delta, d) : d;
        n->pending = true;
    }

    void push_delta(Node *n, int r1, int c1, int r2, int c2) {
        if (n->pending) {
            int rmid = r1 + (r2 - r1) / 2, cmid = c1 + (c2 - c1) / 2;
```

```

    long long cur_area = area(r1, c1, r2, c2);
    n->value = apply_delta(n->value, n->delta, cur_area);
    if (cur_area > 1) {
        int rlen = r2 - r1 + 1, clen = c2 - c1 + 1;
        int rlen1 = rmid - r1 + 1, rlen2 = rlen - rlen1;
        int clen1 = cmid - c1 + 1, clen2 = clen - clen1;
        update_delta(n->child[0], n->delta, static_cast<long long>(rlen1) * clen1);
        update_delta(n->child[1], n->delta, static_cast<long long>(rlen2) * clen1);
        update_delta(n->child[2], n->delta, static_cast<long long>(rlen1) * clen2);
        update_delta(n->child[3], n->delta, static_cast<long long>(rlen2) * clen2);
    }
    n->pending = false;
}

// Helper variables for query() and update().
int tgt_r1, tgt_c1, tgt_r2, tgt_c2;
T res, delta;
bool found;

void append_result(const T &v) {
    res = found ? combine(res, v) : v;
    found = true;
}

T child_value(Node *n, int r1, int c1, int r2, int c2) {
    return (n != nullptr) ? n->value : repeat_value(init, area(r1, c1, r2, c2));
}

void query(Node *n, int r1, int c1, int r2, int c2) {
    if (tgt_r2 < r1 || tgt_r1 > r2 || tgt_c2 < c1 || tgt_c1 > c2) {
        return;
    }
    if (n == nullptr) {
        int rlen = std::min(r2, tgt_r2) - std::max(r1, tgt_r1) + 1;
        int clen = std::min(c2, tgt_c2) - std::max(c1, tgt_c1) + 1;
        append_result(repeat_value(init, static_cast<long long>(rlen) * clen));
        return;
    }
    push_delta(n, r1, c1, r2, c2);
    if (tgt_r1 <= r1 && r2 <= tgt_r2 && tgt_c1 <= c1 && c2 <= tgt_c2) {
        append_result(n->value);
        return;
    }
    int rmid = r1 + (r2 - r1) / 2, cmid = c1 + (c2 - c1) / 2;
    query(n->child[0], r1, c1, rmid, cmid);
    query(n->child[1], rmid + 1, c1, r2, cmid);
    query(n->child[2], r1, cmid + 1, rmid, c2);
    query(n->child[3], rmid + 1, cmid + 1, r2, c2);
}

void update(Node *&n, int r1, int c1, int r2, int c2) {
    if (n == nullptr) {
        if (tgt_r2 < r1 || tgt_r1 > r2 || tgt_c2 < c1 || tgt_c1 > c2) {
            return;
        }
        n = new Node(repeat_value(init, area(r1, c1, r2, c2)));
    } else {
        push_delta(n, r1, c1, r2, c2);
    }
    if (tgt_r2 < r1 || tgt_r1 > r2 || tgt_c2 < c1 || tgt_c1 > c2) {
        return;
    }
    if (tgt_r1 <= r1 && r2 <= tgt_r2 && tgt_c1 <= c1 && c2 <= tgt_c2) {
        n->delta = delta;
        n->pending = true;
        push_delta(n, r1, c1, r2, c2);
        return;
    }
}

```

```

    }
    int rmid = r1 + (r2 - r1) / 2, cmid = c1 + (c2 - c1) / 2;
    update(n->child[0], r1, c1, rmid, cmid);
    update(n->child[1], rmid + 1, c1, r2, cmid);
    update(n->child[2], r1, cmid + 1, rmid, c2);
    update(n->child[3], rmid + 1, cmid + 1, r2, c2);
    n->value = combine(
        combine(
            child_value(n->child[0], r1, c1, rmid, cmid),
            child_value(n->child[1], rmid + 1, c1, r2, cmid)
        ),
        combine(
            child_value(n->child[2], r1, cmid + 1, rmid, c2),
            child_value(n->child[3], rmid + 1, cmid + 1, r2, c2)
        )
    );
}

static void clean_up(Node *n) {
    if (n != nullptr) {
        for (auto *c : n->child) {
            clean_up(c);
        }
        delete n;
    }
}

public:
    explicit Quadtree(const T &v = T()) : root(nullptr), init(v) {}

    ~Quadtree() { clean_up(root); }
    Quadtree(const Quadtree &) = delete;
    Quadtree &operator=(const Quadtree &) = delete;
    T at(int r, int c) { return query(r, c, r, c); }

    T query(int r1, int c1, int r2, int c2) {
        tgt_r1 = r1;
        tgt_c1 = c1;
        tgt_r2 = r2;
        tgt_c2 = c2;
        found = false;
        query(root, 0, 0, R, C);
        return found ? res : repeat_value(init, area(r1, c1, r2, c2));
    }

    void update(int r, int c, const T &d) { update(r, c, r, c, d); }

    void update(int r1, int c1, int r2, int c2, const T &d) {
        tgt_r1 = r1;
        tgt_c1 = c1;
        tgt_r2 = r2;
        tgt_c2 = c2;
        delta = d;
        update(root, 0, 0, R, C);
    }
};

```

Example Usage

```

#include <cassert>
#include <iostream>
using namespace std;

int main() {
    Quadtree<int> t(0);
    t.update(0, 0, 7);
}

```

```

t.update(0, 1, 6);
t.update(1, 0, 5);
t.update(1, 1, 4);
t.update(2, 1, 1);
t.update(2, 2, 9);
cout << "Values:" << endl;
for (int i = 0; i < 3; i++) {
    for (int j = 0; j < 3; j++) {
        cout << t.at(i, j) << " ";
    }
    cout << endl;
}
assert(t.query(0, 0, 0, 1) == 6);
assert(t.query(0, 0, 1, 0) == 5);
assert(t.query(1, 1, 2, 2) == 0);
assert(t.query(0, 0, 1000000000, 1000000000) == 0);
t.update(500000000, 500000000, -100);
assert(t.query(0, 0, 1000000000, 1000000000) == -100);
return 0;
}

```

Example Output

```

Values:
7 6 0
5 4 0
0 1 9

```

2.5.3 2D Segment Tree

Maintain a two-dimensional array while supporting dynamic queries of rectangular sub-arrays and dynamic updates of individual indices. This implementation uses lazy initialization of nodes to conserve memory while supporting large indices.

The query operation is defined by a commutative associative aggregate function `combine(a, b)`. Because untouched regions are implicit, `repeat_value(v, area)` must return the aggregate summary of `area` copies of the initial value `v`. The default code below assumes a numerical array type, defining queries for the “min” of the target range. For rectangle-sum queries, `combine(a, b)` should return $a + b$ and `repeat_value(v, area)` should return $v \cdot area$.

The point update operation is defined by `apply_delta(v, d)`, which returns the new value at one updated cell. The default code below defines updates that “set” the chosen cell to a new value. Another possible update operation is “increment”, in which case `apply_delta(v, d)` should return $v + d$.

- `SegTree2D(v)` constructs a two-dimensional array with rows from 0 to `R` and columns from 0 to `C`, inclusive. All values are implicitly initialized to `v`.
- `at(r, c)` returns the value at row `r`, column `c`.
- `query(r1, c1, r2, c2)` returns the result of `combine()` applied to every value in the rectangular region consisting of rows from `r1` to `r2`, inclusive, and columns from `c1` to `c2`, inclusive.
- `update(r, c, d)` assigns the value `v` at `(r, c)` to `apply_delta(v, d)`.

Time Complexity:

- $O(1)$ per call to the constructor.
- $O(\log(R) \cdot \log(C))$ per call to `at()`, `update()`, and `query()`.

Space Complexity:

- $O(n \log(R) \log(C))$ for storage after n point updates.
- $O(\log(R) + \log(C))$ auxiliary stack space for `update()`, `query()`, and `at()`.

```

#include <algorithm>
#include <cstdint>

```

```

template<class T>
class SegTree2D {
    static const int R = 1000000000;
    static const int C = 1000000000;

    static T combine(const T &a, const T &b) { return std::min(a, b); }
    static T repeat_value(const T &v, long long area) { return v; }
    static T apply_delta(const T &v, const T &d) { return d; }

    struct InnerNode {
        T value;
        int low, high;
        InnerNode *left, *right;

        InnerNode(int lo, int hi, const T &v)
            : value(v), low(lo), high(hi), left(nullptr), right(nullptr) {}
    };

    struct OuterNode {
        InnerNode root;
        int low, high;
        OuterNode *left, *right;

        OuterNode(int lo, int hi, const T &v)
            : root(0, C, v), low(lo), high(hi), left(nullptr), right(nullptr) {}
    } *root;

    T init;

    // Helper variables for query() and update().
    int tgt_r1, tgt_c1, tgt_r2, tgt_c2;
    long long width;

    static long long length(int lo, int hi) { return hi - lo + 1LL; }

    void append_result(T &res, bool &found, const T &v) {
        res = found ? combine(res, v) : v;
        found = true;
    }

    T query(InnerNode *n, int qlo, int qhi, long long rows) {
        if (n == nullptr) {
            return repeat_value(init, rows * length(qlo, qhi));
        }
        int lo = n->low, hi = n->high, mid = lo + (hi - lo) / 2;
        T res;
        bool found = false;
        if (qlo < lo) {
            int missing_hi = std::min(qhi, lo - 1);
            append_result(res, found, repeat_value(init, rows * length(qlo, missing_hi)));
        }
        int overlap_lo = std::max(qlo, lo), overlap_hi = std::min(qhi, hi);
        if (overlap_lo <= overlap_hi) {
            if (overlap_lo == lo && overlap_hi == hi) {
                append_result(res, found, n->value);
            } else {
                if (overlap_lo <= mid) {
                    append_result(res, found, query(n->left, overlap_lo, std::min(overlap_hi, mid), rows));
                }
                if (mid < overlap_hi) {
                    append_result(
                        res, found, query(n->right, std::max(overlap_lo, mid + 1), overlap_hi, rows)
                    );
                }
            }
        }
    }

    if (hi < qhi) {
        int missing_lo = std::max(qlo, hi + 1);

```

```

    append_result(res, found, repeat_value(init, rows * length(missing_lo, qhi)));
}
return res;
}

T query(OuterNode *n, int qlo, int qhi) {
    if (n == nullptr) {
        return repeat_value(init, width * length(qlo, qhi));
    }
    int lo = n->low, hi = n->high, mid = lo + (hi - lo) / 2;
    T res;
    bool found = false;
    if (qlo < lo) {
        int missing_hi = std::min(qhi, lo - 1);
        append_result(res, found, repeat_value(init, width * length(qlo, missing_hi)));
    }
    int overlap_lo = std::max(qlo, lo), overlap_hi = std::min(qhi, hi);
    if (overlap_lo <= overlap_hi) {
        if (overlap_lo == lo && overlap_hi == hi) {
            append_result(res, found, query(&(n->root), tgt_c1, tgt_c2, length(lo, hi)));
        } else {
            if (overlap_lo <= mid) {
                append_result(res, found, query(n->left, overlap_lo, std::min(overlap_hi, mid)));
            }
            if (mid < overlap_hi) {
                append_result(res, found, query(n->right, std::max(overlap_lo, mid + 1), overlap_hi));
            }
        }
    }
    if (hi < qhi) {
        int missing_lo = std::max(qlo, hi + 1);
        append_result(res, found, repeat_value(init, width * length(missing_lo, qhi)));
    }
    return res;
}

void update(InnerNode *n, int c, const T &d, bool leaf_row, long long rows) {
    int lo = n->low, hi = n->high, mid = lo + (hi - lo) / 2;
    if (lo == hi) {
        if (leaf_row) {
            n->value = apply_delta(n->value, d);
        } else {
            n->value = d;
        }
        return;
    }
    InnerNode *&target = (c <= mid) ? n->left : n->right;
    if (target == nullptr) {
        target = new InnerNode(c, c, repeat_value(init, rows));
    }
    if (target->low <= c && c <= target->high) {
        update(target, c, d, leaf_row, rows);
    } else {
        int split_lo = lo, split_hi = hi, split_mid = mid;
        do {
            if (c <= split_mid) {
                split_hi = split_mid;
            } else {
                split_lo = split_mid + 1;
            }
            split_mid = split_lo + (split_hi - split_lo) / 2;
        } while ((c <= split_mid) == (target->low <= split_mid));
        InnerNode *tmp =
            new InnerNode(split_lo, split_hi, repeat_value(init, rows * length(split_lo, split_hi)));
        if (target->low <= split_mid) {
            tmp->left = target;
        } else {
            tmp->right = target;
        }
    }
}

```



```

    }
    target = tmp;
    update(tmp, c, d, leaf_row, rows);
}
T left_value =
    (n->left != nullptr) ? n->left->value : repeat_value(init, rows * length(lo, mid));
T right_value =
    (n->right != nullptr) ? n->right->value : repeat_value(init, rows * length(mid + 1, hi));
n->value = combine(left_value, right_value);
}

void update(OuterNode *n, int r, int c, const T &d) {
    int lo = n->low, hi = n->high, mid = lo + (hi - lo) / 2;
    long long rows = length(lo, hi);
    if (lo == hi) {
        update(&(n->root), c, d, true, 1);
        return;
    }
    if (r <= mid) {
        if (n->left == nullptr) {
            n->left = new OuterNode(lo, mid, repeat_value(init, length(lo, mid) * length(0, C)));
        }
        update(n->left, r, c, d);
    } else {
        if (n->right == nullptr) {
            n->right =
                new OuterNode(mid + 1, hi, repeat_value(init, length(mid + 1, hi) * length(0, C)));
        }
        update(n->right, r, c, d);
    }
    T value = repeat_value(init, rows);
    if (n->left != nullptr || n->right != nullptr) {
        tgt_c1 = tgt_c2 = c;
        T left_value = (n->left != nullptr) ? query(&(n->left->root), c, c, length(lo, mid))
            : repeat_value(init, length(lo, mid));
        T right_value = (n->right != nullptr) ? query(&(n->right->root), c, c, length(mid + 1, hi))
            : repeat_value(init, length(mid + 1, hi));
        value = combine(left_value, right_value);
    }
    update(&(n->root), c, value, false, rows);
}

static void clean_up(InnerNode *n) {
    if (n != nullptr) {
        clean_up(n->left);
        clean_up(n->right);
        delete n;
    }
}

static void clean_up(OuterNode *n) {
    if (n != nullptr) {
        clean_up(n->root.left);
        clean_up(n->root.right);
        clean_up(n->left);
        clean_up(n->right);
        delete n;
    }
}

public:
    explicit SegTree2D(const T &v = T())
        : root(new OuterNode(0, R, repeat_value(v, length(0, R) * length(0, C))), init(v) {}

    ~SegTree2D() { clean_up(root); }
    SegTree2D(const SegTree2D &) = delete;
    SegTree2D &operator=(const SegTree2D &) = delete;
    T at(int r, int c) { return query(r, c, r, c); }

```

```

T query(int r1, int c1, int r2, int c2) {
    tgt_r1 = r1;
    tgt_c1 = c1;
    tgt_r2 = r2;
    tgt_c2 = c2;
    width = length(c1, c2);
    return query(root, r1, r2);
}

void update(int r, int c, const T &d) { update(root, r, c, d); }
};

```

Example Usage

```

#include <cassert>
#include <iostream>
using namespace std;

int main() {
    SegTree2D<int> t(0);
    t.update(0, 0, 7);
    t.update(0, 1, 6);
    t.update(1, 0, 5);
    t.update(1, 1, 4);
    t.update(2, 1, 1);
    t.update(2, 2, 9);
    cout << "Values:" << endl;
    for (int i = 0; i < 3; i++) {
        for (int j = 0; j < 3; j++) {
            cout << t.at(i, j) << " ";
        }
        cout << endl;
    }
    assert(t.query(0, 0, 0, 1) == 6);
    assert(t.query(0, 0, 1, 0) == 5);
    assert(t.query(1, 1, 2, 2) == 0);
    assert(t.query(0, 0, 1000000000, 1000000000) == 0);
    t.update(500000000, 500000000, -100);
    assert(t.query(0, 0, 1000000000, 1000000000) == -100);
    return 0;
}

```

Example Output

```

Values:
7 6 0
5 4 0
0 1 9

```

2.5.4 2D Range Tree

Maintain a set of two-dimensional points while supporting queries for all points that fall inside given rectangular regions. This implementation uses `std::pair` to represent points, requiring operators `<` and `==` to be defined on the numeric template type.

- `RangeTree(lo, hi)` constructs a set from two random-access iterators to `std::pair` as a range `[lo, hi)` of points.
- `query(x1, y1, x2, y2, f)` calls the function `f(i, p)` on each point in the set that falls into the rectangular region consisting of rows from `x1` to `x2`, inclusive, and columns from `y1` to `y2`, inclusive. The first argument to `f` is the zero-based index of the point in the original range given to the constructor. The second argument is the point itself as an `std::pair`.

Time Complexity:

- $O(n \log n)$ per call to the constructor, where n is the number of points.
- $O(\log^2(n) + m)$ per call to `query()`, where m is the number of points that are reported by the query.

Space Complexity:

- $O(n \log n)$ for storage of the points.
- $O(\log^2(n))$ auxiliary stack space for `query()`.

```
#include <algorithm>
#include <iterator>
#include <utility>
#include <vector>

template<class T>
class RangeTree {
    using point = std::pair<T, T>;
    using colindex = std::pair<int, T>;

    std::vector<point> points;
    std::vector<std::vector<colindex>> columns;

    static inline bool comp1(const colindex &a, const colindex &b) { return a.second < b.second; }
    static inline bool comp2(const colindex &a, const T &v) { return a.second < v; }

    void build(int n, int lo, int hi) {
        if (points[lo].first == points[hi].first) {
            for (int i = lo; i <= hi; i++) {
                columns[n].emplace_back(i, points[i].second);
            }
            return;
        }
        int l = n * 2 + 1, r = n * 2 + 2, mid = lo + (hi - lo) / 2;
        build(l, lo, mid);
        build(r, mid + 1, hi);
        columns[n].resize(columns[l].size() + columns[r].size());
        std::merge(
            columns[l].begin(), columns[l].end(), columns[r].begin(), columns[r].end(),
            columns[n].begin(), comp1
        );
    }

    // Helper variables for query().
    T x1, y1, x2, y2;

    template<class Fn>
    void query(int n, int lo, int hi, Fn f) {
        if (points[hi].first < x1 || x2 < points[lo].first) {
            return;
        }
        if (!(points[lo].first < x1 || x2 < points[hi].first)) {
            if (!columns[n].empty() && !(y2 < y1)) {
                auto it = std::lower_bound(columns[n].begin(), columns[n].end(), y1, comp2);
                for (; it != columns[n].end() && it->second <= y2; ++it) {
                    f(it->first, points[it->first]);
                }
            }
        }
        else if (lo != hi) {
            int mid = lo + (hi - lo) / 2;
            query(n * 2 + 1, lo, mid, f);
            query(n * 2 + 2, mid + 1, hi, f);
        }
    }

public:
    template<class It>
    RangeTree(It lo, It hi) : points(lo, hi) {
```

```

    int n = std::distance(lo, hi);
    columns.resize(4 * n + 1);
    std::sort(points.begin(), points.end());
    build(0, 0, n - 1);
}

template<class Fn>
void query(const T &x1, const T &y1, const T &x2, const T &y2, Fn f) {
    this->x1 = x1;
    this->y1 = y1;
    this->x2 = x2;
    this->y2 = y2;
    query(0, 0, points.size() - 1, f);
}
};

```

Example Usage

```

#include <iostream>
using namespace std;

void print(int i, const pair<int, int> &p) {
    cout << "(" << p.first << ", " << p.second << ") ";
}

int main() {
    vector<pair<int, int>> v{{1, 4}, {5, 4}, {2, 2}, {3, 1}, {6, -5},
                          {5, -1}, {3, -3}, {-1, -2}, {-1, -1}, {2, -1}};
    RangeTree<int> t(v.begin(), v.end());
    t.query(-1, -1, 2, 5, print);
    cout << endl;
    t.query(1, 1, 4, 8, print);
    cout << endl;
    return 0;
}

```

Example Output

```

(-1, -1) (2, -1) (2, 2) (1, 4)
(1, 4) (2, 2) (3, 1)

```

2.5.5 K-d Tree (2D Range Query)

Maintain a set of two-dimensional points while supporting queries for all points that fall inside given rectangular regions. This implementation uses `std::pair` to represent points, requiring operators `<` and `==` to be defined on the numeric template type.

- `KDTree(lo, hi)` constructs a set from two random-access iterators to `std::pair` as a range `[lo, hi)` of points.
- `query(x1, y1, x2, y2, f)` calls the function `f(i, p)` on each point in the set that falls into the rectangular region consisting of rows from `x1` to `x2`, inclusive, and columns from `y1` to `y2`, inclusive. The first argument to `f` is the zero-based index of the point in the original range given to the constructor. The second argument is the point itself as an `std::pair`.

Time Complexity:

- $O(n \log n)$ per call to the constructor, where n is the number of points.
- $O(\log(n) + m)$ on average per call to `query()`, where m is the number of points that are reported by the query.

Space Complexity:

- $O(n)$ for storage of the points.

- $O(\log n)$ auxiliary stack space for `query()`.

```
#include <algorithm>
#include <utility>
#include <vector>

template<class T>
class KDTree {
    using point = std::pair<T, T>;

    static inline bool comp1(const point &a, const point &b) { return a.first < b.first; }
    static inline bool comp2(const point &a, const point &b) { return a.second < b.second; }

    std::vector<point> tree, minp, maxp;
    std::vector<int> l_index, h_index;

    void build(int lo, int hi, bool div_x) {
        if (lo >= hi) {
            return;
        }
        int mid = lo + (hi - lo) / 2;
        std::nth_element(
            tree.begin() + lo, tree.begin() + mid, tree.begin() + hi, div_x ? comp1 : comp2
        );
        l_index[mid] = lo;
        h_index[mid] = hi;
        minp[mid].first = maxp[mid].first = tree[lo].first;
        minp[mid].second = maxp[mid].second = tree[lo].second;
        for (int i = lo + 1; i < hi; i++) {
            minp[mid].first = std::min(minp[mid].first, tree[i].first);
            minp[mid].second = std::min(minp[mid].second, tree[i].second);
            maxp[mid].first = std::max(maxp[mid].first, tree[i].first);
            maxp[mid].second = std::max(maxp[mid].second, tree[i].second);
        }
        build(lo, mid, !div_x);
        build(mid + 1, hi, !div_x);
    }

    // Helper variables for query().
    T x1, y1, x2, y2;

    template<class Fn>
    void query(int lo, int hi, Fn f) {
        if (lo >= hi) {
            return;
        }
        int mid = lo + (hi - lo) / 2;
        T ax = minp[mid].first, ay = minp[mid].second;
        T bx = maxp[mid].first, by = maxp[mid].second;
        if (x2 < ax || bx < x1 || y2 < ay || by < y1) {
            return;
        }
        if (!(ax < x1 || x2 < bx || ay < y1 || y2 < by)) {
            for (int i = l_index[mid]; i < h_index[mid]; i++) {
                f(tree[i]);
            }
            return;
        }
        query(lo, mid, f);
        query(mid + 1, hi, f);
        if (tree[mid].first < x1 || x2 < tree[mid].first || tree[mid].second < y1 ||
            y2 < tree[mid].second) {
            return;
        }
        f(tree[mid]);
    }

public:
```

```

template<class It>
KDTree(It lo, It hi) : tree(lo, hi) {
    int n = std::distance(lo, hi);
    l_index.resize(n);
    h_index.resize(n);
    minp.resize(n);
    maxp.resize(n);
    build(0, n, true);
}

template<class Fn>
void query(const T &x1, const T &y1, const T &x2, const T &y2, Fn f) {
    this->x1 = x1;
    this->y1 = y1;
    this->x2 = x2;
    this->y2 = y2;
    query(0, tree.size(), f);
}
};

```

Example Usage

```

#include <iostream>
using namespace std;

void print(const pair<int, int> &p) {
    cout << "(" << p.first << ", " << p.second << ") ";
}

int main() {
    vector<pair<int, int>> v{{1, 4}, {5, 4}, {2, 2}, {3, 1}, {6, -5},
                           {5, -1}, {3, -3}, {-1, -2}, {-1, -1}, {2, -1}};
    KDTree<int> t(v.begin(), v.end());
    t.query(-1, -1, 2, 5, print);
    cout << endl;
    t.query(1, 1, 4, 8, print);
    cout << endl;
    return 0;
}

```

Example Output

```

(2, -1) (1, 4) (2, 2) (-1, -1)
(1, 4) (2, 2) (3, 1)

```

2.5.6 K-d Tree (Nearest Neighbor)

Maintain a set of two-dimensional points while supporting queries for the closest point in the set to a given query point. This implementation uses `std::pair` to represent points, requiring operators `<`, `==`, `-`, and `long double` casting to be defined on the numeric template type.

- `KDTree(lo, hi)` constructs a set from two random-access iterators to `std::pair` as a range `[lo, hi)` of points.
- `nearest(x, y, can_equal)` returns a point in the set that is closest to `(x, y)` by Euclidean distance. This may be equal to `(x, y)` only if `can_equal` is `true`.

Time Complexity:

- $O(n \log n)$ per call to the constructor, where n is the number of points.
- $O(\log n)$ on average per call to `nearest()`.

Space Complexity:

- $O(n)$ for storage of the points.

- $O(\log n)$ auxiliary stack space for `nearest()`.

```

#include <algorithm>
#include <cfloat>
#include <stdexcept>
#include <utility>
#include <vector>

template<class T>
class KDTree {
    using point = std::pair<T, T>;

    static inline bool comp1(const point &a, const point &b) { return a.first < b.first; }
    static inline bool comp2(const point &a, const point &b) { return a.second < b.second; }

    std::vector<point> tree;
    std::vector<bool> div_x;

    void build(int lo, int hi) {
        if (lo >= hi) {
            return;
        }
        int mid = lo + (hi - lo) / 2;
        T minx, maxx, miny, maxy;
        minx = maxx = tree[lo].first;
        miny = maxy = tree[lo].second;
        for (int i = lo + 1; i < hi; i++) {
            minx = std::min(minx, tree[i].first);
            miny = std::min(miny, tree[i].second);
            maxx = std::max(maxx, tree[i].first);
            maxy = std::max(maxy, tree[i].second);
        }
        div_x[mid] = !((maxx - minx) < (maxy - miny));
        std::nth_element(
            tree.begin() + lo, tree.begin() + mid, tree.begin() + hi, div_x[mid] ? comp1 : comp2
        );
        if (lo + 1 == hi) {
            return;
        }
        build(lo, mid);
        build(mid + 1, hi);
    }

    // Helper variables for nearest().
    long double min_dist;
    int id;

    void nearest(int lo, int hi, const T &x, const T &y, bool can_equal) {
        if (lo >= hi) {
            return;
        }
        int mid = lo + (hi - lo) / 2;
        T dx = x - tree[mid].first, dy = y - tree[mid].second;
        long double d = dx * static_cast<long double>(dx) + dy * static_cast<long double>(dy);
        if (d < min_dist && (can_equal || d != 0)) {
            min_dist = d;
            id = mid;
        }
        if (lo + 1 == hi) {
            return;
        }
        d = static_cast<long double>((div_x[mid] ? dx : dy));
        int l1 = lo, r1 = mid, l2 = mid + 1, r2 = hi;
        if (d > 0) {
            std::swap(l1, l2);
            std::swap(r1, r2);
        }
        nearest(l1, r1, x, y, can_equal);
    }
};

```

```

    if (d * static_cast<long double>(d) < min_dist) {
        nearest(l2, r2, x, y, can_equal);
    }
}

public:
template<class It>
KDTree(It lo, It hi) : tree(lo, hi) {
    int n = std::distance(lo, hi);
    if (n <= 1) {
        throw std::runtime_error("K-d tree must have at least 2 points.");
    }
    div_x.resize(n);
    build(0, n);
}

point nearest(const T &x, const T &y, bool can_equal = true) {
    min_dist = LDBL_MAX;
    nearest(0, tree.size(), x, y, can_equal);
    return tree[id];
}
};

```

Example Usage

```

#include <cassert>
using namespace std;

int main() {
    vector<pair<int, int>> p{{0, 2}, {0, 3}, {-1, 0}};
    KDTree<int> t(p.begin(), p.end());
    assert(t.nearest(0, 2, true) == (pair<int, int>{0, 2}));
    assert(t.nearest(0, 2, false) == (pair<int, int>{0, 3}));
    assert(t.nearest(0, 0) == (pair<int, int>{-1, 0}));
    assert(t.nearest(-10000, 0) == (pair<int, int>{-1, 0}));
    return 0;
}

```

2.6 Fenwick Trees

2.6.1 Fenwick Tree (Simple)

Maintain an array of numerical type, allowing for updates of individual indices (point update) and queries for the sum of contiguous sub-arrays (range queries). This implementation assumes that the array is 1-based (i.e. has valid indices from 1 to n , inclusive).

- `initialize(n)` resets the data structure.
- `vals[i]` stores the value at index `i`.
- `add(i, x)` adds `x` to the value at index `i`.
- `set(i, x)` assigns the value at index `i` to `x`.
- `sum(hi)` returns the sum of all values at indices from 1 to `hi`, inclusive.
- `sum(lo, hi)` returns the sum of all values at indices from `lo` to `hi`, inclusive.

Time Complexity:

- $O(n)$ per call to `initialize(n)`, where n is the size of the array.
- $O(\log n)$ per call to all other operations.

Space Complexity:

- $O(n)$ for storage of the array elements.
- $O(1)$ auxiliary for all operations.

```
#include <vector>

std::vector<int> vals, tree;

void initialize(int n) {
    vals.assign(n + 2, 0);
    tree.assign(n + 2, 0);
}

void add(int i, int x) {
    vals[i] += x;
    for (; i < static_cast<int>(tree.size()); i += i & -i) {
        tree[i] += x;
    }
}

void set(int i, int x) {
    add(i, x - vals[i]);
}

int sum(int hi) {
    int res = 0;
    for (; hi > 0; hi -= hi & -hi) {
        res += tree[hi];
    }
    return res;
}

int sum(int lo, int hi) {
    return sum(hi) - sum(lo - 1);
}
```

Example Usage

```
#include <cassert>
#include <iostream>
using namespace std;

int main() {
    vector<int> v{10, 1, 2, 3, 4};
    int n = 5;
    initialize(n);
    for (int i = 1; i <= n; i++) {
        set(i, v[i - 1]);
    }
    add(1, -5);
    cout << "Values: ";
    for (int i = 1; i <= n; i++) {
        cout << vals[i] << " ";
    }
    cout << endl;
    assert(sum(2, 4) == 6);
    return 0;
}
```

Example Output

```
Values: 5 1 2 3 4
```

2.6.2 Fenwick Tree (Range Update, Point Query)

Maintain an array of numerical type, allowing for contiguous sub-arrays to be simultaneously incremented by arbitrary values (range update) and values at individual indices to be queried (point query). This implementation assumes that the array is 0-based (i.e. has valid indices from 0 to `size() - 1`, inclusive).

- `size()` returns the size of the array.
- `at(i)` returns the value at index `i`.
- `add(i, x)` adds `x` to the value at index `i`.
- `add(lo, hi, x)` adds `x` to the values at all indices from `lo` to `hi`, inclusive.

Time Complexity:

- $O(n)$ per call to the constructor, where n is the size of the array.
- $O(1)$ per call to `size()`.
- $O(\log n)$ per call to `at()` and both `add()` functions.

Space Complexity:

- $O(n)$ for storage of the array elements.
- $O(1)$ auxiliary for all operations.

```
#include <vector>

template<class T>
class FenwickTree {
    int len;
    std::vector<T> tree;

public:
    explicit FenwickTree(int n) : len(n), tree(n + 2) {}

    int size() const { return len; }

    T at(int i) const {
        T res = 0;
        for (i++; i > 0; i -= i & -i) {
            res += tree[i];
        }
        return res;
    }

    void add(int i, const T &x) {
        for (i++; i <= len + 1; i += i & -i) {
            tree[i] += x;
        }
    }

    void add(int lo, int hi, const T &x) {
        add(lo, x);
        add(hi + 1, -x);
    }
};
```

Example Usage

```
#include <iostream>
using namespace std;

int main() {
    FenwickTree<int> t(5);
    t.add(0, 1, 5);
    t.add(1, 2, 5);
    t.add(2, 4, 10);
```

```

cout << "Values: ";
for (int i = 0; i < t.size(); i++) {
    cout << t.at(i) << " ";
}
cout << endl;
return 0;
}

```

Example Output

Values: 5 10 15 10 10

2.6.3 Fenwick Tree (Point Update, Range Query)

Maintain an array of numerical type, allowing for updates of individual indices (point update) and queries for the sum of contiguous sub-arrays (range queries). This implementation assumes that the array is 0-based (i.e. has valid indices from 0 to `size() - 1`, inclusive).

- `size()` returns the size of the array.
- `at(i)` returns the value at index `i`.
- `add(i, x)` adds `x` to the value at index `i`.
- `set(i, x)` assigns the value at index `i` to `x`.
- `sum(hi)` returns the sum of all values at indices from 0 to `hi`, inclusive.
- `sum(lo, hi)` returns the sum of all values at indices from `lo` to `hi`, inclusive.

Time Complexity:

- $O(n)$ per call to the constructor, where n is the size of the array.
- $O(1)$ per call to `size()` and `at()`.
- $O(\log n)$ per call to `add()`, `set()`, and both `sum()` functions.

Space Complexity:

- $O(n)$ for storage of the array elements.
- $O(1)$ auxiliary for all operations.

```

#include <vector>

template<class T>
class FenwickTree {
    int len;
    std::vector<T> vals, tree;

public:
    explicit FenwickTree(int n) : len(n), vals(n + 1), tree(n + 1) {}

    int size() const { return len; }
    T at(int i) const { return vals[i + 1]; }

    void add(int i, const T &x) {
        vals[++i] += x;
        for (; i <= len; i += i & -i) {
            tree[i] += x;
        }
    }

    void set(int i, const T &x) {
        T inc = x - vals[i + 1];
        add(i, inc);
    }

    T sum(int hi) {
        T res = 0;

```

```

    for (hi++; hi > 0; hi -= hi & -hi) {
        res += tree[hi];
    }
    return res;
}

T sum(int lo, int hi) { return sum(hi) - sum(lo - 1); }
};

```

Example Usage

```

#include <cassert>
#include <iostream>
using namespace std;

int main() {
    vector<int> a{10, 1, 2, 3, 4};
    FenwickTree<int> t(5);
    for (int i = 0; i < static_cast<int>(a.size()); i++) {
        t.set(i, a[i]);
    }
    t.add(0, -5);
    cout << "Values: ";
    for (int i = 0; i < t.size(); i++) {
        cout << t.at(i) << " ";
    }
    cout << endl;
    assert(t.sum(1, 3) == 6);
    return 0;
}

```

Example Output

Values: 5 1 2 3 4

2.6.4 Fenwick Tree (Range Update, Range Query)

Maintain an array of numerical type, allowing for contiguous sub-arrays to be simultaneously incremented by arbitrary values (range update) and queries for the sum of contiguous sub-arrays (range query). This implementation assumes that the array is 0-based (i.e. has valid indices from 0 to `size() - 1`, inclusive).

- `size()` returns the size of the array.
- `at(i)` returns the value at index `i`.
- `add(i, x)` increments the value at index `i` by `x`.
- `add(lo, hi, x)` adds `x` to the values at all indices from `lo` to `hi`, inclusive.
- `set(i, x)` assigns the value at index `i` to `x`.
- `sum(hi)` returns the sum of all values at indices from 0 to `hi`, inclusive.
- `sum(lo, hi)` returns the sum of all values at indices from `lo` to `hi`, inclusive.

Time Complexity:

- $O(n)$ per call to the constructor, where n is the size of the array.
- $O(1)$ per call to `size()`.
- $O(\log n)$ per call to all other operations.

Space Complexity:

- $O(n)$ for storage of the array elements.
- $O(1)$ auxiliary for all operations.

```
#include <vector>
```

```

template<class T>
class FenwickTree {
    int len;
    std::vector<T> t1, t2;

    T sum(const std::vector<T> &tree, int i) {
        T res = 0;
        for (; i != 0; i -= i & -i) {
            res += tree[i];
        }
        return res;
    }

    void add(std::vector<T> &tree, int i, const T &x) {
        for (; i <= len + 1; i += i & -i) {
            tree[i] += x;
        }
    }

public:
    explicit FenwickTree(int n) : len(n), t1(n + 2), t2(n + 2) {}

    int size() const { return len; }

    void add(int lo, int hi, const T &x) {
        lo++;
        hi++;
        add(t1, lo, x);
        add(t1, hi + 1, -x);
        add(t2, lo, x * (lo - 1));
        add(t2, hi + 1, -x * hi);
    }

    void add(int i, const T &x) { return add(i, i, x); }
    void set(int i, const T &x) { add(i, x - at(i)); }

    T sum(int hi) {
        hi++;
        return hi * sum(t1, hi) - sum(t2, hi);
    }

    T sum(int lo, int hi) { return sum(hi) - sum(lo - 1); }
    T at(int i) { return sum(i, i); }
};

```

Example Usage

```

#include <cassert>
#include <iostream>
using namespace std;

int main() {
    vector<int> a{10, 1, 2, 3, 4};
    FenwickTree<int> t(5);
    for (int i = 0; i < t.size(); i++) {
        t.set(i, a[i]);
    }
    t.add(0, 2, 5);
    t.set(3, -5);
    cout << "Values: ";
    for (int i = 0; i < t.size(); i++) {
        cout << t.at(i) << " ";
    }
    cout << endl;
    assert(t.sum(0, 4) == 27);
}

```

```

return 0;
}

```

Example Output

Values: 15 6 7 -5 4

2.6.5 Fenwick Tree (Compressed)

Maintain an array of numerical type, allowing for contiguous sub-arrays to be simultaneously incremented by arbitrary values (range update) and queries for the sum of contiguous sub-arrays (range query). This implementation uses `std::unordered_map` for coordinate compression, allowing for large indices to be accessed with efficient space complexity. That is, all array indices from 0 to N , inclusive, are accessible.

- `at(i)` returns the value at index `i`.
- `add(i, x)` adds `x` to the value at index `i`.
- `add(lo, hi, x)` adds `x` to the values at all indices from `lo` to `hi`, inclusive.
- `set(i, x)` assigns the value at index `i` to `x`.
- `sum(hi)` returns the sum of all values at indices from 0 to `hi`, inclusive.
- `sum(lo, hi)` returns the sum of all values at indices from `lo` to `hi`, inclusive.

Time Complexity: $O(\log n)$ per call to all member functions, where n is the number of distinct indices that have been accessed.

Space Complexity:

- $O(n \log n)$ for storage of the array elements, where n is the number of distinct indices that have been accessed across all of the operations so far.
- $O(1)$ auxiliary for all operations.

```

#include <unordered_map>

template<class T>
class FenwickTree {
    static const int N = 1000000001;
    std::unordered_map<int, T> tmul, tadd;

    void add_helper(int at, int mul, T add) {
        for (int i = at; i <= N; i |= i + 1) {
            tmul[i] += mul;
            tadd[i] += add;
        }
    }

public:
    void add(int lo, int hi, const T &x) {
        add_helper(lo, x, -x * (lo - 1));
        add_helper(hi, -x, x * hi);
    }

    void add(int i, const T &x) { add(i, i, x); }
    void set(int i, const T &x) { add(i, x - at(i)); }

    T sum(int hi) {
        T mul = 0, add = 0;
        for (int i = hi; i >= 0; i = (i & (i + 1)) - 1) {
            if (auto it = tmul.find(i); it != tmul.end()) {
                mul += it->second;
            }
            if (auto it = tadd.find(i); it != tadd.end()) {
                add += it->second;
            }
        }
    }
}

```

```

    return mul * hi + add;
}

T sum(int lo, int hi) { return sum(hi) - sum(lo - 1); }
T at(int i) { return sum(i, i); }
};

```

Example Usage

```

#include <cassert>
#include <iostream>
#include <vector>
using namespace std;

int main() {
    vector<int> a{10, 1, 2, 3, 4};
    FenwickTree<int> t;
    for (int i = 0; i < static_cast<int>(a.size()); i++) {
        t.set(i, a[i]);
    }
    t.add(0, 2, 5);
    t.set(3, -5);
    cout << "Values: ";
    for (int i = 0; i < 5; i++) {
        cout << t.at(i) << " ";
    }
    cout << endl;
    assert(t.sum(0, 4) == 27);
    t.add(500000001, 500000010, 3);
    t.add(500000011, 500000015, 5);
    t.set(500000000, 10);
    assert(t.sum(0, 1000000000) == 92);
    return 0;
}

```

Example Output

Values: 15 6 7 -5 4

2.6.6 Fenwick Tree (Order Statistic)

Maintains frequencies over a 1-based integer domain and finds order statistics by binary lifting on a Fenwick tree. This is useful for dynamic multisets of bounded integer values, coordinate-compressed *k*th-element queries, and online rank queries.

- `initialize(n)` resets all frequencies.
- `add(i, delta)` adds `delta` to the frequency of value/index `i`.
- `sum(i)` returns the total frequency over indices `[1, i]`.
- `kth(k)` returns the smallest index `i` such that `sum(i) >= k`. The argument `k` is 1-based and must satisfy `1 <= k <= sum(n)`.

Time Complexity:

- $O(n)$ per call to `initialize()`.
- $O(\log n)$ per call to `add(i, delta)`, `sum(i)`, and `kth(k)`.

Space Complexity: $O(n)$ for the Fenwick tree.

```

#include <algorithm>
#include <vector>

std::vector<int> tree;

```

```

void initialize(int n) {
    tree.assign(n + 1, 0);
}

void add(int i, int delta) {
    for (; i < tree.size(); i += i & -i) {
        tree[i] += delta;
    }
}

int sum(int i) {
    int res = 0;
    for (; i > 0; i -= i & -i) {
        res += tree[i];
    }
    return res;
}

int kth(int k) {
    int idx = 0, bits = 1;
    while (bits * 2 < tree.size()) {
        bits *= 2;
    }
    for (; bits > 0; bits >>= 1) {
        int next = idx + bits;
        if (next < tree.size() && tree[next] < k) {
            idx = next;
            k -= tree[next];
        }
    }
    return idx + 1;
}

```

Example Usage

```

#include <cassert>

int main() {
    initialize(100);
    add(2, 1);
    add(4, 3);
    add(7, 1);
    assert(sum(4) == 4);
    assert(kth(1) == 2);
    assert(kth(2) == 4);
    assert(kth(4) == 4);
    assert(kth(5) == 7);
    return 0;
}

```

2.6.7 2D Fenwick Tree (Simple)

Maintain a 2D array of numerical type, allowing for updates of individual cells in the matrix (point update) and queries for the sum of rectangular sub-matrices (range query). This implementation assumes that array dimensions are 1-based (i.e. rows have valid indices from 1 to $\mathbb{R}$, inclusive, and columns have valid indices from 1 to $\mathbb{C}$, inclusive).

- `initialize(R, C)` resets the data structure.
- `vals[r][c]` stores the value at index `(r, c)`.
- `add(r, c, x)` adds `x` to the value at index `(r, c)`.
- `set(r, c, x)` assigns `x` to the value at index `(r, c)`.

- `sum(r, c)` returns the sum of the rectangle with upper-left corner (1, 1) and lower-right corner (r, c).
- `sum(r1, c1, r2, c2)` returns the sum of the rectangle with upper-left corner (r1, c1) and lower-right corner (r2, c2).

Time Complexity:

- $O(R \cdot C)$ per call to `initialize(R, C)`.
- $O(\log(R) \cdot \log(C))$ per call to all other operations.

Space Complexity:

- $O(R \cdot C)$ for storage of the array elements.
- $O(1)$ auxiliary for all operations.

```
#include <algorithm>
#include <vector>

std::vector<std::vector<int>> vals, tree;

void initialize(int R, int C) {
    vals.assign(R + 2, std::vector<int>(C + 2, 0));
    tree.assign(R + 2, std::vector<int>(C + 2, 0));
}

void add(int r, int c, int x) {
    vals[r][c] += x;
    for (int i = r; i < static_cast<int>(tree.size()); i += i & -i) {
        for (int j = c; j < static_cast<int>(tree[0].size()); j += j & -j) {
            tree[i][j] += x;
        }
    }
}

void set(int r, int c, int x) {
    add(r, c, x - vals[r][c]);
}

int sum(int r, int c) {
    int res = 0;
    for (int i = r; i > 0; i -= i & -i) {
        for (int j = c; j > 0; j -= j & -j) {
            res += tree[i][j];
        }
    }
    return res;
}

int sum(int r1, int c1, int r2, int c2) {
    return sum(r2, c2) + sum(r1 - 1, c1 - 1) - sum(r1 - 1, c2) - sum(r2, c1 - 1);
}
```

Example Usage

```
#include <cassert>
#include <iostream>
using namespace std;

int main() {
    initialize(3, 3);
    set(1, 1, 5);
    set(1, 2, 6);
    set(2, 1, 7);
    add(3, 3, 9);
    add(2, 1, -4);
    cout << "Values:" << endl;
    for (int i = 1; i <= 3; i++) {
```

```

    for (int j = 1; j <= 3; j++) {
        cout << vals[i][j] << " ";
    }
    cout << endl;
}
assert(sum(1, 1, 1, 2) == 11);
assert(sum(1, 1, 2, 1) == 8);
assert(sum(1, 1, 3, 3) == 23);
return 0;
}

```

Example Output

```

Values:
5 6 0
3 0 0
0 0 9

```

2.6.8 2D Fenwick Tree (Compressed)

Maintain a 2D array of numerical type, allowing for rectangular sub-matrices to be simultaneously incremented by arbitrary values (range update) and queries for the sum of rectangular sub-matrices (range query). This implementation uses `std::unordered_map` for coordinate compression, allowing for large indices to be accessed with efficient space complexity. That is, rows have valid indices from 0 to $\mathbb{R}$, inclusive, and columns have valid indices from 0 to $\mathbb{C}$, inclusive.

- `add(r, c, x)` adds x to the value at index (r, c) .
- `add(r1, c1, r2, c2, x)` adds x to all indices in the rectangle with upper-left corner $(r1, c1)$ and lower-right corner $(r2, c2)$.
- `set(r, c, x)` assigns x to the value at index (r, c) .
- `sum(r, c)` returns the sum of the rectangle with upper-left corner $(0, 0)$ and lower-right corner (r, c) .
- `sum(r1, c1, r2, c2)` returns the sum of the rectangle with upper-left corner $(r1, c1)$ and lower-right corner $(r2, c2)$.
- `at(r, c)` returns the value at index (r, c) .

Time Complexity: $O(\log(R) \cdot \log(C))$ per call to all member functions.

Space Complexity:

- $O(n \cdot \log(R) \cdot \log(C))$ for storage of the array elements, where n is the number of distinct indices that have been accessed across all of the operations so far.
- $O(1)$ auxiliary for all operations.

```

#include <unordered_map>
#include <utility>

template<class T>
class FenwickTree2D {
    static const int R = 1000000001;
    static const int C = 1000000001;

    std::unordered_map<long long, T> t1, t2, t3, t4;

    template<class Map>
    static T get(const Map &tree, int r, int c) {
        auto it = tree.find(static_cast<long long>(r) * C + c);
        return it == tree.end() ? T() : it->second;
    }

    template<class Map>
    static void add(Map &tree, int r, int c, const T &x) {
        for (int i = r + 1; i <= R; i += i & -i) {

```

```

        for (int j = c + 1; j <= C; j += j & -j) {
            tree[static_cast<long long>(i) * C + j] += x;
        }
    }
}

void add_helper(int r, int c, const T &x) {
    add(t1, 0, 0, x);
    add(t1, 0, c, -x);
    add(t2, 0, c, x * c);
    add(t1, r, 0, -x);
    add(t3, r, 0, x * r);
    add(t1, r, c, x);
    add(t2, r, c, -x * c);
    add(t3, r, c, -x * r);
    add(t4, r, c, x * r * c);
}

public:
void add(int r1, int c1, int r2, int c2, const T &x) {
    add_helper(r2 + 1, c2 + 1, x);
    add_helper(r1, c2 + 1, -x);
    add_helper(r2 + 1, c1, -x);
    add_helper(r1, c1, x);
}

void add(int r, int c, const T &x) { add(r, c, r, c, x); }
void set(int r, int c, const T &x) { add(r, c, x - at(r, c)); }

T sum(int r, int c) {
    r++;
    c++;
    T s1 = 0, s2 = 0, s3 = 0, s4 = 0;
    for (int i = r; i > 0; i -= i & -i) {
        for (int j = c; j > 0; j -= j & -j) {
            s1 += get(t1, i, j);
            s2 += get(t2, i, j);
            s3 += get(t3, i, j);
            s4 += get(t4, i, j);
        }
    }
    return s1 * r * c + s2 * r + s3 * c + s4;
}

T sum(int r1, int c1, int r2, int c2) {
    return sum(r2, c2) + sum(r1 - 1, c1 - 1) - sum(r1 - 1, c2) - sum(r2, c1 - 1);
}

T at(int r, int c) { return sum(r, c, r, c); }
};

```

Example Usage

```

#include <cassert>
#include <iostream>
using namespace std;

int main() {
    FenwickTree2D<int> t;
    t.set(0, 0, 5);
    t.set(0, 1, 6);
    t.set(1, 0, 7);
    t.add(2, 2, 9);
    t.add(1, 0, -4);
    t.add(1, 1, 2, 2, 5);
    cout << "Values:" << endl;
}

```

```

for (int i = 0; i < 3; i++) {
    for (int j = 0; j < 3; j++) {
        cout << t.at(i, j) << " ";
    }
    cout << endl;
}
assert(t.sum(0, 0, 0, 1) == 11);
assert(t.sum(0, 0, 1, 0) == 8);
assert(t.sum(1, 1, 2, 2) == 29);
t.set(500000000, 500000000, 100);
assert(t.sum(0, 0, 1000000000, 1000000000) == 143);
return 0;
}

```

Example Output

```

Values:
5 6 0
3 5 5
0 5 14

```

2.7 Disjoint Sets and Tree Structures

2.7.1 Disjoint Set Forest (Simple)

Maintain a set of elements partitioned into non-overlapping subsets. Each partition is assigned a unique representative known as the parent, or root. The following implements two well-known optimizations known as union-by-rank and path compression. This version is simplified to only work on integer elements.

- `initialize()` resets the data structure.
- `make_set(u)` creates a new partition consisting of the single element `u`, which must not have been previously added to the data structure.
- `find_root(u)` returns the unique representative of the partition containing `u`.
- `is_united(u, v)` returns whether elements `u` and `v` belong to the same partition.
- `unite(u, v)` replaces the partitions containing `u` and `v` with a single new partition consisting of the union of elements in the original partitions.

A precondition to the last three operations is that `make_set()` must have been previously called on their arguments.

Time Complexity:

- $O(1)$ per call to `initialize()` and `make_set()`.
- $O(\alpha(n))$ per call to `find_root()`, `is_united()`, and `unite()`, where n is the number of elements that has been added via `make_set()` so far, and $\alpha(n)$ is the extremely slow growing inverse of the Ackermann function (effectively a very small constant for all practical values of n).

Space Complexity:

- $O(n)$ for storage of the disjoint set forest elements.
- $O(1)$ auxiliary for all operations.

```

#include <vector>

int num_sets;
std::vector<int> root, rank;

void initialize() {
    num_sets = 0;
    root.clear();
    rank.clear();
}

```

```

void make_set(int u) {
    if (u >= static_cast<int>(root.size())) {
        root.resize(u + 1);
        rank.resize(u + 1);
    }
    root[u] = u;
    rank[u] = 0;
    num_sets++;
}

int find_root(int u) {
    if (root[u] != u) {
        root[u] = find_root(root[u]);
    }
    return root[u];
}

bool is_united(int u, int v) {
    return find_root(u) == find_root(v);
}

void unite(int u, int v) {
    int ru = find_root(u), rv = find_root(v);
    if (ru == rv) {
        return;
    }
    num_sets--;
    if (rank[ru] < rank[rv]) {
        root[ru] = rv;
    } else {
        root[rv] = ru;
        if (rank[ru] == rank[rv]) {
            rank[ru]++;
        }
    }
}

```

Example Usage

```

#include <cassert>

int main() {
    initialize();
    for (char c = 'a'; c <= 'g'; c++) {
        make_set(c);
    }
    unite('a', 'b');
    unite('b', 'f');
    unite('d', 'e');
    unite('d', 'g');
    assert(num_sets == 3);
    assert(is_united('a', 'b'));
    assert(!is_united('a', 'c'));
    assert(!is_united('b', 'g'));
    assert(is_united('e', 'g'));
    return 0;
}

```

2.7.2 Disjoint Set Forest (Compressed)

Maintain a set of elements partitioned into non-overlapping subsets using a collection of trees. Each partition is assigned a unique representative known as the parent, or root. The following implements two well-known

optimizations known as union-by-rank and path compression. This version uses an `std::unordered_map` for storage and coordinate compression (thus, element types must meet the requirements of key types for `std::unordered_map`). The order of sets returned by `get_all_sets()` is unspecified.

- `make_set(u)` creates a new partition consisting of the single element `u`, which must not have been previously added to the data structure.
- `is_united(u, v)` returns whether elements `u` and `v` belong to the same partition.
- `unite(u, v)` replaces the partitions containing `u` and `v` with a single new partition consisting of the union of elements in the original partitions.
- `get_all_sets()` returns all current partitions as a vector of vectors.

A precondition to the last three operations is that `make_set()` must have been previously called on their arguments.

Time Complexity:

- $O(1)$ per call to the constructor.
- $O(1)$ on average per call to `make_set()`, where n is the number of elements that have been added via `make_set()` so far.
- $O(\alpha(n))$ on average per call to `is_united()` and `unite()`, where n is the number of elements that have been added via `make_set()` so far, and $\alpha(n)$ is the extremely slow growing inverse of the Ackermann function (effectively a very small constant for all practical values of n).
- $O(n)$ per call to `get_all_sets()`.

Space Complexity:

- $O(n)$ for storage of the disjoint set forest elements.
- $O(n)$ auxiliary heap space for `get_all_sets()`.
- $O(1)$ auxiliary for all other operations.

```
#include <algorithm>
#include <unordered_map>
#include <vector>

template<class T>
class DisjointSetForest {
    int num_elements, num_sets;
    std::unordered_map<T, int> id;
    std::vector<int> root, rank;

    int find_root(int u) {
        if (root[u] != u) {
            root[u] = find_root(root[u]);
        }
        return root[u];
    }
}

public:
    DisjointSetForest() : num_elements(0), num_sets(0) {}

    int size() const { return num_elements; }
    int sets() const { return num_sets; }

    void make_set(const T &u) {
        if (id.find(u) != id.end()) {
            return;
        }
        id[u] = num_elements;
        root.push_back(num_elements++);
        rank.push_back(0);
        num_sets++;
    }

    bool is_united(const T &u, const T &v) { return find_root(id[u]) == find_root(id[v]); }

    void unite(const T &u, const T &v) {
```

```

    int ru = find_root(id[u]), rv = find_root(id[v]);
    if (ru == rv) {
        return;
    }
    num_sets--;
    if (rank[ru] < rank[rv]) {
        root[ru] = rv;
    } else {
        root[rv] = ru;
        if (rank[ru] == rank[rv]) {
            rank[ru]++;
        }
    }
}

std::vector<std::vector<T>> get_all_sets() {
    std::unordered_map<int, std::vector<T>> tmp;
    for (auto &[key, val] : id) {
        tmp[find_root(val)].push_back(key);
    }
    std::vector<std::vector<T>> res;
    for (auto &[key, val] : tmp) {
        res.push_back(val);
    }
    return res;
}
};

```

Example Usage

```

#include <cassert>
#include <iostream>
using namespace std;

int main() {
    DisjointSetForest<char> dsf;
    for (char c = 'a'; c <= 'g'; c++) {
        dsf.make_set(c);
    }
    dsf.unite('a', 'b');
    dsf.unite('b', 'f');
    dsf.unite('d', 'e');
    dsf.unite('d', 'g');
    assert(dsf.size() == 7);
    assert(dsf.sets() == 3);
    auto s = dsf.get_all_sets();
    for (auto &set : s) {
        sort(set.begin(), set.end());
    }
    sort(s.begin(), s.end());
    bool first_set = true;
    for (const auto &set : s) {
        cout << (first_set ? "[" : ", [");
        first_set = false;
        bool first_value = true;
        for (char value : set) {
            cout << (first_value ? "" : ", ") << value;
            first_value = false;
        }
        cout << "]";
    }
    cout << endl;
    return 0;
}

```

Example Output

[a, b, f], [c], [d, e, g]

2.7.3 Disjoint Set Forest (Rollback)

Maintains disjoint sets while supporting rollback to previous snapshots. Rollback DSU is useful for offline dynamic connectivity, divide-and-conquer over time, and backtracking search where unions must be undone.

Path compression is intentionally not used, because it is hard to undo. Union by size keeps tree height logarithmic, and every successful union records enough information to restore the previous state.

- `initialize(n)` creates singleton sets over elements `0` through `n - 1`.
- `find_root(u)` returns the representative of the set containing `u`.
- `is_united(u, v)` returns whether `u` and `v` are in the same set.
- `unite(u, v)` merges two sets and returns whether a merge occurred.
- `snapshot()` returns a token representing the current history size.
- `rollback(s)` undoes all changes made after snapshot `s`.

Time Complexity:

- $O(n)$ per call to `initialize(n)`.
- $O(\log n)$ worst-case per call to `find_root(u)`, `is_united(u, v)`, and `unite(u, v)`.
- $O(1)$ per undone union during `rollback(s)`.

Space Complexity: $O(n + q)$ for q successful unions stored in the rollback history.

```
#include <numeric>
#include <vector>

class RollbackDSU {
    struct Change {
        int child, parent, parent_size;

        Change(int child = -1, int parent = -1, int parent_size = 0)
            : child(child), parent(parent), parent_size(parent_size) {}
    };

    std::vector<int> root, size;
    std::vector<Change> history;
    int sets;

public:
    RollbackDSU() : sets(0) {}

    explicit RollbackDSU(int n) { initialize(n); }

    void initialize(int n) {
        root.resize(n);
        size.assign(n, 1);
        history.clear();
        sets = n;
        std::iota(root.begin(), root.end(), 0);
    }

    int count_sets() const { return sets; }

    int find_root(int u) const {
        while (root[u] != u) {
            u = root[u];
        }
        return u;
    }

    bool is_united(int u, int v) const { return find_root(u) == find_root(v); }
```



```

bool unite(int u, int v) {
    u = find_root(u);
    v = find_root(v);
    if (u == v) {
        return false;
    }
    if (size[u] > size[v]) {
        int tmp = u;
        u = v;
        v = tmp;
    }
    history.emplace_back(u, v, size[v]);
    root[u] = v;
    size[v] += size[u];
    sets--;
    return true;
}

int snapshot() const { return history.size(); }

void rollback(int snapshot) {
    while (static_cast<int>(history.size()) > snapshot) {
        Change c = history.back();
        history.pop_back();
        root[c.child] = c.child;
        size[c.parent] = c.parent_size;
        sets++;
    }
}
};

```

Example Usage

```

#include <cassert>

int main() {
    RollbackDSU dsu(5);
    dsu.unite(0, 1);
    int s = dsu.snapshot();
    dsu.unite(1, 2);
    dsu.unite(3, 4);
    assert(dsu.is_united(0, 2));
    assert(dsu.count_sets() == 2);
    dsu.rollback(s);
    assert(dsu.is_united(0, 1));
    assert(!dsu.is_united(0, 2));
    assert(!dsu.is_united(3, 4));
    assert(dsu.count_sets() == 4);
    return 0;
}

```

2.7.4 Lowest Common Ancestor (Sparse Table)

Given a tree, determine the lowest common ancestor of any two nodes in the tree. The lowest common ancestor of two nodes u and v is the node that has the longest distance from the root while having both u and v as its descendant. A node is considered to be a descendant of itself. `build()` applies to a global, pre-populated adjacency list `adj` which must only consist of nodes numbered with integers between 0 (inclusive) and the total number of nodes (exclusive), as passed in the function argument. The adjacency list must define one connected tree rooted at `root`.

Time Complexity:

- $O(n \log n)$ per call to `build()`, where n is the number of nodes.
- $O(\log n)$ per call to `lca()`.

Space Complexity:

- $O(n \log n)$ to store the sparse table, where n is the number of nodes.
- $O(n)$ auxiliary stack space for `build()`.
- $O(1)$ auxiliary for `lca()`.

```
#include <vector>

std::vector<std::vector<int>> adj, dp;
int len, timer;
std::vector<int> tin, tout;

void dfs(int u, int p) {
    tin[u] = timer++;
    dp[u][0] = p;
    for (int i = 1; i < len; i++) {
        dp[u][i] = dp[dp[u][i - 1]][i - 1];
    }
    for (int v : adj[u]) {
        if (v != p) {
            dfs(v, u);
        }
    }
    tout[u] = timer++;
}

void build(int nodes, int root = 0) {
    len = 1;
    while ((1 << len) <= nodes) {
        len++;
    }
    dp.assign(nodes, std::vector<int>(len));
    tin.assign(nodes, 0);
    tout.assign(nodes, 0);
    timer = 0;
    dfs(root, root);
}

bool is_ancestor(int parent, int child) {
    return (tin[parent] <= tin[child]) && (tout[child] <= tout[parent]);
}

int lca(int u, int v) {
    if (is_ancestor(u, v)) {
        return u;
    }
    if (is_ancestor(v, u)) {
        return v;
    }
    for (int i = len - 1; i >= 0; i--) {
        if (!is_ancestor(dp[u][i], v)) {
            u = dp[u][i];
        }
    }
    return dp[u][0];
}
```

Example Usage

```
#include <cassert>
using namespace std;
```

```

int main() {
    int nodes = 5;
    adj.resize(nodes);
    adj[0].push_back(1);
    adj[1].push_back(0);
    adj[1].push_back(2);
    adj[2].push_back(1);
    adj[3].push_back(1);
    adj[1].push_back(3);
    adj[0].push_back(4);
    adj[4].push_back(0);
    build(nodes, 0);
    assert(lca(3, 2) == 1);
    assert(lca(2, 4) == 0);
    return 0;
}

```

2.7.5 Lowest Common Ancestor (Segment Tree)

Given a tree, determine the lowest common ancestor of any two nodes in the tree. The lowest common ancestor of two nodes u and v is the node that has the longest distance from the root while having both u and v as its descendant. A node is considered to be a descendant of itself. `build()` applies to a global, pre-populated adjacency list `adj` which must only consist of nodes numbered with integers between 0 (inclusive) and the total number of nodes (exclusive), as passed in the function argument. The adjacency list must define one connected tree rooted at `root`.

Time Complexity:

- $O(n \log n)$ per call to `build()`, where n is the number of nodes.
- $O(\log n)$ per call to `lca()`.

Space Complexity:

- $O(n)$ for storage of the segment tree, where n is the number of nodes.
- $O(n)$ auxiliary stack space for `build()`.
- $O(\log n)$ auxiliary stack space for `lca()`.

```

#include <algorithm>
#include <vector>

std::vector<std::vector<int>> adj;
int len, counter;
std::vector<int> depth, dfs_order, first, minpos;

void dfs(int u, int d) {
    depth[u] = d;
    dfs_order[counter++] = u;
    for (int v : adj[u]) {
        if (depth[v] == -1) {
            dfs(v, d + 1);
            dfs_order[counter++] = u;
        }
    }
}

void build(int n, int lo, int hi) {
    if (lo == hi) {
        minpos[n] = dfs_order[lo];
        return;
    }
    int lchild = 2 * n + 1, rchild = 2 * n + 2, mid = lo + (hi - lo) / 2;
    build(lchild, lo, mid);

```

```

    build(rchild, mid + 1, hi);
    minpos[n] = depth[minpos[lchild]] < depth[minpos[rchild]] ? minpos[lchild] : minpos[rchild];
}

void build(int nodes, int root) {
    depth.assign(nodes, -1);
    dfs_order.assign(2 * nodes, 0);
    first.assign(nodes, -1);
    minpos.assign(8 * nodes, 0);
    len = 2 * nodes - 1;
    counter = 0;
    dfs(root, 0);
    build(0, 0, len - 1);
    for (int i = 0; i < len; i++) {
        if (first[dfs_order[i]] == -1) {
            first[dfs_order[i]] = i;
        }
    }
}

int get_minpos(int a, int b, int n, int lo, int hi) {
    if (a == lo && b == hi) {
        return minpos[n];
    }
    int mid = lo + (hi - lo) / 2;
    if (a <= mid && mid < b) {
        int p1 = get_minpos(a, std::min(b, mid), 2 * n + 1, lo, mid);
        int p2 = get_minpos(std::max(a, mid + 1), b, 2 * n + 2, mid + 1, hi);
        return depth[p1] < depth[p2] ? p1 : p2;
    }
    if (a <= mid) {
        return get_minpos(a, std::min(b, mid), 2 * n + 1, lo, mid);
    }
    return get_minpos(std::max(a, mid + 1), b, 2 * n + 2, mid + 1, hi);
}

int lca(int u, int v) {
    return get_minpos(std::min(first[u], first[v]), std::max(first[u], first[v]), 0, 0, len - 1);
}

```

Example Usage

```

#include <cassert>
using namespace std;

int main() {
    int nodes = 5;
    adj.resize(nodes);
    adj[0].push_back(1);
    adj[1].push_back(0);
    adj[1].push_back(2);
    adj[2].push_back(1);
    adj[3].push_back(1);
    adj[1].push_back(3);
    adj[0].push_back(4);
    adj[4].push_back(0);
    build(nodes, 0);
    assert(lca(3, 2) == 1);
    assert(lca(2, 4) == 0);
    return 0;
}

```

2.7.6 Heavy Light Decomposition

Maintain a tree with values associated with either its edges or nodes, while supporting both dynamic queries and dynamic updates of all values on a given path between two nodes in the tree. Heavy-light decomposition partitions the nodes of the tree into disjoint paths where all nodes have degree two, except the endpoints of a path which has degree one.

The query operation is defined by an associative `combine()` function which satisfies `combine(x, combine(y, z)) = combine(combine(x, y), z)` for all values `x`, `y`, and `z` in the tree. The default code below assumes a numerical tree type, defining queries for the “min” of the target range. Another possible query operation is “sum”, in which case `combine(a, b)` should return `a + b`. For direction-independent path queries, `combine()` should also be commutative; otherwise, store enough information in each aggregate to combine paths in the required order.

The update operation is defined by `apply_delta()` and `compose_deltas()`. A delta must act on an aggregate summary of a path of length `len`: `apply_delta(v, d, len)` returns the aggregate after applying update `d` to every element represented by aggregate `v`. Pending deltas are combined in chronological order by `compose_deltas(old, d)`, meaning “apply `old`, then apply `d`”. These hooks do not support arbitrary query/update pairings; the delta operation must distribute over `combine()`, and composed deltas must have the same effect as applying their updates sequentially. The default code below defines updates that “set” a path’s edges or nodes to a new value. For range increment updates, `apply_delta(v, d, len)` would return `v + d` for min/max queries, or `v + d * len` for sum queries, and `compose_deltas(old, d)` would return `old + d`.

- `HeavyLight(adj, v)` constructs a new heavy light decomposition on a tree defined by the adjacency list `adj`, with all values initialized to `v`. The adjacency list must consist of only the integers from 0 to `adj.size() - 1`, inclusive. No duplicate edges should exist, and the graph must be connected.
- `query(u, v)` returns the result of `combine()` applied to all values on the path from node `u` to node `v`.
- `update(u, v, d)` modifies all values on the path from node `u` to node `v` by respectively applying the delta `d`.

Time Complexity:

- $O(n)$ per call to the constructor, where n is the number of nodes.
- $O(\log^2 n)$ per call to `query()` and `update()`;

Space Complexity:

- $O(n)$ for storage of the decomposition.
- $O(n)$ auxiliary stack space for the constructor.
- $O(1)$ auxiliary for `query()` and `update()`.

```
#include <algorithm>
#include <stdexcept>
#include <vector>

template<class T>
class HeavyLight {
    // Set this to true to store values on edges, false to store values on nodes.
    static const bool VALUES_ON_EDGES = true;

    static T combine(const T &a, const T &b) { return std::min(a, b); }
    static T apply_delta(const T &v, const T &d, int len) { return d; }
    static T compose_deltas(const T &old, const T &d) { return d; }

    int counter, paths;
    std::vector<std::vector<T>> value, delta;
    std::vector<std::vector<bool>> pending;
    std::vector<std::vector<int>> len;
    std::vector<int> size, parent, tin, tout, path, pathlen, pathpos, pathroot;
    const std::vector<std::vector<int>> &adj;

    void dfs(int u, int p) {
        tin[u] = counter++;
        parent[u] = p;
```

```

    size[u] = 1;
    for (int v : adj[u]) {
        if (v != p) {
            dfs(v, u);
            size[u] += size[v];
        }
    }
    tout[u] = counter++;
}

int new_path(int u) {
    pathroot[paths] = u;
    return paths++;
}

void build_paths(int u, int path) {
    this->path[u] = path;
    pathpos[u] = pathlen[path]++;
    for (int v : adj[u]) {
        if (v != parent[u]) {
            build_paths(v, (2 * size[v] >= size[u]) ? path : new_path(v));
        }
    }
}

inline T applied_value(int path, int i) {
    return pending[path][i] ? apply_delta(value[path][i], delta[path][i], len[path][i])
        : value[path][i];
}

void push_delta(int path, int i) {
    int d = 0;
    while ((i >> d) > 0) {
        d++;
    }
    for (d -= 2; d >= 0; d--) {
        int l = (i >> d), r = (l ^ 1), n = l / 2;
        if (pending[path][n]) {
            value[path][n] = applied_value(path, n);
            delta[path][l] =
                pending[path][l] ? compose_deltas(delta[path][l], delta[path][n]) : delta[path][n];
            delta[path][r] =
                pending[path][r] ? compose_deltas(delta[path][r], delta[path][n]) : delta[path][n];
            pending[path][l] = pending[path][r] = true;
            pending[path][n] = false;
        }
    }
}

bool query(int path, int u, int v, T *res) {
    push_delta(path, u += value[path].size() / 2);
    push_delta(path, v += value[path].size() / 2);
    bool found = false;
    for (; u <= v; u = (u + 1) / 2, v = (v - 1) / 2) {
        if ((u & 1) != 0) {
            T value = applied_value(path, u);
            *res = found ? combine(*res, value) : value;
            found = true;
        }
        if ((v & 1) == 0) {
            T value = applied_value(path, v);
            *res = found ? combine(*res, value) : value;
            found = true;
        }
    }
    return found;
}

```

```

void update(int path, int u, int v, const T &d) {
    push_delta(path, u += value[path].size() / 2);
    push_delta(path, v += value[path].size() / 2);
    int tu = -1, tv = -1;
    for (; u <= v; u = (u + 1) / 2, v = (v - 1) / 2) {
        if ((u & 1) != 0) {
            delta[path][u] = pending[path][u] ? compose_deltas(delta[path][u], d) : d;
            pending[path][u] = true;
            if (tu == -1) {
                tu = u;
            }
        }
        if ((v & 1) == 0) {
            delta[path][v] = pending[path][v] ? compose_deltas(delta[path][v], d) : d;
            pending[path][v] = true;
            if (tv == -1) {
                tv = v;
            }
        }
    }
    for (int i = tu; i > 1; i /= 2) {
        value[path][i / 2] = combine(applied_value(path, i), applied_value(path, i ^ 1));
    }
    for (int i = tv; i > 1; i /= 2) {
        value[path][i / 2] = combine(applied_value(path, i), applied_value(path, i ^ 1));
    }
}

inline bool is_ancestor(int parent, int child) {
    return (tin[parent] <= tin[child]) && (tout[child] <= tout[parent]);
}

public:
    explicit HeavyLight(const std::vector<std::vector<int>> &adj, const T &v = T())
        : counter(0),
          paths(0),
          size(adj.size()),
          parent(adj.size()),
          tin(adj.size()),
          tout(adj.size()),
          path(adj.size()),
          pathlen(adj.size()),
          pathpos(adj.size()),
          pathroot(adj.size()),
          adj(adj) {
        dfs(0, -1);
        build_paths(0, new_path(0));
        value.resize(paths);
        delta.resize(paths);
        pending.resize(paths);
        len.resize(paths);
        for (int i = 0; i < paths; i++) {
            int m = pathlen[i];
            value[i].assign(2 * m, v);
            delta[i].resize(2 * m);
            pending[i].assign(2 * m, false);
            len[i].assign(2 * m, 1);
            for (int j = 2 * m - 1; j > 1; j -= 2) {
                value[i][j / 2] = combine(value[i][j], value[i][j ^ 1]);
                len[i][j / 2] = len[i][j] + len[i][j ^ 1];
            }
        }
    }

    T query(int u, int v) {
        if (VALUES_ON_EDGES && u == v) {
            throw std::runtime_error("No edge between u and v to be queried.");
        }
    }

```

```

bool found = false;
T res = T(), value;
int root;
while (!is_ancestor(root = pathroot[path[u]], v)) {
    if (query(path[u], 0, pathpos[u], &value)) {
        res = found ? combine(res, value) : value;
        found = true;
    }
    u = parent[root];
}
while (!is_ancestor(root = pathroot[path[v]], u)) {
    if (query(path[v], 0, pathpos[v], &value)) {
        res = found ? combine(res, value) : value;
        found = true;
    }
    v = parent[root];
}
if (query(
    path[u], std::min(pathpos[u], pathpos[v]) + static_cast<int>(VALUES_ON_EDGES),
    std::max(pathpos[u], pathpos[v]), &value
)) {
    res = found ? combine(res, value) : value;
    found = true;
}
if (!found) {
    throw std::runtime_error("Unexpected error: No values found.");
}
return res;
}

void update(int u, int v, const T &d) {
    if (VALUES_ON_EDGES && u == v) {
        return;
    }
    int root;
    while (!is_ancestor(root = pathroot[path[u]], v)) {
        update(path[u], 0, pathpos[u], d);
        u = parent[root];
    }
    while (!is_ancestor(root = pathroot[path[v]], u)) {
        update(path[v], 0, pathpos[v], d);
        v = parent[root];
    }
    update(
        path[u], std::min(pathpos[u], pathpos[v]) + static_cast<int>(VALUES_ON_EDGES),
        std::max(pathpos[u], pathpos[v]), d
    );
}
};

```

Example Usage

```

#include <cassert>
using namespace std;

int main() {
    //      w=40      w=20      w=10
    // 0-----1-----2-----3
    //           |
    //           -----4
    //           w=30
    vector<vector<int>> adj(5);
    adj[0].push_back(1);
    adj[1].push_back(0);
    adj[1].push_back(2);
    adj[2].push_back(1);
}

```



```

adj[2].push_back(3);
adj[3].push_back(2);
adj[2].push_back(4);
adj[4].push_back(2);
HeavyLight<int> hld(adj, 0);
hld.update(0, 1, 40);
hld.update(1, 2, 20);
hld.update(2, 3, 10);
hld.update(2, 4, 30);
assert(hld.query(0, 3) == 10);
assert(hld.query(2, 4) == 30);
hld.update(3, 4, 5);
assert(hld.query(1, 4) == 5);
return 0;
}

```

2.7.7 Link-Cut Tree

Maintain a forest of trees with values associated with its nodes, while supporting both dynamic queries and dynamic updates of all values on any path between two nodes in a given tree. In addition, support testing of whether two nodes are connected in the forest, as well as the merging and splitting of trees by adding or removing specific edges. Link/cut forests divide each of its trees into vertex-disjoint paths, each represented by a splay tree.

The query operation is defined by an associative `combine()` function which satisfies

`combine(x, combine(y, z)) = combine(combine(x, y), z)` for all values `x`, `y`, and `z` in the forest. The default code below assumes a numerical forest type, defining queries for the “min” of the target range. Another possible query operation is “sum”, in which case `combine(a, b)` should return $a + b$. For direction-independent path queries, `combine()` should also be commutative; otherwise, store enough information in each aggregate to combine paths in the required order.

The update operation is defined by `apply_delta()` and `compose_deltas()`. A delta must act on an aggregate summary of a path of length `len`: `apply_delta(v, d, len)` returns the aggregate after applying update `d` to every element represented by aggregate `v`. Pending deltas are combined in chronological order by `compose_deltas(old, d)`, meaning “apply `old`, then apply `d`”. These hooks do not support arbitrary query/update pairings; the delta operation must distribute over `combine()`, and composed deltas must have the same effect as applying their updates sequentially. The default code below defines updates that “set” a path’s nodes to a new value. For range increment updates, `apply_delta(v, d, len)` would return $v + d$ for min/max queries, or $v + d * len$ for sum queries, and `compose_deltas(old, d)` would return `old + d`.

- `LinkCutForest()` constructs an empty forest with no trees.
- `size()` returns the number of nodes in the forest.
- `trees()` returns the number of trees in the forest.
- `make_root(i, v)` creates a new tree in the forest consisting of a single node labeled with the integer `i` and value initialized to `v`.
- `is_connected(a, b)` returns whether nodes `a` and `b` are connected.
- `link(a, b)` adds an edge between the nodes `a` and `b`, both of which must exist and not be connected.
- `cut(a, b)` removes the edge between the nodes `a` and `b`, both of which must exist and be connected.
- `query(a, b)` returns the result of `combine()` applied to all values on the path from the node `a` to node `b`.
- `update(a, b, d)` modifies all the values on the path from node `a` to node `b` by respectively applying the delta `d`.

Time Complexity:

- $O(1)$ per call to the constructor, `size()`, and `trees()`.
- $O(\log n)$ amortized per call to all other operations, where n is the number of nodes.

Space Complexity:

- $O(n)$ for storage of the forest, where n is the number of nodes.

- $O(1)$ auxiliary for all operations.

```

#include <algorithm>
#include <cstdint>
#include <stdexcept>
#include <unordered_map>

template<class T>
class LinkCutForest {
    static T combine(const T &a, const T &b) { return std::min(a, b); }
    static T apply_delta(const T &v, const T &d, int len) { return d; }
    static T compose_deltas(const T &old, const T &d) { return d; }

    struct Node {
        T value, subtree_value, delta;
        int size;
        bool rev, pending;
        Node *left, *right, *parent;

        explicit Node(const T &v)
            : value(v),
              subtree_value(v),
              size(1),
              rev(false),
              pending(false),
              left(nullptr),
              right(nullptr),
              parent(nullptr) {}

        inline bool is_root() const {
            return parent == nullptr || (parent->left != this && parent->right != this);
        }

        inline T get_subtree_value() const {
            return pending ? apply_delta(subtree_value, delta, size) : subtree_value;
        }

        void push() {
            if (rev) {
                rev = false;
                std::swap(left, right);
                if (left != nullptr) {
                    left->rev = !left->rev;
                }
                if (right != nullptr) {
                    right->rev = !right->rev;
                }
            }
            if (pending) {
                value = apply_delta(value, delta, 1);
                subtree_value = apply_delta(subtree_value, delta, size);
                if (left != nullptr) {
                    left->delta = left->pending ? compose_deltas(left->delta, delta) : delta;
                    left->pending = true;
                }
                if (right != nullptr) {
                    right->delta = right->pending ? compose_deltas(right->delta, delta) : delta;
                    right->pending = true;
                }
                pending = false;
            }
        }

        void update() {
            size = 1;
            subtree_value = value;
            if (left != nullptr) {
                subtree_value = combine(subtree_value, left->get_subtree_value());
            }
        }
    };
};

```

```

        size += left->size;
    }
    if (right != nullptr) {
        subtree_value = combine(subtree_value, right->get_subtree_value());
        size += right->size;
    }
}
};

int num_trees;
std::unordered_map<int, Node *> nodes;

static void connect(Node *child, Node *parent, bool is_left) {
    if (child != nullptr) {
        child->parent = parent;
    }
    if (is_left) {
        parent->left = child;
    } else {
        parent->right = child;
    }
}

static void rotate(Node *n) {
    Node *parent = n->parent, *grandparent = parent->parent;
    bool parent_is_root = parent->is_root(), is_left = (n == parent->left);
    connect(is_left ? n->right : n->left, parent, is_left);
    connect(parent, n, !is_left);
    if (parent_is_root) {
        if (n != nullptr) {
            n->parent = grandparent;
        }
    } else {
        connect(n, grandparent, parent == grandparent->left);
    }
    parent->update();
}

static void splay(Node *n) {
    while (!n->is_root()) {
        Node *parent = n->parent, *grandparent = parent->parent;
        if (!parent->is_root()) {
            grandparent->push();
        }
        parent->push();
        n->push();
        if (!parent->is_root()) {
            if ((n == parent->left) == (parent == grandparent->left)) {
                rotate(parent);
            } else {
                rotate(n);
            }
        }
        rotate(n);
    }
    n->push();
    n->update();
}

static Node *expose(Node *n) {
    Node *prev = nullptr;
    for (Node *curr = n; curr != nullptr; curr = curr->parent) {
        splay(curr);
        curr->left = prev;
        prev = curr;
    }
    splay(n);
    return prev;
}

```

```

}

// Helper variables.
Node *u, *v;

void get_uv(int a, int b) {
    auto it1 = nodes.find(a);
    auto it2 = nodes.find(b);
    if (it1 == nodes.end() || it2 == nodes.end()) {
        throw std::runtime_error("Queried node ID does not exist in forest.");
    }
    u = it1->second;
    v = it2->second;
}

public:
LinkCutForest() : num_trees(0) {}

~LinkCutForest() {
    for (auto &[key, node] : nodes) {
        delete node;
    }
}
LinkCutForest(const LinkCutForest &) = delete;
LinkCutForest &operator=(const LinkCutForest &) = delete;

int size() const { return nodes.size(); }
int trees() const { return num_trees; }

void make_root(int i, const T &v = T()) {
    if (auto it = nodes.find(i); it != nodes.end()) {
        throw std::runtime_error("Cannot make a root with an existing ID.");
    }
    Node *n = new Node(v);
    expose(n);
    n->rev = !n->rev;
    nodes[i] = n;
    num_trees++;
}

bool is_connected(int a, int b) {
    get_uv(a, b);
    if (a == b) {
        return true;
    }
    expose(u);
    expose(v);
    return u->parent != nullptr;
}

void link(int a, int b) {
    if (is_connected(a, b)) {
        throw std::runtime_error("Cannot link nodes that are already connected.");
    }
    get_uv(a, b);
    expose(u);
    u->rev = !u->rev;
    u->parent = v;
    num_trees--;
}

void cut(int a, int b) {
    get_uv(a, b);
    expose(u);
    u->rev = !u->rev;
    expose(v);
    if (v->right != u || u->left != nullptr) {
        throw std::runtime_error("Cannot cut edge that does not exist.");
    }
}

```

```

    }
    v->right->parent = nullptr;
    v->right = nullptr;
    num_trees++;
}

T query(int a, int b) {
    if (!is_connected(a, b)) {
        throw std::runtime_error("Cannot query nodes that are not connected.");
    }
    get_uv(a, b);
    expose(u);
    u->rev = !u->rev;
    expose(v);
    return v->get_subtree_value();
}

void update(int a, int b, const T &d) {
    if (!is_connected(a, b)) {
        throw std::runtime_error("Cannot update nodes that are not connected.");
    }
    get_uv(a, b);
    expose(u);
    u->rev = !u->rev;
    expose(v);
    v->delta = v->pending ? compose_deltas(v->delta, d) : d;
    v->pending = true;
}
};

```

Example Usage

```

#include <cassert>
using namespace std;

int main() {
    LinkCutForest<int> lcf;
    lcf.make_root(0, 10);
    lcf.make_root(1, 40);
    lcf.make_root(2, 20);
    lcf.make_root(3, 10);
    lcf.make_root(4, 30);
    assert(lcf.size() == 5);
    assert(lcf.trees() == 5);
    lcf.link(0, 1);
    lcf.link(1, 2);
    lcf.link(2, 3);
    lcf.link(2, 4);
    assert(lcf.trees() == 1);

    // v=10      v=40      v=20      v=10
    // 0-----1-----2-----3
    //                |
    //                -----4
    //                      v=30
    assert(lcf.query(1, 4) == 20);
    lcf.update(1, 1, 100);
    lcf.update(2, 4, 100);

    // v=10      v=100      v=100      v=10
    // 0-----1-----2-----3
    //                |
    //                -----4
    //                      v=100
    assert(lcf.query(4, 4) == 100);
    assert(lcf.query(0, 4) == 10);
}

```

```

assert(lcf.query(3, 4) == 10);
lcf.cut(1, 2);

// v=10      v=100      v=100      v=0
// 0-----1      2-----3
//              |
//              -----4
//                      v=100
assert(lcf.trees() == 2);
assert(!lcf.is_connected(1, 2));
assert(!lcf.is_connected(0, 4));
assert(lcf.is_connected(2, 3));
return 0;
}

```

Chapter 3

Strings

3.1 String Utilities

Common string functions, many of which already have standard STL equivalents. These are written for educational purposes, without heavy optimizations (often depending on certain `std::string` functions that have unspecified complexity).

Time Complexity: $O(n)$ per call to most operations, where n is the length of the input string or total length of all input strings. Exceptions are noted with the individual helpers below.

Space Complexity:

- $O(n)$ auxiliary heap space per call to operations that return a new string or vector of strings.
- $O(1)$ auxiliary space per call to in-place operations.

```
#include <cctype>
#include <sstream>
#include <stdexcept>
#include <string>
#include <vector>
using std::string;
```

Integer Conversion:

- `to_str(i)` returns the string representation of integer `i`, much like `std::to_string()`.
- `to_int(s)` returns the integer representation of string `s`, much like `std::atoi()`, except here we handle special cases of overflow by throwing an exception.
- `itoa(value, &str, base)` implements the non-standard C function which converts `value` into a C string, storing it into pointer `str` in the given `base`. For more generalized base conversion, see the math utilities section.

```
template<class Int>
string to_str(Int i) {
    std::ostringstream oss;
    oss << i;
    return oss.str();
}

int to_int(const string &s) {
    std::istringstream iss(s);
    int res;
    if (!(iss >> res)) {
        throw std::runtime_error("to_int failed");
    }
}
```

```

    }
    return res;
}

char *itoa(int value, char *str, int base = 10) {
    if (base < 2 || base > 36) {
        *str = '\0';
        return str;
    }
    char *ptr = str, *ptr1 = str, tmp_c;
    int tmp_v;
    do {
        tmp_v = value;
        value /= base;
        *ptr++ =
            "zyxwvutsrqponmlkjihgfedcba9876543210123456789"
            "abcdefghijklmnopqrstuvwxyz"[35 + (tmp_v - value * base)];
    } while (value);
    if (tmp_v < 0) {
        *ptr++ = '-';
    }
    for (*ptr-- = '\0'; ptr1 < ptr; *ptr1++ = tmp_c) {
        tmp_c = *ptr;
        *ptr-- = *ptr1;
    }
    return str;
}

```

Case Conversion:

- `to_upper(s)` returns `s` with all alphabetical characters converted to uppercase.
- `to_lower(s)` returns `s` with all alphabetical characters converted to lowercase.
- `to_title(s)` returns the title case representation of string `s`, where the first letter of every word (consecutive alphabetical characters) is capitalized.

```

string to_upper(const string &s) {
    string res;
    auto upper = [](unsigned char c) { return static_cast<char>(toupper(c)); };
    for (char c : s) {
        res.push_back(upper(c));
    }
    return res;
}

string to_lower(const string &s) {
    string res;
    auto lower = [](unsigned char c) { return static_cast<char>(tolower(c)); };
    for (char c : s) {
        res.push_back(lower(c));
    }
    return res;
}

string to_title(const string &s) {
    string res;
    auto lower = [](unsigned char c) { return static_cast<char>(tolower(c)); };
    auto upper = [](unsigned char c) { return static_cast<char>(toupper(c)); };
    char prev = '\0';
    for (char c : s) {
        res.push_back(isalpha(static_cast<unsigned char>(prev)) ? lower(c) : upper(c));
        prev = res.back();
    }
    return res;
}

```


Stripping:

- `lstrip(s)` strips the left side of `s` in-place (that is, the input is modified) using the given delimiters and returns a reference to the stripped string.
- `rstrip(s)` strips the right side of `s` in-place using the given delimiters and returns a reference to the stripped string.
- `strip(s)` strips both sides of `s` in-place and returns a reference to the stripped string.

```
string &lstrip(string &s, const string &delim = " \n\t\v\f\r") {
    size_t pos = s.find_first_not_of(delim);
    if (pos != string::npos) {
        s.erase(0, pos);
    }
    return s;
}

string &rstrip(string &s, const string &delim = " \n\t\v\f\r") {
    size_t pos = s.find_last_not_of(delim);
    if (pos != string::npos) {
        s.erase(pos + 1);
    }
    return s;
}

string &strip(string &s, const string &delim = " \n\t\v\f\r") {
    return lstrip(rstrip(s));
}
```

Find and Replace:

- `find_all(haystack, needle)` returns a vector of all positions where the string `needle` appears in the string `haystack`.
- `replace(s, old, replacement)` returns a copy of `s` with all occurrences of the string `old` replaced with the given `replacement`.

```
std::vector<int> find_all(const string &haystack, const string &needle) {
    std::vector<int> res;
    size_t pos = haystack.find(needle, 0);
    while (pos != string::npos) {
        res.push_back(pos);
        pos = haystack.find(needle, pos + 1);
    }
    return res;
}

string replace(const string &s, const string &old, const string &replacement) {
    if (old.empty()) {
        return s;
    }
    string res(s);
    size_t pos = 0;
    while ((pos = res.find(old, pos)) != string::npos) {
        res.replace(pos, old.length(), replacement);
        pos += replacement.length();
    }
    return res;
}
```

Joining and Splitting:

- `join(v, delim)` returns the strings in vector `v` concatenated, separated by the given delimiter.
- `split(s, char delim)` returns a vector of tokens of `s`, split on a single character delimiter. Note that this version will not skip empty tokens. For example, `split("a::b", ":")` returns `{"a", "b"}`, not `{"a", "", "b"}`.

- `split(s, string delim)` returns a vector of tokens of `s`, split on a set of many possible single character delimiters. All characters of `delim` will be removed from `s`, and the remaining token(s) of `s` will be added sequentially to a vector and returned. Unlike the first version, empty tokens are skipped. For example, `split("a::b", ":")` returns `{"a", "b"}`, not `{"a", "", "b"}`.
- `explode(s, delim)` returns a vector of tokens of `s`, split on the entire delimiter string `delim`. Unlike the `split()` functions above, `delim` is treated as a contiguous boundary string, not merely a set of possible boundary characters. This will not skip empty tokens. For example, `explode("a:::b", ":::")` yields `{"a", "", "b"}`, not `{"a", "b"}`.

```
string join(const std::vector<string> &v, const string &delim = " ") {
    string res;
    for (int i = 0; i < static_cast<int>(v.size()); i++) {
        if (i > 0) res += delim;
        res += v[i];
    }
    return res;
}

std::vector<string> split(const string &s, char delim) {
    std::vector<string> res;
    std::stringstream ss(s);
    string curr;
    while (std::getline(ss, curr, delim)) {
        res.push_back(curr);
    }
    return res;
}

std::vector<string> split(const string &s, const string &delim = " \n\t\v\f\r") {
    std::vector<string> res;
    string curr;
    for (char c : s) {
        if (delim.find(c) == string::npos) {
            curr += c;
        } else if (!curr.empty()) {
            res.push_back(curr);
            curr = "";
        }
    }
    if (!curr.empty()) {
        res.push_back(curr);
    }
    return res;
}

std::vector<string> explode(const string &s, const string &delim) {
    std::vector<string> res;
    size_t last = 0, next = 0;
    while ((next = s.find(delim, last)) != string::npos) {
        res.push_back(s.substr(last, next - last));
        last = next + delim.size();
    }
    res.push_back(s.substr(last));
    return res;
}
```

Example Usage

```
#include <cassert>
using namespace std;

int main() {
    assert(to_str(123) + "4" == "1234");
    assert(to_int("1234") == 1234);
}
```

```

vector<char> buffer(50);
assert(string(itoa(1750, buffer.data(), 10)) == "1750");
assert(string(itoa(1750, buffer.data(), 16)) == "6d6");
assert(string(itoa(1750, buffer.data(), 2)) == "11011010110");

assert(to_upper("Hello world") == "HELLO WORLD");
assert(to_lower("Hello World") == "hello world");
assert(to_title("hello world") == "Hello World");

string s("  abc \n");
string t = s;
assert(lstrip(s) == "abc \n");
assert(rstrip(s) == strip(t));

vector<int> pos;
pos.push_back(0);
pos.push_back(7);
assert(find_all("abracadabra", "ab") == pos);
assert(replace("abcdabba", "ab", "00") == "00cd00ba");

assert(join(split("a\nb\ncde\nf", '\n'), "|") == "a|b|cde|f"); // split v1
assert(join(split("a::b,cde:,f", ":", "|"), "|") == "a|b|cde|f"); // split v2
assert(join(explode("a..b.cde...f", "."), "|") == "a|b.cde|f");
return 0;
}

```

3.2 Hashing

3.2.1 Hash Functions

Provides a small library of non-cryptographic hash functions and hash functors for common contest keys. These helpers are useful for fingerprinting values, combining keys, seeding randomized algorithms, and defining `std::unordered_map` keys for pairs and vectors. They are not suitable for passwords, signatures, or adversarial security.

The standalone functions below are deterministic. The `IntHasher` functor adds a per-run random seed before mixing, which is useful for making integer-keyed hash tables harder to hack in open-test contests.

- `mix32(x)` mixes a 32-bit unsigned integer `x`, using the MurmurHash3 finalizer.
- `mix64(x)` mixes a 64-bit unsigned integer `x`, using the SplitMix64 finalizer.
- `hash32(x)` and `hash64(x)` hash any integer type that can be converted to `unsigned int` or `uint64`, respectively.
- `hash_combine32(h, x)` and `hash_combine64(h, x)` fold another already-hashed value into an existing hash accumulator.
- `hash_float(x)` and `hash_double(x)` hash floating-point bit patterns, normalizing `-0.0` to `0.0`. Different NaN representations may still hash differently.
- `hash_string_fnv1a32(s)` and `hash_string_fnv1a64(s)` compute FNV-1a hashes of string `s`.
- `hash_range32(first, last)` and `hash_range64(first, last)` hash a sequence of integer-like values.
- `IntHasher<Int>`, `PairHasher<A, B>`, and `VectorHasher<T>` are hash functors for `std::unordered_map` and `std::unordered_set`.
- `GenericHasher<T>` recursively handles integer, pair, and vector keys, and falls back to `std::hash<T>` for other hashable types.

Time Complexity:

- $O(1)$ per call to the integer, combiner, and floating-point hash functions.
- $O(n)$ per call to string and range hashing functions, where n is the number of elements processed.
- $O(n)$ per call to `VectorHasher<T>::operator()`, where n is the vector size.

Space Complexity: $O(1)$ auxiliary.

```

#include <chrono>
#include <cstdint>
#include <cstring>
#include <functional>
#include <string>
#include <type_traits>
#include <unordered_map>
#include <utility>
#include <vector>

using uint64 = unsigned long long;

unsigned int mix32(unsigned int x) {
    x ^= x >> 16;
    x *= 0x85ebca6bU;
    x ^= x >> 13;
    x *= 0xc2b2ae35U;
    return x ^ (x >> 16);
}

uint64 mix64(uint64 x) {
    x += 0x9e3779b97f4a7c15ULL;
    x = (x ^ (x >> 30)) * 0xbf58476d1ce4e5b9ULL;
    x = (x ^ (x >> 27)) * 0x94d049bb133111ebULL;
    return x ^ (x >> 31);
}

template<class Int>
unsigned int hash32(Int x) {
    return mix32(static_cast<unsigned int>(x));
}

template<class Int>
uint64 hash64(Int x) {
    return mix64(static_cast<uint64>(x));
}

unsigned int hash_combine32(unsigned int h, unsigned int x) {
    return mix32(h ^ (x + 0x9e3779b9U + (h << 6) + (h >> 2)));
}

uint64 hash_combine64(uint64 h, uint64 x) {
    return mix64(h ^ (x + 0x9e3779b97f4a7c15ULL + (h << 6) + (h >> 2)));
}

unsigned int hash_float(float x) {
    if (x == 0.0f) {
        x = 0.0f; // Normalize -0.0.
    }
    unsigned int bits = 0;
    std::memcpy(&bits, &x, sizeof(bits));
    return mix32(bits);
}

uint64 hash_double(double x) {
    if (x == 0.0) {
        x = 0.0; // Normalize -0.0.
    }
    uint64 bits = 0;
    std::memcpy(&bits, &x, sizeof(bits));
    return mix64(bits);
}

unsigned int hash_string_fnv1a32(const std::string &s) {
    unsigned int h = 2166136261U;
    for (unsigned char c : s) {
        h ^= c;
        h *= 16777619U;
    }
}

```

```

    }
    return h;
}

uint64 hash_string_fnv1a64(const std::string &s) {
    uint64 h = 14695981039346656037ULL;
    for (unsigned char c : s) {
        h ^= c;
        h *= 1099511628211ULL;
    }
    return h;
}

template<class It>
unsigned int hash_range32(It first, It last) {
    unsigned int h = 0;
    for (It it = first; it != last; ++it) {
        h = hash_combine32(h, hash32(*it));
    }
    return h;
}

template<class It>
uint64 hash_range64(It first, It last) {
    uint64 h = 0;
    for (It it = first; it != last; ++it) {
        h = hash_combine64(h, hash64(*it));
    }
    return h;
}

template<class Int>
struct IntHasher {
    std::size_t operator()(Int x) const {
        static const uint64 RANDOM = std::chrono::steady_clock::now().time_since_epoch().count();
        return static_cast<std::size_t>(mix64(static_cast<uint64>(x) + RANDOM));
    }
};

template<class T>
struct GenericHasher;

template<class T, bool IsInteger>
struct ScalarHasher {
    std::size_t operator()(const T &x) const { return std::hash<T>()(x); }
};

template<class T>
struct ScalarHasher<T, true> {
    std::size_t operator()(T x) const { return IntHasher<T>{}(x); }
};

template<class A, class B>
struct PairHasher {
    std::size_t operator()(const std::pair<A, B> &p) const {
        uint64 h = 0;
        h = hash_combine64(h, static_cast<uint64>(GenericHasher<A>()(p.first)));
        h = hash_combine64(h, static_cast<uint64>(GenericHasher<B>()(p.second)));
        return static_cast<std::size_t>(h);
    }
};

template<class T>
struct VectorHasher {
    std::size_t operator()(const std::vector<T> &v) const {
        uint64 h = 0;
        for (const auto &x : v) {
            h = hash_combine64(h, static_cast<uint64>(GenericHasher<T>()(x)));
        }
    }
};

```

```

    }
    return static_cast<std::size_t>(h);
}
};

template<class T>
struct GenericHasher : ScalarHasher<T, std::is_integral<T>::value> {};

template<class A, class B>
struct GenericHasher<std::pair<A, B>> : PairHasher<A, B> {};

template<class T>
struct GenericHasher<std::vector<T>> : VectorHasher<T> {};

```

Example Usage

```

#include <cassert>
using namespace std;

int main() {
    assert(mix32(123U) == mix32(123U));
    assert(mix32(123U) != mix32(124U));
    assert(hash32(-1) == hash32(-1));
    assert(hash64(123) == mix64(123ULL));
    assert(hash_float(-0.0f) == hash_float(0.0f));
    assert(hash_double(-0.0) == hash_double(0.0));
    assert(hash_string_fnv1a32("abc") == hash_string_fnv1a32("abc"));
    assert(hash_string_fnv1a64("abc") != hash_string_fnv1a64("acb"));

    vector<int> v;
    v.push_back(1);
    v.push_back(2);
    v.push_back(3);
    assert(hash_range32(v.begin(), v.end()) == hash_range32(v.begin(), v.end()));
    assert(hash_range64(v.begin(), v.end()) == hash_range64(v.begin(), v.end()));

    unordered_map<long long, int, IntHasher<long long>> count;
    count[1000000000000LL] = 7;
    assert(count[1000000000000LL] == 7);

    using point = pair<int, int>;
    unordered_map<point, int, PairHasher<int, int>> dist;
    dist[point(3, 4)] = 5;
    assert(dist[point(3, 4)] == 5);

    unordered_map<vector<int>, int, VectorHasher<int>> seen;
    seen[v] = 1;
    assert(seen[v] == 1);

    using state = pair<vector<int>, int>;
    unordered_map<state, int, GenericHasher<state>> dp;
    dp[state(v, 4)] = 9;
    assert(dp[state(v, 4)] == 9);
    return 0;
}

```

3.2.2 Rolling Hash

Computes polynomial rolling hashes for sequences, supporting $O(1)$ contiguous subsequence hash queries after preprocessing. This is useful for probabilistic string matching, substring equality checks, repeated subarray detection, and binary-searching over candidate lengths.

The implementation works modulo the Mersenne prime $2^{61} - 1$ and uses `__uint128_t` for multiplication. The base `HASH_BASE` should be changed or chosen randomly for open-hacking environments. With a fixed base, hashing remains fast and practical, but it is still probabilistic and should not be used as proof of equality when exact verification is required.

By default, each sequence value is cast to `uint64` and mixed. For non-integer element types, pass a custom value hasher that maps each element to a stable nonzero value in `[1, HASH_MOD)`.

- `RollingHash<T>(first, last)` constructs prefix hashes for any iterator range of values accepted by the value hasher.
- `RollingHash<T>(v)` constructs prefix hashes for vector `v`.
- `get(l, r)` returns the hash of the half-open subsequence `[l, r)`.
- `hash(first, last)` returns the hash of an iterator range.
- `hash(v)` returns the hash of vector `v`.
- `concat(left, right, right_len)` returns the hash of the concatenation of a sequence with hash `left` and a sequence with hash `right` and length `right_len`.

Time Complexity:

- $O(n)$ per constructor call and whole-sequence hash, where n is the sequence length.
- $O(1)$ per call to `get(l, r)` and `concat(left, right, right_len)`.

Space Complexity:

- $O(n)$ for storage of prefix hashes and powers, where n is the maximum sequence length processed so far.
- $O(1)$ auxiliary per query.

```
#include <string>
#include <utility>
#include <vector>

using uint64 = unsigned long long;

const uint64 HASH_MOD = (1ULL << 61) - 1;
const uint64 HASH_BASE = 911382323ULL; // Change or randomize.

uint64 hash_add(uint64 a, uint64 b) {
    uint64 c = a + b;
    return c >= HASH_MOD ? c - HASH_MOD : c;
}

uint64 hash_sub(uint64 a, uint64 b) {
    return a >= b ? a - b : a + HASH_MOD - b;
}

uint64 hash_mul(uint64 a, uint64 b) {
    __uint128_t p = (__uint128_t)a * b;
    uint64 low = (uint64)p & HASH_MOD;
    uint64 high = (uint64)(p >> 61);
    uint64 res = low + high;
    if (res >= HASH_MOD) {
        res -= HASH_MOD;
    }
    return res;
}

uint64 hash_mix(uint64 x) {
    x += 0x9e3779b97f4a7c15ULL;
    x = (x ^ (x >> 30)) * 0xbf58476d1ce4e5b9ULL;
    x = (x ^ (x >> 27)) * 0x94d049bb133111ebULL;
    return x ^ (x >> 31);
}

template<class T>
struct RollingValueHasher {
```

```

// +1 ensures the result is in [1, HASH_MOD) and never zero; a zero element in a polynomial
// hash is invisible and enables trivial collisions.
uint64 operator()(const T &x) const { return hash_mix((uint64)x) % (HASH_MOD - 1) + 1; }
};

template<class T, class ValueHasher = RollingValueHasher<T>>
class RollingHash {
    static std::vector<uint64> pow_base;
    std::vector<uint64> pref;
    ValueHasher value_hasher;

    static void ensure_powers(int n) {
        while (static_cast<int>(pow_base.size()) <= n) {
            pow_base.push_back(hash_mul(pow_base.back(), HASH_BASE));
        }
    }

    template<class It>
    void build(It first, It last) {
        pref.clear();
        pref.push_back(0);
        for (; first != last; ++first) {
            pref.push_back(hash_add(hash_mul(pref.back(), HASH_BASE), value_hasher(*first)));
        }
        ensure_powers(pref.size() - 1);
    }

public:
    explicit RollingHash(const ValueHasher &hasher = ValueHasher()) : value_hasher(hasher) {
        ensure_powers(0);
        pref.push_back(0);
    }

    template<class It>
    RollingHash(It first, It last, const ValueHasher &hasher = ValueHasher()) : value_hasher(hasher) {
        build(first, last);
    }

    explicit RollingHash(const std::vector<T> &v, const ValueHasher &hasher = ValueHasher())
        : value_hasher(hasher) {
        build(v.begin(), v.end());
    }

    int size() const { return static_cast<int>(pref.size()) - 1; }

    uint64 get(int lo, int hi) const {
        return hash_sub(pref[hi], hash_mul(pref[lo], pow_base[hi - lo]));
    }

    template<class It>
    static uint64 hash(It first, It last, const ValueHasher &hasher = ValueHasher()) {
        RollingHash<T, ValueHasher> h(first, last, hasher);
        return h.get(0, h.size());
    }

    static uint64 hash(const std::vector<T> &v, const ValueHasher &hasher = ValueHasher()) {
        RollingHash<T, ValueHasher> h(v, hasher);
        return h.get(0, h.size());
    }

    static uint64 concat(uint64 left, uint64 right, int right_len) {
        ensure_powers(right_len);
        return hash_add(hash_mul(left, pow_base[right_len]), right);
    }
};

template<class T, class ValueHasher>
std::vector<uint64> RollingHash<T, ValueHasher>::pow_base(1, 1);

```


Example Usage

```
#include <cassert>
using namespace std;

using point = pair<int, int>;

struct PointHasher {
    uint64 operator()(const point &p) const {
        return hash_mix((uint64)p.first * 1000003ULL + (uint64)p.second) % (HASH_MOD - 1) + 1;
    }
};

int main() {
    string s = "abracadabra";
    RollingHash<char> hs(s.begin(), s.end());
    assert(hs.get(0, 4) == hs.get(7, 11)); // "abra" == "abra"
    assert(hs.get(0, 4) != hs.get(3, 7));  // "abra" != "acad"

    string a = "abc", b = "def";
    uint64 ab = RollingHash<char>::concat(
        RollingHash<char>::hash(a.begin(), a.end()), RollingHash<char>::hash(b.begin(), b.end()),
        b.size()
    );
    string c = a + b;
    assert(ab == RollingHash<char>::hash(c.begin(), c.end()));

    vector<int> v;
    v.push_back(1);
    v.push_back(2);
    v.push_back(3);
    v.push_back(1);
    v.push_back(2);
    RollingHash<int> hv(v);
    assert(hv.get(0, 2) == hv.get(3, 5)); // [1, 2] == [1, 2]
    assert(hv.get(0, 3) == RollingHash<int>::hash(v.begin(), v.begin() + 3));

    vector<point> poly{{1, 2}, {3, 4}};
    RollingHash<point, PointHasher> hp(poly);
    assert((hp.get(0, 2) == RollingHash<point, PointHasher>::hash(poly)));
    return 0;
}
```

3.2.3 Zobrist Hashing

Computes randomized XOR fingerprints for keys and sets of keys using Zobrist hashing. Each distinct key is assigned a pseudorandom 64-bit token, and a set is represented by the XOR of its members' tokens. This makes insertion and deletion the same operation: XOR the key's token once.

This technique is useful for probabilistic equality checks on sets, distinct-prefix queries, board-game states, and other unordered collections. It is not cryptographic, and collisions are possible with probability roughly $q^2/2^{64}$ over q compared fingerprints. For multisets, plain XOR only records element parity; use a summed hash or pair it with counts if multiplicity matters.

- `ZobristHash(seed)` constructs a token generator with initial value `seed`.
- `get(x)` returns the stable token assigned to key `x`, creating it if needed.
- `toggle(h, x)` returns the hash obtained by inserting `x` into set hash `h` if absent, or removing `x` from `h` if present.
- `distinct_prefix_hashes(lo, hi)` returns prefix hashes where each distinct key in the range `[lo, hi)` contributes only on its first occurrence.

Time Complexity:

- $O(1)$ expected per call to `get(x)` and `toggle(h, x)`.
- $O(n)$ expected per call to `distinct_prefix_hashes(lo, hi)`, where n is the distance between `lo` and `hi`.

Space Complexity:

- $O(n)$ for stored key tokens, where n is the number of distinct keys seen so far.
- $O(n)$ auxiliary for `distinct_prefix_hashes(lo, hi)`.

```
#include <unordered_map>
#include <unordered_set>
#include <vector>

using uint64 = unsigned long long;

template<class T>
class ZobristHash {
    std::unordered_map<T, uint64> token;
    uint64 state;

    static uint64 splitmix64(uint64 x) {
        x += 0x9e3779b97f4a7c15ULL;
        x = (x ^ (x >> 30)) * 0xbf58476d1ce4e5b9ULL;
        x = (x ^ (x >> 27)) * 0x94d049bb133111ebULL;
        return x ^ (x >> 31);
    }

public:
    explicit ZobristHash(uint64 seed = 0) : state(seed) {}

    uint64 get(const T &x) {
        if (auto it = token.find(x); it != token.end()) {
            return it->second;
        }
        state = splitmix64(state);
        token[x] = state;
        return state;
    }

    uint64 toggle(uint64 h, const T &x) { return h ^ get(x); }

    template<class It>
    std::vector<uint64> distinct_prefix_hashes(It lo, It hi) {
        std::unordered_set<T> seen;
        std::vector<uint64> res(1, 0);
        uint64 h = 0;
        for (It it = lo; it != hi; ++it) {
            if (seen.insert(*it).second) {
                h ^= get(*it);
            }
            res.push_back(h);
        }
        return res;
    }
};
```

Example Usage

```
#include <cassert>
using namespace std;

int main() {
    vector<int> a{1, 2, 1, 3};
    vector<int> b{2, 1, 3};
    ZobristHash<int> zh(123);
```

```

vector<uint64> pa = zh.distinct_prefix_hashes(a.begin(), a.end());
vector<uint64> pb = zh.distinct_prefix_hashes(b.begin(), b.end());

assert(pa[2] == pb[2]); // Distinct sets {1, 2} and {2, 1}.
assert(pa[4] == pb[3]); // Distinct sets {1, 2, 3} and {2, 1, 3}.

uint64 h = 0;
h = zh.toggle(h, 5);
assert(h != 0);
h = zh.toggle(h, 5);
assert(h == 0);
return 0;
}

```

3.3 Pattern Matching

3.3.1 String Searching (KMP)

Given a single string (needle) and subsequent queries of texts (haystacks) to be searched, determine the first positions in which the needle occurs within the given haystacks in linear time using the Knuth-Morris-Pratt algorithm. In comparison, `std::string::find` runs in $O(n^2)$ in the worst case.

- `KMP(needle)` constructs the partial match table for a string `needle` that is to be searched for subsequently in `haystack` queries.
- `find_in(haystack)` returns the first position that `needle` occurs in `haystack`, or `std::string::npos` if it cannot be found. Note that the function can be modified to return all matches by simply letting the loop run and storing the results instead of returning early.

Time Complexity:

- $O(m)$ per call to the constructor, where m is the length of `needle`.
- $O(n)$ per call to `find_in(haystack)`, where n is the length of `haystack`.

Space Complexity:

- $O(m)$ for storage of the partial match table, where m is the length of `needle`.
- $O(1)$ auxiliary space per call to `find_in(haystack)`.

```

#include <string>
#include <vector>
using std::string;

class KMP {
    string needle;
    std::vector<int> table;

public:
    explicit KMP(const string &needle) : needle(needle), table(needle.size()) {
        for (int i = 1, j = 0; i < static_cast<int>(needle.size()); i++) {
            while (j > 0 && needle[i] != needle[j]) {
                j = table[j - 1];
            }
            if (needle[i] == needle[j]) {
                j++;
            }
            table[i] = j;
        }
    }

    size_t find_in(const string &haystack) {
        int m = needle.size();

```

```

    if (m == 0) {
        return 0;
    }
    for (int i = 0, j = 0; i < static_cast<int>(haystack.size()); i++) {
        while (j > 0 && needle[j] != haystack[i]) {
            j = table[j - 1];
        }
        if (needle[j] == haystack[i]) {
            j++;
        }
        if (j == m) {
            return i + 1 - m;
        }
    }
    return string::npos;
}
};

```

Example Usage

```

#include <cassert>

int main() {
    assert(15 == KMP("ABCDABD").find_in("ABC ABCDAB ABCDABCDABDE"));
    return 0;
}

```

3.3.2 String Searching (Z Algorithm)

Given a single string (needle) and a single text (haystack) to be searched, determine the first position in which the needle occurs within the haystack in linear time using the Z algorithm. In comparison, `std::string::find` runs in quadratic time.

The `find()` function below calls the Z algorithm on the concatenation of `needle` and `haystack`, separated by a sentinel value that is guaranteed not to collide with any byte in either input.

- `z_array(s)` constructs the Z array for a string `needle` that can be used for string searching. The Z array on an input string `s` is an array `z` where `z[i]` is the length of the longest substring starting from `s[i]` which is also a prefix of `s`.
- `find(haystack, needle)` returns the first position that `needle` occurs in `haystack`, or `std::string::npos` if it cannot be found. Note that the function can be modified to return all matches by simply letting the loop run and storing the results instead of returning early.

Time Complexity:

- $O(n)$ per call to `z_array(s)`, where n is the length of `s`.
- $O(n + m)$ per call to `find(haystack, needle)`, where n is the length of `haystack` and m is the length of `needle`.

Space Complexity:

- $O(n)$ auxiliary heap space for `z_array(s)`, where n is the length of `s`.
- $O(n + m)$ auxiliary heap space for `find(haystack, needle)` where n is the length of `haystack` and m is the length of `needle`.

```

#include <algorithm>
#include <string>
#include <vector>
using std::string;

template<class Seq>
std::vector<int> z_array(const Seq &s) {

```

```

std::vector<int> z(s.size());
for (int i = 1, l = 0, r = 0; i < static_cast<int>(z.size()); i++) {
    if (i <= r) {
        z[i] = std::min(r - i + 1, z[i - 1]);
    }
    while (i + z[i] < static_cast<int>(z.size()) && s[z[i]] == s[i + z[i]]) {
        z[i]++;
    }
    if (r < i + z[i] - 1) {
        l = i;
        r = i + z[i] - 1;
    }
}
return z;
}

size_t find(const string &haystack, const string &needle) {
    if (needle.empty()) {
        return 0;
    }
    std::vector<int> s;
    s.reserve(needle.size() + haystack.size() + 1);
    for (unsigned char c : needle) {
        s.push_back(c + 1);
    }
    s.push_back(0);
    for (unsigned char c : haystack) {
        s.push_back(c + 1);
    }
    auto z = z_array(s);
    int m = static_cast<int>(needle.size());
    for (int i = m + 1; i < static_cast<int>(z.size()); i++) {
        if (z[i] == m) {
            return i - m - 1;
        }
    }
    return string::npos;
}

```

Example Usage

```

#include <cassert>

int main() {
    assert(15 == find("ABC ABCDAB ABCDABCDABDE", "ABCDABD"));
    return 0;
}

```

3.3.3 String Searching (Aho-Corasick)

Given a set of strings (needles) and subsequent queries of texts (haystacks) to be searched, determine all positions in which needles occur within the given haystacks in linear time using the Aho-Corasick algorithm.

This implementation stores transitions in hash tables, which keeps lookups expected constant time without assuming a fixed alphabet size. The output tables are stored as ordered sets so matches ending at the same position are reported in deterministic needle order.

- `AhoCorasick(needles)` constructs the finite-state automaton for a set of needle strings that are to be searched for subsequently in haystack queries. Empty needles are ignored.
- `find_all_in(haystack, report_match)` calls the function `report_match(s, pos)` once on each occurrence of each needle that occurs in the `haystack`, where `pos` is the starting position in the `haystack` at which string `s` (a

matched needle) occurs. The matches will be reported in increasing order of their ending positions within the `haystack`.

Time Complexity:

- $O(m \cdot l + z \log m)$ expected per call to the constructor, where m is the number of needles, l is the maximum length of any needle, and z is the total number of inherited output entries.
- $O(n + z)$ expected per call to `find_all_in(haystack, report_match)`, where n is the length of `haystack` and z is the number of matches.

Space Complexity:

- $O(m \cdot l)$ for storage of the automaton, where m is the number of needles and l is the maximum length for any needle.
- $O(1)$ auxiliary space per call to `find_all_in(haystack, report_match)`.

```
#include <queue>
#include <set>
#include <string>
#include <unordered_map>
#include <vector>
using std::string;

class AhoCorasick {
    std::vector<string> needles;
    std::vector<int> fail;
    std::vector<std::unordered_map<char, int>> adj;
    std::vector<std::set<int>> out;

    int next_state(int curr, char c) {
        int next = curr;
        while (next != 0 && adj[next].find(c) == adj[next].end()) {
            next = fail[next];
        }
        auto it = adj[next].find(c);
        return (it != adj[next].end()) ? it->second : 0;
    }

public:
    explicit AhoCorasick(const std::vector<string> &needles) : needles(needles) {
        int total_len = 0;
        for (const auto &needle : needles) {
            total_len += needle.size();
        }
        // The trie has at most total_len + 1 states: the root plus one per character when no prefixes
        // are shared. Sizing to total_len alone overflows for low-sharing needle sets (e.g. a single
        // needle, which still needs the root plus one node).
        int max_states = total_len + 1;
        fail.resize(max_states, -1);
        adj.resize(max_states);
        out.resize(max_states);
        int states = 1;
        for (int i = 0; i < static_cast<int>(needles.size()); i++) {
            if (needles[i].empty()) {
                continue;
            }
            int curr = 0;
            for (char c : needles[i]) {
                if (auto it = adj[curr].find(c); it != adj[curr].end()) {
                    curr = it->second;
                } else {
                    curr = adj[curr][c] = states++;
                }
            }
            out[curr].insert(i);
        }
        std::queue<int> q;
```

```

    for (auto &[c, v] : adj[0]) {
        if (v != 0) {
            fail[v] = 0;
            q.push(v);
        }
    }
    while (!q.empty()) {
        int u = q.front();
        q.pop();
        for (auto &[c, v] : adj[u]) {
            int f = fail[u];
            while (f != 0 && adj[f].find(c) == adj[f].end()) {
                f = fail[f];
            }
            auto fit = adj[f].find(c);
            f = (fit != adj[f].end()) ? fit->second : 0;
            fail[v] = f;
            out[v].insert(out[f].begin(), out[f].end());
            q.push(v);
        }
    }
}

template<class Fn>
void find_all_in(const string &haystack, Fn report_match) {
    int state = 0;
    for (int i = 0; i < static_cast<int>(haystack.size()); i++) {
        state = next_state(state, haystack[i]);
        for (int idx : out[state]) {
            report_match(needles[idx], i - static_cast<int>(needles[idx].size()) + 1);
        }
    }
}
};

```

Example Usage

```

#include <iostream>
using namespace std;

void report_match(const string &needle, int pos) {
    cout << "Matched \"" << needle << "\" at position " << pos << "." << endl;
}

int main() {
    vector<string> needles;
    needles.push_back("a");
    needles.push_back("ab");
    needles.push_back("bab");
    needles.push_back("bc");
    needles.push_back("bca");
    needles.push_back("c");
    needles.push_back("caa");
    needles.push_back("abccab");

    AhoCorasick(needles).find_all_in("abccab", report_match);
    return 0;
}

```

Example Output

```

Matched "a" at position 0.
Matched "ab" at position 0.
Matched "bc" at position 1.
Matched "c" at position 2.

```

```

Matched "c" at position 3.
Matched "a" at position 4.
Matched "ab" at position 4.
Matched "abccab" at position 0.

```

3.3.4 Palindrome Searching (Manacher)

Computes all odd- and even-length palindromic radii of a string in linear time using Manacher's algorithm. These radii can be used to test whether any substring is a palindrome in $O(1)$, find the longest palindromic substring, or count all palindromic substrings.

For odd palindromes, `odd[i]` is the radius centered at character `s[i]`, including the center character. Thus the palindrome spans `[i - odd[i] + 1, i + odd[i]]`. For even palindromes, `even[i]` is the radius centered between `s[i - 1]` and `s[i]`. Thus the palindrome spans `[i - even[i], i + even[i]]`.

- `Manacher(s)` constructs the palindromic radii arrays for string `s`.
- `is_palindrome(lo, hi)` returns whether substring `[lo, hi)` is a palindrome.
- `longest_palindrome()` returns one longest palindromic substring of `s`.
- `count_palindromes()` returns the total number of palindromic substrings of `s`, counted with multiplicity by position.

Time Complexity:

- $O(n)$ per constructor call, where n is the length of `s`.
- $O(1)$ per call to `is_palindrome(lo, r)`.
- $O(n)$ per call to `longest_palindrome()` and `count_palindromes()`.

Space Complexity:

- $O(n)$ for storage of the radii arrays.
- $O(1)$ auxiliary per query.

```

#include <algorithm>
#include <string>
#include <vector>
using std::string;

class Manacher {
    string s;
    std::vector<int> odd, even;

public:
    explicit Manacher(const string &s) : s(s), odd(s.size()), even(s.size()) {
        int n = s.size();
        for (int i = 0, l = 0, r = -1; i < n; i++) {
            int k = (i > r) ? 1 : std::min(odd[l + r - i], r - i + 1);
            while (0 <= i - k && i + k < n && s[i - k] == s[i + k]) {
                k++;
            }
            odd[i] = k;
            if (i + k - 1 > r) {
                l = i - k + 1;
                r = i + k - 1;
            }
        }
        for (int i = 0, l = 0, r = -1; i < n; i++) {
            int k = (i > r) ? 0 : std::min(even[l + r - i + 1], r - i + 1);
            while (0 <= i - k - 1 && i + k < n && s[i - k - 1] == s[i + k]) {
                k++;
            }
            even[i] = k;
            if (i + k - 1 > r) {
                l = i - k;
                r = i + k - 1;
            }
        }
    }
};

```



```

    }
}

bool is_palindrome(int lo, int hi) const {
    int len = hi - lo;
    if (len <= 0) {
        return true;
    }
    int mid = lo + (hi - lo) / 2;
    return (len % 2) ? odd[mid] >= len / 2 + 1 : even[mid] >= len / 2;
}

string longest_palindrome() const {
    int best_l = 0, best_len = 0;
    int n = static_cast<int>(s.size());
    for (int i = 0; i < n; i++) {
        int len = 2 * odd[i] - 1;
        if (len > best_len) {
            best_len = len;
            best_l = i - odd[i] + 1;
        }
        len = 2 * even[i];
        if (len > best_len) {
            best_len = len;
            best_l = i - even[i];
        }
    }
    return s.substr(best_l, best_len);
}

long long count_palindromes() const {
    long long res = 0;
    for (int i = 0; i < static_cast<int>(s.size()); i++) {
        res += odd[i] + even[i];
    }
    return res;
}
};

```

Example Usage

```

#include <cassert>

int main() {
    Manacher m("abacaba");
    assert(m.is_palindrome(0, 7));
    assert(m.is_palindrome(1, 6));
    assert(!m.is_palindrome(0, 6));
    assert(m.longest_palindrome() == "abacaba");
    assert(m.count_palindromes() == 12);

    Manacher e("cabbad");
    assert(e.is_palindrome(1, 5));
    assert(e.longest_palindrome() == "abba");
    return 0;
}

```

3.4 Expression Parsing

3.4.1 Recursive Descent Parsing (Simple)

Evaluate an expression in accordance to the order of operations (parentheses, unary plus and minus signs, multiplication/division, addition/subtraction). The following is a minimalistic recursive descent implementation using iterators.

- `eval(s)` returns an evaluation of the arithmetic expression `s`.

Time Complexity: $O(n)$ per call to `eval(s)`, where n is the length of `s`.

Space Complexity: $O(n)$ auxiliary stack space for `eval(s)`, where n is the length of `s`.

```
#include <string>

template<class It>
int eval(It &it, int prec) {
    if (prec == 0) {
        int sign = 1, res = 0;
        for (; *it == '-'; it++) {
            sign *= -1;
        }
        if (*it == '(') {
            res = eval(++it, 2);
            it++;
        } else
            while (*it >= '0' && *it <= '9') {
                res = 10 * res + (*(it++) - '0');
            }
        return sign * res;
    }
    int num = eval(it, prec - 1);
    while (!(prec == 2 && *it != '+' && *it != '-' || (prec == 1 && *it != '*' && *it != '/'))) {
        switch (*(it++)) {
            case '+':
                num += eval(it, prec - 1);
                break;
            case '-':
                num -= eval(it, prec - 1);
                break;
            case '*':
                num *= eval(it, prec - 1);
                break;
            case '/':
                num /= eval(it, prec - 1);
                break;
        }
    }
    return num;
}

int eval(const std::string &s) {
    std::string expr = s + '\0';
    auto it = expr.begin();
    return eval(it, 2);
}
```

Example Usage

```
#include <cassert>

int main() {
    assert(eval("1++1") == 2);
}
```

```

assert(eval("1+2*3*4+3*(2+2)-100") == -63);
return 0;
}

```

3.4.2 Recursive Descent Parsing (Generic)

Evaluate an expression using a generalized parser class for custom-defined operand types, prefix unary operators, binary operators, and precedences. Typical parentheses behavior is supported, but multiplication by juxtaposition is not. Evaluation is performed using the recursive descent algorithm.

An arbitrary operand type is supported, with its string representation defined by a user-specified `is_operand()` and `eval_operand()` functions. For maximum reliability, the string representation of operands should not use characters shared by any operator. For instance, the best practice instead of accepting `-1` as a valid operand (since the `-` sign may conflict with the identical binary operator), is to specify non-negative number as operands alongside the unary operator `-`.

Operators may be non-empty strings of any length, but should not contain any parentheses or shared characters with the string representations of operands. Ideally, operators should not be prefixes or suffixes of one another, else the tokenization process may be ambiguous. For example, if `++` and `+` are both operators, then `++` may be split into either `["+", "+"]` or `["++"]` depending on the lexicographical ordering of conflicting operators.

- `parser(unary_op, binary_op)` initializes a parser with operators specified by hash tables `unary_op` (of operator to function pointer) and `binary_op` (of operator to pair of function pointer and operator precedence). Operator precedences should be numbered upwards starting at 0 (lowest precedence, evaluated last).
- `split(s)` returns a vector of tokens for the expression `s`, split on the given operators during construction. Each parenthesis, operator, and operand satisfying `is_operand()` will be split into a separate token. The algorithm is naive, matching operators lazily in the case of overlapping operators as mentioned above. Under these circumstances, the parse may not always succeed.
- `eval(lo, hi)` returns the evaluation of a range `[lo, hi)` of already split-up expression tokens, where `lo` and `hi` must be random-access iterators.
- `eval(s)` returns the evaluation of expression `s`, after first calling `split(s)` to obtain the tokens.

Time Complexity:

- $O(m)$ per call to the constructor, where m is the total number of operators.
- $O(n \cdot m \cdot k)$ per call to `split(s)`, where n is the length of `s`, m is the total number of operators defined for the parser instance, and k is the maximum length for any operator representation.
- $O(n)$ expected per call to `eval(lo, hi)`, where n is the distance between `lo` and `hi`.
- $O(n \cdot m \cdot k + n)$ expected per call to `eval(s)`, where n is the distance between `lo` and `hi`, and m and k are as defined previous.

Space Complexity:

- $O(m \cdot k)$ for storage of the m operators, of maximum length k .
- $O(n)$ auxiliary stack space for `split(s)`, `eval(lo, hi)`, and `eval(s)`, where n is the length of the argument.

```

#include <algorithm>
#include <cctype>
#include <set>
#include <sstream>
#include <stdexcept>
#include <string>
#include <unordered_map>
#include <utility>
#include <vector>
using std::string;

// Define the custom operand type and representation below.
using Operand = double;

```

```

using UnaryOp = Operand (*)(Operand a);
using BinaryOp = Operand (*)(Operand a, Operand b);

struct BinaryRule {
    BinaryOp op;
    int precedence;
};

bool is_operand(const string &s) {
    int npoints = 0;
    for (char c : s) {
        if (c == '.') {
            if (++npoints > 1) {
                return false;
            }
        } else if (!isdigit(static_cast<unsigned char>(c))) {
            return false;
        }
    }
    return !s.empty();
}

Operand eval_operand(const string &s) {
    Operand res;
    std::stringstream ss(s);
    ss >> res;
    return res;
}

class parser {
    using unary_op_map = std::unordered_map<string, UnaryOp>;
    using binary_op_map = std::unordered_map<string, BinaryRule>;
    unary_op_map unary_ops;
    binary_op_map binary_ops;
    std::set<string> ops;
    int max_precedence;

    template<class StrIt>
    Operand eval_unary(StrIt &lo, StrIt hi) {
        if (is_operand(*lo)) {
            return eval_operand(*lo++);
        }
        if (auto it = unary_ops.find(*lo); it != unary_ops.end()) {
            return (it->second)(eval_unary(++lo, hi));
        }
        if (*lo != "(") {
            throw std::runtime_error("Expected \"(\" during eval.");
        }
        Operand res = eval_binary(++lo, hi, 0);
        if (*lo != ")") {
            throw std::runtime_error("Expected \")\" during eval.");
        }
        ++lo;
        return res;
    }

    template<class StrIt>
    Operand eval_binary(StrIt &lo, StrIt hi, Operand precedence) {
        if (precedence > max_precedence) {
            return eval_unary(lo, hi);
        }
        Operand v = eval_binary(lo, hi, precedence + 1);
        while (lo != hi) {
            auto it = binary_ops.find(*lo);
            if (it == binary_ops.end() || it->second.precedence != precedence) {
                return v;
            }
            v = (it->second.op)(v, eval_binary(++lo, hi, precedence + 1));
        }
    }
};

```

```

    }
    return v;
}

static string strip(string s) {
    auto not_space = [](unsigned char c) { return !std::isspace(c); };
    s.erase(s.begin(), std::find_if(s.begin(), s.end(), not_space));
    s.erase(std::find_if(s.rbegin(), s.rend(), not_space).base(), s.end());
    return s;
}

public:
parser(const unary_op_map &unary_ops, const binary_op_map &binary_ops)
    : unary_ops(unary_ops), binary_ops(binary_ops) {
    for (const auto &[op, _] : unary_ops) {
        ops.insert(op);
    }
    max_precedence = 0;
    for (const auto &[op, fn_prec] : binary_ops) {
        ops.insert(op);
        max_precedence = std::max(max_precedence, fn_prec.precedence);
    }
}

std::vector<string> split(const string &s) {
    std::vector<string> res;
    for (int i = 0; i < static_cast<int>(s.size()); i++) {
        if (s[i] == ' ') {
            continue;
        }
        int next_paren = static_cast<int>(s.size());
        for (int j = i; j < static_cast<int>(s.size()); j++) {
            if (s[j] == '(' || s[j] == ')') {
                next_paren = j;
                break;
            }
        }
        while (i < next_paren) {
            int found = next_paren;
            string found_op;
            for (int j = i; j < next_paren && found == next_paren; j++) {
                for (const auto &op : ops) {
                    if (s.substr(j, op.size()) == op) {
                        found = j;
                        found_op = op;
                        break;
                    }
                }
            }
            string term = strip(s.substr(i, found - i));
            if (!term.empty()) {
                res.push_back(term);
                if (!is_operand(term)) {
                    throw std::runtime_error("Failed to split term: \"" + term + "\".");
                }
            }
            if (found < next_paren) {
                res.push_back(found_op);
                i = found + static_cast<int>(found_op.size());
            } else {
                i = next_paren;
            }
        }
        if (next_paren < static_cast<int>(s.size())) {
            res.emplace_back(1, s[next_paren]);
        }
    }
    return res;
}

```

```

}

template<class StrIt>
Operand eval(StrIt lo, StrIt hi) {
    Operand res = eval_binary(lo, hi, 0);
    if (lo != hi) {
        throw std::runtime_error("Eval failed at token " + *lo + ".");
    }
    return res;
}

Operand eval(const string &s) {
    std::vector<string> tokens = split(s);
    return eval(tokens.begin(), tokens.end());
}
};

```

Example Usage

```

#include <cassert>
#include <cmath>
using namespace std;

#define EQ(a, b) (fabs((a) - (b)) < 1e-7)

double pos(double a) { return +a; }
double neg(double a) { return -a; }
double add(double a, double b) { return a + b; }
double sub(double a, double b) { return a - b; }
double mul(double a, double b) { return a * b; }
double div(double a, double b) { return a / b; }

int main() {
    unordered_map<string, UnaryOp> unary_ops;
    unary_ops["+"] = pos;
    unary_ops["-"] = neg;

    unordered_map<string, BinaryRule> binary_ops;
    binary_ops["+"] = {static_cast<BinaryOp>(add), 0};
    binary_ops["-"] = {static_cast<BinaryOp>(sub), 0};
    binary_ops["*"] = {static_cast<BinaryOp>(mul), 1};
    binary_ops["/"] = {static_cast<BinaryOp>(div), 1};
    binary_ops["^"] = {static_cast<BinaryOp>(pow), 2};

    parser p(unary_ops, binary_ops);
    assert(EQ(p.eval("-+(-(+(+1)))"), -1));
    assert(EQ(p.eval("5*(3+3)-2-2"), 26));
    assert(EQ(p.eval("1+2*3*4+3*(+2)-100"), -69));
    assert(EQ(p.eval("3*3*3*3*3-2*2*2*2*2*2"), 473));
    assert(EQ(p.eval("3.14 + 3 * (7.7/9.8^32.9 )"), 3.14));
    assert(EQ(p.eval("5*(3+2)/-1*-2+(-2-2+3)-3-(-2)+15/2/2+(-2)"), 45.875));
    assert(EQ(p.eval("123456789./3/3/3*2*2+456/6-23/3"), 36579857.666666666667));
    assert(EQ(p.eval("10/3+10/4+10/5+10/6+10/7+10/8+10/9+10/10+15*23456"), 351854.28968253968));
    assert(
        EQ(p.eval(
            "-(5-(5-(5-(5-(5-2)))))+(3-(3-(3-(3-(3+3)))))*"
            "(7-(7-(7-(7-(7-7+4*5)))))"
        ),
        117)
    );
    return 0;
}

```

3.4.3 Shunting Yard Parsing

Evaluate an expression using a generalized parser class for custom-defined operand types, prefix unary operators, binary operators, and precedences. Typical parentheses behavior is supported, but multiplication by juxtaposition is not. Evaluation is performed using the shunting yard algorithm.

An arbitrary operand type is supported, with its string representation defined by a user-specified `is_operand()` and `eval_operand()` functions. For maximum reliability, the string representation of operands should not use characters shared by any operator. For instance, the best practice instead of accepting `-1` as a valid operand (since the `-` sign may conflict with the identical binary operator), is to specify non-negative number as operands alongside the unary operator `-`.

Operators may be non-empty strings of any length, but should not contain any parentheses or shared characters with the string representations of operands. Ideally, operators should not be prefixes or suffixes of one another, else the tokenization process may be ambiguous. For example, if `++` and `+` are both operators, then `++` may be split into either `["+", "+"]` or `["++"]` depending on the lexicographical ordering of conflicting operators.

- `parser(unary_op, binary_op)` initializes a parser with operators specified by hash tables `unary_op` (of operator to function pointer) and `binary_op` (of operator to pair of function pointer and operator precedence). Operator precedences should be numbered upwards starting at 0 (lowest precedence, evaluated last).
- `split(s)` returns a vector of tokens for the expression `s`, split on the given operators during construction. Each parenthesis, operator, and operand satisfying `is_operand()` will be split into a separate token. The algorithm is naive, matching operators lazily in the case of overlapping operators as mentioned above. Under these circumstances, the parse may not always succeed.
- `eval(lo, hi)` returns the evaluation of a range `[lo, hi)` of already split-up expression tokens, where `lo` and `hi` must be random-access iterators.
- `eval(s)` returns the evaluation of expression `s`, after first calling `split(s)` to obtain the tokens.

Time Complexity:

- $O(m)$ per call to the constructor, where m is the total number of operators.
- $O(n \cdot m \cdot k)$ per call to `split(s)`, where n is the length of `s`, m is the total number of operators defined for the parser instance, and k is the maximum length for any operator representation.
- $O(n)$ expected per call to `eval(lo, hi)`, where n is the distance between `lo` and `hi`.
- $O(n \cdot m \cdot k + n)$ expected per call to `eval(s)`, where n is the distance between `lo` and `hi`, and m and k are as defined previous.

Space Complexity:

- $O(m \cdot k)$ for storage of the m operators, of maximum length k .
- $O(n)$ auxiliary stack space for `split(s)`, `eval(lo, hi)`, and `eval(s)`, where n is the length of the argument.

```
#include <algorithm>
#include <cctype>
#include <set>
#include <sstream>
#include <stack>
#include <stdexcept>
#include <string>
#include <unordered_map>
#include <utility>
#include <vector>
using std::string;

// Define the custom operand type and representation below.
using Operand = double;
using UnaryOp = Operand (*)(Operand a);
using BinaryOp = Operand (*)(Operand a, Operand b);

struct BinaryRule {
    BinaryOp op;
    int precedence;
```

```

};

bool is_operand(const string &s) {
    int npoints = 0;
    for (char c : s) {
        if (c == '.') {
            if (++npoints > 1) {
                return false;
            }
        } else if (!isdigit(static_cast<unsigned char>(c))) {
            return false;
        }
    }
    return !s.empty();
}

Operand eval_operand(const string &s) {
    Operand res;
    std::stringstream ss(s);
    ss >> res;
    return res;
}

class parser {
    using unary_op_map = std::unordered_map<string, UnaryOp>;
    using binary_op_map = std::unordered_map<string, BinaryRule>;
    unary_op_map unary_ops;
    binary_op_map binary_ops;
    std::set<string> ops;

    static string strip(string s) {
        auto not_space = [](unsigned char c) { return !std::isspace(c); };
        s.erase(s.begin(), std::find_if(s.begin(), s.end(), not_space));
        s.erase(std::find_if(s.rbegin(), s.rend(), not_space).base(), s.end());
        return s;
    }

public:
    parser(const unary_op_map &unary_ops, const binary_op_map &binary_ops)
        : unary_ops(unary_ops), binary_ops(binary_ops) {
        for (const auto &[op, fn] : unary_ops) {
            ops.insert(op);
        }
        for (const auto &[op, fn_prec] : binary_ops) {
            ops.insert(op);
        }
    }

    std::vector<string> split(const string &s) {
        std::vector<string> res;
        for (int i = 0; i < static_cast<int>(s.size()); i++) {
            if (s[i] == ' ') {
                continue;
            }
            int next_paren = static_cast<int>(s.size());
            for (int j = i; j < static_cast<int>(s.size()); j++) {
                if (s[j] == '(' || s[j] == ')') {
                    next_paren = j;
                    break;
                }
            }
            while (i < next_paren) {
                int found = next_paren;
                string found_op;
                for (int j = i; j < next_paren && found == next_paren; j++) {
                    for (const auto &op : ops) {
                        if (s.substr(j, op.size()) == op) {
                            found = j;
                        }
                    }
                }
            }
        }
    }

```



```

        found_op = op;
        break;
    }
}
}
string term = strip(s.substr(i, found - i));
if (!term.empty()) {
    res.push_back(term);
    if (!is_operand(term)) {
        throw std::runtime_error("Failed to split term: \"" + term + "\".");
    }
}
if (found < next_paren) {
    res.push_back(found_op);
    i = found + static_cast<int>(found_op.size());
} else {
    i = next_paren;
}
}
if (next_paren < static_cast<int>(s.size())) {
    res.emplace_back(1, s[next_paren]);
}
}
return res;
}

template<class StrIt>
Operand eval(StrIt lo, StrIt hi) {
    std::stack<Operand> vals;
    std::stack<std::pair<string, bool>> ops;
    ops.emplace("(", false);
    StrIt prev = hi;
    do {
        string curr = (lo == hi) ? ")" : *lo;
        if (is_operand(curr)) {
            vals.push(eval_operand(curr));
        } else if (curr == "(") {
            ops.emplace(curr, false);
        } else if (
            unary_ops.find(curr) != unary_ops.end() &&
            (prev == hi || *prev == "(" || binary_ops.find(*prev) != binary_ops.end())
        ) {
            ops.emplace(curr, true);
        } else {
            for (;;) {
                auto [op, is_unary] = ops.top();
                auto it1 = binary_ops.find(op);
                auto it2 = binary_ops.find(curr);
                if (!is_unary && (it1 == binary_ops.end() ? -1 : it1->second.precedence) <
                    (it2 == binary_ops.end() ? -1 : it2->second.precedence)) {
                    break;
                }
            }
            ops.pop();
            if (op == "(") {
                break;
            }
            Operand b = vals.top();
            vals.pop();
            if (is_unary) {
                if (auto it = unary_ops.find(op); it != unary_ops.end()) {
                    vals.push((it->second)(b));
                } else {
                    throw std::runtime_error("Failed to eval unary op: " + op);
                }
            } else {
                Operand a = vals.top();
                vals.pop();
                if (it1 == binary_ops.end()) {

```

```

        throw std::runtime_error("Failed to eval binary op: " + op);
    }
    vals.push((it1->second.op)(a, b));
}
}
if (curr != ")") {
    ops.emplace(*lo, false);
}
}
prev = lo;
} while (lo++ != hi);
return vals.top();
}

Operand eval(const string &s) {
    std::vector<string> tokens = split(s);
    return eval(tokens.begin(), tokens.end());
}
};

```

Example Usage

```

#include <cassert>
#include <cmath>
using namespace std;

#define EQ(a, b) (fabs((a) - (b)) < 1e-7)

double pos(double a) { return +a; }
double neg(double a) { return -a; }
double add(double a, double b) { return a + b; }
double sub(double a, double b) { return a - b; }
double mul(double a, double b) { return a * b; }
double div(double a, double b) { return a / b; }

int main() {
    unordered_map<string, UnaryOp> unary_ops;
    unary_ops["+"] = pos;
    unary_ops["-"] = neg;

    unordered_map<string, BinaryRule> binary_ops;
    binary_ops["+"] = {static_cast<BinaryOp>(add), 0};
    binary_ops["-"] = {static_cast<BinaryOp>(sub), 0};
    binary_ops["*"] = {static_cast<BinaryOp>(mul), 1};
    binary_ops["/"] = {static_cast<BinaryOp>(div), 1};
    binary_ops["^"] = {static_cast<BinaryOp>(pow), 2};

    parser p(unary_ops, binary_ops);
    assert(EQ(p.eval("-+-(--(+1))"), -1));
    assert(EQ(p.eval("5*(3+3)-2-2"), 26));
    assert(EQ(p.eval("1+2*3*4+3*(+2)-100"), -69));
    assert(EQ(p.eval("3*3*3*3*3-2*2*2*2*2*2"), 473));
    assert(EQ(p.eval("3.14 + 3 * (7.7/9.8^32.9 )"), 3.14));
    assert(EQ(p.eval("5*(3+2)/-1*-2+(-2-2-2+3)-3-(-2)+15/2/2/2+(-2)"), 45.875));
    assert(EQ(p.eval("123456789./3/3/3*2*2*2+456/6-23/3"), 36579857.666666666));
    assert(EQ(p.eval("10/3+10/4+10/5+10/6+10/7+10/8+10/9+10/10+15*23456"), 351854.28968253968));
    assert(
        EQ(p.eval(
            "-(5-(5-(5-(5-(5-2)))))+(3-(3-(3-(3-(3+3)))))*"
            "(7-(7-(7-(7-(7-7+4*5)))))"
        ),
        117)
    );
    return 0;
}

```

3.5 Sequence Dynamic Programming

3.5.1 Longest Common Substring

Given two strings, determine their longest common substring (i.e. consecutive subsequence) using dynamic programming.

Time Complexity: $O(n \cdot m)$ per call to `longest_common_substring(s1, s2)`, where n and m are the lengths of `s1` and `s2`, respectively.

Space Complexity: $O(\min(n, m))$ auxiliary heap space, where n and m are the lengths of `s1` and `s2`, respectively.

```
#include <string>
#include <vector>
using std::string;

string longest_common_substring(const string &s1, const string &s2) {
    int n = s1.size(), m = s2.size();
    if (n == 0 || m == 0) {
        return "";
    }
    if (n < m) {
        return longest_common_substring(s2, s1);
    }
    std::vector<int> curr(m), prev(m);
    int pos = 0, len = 0;
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < m; j++) {
            if (s1[i] == s2[j]) {
                curr[j] = (i > 0 && j > 0) ? prev[j - 1] + 1 : 1;
                if (len < curr[j]) {
                    len = curr[j];
                    pos = i - curr[j] + 1;
                }
            } else {
                curr[j] = 0;
            }
        }
        curr.swap(prev);
    }
    return s1.substr(pos, len);
}
```

Example Usage

```
#include <cassert>

int main() {
    assert(longest_common_substring("bbbabca", "aababcd") == "babc");
    return 0;
}
```

3.5.2 Longest Common Subsequence

Given two strings, determine their longest common subsequence. A subsequence is a string that can be derived from the original string by deleting some elements without changing the order of the remaining elements (e.g.

`"ACE"` is a subsequence of `"ABCDE"`, but `"BAE"` is not).

- `longest_common_subsequence(s1, s2)` returns the longest common subsequence of strings `s1` and `s2` using a classic dynamic programming approach. This implementation computes `dp[i][j]` (the length of the longest common subsequence for the length i prefix of `s1` and the length j prefix of `s2`) before following the path backwards to construct the answer.
- `hirschberg_lcs(s1, s2)` returns the longest common subsequence of strings `s1` and `s2` using the more memory efficient Hirschberg's algorithm.

Time Complexity: $O(n \cdot m)$ per call to `longest_common_subsequence(s1, s2)` as well as `hirschberg_lcs(s1, s2)`, where n and m are the lengths of `s1` and `s2`, respectively.

Space Complexity:

- $O(n \cdot m)$ auxiliary heap space for `longest_common_subsequence(s1, s2)`, where n and m are the lengths of `s1` and `s2`, respectively.
- $O(\log \max(n, m))$ auxiliary stack space and $O(\min(n, m))$ auxiliary heap space for `hirschberg_lcs(s1, s2)`, where n and m are the lengths of `s1` and `s2`, respectively.

```
#include <algorithm>
#include <iterator>
#include <string>
#include <vector>
using std::string;

string longest_common_subsequence(const string &s1, const string &s2) {
    int n = s1.size(), m = s2.size();
    std::vector<std::vector<int>> dp(n + 1, std::vector<int>(m + 1, 0));
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= m; j++) {
            if (s1[i - 1] == s2[j - 1]) {
                dp[i][j] = dp[i - 1][j - 1] + 1;
            } else {
                dp[i][j] = std::max(dp[i][j - 1], dp[i - 1][j]);
            }
        }
    }
    string res;
    for (int i = n, j = m; i > 0 && j > 0;) {
        if (s1[i - 1] == s2[j - 1]) {
            res += s1[i - 1];
            i--;
            j--;
        } else if (dp[i - 1][j] >= dp[i][j - 1]) {
            i--;
        } else {
            j--;
        }
    }
    std::reverse(res.begin(), res.end());
    return res;
}

template<class It>
std::vector<int> lcs_len(It lo1, It hi1, It lo2, It hi2) {
    std::vector<int> res(std::distance(lo2, hi2) + 1, prev(res));
    for (It it1 = lo1; it1 != hi1; ++it1) {
        res.swap(prev);
        int i = 0;
        for (It it2 = lo2; it2 != hi2; ++it2) {
            res[i + 1] = (*it1 == *it2) ? prev[i] + 1 : std::max(res[i], prev[i + 1]);
            i++;
        }
    }
    return res;
}

template<class It>
```

```

void hirschberg_rec(It lo1, It hi1, It lo2, It hi2, string *res) {
    if (lo1 == hi1) {
        return;
    }
    if (lo1 + 1 == hi1) {
        if (std::find(lo2, hi2, *lo1) != hi2) {
            *res += *lo1;
        }
        return;
    }
    It mid1 = lo1 + (hi1 - lo1) / 2;
    std::reverse_iterator<It> rlo1(hi1), rmid1(mid1), rlo2(hi2), rhi2(lo2);
    std::vector<int> fwd = lcs_len(lo1, mid1, lo2, hi2);
    std::vector<int> rev = lcs_len(rlo1, rmid1, rlo2, rhi2);
    It mid2 = lo2;
    int maxlen = -1;
    for (int i = 0, j = static_cast<int>(rev.size()) - 1; i < static_cast<int>(fwd.size());
        i++, j--) {
        if (fwd[i] + rev[j] > maxlen) {
            maxlen = fwd[i] + rev[j];
            mid2 = lo2 + i;
        }
    }
    hirschberg_rec(lo1, mid1, lo2, mid2, res);
    hirschberg_rec(mid1, hi1, mid2, hi2, res);
}

string hirschberg_lcs(const string &s1, const string &s2) {
    if (s1.size() < s2.size()) {
        return hirschberg_lcs(s2, s1);
    }
    string res;
    hirschberg_rec(s1.begin(), s1.end(), s2.begin(), s2.end(), &res);
    return res;
}

```

Example Usage

```

#include <cassert>

int main() {
    assert(longest_common_subsequence("xmjyauz", "mzjawxu") == "mjau");
    assert(hirschberg_lcs("xmjyauz", "mzjawxu") == "mjau");
    return 0;
}

```

3.5.3 Sequence Alignment

Given two strings, determine their minimum-cost alignment. An alignment of two strings is a transformation of both strings by inserting gap characters `_` in some way to make the final lengths equal. The total cost of an alignment given a `gap_cost` (insertion or deletion cost) and a `sub_cost` (substitution, i.e. mismatch cost) is `gap_cost` times the number of inserted gaps, plus `sub_cost` times the number of indices at which the two aligned strings differ.

- `align_sequences(s1, s2, gap_cost, sub_cost)` returns a pair of aligned strings for strings `s1` and `s2`, using a classic dynamic programming approach. This implementation first computes `dp[i][j]` (the cost of aligning the length `i` prefix of `s1` with the length `j` prefix of `s2`) before following the path backwards to construct the answer. For `gap_cost = sub_cost = 1`, `dp[n][m]` will be the Levenshtein edit distance, where `n` and `m` are the lengths of `s1` and `s2`, respectively.
- `hirschberg_align_sequences(s1, s2, gap_cost, sub_cost)` returns the sequence alignment of strings `s1` and `s2` using the more memory efficient Hirschberg's algorithm.

Time Complexity: $O(n \cdot m)$ per call to `align_sequences(s1, s2)` as well as `hirschberg_align_sequences(s1, s2)`, where n and m are the lengths of `s1` and `s2`, respectively.

Space Complexity:

- $O(n \cdot m)$ auxiliary heap space for `align_sequences(s1, s2)`, where n and m are the lengths of `s1` and `s2`, respectively.
- $O(\log \max(n, m))$ auxiliary stack space and $O(\min(n, m))$ auxiliary heap space for `hirschberg_align_sequences(s1, s2)`, where n and m are the lengths of `s1` and `s2`, respectively.

```
#include <algorithm>
#include <string>
#include <utility>
#include <vector>
using std::string;

std::pair<string, string> align_sequences(
    const string &s1, const string &s2, int gap_cost = 1, int sub_cost = 1
) {
    int n = s1.size(), m = s2.size();
    std::vector<std::vector<int>>> dp(n + 1, std::vector<int>(m + 1, 0));
    for (int i = 0; i <= n; i++) {
        dp[i][0] = i * gap_cost;
    }
    for (int j = 0; j <= m; j++) {
        dp[0][j] = j * gap_cost;
    }
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= m; j++) {
            dp[i][j] = (s1[i - 1] == s2[j - 1]) ? dp[i - 1][j - 1]
                : std::min(
                    dp[i - 1][j - 1] + sub_cost,
                    std::min(dp[i - 1][j], dp[i][j - 1]) + gap_cost
                );
        }
    }
    string res1, res2;
    int i = n, j = m;
    while (i > 0 && j > 0) {
        if (s1[i - 1] == s2[j - 1] || dp[i][j] == dp[i - 1][j - 1] + sub_cost) {
            res1 += s1[--i];
            res2 += s2[--j];
        } else if (dp[i][j] == dp[i - 1][j] + gap_cost) {
            res1 += s1[--i];
            res2 += '_';
        } else if (dp[i][j] == dp[i][j - 1] + gap_cost) {
            res1 += '_';
            res2 += s2[--j];
        }
    }
    while (i > 0 || j > 0) {
        res1 += (i > 0) ? s1[--i] : '_';
        res2 += (j > 0) ? s2[--j] : '_';
    }
    std::reverse(res1.begin(), res1.end());
    std::reverse(res2.begin(), res2.end());
    return {res1, res2};
}

template<class It>
std::vector<int> row_cost(It lo1, It hi1, It lo2, It hi2, int gap_cost, int sub_cost) {
    std::vector<int> res(std::distance(lo2, hi2) + 1, prev(res));
    for (It it1 = lo1; it1 != hi1; ++it1) {
        res.swap(prev);
        int i = 0;
        for (It it2 = lo2; it2 != hi2; ++it2) {
            res[i + 1] = (*it1 == *it2) ? prev[i] : std::min(prev[i] + sub_cost, res[i] + gap_cost);
        }
    }
}
```

```

        i++;
    }
}
return res;
}

template<class It>
void hirschberg_rec(
    It lo1, It hi1, It lo2, It hi2, string *res1, string *res2, int gap_cost, int sub_cost
) {
    if (lo1 == hi1) {
        for (It it2 = lo2; it2 != hi2; ++it2) {
            *res1 += '_';
            *res2 += *it2;
        }
        return;
    }
    if (lo1 + 1 == hi1) {
        It pos = std::find(lo2, hi2, *lo1);
        bool insert = (pos == hi2) && (gap_cost * (hi2 - lo2 + 1) < sub_cost);
        if (lo2 == hi2 || insert) {
            *res1 += *lo1;
            *res2 += '_';
        }
        for (It it2 = lo2; it2 != hi2; ++it2) {
            *res1 += (pos == it2 || (!insert && it2 == lo2)) ? *lo1 : '_';
            *res2 += *it2;
        }
        return;
    }
    It mid1 = lo1 + (hi1 - lo1) / 2;
    std::reverse_iterator<It> rlo1(hi1), rmid1(mid1), rlo2(hi2), rhi2(lo2);
    std::vector<int> fwd = row_cost(lo1, mid1, lo2, hi2, gap_cost, sub_cost);
    std::vector<int> rev = row_cost(rlo1, rmid1, rlo2, rhi2, gap_cost, sub_cost);
    It mid2 = lo2;
    int mincost = -1;
    for (int i = 0, j = static_cast<int>(rev.size()) - 1; i < static_cast<int>(fwd.size());
        i++, j--) {
        if (mincost < 0 || fwd[i] + rev[j] < mincost) {
            mincost = fwd[i] + rev[j];
            mid2 = lo2 + i;
        }
    }
    hirschberg_rec(lo1, mid1, lo2, mid2, res1, res2, gap_cost, sub_cost);
    hirschberg_rec(mid1, hi1, mid2, hi2, res1, res2, gap_cost, sub_cost);
}

std::pair<string, string> hirschberg_align_sequences(
    const string &s1, const string &s2, int gap_cost = 1, int sub_cost = 1
) {
    if (s1.size() < s2.size()) {
        return hirschberg_align_sequences(s2, s1, gap_cost, sub_cost);
    }
    string res1, res2;
    hirschberg_rec(s1.begin(), s1.end(), s2.begin(), s2.end(), &res1, &res2, gap_cost, sub_cost);
    return {res1, res2};
}

```

Example Usage

```

#include <cassert>

int main() {
    assert(
        align_sequences("AGGGCT", "AGGCA", 2, 3) == (std::pair<string, string>{"AGGGCT", "A_GGCA"})
    );
}

```

```

assert(
    hirschberg_align_sequences("AGGGCT", "AGGCA", 2, 3) ==
    (std::pair<string, string>{"AGGGCT", "A_GGCA"});
);
return 0;
}

```

3.6 Suffix Arrays and LCP

3.6.1 Suffix Array and LCP (Manber-Myers)

Given a string `s`, a suffix array is the array of the smallest starting positions for the sorted suffixes of `s`. That is, the i -th position of the suffix array stores the starting position of the i -th lexicographically smallest suffix of `s`. For example, `s = "cab"` has the suffixes `"cab"`, `"ab"`, and `"b"`. When sorted, the indices of the suffixes are `"ab"`, `"b"`, and `"cab"`, so the suffix array (assuming zero-based indices) is `[1, 2, 0]`.

For a string `s` of length n , the longest common prefix (LCP) array of length $n - 1$ stores the lengths of the longest common prefixes between all pairs of lexicographically adjacent suffixes in `s`. For example, `"baa"` has the sorted suffixes `"a"`, `"aa"`, and `"baa"`, with an LCP array of `[1, 0]`.

- `SuffixArray(s)` constructs a suffix array from the given string `s` using the original Manber-Myers gap partitioning algorithm with a comparison-based sort.
- `get_sa()` returns the constructed suffix array.
- `get_lcp()` returns the corresponding LCP array for the suffix array.
- `find(needle)` returns one position that `needle` occurs in `s` (not necessarily the first), or `std::string::npos` if it cannot be found. For a `needle` of length m , this implementation uses an $O(m \log n)$ binary search, but can be optimized to $O(m + \log n)$ by first computing the LCP-LR array using the LCP array.

Time Complexity:

- $O(n \log^2 n)$ per call to the constructor, where n is the length of `s`.
- $O(1)$ per call to `get_sa()`.
- $O(n)$ per call to `get_lcp()`, where n is the length of `s`.
- $O(m \log n)$ per call to `find(needle)`, where m is the length of `needle` and n is the length of `s`.

Space Complexity:

- $O(n)$ auxiliary for storage of the suffix and LCP arrays, where n is the length of `s`.
- $O(n)$ auxiliary heap space for the constructor.
- $O(1)$ auxiliary space for all other operations.

```

#include <algorithm>
#include <numeric>
#include <string>
#include <utility>
#include <vector>
using std::string;

class SuffixArray {
    string s;
    std::vector<int> sa, rk;

public:
    explicit SuffixArray(const string &s) : s(s), sa(s.size()), rk(s.size()) {
        int n = s.size();
        std::iota(sa.begin(), sa.end(), 0);
        for (int i = 0; i < n; i++) {
            rk[i] = static_cast<unsigned char>(s[i]);
        }
    }
}

```



```

std::vector<std::pair<int, int>> rk2(n);
for (int gap = 1; gap < n; gap *= 2) {
    for (int i = 0; i < n; i++) {
        rk2[i] = {rk[i], i + gap < n ? rk[i + gap] + 1 : 0};
    }
    std::sort(sa.begin(), sa.end(), [&](int i, int j) { return rk2[i] < rk2[j]; });
    for (int i = 0; i < n; i++) {
        rk[sa[i]] = (i > 0 && rk2[sa[i - 1]] == rk2[sa[i]]) ? rk[sa[i - 1]] : i;
    }
}
}

const std::vector<int> &get_sa() const { return sa; }

std::vector<int> get_lcp() const {
    int n = s.size();
    if (n == 0) {
        return {}; // Avoid constructing a vector of size (size_t)(-1).
    }
    std::vector<int> lcp(n - 1);
    for (int i = 0, k = 0; i < n; i++) {
        if (rk[i] < n - 1) {
            int j = sa[rk[i] + 1];
            while (std::max(i, j) + k < n && s[i + k] == s[j + k]) {
                k++;
            }
            lcp[rk[i]] = k;
            if (k > 0) {
                k--;
            }
        }
    }
    return lcp;
}

size_t find(const string &needle) {
    if (needle.empty()) {
        return 0;
    }
    int lo = 0, hi = static_cast<int>(s.size()) - 1;
    while (lo <= hi) {
        int mid = lo + (hi - lo) / 2;
        int cmp = s.compare(sa[mid], needle.size(), needle);
        if (cmp < 0) {
            lo = mid + 1;
        } else if (cmp > 0) {
            hi = mid - 1;
        } else {
            return sa[mid];
        }
    }
    return string::npos;
}
};

```

Example Usage

```

#include <cassert>
using namespace std;

int main() {
    SuffixArray sa("banana");
    vector<int> sarr = sa.get_sa(), lcp = sa.get_lcp();
    vector<int> sarr_expected{5, 3, 1, 0, 4, 2};
    vector<int> lcp_expected{1, 3, 0, 0, 2};
    assert(sarr == sarr_expected);
}

```

```

assert(lcp == lcp_expected);
assert(sa.find("ana") == 1);
assert(sa.find("x") == string::npos);
return 0;
}

```

3.6.2 Suffix Array and LCP (Counting Sort)

Given a string `s`, a suffix array is the array of the smallest starting positions for the sorted suffixes of `s`. That is, the i -th position of the suffix array stores the starting position of the i -th lexicographically smallest suffix of `s`. For example, `s = "cab"` has the suffixes `"cab"`, `"ab"`, and `"b"`. When sorted, the indices of the suffixes are `"ab"`, `"b"`, and `"cab"`, so the suffix array (assuming zero-based indices) is `[1, 2, 0]`.

For a string `s` of length n , the longest common prefix (LCP) array of length $n - 1$ stores the lengths of the longest common prefixes between all pairs of lexicographically adjacent suffixes in `s`. For example, `"baa"` has the sorted suffixes `"a"`, `"aa"`, and `"baa"`, with an LCP array of `[1, 0]`.

- `SuffixArray(s)` constructs a suffix array from the given string `s` using the original Manber-Myers gap partitioning algorithm with a counting sort instead of a comparison-based sort to reduce the running time to $O(n \log n)$.
- `get_sa()` returns the constructed suffix array.
- `get_lcp()` returns the corresponding LCP array for the suffix array.
- `find(needle)` returns one position that `needle` occurs in `s` (not necessarily the first), or `std::string::npos` if it cannot be found. For a `needle` of length m , this implementation uses an $O(m \log n)$ binary search, but can be optimized to $O(m + \log n)$ by first computing the LCP-LR array using the LCP array.

Time Complexity:

- $O(n \log n)$ per call to the constructor, where n is the length of `s`.
- $O(1)$ per call to `get_sa()`.
- $O(n)$ per call to `get_lcp()`, where n is the length of `s`.
- $O(m \log n)$ per call to `find(needle)`, where m is the length of `needle` and n is the length of `s`.

Space Complexity:

- $O(n)$ auxiliary for storage of the suffix and LCP arrays, where n is the length of `s`.
- $O(n)$ auxiliary heap space for the constructor.
- $O(1)$ auxiliary space for all other operations.

```

#include <algorithm>
#include <numeric>
#include <string>
#include <utility>
#include <vector>
using std::string;

class SuffixArray {
    string s;
    std::vector<int> sa, rk;

public:
    explicit SuffixArray(const string &s) : s(s), sa(s.size()), rk(s.size()) {
        int n = s.size();
        std::iota(sa.rbegin(), sa.rend(), 0);
        for (int i = 0; i < n; i++) {
            rk[i] = static_cast<unsigned char>(s[i]);
        }
        std::stable_sort(sa.begin(), sa.end(), [&](int i, int j) {
            return static_cast<unsigned char>(s[i]) < static_cast<unsigned char>(s[j]);
        });
        for (int gap = 1; gap < n; gap *= 2) {

```

```

std::vector<int> prev_rk(rk), prev_sa(sa), cnt(n);
std::iota(cnt.begin(), cnt.end(), 0);
for (int i = 0; i < n; i++) {
    rk[sa[i]] = (i > 0 && prev_rk[sa[i - 1]] == prev_rk[sa[i]] && sa[i - 1] + gap < n &&
        prev_rk[sa[i - 1] + gap / 2] == prev_rk[sa[i] + gap / 2])
        ? rk[sa[i - 1]]
        : i;
}
for (int i = 0; i < n; i++) {
    int s1 = prev_sa[i] - gap;
    if (s1 >= 0) {
        sa[cnt[rk[s1]]++] = s1;
    }
}
}
}

const std::vector<int> &get_sa() const { return sa; }

std::vector<int> get_lcp() const {
    int n = s.size();
    if (n == 0) {
        return {}; // Avoid constructing a vector of size (size_t)(-1).
    }
    std::vector<int> lcp(n - 1);
    for (int i = 0, k = 0; i < n; i++) {
        if (rk[i] < n - 1) {
            int j = sa[rk[i] + 1];
            while (std::max(i, j) + k < n && s[i + k] == s[j + k]) {
                k++;
            }
            lcp[rk[i]] = k;
            if (k > 0) {
                k--;
            }
        }
    }
    return lcp;
}

size_t find(const string &needle) {
    if (needle.empty()) {
        return 0;
    }
    int lo = 0, hi = static_cast<int>(s.size()) - 1;
    while (lo <= hi) {
        int mid = lo + (hi - lo) / 2;
        int cmp = s.compare(sa[mid], needle.size(), needle);
        if (cmp < 0) {
            lo = mid + 1;
        } else if (cmp > 0) {
            hi = mid - 1;
        } else {
            return sa[mid];
        }
    }
    return string::npos;
}
};

```

Example Usage

```

#include <cassert>
using namespace std;

int main() {

```

```

SuffixArray sa("banana");
vector<int> sarr = sa.get_sa(), lcp = sa.get_lcp();
vector<int> sarr_expected{5, 3, 1, 0, 4, 2};
vector<int> lcp_expected{1, 3, 0, 0, 2};
assert(sarr == sarr_expected);
assert(lcp == lcp_expected);
assert(sa.find("ana") == 1);
assert(sa.find("x") == string::npos);
return 0;
}

```

3.6.3 Suffix Array and LCP (Linear DC3)

Given a string s , a suffix array is the array of the smallest starting positions for the sorted suffixes of s . That is, the i -th position of the suffix array stores the starting position of the i -th lexicographically smallest suffix of s . For example, $s = \text{"cab"}$ has the suffixes "cab" , "ab" , and "b" . When sorted, the indices of the suffixes are "ab" , "b" , and "cab" , so the suffix array (assuming zero-based indices) is $[1, 2, 0]$.

For a string s of length n , the longest common prefix (LCP) array of length $n - 1$ stores the lengths of the longest common prefixes between all pairs of lexicographically adjacent suffixes in s . For example, "baa" has the sorted suffixes "a" , "aa" , and "baa" , with an LCP array of $[1, 0]$.

- `SuffixArray(s)` constructs a suffix array from the given string s using the linear time DC3/skew algorithm by Karkkainen & Sanders (2003) with radix sort.
- `get_sa()` returns the constructed suffix array.
- `get_lcp()` returns the corresponding LCP array for the suffix array.
- `find(needle)` returns one position that `needle` occurs in s (not necessarily the first), or `std::string::npos` if it cannot be found. For a `needle` of length m , this implementation uses an $O(m \log n)$ binary search, but can be optimized to $O(m + \log n)$ by first computing the LCP-LR array using the LCP array.

Time Complexity:

- $O(n)$ per call to the constructor, where n is the length of s .
- $O(1)$ per call to `get_sa()`.
- $O(n)$ per call to `get_lcp()`, where n is the length of s .
- $O(m \log n)$ per call to `find(needle)`, where m is the length of `needle` and n is the length of s .

Space Complexity:

- $O(n)$ auxiliary for storage of the suffix and LCP arrays, where n is the length of s .
- $O(n)$ auxiliary heap space for the constructor.
- $O(1)$ auxiliary space for all other operations.

```

#include <algorithm>
#include <string>
#include <utility>
#include <vector>
using std::string;

class SuffixArray {
    static bool leq(int a1, int a2, int b1, int b2) { return (a1 < b1) || (a1 == b1 && a2 <= b2); }

    static bool leq(int a1, int a2, int a3, int b1, int b2, int b3) {
        return (a1 < b1) || (a1 == b1 && leq(a2, a3, b2, b3));
    }
}

template<class It>
static void radix_pass(It a, It b, It r, int n, int K) {
    std::vector<int> cnt(K + 1);
    for (int i = 0; i < n; i++) {
        cnt[r[a[i]]]++;
    }
}

```

```

    }
    for (int i = 1; i <= K; i++) {
        cnt[i] += cnt[i - 1];
    }
    for (int i = n - 1; i >= 0; i--) {
        b[--cnt[r[a[i]]]] = a[i];
    }
}

template<class It>
static void suffix_array_dc3(It s, It sa, int n, int K) {
    int n0 = (n + 2) / 3, n1 = (n + 1) / 3, n2 = n / 3, n02 = n0 + n2;
    std::vector<int> s12(n02 + 3), sa12(n02 + 3), s0(n0), sa0(n0);
    s12[n02] = s12[n02 + 1] = s12[n02 + 2] = 0;
    sa12[n02] = sa12[n02 + 1] = sa12[n02 + 2] = 0;
    for (int i = 0, j = 0; i < n + n0 - n1; i++) {
        if (i % 3 != 0) {
            s12[j++] = i;
        }
    }
    radix_pass(s12.begin(), sa12.begin(), s + 2, n02, K);
    radix_pass(sa12.begin(), s12.begin(), s + 1, n02, K);
    radix_pass(s12.begin(), sa12.begin(), s, n02, K);
    int name = 0, c0 = -1, c1 = -1, c2 = -1;
    for (int i = 0; i < n02; i++) {
        if (s[sa12[i]] != c0 || s[sa12[i] + 1] != c1 || s[sa12[i] + 2] != c2) {
            name++;
            c0 = s[sa12[i]];
            c1 = s[sa12[i] + 1];
            c2 = s[sa12[i] + 2];
        }
        (sa12[i] % 3 == 1 ? s12[sa12[i] / 3] : s12[sa12[i] / 3 + n0]) = name;
    }
    if (name < n02) {
        suffix_array_dc3(s12.begin(), sa12.begin(), n02, name);
        for (int i = 0; i < n02; i++) {
            s12[sa12[i]] = i + 1;
        }
    } else {
        for (int i = 0; i < n02; i++) {
            sa12[s12[i] - 1] = i;
        }
    }
    for (int i = 0, j = 0; i < n02; i++) {
        if (sa12[i] < n0) {
            s0[j++] = 3 * sa12[i];
        }
    }
    radix_pass(s0.begin(), sa0.begin(), s, n0, K);
    for (int p = 0, t = n0 - n1, k = 0; k < n; k++) {
        int i = (sa12[t] < n0) ? 3 * sa12[t] + 1 : 3 * (sa12[t] - n0) + 2, j = sa0[p];
        if (sa12[t] < n0
            ? leq(s[i], s12[sa12[t] + n0], s[j], s12[j / 3])
            : leq(s[i], s[i + 1], s12[sa12[t] - n0 + 1], s[j], s[j + 1], s12[j / 3 + n0])) {
            sa[k] = i;
            if (++t == n02) {
                for (k++; p < n0; p++, k++) {
                    sa[k] = sa0[p];
                }
            }
        } else {
            sa[k] = j;
            if (++p == n0) {
                for (k++; t < n02; t++, k++) {
                    sa[k] = (sa12[t] < n0) ? 3 * sa12[t] + 1 : 3 * (sa12[t] - n0) + 2;
                }
            }
        }
    }
}

```

```

    }
}

string s;
std::vector<int> sa;

public:
explicit SuffixArray(const string &s) : s(s), sa(s.size() + 1) {
    int n = s.size();
    std::vector<int> scopy(n);
    for (int i = 0; i < n; i++) {
        scopy[i] = static_cast<unsigned char>(s[i]) + 1;
    }
    scopy.resize(n + 4);
    suffix_array_dc3(scopy.begin(), sa.begin(), n + 1, 256);
    sa.erase(sa.begin());
}

const std::vector<int> &get_sa() const { return sa; }

std::vector<int> get_lcp() const {
    int n = s.size();
    if (n == 0) {
        return {};
    }
    std::vector<int> rank(n), lcp(n - 1);
    for (int i = 0; i < n; i++) {
        rank[sa[i]] = i;
    }
    for (int i = 0, k = 0; i < n; i++) {
        if (rank[i] < n - 1) {
            int j = sa[rank[i] + 1];
            while (std::max(i, j) + k < n && s[i + k] == s[j + k]) {
                k++;
            }
            lcp[rank[i]] = k;
            if (k > 0) {
                k--;
            }
        }
    }
    return lcp;
}

size_t find(const string &needle) {
    if (needle.empty()) {
        return 0;
    }
    int lo = 0, hi = static_cast<int>(s.size()) - 1;
    while (lo <= hi) {
        int mid = lo + (hi - lo) / 2;
        int cmp = s.compare(sa[mid], needle.size(), needle);
        if (cmp < 0) {
            lo = mid + 1;
        } else if (cmp > 0) {
            hi = mid - 1;
        } else {
            return sa[mid];
        }
    }
    return string::npos;
}
};

```

Example Usage

```
#include <cassert>
using namespace std;

int main() {
    SuffixArray sa("banana");
    vector<int> sarr = sa.get_sa(), lcp = sa.get_lcp();
    vector<int> sarr_expected{5, 3, 1, 0, 4, 2};
    vector<int> lcp_expected{1, 3, 0, 0, 2};
    assert(sarr == sarr_expected);
    assert(lcp == lcp_expected);
    assert(sa.find("ana") == 1);
    assert(sa.find("x") == string::npos);
    return 0;
}
```

3.7 String Data Structures

3.7.1 Trie

Maintain a map of strings to values using a tree data structure. Each node corresponds to a character, and each inserted string corresponds to a path from the root to a node that is flagged as a terminal node. Children are stored in hash tables, and `walk()` sorts child labels as needed to preserve lexicographic traversal.

- `Trie()` constructs an empty map.
- `size()` returns the size of the map.
- `empty()` returns whether the map is empty.
- `insert(s, v)` adds an entry with string key `s` and value `v` to the map, returning `true` if a new entry was added or `false` if the string already exists (in which case the map is unchanged and the old value associated with the string key is preserved).
- `erase(s)` removes the entry with string key `s` from the map, returning `true` if the removal was successful or `false` if the string to be removed was not found.
- `find(s)` returns a pointer to a const value associated with string key `s`, or `nullptr` if the key was not found.
- `walk(f)` calls the function `f(s, v)` on each entry of the map, in lexicographically ascending order of the string keys.

Time Complexity:

- $O(n)$ expected per call to `insert(s, v)`, `erase(s)`, and `find(s)`, where n is the length of `s`.
- $O(l)$ per call to `walk()`, where l is the total length of string keys that are currently in the map, treating the character alphabet as constant.
- $O(1)$ per call to all other operations.

Space Complexity:

- $O(l)$ for storage of the Trie, where l is the total length of string keys that are currently in the map.
- $O(n)$ auxiliary stack space for construction, destruction, `walk()`, where n is the maximum length of any string that has been inserted so far.
- $O(n)$ auxiliary stack space for `erase(s)`, where n is the length of `s`.
- $O(1)$ auxiliary for all other operations.

```
#include <algorithm>
#include <cstdint>
#include <string>
#include <unordered_map>
#include <utility>
#include <vector>
```

```

using std::string;

template<class V>
class Trie {
    struct Node {
        V value;
        bool is_terminal;
        std::unordered_map<char, Node *> children;

        Node() : is_terminal(false) {}
    } *root;

    static bool erase(Node *n, const string &s, int i) {
        if (i == static_cast<int>(s.size())) {
            if (!n->is_terminal) {
                return false;
            }
            n->is_terminal = false;
            return true;
        }
        auto it = n->children.find(s[i]);
        if (it == n->children.end() || !erase(it->second, s, i + 1)) {
            return false;
        }
        if (it->second->children.empty()) {
            delete it->second;
            n->children.erase(it);
        }
        return true;
    }

    template<class Fn>
    static void walk(Node *n, string &s, Fn f) {
        if (n->is_terminal) {
            f(s, n->value);
        }
        std::vector<std::pair<char, Node *>> children(n->children.begin(), n->children.end());
        std::sort(children.begin(), children.end());
        for (auto &[c, child] : children) {
            s += c;
            walk(child, s, f);
            s.pop_back();
        }
    }

    static void clean_up(Node *n) {
        for (auto &[c, child] : n->children) {
            clean_up(child);
        }
        delete n;
    }

    int num_terminals;

public:
    Trie() : root(new Node()), num_terminals(0) {}

    ~Trie() { clean_up(root); }
    Trie(const Trie &) = delete;
    Trie &operator=(const Trie &) = delete;
    int size() const { return num_terminals; }
    bool empty() const { return num_terminals == 0; }

    bool insert(const string &s, const V &v) {
        Node *n = root;
        for (char c : s) {
            auto [it, inserted] = n->children.try_emplace(c, nullptr);
            if (inserted) {

```



```

        it->second = new Node();
    }
    n = it->second;
}
if (n->is_terminal) {
    return false;
}
num_terminals++;
n->is_terminal = true;
n->value = v;
return true;
}

bool erase(const string &s) {
    if (erase(root, s, 0)) {
        num_terminals--;
        return true;
    }
    return false;
}

const V *find(const string &s) const {
    Node *n = root;
    for (char c : s) {
        if (auto it = n->children.find(c); it != n->children.end()) {
            n = it->second;
        } else {
            return nullptr;
        }
    }
    return n->is_terminal ? &(n->value) : nullptr;
}

template<class Fn>
void walk(Fn f) const {
    string s;
    walk(root, s, f);
}
};

```

Example Usage

```

#include <cassert>
#include <iostream>
using namespace std;

void print_entry(const string &k, int v) {
    cout << "(" << k << ", " << v << ")" << endl;
}

int main() {
    vector<string> s{"", "a", "to", "tea", "ted", "ten", "i", "in", "inn"};
    Trie<int> t;
    assert(t.empty());
    for (int i = 0; i < static_cast<int>(s.size()); i++) {
        assert(t.insert(s[i], i));
    }
    t.walk(print_entry);
    assert(!t.empty());
    assert(t.size() == 9);
    assert(!t.insert(s[0], 2));
    assert(t.size() == 9);
    assert(*t.find("") == 0);
    assert(*t.find("ten") == 5);
    assert(t.erase("tea"));
    assert(t.size() == 8);
}

```

```

assert(t.find("tea") == nullptr);
assert(t.erase(""));
assert(t.find("") == nullptr);
return 0;
}

```

Example Output

```

("", 0)
("a", 1)
("i", 6)
("in", 7)
("inn", 8)
("tea", 3)
("ted", 4)
("ten", 5)
("to", 2)

```

3.7.2 Radix Tree

Maintain a map of strings to values using a compressed tree data structure. Each node corresponds to a substring of an inserted string, and each inserted string corresponds to a path from the root to a node that is flagged as a terminal node. Contrary to a regular trie, a radix tree is more space efficient as it combines chains of nodes with only a single child. Children are stored in hash tables, and `walk()` sorts child labels as needed to preserve lexicographic traversal.

- `RadixTree()` constructs an empty map.
- `size()` returns the size of the map.
- `empty()` returns whether the map is empty.
- `insert(s, v)` adds an entry with string key `s` and value `v` to the map, returning `true` if a new entry was added or `false` if the string already exists (in which case the map is unchanged and the old value associated with the string key is preserved).
- `erase(s)` removes the entry with string key `s` from the map, returning `true` if the removal was successful or `false` if the string to be removed was not found.
- `find(s)` returns a pointer to a const value associated with string key `s`, or `nullptr` if the key was not found.
- `walk(f)` calls the function `f(s, v)` on each entry of the map, in lexicographically ascending order of the string keys.

Time Complexity:

- $O(n)$ expected per call to `insert(s, v)`, `erase(s)`, and `find(s)`, where n is the length of `s`, assuming the number of outgoing compressed edges checked at each node is small.
- $O(l \log l)$ per call to `walk()`, where l is the total length of string keys that are currently in the map.
- $O(1)$ per call to all other operations.

Space Complexity:

- $O(l)$ for storage of the radix tree, where l is the total length of string keys that are currently in the map.
- $O(n)$ auxiliary stack space for construction, destruction, `walk()`, where n is the maximum length of any string that has been inserted so far.
- $O(n)$ auxiliary stack space for `erase(s)`, where n is the length of `s`.
- $O(1)$ auxiliary for all other operations.

```

#include <algorithm>
#include <cstdint>
#include <string>
#include <unordered_map>
#include <utility>
#include <vector>
using std::string;

```

```

template<class V>
class RadixTree {
    struct Node {
        V value;
        bool is_terminal;
        std::unordered_map<string, Node *> children;

        Node(const V &value = V(), bool is_terminal = false) : value(value), is_terminal(is_terminal) {}
    } *root;

    static int lcp_len(const string &s1, const string &s2, int s2start) {
        int i = 0;
        for (int j = s2start; i < static_cast<int>(s1.size()) && j < static_cast<int>(s2.size());
            i++, j++) {
            if (s1[i] != s2[j]) {
                break;
            }
        }
        return i;
    }

    static bool insert(Node *n, const string &s, int i, const V &v) {
        if (i == static_cast<int>(s.size())) {
            if (n->is_terminal) {
                return false;
            }
            n->is_terminal = true;
            return true;
        }
        for (auto it = n->children.begin(); it != n->children.end(); ++it) {
            int len = lcp_len(it->first, s, i);
            if (len == 0) {
                continue;
            }
            if (len == static_cast<int>(it->first.size())) {
                return insert(it->second, s, i + len, v);
            }
            string left = it->first.substr(0, len);
            string right = it->first.substr(len);
            Node *tmp = new Node();
            tmp->children[right] = it->second;
            n->children.erase(it);
            n->children[left] = tmp;
            if (len == static_cast<int>(s.size()) - i) {
                tmp->value = v;
                tmp->is_terminal = true;
                return true;
            }
            return insert(tmp, s, i + len, v);
        }
        n->children[s.substr(i)] = new Node(v, true);
        return true;
    }

    static bool erase(Node *n, const string &s, int i) {
        if (i == static_cast<int>(s.size())) {
            if (!n->is_terminal) {
                return false;
            }
            n->is_terminal = false;
            return true;
        }
        for (auto it = n->children.begin(); it != n->children.end(); ++it) {
            int len = lcp_len(it->first, s, i);
            if (len == 0) {
                continue;
            }

```

```

// The entire edge label must be consumed; a partial match means the key is not present, so
// erasing must fail rather than descend (otherwise erase("te") would remove "tea").
if (len < static_cast<int>(it->first.size())) {
    return false;
}
Node *child = it->second;
if (!erase(child, s, i + len)) {
    return false;
}
if (child->children.empty()) {
    delete child;
    n->children.erase(it);
} else if (child->children.size() == 1) {
    Node *grandchild = child->children.begin()->second;
    if (!child->is_terminal) {
        string merged_key(it->first + child->children.begin()->first);
        child->value = grandchild->value;
        child->is_terminal = grandchild->is_terminal;
        child->children = grandchild->children;
        delete grandchild;
        n->children.erase(it);
        n->children[merged_key] = child;
    }
}
return true;
}
return false;
}

template<class Fn>
static void walk(Node *n, string &s, Fn f) {
    if (n->is_terminal) {
        f(s, n->value);
    }
    std::vector<std::pair<string, Node *>> children(n->children.begin(), n->children.end());
    std::sort(children.begin(), children.end());
    for (auto &[key, child] : children) {
        s += key;
        walk(child, s, f);
        s.erase(s.size() - key.size());
    }
}

static void clean_up(Node *n) {
    for (auto &[key, child] : n->children) {
        clean_up(child);
    }
    delete n;
}

int num_terminals;

public:
RadixTree() : root(new Node()), num_terminals(0) {}

~RadixTree() { clean_up(root); }
RadixTree(const RadixTree &) = delete;
RadixTree &operator=(const RadixTree &) = delete;
int size() const { return num_terminals; }
bool empty() const { return num_terminals == 0; }

bool insert(const string &s, const V &v) {
    if (insert(root, s, 0, v)) {
        num_terminals++;
        return true;
    }
    return false;
}
}

```

```

bool erase(const string &s) {
    if (erase(root, s, 0)) {
        num_terminals--;
        return true;
    }
    return false;
}

const V *find(const string &s) const {
    Node *n = root;
    int i = 0;
    while (i < static_cast<int>(s.size())) {
        bool found = false;
        for (auto &[key, child] : n->children) {
            if (key[0] == s[i]) {
                // The entire edge label must be consumed; a partial match (s ends mid-edge or diverges
                // from it) means the key is not present.
                if (lcp_len(key, s, i) < static_cast<int>(key.size())) {
                    return nullptr;
                }
                i += static_cast<int>(key.size());
                n = child;
                found = true;
                break;
            }
        }
        if (!found) {
            return nullptr;
        }
    }
    return n->is_terminal ? &(n->value) : nullptr;
}

template<class Fn>
void walk(Fn f) const {
    string s;
    walk(root, s, f);
}
};

```

Example Usage

```

#include <cassert>
#include <iostream>
using namespace std;

void print_entry(const string &k, int v) {
    cout << "(" << k << ", " << v << ")" << endl;
}

int main() {
    vector<string> s{"", "a", "to", "tea", "ted", "ten", "i", "in", "inn"};
    RadixTree<int> t;
    assert(t.empty());
    for (int i = 0; i < static_cast<int>(s.size()); i++) {
        assert(t.insert(s[i], i));
    }
    t.walk(print_entry);
    assert(!t.empty());
    assert(t.size() == 9);
    assert(!t.insert(s[0], 2));
    assert(t.size() == 9);
    assert(t.find("") && *t.find("") == 0);
    assert(*t.find("ten") == 5);
    assert(t.erase("tea"));
}

```

```

assert(t.size() == 8);
assert(t.find("tea") == nullptr);
assert(t.erase(""));
assert(t.find("") == nullptr);
return 0;
}

```

Example Output

```

("", 0)
("a", 1)
("i", 6)
("in", 7)
("inn", 8)
("tea", 3)
("ted", 4)
("ten", 5)
("to", 2)

```

3.7.3 Suffix Automaton

Maintains a compact deterministic automaton representing all substrings of a string. A suffix automaton is useful for substring membership queries, counting distinct substrings, finding longest common substrings, and many other string dynamic programming tasks.

Each state represents an equivalence class of substrings with the same set of ending positions. The value `st[v].len` is the maximum length represented by state `v`, while `st[v].link` points to the state representing the longest proper suffix of that class.

- `SuffixAutomaton()` constructs an empty automaton.
- `SuffixAutomaton(s)` constructs the automaton for string `s`.
- `extend(c)` appends character `c` to the current string.
- `contains(t)` returns whether string `t` occurs as a substring.
- `first_occurrence(t)` returns the starting index of the first occurrence of `t`, or `-1` if absent.
- `count_distinct_substrings()` returns the number of distinct nonempty substrings.
- `longest_common_substring(t)` returns one longest substring common to the built string and string `t`.

Time Complexity:

- $O(n)$ expected to construct the automaton for a string of length n .
- $O(m)$ expected per call to `contains(t)`, `first_occurrence(t)`, or `longest_common_substring(t)`, where m is the length of `t`.
- $O(n)$ per call to `count_distinct_substrings()`.

Space Complexity:

- $O(n)$ transition storage and $O(n)$ states.
- $O(1)$ auxiliary per character processed.

```

#include <string>
#include <unordered_map>
#include <vector>
using std::string;

class SuffixAutomaton {
    struct State {
        int len, link, first_pos;
        std::unordered_map<char, int> next;

        State() : len(0), link(-1), first_pos(-1) {}
    };

    std::vector<State> st;

```

```

int last;

public:
    SuffixAutomaton() : st(1), last(0) {}

    explicit SuffixAutomaton(const string &s) : st(1), last(0) {
        for (char c : s) {
            extend(c);
        }
    }

    void extend(char c) {
        int cur = st.size();
        st.emplace_back();
        st[cur].len = st[last].len + 1;
        st[cur].first_pos = st[cur].len - 1;

        int p = last;
        while (p != -1 && st[p].next.find(c) == st[p].next.end()) {
            st[p].next[c] = cur;
            p = st[p].link;
        }
        if (p == -1) {
            st[cur].link = 0;
        } else {
            int q = st[p].next[c];
            if (st[p].len + 1 == st[q].len) {
                st[cur].link = q;
            } else {
                int clone = st.size();
                st.push_back(st[q]);
                st[clone].len = st[p].len + 1;
                while (p != -1 && st[p].next[c] == q) {
                    st[p].next[c] = clone;
                    p = st[p].link;
                }
                st[q].link = st[cur].link = clone;
            }
        }
        last = cur;
    }

    bool contains(const string &t) const {
        int v = 0;
        for (char c : t) {
            if (auto it = st[v].next.find(c); it != st[v].next.end()) {
                v = it->second;
            } else {
                return false;
            }
        }
        return true;
    }

    // Returns the starting index of the first (leftmost) occurrence of `t` as a substring, or -1 if
    // `t` does not occur. `first_pos` stores the end index of the first occurrence of each state's
    // class, so subtracting the length of `t` recovers its start.
    int first_occurrence(const string &t) const {
        int v = 0;
        for (char c : t) {
            auto it = st[v].next.find(c);
            if (it == st[v].next.end()) {
                return -1;
            }
            v = it->second;
        }
        return st[v].first_pos - static_cast<int>(t.size()) + 1;
    }
}

```

```

long long count_distinct_substrings() const {
    long long res = 0;
    for (int v = 1; v < static_cast<int>(st.size()); v++) {
        res += st[v].len - st[st[v].link].len;
    }
    return res;
}

string longest_common_substring(const string &t) const {
    int v = 0, len = 0, best = 0, best_pos = -1;
    for (int i = 0; i < static_cast<int>(t.size()); i++) {
        while (v && !st[v].next.count(t[i])) {
            v = st[v].link;
            len = st[v].len;
        }
        if (auto it = st[v].next.find(t[i]); it != st[v].next.end()) {
            v = it->second;
            len++;
        } else {
            v = 0;
            len = 0;
        }
        if (len > best) {
            best = len;
            best_pos = i;
        }
    }
    return best == 0 ? "" : t.substr(best_pos - best + 1, best);
}
};

```

Example Usage

```

#include <cassert>

int main() {
    SuffixAutomaton sa("ababa");
    assert(sa.contains("aba"));
    assert(sa.contains("bab"));
    assert(!sa.contains("abba"));
    assert(sa.count_distinct_substrings() == 9);
    assert(sa.longest_common_substring("zzbabab") == "baba");
    assert(sa.first_occurrence("aba") == 0);
    assert(sa.first_occurrence("bab") == 1);
    assert(sa.first_occurrence("abba") == -1);

    SuffixAutomaton online;
    online.extend('a');
    online.extend('b');
    online.extend('c');
    assert(online.contains("bc"));
    return 0;
}

```

3.7.4 Eertree

Maintains all distinct palindromic substrings of a string online using an Eertree, also known as a palindromic tree. Each non-root node represents one distinct palindrome, and suffix links connect each palindrome to its longest proper palindromic suffix.

The structure has two roots: one of length -1 , which simplifies boundary cases, and one of length 0 , representing the empty palindrome. After each appended character, `last` points to the node for the longest palindromic suffix of the current string.

- `Eertree()` constructs an empty palindromic tree.
- `Eertree(s)` constructs the tree for string `s`.
- `add(c)` appends character `c` and returns `true` if this creates a new distinct palindrome.
- `count_distinct_palindromes()` returns the number of distinct nonempty palindromic substrings.
- `longest_suffix_length()` returns the length of the current longest palindromic suffix.
- `count_occurrences()` propagates occurrence counts through suffix links and returns `occ[v]` for each node `v`.

Time Complexity:

- $O(n)$ expected to build the tree for a string of length n .
- $O(1)$ expected amortized per call to `add(c)`.
- $O(n)$ per call to `count_occurrences()`.

Space Complexity:

- $O(n)$ transition storage and $O(n)$ nodes.
- $O(1)$ auxiliary per appended character.

```
#include <string>
#include <unordered_map>
#include <vector>
using std::string;

class Eertree {
    struct Node {
        int len, link, occ;
        std::unordered_map<char, int> next;

        Node(int len = 0) : len(len), link(0), occ(0) {}
    };

    std::vector<Node> tree;
    string s;
    int last;

    int get_suffix(int v, int pos, char c) const {
        while (pos - 1 - tree[v].len < 0 || s[pos - 1 - tree[v].len] != c) {
            v = tree[v].link;
        }
        return v;
    }

public:
    Eertree() : tree(), s(), last(1) {
        tree.emplace_back(-1);
        tree.emplace_back(0);
        tree[0].link = 0;
        tree[1].link = 0;
    }

    explicit Eertree(const string &text) : tree(), s(), last(1) {
        tree.emplace_back(-1);
        tree.emplace_back(0);
        tree[0].link = 0;
        tree[1].link = 0;
        for (char c : text) {
            add(c);
        }
    }

    bool add(char c) {
        s += c;
```

```

int pos = s.size() - 1;
int cur = get_suffix(last, pos, c);
auto it = tree[cur].next.find(c);
if (it != tree[cur].next.end()) {
    last = it->second;
    tree[last].occ++;
    return false;
}

last = tree.size();
tree.emplace_back(tree[cur].len + 2);
tree[last].occ = 1;
tree[cur].next[c] = last;

if (tree[last].len == 1) {
    tree[last].link = 1;
    return true;
}
int link_parent = get_suffix(tree[cur].link, pos, c);
tree[last].link = tree[link_parent].next[c];
return true;
}

int count_distinct_palindromes() const { return static_cast<int>(tree.size()) - 2; }
int longest_suffix_length() const { return tree[last].len; }

std::vector<int> count_occurrences() const {
    std::vector<int> occ(tree.size());
    for (int i = 0; i < static_cast<int>(tree.size()); i++) {
        occ[i] = tree[i].occ;
    }
    for (int i = static_cast<int>(tree.size()) - 1; i >= 2; i--) {
        occ[tree[i].link] += occ[i];
    }
    return occ;
}
};

```

Example Usage

```

#include <cassert>

int main() {
    Eertree t("abacaba");
    assert(t.count_distinct_palindromes() == 7);
    assert(t.longest_suffix_length() == 7);

    Eertree online;
    assert(online.add('a'));
    assert(online.add('a'));
    assert(online.add('b'));
    assert(online.add('a'));
    assert(online.count_distinct_palindromes() == 4);
    assert(online.longest_suffix_length() == 3);

    Eertree duplicate;
    assert(duplicate.add('a'));
    assert(duplicate.add('b'));
    assert(duplicate.add('c'));
    assert(!duplicate.add('a'));

    std::vector<int> occ = t.count_occurrences();
    assert(static_cast<int>(occ.size()) == t.count_distinct_palindromes() + 2);
    return 0;
}

```

3.8 Encoding and Compression

3.8.1 Entropy and Frequency Tables

Computes byte frequency tables and empirical information-theoretic quantities for strings. These primitives are useful when analyzing compression, building coding schemes such as Huffman coding, or comparing symbol distributions.

For a string with symbol probabilities p_i , empirical entropy is $H = -\sum_i p_i \log_2 p_i$ bits per symbol. This is a lower bound on the average number of bits per symbol achievable by any uniquely decodable code for the same independent symbol model.

- `byte_frequencies(s)` returns a length-256 table of byte counts in string `s`.
- `entropy(freq)` returns empirical entropy in bits per symbol for a frequency table.
- `entropy(s)` returns empirical entropy in bits per symbol for string `s`.
- `expected_code_length(freq, length)` returns the average encoded bits per symbol for code lengths `length[c]`.

Time Complexity:

- $O(n + A)$ per call to `entropy(s)`, where n is the string length and $A = 256$.
- $O(A)$ per call to `entropy(freq)` and `expected_code_length(freq, length)`.

Space Complexity:

- $O(A)$ auxiliary heap space for `byte_frequencies(s)`.
- $O(1)$ auxiliary for the other operations.

```
#include <cmath>
#include <string>
#include <vector>
using std::string;

std::vector<int> byte_frequencies(const string &s) {
    std::vector<int> freq(256, 0);
    for (int i = 0; i < static_cast<int>(s.size()); i++) {
        freq[static_cast<unsigned char>(s[i])]++;
    }
    return freq;
}

double entropy(const std::vector<int> &freq) {
    int total = 0;
    for (int i = 0; i < static_cast<int>(freq.size()); i++) {
        total += freq[i];
    }
    if (total == 0) {
        return 0;
    }
    double res = 0;
    for (int i = 0; i < static_cast<int>(freq.size()); i++) {
        if (freq[i] == 0) {
            continue;
        }
        double p = static_cast<double>(freq[i]) / total;
        res -= p * (log(p) / log(2.0));
    }
    return res;
}

double entropy(const string &s) {
    return entropy(byte_frequencies(s));
}

double expected_code_length(const std::vector<int> &freq, const std::vector<int> &length) {
```

```

int total = 0;
for (int i = 0; i < static_cast<int>(freq.size()); i++) {
    total += freq[i];
}
if (total == 0) {
    return 0;
}
double res = 0;
for (int i = 0; i < static_cast<int>(freq.size()); i++) {
    res += static_cast<double>(freq[i]) * length[i] / total;
}
return res;
}

```

Example Usage

```

#include <cassert>
#include <cmath>
using namespace std;

bool close(double a, double b) {
    return fabs(a - b) < 1e-9;
}

int main() {
    vector<int> freq = byte_frequencies("aabb");
    assert(freq['a'] == 2);
    assert(freq['b'] == 2);
    assert(close(entropy(freq), 1.0));
    assert(close(entropy("aaaa"), 0.0));

    vector<int> length(256, 0);
    length['a'] = 1;
    length['b'] = 1;
    assert(close(expected_code_length(freq, length), 1.0));
    return 0;
}

```

3.8.2 Run-Length Encoding

Encodes consecutive equal characters as (`character`, `count`) runs. Run-length encoding is a simple lossless compression technique that works well when strings contain long repeated blocks and poorly when repetitions are rare.

The representation below stores runs in a vector of pairs rather than formatting them as a string. This avoids ambiguity when the original text contains digits or separator characters.

- `run_length_encode(s)` returns the sequence of runs in string `s`.
- `run_length_decode(runs)` reconstructs the original string from a sequence of runs.

Time Complexity: $O(n)$ per call, where n is the input or output length.

Space Complexity:

- $O(r)$ for encoded output, where r is the number of runs.
- $O(n)$ for decoded output.

```

#include <string>
#include <utility>
#include <vector>
using std::string;

```

```

std::vector<std::pair<char, int>> run_length_encode(const string &s) {
    std::vector<std::pair<char, int>> res;
    for (int i = 0; i < static_cast<int>(s.size()); i++) {
        if (res.empty() || res.back().first != s[i]) {
            res.emplace_back(s[i], 1);
        } else {
            res.back().second++;
        }
    }
    return res;
}

string run_length_decode(const std::vector<std::pair<char, int>> &runs) {
    string res;
    for (int i = 0; i < static_cast<int>(runs.size()); i++) {
        res.append(runs[i].second, runs[i].first);
    }
    return res;
}

```

Example Usage

```

#include <cassert>
using namespace std;

int main() {
    string s = "aaabcccd";
    vector<pair<char, int>> runs = run_length_encode(s);
    assert(runs.size() == 4);
    assert(runs[0] == (pair<char, int>{'a', 3}));
    assert(runs[2] == (pair<char, int>{'c', 4}));
    assert(run_length_decode(runs) == s);
    return 0;
}

```

3.8.3 Base64 Encoding

Encodes binary data as printable ASCII using Base64, and decodes Base64 text back to bytes. Base64 is an encoding, not encryption or compression: it preserves all input data exactly, but usually increases size by about one third.

This implementation uses the standard alphabet with `+`, `/`, and `=` padding. Whitespace is not skipped by the decoder; clean the input first if line breaks or spaces may appear.

- `base64_encode(bytes)` returns the Base64 representation of string `bytes`.
- `base64_decode(text)` returns the decoded byte string. It assumes `text` is valid padded Base64.

Time Complexity: $O(n)$ per call, where n is the input length.

Space Complexity: $O(n)$ for the output string.

```

#include <string>
#include <vector>
using std::string;

const string BASE64_ALPHABET = "ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz0123456789+/";

string base64_encode(const string &bytes) {
    string res;
    int val = 0, bits = -6;
    for (int i = 0; i < static_cast<int>(bytes.size()); i++) {
        val = (val << 8) + static_cast<unsigned char>(bytes[i]);

```

```

    bits += 8;
    while (bits >= 0) {
        res.push_back(BASE64_ALPHABET[(val >> bits) & 63]);
        bits -= 6;
    }
}
if (bits > -6) {
    res.push_back(BASE64_ALPHABET[((val << 8) >> (bits + 8)) & 63]);
}
while (res.size() % 4) {
    res.push_back('=');
}
return res;
}

string base64_decode(const string &text) {
    std::vector<int> table(256, -1);
    for (int i = 0; i < 64; i++) {
        table[static_cast<unsigned char>(BASE64_ALPHABET[i])] = i;
    }
    string res;
    int val = 0, bits = -8;
    for (int i = 0; i < static_cast<int>(text.size()); i++) {
        if (text[i] == '=') {
            break;
        }
        int x = table[static_cast<unsigned char>(text[i])];
        if (x < 0) {
            break;
        }
        val = (val << 6) + x;
        bits += 6;
        if (bits >= 0) {
            res.push_back(static_cast<char>(((val >> bits) & 255)));
            bits -= 8;
        }
    }
    return res;
}

```

Example Usage

```

#include <cassert>

int main() {
    assert(base64_encode("hello") == "aGVsbG8=");
    assert(base64_decode("aGVsbG8=") == "hello");
    assert(base64_decode(base64_encode("abc123!?")) == "abc123!?");
    return 0;
}

```

3.8.4 Huffman Coding

Builds a Huffman code for a string and uses it to encode and decode that string. Huffman coding is a lossless prefix-code compression algorithm: more frequent characters receive shorter bit strings, and no code is a prefix of another code.

The implementation below stores encoded bits as a string of `'0'` and `'1'` characters for clarity. For real compression, pack those bits into bytes. The tree is also needed to decode the bit string, so compressed data normally stores enough metadata to reconstruct the same tree.

- `HuffmanTree(text)` constructs a Huffman tree from the character frequencies in `text`.

- `codes()` returns a table mapping each byte value to its bit string.
- `encode(text)` returns the encoded bit string.
- `decode(bits)` returns the decoded text.

Time Complexity:

- $O(n + A \log A)$ to construct the tree, where n is the input length and A is the number of distinct characters.
- $O(n)$ to encode or decode, plus the number of produced or consumed bits.

Space Complexity:

- $O(A)$ for the tree and code table.
- $O(n)$ for the encoded or decoded output.

```
#include <queue>
#include <string>
#include <vector>
using std::string;

class HuffmanTree {
    struct Node {
        int freq;
        unsigned char ch;
        int left, right;

        Node(int freq = 0, unsigned char ch = 0, int left = -1, int right = -1)
            : freq(freq), ch(ch), left(left), right(right) {}
    };

    struct ByFrequency {
        const std::vector<Node> *nodes;
        explicit ByFrequency(const std::vector<Node> *nodes = nullptr) : nodes(nodes) {}
        bool operator()(int a, int b) const { return (*nodes)[a].freq > (*nodes)[b].freq; }
    };

    std::vector<Node> nodes;
    int root;
    std::vector<string> code;

    void build_codes(int u, const string &path) {
        if (nodes[u].left == -1 && nodes[u].right == -1) {
            code[nodes[u].ch] = path.empty() ? "0" : path;
            return;
        }
        build_codes(nodes[u].left, path + '0');
        build_codes(nodes[u].right, path + '1');
    }

public:
    explicit HuffmanTree(const string &text) : root(-1), code(256) {
        std::vector<int> freq(256, 0);
        for (int i = 0; i < static_cast<int>(text.size()); i++) {
            freq[static_cast<unsigned char>(text[i])]++;
        }
        ByFrequency cmp(&nodes);
        std::priority_queue<int, std::vector<int>, ByFrequency> pq(cmp);
        for (int c = 0; c < 256; c++) {
            if (freq[c] > 0) {
                nodes.emplace_back(freq[c], static_cast<unsigned char>(c));
                pq.push(static_cast<int>(nodes.size()) - 1);
            }
        }
        if (pq.empty()) {
            return;
        }
        while (pq.size() > 1) {
            int a = pq.top();
```

```

    pq.pop();
    int b = pq.top();
    pq.pop();
    nodes.emplace_back(nodes[a].freq + nodes[b].freq, 0, a, b);
    pq.push(static_cast<int>(nodes.size() - 1));
}
root = pq.top();
build_codes(root, "");
}

std::vector<string> codes() const { return code; }

string encode(const string &text) const {
    string bits;
    for (int i = 0; i < static_cast<int>(text.size()); i++) {
        bits += code[static_cast<unsigned char>(text[i])];
    }
    return bits;
}

string decode(const string &bits) const {
    string text;
    if (root == -1) {
        return text;
    }
    // Degenerate tree: the input had a single distinct character, so the root is itself a leaf and
    // each symbol was encoded as one bit. Walking children would dereference node -1.
    if (nodes[root].left == -1 && nodes[root].right == -1) {
        return string(bits.size(), static_cast<char>(nodes[root].ch));
    }
    int u = root;
    for (int i = 0; i < static_cast<int>(bits.size()); i++) {
        u = bits[i] == '0' ? nodes[u].left : nodes[u].right;
        if (nodes[u].left == -1 && nodes[u].right == -1) {
            text.push_back(static_cast<char>(nodes[u].ch));
            u = root;
        }
    }
    return text;
}
};

```

Example Usage

```

#include <cassert>

int main() {
    string text = "banana bandana";
    HuffmanTree h(text);
    string bits = h.encode(text);
    assert(h.decode(bits) == text);
    assert(bits.size() < text.size() * 8);
    return 0;
}

```


Chapter 4

Graphs

4.1 DFS and Tree Algorithms

4.1.1 Graph Class and Depth-First Search

A graph consists of a set of objects (a.k.a. vertices, or nodes) and a set of connections (a.k.a. edges) between pairs of said objects. A graph may be stored as an adjacency list, which is a space efficient representation that is also time-efficient for traversals.

The following class implements a simple graph using adjacency lists, along with depth-first search and a few other applications. The constructor takes a Boolean argument which specifies whether the instance is a directed or undirected graph. The nodes of the graph are identified by integers indices numbered consecutively starting from 0. The total number of nodes will automatically increase based on the maximum node index passed to `add_edge()` so far. For undirected graphs, `has_cycle()` assumes a simple graph; parallel edges are not distinguished from the tree edge back to the parent.

Time Complexity:

- $O(1)$ amortized per call to `add_edge()`, or $O(\max(n, m))$ for n calls where the maximum node index passed as an argument is m .
- $O(\max(n, m))$ per call for `dfs()`, `has_cycle()`, `is_forest()`, or `is_dag()`, where n is the number of nodes and m is the number of edges.
- $O(1)$ per call to all other public member functions.

Space Complexity:

- $O(\max(n, m))$ for storage of the graph, where n is the number of nodes and m is the number of edges.
- $O(n)$ auxiliary stack space for `dfs()`, `has_cycle()`, `is_forest()`, and `is_dag()`.
- $O(1)$ auxiliary for all other public member functions.

```
#include <algorithm>
#include <vector>

class Graph {
    std::vector<std::vector<int>>> adj;
    bool directed;

    template<class Fn>
    void dfs(int n, std::vector<bool> &visit, Fn f) const {
        f(n);
        visit[n] = true;
        for (int v : adj[n]) {
            if (!visit[v]) {
```

```

        dfs(v, visit, f);
    }
}

bool has_cycle(int n, int prev, std::vector<bool> &visit, std::vector<bool> &onstack) const {
    visit[n] = true;
    onstack[n] = true;
    for (int v : adj[n]) {
        if (directed && onstack[v]) {
            return true;
        }
        if (!directed && visit[v] && v != prev) {
            return true;
        }
        if (!visit[v] && has_cycle(v, n, visit, onstack)) {
            return true;
        }
    }
    onstack[n] = false;
    return false;
}

public:
    Graph(bool directed = true) : directed(directed) {}

    int nodes() const { return static_cast<int>(adj.size()); }
    std::vector<int> &operator[](int n) { return adj[n]; }

    void add_edge(int u, int v) {
        int n = adj.size();
        if (u >= n || v >= n) {
            adj.resize(std::max(u, v) + 1);
        }
        adj[u].push_back(v);
        if (!directed) {
            adj[v].push_back(u);
        }
    }

    bool has_cycle() const {
        int n = adj.size();
        std::vector<bool> visit(n, false), onstack(n, false);
        for (int i = 0; i < n; i++) {
            if (!visit[i] && has_cycle(i, -1, visit, onstack)) {
                return true;
            }
        }
        return false;
    }

    bool is_directed() const { return directed; }
    bool is_forest() const { return !directed && !has_cycle(); }
    bool is_dag() const { return directed && !has_cycle(); }

    template<class Fn>
    void dfs(int start, Fn f) const {
        std::vector<bool> visit(adj.size(), false);
        dfs(start, visit, f);
    }
};

```

Example Usage

```

#include <cassert>
#include <iostream>

```

```
using namespace std;

int main() {
    {
        Graph g;
        g.add_edge(0, 1);
        g.add_edge(0, 6);
        g.add_edge(0, 7);
        g.add_edge(1, 2);
        g.add_edge(1, 5);
        g.add_edge(2, 3);
        g.add_edge(2, 4);
        g.add_edge(7, 8);
        g.add_edge(7, 11);
        g.add_edge(8, 9);
        g.add_edge(8, 10);
        cout << "DFS order: ";
        g.dfs(0, [](int n) { cout << n << " "; });
        cout << endl;
        assert(g[0].size() == 3);
        assert(g.is_dag());
        assert(!g.has_cycle());
    }
    {
        Graph tree(false);
        tree.add_edge(0, 1);
        tree.add_edge(0, 2);
        tree.add_edge(1, 3);
        tree.add_edge(1, 4);
        assert(tree.is_forest());
        assert(!tree.is_dag());
        tree.add_edge(2, 3);
        assert(!tree.is_forest());
    }
    return 0;
}
```

Example Output

```
DFS order: 0 1 2 3 4 5 6 7 8 9 10 11
```

4.1.2 Topological Sorting

Given a directed acyclic graph, find one of possibly many orderings of the nodes such that for every edge from node u to v , u comes before v in the ordering. Depth-first search is used to traverse all nodes in post-order.

- `toposort()` returns a valid topological ordering, or throws if the graph contains a cycle.

Time Complexity: $O(\max(n, m))$ per call to `toposort()`, where n is the number of nodes and m is the number of edges.

Space Complexity:

- $O(\max(n, m))$ for storage of the graph, where n is the number of nodes and m is the number of edges.
- $O(n)$ auxiliary stack space for `toposort()`.

```
#include <algorithm>
#include <stdexcept>
#include <vector>

std::vector<std::vector<int>> adj;
std::vector<bool> visit, done;

void dfs(int u, std::vector<int> &res) {
    if (visit[u]) {
```

```

    throw std::runtime_error("Not a directed acyclic graph.");
}
if (done[u]) {
    return;
}
visit[u] = true;
for (int v : adj[u]) {
    dfs(v, res);
}
visit[u] = false;
done[u] = true;
res.push_back(u);
}

std::vector<int> toposort() {
    int nodes = adj.size();
    visit.assign(nodes, false);
    done.assign(nodes, false);
    std::vector<int> res;
    for (int i = 0; i < nodes; i++) {
        if (!done[i]) {
            dfs(i, res);
        }
    }
    std::reverse(res.begin(), res.end());
    return res;
}

```

Example Usage

```

#include <iostream>
using namespace std;

int main() {
    adj.assign(8, {});
    adj[0].push_back(3);
    adj[0].push_back(4);
    adj[1].push_back(3);
    adj[2].push_back(4);
    adj[2].push_back(7);
    adj[3].push_back(5);
    adj[3].push_back(6);
    adj[3].push_back(7);
    adj[4].push_back(6);
    auto res = toposort();
    cout << "The topological order:";
    for (int v : res) {
        cout << " " << v;
    }
    cout << endl;
    return 0;
}

```

Example Output

```
The topological order: 2 1 0 4 3 7 6 5
```

4.1.3 Tree Centers and Diameter

An unweighted tree possesses a center, centroid, and diameter. The following functions apply to a global, bidirectionally pre-populated adjacency list `adj` which must form a valid tree with nodes numbered from 0 to `adj.size() - 1`.

- `find_centers()` returns a vector of either one or two tree Jordan centers. The Jordan center of a tree is the set of all nodes with minimum eccentricity, that is, the set of all nodes where the maximum distance to all other nodes in the tree is minimal.
- `find_centroid()` returns the node where all of its subtrees have a size less than or equal to $n/2$, where n is the number of nodes in the tree.
- `diameter()` returns the maximum distance between any two nodes in the tree, using a well-known double depth-first search technique.

Time Complexity: $O(\max(n, m))$ per call to `find_centers()`, `find_centroid()`, and `diameter()`, where n is the number of nodes and m is the number of edges.

Space Complexity: $O(n)$ auxiliary stack space for `find_centers()`, `find_centroid()`, and `diameter()`, where n is the number of nodes.

```
#include <algorithm>
#include <utility>
#include <vector>

std::vector<std::vector<int>> adj;

std::vector<int> find_centers() {
    int nodes = adj.size();
    std::vector<int> leaves, degree(nodes);
    for (int i = 0; i < nodes; i++) {
        degree[i] = adj[i].size();
        if (degree[i] <= 1) {
            leaves.push_back(i);
        }
    }
    int removed = leaves.size();
    while (removed < nodes) {
        std::vector<int> nleaves;
        for (int u : leaves) {
            for (int v : adj[u]) {
                if (--degree[v] == 1) {
                    nleaves.push_back(v);
                }
            }
        }
        leaves = nleaves;
        removed += leaves.size();
    }
    return leaves;
}

// Returns the centroid node index if found in this subtree, or -(subtree size) to propagate
// the size up to the parent so it can check the complementary component's size.
int find_centroid(int u = 0, int p = -1) {
    int nodes = adj.size();
    int count = 1;
    bool good_center = true;
    for (int v : adj[u]) {
        if (v == p) {
            continue;
        }
        int res = find_centroid(v, u);
        if (res >= 0) {
            return res;
        }
        int size = -res;
        good_center &= (size <= nodes / 2);
        count += size;
    }
    good_center &= (nodes - count <= nodes / 2);
    return good_center ? u : -count;
}
```

```

std::pair<int, int> dfs(int u, int p, int depth) {
    std::pair<int, int> res{depth, u};
    for (int v : adj[u]) {
        if (v != p) {
            res = std::max(res, dfs(v, u, depth + 1));
        }
    }
    return res;
}

int diameter() {
    int furthest_node = dfs(0, -1, 0).second;
    return dfs(furthest_node, -1, 0).first;
}

```

Example Usage

```

#include <cassert>
using namespace std;

void add_edge(int u, int v) {
    adj[u].push_back(v);
    adj[v].push_back(u);
}

int main() {
    adj.assign(6, {});
    add_edge(0, 1);
    add_edge(1, 2);
    add_edge(1, 4);
    add_edge(3, 4);
    add_edge(4, 5);
    vector<int> centers = find_centers();
    assert(centers.size() == 2 && centers[0] == 1 && centers[1] == 4);
    assert(find_centroid() == 4);
    assert(diameter() == 3);
    return 0;
}

```

4.1.4 Centroid Decomposition

Given a tree, build its centroid decomposition. A centroid is a node whose removal splits the current subtree into connected components of size at most half of that subtree. Recursively choosing centroids creates a decomposition tree of height $O(\log n)$, which is useful for queries based on distances to marked nodes, nearest special node, and other “path through a centroid” problems.

- `build_centroid_tree()` populates `subtree_size`, `centroid_parent`, and `subtree_size` for the global, bidirectionally pre-populated adjacency list `adj` which must form a valid tree with nodes numbered from 0 to `adj.size() - 1`.
- `centroid_parent[u]` stores the parent of node u in the centroid tree.
- The root centroid has parent -1 .

Time Complexity: $O(n \log n)$ per call to `build_centroid_tree()`, where n is the number of nodes.

Space Complexity:

- $O(n)$ for `subtree_size`, `removed`, and `centroid_parent`.
- $O(n)$ auxiliary stack space for the recursive searches.

```

#include <algorithm>

```

```

#include <vector>

std::vector<std::vector<int>> adj;
std::vector<int> subtree_size, centroid_parent;
std::vector<bool> removed;

int get_size(int u, int p) {
    subtree_size[u] = 1;
    for (int v : adj[u]) {
        if (v != p && !removed[v]) {
            subtree_size[u] += get_size(v, u);
        }
    }
    return subtree_size[u];
}

int get_centroid(int u, int p, int total) {
    for (int v : adj[u]) {
        if (v != p && !removed[v] && subtree_size[v] > total / 2) {
            return get_centroid(v, u, total);
        }
    }
    return u;
}

void decompose(int entry, int parent) {
    int total = get_size(entry, -1);
    int centroid = get_centroid(entry, -1, total);
    centroid_parent[centroid] = parent;
    removed[centroid] = true;
    for (int v : adj[centroid]) {
        if (!removed[v]) {
            decompose(v, centroid);
        }
    }
}

void build_centroid_tree() {
    int nodes = adj.size();
    subtree_size.assign(nodes, 0);
    centroid_parent.assign(nodes, -1);
    removed.assign(nodes, false);
    decompose(0, -1);
}

```

Example Usage

```

#include <cassert>
using namespace std;

void add_edge(int u, int v) {
    adj[u].push_back(v);
    adj[v].push_back(u);
}

int main() {
    adj.assign(7, {});
    add_edge(0, 1);
    add_edge(1, 2);
    add_edge(1, 3);
    add_edge(3, 4);
    add_edge(3, 5);
    add_edge(5, 6);
    build_centroid_tree();
    assert(centroid_parent[3] == -1);
    assert(centroid_parent[1] == 3);
}

```

```

    assert(centroid_parent[5] == 3);
    return 0;
}

```

4.1.5 Prufer Code

Encode and decode labeled trees using Prufer codes. A Prufer code is a sequence of length $n - 2$ that uniquely represents a labeled tree on nodes $0, 1, \dots, n - 1$. It is useful for counting labeled trees, generating test cases, and converting between trees and compact sequences. Both functions choose the smallest available leaf at each step. This makes the implementation deterministic and matches the usual textbook convention.

- `encode_prufer()` returns the prufer code for the global, bidirectionally pre-populated adjacency list `adj` which must form a valid tree with nodes numbered from 0 to `adj.size() - 1`.
- `decode_prufer()` takes a Prufer code and returns the corresponding tree edges.

Time Complexity: $O(n \log n)$ per call to `encode_prufer()` or `decode_prufer()`, where n is the number of nodes in the tree.

Space Complexity: $O(n)$ auxiliary space per call to `encode_prufer()` or `decode_prufer()`.

```

#include <set>
#include <utility>
#include <vector>

std::vector<std::vector<int>> adj;

std::vector<int> encode_prufer() {
    int nodes = adj.size();
    std::vector<int> degree(nodes), parent(nodes, -1), code;
    std::set<int> leaves;
    if (nodes <= 2) {
        return code;
    }
    // Root at node n - 1, which is guaranteed to survive smallest-leaf removal (it is never the
    // smallest leaf while other nodes remain). Rooting elsewhere risks recording parent[root] == -2 if
    // the chosen root itself gets stripped to a leaf (e.g. when n - 1 happens to be a leaf).
    int root = nodes - 1;
    for (int u = 0; u < nodes; u++) {
        degree[u] = adj[u].size();
        if (degree[u] == 1) {
            leaves.insert(u);
        }
    }
    std::vector<int> stack(1, root);
    parent[root] = -2;
    while (!stack.empty()) {
        int u = stack.back();
        stack.pop_back();
        for (int v : adj[u]) {
            if (parent[v] == -1) {
                parent[v] = u;
                stack.push_back(v);
            }
        }
    }
    for (int i = 0; i < nodes - 2; i++) {
        int leaf = *leaves.begin();
        leaves.erase(leaves.begin());
        int p = parent[leaf];
        code.push_back(p);
        if (--degree[p] == 1) {
            leaves.insert(p);
        }
    }
}

```



```

    }
    return code;
}

std::vector<std::pair<int, int>> decode_prufer(const std::vector<int> &code) {
    int nodes = code.size() + 2;
    std::vector<int> degree(nodes, 1);
    std::set<int> leaves;
    std::vector<std::pair<int, int>> edges;
    for (int c : code) {
        degree[c]++;
    }
    for (int u = 0; u < nodes; u++) {
        if (degree[u] == 1) {
            leaves.insert(u);
        }
    }
    for (int u : code) {
        int leaf = *leaves.begin();
        leaves.erase(leaves.begin());
        edges.emplace_back(leaf, u);
        if (--degree[u] == 1) {
            leaves.insert(u);
        }
    }
    edges.emplace_back(*leaves.begin(), *leaves.rbegin());
    return edges;
}

```

Example Usage

```

#include <cassert>
using namespace std;

void add_edge(int u, int v) {
    adj[u].push_back(v);
    adj[v].push_back(u);
}

int main() {
    adj.assign(5, {});
    add_edge(0, 3);
    add_edge(1, 3);
    add_edge(2, 3);
    add_edge(3, 4);
    vector<int> code = encode_prufer();
    assert(code.size() == 3 && code[0] == 3 && code[1] == 3 && code[2] == 3);
    vector<pair<int, int>> edges = decode_prufer(code);
    assert(edges.size() == 4);
    return 0;
}

```

4.1.6 Eulerian Cycles

A Eulerian trail is a path in a graph which contains every edge exactly once. An Eulerian cycle or circuit is an Eulerian trail which begins and ends on the same node. A directed graph has an Eulerian cycle if and only if every node has an in-degree equal to its out-degree, and all of its nodes with nonzero degree belong to a single strongly connected component. An undirected graph has an Eulerian cycle if and only if every node has even degree, and all of its nodes with nonzero degree belong to a single connected component.

- `euler_cycle_directed(adj, u)` returns a vector of all nodes in a directed graph that are reachable from the starting node `u` in an order which forms an Eulerian cycle. The first node will be repeated as the last

element of the vector. The directed graph given by adjacency list `adj` must consist of nodes numbered from 0 to `adj.size() - 1`.

- `euler_cycle_undirected(adj, u)` returns a vector of all nodes in a undirected graph that are reachable from the starting node `u` in an order which forms an Eulerian cycle. The first node will be repeated as the last element of the vector. The undirected graph given by adjacency list `adj` must consist of bidirectional edges with nodes numbered from 0 to `adj.size() - 1`.

Limitation: `euler_cycle_undirected()` marks used edges by their endpoint pair, so it does not support multigraphs (parallel edges between the same pair of nodes) – the duplicate edge would be seen as already used. Supporting parallel edges requires marking edges by a unique edge ID instead. The directed version has no such restriction.

Time Complexity: $O(\max(n, m))$ per call to either function, where n and m are the numbers of nodes and edges respectively.

Space Complexity:

- $O(n)$ auxiliary heap space for `euler_cycle_directed()`, where n is the number of nodes.
- $O(n^2)$ auxiliary heap space for `euler_cycle_undirected()`, where n is the number of nodes. This can be reduced to $O(m)$ auxiliary heap space on the number of edges if the `used` bit matrix is replaced with an `std::unordered_set<std::pair<int, int>>`.

```
#include <algorithm>
#include <vector>

std::vector<int> euler_cycle_directed(const std::vector<std::vector<int>>& adj, int u) {
    int nodes = adj.size();
    std::vector<int> stack, curr_edge(nodes), res;
    stack.push_back(u);
    while (!stack.empty()) {
        u = stack.back();
        stack.pop_back();
        while (curr_edge[u] < static_cast<int>(adj[u].size())) {
            stack.push_back(u);
            u = adj[u][curr_edge[u]++];
        }
        res.push_back(u);
    }
    std::reverse(res.begin(), res.end());
    return res;
}

std::vector<int> euler_cycle_undirected(const std::vector<std::vector<int>>& adj, int u) {
    int nodes = adj.size();
    std::vector<std::vector<bool>> used(nodes, std::vector<bool>(nodes));
    std::vector<int> stack, curr_edge(nodes), res;
    stack.push_back(u);
    while (!stack.empty()) {
        u = stack.back();
        stack.pop_back();
        while (curr_edge[u] < static_cast<int>(adj[u].size())) {
            int v = adj[u][curr_edge[u]++];
            int mn = std::min(u, v), mx = std::max(u, v);
            if (!used[mn][mx]) {
                used[mn][mx] = true;
                stack.push_back(u);
                u = v;
            }
        }
        res.push_back(u);
    }
    std::reverse(res.begin(), res.end());
    return res;
}
```

Example Usage

```
#include <iostream>
using namespace std;

int main() {
    {
        vector<vector<int>> g(5);
        g[0].push_back(1);
        g[1].push_back(2);
        g[2].push_back(0);
        g[1].push_back(3);
        g[3].push_back(4);
        g[4].push_back(1);
        vector<int> cycle = euler_cycle_directed(g, 0);
        cout << "Eulerian cycle from 0 (directed):";
        for (int v : cycle) {
            cout << " " << v;
        }
        cout << endl;
    }
    {
        vector<vector<int>> g(5);
        g[0].push_back(1);
        g[1].push_back(0);
        g[1].push_back(2);
        g[2].push_back(1);
        g[2].push_back(0);
        g[0].push_back(2);
        g[1].push_back(3);
        g[3].push_back(1);
        g[3].push_back(4);
        g[4].push_back(3);
        g[4].push_back(1);
        g[1].push_back(4);
        vector<int> cycle = euler_cycle_undirected(g, 2);
        cout << "Eulerian cycle from 2 (undirected):";
        for (int v : cycle) {
            cout << " " << v;
        }
        cout << endl;
    }
    return 0;
}
```

Example Output

```
Eulerian cycle from 0 (directed): 0 1 3 4 1 2 0
Eulerian cycle from 2 (undirected): 2 1 3 4 1 0 2
```

4.2 Connectivity

4.2.1 Connected Components

Given an undirected graph, label every node with its connected component and collect the nodes in each component. This is the standard graph reachability primitive behind many larger routines: run one depth-first search from every unseen node, and all nodes visited by that search belong to the same component. For directed graphs, use a strongly connected components algorithm instead.

- `connected_components()` populates `comp_id[]` (component ID for each node) and `components[]` (nodes for each component ID) given a global, bidirectionally pre-populated adjacency list `adj` which must consist of nodes numbered from 0 to `adj.size() - 1`.

Time Complexity: $O(\max(n, m))$ per call to `connected_components()`, where n is the number of nodes and m is the number of edges.

Space Complexity:

- $O(\max(n, m))$ for storage of the graph, where n is the number of nodes and m is the number of edges.
- $O(n)$ auxiliary stack space for `dfs()`.

```
#include <algorithm>
#include <vector>

std::vector<std::vector<int>> adj;
std::vector<std::vector<int>> components;
std::vector<int> comp_id;

void dfs(int u, int id) {
    comp_id[u] = id;
    components[id].push_back(u);
    for (int v : adj[u]) {
        if (comp_id[v] == -1) {
            dfs(v, id);
        }
    }
}

void connected_components() {
    int nodes = adj.size();
    components.clear();
    comp_id.assign(nodes, -1);
    for (int u = 0; u < nodes; u++) {
        if (comp_id[u] == -1) {
            components.push_back({});
            dfs(u, static_cast<int>(components.size() - 1));
        }
    }
}
```

Example Usage

```
#include <cassert>
using namespace std;

void add_edge(int u, int v) {
    adj[u].push_back(v);
    adj[v].push_back(u);
}

int main() {
    adj.assign(6, {});
    add_edge(0, 1);
    add_edge(1, 2);
    add_edge(3, 4);
    connected_components();
    assert(components.size() == 3);
    assert(comp_id[0] == comp_id[2]);
    assert(comp_id[0] != comp_id[3]);
    assert(comp_id[5] == 2);
    return 0;
}
```

4.2.2 Strongly Connected Components (Kosaraju)

Given a directed graph, determine the strongly connected components (SCCs) using the Kosaraju-Sharir algorithm. A strongly connected component is a maximal set of vertices where every vertex can reach every other vertex. Condensing each SCC into one node produces a directed acyclic graph.

- `KosarajuSCC(n)` constructs a directed graph on nodes numbered from 0 to `n - 1`.
- `add_edge(u, v)` adds the directed edge from `u` to `v`.
- `build_scc()` populates `scc` with the strongly connected components.

Time Complexity: $O(\max(n, m))$ per call to `build_scc()`, where n is the number of nodes and m is the number of edges.

Space Complexity:

- $O(\max(n, m))$ for storage of the graph, reverse graph, and SCCs.
- $O(n)$ auxiliary stack space.

```
#include <algorithm>
#include <vector>

struct KosarajuSCC {
    std::vector<std::vector<int>>> adj, rev, scc;
    std::vector<bool> visited;

    KosarajuSCC(int nodes = 0) : adj(nodes), rev(nodes) {}

    void add_edge(int u, int v) {
        adj[u].push_back(v);
        rev[v].push_back(u);
    }

    void dfs(const std::vector<std::vector<int>>> &g, std::vector<int> &res, int u) {
        visited[u] = true;
        for (int v : g[u]) {
            if (!visited[v]) {
                dfs(g, res, v);
            }
        }
        res.push_back(u);
    }

    void build_scc() {
        int nodes = adj.size();
        visited.assign(nodes, false);
        std::vector<int> order;
        for (int i = 0; i < nodes; i++) {
            if (!visited[i]) {
                dfs(adj, order, i);
            }
        }
        std::reverse(order.begin(), order.end());
        visited.assign(nodes, false);
        scc.clear();
        for (int u : order) {
            if (visited[u]) {
                continue;
            }
            std::vector<int> component;
            dfs(rev, component, u);
            scc.push_back(component);
        }
    }
};
```

Example Usage

```
#include <iostream>
using namespace std;

int main() {
    KosarajuSCC g(8);
    g.add_edge(0, 1);
    g.add_edge(1, 2);
    g.add_edge(1, 4);
    g.add_edge(1, 5);
    g.add_edge(2, 3);
    g.add_edge(2, 6);
    g.add_edge(3, 2);
    g.add_edge(3, 7);
    g.add_edge(4, 0);
    g.add_edge(4, 5);
    g.add_edge(5, 6);
    g.add_edge(6, 5);
    g.add_edge(7, 3);
    g.add_edge(7, 6);
    g.build_scc();
    cout << "Components:" << endl;
    for (auto &component : g.scc) {
        for (int v : component) {
            cout << v << " ";
        }
        cout << endl;
    }
    return 0;
}
```

Example Output

```
Components:
1 4 0
7 3 2
5 6
```

4.2.3 Strongly Connected Components (Tarjan)

Given a directed graph, determine the strongly connected components (SCCs) using Tarjan's algorithm. A strongly connected component is a maximal set of vertices where every vertex can reach every other vertex. Condensing each SCC into one node produces a directed acyclic graph.

- `TarjanSCC(n)` constructs a directed graph on nodes numbered from 0 to `n - 1`.
- `add_edge(u, v)` adds the directed edge from `u` to `v`.
- `build_scc()` populates `scc` with the strongly connected components.

Time Complexity: $O(\max(n, m))$ per call to `build_scc()`, where n is the number of nodes and m is the number of edges.

Space Complexity:

- $O(\max(n, m))$ for storage of the graph and SCCs.
- $O(n)$ auxiliary stack space.

```
#include <algorithm>
#include <climits>
#include <vector>

struct TarjanSCC {
    static const int INF = INT_MAX / 2;
    std::vector<std::vector<int>>> adj, scc;
```

```

std::vector<int> currstack, lowlink;
std::vector<bool> visited;
int timer;

TarjanSCC(int nodes = 0) : adj(nodes) {}

void add_edge(int u, int v) { adj[u].push_back(v); }

void dfs(int u) {
    lowlink[u] = timer++;
    visited[u] = true;
    currstack.push_back(u);
    bool is_component_root = true;
    for (int v : adj[u]) {
        if (!visited[v]) {
            dfs(v);
        }
        if (lowlink[u] > lowlink[v]) {
            lowlink[u] = lowlink[v];
            is_component_root = false;
        }
    }
    if (!is_component_root) {
        return;
    }
    std::vector<int> component;
    int v;
    do {
        v = currstack.back();
        currstack.pop_back();
        lowlink[v] = INF; // marks v as removed from the stack
        component.push_back(v);
    } while (u != v);
    scc.push_back(component);
}

void build_scc() {
    int nodes = adj.size();
    scc.clear();
    currstack.clear();
    lowlink.assign(nodes, 0);
    visited.assign(nodes, false);
    timer = 0;
    for (int i = 0; i < nodes; i++) {
        if (!visited[i]) {
            dfs(i);
        }
    }
}
};

```

Example Usage

```

#include <iostream>
using namespace std;

int main() {
    TarjanSCC g(8);
    g.add_edge(0, 1);
    g.add_edge(1, 2);
    g.add_edge(1, 4);
    g.add_edge(1, 5);
    g.add_edge(2, 3);
    g.add_edge(2, 6);
    g.add_edge(3, 2);
    g.add_edge(3, 7);
}

```

```

g.add_edge(4, 0);
g.add_edge(4, 5);
g.add_edge(5, 6);
g.add_edge(6, 5);
g.add_edge(7, 3);
g.add_edge(7, 6);
g.build_scc();
cout << "Components:" << endl;
for (auto &component : g.scc) {
    for (int v : component) {
        cout << v << " ";
    }
    cout << endl;
}
return 0;
}

```

Example Output

```

Components:
5 6
7 3 2
4 1 0

```

4.2.4 Articulation Points, Biconnected Components, Block-Cut Forest

Given an undirected graph, compute articulation points, vertex-biconnected components (BCCs), and the block-cut forest using Tarjan's algorithm. An articulation point (a.k.a. cut vertex) is a vertex whose removal increases the number of connected components in the graph. A vertex-biconnected component is a maximal subgraph that cannot be disconnected by removing one vertex, with single-edge and isolated-vertex components handled as degenerate cases.

The block-cut forest is a bipartite forest with one node for each BCC and one node for each articulation point, with an edge whenever an articulation point belongs to a BCC. Note that this differs from a bridge forest: the block-cut forest describes vertex connectivity, while a bridge forest describes edge connectivity after compressing 2-edge-connected components.

- `BiconnectedComponents(n)` constructs an undirected graph on nodes numbered from 0 to `n - 1`.
- `add_edge(u, v)` adds the undirected edge `u - v`.
- `build_bcc()` populates `articulation_points` and `biconnected_components`.
- `build_block_cut_forest()` populates `block_cut_forest` and `block_cut_id` using the results of the previous `build_bcc()` call.
- BCC nodes in `block_cut_forest` are numbered `[0, biconnected_components.size())`.
- For an articulation point `v`, `block_cut_id[v]` stores its node ID in the block-cut forest.
- For a non-articulation point `v`, `block_cut_id[v] == -1`.

Limitation: the DFS skips the parent by node ID (`v == p`), so parallel edges (multigraphs) are not supported. To support multigraphs, skip the specific edge used to reach a node by its edge ID instead.

Time Complexity: $O(\max(n, m))$ per call to `build_bcc()` and `build_block_cut_forest()`, where n is the number of nodes and m is the number of edges.

Space Complexity:

- $O(\max(n, m))$ for storage of the graph, edge stack, BCCs, and block-cut forest.
- $O(n)$ auxiliary stack space for `build_bcc()`.

```

#include <algorithm>
#include <utility>
#include <vector>

struct BiconnectedComponents {

```



```

std::vector<std::vector<int>> adj, biconnected_components, block_cut_forest;
std::vector<int> lowlink, tin, block_cut_id, articulation_points;
std::vector<bool> visited, is_articulation;
std::vector<std::pair<int, int>> edge_stack;
int timer;

BiconnectedComponents(int nodes = 0) : adj(nodes) {}

void add_edge(int u, int v) {
    adj[u].push_back(v);
    adj[v].push_back(u);
}

void add_component_until(int u, int v) {
    std::vector<int> component;
    for (;;) {
        auto e = edge_stack.back();
        edge_stack.pop_back();
        component.push_back(e.first);
        component.push_back(e.second);
        if (e.first == u && e.second == v) {
            break;
        }
    }
    std::sort(component.begin(), component.end());
    component.erase(std::unique(component.begin(), component.end()), component.end());
    biconnected_components.push_back(component);
}

void dfs(int u, int p) {
    visited[u] = true;
    lowlink[u] = tin[u] = timer++;
    int children = 0;
    for (int v : adj[u]) {
        if (v == p) {
            continue;
        }
        if (visited[v]) {
            if (tin[v] < tin[u]) {
                edge_stack.emplace_back(u, v);
                lowlink[u] = std::min(lowlink[u], tin[v]);
            }
        } else {
            edge_stack.emplace_back(u, v);
            dfs(v, u);
            lowlink[u] = std::min(lowlink[u], lowlink[v]);
            children++;
            if (lowlink[v] >= tin[u]) {
                if (p != -1 || children > 1) {
                    is_articulation[u] = true;
                }
                add_component_until(u, v);
            }
        }
    }
    if (p == -1 && children == 0) {
        biconnected_components.push_back({u});
    }
}

void build_bcc() {
    int nodes = adj.size();
    articulation_points.clear();
    biconnected_components.clear();
    edge_stack.clear();
    lowlink.assign(nodes, 0);
    tin.assign(nodes, 0);
    visited.assign(nodes, false);
}

```

```

    is_articulation.assign(nodes, false);
    timer = 0;
    for (int i = 0; i < nodes; i++) {
        if (!visited[i]) {
            dfs(i, -1);
        }
    }
    for (int i = 0; i < nodes; i++) {
        if (is_articulation[i]) {
            articulation_points.push_back(i);
        }
    }
}

void build_block_cut_forest() {
    int nodes = adj.size();
    int blocks = biconnected_components.size();
    block_cut_id.assign(nodes, -1);
    int total = blocks;
    for (int v : articulation_points) {
        block_cut_id[v] = total++;
    }
    block_cut_forest.assign(total, std::vector<int>());
    for (int b = 0; b < blocks; b++) {
        for (int v : biconnected_components[b]) {
            if (block_cut_id[v] != -1) {
                block_cut_forest[b].push_back(block_cut_id[v]);
                block_cut_forest[block_cut_id[v]].push_back(b);
            }
        }
    }
}
};

```

Example Usage

```

#include <iostream>
using namespace std;

int main() {
    BiconnectedComponents g(8);
    g.add_edge(0, 1);
    g.add_edge(0, 5);
    g.add_edge(1, 2);
    g.add_edge(1, 5);
    g.add_edge(3, 7);
    g.add_edge(4, 5);
    g.build_bcc();
    g.build_block_cut_forest();
    cout << "Articulation points:";
    for (int v : g.articulation_points) {
        cout << " " << v;
    }
    cout << endl << "Biconnected components:" << endl;
    for (auto &component : g.biconnected_components) {
        for (int v : component) {
            cout << v << " ";
        }
        cout << endl;
    }
    cout << "Adjacency List for Block-Cut Forest:" << endl;
    for (int i = 0, n = static_cast<int>(g.block_cut_forest.size()); i < n; i++) {
        cout << i << " =>";
        for (int v : g.block_cut_forest[i]) {
            cout << " " << v;
        }
    }
}

```

```

    cout << endl;
}
return 0;
}

```

Example Output

```

Articulation points: 1 5
Biconnected components:
1 2
4 5
0 1 5
3 7
6
Adjacency List for Block-Cut Forest:
0 => 5
1 => 6
2 => 5 6
3 =>
4 =>
5 => 0 2
6 => 1 2

```

4.2.5 Bridges, 2-Edge-Connected Components, Bridge Forest

Given an undirected graph, compute bridges, 2-edge-connected components, and the bridge forest using Tarjan's algorithm. A bridge is an edge whose removal increases the number of connected components in the graph. A 2-edge-connected component is a maximal set of vertices connected without crossing any bridge.

After each 2-edge-connected component is condensed into one node, the original bridges connect those nodes into a bridge tree, or a bridge forest when the original graph is disconnected. This differs from a block-cut forest: the bridge forest describes edge connectivity, while the block-cut forest describes vertex connectivity using articulation points and vertex-biconnected components.

- `BridgeDecomposition(n)` constructs an undirected graph on nodes numbered from 0 to `n - 1`.
- `add_edge(u, v)` adds the undirected edge `u - v`.
- `build_bridges()` populates `bridges`.
- `build_bridge_forest()` populates `component`, `two_edge_components`, and `bridge_forest` using the results of the previous `build_bridges()` call.
- `component[u]` stores the 2-edge-connected component ID containing vertex `u`.

Limitation: the DFS skips the parent by node ID (`v == p`), so parallel edges (multigraphs) are not supported – a doubled edge `u - v` would be misreported as a bridge even though it lies on a cycle. To support multigraphs, skip the specific edge used to reach a node by its edge ID instead.

Time Complexity: $O((n + m) \log m)$ per call to `build_bridges()` followed by `build_bridge_forest()`, where n is the number of nodes and m is the number of edges. The logarithmic factor comes from bridge lookups.

Space Complexity:

- $O(\max(n, m))$ for storage of the graph, bridges, 2-edge-connected components, and bridge forest.
- $O(n)$ auxiliary stack space for the DFS calls.

```

#include <algorithm>
#include <set>
#include <utility>
#include <vector>

struct BridgeDecomposition {
    std::vector<std::vector<int>>> adj, two_edge_components, bridge_forest;
    std::vector<int> lowlink, tin, component;
    std::vector<bool> visited;
    std::vector<std::pair<int, int>>> bridges;

```

```

std::set<std::pair<int, int>> bridge_edges;
int timer;

BridgeDecomposition(int nodes = 0) : adj(nodes) {}

void add_edge(int u, int v) {
    adj[u].push_back(v);
    adj[v].push_back(u);
}

std::pair<int, int> edge_key(int u, int v) { return std::minmax(u, v); }

bool is_bridge(int u, int v) { return bridge_edges.count(edge_key(u, v)) > 0; }

void dfs_bridges(int u, int p) {
    visited[u] = true;
    lowlink[u] = tin[u] = timer++;
    for (int v : adj[u]) {
        if (v == p) {
            continue;
        }
        if (visited[v]) {
            lowlink[u] = std::min(lowlink[u], tin[v]);
        } else {
            dfs_bridges(v, u);
            lowlink[u] = std::min(lowlink[u], lowlink[v]);
            if (lowlink[v] > tin[u]) {
                bridges.emplace_back(u, v);
                bridge_edges.insert(edge_key(u, v));
            }
        }
    }
}

void build_bridges() {
    int nodes = adj.size();
    bridges.clear();
    bridge_edges.clear();
    lowlink.assign(nodes, 0);
    tin.assign(nodes, 0);
    visited.assign(nodes, false);
    timer = 0;
    for (int i = 0; i < nodes; i++) {
        if (!visited[i]) {
            dfs_bridges(i, -1);
        }
    }
}

void dfs_component(int u, int id) {
    component[u] = id;
    two_edge_components[id].push_back(u);
    for (int v : adj[u]) {
        if (component[v] == -1 && !is_bridge(u, v)) {
            dfs_component(v, id);
        }
    }
}

void build_bridge_forest() {
    int nodes = adj.size();
    component.assign(nodes, -1);
    two_edge_components.clear();
    for (int i = 0; i < nodes; i++) {
        if (component[i] == -1) {
            two_edge_components.push_back(std::vector<int>());
            dfs_component(i, static_cast<int>(two_edge_components.size() - 1));
        }
    }
}

```

```

    }
    bridge_forest.assign(two_edge_components.size(), std::vector<int>());
    for (auto &[u, v] : bridges) {
        int cu = component[u], cv = component[v];
        bridge_forest[cu].push_back(cv);
        bridge_forest[cv].push_back(cu);
    }
}
};

```

Example Usage

```

#include <iostream>
using namespace std;

int main() {
    BridgeDecomposition g(8);
    g.add_edge(0, 1);
    g.add_edge(0, 5);
    g.add_edge(1, 2);
    g.add_edge(1, 5);
    g.add_edge(3, 7);
    g.add_edge(4, 5);
    g.build_bridges();
    g.build_bridge_forest();
    cout << "Bridges:" << endl;
    for (auto &[u, v] : g.bridges) {
        cout << u << " " << v << endl;
    }
    cout << "2-edge-connected components:" << endl;
    for (auto &component : g.two_edge_components) {
        for (int v : component) {
            cout << v << " ";
        }
        cout << endl;
    }
    cout << "Adjacency List for Bridge Forest:" << endl;
    for (int i = 0, n = static_cast<int>(g.bridge_forest.size()); i < n; i++) {
        cout << i << " =>";
        for (int v : g.bridge_forest[i]) {
            cout << " " << v;
        }
        cout << endl;
    }
    return 0;
}

```

Example Output

```

Bridges:
1 2
5 4
3 7
2-edge-connected components:
0 1 5
2
3
4
6
7
Adjacency List for Bridge Forest:
0 => 1 3
1 => 0
2 => 5
3 => 0

```

```
4 =>
5 => 2
```

4.2.6 Bridge Finding (Online)

Maintain the number of bridges in an undirected graph while edges are inserted one at a time.

- `OnlineBridges(nodes)` constructs a graph with `nodes` isolated nodes.
- `add_edge()` adds an undirected edge and updates `bridges` to store the current number of bridges.

This data structure maintains two disjoint-set forests:

- `dsu_2ecc` tracks the current 2-edge-connected components.
- `dsu_cc` tracks the ordinary connected components.

When an inserted edge joins two different connected components, it creates a new bridge. When it joins two nodes already in the same connected component, it creates a cycle and every bridge on the path between the two endpoints stops being a bridge.

Time Complexity: $O(\log n)$ amortized per call to `add_edge()` in this implementation, where n is the number of nodes.

Space Complexity: $O(n)$ for the disjoint-set arrays and auxiliary parent links.

```
#include <algorithm>
#include <vector>

struct OnlineBridges {
    std::vector<int> dsu_2ecc, dsu_cc, dsu_cc_size, parent, last_visit;
    int lca_iteration, bridges;

    OnlineBridges(int nodes = 0) {
        dsu_2ecc.resize(nodes);
        dsu_cc.resize(nodes);
        dsu_cc_size.assign(nodes, 1);
        parent.assign(nodes, -1);
        last_visit.assign(nodes, 0);
        lca_iteration = 0;
        bridges = 0;
        for (int i = 0; i < nodes; i++) {
            dsu_2ecc[i] = dsu_cc[i] = i;
        }
    }

    int find_2ecc(int v) {
        if (v == -1) {
            return -1;
        }
        if (dsu_2ecc[v] == v) {
            return v;
        }
        return dsu_2ecc[v] = find_2ecc(dsu_2ecc[v]);
    }

    int find_cc(int v) {
        v = find_2ecc(v);
        if (dsu_cc[v] == v) {
            return v;
        }
        return dsu_cc[v] = find_cc(dsu_cc[v]);
    }

    void make_root(int v) {
        v = find_2ecc(v);
        int root = v, child = -1;
        while (v != -1) {
```

```

    int p = find_2ecc(parent[v]);
    parent[v] = child;
    dsu_cc[v] = root;
    child = v;
    v = p;
}
dsu_cc_size[root] = dsu_cc_size[child];
}

void merge_path(int u, int v) {
    lca_iteration++;
    std::vector<int> path_a, path_b;
    int lca = -1;
    while (lca == -1) {
        if (u != -1) {
            u = find_2ecc(u);
            path_a.push_back(u);
            if (last_visit[u] == lca_iteration) {
                lca = u;
                break;
            }
            last_visit[u] = lca_iteration;
            u = parent[u];
        }
        if (v != -1) {
            v = find_2ecc(v);
            path_b.push_back(v);
            if (last_visit[v] == lca_iteration) {
                lca = v;
                break;
            }
            last_visit[v] = lca_iteration;
            v = parent[v];
        }
    }
    for (int v : path_a) {
        dsu_2ecc[v] = lca;
        if (v == lca) {
            break;
        }
    }
    bridges--;
    for (int v : path_b) {
        dsu_2ecc[v] = lca;
        if (v == lca) {
            break;
        }
    }
    bridges--;
}

void add_edge(int u, int v) {
    u = find_2ecc(u);
    v = find_2ecc(v);
    if (u == v) {
        return;
    }
    int ca = find_cc(u), cb = find_cc(v);
    if (ca != cb) {
        bridges++;
        if (dsu_cc_size[ca] > dsu_cc_size[cb]) {
            std::swap(u, v);
            std::swap(ca, cb);
        }
        make_root(u);
        parent[u] = v;
        dsu_cc[u] = v;
        dsu_cc_size[cb] += dsu_cc_size[u];
    }
}

```

```

    } else {
        merge_path(u, v);
    }
}
};

```

Example Usage

```

#include <cassert>

int main() {
    OnlineBridges g(4);
    g.add_edge(0, 1);
    assert(g.bridges == 1);
    g.add_edge(1, 2);
    assert(g.bridges == 2);
    g.add_edge(2, 0);
    assert(g.bridges == 0);
    g.add_edge(2, 3);
    assert(g.bridges == 1);
    return 0;
}

```

4.2.7 2-SAT

Solve a Boolean formula in 2-CNF, where each clause contains at most two literals. A clause like $(a \vee b)$ is represented by the implications $\neg a \rightarrow b$ and $\neg b \rightarrow a$. The formula is satisfiable if and only if no variable and its negation belong to the same strongly connected component of the implication graph.

Variables are numbered from 0 to $n - 1$. A literal is represented by `literal(variable, value)`, where `value == true` means the variable itself and `value == false` means its negation.

- `add_or(a, b)` adds the clause $(a \vee b)$.
- `add_true(a)` forces literal `a` to be true.
- `satisfiable()` returns whether all added clauses can be satisfied and stores one valid assignment in `answer`.

Time Complexity: $O(\max(n, m))$ per call to `satisfiable()`, where n is the number of variables and m is the number of clauses.

Space Complexity: $O(\max(n, m))$ for the implication graph and DFS stacks.

```

#include <algorithm>
#include <vector>

struct TwoSAT {
    int variables;
    std::vector<std::vector<int>>> adj, rev;
    std::vector<int> order, component, answer;

    TwoSAT(int n = 0) { init(n); }

    void init(int n) {
        variables = n;
        adj.assign(2 * n, std::vector<int>());
        rev.assign(2 * n, std::vector<int>());
        answer.assign(n, false);
    }

    int literal(int variable, bool value) { return 2 * variable + (value ? 0 : 1); }
    int neg(int x) { return x ^ 1; }

    void add_implication(int a, int b) {

```



```

    adj[a].push_back(b);
    rev[b].push_back(a);
}

void add_or(int a, int b) {
    add_implication(neg(a), b);
    add_implication(neg(b), a);
}

void add_true(int a) { add_implication(neg(a), a); }
void add_false(int a) { add_implication(a, neg(a)); }

void dfs_order(int u, std::vector<bool> &visited) {
    visited[u] = true;
    for (int v : adj[u]) {
        if (!visited[v]) {
            dfs_order(v, visited);
        }
    }
    order.push_back(u);
}

void dfs_component(int u, int id) {
    component[u] = id;
    for (int v : rev[u]) {
        if (component[v] == -1) {
            dfs_component(v, id);
        }
    }
}

bool satisfiable() {
    order.clear();
    component.assign(2 * variables, -1);
    std::vector<bool> visited(2 * variables, false);
    for (int i = 0; i < 2 * variables; i++) {
        if (!visited[i]) {
            dfs_order(i, visited);
        }
    }
    std::reverse(order.begin(), order.end());
    int id = 0;
    for (int u : order) {
        if (component[u] == -1) {
            dfs_component(u, id++);
        }
    }
    for (int i = 0; i < variables; i++) {
        if (component[2 * i] == component[2 * i + 1]) {
            return false;
        }
        answer[i] = component[2 * i] > component[2 * i + 1];
    }
    return true;
}
};

```

Example Usage

```

#include <cassert>
using namespace std;

int main() {
    TwoSAT solver(3);
    int x0 = solver.literal(0, true);
    int not_x0 = solver.literal(0, false);

```

```

int x1 = solver.literal(1, true);
int x2 = solver.literal(2, true);
solver.add_or(x0, x1);
solver.add_or(not_x0, x2);
solver.add_true(x0);
assert(solver.satisfiable());
assert(solver.answer[0]);
return 0;
}

```

4.3 BFS and Shortest Paths

4.3.1 Shortest Path (BFS)

Given a starting node in an unweighted, directed graph, visit every connected node and determine the minimum distance to each such node. Optionally, output the shortest path to a specific destination node using the shortest-path tree from the predecessor vector `pred`.

- `bfs(start)` populates `dist` and `pred` for a global, pre-populated adjacency list `adj` which must consist of nodes numbered from 0 to `adj.size() - 1`.

Time Complexity: $O(\max(n, m))$ per call to `bfs()`, where n is the number of nodes and m is the number of edges.

Space Complexity:

- $O(\max(n, m))$ for storage of the graph, where n is the number of nodes and m is the number of edges.
- $O(n)$ auxiliary heap space for `bfs()`.

```

#include <algorithm>
#include <climits>
#include <queue>
#include <vector>

const int INF = INT_MAX / 2;
std::vector<std::vector<int>>> adj;
std::vector<int> dist, pred;

void bfs(int start) {
    int nodes = adj.size();
    dist.assign(nodes, INF);
    pred.assign(nodes, -1);
    std::queue<int> q;
    dist[start] = 0;
    q.push(start);
    while (!q.empty()) {
        int u = q.front();
        q.pop();
        for (int v : adj[u]) {
            if (dist[v] != INF) {
                continue;
            }
            dist[v] = dist[u] + 1;
            pred[v] = u;
            q.push(v);
        }
    }
}

```

Example Usage

```
#include <iostream>
using namespace std;

void print_path(int dest) {
    vector<int> path;
    for (int j = dest; pred[j] != -1; j = pred[j]) {
        path.push_back(pred[j]);
    }
    cout << "Take the path: ";
    while (!path.empty()) {
        cout << path.back() << "->";
        path.pop_back();
    }
    cout << dest << "." << endl;
}

int main() {
    int nodes = 4, start = 0, dest = 3;
    adj.assign(nodes, {});
    adj[0].push_back(1);
    adj[1].push_back(2);
    adj[1].push_back(3);
    adj[2].push_back(3);
    bfs(start);
    cout << "The shortest distance from " << start << " to " << dest << " is " << dist[dest] << "."
         << endl;
    print_path(dest);
    return 0;
}
```

Example Output

```
The shortest distance from 0 to 3 is 2.
Take the path: 0->1->3.
```

4.3.2 Shortest Path (0-1 BFS)

Given a starting node in a weighted graph whose edge weights are only 0 or 1, compute the shortest distance to every reachable node.

- `bfs_zero_one(start)` populates `dist` and `pred` for a global, pre-populated adjacency list `adj` which must consist of nodes numbered from 0 to `adj.size() - 1`. Each edge is stored as `(neighbor, weight)`, where `weight` is either 0 or 1.

This is a specialized version of Dijkstra's algorithm. Because every relaxation changes the distance by either 0 or 1, a deque maintains nodes in nondecreasing distance order: push weight-0 relaxations to the front and weight-1 relaxations to the back.

Time Complexity: $O(\max(n, m))$ per call to `bfs_zero_one()`, where n is the number of nodes and m is the number of edges.

Space Complexity:

- $O(\max(n, m))$ for storage of the graph, where n is the number of nodes and m is the number of edges.
- $O(n)$ auxiliary deque space for `bfs_zero_one()`.

```
#include <algorithm>
#include <climits>
#include <deque>
#include <utility>
#include <vector>
```

```

const int INF = INT_MAX / 2;
std::vector<std::vector<std::pair<int, int>>> adj;
std::vector<int> dist, pred;

void bfs_zero_one(int start) {
    int nodes = adj.size();
    dist.assign(nodes, INF);
    pred.assign(nodes, -1);
    std::deque<int> dq;
    dist[start] = 0;
    dq.push_front(start);
    while (!dq.empty()) {
        int u = dq.front();
        dq.pop_front();
        for (auto &[v, w] : adj[u]) {
            if (dist[v] > dist[u] + w) {
                dist[v] = dist[u] + w;
                pred[v] = u;
                if (w == 0) {
                    dq.push_front(v);
                } else {
                    dq.push_back(v);
                }
            }
        }
    }
}

```

Example Usage

```

#include <cassert>
using namespace std;

void add_edge(int u, int v, int w) {
    adj[u].push_back({v, w});
}

int main() {
    int nodes = 4;
    adj.assign(nodes, {});
    add_edge(0, 1, 1);
    add_edge(0, 2, 0);
    add_edge(2, 1, 0);
    add_edge(1, 3, 1);
    add_edge(2, 3, 1);
    bfs_zero_one(0);
    assert(dist[1] == 0);
    assert(dist[3] == 1);
    assert(pred[1] == 2);
    return 0;
}

```

4.3.3 Shortest Path (Dijkstra)

Given a starting node in a weighted, directed graph with nonnegative weights only, visit every connected node and determine the minimum distance to each such node. Optionally, output the shortest path to a specific destination node using the shortest-path tree from the predecessor array `pred`.

- `dijkstra(start)` populates `dist` and `pred` for a global, pre-populated adjacency list `adj` which must consist of nodes numbered from 0 to `adj.size() - 1`. Each edge is stored as `(neighbor, weight)`, where `weight` is nonnegative.

Dijkstra's algorithm requires nonnegative edge weights. Use Bellman-Ford or SPFA instead when negative edges are present.

Time Complexity: $O(m \log n)$ for `dijkstra()`, where m is the number of edges and n is the number of nodes.

Space Complexity:

- $O(\max(n, m))$ for storage of the graph, where n is the number of nodes and m is the number of edges.
- $O(m)$ auxiliary heap space for `dijkstra()`.

```
#include <climits>
#include <functional>
#include <queue>
#include <utility>
#include <vector>

const long long INF = LLONG_MAX / 4;
std::vector<std::vector<std::pair<int, int>>> adj;
std::vector<long long> dist;
std::vector<int> pred;

void dijkstra(int start) {
    int nodes = adj.size();
    std::vector<bool> visit(nodes, false);
    dist.assign(nodes, INF);
    pred.assign(nodes, -1);
    dist[start] = 0;
    std::priority_queue<
        std::pair<long long, int>, std::vector<std::pair<long long, int>>,
        std::greater<std::pair<long long, int>>>
    > pq;
    pq.emplace(0, start);
    while (!pq.empty()) {
        auto [du, u] = pq.top();
        pq.pop();
        if (visit[u] || du != dist[u]) {
            continue;
        }
        visit[u] = true;
        for (auto &[v, w] : adj[u]) {
            if (visit[v]) {
                continue;
            }
            if (dist[v] > dist[u] + w) {
                dist[v] = dist[u] + w;
                pred[v] = u;
                pq.emplace(dist[v], v);
            }
        }
    }
}
```

Example Usage

```
#include <iostream>
using namespace std;

void print_path(int dest) {
    vector<int> path;
    for (int j = dest; pred[j] != -1; j = pred[j]) {
        path.push_back(pred[j]);
    }
    cout << "Take the path: ";
    while (!path.empty()) {
        cout << path.back() << "->";
        path.pop_back();
    }
}
```

```

    }
    cout << dest << "." << endl;
}

int main() {
    int nodes = 4, start = 0, dest = 3;
    adj.assign(nodes, {});
    adj[0].push_back({1, 2});
    adj[0].push_back({3, 8});
    adj[1].push_back({2, 2});
    adj[1].push_back({3, 4});
    adj[2].push_back({3, 1});
    dijkstra(start);
    cout << "The shortest distance from " << start << " to " << dest << " is " << dist[dest] << "."
        << endl;
    print_path(dest);
    return 0;
}

```

Example Output

The shortest distance from 0 to 3 is 5.
Take the path: 0->1->2->3.

4.3.4 Shortest Path (Bellman-Ford)

Given a starting node in a weighted, directed graph with possibly negative weights, visit every connected node and determine the minimum distance to each such node. Optionally, output the shortest path to a specific destination node using the shortest-path tree from the predecessor array `pred`.

- `bellman_ford(nodes, start)` populates `dist` and `pred` for a global, pre-populated edge list `edges` whose endpoints must be numbered from 0 to `nodes - 1`.

This function will also detect whether the graph contains a negative-weight cycle reachable from the start node, in which case the affected shortest paths are undefined and an error will be thrown. (To detect a negative cycle anywhere in the graph, add a virtual source with zero-weight edges to every node and start from it.)

Time Complexity: $O(n \cdot m)$ per call to `bellman_ford()`, where n is the number of nodes and m is the number of edges.

Space Complexity:

- $O(\max(n, m))$ for storage of the graph, where n is the number of nodes and m is the number of edges.
- $O(n)$ auxiliary heap space for `bellman_ford()`, where n is the number of nodes.

```

#include <climits>
#include <stdexcept>
#include <vector>

struct Edge {
    int u, v, w;
};

const long long INF = LLONG_MAX / 4;
std::vector<Edge> edges;
std::vector<long long> dist;
std::vector<int> pred;

void bellman_ford(int nodes, int start) {
    dist.assign(nodes, INF);
    pred.assign(nodes, -1);
    dist[start] = 0;
    for (int i = 0; i < nodes - 1; i++) {
        for (auto &[u, v, w] : edges) {

```

```

    // The dist[u] != INF guard avoids relaxing out of unreachable nodes: a negative edge from an
    // unreachable u would otherwise give v a bogus finite distance (INF + w < INF).
    if (dist[u] != INF && dist[v] > dist[u] + w) {
        dist[v] = dist[u] + w;
        pred[v] = u;
    }
}
}
// Optional: report a negative-weight cycle reachable from the start node.
for (auto &[u, v, w] : edges) {
    if (dist[u] != INF && dist[v] > dist[u] + w) {
        throw std::runtime_error("Negative-weight cycle found.");
    }
}
}
}

```

Example Usage

```

#include <iostream>
using namespace std;

void print_path(int dest) {
    vector<int> path;
    for (int j = dest; pred[j] != -1; j = pred[j]) {
        path.push_back(pred[j]);
    }
    cout << "Take the path: ";
    while (!path.empty()) {
        cout << path.back() << "->";
        path.pop_back();
    }
    cout << dest << "." << endl;
}

int main() {
    int start = 0, dest = 2;
    edges.push_back(Edge{0, 1, 1});
    edges.push_back(Edge{1, 2, 2});
    edges.push_back(Edge{0, 2, 5});
    bellman_ford(3, start);
    cout << "The shortest distance from " << start << " to " << dest << " is " << dist[dest] << "."
        << endl;
    print_path(dest);
    return 0;
}

```

Example Output

```

The shortest distance from 0 to 2 is 3.
Take the path: 0->1->2.

```

4.3.5 Shortest Path (SPFA)

Given a starting node in a weighted directed graph, compute shortest paths even when some edge weights are negative.

- `spfa(start)` populates `dist` and `pred` for a global, pre-populated adjacency list `adj` which must consist of nodes numbered from 0 to `adj.size() - 1`. Each edge is stored as `(neighbor, weight)`. The function returns `false` if it detects a reachable negative cycle, and returns `true` otherwise.

The Shortest Path Faster Algorithm is a queue-based optimization of Bellman-Ford. It is often fast on benign inputs, but it still has Bellman-Ford's worst-case behavior and can be forced to run in $O(n \cdot m)$. Prefer Dijkstra for nonnegative weights, and use SPFA mainly when negative edges are present and the input is not adversarial.

Time Complexity: $O(n \cdot m)$ in the worst case for `spfa()`, where n is the number of nodes and m is the number of edges.

Space Complexity:

- $O(\max(n, m))$ for storage of the graph, where n is the number of nodes and m is the number of edges.
- $O(n)$ auxiliary queue space for `spfa()`.

```
#include <algorithm>
#include <climits>
#include <queue>
#include <utility>
#include <vector>

const long long INF = LLONG_MAX / 4;
std::vector<std::vector<std::pair<int, int>>> adj;
std::vector<long long> dist;
std::vector<int> pred, relax_count;
std::vector<bool> in_queue;

bool spfa(int start) {
    int nodes = adj.size();
    dist.assign(nodes, INF);
    pred.assign(nodes, -1);
    relax_count.assign(nodes, 0);
    in_queue.assign(nodes, false);
    std::queue<int> q;
    dist[start] = 0;
    q.push(start);
    in_queue[start] = true;
    while (!q.empty()) {
        int u = q.front();
        q.pop();
        in_queue[u] = false;
        for (auto &[v, w] : adj[u]) {
            if (dist[u] != INF && dist[v] > dist[u] + w) {
                dist[v] = dist[u] + w;
                pred[v] = u;
                if (!in_queue[v]) {
                    q.push(v);
                    in_queue[v] = true;
                    if (++relax_count[v] >= nodes) {
                        return false;
                    }
                }
            }
        }
    }
    return true;
}
```

Example Usage

```
#include <cassert>
using namespace std;

void add_edge(int u, int v, int w) {
    adj[u].push_back({v, w});
}

int main() {
    int nodes = 4;
    adj.assign(nodes, {});
    add_edge(0, 1, 4);
    add_edge(0, 2, 5);
    add_edge(1, 2, -2);
}
```



```

add_edge(2, 3, 3);
assert(spfa(0));
assert(dist[3] == 5);
assert(pred[2] == 1);
return 0;
}

```

4.3.6 Shortest Path (Floyd-Warshall)

Given a weighted, directed graph with possibly negative weights, determine the minimum distance between all pairs of start and destination nodes in the graph. Optionally, output the shortest path between two nodes using the shortest-path tree precomputed into the `parent` matrix.

- `initialize(nodes)` initializes `dist` and `parent` for nodes numbered from 0 to `nodes - 1`.
- `floyd_warshall()` updates the global adjacency matrix `dist` so `dist[u][v]` stores the shortest path from u to v , and updates `parent` for path reconstruction. If the graph contains negative-weighted cycles, there is no shortest path and an error will be thrown.

Time Complexity:

- $O(n^2)$ per call to `initialize()`, where n is the number of nodes.
- $O(n^3)$ per call to `floyd_warshall()`.

Space Complexity:

- $O(n^2)$ for storage of the graph, where n is the number of nodes.
- $O(n^2)$ auxiliary heap space for `initialize()` and `floyd_warshall()`.

```

#include <climits>
#include <stdexcept>
#include <vector>

const long long INF = LLONG_MAX / 4;
std::vector<std::vector<long long>> dist;
std::vector<std::vector<int>> parent;

void initialize(int nodes) {
    dist.assign(nodes, std::vector<long long>(nodes));
    parent.assign(nodes, std::vector<int>(nodes));
    for (int i = 0; i < nodes; i++) {
        for (int j = 0; j < nodes; j++) {
            dist[i][j] = (i == j) ? 0 : INF;
            parent[i][j] = j;
        }
    }
}

void floyd_warshall() {
    int nodes = dist.size();
    for (int k = 0; k < nodes; k++) {
        for (int i = 0; i < nodes; i++) {
            for (int j = 0; j < nodes; j++) {
                // The INF guards avoid relaxing through an unreachable intermediate: with a negative edge,
                // INF + w < INF would otherwise give an unreachable pair a bogus finite distance.
                if (dist[i][k] != INF && dist[k][j] != INF && dist[i][k] + dist[k][j] < dist[i][j]) {
                    dist[i][j] = dist[i][k] + dist[k][j];
                    parent[i][j] = parent[i][k];
                }
            }
        }
    }
}

// Optional: Report negative-weighted cycles.
for (int i = 0; i < nodes; i++) {
    if (dist[i][i] < 0) {

```

```

        throw std::runtime_error("Negative-weight cycle found.");
    }
}
}

```

Example Usage

```

#include <iostream>
using namespace std;

void print_path(int u, int v) {
    cout << "Take the path: " << u;
    while (u != v) {
        u = parent[u][v];
        cout << "->" << u;
    }
    cout << "." << endl;
}

int main() {
    initialize(3);
    int start = 0, dest = 2;
    dist[0][1] = 1;
    dist[1][2] = 2;
    dist[0][2] = 5;
    floyd_warshall();
    cout << "The shortest distance from " << start << " to " << dest << " is " << dist[start][dest]
        << "." << endl;
    print_path(start, dest);
    return 0;
}

```

Example Output

```

The shortest distance from 0 to 2 is 3.
Take the path: 0->1->2.

```

4.4 Spanning Trees

4.4.1 Minimum Spanning Tree (Prim)

Given a connected, undirected, weighted graph with possibly negative weights, its minimum spanning tree is a subgraph which is a tree that connects all nodes with a subset of its edges such that their total weight is minimized. If the input graph is not connected, then this implementation will find the minimum spanning forest.

- `prim_mst()` populates `mst` and returns total weight for a global, bidirectionally pre-populated adjacency list `adj` which must consist of nodes numbered from 0 to `adj.size() - 1`. Each edge is stored as `(neighbor, weight)`.

Since `std::priority_queue` is by default a max-heap, we simulate a min-heap by negating node distances before pushing them and negating them again after popping them. To modify this implementation to find the maximum spanning tree, the two negation steps can be skipped to prioritize the max edges.

Time Complexity: $O(m \log n)$ per call to `prim_mst()`, where m is the number of edges and n is the number of nodes.

Space Complexity:

- $O(\max(n, m))$ for storage of the graph, where n is the number of nodes and m is the number of edges.
- $O(m)$ auxiliary heap space for `prim_mst()`.

```

#include <queue>
#include <tuple>
#include <utility>
#include <vector>

std::vector<std::vector<std::pair<int, int>>> adj;
std::vector<std::pair<int, int>> mst;

int prim_mst() {
    int nodes = adj.size();
    mst.clear();
    std::vector<bool> visit(nodes);
    int total_dist = 0;
    for (int i = 0; i < nodes; i++) {
        if (visit[i]) {
            continue;
        }
        visit[i] = true;
        std::priority_queue<std::tuple<int, int, int>> pq; // (-weight, from, to)
        for (auto &[v, w] : adj[i]) {
            pq.emplace(-w, i, v);
        }
        while (!pq.empty()) {
            auto [neg_w, u, v] = pq.top();
            int w = -neg_w;
            pq.pop();
            if (visit[u] && !visit[v]) {
                visit[v] = true;
                if (v != i) {
                    mst.emplace_back(u, v);
                    total_dist += w;
                }
                for (auto &[nb, ew] : adj[v]) {
                    pq.emplace(-ew, v, nb);
                }
            }
        }
    }
    return total_dist;
}

```

Example Usage

```

#include <iostream>
using namespace std;

void add_edge(int u, int v, int w) {
    adj[u].push_back({v, w});
    adj[v].push_back({u, w});
}

int main() {
    int nodes = 7;
    adj.assign(nodes, {});
    add_edge(0, 1, 4);
    add_edge(1, 2, 6);
    add_edge(2, 0, 3);
    add_edge(3, 4, 1);
    add_edge(4, 5, 2);
    add_edge(5, 6, 3);
    add_edge(6, 4, 4);
    cout << "Total distance: " << prim_mst() << endl;
    for (auto &[u, v] : mst) {
        cout << u << " <-> " << v << endl;
    }
    return 0;
}

```

```
}
```

Example Output

```
Total distance: 13
0 <-> 2
0 <-> 1
3 <-> 4
4 <-> 5
5 <-> 6
```

4.4.2 Minimum Spanning Tree (Kruskal)

Given a connected, undirected, weighted graph with possibly negative weights, its minimum spanning tree is a subgraph which is a tree that connects all nodes with a subset of its edges such that their total weight is minimized. If the input graph is not connected, then this implementation will find the minimum spanning forest.

- `kruskal_mst(nodes)` populates `mst` and returns total weight for a global, pre-populated edge list `edges` whose endpoints must be numbered from 0 to `nodes - 1`. Each edge is stored as `(weight, u, v)`.

Time Complexity: $O(m \log n)$ per call to `kruskal_mst()`, where m is the number of edges and n is the number of nodes. The internal DSU uses path compression but not union-by-rank (simplified for brevity); `find_root()` is $O(\log n)$ amortized rather than $O(\alpha(n))$.

Space Complexity:

- $O(\max(n, m))$ for storage of the graph, where n is the number of nodes and m is the number of edges
- $O(n)$ auxiliary stack space for `kruskal_mst()`.

```
#include <algorithm>
#include <numeric>
#include <tuple>
#include <utility>
#include <vector>

std::vector<std::tuple<int, int, int>> edges;
std::vector<int> root;
std::vector<std::pair<int, int>> mst;

int find_root(int u) {
    if (root[u] != u) {
        root[u] = find_root(root[u]);
    }
    return root[u];
}

int kruskal_mst(int nodes) {
    mst.clear();
    std::sort(edges.begin(), edges.end());
    int total_dist = 0;
    root.assign(nodes, 0);
    std::iota(root.begin(), root.end(), 0);
    for (auto &[w, a, b] : edges) {
        int u = find_root(a);
        int v = find_root(b);
        if (u != v) {
            root[u] = root[v];
            mst.emplace_back(a, b);
            total_dist += w;
        }
    }
    return total_dist;
}
```

Example Usage

```
#include <iostream>
using namespace std;

void add_edge(int u, int v, int w) {
    edges.emplace_back(w, u, v);
}

int main() {
    add_edge(0, 1, 4);
    add_edge(1, 2, 6);
    add_edge(2, 0, 3);
    add_edge(3, 4, 1);
    add_edge(4, 5, 2);
    add_edge(5, 6, 3);
    add_edge(6, 4, 4);
    cout << "Total distance: " << kruskal_mst(7) << endl;
    for (auto &[u, v] : mst) {
        cout << u << " <-> " << v << endl;
    }
    return 0;
}
```

Example Output

```
Total distance: 13
3 <-> 4
4 <-> 5
2 <-> 0
5 <-> 6
0 <-> 1
```

4.5 Flows and Cuts

4.5.1 Maximum Flow (Ford-Fulkerson)

Given a flow network with integer capacities, find the maximum flow from a given source node to a given sink node. The flow along each edge may not exceed its capacity, and flow is conserved at every node other than the source and sink.

- `ford_fulkerson()` uses global `source` and `sink`, modifies the global residual capacity matrix `cap`, and returns maximum flow. Nodes are numbered from 0 to `cap.size() - 1`.

The Ford-Fulkerson algorithm is only optimal on graphs with integer capacities, as there exists certain real-valued flow inputs for which the algorithm never terminates. The Edmonds-Karp algorithm is an improvement using breadth-first search, addressing this problem.

Time Complexity: $O(n^2 \cdot f)$ per call to `ford_fulkerson()`, where n is the number of nodes and f is the maximum flow.

Space Complexity:

- $O(n^2)$ for storage of the flow network, where n is the number of nodes.
- $O(n)$ auxiliary stack space for `ford_fulkerson()`.

```
#include <algorithm>
#include <climits>
#include <vector>

const int INF = INT_MAX / 2;
int source, sink;
```

```

std::vector<std::vector<int>> cap;
std::vector<bool> visit;

int dfs(int u, int f) {
    int nodes = cap.size();
    if (u == sink) {
        return f;
    }
    visit[u] = true;
    for (int v = 0; v < nodes; v++) {
        if (!visit[v] && cap[u][v] > 0) {
            int flow = dfs(v, std::min(f, cap[u][v]));
            if (flow > 0) {
                cap[u][v] -= flow;
                cap[v][u] += flow;
                return flow;
            }
        }
    }
    return 0;
}

long long ford_fulkerson() {
    int nodes = cap.size();
    long long max_flow = 0;
    for (;;) {
        visit.assign(nodes, false);
        int flow = dfs(source, INF);
        if (flow == 0) {
            break;
        }
        max_flow += flow;
    }
    return max_flow;
}

```

Example Usage

```

#include <cassert>

int main() {
    int nodes = 6;
    source = 0;
    sink = 5;
    cap.assign(nodes, std::vector<int>(nodes));
    cap[0][1] = 3;
    cap[0][2] = 3;
    cap[1][2] = 2;
    cap[1][3] = 3;
    cap[2][4] = 2;
    cap[3][4] = 1;
    cap[3][5] = 2;
    cap[4][5] = 3;
    assert(ford_fulkerson() == 5);
    return 0;
}

```

4.5.2 Maximum Flow (Edmonds-Karp)

Given a flow network with integer capacities, find the maximum flow from a given source node to a given sink node. The flow along each edge may not exceed its capacity, and flow is conserved at every node other than the source and sink.

- `add_edge(u, v, cap)` adds a directed residual-network edge.
- `edmonds_karp(source, sink)` modifies the global adjacency list `adj` and returns maximum flow. Nodes are numbered from 0 to `adj.size() - 1`.

The Edmonds-Karp algorithm will also support real-valued flow capacities. As such, this implementation will work as intended upon changing the appropriate variables to doubles.

Time Complexity: $O(\min(n \cdot m^2, m \cdot f))$ per call to `edmonds_karp()`, where n is the number of nodes, m is the number of edges, and f is the maximum flow.

Space Complexity: $O(\max(n, m))$ for storage of the flow network, where n is the number of nodes and m is the number of edges.

```
#include <algorithm>
#include <climits>
#include <queue>
#include <vector>

struct edge {
    int u, v, rev, cap, f;
};

const int INF = INT_MAX / 2;
std::vector<std::vector<edge>> adj;

void add_edge(int u, int v, int cap) {
    adj[u].push_back(edge{u, v, static_cast<int>(adj[v].size()), cap, 0});
    adj[v].push_back(edge{v, u, static_cast<int>(adj[u].size()) - 1, 0, 0});
}

long long edmonds_karp(int source, int sink) {
    int nodes = adj.size();
    long long max_flow = 0;
    for (;;) {
        std::vector<edge*> pred(nodes, nullptr);
        std::vector<bool> visit(nodes, false);
        std::queue<int> q;
        q.push(source);
        visit[source] = true;
        while (!q.empty() && !pred[sink]) {
            int u = q.front();
            q.pop();
            for (edge &e : adj[u]) {
                if (!visit[e.v] && e.cap > e.f) {
                    visit[e.v] = true;
                    pred[e.v] = &e;
                    q.push(e.v);
                }
            }
        }
        if (!pred[sink]) {
            break;
        }
        int flow = INF;
        for (int u = sink; u != source; u = pred[u]->u) {
            flow = std::min(flow, pred[u]->cap - pred[u]->f);
        }
        for (int u = sink; u != source; u = pred[u]->u) {
            pred[u]->f += flow;
            adj[pred[u]->v][pred[u]->rev].f -= flow;
        }
        max_flow += flow;
    }
    return max_flow;
}
```

Example Usage

```
#include <cassert>

int main() {
    int nodes = 6;
    adj.assign(nodes, {});
    add_edge(0, 1, 3);
    add_edge(0, 2, 3);
    add_edge(1, 2, 2);
    add_edge(1, 3, 3);
    add_edge(2, 4, 2);
    add_edge(3, 4, 1);
    add_edge(3, 5, 2);
    add_edge(4, 5, 3);
    assert(edmonds_karp(0, 5) == 5);
    return 0;
}
```

4.5.3 Maximum Flow (Dinic)

Given a flow network with integer capacities, find the maximum flow from a given source node to a given sink node. The flow along each edge may not exceed its capacity, and flow is conserved at every node other than the source and sink.

- `add_edge(u, v, cap)` adds a directed residual-network edge.
- `dinic(source, sink)` modifies the global adjacency list `adj` and returns maximum flow. Nodes are numbered from 0 to `adj.size() - 1`.

Dinic's algorithm will also support real-valued flow capacities. As such, this implementation will work as intended upon changing the appropriate variables to doubles.

Time Complexity: $O(n^2 \cdot m)$ per call to `dinic()`, where n is the number of nodes and m is the number of edges.

Space Complexity:

- $O(\max(n, m))$ for storage of the flow network, where n is the number of nodes and m is the number of edges.
- $O(n)$ auxiliary stack and heap space for `dinic()`.

```
#include <algorithm>
#include <climits>
#include <queue>
#include <vector>

struct Edge {
    int v, rev, cap, f;
};

const int INF = INT_MAX / 2;
std::vector<std::vector<Edge>> adj;
std::vector<int> dist, ptr;

void add_edge(int u, int v, int cap) {
    adj[u].push_back(Edge{v, static_cast<int>(adj[v].size()), cap, 0});
    adj[v].push_back(Edge{u, static_cast<int>(adj[u].size()) - 1, 0, 0});
}

bool dinic_bfs(int source, int sink) {
    int nodes = adj.size();
    dist.assign(nodes, -1);
    dist[source] = 0;
    std::queue<int> q;
    q.push(source);
```



```

while (!q.empty()) {
    int u = q.front();
    q.pop();
    for (Edge &e : adj[u]) {
        if (dist[e.v] < 0 && e.f < e.cap) {
            dist[e.v] = dist[u] + 1;
            q.push(e.v);
        }
    }
}
return dist[sink] >= 0;
}

int dinic_dfs(int u, int f, int sink) {
    if (u == sink) {
        return f;
    }
    for (; ptr[u] < static_cast<int>(adj[u].size()); ptr[u]++) {
        Edge &e = adj[u][ptr[u]];
        if (dist[e.v] == dist[u] + 1 && e.f < e.cap) {
            int flow = dinic_dfs(e.v, std::min(f, e.cap - e.f), sink);
            if (flow > 0) {
                e.f += flow;
                adj[e.v][e.rev].f -= flow;
                return flow;
            }
        }
    }
    return 0;
}

long long dinic(int source, int sink) {
    int nodes = adj.size();
    int flow;
    long long max_flow = 0;
    while (dinic_bfs(source, sink)) {
        ptr.assign(nodes, 0);
        while ((flow = dinic_dfs(source, INF, sink)) != 0) {
            max_flow += flow;
        }
    }
    return max_flow;
}

```

Example Usage

```

#include <cassert>

int main() {
    int nodes = 6;
    adj.assign(nodes, {});
    add_edge(0, 1, 3);
    add_edge(0, 2, 3);
    add_edge(1, 2, 2);
    add_edge(1, 3, 3);
    add_edge(2, 4, 2);
    add_edge(3, 4, 1);
    add_edge(3, 5, 2);
    add_edge(4, 5, 3);
    assert(dinic(0, 5) == 5);
    return 0;
}

```

4.5.4 Maximum Flow (Push-Relabel)

Given a flow network with integer capacities, find the maximum flow from a given source node to a given sink node. The flow along each edge may not exceed its capacity, and flow is conserved at every node other than the source and sink.

- `push_relabel(source, sink)` returns maximum flow for a global capacity matrix `cap` whose nodes are numbered from 0 to `cap.size() - 1`.

Although the push-relabel algorithm is considered one of the most efficient maximum flow algorithms, it cannot take advantage of the magnitude of the maximum flow being less than n^3 (in which case the Ford-Fulkerson or Edmonds-Karp algorithms may be more efficient).

Time Complexity: $O(n^3)$ per call to `push_relabel()`, where n is the number of nodes.

Space Complexity:

- $O(n^2)$ for storage of the flow network, where n is the number of nodes.
- $O(n)$ auxiliary heap space for `push_relabel()`.

```
#include <algorithm>
#include <climits>
#include <vector>

const int INF = INT_MAX / 2;
std::vector<std::vector<int>>> cap;
std::vector<std::vector<int>>> f;

long long push_relabel(int source, int sink) {
    int nodes = cap.size();
    f.assign(nodes, std::vector<int>(nodes, 0));
    std::vector<int> e(nodes, 0), h(nodes, 0), maxh(nodes, 0);
    h[source] = nodes - 1;
    for (int i = 0; i < nodes; i++) {
        f[source][i] = cap[source][i];
        f[i][source] = -f[source][i];
        e[i] = cap[source][i];
    }
    int size = 0;
    for (;;) {
        if (size == 0) {
            for (int i = 0; i < nodes; i++) {
                if (i != source && i != sink && e[i] > 0) {
                    if (size != 0 && h[i] > h[maxh[0]]) {
                        size = 0;
                    }
                    maxh[size++] = i;
                }
            }
        }
        if (size == 0) {
            break;
        }
        while (size != 0) {
            int i = maxh[size - 1];
            bool pushed = false;
            for (int j = 0; j < nodes && e[i] != 0; j++) {
                if (h[i] == h[j] + 1 && cap[i][j] - f[i][j] > 0) {
                    int df = std::min(cap[i][j] - f[i][j], e[i]);
                    f[i][j] += df;
                    f[j][i] -= df;
                    e[i] -= df;
                    e[j] += df;
                    if (e[i] == 0) {
                        size--;
                    }
                    pushed = true;
                }
            }
        }
    }
}
```

```

    }
}
if (pushed) {
    continue;
}
h[i] = INF;
for (int j = 0; j < nodes; j++) {
    if (h[i] > h[j] + 1 && cap[i][j] - f[i][j] > 0) {
        h[i] = h[j] + 1;
    }
}
if (h[i] > h[maxh[0]]) {
    size = 0;
    break;
}
}
}
long long max_flow = 0;
for (int i = 0; i < nodes; i++) {
    max_flow += f[source][i];
}
return max_flow;
}

```

Example Usage

```

#include <cassert>

int main() {
    int nodes = 6;
    cap.assign(nodes, std::vector<int>(nodes));
    cap[0][1] = 3;
    cap[0][2] = 3;
    cap[1][2] = 2;
    cap[1][3] = 3;
    cap[2][4] = 2;
    cap[3][4] = 1;
    cap[3][5] = 2;
    cap[4][5] = 3;
    assert(push_relabel(0, 5) == 5);
    return 0;
}

```

4.5.5 Minimum-Cost Maximum-Flow

Given a flow network with integer capacities and edge costs, find a minimum-cost flow from a source node to a sink node.

- `MinCostMaxFlow(nodes)` constructs an empty residual network whose nodes are numbered from 0 to `nodes - 1`.
- `add_edge(u, v, cap, cost)` adds a directed residual-network edge.
- `min_cost_flow(source, sink, target_flow)` sends up to `target_flow` units of flow and returns `(flow, cost)`. If the returned flow is smaller than `target_flow`, the network cannot carry the requested amount.

This implementation uses shortest augmenting paths with SPFA, so it supports negative edge costs as long as the residual graph has no reachable negative-cost cycle. For dense or adversarial inputs, a potential-based Dijkstra version is usually faster when reduced costs are nonnegative.

Time Complexity: $O(F \cdot n \cdot m)$ per call to `min_cost_flow()`, where F is the amount of flow sent, n is the number of nodes, and m is the number of edges.

Space Complexity: $O(\max(n, m))$ for storage of the residual network and auxiliary arrays.

```

#include <algorithm>
#include <climits>
#include <queue>
#include <utility>
#include <vector>

struct Edge {
    int v, rev, cap, cost;
    Edge(int v_, int rev_, int cap_, int cost_) : v(v_), rev(rev_), cap(cap_), cost(cost_) {}
};

struct FlowResult {
    int flow;
    long long cost; // Total cost can exceed 32 bits even with int capacities and edge costs.
};

const long long MCMF_INF = LLONG_MAX / 4;

struct MinCostMaxFlow {
    std::vector<std::vector<Edge>> adj;

    MinCostMaxFlow(int nodes = 0) { init(nodes); }

    void init(int nodes) { adj.assign(nodes, std::vector<Edge>()); }

    void add_edge(int u, int v, int cap, int cost) {
        adj[u].emplace_back(v, static_cast<int>(adj[v].size()), cap, cost);
        adj[v].emplace_back(u, static_cast<int>(adj[u].size()) - 1, 0, -cost);
    }

    FlowResult min_cost_flow(int source, int sink, int target_flow) {
        int nodes = adj.size(), flow = 0;
        long long cost = 0;
        std::vector<long long> dist(nodes);
        std::vector<int> parent_node(nodes), parent_edge(nodes);
        std::vector<bool> in_queue(nodes);
        while (flow < target_flow) {
            std::fill(dist.begin(), dist.end(), MCMF_INF);
            std::fill(in_queue.begin(), in_queue.end(), false);
            std::queue<int> q;
            dist[source] = 0;
            q.push(source);
            in_queue[source] = true;
            while (!q.empty()) {
                int u = q.front();
                q.pop();
                in_queue[u] = false;
                for (int i = 0, n = static_cast<int>(adj[u].size()); i < n; i++) {
                    Edge &e = adj[u][i];
                    if (e.cap > 0 && dist[e.v] > dist[u] + e.cost) {
                        dist[e.v] = dist[u] + e.cost;
                        parent_node[e.v] = u;
                        parent_edge[e.v] = i;
                        if (!in_queue[e.v]) {
                            q.push(e.v);
                            in_queue[e.v] = true;
                        }
                    }
                }
            }
        }
        if (dist[sink] == MCMF_INF) {
            break;
        }
        int aug = target_flow - flow;
        for (int v = sink; v != source; v = parent_node[v]) {
            aug = std::min(aug, adj[parent_node[v]][parent_edge[v]].cap);
        }
        for (int v = sink; v != source; v = parent_node[v]) {

```

```

    Edge &e = adj[parent_node[v]][parent_edge[v]];
    e.cap -= aug;
    adj[v][e.rev].cap += aug;
    cost += static_cast<long long>(aug) * e.cost;
}
flow += aug;
}
return {flow, cost};
}
};

```

Example Usage

```

#include <cassert>
using namespace std;

int main() {
    MinCostMaxFlow g(4);
    g.add_edge(0, 1, 2, 1);
    g.add_edge(0, 2, 1, 5);
    g.add_edge(1, 2, 1, 1);
    g.add_edge(1, 3, 1, 3);
    g.add_edge(2, 3, 2, 1);
    FlowResult result = g.min_cost_flow(0, 3, 3);
    assert(result.flow == 3);
    assert(result.cost == 13);
    return 0;
}

```

4.6 Matching and Assignment

4.6.1 Maximum Bipartite Matching (Kuhn)

Given two sets of nodes $A = \{0, 1, \dots, n_1 - 1\}$ and $B = \{0, 1, \dots, n_2 - 1\}$, as well as a set of edges E mapping nodes from set A to set B , find the largest possible subset of E containing no edges that share the same node.

- `kuhn(n2)` populates `match` and returns maximum matching size for a global, pre-populated adjacency list `adj` whose left-side nodes are numbered from 0 to `adj.size() - 1` and whose right-side neighbors are numbered from 0 to `n2 - 1`.

Time Complexity: $O(m \cdot (n_1 + n_2))$ per call to `kuhn()`, where m is the number of edges.

Space Complexity: $O(n_1 + n_2)$ auxiliary stack space for `kuhn()`.

```

#include <algorithm>
#include <vector>

std::vector<int> match;
std::vector<bool> visit;
std::vector<std::vector<int>> adj;

bool dfs(int u) {
    visit[u] = true;
    for (int nb : adj[u]) {
        int v = match[nb];
        if (v == -1 || (!visit[v] && dfs(v))) {
            match[nb] = u;
            return true;
        }
    }
}

```

```

    return false;
}

int kuhn(int n2) {
    int n1 = adj.size();
    visit.assign(n1, false);
    match.assign(n2, -1);
    int matches = 0;
    for (int i = 0; i < n1; i++) {
        visit.assign(n1, false);
        if (dfs(i)) {
            matches++;
        }
    }
    return matches;
}

```

Example Usage

```

#include <iostream>
using namespace std;

int main() {
    int n1 = 3, n2 = 4;
    adj.assign(n1, {});
    adj[0].push_back(1);
    adj[1].push_back(0);
    adj[1].push_back(1);
    adj[1].push_back(2);
    adj[2].push_back(2);
    adj[2].push_back(3);
    cout << "Matched " << kuhn(n2) << " pair(s):" << endl;
    for (int i = 0; i < n2; i++) {
        if (match[i] != -1) {
            cout << match[i] << " " << i << endl;
        }
    }
    return 0;
}

```

Example Output

```

Matched 3 pair(s):
1 0
0 1
2 2

```

4.6.2 Maximum Bipartite Matching (Hopcroft-Karp)

Given two sets of nodes $A = \{0, 1, \dots, n_1 - 1\}$ and $B = \{0, 1, \dots, n_2 - 1\}$, as well as a set of edges E mapping nodes from set A to set B , find the largest possible subset of E containing no edges that share the same node.

- `hopcroft_karp(n2)` populates `match` and returns maximum matching size for a global, pre-populated adjacency list `adj` whose left-side nodes are numbered from 0 to `adj.size() - 1` and whose right-side neighbors are numbered from 0 to `n2 - 1`.

Time Complexity: $O(m \cdot \sqrt{n_1 + n_2})$ per call to `hopcroft_karp()`, where m is the number of edges.

Space Complexity:

- $O(\max(n, m))$ for storage of the graph, where n is the number of nodes and m is the number of edges.
- $O(n_1 + n_2)$ auxiliary stack and heap space for `hopcroft_karp()`.

```

#include <algorithm>
#include <queue>
#include <vector>

std::vector<std::vector<int>>> adj;
std::vector<bool> used, visit;
std::vector<int> match, dist;

void bfs() {
    int n1 = adj.size();
    dist.assign(n1, -1);
    std::queue<int> q;
    for (int u = 0; u < n1; u++) {
        if (!used[u]) {
            q.push(u);
            dist[u] = 0;
        }
    }
    while (!q.empty()) {
        int u = q.front();
        q.pop();
        for (int nb : adj[u]) {
            int v = match[nb];
            if (v >= 0 && dist[v] < 0) {
                dist[v] = dist[u] + 1;
                q.push(v);
            }
        }
    }
}

bool dfs(int u) {
    visit[u] = true;
    for (int nb : adj[u]) {
        int v = match[nb];
        if (v < 0 || (!visit[v] && dist[v] == dist[u] + 1 && dfs(v))) {
            match[nb] = u;
            used[u] = true;
            return true;
        }
    }
    return false;
}

int hopcroft_karp(int n2) {
    int n1 = adj.size();
    match.assign(n2, -1);
    used.assign(n1, false);
    int res = 0;
    for (;;) {
        bfs();
        visit.assign(n1, false);
        int f = 0;
        for (int u = 0; u < n1; u++) {
            if (!used[u] && dfs(u)) {
                f++;
            }
        }
        if (f == 0) {
            return res;
        }
        res += f;
    }
    return res;
}

```

Example Usage

```
#include <iostream>
using namespace std;

int main() {
    int n1 = 3, n2 = 4;
    adj.assign(n1, {});
    adj[0].push_back(1);
    adj[1].push_back(0);
    adj[1].push_back(1);
    adj[1].push_back(2);
    adj[2].push_back(2);
    adj[2].push_back(3);
    cout << "Matched " << hopcroft_karp(n2) << " pair(s):" << endl;
    for (int i = 0; i < n2; i++) {
        if (match[i] != -1) {
            cout << match[i] << " " << i << endl;
        }
    }
    return 0;
}
```

Example Output

```
Matched 3 pair(s):
1 0
0 1
2 2
```

4.6.3 Assignment (Hungarian Algorithm)

Solves the minimum-cost assignment problem for a rectangular cost matrix. Given n workers and m jobs with $n \leq m$, choose a distinct job for each worker so that the total assignment cost is minimized.

The implementation below is the classical Hungarian algorithm using potentials. It is 1-indexed internally, but the input matrix `cost` is 0-indexed. The returned assignment vector has length n , where `assignment[i]` is the chosen job for worker `i`, using 0-indexed job numbers.

- `hungarian(cost, &assignment)` returns the minimum total assignment cost.
- `cost[i][j]` is the cost of assigning worker `i` to job `j`.

Time Complexity: $O(n^2 \cdot m)$, where `cost` has n rows and m columns.

Space Complexity: $O(n + m)$ auxiliary heap space, not counting the input matrix and returned assignment.

```
#include <algorithm>
#include <cassert>
#include <climits>
#include <vector>

long long hungarian(const std::vector<std::vector<long long>> &cost, std::vector<int> *assignment) {
    int n = cost.size(), m = cost.empty() ? 0 : cost[0].size();
    assert(n <= m);
    const long long INF = LLONG_MAX / 4;
    std::vector<long long> u(n + 1), v(m + 1);
    std::vector<int> p(m + 1), way(m + 1);

    for (int i = 1; i <= n; i++) {
        p[0] = i;
        int j0 = 0;
        std::vector<long long> minv(m + 1, INF);
        std::vector<char> used(m + 1, false);
        do {
```



```

    used[j0] = true;
    int i0 = p[j0], j1 = 0;
    long long delta = INF;
    for (int j = 1; j <= m; j++) {
        if (used[j]) {
            continue;
        }
        long long cur = cost[i0 - 1][j - 1] - u[i0] - v[j];
        if (cur < minv[j]) {
            minv[j] = cur;
            way[j] = j0;
        }
        if (minv[j] < delta) {
            delta = minv[j];
            j1 = j;
        }
    }
    for (int j = 0; j <= m; j++) {
        if (used[j]) {
            u[p[j]] += delta;
            v[j] -= delta;
        } else {
            minv[j] -= delta;
        }
    }
    j0 = j1;
} while (p[j0] != 0);

do {
    int j1 = way[j0];
    p[j0] = p[j1];
    j0 = j1;
} while (j0 != 0);
}

assignment->assign(n, -1);
for (int j = 1; j <= m; j++) {
    if (p[j] != 0) {
        (*assignment)[p[j] - 1] = j - 1;
    }
}
return -v[0];
}

```

Example Usage

```

#include <cassert>
using namespace std;

int main() {
    vector<vector<long long>> cost{
        {9, 2, 7, 8},
        {6, 4, 3, 7},
        {5, 8, 1, 8},
    };
    vector<int> assignment;
    assert(hungarian(cost, &assignment) == 9);
    assert(assignment[0] == 1);
    assert(assignment[1] == 0);
    assert(assignment[2] == 2);
    return 0;
}

```

4.6.4 Maximum Graph Matching (Edmonds)

Given an undirected graph, determine a maximum matching: a maximum subset of its edges such that no node is shared between different edges in the resulting subset.

- `edmonds()` populates `match` and returns matching size for a global, bidirectionally pre-populated adjacency list `adj` which must consist of nodes numbered from 0 to `adj.size() - 1`.

Time Complexity: $O(n^3)$ per call to `edmonds()`, where n is the number of nodes.

Space Complexity:

- $O(\max(n, m))$ for storage of the graph, where n is the number of nodes and m is the number of edges.
- $O(n)$ auxiliary heap space for `edmonds()`, where n is the number of nodes.

```
#include <algorithm>
#include <numeric>
#include <queue>
#include <vector>

std::vector<std::vector<int>> adj;
std::vector<int> p, base, match;

int lca(int u, int v) {
    int nodes = adj.size();
    std::vector<bool> used(nodes);
    for (;;) {
        u = base[u];
        used[u] = true;
        if (match[u] == -1) {
            break;
        }
        u = p[match[u]];
    }
    for (;;) {
        v = base[v];
        if (used[v]) {
            return v;
        }
        v = p[match[v]];
    }
}

void mark_path(std::vector<bool> &blossom, int u, int b, int child) {
    for (; base[u] != b; u = p[match[u]]) {
        blossom[base[u]] = true;
        blossom[base[match[u]]] = true;
        p[u] = child;
        child = match[u];
    }
}

int find_path(int root) {
    int nodes = adj.size();
    std::vector<bool> used(nodes);
    p.assign(nodes, -1);
    base.assign(nodes, 0);
    std::iota(base.begin(), base.end(), 0);
    used[root] = true;
    std::queue<int> q;
    q.push(root);
    while (!q.empty()) {
        int u = q.front();
        q.pop();
        for (int v : adj[u]) {
            if (base[u] == base[v] || match[u] == v) {
                continue;
            }

```

```

    }
    if (v == root || (match[v] != -1 && p[match[v]] != -1)) {
        int curr_base = lca(u, v);
        std::vector<bool> blossom(nodes);
        mark_path(blossom, u, curr_base, v);
        mark_path(blossom, v, curr_base, u);
        for (int i = 0; i < nodes; i++) {
            if (blossom[base[i]]) {
                base[i] = curr_base;
                if (!used[i]) {
                    used[i] = true;
                    q.push(i);
                }
            }
        }
    }
    else if (p[v] == -1) {
        p[v] = u;
        if (match[v] == -1) {
            return v;
        }
        v = match[v];
        used[v] = true;
        q.push(v);
    }
}
}
return -1;
}

int edmonds() {
    int nodes = adj.size();
    match.assign(nodes, -1);
    for (int i = 0; i < nodes; i++) {
        if (match[i] == -1) {
            int u, pu, ppu;
            for (u = find_path(i); u != -1; u = ppu) {
                pu = p[u];
                ppu = match[pu];
                match[u] = pu;
                match[pu] = u;
            }
        }
    }
    int matches = 0;
    for (int i = 0; i < nodes; i++) {
        if (match[i] != -1) {
            matches++;
        }
    }
    return matches / 2;
}

```

Example Usage

```

#include <iostream>
using namespace std;

int main() {
    int nodes = 4;
    adj.assign(nodes, {});
    adj[0].push_back(1);
    adj[1].push_back(0);
    adj[1].push_back(2);
    adj[2].push_back(1);
    adj[2].push_back(3);
    adj[3].push_back(2);
}

```

```

adj[3].push_back(0);
adj[0].push_back(3);
cout << "Matched " << edmonds() << " pair(s):" << endl;
for (int i = 0; i < nodes; i++) {
    if (match[i] != -1 && i < match[i]) {
        cout << i << " " << match[i] << endl;
    }
}
return 0;
}

```

Example Output

```

Matched 2 pair(s):
0 1
2 3

```

4.7 Exponential Graph Problems

4.7.1 Maximum Clique (Bron-Kerbosch)

Given an undirected graph, `max_clique()` returns the size of the maximum clique, that is, the largest subset of nodes such that all pairs of nodes in the subset are connected by an edge. `max_clique_weighted()` additionally uses a global array `w` specifying a weight value for each node, returning the clique in the graph that has maximum total weight.

- `max_clique()` returns the maximum clique size for a global, bidirectionally pre-populated adjacency matrix `adj` which must consist of nodes numbered from 0 to `adj.size() - 1`.
- `max_clique_weighted()` additionally uses global `w` and returns the maximum clique weight.

These implementations use bitmasks of unsigned 64-bit integers, so the number of nodes must be less than 64.

Time Complexity: $O(3^{n/3})$ per call to `max_clique()` and `max_clique_weighted()`, where n is the number of nodes.

Space Complexity:

- $O(n^2)$ for storage of the graph, where n is the number of nodes.
- $O(n)$ auxiliary stack space for `max_clique()` and `max_clique_weighted()`.

```

#include <algorithm>
#include <vector>

using uint64 = unsigned long long;

std::vector<std::vector<bool>> adj;
std::vector<int> w;

std::vector<uint64> build_mask_graph() {
    int nodes = adj.size();
    std::vector<uint64> g(nodes);
    for (int i = 0; i < nodes; i++) {
        for (int j = 0; j < nodes; j++) {
            if (adj[i][j]) {
                g[i] |= 1ULL << j;
            }
        }
    }
    return g;
}

int max_clique_rec(const std::vector<uint64> &g, uint64 curr, uint64 pool, uint64 excl) {
    if (pool == 0) {

```

```

    return __builtin_popcountll(curr);
}
int res = 0;
int pivot = __builtin_ctzll(pool | excl);
uint64 candidates = pool & ~g[pivot];
while (candidates != 0) {
    int u = __builtin_ctzll(candidates);
    res = std::max(res, max_clique_rec(g, curr | (1ULL << u), pool & g[u], excl & g[u]));
    pool ^= 1ULL << u;
    excl |= 1ULL << u;
    candidates &= candidates - 1;
}
return res;
}

int max_clique() {
    int nodes = adj.size();
    std::vector<uint64> g = build_mask_graph();
    return max_clique_rec(g, 0, (1ULL << nodes) - 1, 0);
}

int max_clique_weighted_rec(const std::vector<uint64> &g, uint64 curr, uint64 pool, uint64 excl) {
    if (pool == 0) {
        int res = 0;
        while (curr != 0) {
            int u = __builtin_ctzll(curr);
            res += w[u];
            curr &= curr - 1;
        }
        return res;
    }
    int res = -1, pivot = __builtin_ctzll(pool | excl);
    uint64 z = pool & ~g[pivot];
    while (z != 0) {
        int u = __builtin_ctzll(z);
        res = std::max(res, max_clique_weighted_rec(g, curr | (1ULL << u), pool & g[u], excl & g[u]));
        pool ^= 1ULL << u;
        excl |= 1ULL << u;
        z &= z - 1;
    }
    return res;
}

int max_clique_weighted() {
    int nodes = adj.size();
    std::vector<uint64> g = build_mask_graph();
    return max_clique_weighted_rec(g, 0, (1ULL << nodes) - 1, 0);
}

```

Example Usage

```

#include <cassert>

void add_edge(int u, int v) {
    adj[u][v] = true;
    adj[v][u] = true;
}

int main() {
    int nodes = 5;
    adj.assign(nodes, std::vector<bool>(nodes));
    w.assign(nodes, 0);
    add_edge(0, 1);
    add_edge(0, 2);
    add_edge(0, 3);
    add_edge(1, 2);
}

```

```

add_edge(1, 3);
add_edge(2, 3);
add_edge(3, 4);
add_edge(4, 2);
w[0] = 10;
w[1] = 20;
w[2] = 30;
w[3] = 40;
w[4] = 50;
assert(max_clique() == 4);
assert(max_clique_weighted() == 120);
return 0;
}

```

4.7.2 Graph Coloring

Given an undirected graph, assign a color to every node such that no pair of adjacent nodes have the same color, and that the total number of colors used is minimized.

- `color_graph()` populates `color` and returns minimum color count for a global, bidirectionally pre-populated adjacency matrix `adj` which must consist of nodes numbered from 0 to `adj.size() - 1`.

Time Complexity: Exponential on the number of nodes per call to `color_graph()`.

Space Complexity:

- $O(n^2)$ for storage of the graph, where n is the number of nodes.
- $O(n)$ auxiliary stack and heap space for `color_graph()`.

```

#include <algorithm>
#include <vector>

std::vector<std::vector<bool>>> adj;
int min_colors;
std::vector<int> color, curr, id, degree;

void rec(int lo, int hi, int n, int used_colors) {
    if (used_colors >= min_colors) {
        return;
    }
    if (n == hi) {
        for (int i = lo; i < hi; i++) {
            color[id[i]] = curr[i];
        }
        min_colors = used_colors;
        return;
    }
    std::vector<bool> used(used_colors + 1);
    for (int i = 0; i < n; i++) {
        if (adj[id[n]][id[i]]) {
            used[curr[i]] = true;
        }
    }
    for (int i = 0; i <= used_colors; i++) {
        if (!used[i]) {
            int tmp = curr[n];
            curr[n] = i;
            rec(lo, hi, n + 1, std::max(used_colors, i + 1));
            curr[n] = tmp;
        }
    }
}

int color_graph() {

```

```

int nodes = adj.size();
color.assign(nodes, 0);
curr.assign(nodes, 0);
id.assign(nodes + 1, 0);
degree.assign(nodes + 1, 0);
for (int i = 0; i <= nodes; i++) {
    id[i] = i;
    degree[i] = 0;
}
int res = 1, lo = 0;
for (int hi = 1; hi <= nodes; hi++) {
    int best = hi;
    for (int i = hi; i < nodes; i++) {
        if (adj[id[hi] - 1][id[i]]) {
            degree[id[i]]++;
        }
        if (degree[id[best]] < degree[id[i]]) {
            best = i;
        }
    }
    std::swap(id[hi], id[best]);
    if (degree[id[hi]] == 0) {
        min_colors = nodes + 1;
        curr.assign(nodes, 0);
        rec(lo, hi, lo, 0);
        lo = hi;
        res = std::max(res, min_colors);
    }
}
return res;
}

```

Example Usage

```

#include <cassert>
#include <iostream>
using namespace std;

void add_edge(int u, int v) {
    adj[u][v] = true;
    adj[v][u] = true;
}

int main() {
    int nodes = 5;
    adj.assign(nodes, std::vector<bool>(nodes));
    add_edge(0, 1);
    add_edge(0, 4);
    add_edge(1, 3);
    add_edge(1, 4);
    add_edge(2, 3);
    add_edge(2, 4);
    add_edge(3, 4);
    int colors = color_graph();
    cout << "Colored using " << colors << " color(s):" << endl;
    for (int i = 0; i < colors; i++) {
        cout << "Color " << i + 1 << ":";
        for (int j = 0; j < nodes; j++) {
            if (color[j] == i) {
                cout << " " << j;
            }
        }
        cout << endl;
    }
    return 0;
}

```

Example Output

```
Colored using 3 color(s):
Color 1: 0 3
Color 2: 1 2
Color 3: 4
```

4.7.3 Shortest Hamiltonian Cycle (TSP)

Given a complete weighted graph, determine a cycle of minimum total distance which visits each node exactly once and returns to the starting node. This is known as the traveling salesman problem (TSP). Since this implementation uses bitmasks with signed 32-bit integers, the maximum number of nodes must be less than 31.

- `shortest_hamiltonian_cycle()` populates `order` and returns minimum cycle length for a global, bidirectionally pre-populated adjacency matrix `adj` which must represent a complete graph on nodes numbered from 0 to `adj.size() - 1`.

Time Complexity: $O(2^n \cdot n^2)$ per call to `shortest_hamiltonian_cycle()`, where n is the number of nodes.

Space Complexity:

- $O(n^2)$ for storage of the graph, where n is the number of nodes.
- $O(2^n \cdot n)$ auxiliary heap space for `shortest_hamiltonian_cycle()`.

```
#include <algorithm>
#include <climits>
#include <vector>

const long long INF = LLONG_MAX / 4;
std::vector<std::vector<long long>>> adj, dp;
std::vector<int> order;

long long shortest_hamiltonian_cycle() {
    int nodes = adj.size();
    int max_mask = (1 << nodes) - 1;
    dp.assign(max_mask + 1, std::vector<long long>(nodes, INF));
    order.assign(nodes, 0);
    dp[1][0] = 0;
    for (int mask = 1; mask <= max_mask; mask += 2) {
        for (int i = 1; i < nodes; i++) {
            if ((mask & 1 << i) != 0) {
                for (int j = 0; j < nodes; j++) {
                    if ((mask & 1 << j) != 0 && dp[mask ^ (1 << i)][j] != INF) {
                        dp[mask][i] = std::min(dp[mask][i], dp[mask ^ (1 << i)][j] + adj[j][i]);
                    }
                }
            }
        }
    }
    long long res = INF;
    for (int i = 1; i < nodes; i++) {
        res = std::min(res, dp[max_mask][i] + adj[i][0]);
    }
    int mask = max_mask, old = 0;
    for (int i = nodes - 1; i >= 1; i--) {
        int bj = -1;
        for (int j = 1; j < nodes; j++) {
            if ((mask & 1 << j) != 0 &&
                (bj == -1 || dp[mask][bj] + adj[bj][old] > dp[mask][j] + adj[j][old])) {
                bj = j;
            }
        }
        order[i] = bj;
        mask ^= 1 << bj;
        old = bj;
    }
}
```



```
    return res;
}
```

Example Usage

```
#include <iostream>
using namespace std;

void add_edge(int u, int v, long long w) {
    adj[u][v] = w;
    adj[v][u] = w; // Remove this line if the graph is directed.
}

int main() {
    int nodes = 5;
    adj.assign(nodes, std::vector<long long>(nodes));
    add_edge(0, 1, 1);
    add_edge(0, 2, 10);
    add_edge(0, 3, 1);
    add_edge(0, 4, 10);
    add_edge(1, 2, 10);
    add_edge(1, 3, 10);
    add_edge(1, 4, 1);
    add_edge(2, 3, 1);
    add_edge(2, 4, 1);
    add_edge(3, 4, 10);
    cout << "The shortest hamiltonian cycle has length " << shortest_hamiltonian_cycle() << "."
        << endl
        << "Take the path: ";
    for (int i = 0; i < nodes; i++) {
        cout << order[i] << "->";
    }
    cout << order[0] << "." << endl;
    return 0;
}
```

Example Output

```
The shortest hamiltonian cycle has length 5.
Take the path: 0->3->2->4->1->0.
```

4.7.4 Shortest Hamiltonian Path

Given a complete weighted, directed graph, determine a path of minimum total distance which visits each node exactly once. Unlike the traveling salesman problem, we do not have to return to the starting vertex. Since this implementation uses bitmasks with signed 32-bit integers, the maximum number of nodes must be less than 31.

- `shortest_hamiltonian_path()` populates `order` and returns minimum path length for a global, pre-populated adjacency matrix `adj` which must represent a complete directed graph on nodes numbered from 0 to `adj.size() - 1`.

Time Complexity: $O(2^n \cdot n^2)$ per call to `shortest_hamiltonian_path()`, where n is the number of nodes.

Space Complexity:

- $O(n^2)$ for storage of the graph, where n is the number of nodes.
- $O(2^n \cdot n)$ auxiliary heap space for `shortest_hamiltonian_path()`.

```
#include <algorithm>
#include <climits>
#include <vector>

const long long INF = LLONG_MAX / 4;
```

```

std::vector<std::vector<long long>> adj, dp;
std::vector<int> order;

long long shortest_hamiltonian_path() {
    int nodes = adj.size();
    int max_mask = (1 << nodes) - 1;
    dp.assign(max_mask + 1, std::vector<long long>(nodes, INF));
    order.assign(nodes, 0);
    for (int i = 0; i < nodes; i++) {
        dp[1 << i][i] = 0;
    }
    for (int mask = 1; mask <= max_mask; mask++) {
        for (int i = 0; i < nodes; i++) {
            if ((mask & (1 << i)) != 0) {
                for (int j = 0; j < nodes; j++) {
                    if ((mask & (1 << j)) != 0 && dp[mask ^ (1 << i)][j] != INF) {
                        dp[mask][i] = std::min(dp[mask][i], dp[mask ^ (1 << i)][j] + adj[j][i]);
                    }
                }
            }
        }
    }
    long long res = INF;
    for (int i = 0; i < nodes; i++) {
        res = std::min(res, dp[max_mask][i]);
    }
    int mask = max_mask, old = -1;
    for (int i = nodes - 1; i >= 0; i--) {
        int bj = -1;
        for (int j = 0; j < nodes; j++) {
            if ((mask & (1 << j)) != 0 && (bj == -1 || dp[mask][bj] + (old == -1 ? 0 : adj[bj][old]) >
                dp[mask][j] + (old == -1 ? 0 : adj[j][old]))) {
                bj = j;
            }
        }
        order[i] = bj;
        mask ^= (1 << bj);
        old = bj;
    }
    return res;
}

```

Example Usage

```

#include <iostream>
using namespace std;

int main() {
    int nodes = 3;
    adj.assign(nodes, std::vector<long long>(nodes));
    adj[0][1] = 1;
    adj[0][2] = 1;
    adj[1][0] = 7;
    adj[1][2] = 2;
    adj[2][0] = 3;
    adj[2][1] = 5;
    cout << "The shortest hamiltonian path has length " << shortest_hamiltonian_path() << "." << endl;
    cout << "Take the path: " << order[0];
    for (int i = 1; i < nodes; i++) {
        cout << "->" << order[i];
    }
    cout << "." << endl;
    return 0;
}

```

Example Output

The shortest hamiltonian path has length 3.
Take the path: 0->1->2.

Chapter 5

Optimization

5.1 Monotone Predicate Search

5.1.1 Binary Search

Binary search can be generally used to find the input value corresponding to any output value of a monotonic (strictly non-increasing or strictly non-decreasing) function in $O(\log n)$ time with respect to the domain size. This is a special case of finding the exact point at which any given monotonic Boolean function changes from true to false or vice versa. Unlike searching through an array, discrete binary search is not restricted by available memory, making it useful for handling infinitely large search spaces such as real number intervals.

- `binary_search_first_true()` takes two integers `lo` and `hi` as boundaries for the search space `[lo, hi)` (i.e. including `lo`, but excluding `hi`) and returns the smallest integer `k` in `[lo, hi)` for which the predicate `pred(k)` tests true. If `pred(k)` tests false for every `k` in `[lo, hi)`, then `hi` is returned. This function must be used on a range in which there exists a constant `k` such that `pred(x)` tests false for every `x` in `[lo, k)` and true for every `x` in `[k, hi)`.
- `binary_search_last_true()` takes two integers `lo` and `hi` as boundaries for the search space `[lo, hi)` (i.e. including `lo`, but excluding `hi`) and returns the largest integer `k` in `[lo, hi)` for which the predicate `pred(k)` tests true. If `pred(k)` tests false for every `k` in `[lo, hi)`, then `hi` is returned. This function must be used on a range in which there exists a constant `k` such that `pred(x)` tests true for every `x` in `[lo, k)` and false for every `x` in `[k, hi)`.
- `fbinary_search()` is the equivalent of `binary_search_first_true()` on floating point predicates. Since any interval of real numbers is dense, the exact target cannot be found due to floating point error. Instead, a value that is very close to the border between false and true is returned. The precision of the answer depends on the number of repetitions the function performs. Since each repetition bisects the search space, the absolute error of the answer is $1/(2^n)$ times the distance between `lo` and `hi` after n repetitions. Although it is possible to control the error by looping while `hi - lo` is greater than an arbitrary epsilon, it is simpler to let the loop run for a desired number of iterations until floating point arithmetic break down. 100 iterations is usually sufficient, since the search space will be reduced to 2^{-100} (roughly 10^{-30}) times its original size. This implementation can be modified to find the “last true” point in the range by simply interchanging the assignments of `lo` and `hi` in the if-else statements.

Time Complexity:

- $O(\log n)$ calls will be made to `pred()` in `binary_search_first_true()` and `binary_search_last_true()`, where n is the distance between `lo` and `hi`.
- $O(n)$ calls will be made to `pred()` in `fbinary_search()`, where n is the number of iterations performed.

Space Complexity: $O(1)$ auxiliary for all operations.

```

template<class Int, class Pred>
Int binary_search_first_true(Int lo, Int hi, Pred pred) { // 000[1]11
    while (lo < hi) {
        Int mid = lo + (hi - lo) / 2;
        if (pred(mid)) {
            hi = mid;
        } else {
            lo = mid + 1;
        }
    }
    return lo;
}

template<class Int, class Pred>
Int binary_search_last_true(Int lo, Int hi, Pred pred) { // 11[1]000
    Int _hi = hi;
    if (lo == hi) {
        return _hi;
    }
    hi--;
    while (lo < hi) {
        Int mid = lo + (hi - lo + 1) / 2;
        if (pred(mid)) {
            lo = mid;
        } else {
            hi = mid - 1;
        }
    }
    if (!pred(lo)) {
        return _hi; // All false.
    }
    return lo;
}

template<class Pred>
double fbinary_search(double lo, double hi, Pred pred) { // 000[1]11
    double mid;
    for (int i = 0; i < 100; i++) {
        mid = (lo + hi) / 2.0;
        if (pred(mid)) {
            hi = mid;
        } else {
            lo = mid;
        }
    }
    return lo;
}

```

Example Usage

```

#include <cassert>
#include <cmath>

int main() {
    assert(binary_search_first_true(0, 7, [](int x) { return x >= 3; }) == 3);
    assert(binary_search_first_true(0, 7, [](int x) { return false; }) == 7);

    assert(binary_search_last_true(0, 7, [](int x) { return x <= 5; }) == 5);
    assert(binary_search_last_true(0, 7, [](int x) { return true; }) == 6);

    double res = fbinary_search(-10.0, 10.0, [](double x) { return x >= 1.2345; });
    assert(fabs(res - 1.2345) < 1e-15);
    return 0;
}

```

5.1.2 Exponential Search

Finds the first value where a monotone Boolean predicate becomes true when an upper bound is not known in advance. Exponential search first grows the search range by powers of two until it brackets the transition point, then finishes with ordinary binary search.

This is useful for answer-search problems over unbounded or very large integer domains. The predicate must eventually become true, or the search may overflow or run forever unless an explicit limit is added.

- `exponential_search_first_true(lo, pred)` returns the smallest integer `x >= lo` such that `pred(x)` is true.

Time Complexity: $O(\log n)$ calls to `pred()`, where n is the distance from `lo` to the first true value.

Space Complexity: $O(1)$ auxiliary.

```
template<class Int, class Pred>
Int exponential_search_first_true(Int lo, Pred pred) { // 000[1]11
    if (pred(lo)) {
        return lo;
    }
    Int step = 1;
    while (!pred(lo + step)) {
        step *= 2;
    }
    Int l = lo + step / 2, r = lo + step;
    while (l < r) {
        Int mid = l + (r - l) / 2;
        if (pred(mid)) {
            r = mid;
        } else {
            l = mid + 1;
        }
    }
    return l;
}
```

Example Usage

```
#include <cassert>

bool at_least_1000(int x) {
    return x >= 1000;
}

bool square_large_enough(long long x) {
    return x * x >= 123456789LL;
}

int main() {
    assert(exponential_search_first_true(0, at_least_1000) == 1000);
    assert(exponential_search_first_true(5, at_least_1000) == 1000);
    assert(exponential_search_first_true(0LL, square_large_enough) == 11112);
    return 0;
}
```

5.2 Unimodal and Continuous Search

5.2.1 Ternary Search

Given a unimodal function $f(x)$ taking a single `double` argument, find its global maximum or minimum point to a specified absolute error.

- `ternary_search_min()` takes the domain `[lo, hi]` of a continuous function $f(x)$ and returns a number `x` such that f is strictly decreasing on the interval `[lo, x]` and strictly increasing on the interval `[x, hi]`. For the function to be correct and deterministic, such an `x` must exist and be unique.
- `ternary_search_max()` takes the domain `[lo, hi]` of a continuous function $f(x)$ and returns a number `x` such that f is strictly increasing on the interval `[lo, x]` and strictly decreasing on the interval `[x, hi]`. For the function to be correct and deterministic, such an `x` must exist and be unique.

Time Complexity: $O(\log n)$ calls will be made to $f()$, where n is the distance between `lo` and `hi` divided by the specified absolute error (epsilon).

Space Complexity: $O(1)$ auxiliary for both operations.

```
template<class Fn>
double ternary_search_min(double lo, double hi, Fn f, const double EPS = 1e-12) {
    double lthird, hthird;
    while (hi - lo > EPS) {
        lthird = lo + (hi - lo) / 3;
        hthird = hi - (hi - lo) / 3;
        if (f(lthird) < f(hthird)) {
            hi = hthird;
        } else {
            lo = lthird;
        }
    }
    return lo;
}

template<class Fn>
double ternary_search_max(double lo, double hi, Fn f, const double EPS = 1e-12) {
    double lthird, hthird;
    while (hi - lo > EPS) {
        lthird = lo + (hi - lo) / 3;
        hthird = hi - (hi - lo) / 3;
        if (f(lthird) < f(hthird)) {
            lo = lthird;
        } else {
            hi = hthird;
        }
    }
    return hi;
}
```

Example Usage

```
#include <cassert>
#include <cmath>

bool equal(double a, double b) {
    return fabs(a - b) < 1e-7;
}

// Parabola opening up with vertex at (-2, -24).
double f1(double x) {
    return 3 * x * x + 12 * x - 12;
}
```

```

// Parabola opening down with vertex at (2/19, 8366/95) ~ (0.105, 88.063).
double f2(double x) {
    return -5.7 * x * x + 1.2 * x + 88;
}

// Absolute value function shifted to the right by 30 units.
double f3(double x) {
    return fabs(x - 30);
}

int main() {
    assert(equal(ternary_search_min(-1000, 1000, f1), -2));
    assert(equal(ternary_search_max(-1000, 1000, f2), 2.0 / 19));
    assert(equal(ternary_search_min(-1000, 1000, f3), 30));
    return 0;
}

```

5.2.2 Hill Climbing

Given a continuous function $f(x, y)$ to `double` and a (possibly arbitrary) starting guess (x_0, y_0) , return a potential global minimum found through hill-climbing. Optionally, two `double` pointers `critical_x` and `critical_y` may be passed to store the input points to f at which the returned minimum value is attained.

Hill-climbing is a heuristic which starts at the guess, then considers taking a single step in each of a fixed number of directions. The direction with the best (in this case, minimum) value is chosen, and further steps are taken in it until the answer no longer improves. When this happens, the step size is reduced and the same process repeats until a desired absolute error is reached. The technique's success heavily depends on the behavior of f and the initial guess. Therefore, the result is not guaranteed to be the global minimum.

Time Complexity: $O(d \log n)$ call will be made to f , where d is the number of directions considered at each position and n is the search space that is approximately proportional to the maximum possible step size divided by the minimum possible step size.

Space Complexity: $O(1)$ auxiliary.

```

#include <cmath>
#include <cstdint>

template<class Fn>
double find_min(
    Fn f, double x0, double y0, double *critical_x = nullptr, double *critical_y = nullptr,
    const double STEP_MIN = 1e-9, const double STEP_MAX = 1e6, const int NUM_DIRECTIONS = 6
) {
    static const double PI = acos(-1.0);
    double x = x0, y = y0, res = f(x0, y0);
    for (double step = STEP_MAX; step > STEP_MIN;) {
        double best = res, best_x = x, best_y = y;
        bool found = false;
        for (int i = 0; i < NUM_DIRECTIONS; i++) {
            double a = 2.0 * PI * i / NUM_DIRECTIONS;
            double x2 = x + step * cos(a), y2 = y + step * sin(a);
            double value = f(x2, y2);
            if (best > value) {
                best_x = x2;
                best_y = y2;
                best = value;
                found = true;
            }
        }
        if (!found) {
            step /= 2.0;
        } else {

```



```

        x = best_x;
        y = best_y;
        res = best;
    }
}
if (critical_x != nullptr && critical_y != nullptr) {
    *critical_x = x;
    *critical_y = y;
}
return res;
}

```

Example Usage

```

#include <cassert>
#include <cmath>

bool eq(double a, double b) {
    return fabs(a - b) < 1e-8;
}

// Paraboloid with global minimum at f(2, 3) = 0.
double f(double x, double y) {
    return (x - 2) * (x - 2) + (y - 3) * (y - 3);
}

int main() {
    double x, y;
    assert(eq(find_min(f, 0, 0, &x, &y), 0));
    assert(eq(x, 2) && eq(y, 3));
    return 0;
}

```

5.2.3 Simulated Annealing

Given a function `f(x, y)` to minimize over two real variables, return a good approximate minimum found by simulated annealing. This is a randomized heuristic for difficult search spaces where gradients, convexity, and unimodality are not available.

At each temperature, the algorithm proposes a random nearby point. Better moves are always accepted, while worse moves are accepted with probability `exp((current - candidate) / temperature)`. This lets the search sometimes escape local minima early on, then become greedier as the temperature decreases.

Simulated annealing is not a proof of optimality. Its quality depends heavily on the objective function, the starting point, the initial temperature, the cooling rate, and the number of independent restarts.

- `anneal_min(f, x0, y0, &best_x, &best_y)` returns a small value found by the randomized search starting from `(x0, y0)`.
- `TEMPERATURE_START` controls the initial move scale and willingness to accept worse states.
- `TEMPERATURE_END` controls when the search stops.
- `COOLING_RATE` controls how quickly the temperature decreases.

Time Complexity: $O(r \log n)$ calls to `f()`, where r is the number of restarts and n is roughly `TEMPERATURE_START / TEMPERATURE_END`.

Space Complexity: $O(1)$ auxiliary.

```

#include <cmath>
#include <random>

const double TEMPERATURE_START = 10.0;

```

```

const double TEMPERATURE_END = 1e-7;
const double COOLING_RATE = 0.997;
const int NUM_RESTARTS = 8;

double random_unit() {
    static std::mt19937 rng(1234567);
    return std::uniform_real_distribution<double>(0.0, 1.0)(rng);
}

template<class Fn>
double anneal_min(Fn f, double x0, double y0, double *best_x = nullptr, double *best_y = nullptr) {
    double sol_x = x0, sol_y = y0, best = f(x0, y0);
    for (int restart = 0; restart < NUM_RESTARTS; restart++) {
        double x = x0, y = y0, cur = f(x, y);
        double temperature = TEMPERATURE_START;
        while (temperature > TEMPERATURE_END) {
            double nx = x + (2.0 * random_unit() - 1.0) * temperature;
            double ny = y + (2.0 * random_unit() - 1.0) * temperature;
            double next = f(nx, ny);
            double probability = exp((cur - next) / temperature);
            if (next < cur || random_unit() < probability) {
                x = nx;
                y = ny;
                cur = next;
                if (cur < best) {
                    best = cur;
                    sol_x = x;
                    sol_y = y;
                }
            }
            temperature *= COOLING_RATE;
        }
    }

    if (best_x != nullptr) {
        *best_x = sol_x;
    }
    if (best_y != nullptr) {
        *best_y = sol_y;
    }
    return best;
}

```

Example Usage

```

#include <cassert>

bool close(double a, double b) {
    return fabs(a - b) < 1e-2;
}

// Smooth bowl with global minimum at f(2, -3) = 0.
double f(double x, double y) {
    return (x - 2) * (x - 2) + (y + 3) * (y + 3);
}

int main() {
    double x, y;
    double value = anneal_min(f, 0, 0, &x, &y);
    assert(value < 1e-4);
    assert(close(x, 2));
    assert(close(y, -3));
    return 0;
}

```

5.3 Dynamic Programming Optimization

5.3.1 Monotone Queue DP Optimization

Maintains the minimum or maximum value in a sliding window using a monotone queue. This is useful for dynamic programming recurrences where each transition may only come from one of the last w states, such as `dp[i] = a[i] + min(dp[j])` for $j \in [i - w, i - 1]$.

The queue stores candidate `(index, value)` pairs in monotone order. Expired indices are removed from the front, and dominated values are removed from the back before inserting a new candidate.

- `MonotoneQueue<Compare>()` constructs an empty queue. Use `std::less<T>` for minimum queries and `std::greater<T>` for maximum queries.
- `push(index, value)` inserts candidate `value` at position `index`, removing dominated candidates.
- `expire(first_valid)` removes candidates with index less than `first_valid`.
- `top()` returns the best active `(index, value)` pair.
- `empty()` returns whether the queue has no active candidates.

Time Complexity:

- $O(n)$ for any sequence of n calls to `push(index, value)` and `expire(first_valid)`.
- $O(1)$ amortized per call to `push(index, value)` and `expire(first_valid)`.
- $O(1)$ worst-case per call to `top()` and `empty()`.

Space Complexity: $O(w)$ for a sliding window of width w .

```
#include <deque>
#include <functional>
#include <utility>

template<class T, class Compare>
class MonotoneQueue {
    using entry = std::pair<int, T>;

    std::deque<entry> q;
    Compare better;

public:
    bool empty() const { return q.empty(); }

    void push(int index, const T &value) {
        while (!q.empty() && !better(q.back().second, value)) {
            q.pop_back();
        }
        q.emplace_back(index, value);
    }

    void expire(int first_valid) {
        while (!q.empty() && q.front().first < first_valid) {
            q.pop_front();
        }
    }

    entry top() const { return q.front(); }
};
```

Example Usage

```
#include <cassert>
#include <vector>
using namespace std;
```

```

int main() {
    vector<int> a{4, 2, 7, 1, 3, 6};

    MonotoneQueue<int, less<int>> minq;
    vector<int> window_min;
    for (int i = 0; i < static_cast<int>(a.size()); i++) {
        minq.push(i, a[i]);
        minq.expire(i - 2);
        if (i >= 2) {
            window_min.push_back(minq.top().second);
        }
    }
    assert(window_min[0] == 2);
    assert(window_min[1] == 1);
    assert(window_min[2] == 1);
    assert(window_min[3] == 1);

    // dp[i] = a[i] + min(dp[j]) over max(0, i - 2) <= j < i.
    vector<int> dp(a.size());
    MonotoneQueue<int, less<int>> best;
    dp[0] = a[0];
    best.push(0, dp[0]);
    for (int i = 1; i < static_cast<int>(a.size()); i++) {
        best.expire(i - 2);
        dp[i] = a[i] + best.top().second;
        best.push(i, dp[i]);
    }
    assert(dp[5] == 13);
    return 0;
}

```

5.3.2 Divide-and-Conquer DP Optimization

Computes one layer of a dynamic programming recurrence whose optimal transition indices are monotone. This optimization applies to recurrences of the form $dp_cur[i] = \min(dp_prev[k] + cost(k, i))$, where the minimizing index for i never decreases as i increases.

The function below computes $dp_cur[l..r]$ by evaluating the midpoint first, then recursing on the left half with a restricted upper bound on the optimal k , and on the right half with a restricted lower bound. The caller is responsible for checking that the recurrence satisfies the required monotonicity condition.

- `compute_dc_layer(dp_prev, dp_cur, l, r, opt_l, opt_r, cost)` fills $dp_cur[l..r]$ using candidate transition indices in $[opt_l, opt_r]$.
- `cost(k, i)` must return the transition cost from previous state k to current state i .

Time Complexity: $O(n \log n)$ calls to `cost(k, i)` per layer when each state has $O(n)$ possible transitions.

Space Complexity: $O(\log n)$ auxiliary stack space.

```

#include <algorithm>
#include <vector>

const long long INF = (1LL << 62);

template<class Cost>
void compute_dc_layer(
    const std::vector<long long> &dp_prev, std::vector<long long> &dp_cur, int l, int r, int opt_l,
    int opt_r, Cost cost
) {
    if (l > r) {
        return;
    }
    int mid = l + (r - l) / 2;
    long long best = INF;

```

```

int best_k = opt_l;
int upper = std::min(mid, opt_r);
for (int k = opt_l; k <= upper; k++) {
    long long candidate = dp_prev[k] + cost(k, mid);
    if (candidate < best) {
        best = candidate;
        best_k = k;
    }
}
dp_cur[mid] = best;
compute_dc_layer(dp_prev, dp_cur, l, mid - 1, opt_l, best_k, cost);
compute_dc_layer(dp_prev, dp_cur, mid + 1, r, best_k, opt_r, cost);
}

```

Example Usage

```

#include <cassert>
using namespace std;

struct SquareSegmentCost {
    vector<long long> prefix;

    explicit SquareSegmentCost(const vector<int> &a) : prefix(a.size() + 1) {
        for (int i = 0; i < static_cast<int>(a.size()); i++) {
            prefix[i + 1] = prefix[i] + a[i];
        }
    }

    long long operator()(int k, int i) const {
        long long sum = prefix[i] - prefix[k];
        return sum * sum;
    }
};

int main() {
    vector<int> a{1, 2, 3, 4};
    int n = a.size();
    SquareSegmentCost cost(a);

    vector<long long> dp_prev(n + 1, INF), dp_cur(n + 1, INF);
    dp_prev[0] = 0;
    compute_dc_layer(dp_prev, dp_cur, 1, n, 0, n - 1, cost);
    assert(dp_cur[4] == 100); // One group: (1 + 2 + 3 + 4)^2.

    vector<long long> dp_two(n + 1, INF);
    compute_dc_layer(dp_cur, dp_two, 1, n, 0, n - 1, cost);
    assert(dp_two[4] == 52); // Split as [1, 2] and [3, 4].
    return 0;
}

```

5.3.3 Knuth DP Optimization

Computes interval dynamic programming recurrences whose optimal split points are monotone. Knuth optimization applies to recurrences of the form $dp[l][r] = \min(dp[l][k] + dp[k][r] + cost(l, r))$, where the optimal split $opt[l][r]$ satisfies $opt[l][r - 1] \leq opt[l][r] \leq opt[l + 1][r]$.

Common applications include optimal binary search trees and some range merging problems. The caller is responsible for verifying the quadrangle inequality and monotonicity assumptions for the chosen $cost(l, r)$.

- `knuth_interval_dp(n, cost, &opt)` computes minimum costs for all half-open intervals $[l, r)$ over n items.
- `cost(l, r)` must return the interval cost added after choosing the best split.
- If `opt` is not `nullptr`, it is filled with the chosen split points.

Time Complexity: $O(n^2)$ calls to `cost(l, r)` and $O(n^2)$ candidate split checks.

Space Complexity: $O(n^2)$ for the `dp` and `opt` tables.

```
#include <algorithm>
#include <cstdint>
#include <vector>

const long long INF = (1LL << 62);

template<class Cost>
std::vector<std::vector<long long>> knuth_interval_dp(
    int n, Cost cost, std::vector<std::vector<int>> *opt_out = nullptr
) {
    std::vector<std::vector<long long>> dp(n + 1, std::vector<long long>(n + 1, 0));
    std::vector<std::vector<int>> opt(n + 1, std::vector<int>(n + 1, 0));
    for (int i = 0; i <= n; i++) {
        opt[i][i] = i;
        if (i < n) {
            opt[i][i + 1] = i + 1;
        }
    }
    for (int len = 2; len <= n; len++) {
        for (int l = 0; l + len <= n; l++) {
            int r = l + len;
            dp[l][r] = INF;
            int start = std::max(opt[l][r - 1], l + 1);
            int finish = std::min(opt[l + 1][r], r - 1);
            for (int k = start; k <= finish; k++) {
                long long candidate = dp[l][k] + dp[k][r] + cost(l, r);
                if (candidate < dp[l][r]) {
                    dp[l][r] = candidate;
                    opt[l][r] = k;
                }
            }
        }
    }
    if (opt_out != nullptr) {
        *opt_out = opt;
    }
    return dp;
}
```

Example Usage

```
#include <cassert>
using namespace std;

struct MergeCost {
    vector<long long> prefix;

    explicit MergeCost(const vector<int> &a) : prefix(a.size() + 1) {
        for (int i = 0; i < static_cast<int>(a.size()); i++) {
            prefix[i + 1] = prefix[i] + a[i];
        }
    }

    long long operator()(int l, int r) const { return prefix[r] - prefix[l]; }
};

int main() {
    vector<int> a{1, 2, 3, 4};
    vector<vector<long long>> dp = knuth_interval_dp(a.size(), MergeCost(a));
    assert(dp[0][4] == 19);
    return 0;
}
```

5.3.4 Convex Hull Trick (Semi-Dynamic)

Given a set of pairs (m, b) specifying lines of the form $y = mx + b$, process a set of x -coordinate queries each asking to find the minimum y -value when any of the given lines are evaluated at the specified x . This is useful for dynamic programming recurrences of the form `dp[i] = min(m[j] * x[i] + b[j])`.

The following implementation is a concise, semi-dynamic version of the convex hull optimization technique. It supports an interlaced sequence of `add_line()` and `query()` calls, as long as the preconditions of descending `m` and ascending `x` are satisfied. As a result, it may be necessary to sort the lines and queries before calling the functions. In that case, the overall time complexity will be dominated by the sorting step.

- `SemiDynamicCHT()` constructs an empty hull.
- `add_line(m, b)` inserts the line $y = mx + b$. The slope `m` must be less than or equal to the slope of every line added so far.
- `query(x)` returns the minimum y -value among all inserted lines at coordinate `x`. Query coordinates must be nondecreasing across calls.

Overflow warning: `add_line()` compares intersections by cross-multiplying slope and intercept differences, a product on the order of the squared coefficient magnitude. For large `m/b` (roughly beyond 10^9 with 64-bit `long long`), cast that comparison to `__int128`. `query()` only forms the single product `m * x`, which simply needs to fit in `long long`.

Time Complexity: $O(n)$ for any interlaced sequence of `add_line()` and `query()` calls, where n is the number of lines added. This is because the overall number of steps taken by `add_line()` and `query()` are respectively bounded by the number of lines. Thus a single call to either `add_line()` or `query()` will have an amortized $O(1)$ running time.

Space Complexity:

- $O(n)$ for storage of the lines.
- $O(1)$ auxiliary for `add_line()` and `query()`.

```
#include <vector>

struct SemiDynamicCHT {
    std::vector<long long> M, B;
    int ptr = 0;

    void add_line(long long m, long long b) {
        int len = M.size();
        if (len > 0 && M.back() == m) {
            if (B.back() <= b) {
                return;
            }
            M.pop_back();
            B.pop_back();
            len--;
            if (ptr > len) {
                ptr = len;
            }
        }
        // Overflow risk: this cross-multiplication is ~O(coeff^2); cast to __int128 for large m/b.
        while (len > 1 && (B[len - 2] - B[len - 1]) * (m - M[len - 1]) >=
                    (B[len - 1] - b) * (M[len - 1] - M[len - 2])) {
            len--;
        }
        M.resize(len);
        B.resize(len);
        M.push_back(m);
        B.push_back(b);
    }

    long long query(long long x) {
        if (ptr >= static_cast<int>(M.size())) {
```

```

    ptr = static_cast<int>(M.size()) - 1;
}
while (ptr + 1 < static_cast<int>(M.size()) &&
       M[ptr + 1] * x + B[ptr + 1] <= M[ptr] * x + B[ptr]) {
    ptr++;
}
return M[ptr] * x + B[ptr];
}
};

```

Example Usage

```

#include <cassert>

int main() {
    SemiDynamicCHT cht;
    cht.add_line(3, 0);
    cht.add_line(2, 1);
    cht.add_line(1, 2);
    cht.add_line(0, 6);
    assert(cht.query(0) == 0);
    assert(cht.query(1) == 3);
    assert(cht.query(2) == 4);
    assert(cht.query(3) == 5);
    return 0;
}

```

5.3.5 Convex Hull Trick (Fully-Dynamic)

Given a set of pairs (m, b) specifying lines of the form $y = mx + b$, process a set of x -coordinate queries each asking to find the minimum y -value when any of the given lines are evaluated at the specified x . To instead have the queries optimize for maximum y -value, call the constructor with `query_max = true`. This is useful for dynamic programming recurrences of the form `dp[i] = min(m[j] * x[i] + b[j])` when line slopes and query coordinates are not monotone.

The following implementation is a fully dynamic variant of the convex hull optimization technique, using a self-balancing binary search tree (`std::set`) to support the ability to call `add_line()` and `query()` in any desired order.

- `HullOptimizer(query_max)` constructs an empty hull. By default, `query(x)` minimizes; if `query_max` is true, `query(x)` maximizes.
- `add_line(m, b)` inserts line $y = mx + b$ (can be called in any order).
- `query(x)` returns the best y -value among all inserted lines at coordinate x . At least one line must have been inserted.

Time Complexity: $O(\log n)$ amortized per call to `add_line()` and $O(\log n)$ per call to `query()`, where n is the number of lines added.

Space Complexity:

- $O(n)$ for storage of the lines.
- $O(1)$ auxiliary for `add_line()` and `query()`.

```

#include <cassert>
#include <climits>
#include <set>

class HullOptimizer {
    struct Line {
        long long m, b;
    };

```



```

mutable long long xhi;
bool is_query;

Line(long long m, long long b, long long xhi = 0, bool is_query = false)
: m(m), b(b), xhi(xhi), is_query(is_query) {}

bool operator< (const Line &l) const { return l.is_query ? xhi < l.xhi : m < l.m; }
};

std::multiset<Line> hull;
bool query_max;

using hulliter = std::multiset<Line>::iterator;

static long long div_floor(long long a, long long b) { return a / b - ((a ^ b) < 0 && a % b); }

bool update_border(hulliter x, hulliter y) {
    if (y == hull.end()) {
        x->xhi = LLONG_MAX;
        return false;
    }
    if (x->m == y->m) {
        x->xhi = (x->b > y->b) ? LLONG_MAX : LLONG_MIN;
    } else {
        x->xhi = div_floor(y->b - x->b, x->m - y->m);
    }
    return x->xhi >= y->xhi;
}

public:
explicit HullOptimizer(bool query_max = false) : query_max(query_max) {}

void add_line(long long m, long long b) {
    if (!query_max) {
        m = -m;
        b = -b;
    }
    auto z = hull.insert(Line(m, b));
    auto y = z++;
    auto x = y;
    while (update_border(y, z)) {
        z = hull.erase(z);
    }
    if (x != hull.begin() && update_border(--x, y)) {
        update_border(x, y = hull.erase(y));
    }
    while ((y = x) != hull.begin() && (--x)->xhi >= y->xhi) {
        update_border(x, hull.erase(y));
    }
}

long long query(long long x) const {
    assert(!hull.empty());
    Line q(0, 0, x, true);
    auto it = hull.lower_bound(q);
    long long res = it->m * x + it->b;
    return query_max ? res : -res;
}
};

```

Example Usage

```

#include <cassert>

int main() {
    HullOptimizer h;

```

```

h.add_line(3, 0);
h.add_line(0, 6);
h.add_line(1, 2);
h.add_line(2, 1);
assert(h.query(0) == 0);
assert(h.query(2) == 4);
assert(h.query(1) == 3);
assert(h.query(3) == 5);
return 0;
}

```

5.3.6 Li Chao Tree

Maintains a dynamic set of lines and answers minimum value queries at points in a fixed integer domain. A Li Chao tree is useful for dynamic programming recurrences of the form `dp[i] = min(m[j] * x[i] + b[j])`, especially when line slopes and query points arrive in arbitrary order.

The implementation below stores the lower envelope of lines over the inclusive domain `[x_MIN, x_MAX]`. It supports adding arbitrary lines and querying arbitrary integer x-coordinates in $O(\log C)$, where $C = x_MAX - x_MIN + 1$.

- `LiChaoTree(lo, hi)` constructs an empty tree over integer domain `[lo, hi]`.
- `add_line(m, b)` inserts line $y = mx + b$ (can be called in any order).
- `query(x)` returns the minimum y -value among all inserted lines at coordinate `x`.

Time Complexity: $O(\log C)$ per call to `add_line(m, b)` and `query(x)`, where C is the domain size.

Space Complexity:

- $O(n \log C)$ worst-case node storage for n inserted lines, usually much less.
- $O(\log C)$ auxiliary stack space per operation.

```

#include <algorithm>
#include <cassert>
#include <climits>

const long long INF = LLONG_MAX / 4;

struct Line {
    long long m, b;

    Line(long long m = 0, long long b = INF) : m(m), b(b) {}
    long long eval(long long x) const { return m * x + b; }
};

class LiChaoTree {
    struct node {
        Line f;
        node *left, *right;

        explicit node(const Line &f) : f(f), left(nullptr), right(nullptr) {}
    };

    node *root;
    long long lo, hi;

    static void clean_up(node *n) {
        if (n == nullptr) {
            return;
        }
        clean_up(n->left);
        clean_up(n->right);
        delete n;
    }
};

```

```

static void add_line(node *&n, long long l, long long r, Line f) {
    if (n == nullptr) {
        n = new node(f);
        return;
    }
    long long mid = l + (r - l) / 2;
    bool left_better = f.eval(l) < n->f.eval(l);
    bool mid_better = f.eval(mid) < n->f.eval(mid);
    if (mid_better) {
        std::swap(f, n->f);
    }
    if (l == r) {
        return;
    }
    if (left_better != mid_better) {
        add_line(n->left, l, mid, f);
    } else {
        add_line(n->right, mid + 1, r, f);
    }
}

static long long query(node *n, long long l, long long r, long long x) {
    if (n == nullptr) {
        return INF;
    }
    long long res = n->f.eval(x);
    if (l == r) {
        return res;
    }
    long long mid = l + (r - l) / 2;
    if (x <= mid) {
        return std::min(res, query(n->left, l, mid, x));
    }
    return std::min(res, query(n->right, mid + 1, r, x));
}

public:
    LiChaoTree(long long lo, long long hi) : root(nullptr), lo(lo), hi(hi) {}

    ~LiChaoTree() { clean_up(root); }

    LiChaoTree(const LiChaoTree &) = delete;
    LiChaoTree &operator=(const LiChaoTree &) = delete;

    void add_line(long long m, long long b) { add_line(root, lo, hi, Line(m, b)); }

    long long query(long long x) const {
        assert(lo <= x && x <= hi);
        return query(root, lo, hi, x);
    }
};

```

Example Usage

```

#include <cassert>

int main() {
    LiChaoTree cht(-10, 10);
    cht.add_line(3, 0);
    cht.add_line(1, 2);
    cht.add_line(-1, 10);
    assert(cht.query(0) == 0);
    assert(cht.query(2) == 4);
    assert(cht.query(10) == 0);
    return 0;
}

```

5.4 Numerical Methods

5.4.1 Root Finding (Bracketing)

Finds an x in an interval $[a, b]$ for a continuous function f such that $f(x) = 0$. By the intermediate value theorem, a root must exist in $[a, b]$ if the signs of $f(a)$ and $f(b)$ differ. The answer is found with an absolute error of roughly $1/2^n$, where n is the number of iterations. Although it is possible to control the error by looping while $b - a$ is greater than an arbitrary epsilon, it is simpler to let the loop run for a desired number of iterations until floating point arithmetic break down. 100 iterations is usually sufficient, since the search space will be reduced to 2^{-100} (roughly 10^{-30}) times its original size.

- `bisection_root(f, a, b)` returns a root in an interval `[a, b]` for a continuous function f where $\text{sgn}(f(a)) \neq \text{sgn}(f(b))$, using the bisection method.
- `falsi_illinois_root(f, a, b)` returns a root in an interval `[a, b]` for a continuous function f where $\text{sgn}(f(a)) \neq \text{sgn}(f(b))$, using the Illinois algorithm variant of the false position (a.k.a. regula falsi) method.

Time Complexity: $O(n)$ calls will be made to `f()` in `bisection_root()` and `falsi_illinois_root()`, where n is the number of iterations performed.

Space Complexity: $O(1)$ auxiliary space for both operations.

```
#include <stdexcept>

template<class Fn>
double bisection_root(Fn f, double a, double b, const int ITERATIONS = 100) {
    if (a > b || f(a) * f(b) > 0) {
        throw std::runtime_error("Must give [a, b] where sgn(f(a)) != sgn(f(b)).");
    }
    double m;
    for (int i = 0; i < ITERATIONS; i++) {
        m = a + (b - a) / 2;
        if (f(a) * f(m) >= 0) {
            a = m;
        } else {
            b = m;
        }
    }
    return m;
}

template<class Fn>
double falsi_illinois_root(Fn f, double a, double b, const int ITERATIONS = 100) {
    if (a > b || f(a) * f(b) > 0) {
        throw std::runtime_error("Must give [a, b] where sgn(f(a)) != sgn(f(b)).");
    }
    double m, fm, fa = f(a), fb = f(b);
    int side = 0;
    for (int i = 0; i < ITERATIONS; i++) {
        m = (fa * b - fb * a) / (fa - fb);
        fm = f(m);
        if (fb * fm > 0) {
            b = m;
            fb = fm;
            if (side < 0) {
                fa /= 2;
            }
            side = -1;
        } else if (fa * fm > 0) {
            a = m;
            fa = fm;
            if (side > 0) {
                fb /= 2;
            }
            side = 1;
        }
    }
}
```

```

    } else {
        break;
    }
}
return m;
}

```

Example Usage

```

#include <cassert>
#include <cmath>

double f(double x) {
    return x * x - 4 * sin(x);
}

int main() {
    assert(fabs(f(bisection_root(f, 1, 3))) < 1e-10);
    assert(fabs(f(falsi_illinois_root(f, 1, 3))) < 1e-10);
    return 0;
}

```

5.4.2 Root Finding (Iteration)

Finds an x for a continuous function f such that $f(x) = 0$ using iterative approximation by an initial guess that is close to the answer. Newton's method requires an explicit definition of the function's derivative while the secant method starts with two initial guesses and approximates the derivative using the secant slope from the previous iteration. For n iterations and a good initial guess, the methods below compute approximately 2^n digits of precision, with the secant method converging approximately 1.6 times slower than Newton's.

- `newton_root(f, fprime, x0)` returns a root x for a function `f` with derivative `fprime` using an initial guess `x0` which should be relatively close to x .
- `secant_root(f, x0, x1)` returns a root x for a function `f` using two initial guesses `x0` and `x1` which should be relatively close to x .

Time Complexity: $O(n)$ calls will be made to `f()` in `newton_root()` and `secant_root()`, where n is the number of iterations performed.

Space Complexity: $O(1)$ auxiliary space for both operations.

```

#include <cmath>
#include <stdexcept>

template<class Fn>
double newton_root(
    Fn f, Fn fprime, double x0, const double EPS = 1e-15, const int ITERATIONS = 100
) {
    double x = x0, error = EPS + 1;
    for (int i = 0; error > EPS && i < ITERATIONS; i++) {
        double xnew = x - f(x) / fprime(x);
        error = fabs(xnew - x);
        x = xnew;
    }
    if (error > EPS) {
        throw std::runtime_error("Newton's method failed to converge.");
    }
    return x;
}

template<class Fn>
double secant_root(

```

```

    Fn f, double x0, double x1, const double EPS = 1e-15, const int ITERATIONS = 100
) {
    double xold = x0, fxold = f(x0), x = x1, error = EPS + 1;
    for (int i = 0; error > EPS && i < ITERATIONS; i++) {
        double fx = f(x);
        double xnew = x - fx * ((x - xold) / (fx - fxold));
        xold = x;
        fxold = fx;
        error = fabs(xnew - x);
        x = xnew;
    }
    if (error > EPS) {
        throw std::runtime_error("Secant method failed to converge.");
    }
    return x;
}

```

Example Usage

```

#include <cassert>

double f(double x) {
    return x * x - 4 * sin(x);
}

double fprime(double x) {
    return 2 * x - 4 * cos(x);
}

int main() {
    assert(fabs(f(newton_root(f, fprime, 3))) < 1e-10);
    assert(fabs(f(secant_root(f, 3, 2))) < 1e-10);
    return 0;
}

```

5.4.3 Polynomial Root Finding (Differentiation)

Finds every root x for a polynomial p such that $p(x) = 0$ by differentiation. Each adjacent pair of local extrema is searched using the bisection method, where local extrema are recursively found by finding the root of the derivative.

- `horner_eval(p, x)` evaluates the polynomial `p` of degree d (represented as a vector of size $d + 1$ where `p[i]` stores the coefficient for the x^i term) at `x`, using Horner's method.
- `find_one_root(p, a, b, EPS)` returns a root in the interval `[a, b]` for a polynomial `p` where $\text{sgn}(f(a)) \neq \text{sgn}(f(b))$, using the bisection method. If this precondition is not satisfied, then `NaN` is returned. The root is found to a tolerance of `EPS` in absolute or relative error (whichever is reached first).
- `find_all_roots(p, a, b, EPS)` returns a vector of all roots in the interval `[a, b]` for a polynomial `p` using the bisection method. The roots are found to a tolerance of `EPS` in absolute or relative error (whichever is reached first).

Time Complexity:

- $O(n)$ per call to `horner_eval()`, where n is the degree of the polynomial.
- $O(n \log p)$ per call to `find_one_root()`, where n is the degree of the polynomial and $p = -\log_{10}(\text{EPS})$ is the number of digits of absolute or relative precision that is desired.
- $O(n^3 \log p)$ per call to `find_all_roots()`, where n is the degree of the polynomial and $p = -\log_{10}(\text{EPS})$ is the number of digits of absolute or relative precision that is desired.

Space Complexity:

- $O(1)$ auxiliary space for `horner_eval()` and `find_one_root()`.
- $O(n^2)$ auxiliary heap and $O(n)$ auxiliary stack space for `find_all_roots()`, where n is the degree of the polynomial.

```
#include <cmath>
#include <limits>
#include <utility>
#include <vector>

double horner_eval(const std::vector<double> &p, double x) {
    double res = p.back();
    for (int i = static_cast<int>(p.size()) - 2; i >= 0; i--) {
        res = res * x + p[i];
    }
    return res;
}

double find_one_root(const std::vector<double> &p, double a, double b, const double EPS = 1e-15) {
    double pa = horner_eval(p, a), pb = horner_eval(p, b);
    bool paneg = pa < 0, pbneg = pb < 0;
    if (paneg == pbneg) {
        return std::numeric_limits<double>::quiet_NaN();
    }
    while (b - a > EPS && a * (1 + EPS) < b && a < b * (1 + EPS)) {
        double m = a + (b - a) / 2;
        if ((horner_eval(p, m) < 0) == paneg) {
            a = m;
        } else {
            b = m;
        }
    }
    return a;
}

std::vector<double> find_all_roots(
    const std::vector<double> &p, double a = -1e20, double b = 1e20, const double EPS = 1e-15
) {
    std::vector<double> pprime;
    pprime.reserve(p.size() > 0 ? p.size() - 1 : 0);
    for (int i = 1; i < static_cast<int>(p.size()); i++) {
        pprime.push_back(p[i] * i);
    }
    if (pprime.empty()) {
        return {};
    }
    std::vector<double> res, r = find_all_roots(pprime, a, b, EPS);
    r.push_back(b);
    for (int i = 0; i < static_cast<int>(r.size()); i++) {
        double root = find_one_root(p, i == 0 ? a : r[i - 1], r[i], EPS);
        if (!std::isnan(root) && (res.empty() || root != res.back())) {
            res.push_back(root);
        }
    }
    return res;
}
```

Example Usage

```
#include <cassert>
using namespace std;

int main() {
    { // -1 + 2x - 6x^2 + 2x^3
        vector<double> p{-1.0, 2.0, -6.0, 2.0};
```

```

    vector<double> roots = find_all_roots(p);
    assert(roots.size() == 1 && fabs(horner_eval(p, roots[0])) < 1e-10);
}
{ // -20 + 4x + 3x^2
    vector<double> p{-20.0, 4.0, 3.0};
    vector<double> roots = find_all_roots(p);
    assert(roots.size() == 2);
    assert(fabs(horner_eval(p, roots[0])) < 1e-10);
    assert(fabs(horner_eval(p, roots[1])) < 1e-10);
}
return 0;
}

```

5.4.4 Polynomial Root Finding (Laguerre)

Finds every complex root x for a polynomial p with complex coefficients such that $p(x) = 0$ using Laguerre's method.

- `horner_eval(p, x)` evaluates the complex polynomial p of degree d (represented as a vector of size $d + 1$ where $p[i]$ stores the complex coefficient for the x^i term) at x , using Horner's method, returning a pair where the first value is a vector of sub-evaluations and the second value is the final result $p(x)$.
- `find_one_root(p, x0)` returns a complex root x for a polynomial p (represented as a vector of size $d + 1$ where $p[i]$ stores the complex coefficient for the x^i term) using an initial guess $x0$ which should be relatively close to x . The root is found to a tolerance of `EPS` in absolute or relative error (whichever is reached first).
- `find_all_roots(p)` returns a vector of all complex roots for a complex polynomial p . The roots are found to a tolerance of `EPS` in absolute or relative error (whichever is reached first).

Time Complexity:

- $O(n)$ per call to `horner_eval()`, where n is the degree of the polynomial.
- $O(n \log p)$ per call to `find_one_root()`, where n is the degree of the polynomial and $p = -\log_{10}(\text{EPS})$ is the number of digits of absolute or relative precision that is desired.
- $O(n^2 \log p)$ per call to `find_all_roots()`, where n is the degree of the polynomial and $p = -\log_{10}(\text{EPS})$ is the number of digits of absolute or relative precision that is desired.

Space Complexity:

- $O(n)$ auxiliary heap space and $O(1)$ auxiliary stack space for `horner_eval()` and `find_one_root()`, where n is the degree of the polynomial.
- $O(n)$ auxiliary heap and $O(1)$ auxiliary stack space per for `find_one_root()` and `find_all_roots()`, where n is the degree of the polynomial.

```

#include <complex>
#include <random>
#include <vector>

using cdouble = std::complex<double>;
using cpoly = std::vector<cdouble>;

std::pair<cdouble, cpoly> horner_eval(const cpoly &p, const cdouble &x) {
    int n = p.size();
    cpoly b(std::max(1, n - 1));
    for (int i = n - 1; i > 0; i--) {
        b[i - 1] = p[i] + (i < n - 1 ? b[i] * x : 0);
    }
    return {p[0] + b[0] * x, b};
}

cpoly derivative(const cpoly &p) {
    int n = p.size();
    cpoly res(std::max(1, n - 1));

```



```

    for (int i = 1; i < n; i++) {
        res[i - 1] = p[i] * cdouble(i);
    }
    return res;
}

int comp(const cdouble &a, const cdouble &b, const double EPS = 1e-15) {
    double diff = std::abs(a) - std::abs(b);
    return (diff < -EPS) ? -1 : (diff > EPS ? 1 : 0);
}

cdouble find_one_root(
    const cpoly &p, const cdouble &x0, const double EPS = 1e-15, const int ITERATIONS = 10000
) {
    cdouble x = x0;
    int n = p.size() - 1;
    cpoly p1 = derivative(p), p2 = derivative(p1);
    for (int i = 0; i < ITERATIONS; i++) {
        cdouble y0 = horner_eval(p, x).first;
        if (comp(y0, 0, EPS) == 0) {
            break;
        }
        cdouble g = horner_eval(p1, x).first / y0;
        cdouble h = g * g - horner_eval(p2, x).first / y0;
        cdouble r = std::sqrt(cdouble(n - 1) * (h * cdouble(n) - g * g));
        cdouble d1 = g + r, d2 = g - r;
        cdouble a = cdouble(n) / (comp(d1, d2, EPS) > 0 ? d1 : d2);
        x -= a;
        if (comp(a, 0, EPS) == 0) {
            break;
        }
    }
    return x;
}

std::vector<cdouble> find_all_roots(
    const cpoly &p, const double EPS = 1e-15, const int ITERATIONS = 10000
) {
    std::vector<cdouble> res;
    cpoly q = p;
    static std::mt19937 rng(std::random_device{}());
    std::uniform_real_distribution<double> unit(0.0, 1.0);
    while (q.size() > 2) {
        cdouble z(unit(rng), unit(rng));
        z = find_one_root(p, find_one_root(q, z, EPS, ITERATIONS), EPS, ITERATIONS);
        q = horner_eval(q, z).second;
        res.push_back(z);
    }
    res.push_back(-q[0] / q[1]);
    return res;
}

```

Example Usage

```

#include <algorithm>
#include <cstdio>
#include <iostream>
using namespace std;

void print_roots(vector<cdouble> x) {
    sort(x.begin(), x.end(), [](const cdouble &a, const cdouble &b) {
        return abs(a.real() - b.real()) > 0.5e-5 ? a.real() < b.real() : a.imag() < b.imag();
    });
    for (const auto &z : x) {
        double real = abs(z.real()) < 0.5e-5 ? 0 : z.real();
        double imag = abs(z.imag()) < 0.5e-5 ? 0 : z.imag();
    }
}

```

```

    printf("%.5lf, %.5lf)\n", real, imag);
}
}

int main() {
{ // 140 - 13x - 8x^2 + x^3 = (x + 4)(x - 5)(x - 7)
    printf("Roots of 140 - 13x - 8x^2 + x^3:\n");
    cpoly p;
    p.push_back(140);
    p.push_back(-13);
    p.push_back(-8);
    p.push_back(1);
    print_roots(find_all_roots(p));
}
{ // (-24+36i) + (-26+12i)x + (-30+40i)x^2 + (-26+12i)x^3 + (-6+4i)x^4
  // = ((2 + 3i)x + 6)(x + i)(2x + (6 + 4i))(xi + 1):
    printf("Roots of ((2 + 3i)x + 6)(x + i)(2x + (6 + 4i))(xi + 1):\n");
    cpoly p;
    p.emplace_back(-24, 36);
    p.emplace_back(-26, 12);
    p.emplace_back(-30, 40);
    p.emplace_back(-26, 12);
    p.emplace_back(-6, 4);
    print_roots(find_all_roots(p));
}
return 0;
}

```

Example Output

```

Roots of 140 - 13x - 8x^2 + x^3:
(-4.00000, 0.00000)
(5.00000, 0.00000)
(7.00000, 0.00000)
Roots of ((2 + 3i)x + 6)(x + i)(2x + (6 + 4i))(xi + 1):
(-3.00000, -2.00000)
(-0.92308, 1.38462)
(0.00000, -1.00000)
(0.00000, 1.00000)

```

5.4.5 Polynomial Root Finding (Ehrlich-Aberth)

Finds every complex root x for a polynomial p such that $p(x) = 0$ using the Ehrlich-Aberth method. This is a simultaneous iteration: every root estimate is updated using both the Newton correction and a repulsion term from the other estimates.

This routine is intended for well-scaled polynomials when all complex roots are wanted. If only real roots are needed, a real-root isolator such as derivative recursion with bisection is usually more reliable. Multiple or tightly clustered roots may converge slowly, and very large or very small root scales may require rescaling the input or using multiprecision arithmetic.

- `eval_with_derivative(p, x)` returns `p(x)` and `p'(x)` for a polynomial `p` represented as a vector where `p[i]` stores the coefficient for the x^i term.
- `find_all_roots(p, EPS, ITERATIONS)` returns a vector of all complex roots for a complex polynomial `p`. A `vector<LD>` overload is provided for polynomials with real coefficients. The roots are found to a tolerance of `EPS` in absolute or relative error (whichever is reached first), and zero roots are removed exactly before the simultaneous iteration starts.

Time Complexity:

- $O(n)$ per call to `eval_with_derivative(p, x)`, where n is the degree of the polynomial.
- $O(n^2p)$ per call to `find_all_roots(p, EPS, ITERATIONS)`, where n is the degree of the polynomial and p is the number of iterations required to reach the desired precision.

Space Complexity:

- $O(1)$ auxiliary space for `eval_with_derivative(p, x)`.
- $O(n)$ auxiliary heap space for `find_all_roots(p, EPS, ITERATIONS)`, where n is the degree of the polynomial.

```

#include <algorithm>
#include <cmath>
#include <complex>
#include <vector>

using LD = long double;
using cdouble = std::complex<LD>;
using cpoly = std::vector<cdouble>;

const LD PI = acosl(-1.0L);
const LD ZERO_EPS = 1e-30L; // Treat coefficients and denominators this small as zero.
const LD ROOT_EPS = 1e-18L; // Stop once root updates are this small relative to the root.
const LD CHECK_EPS = 1e-12L; // Residual tolerance used by the example assertions.

bool is_zero(const cdouble &z, const LD EPS = ZERO_EPS) {
    return std::abs(z) <= EPS;
}

bool is_finite(const cdouble &z) {
    return std::isfinite(static_cast<double>(z.real())) &&
        std::isfinite(static_cast<double>(z.imag()));
}

std::pair<cdouble, cdouble> eval_with_derivative(const cpoly &p, const cdouble &x) {
    cdouble value = p.back(), derivative = 0;
    for (int i = static_cast<int>(p.size()) - 2; i >= 0; i--) {
        derivative = derivative * x + value;
        value = value * x + p[i];
    }
    return {value, derivative};
}

LD root_bound(const cpoly &p) {
    LD res = 0;
    int n = p.size() - 1;
    for (int i = 0; i + 1 < static_cast<int>(p.size()); i++) {
        LD ratio = std::abs(p[i] / p.back());
        if (ratio > 0) {
            res = std::max(res, powl(ratio, 1.0L / (n - i)));
        }
    }
    return 2 * std::max((LD)1, res);
}

bool root_less(const cdouble &a, const cdouble &b) {
    if (a.real() != b.real()) {
        return a.real() < b.real();
    }
    return a.imag() < b.imag();
}

cpoly find_all_roots(cpoly p, const LD EPS = ROOT_EPS, const int ITERATIONS = 2000) {
    while (!p.empty() && is_zero(p.back())) {
        p.pop_back();
    }
    cpoly roots;
    while (p.size() > 1 && is_zero(p[0])) {
        roots.push_back(0);
        p.erase(p.begin());
    }
    if (p.size() <= 1) {
        return roots;
    }
}

```

```

LD scale = 0;
for (int i = 0; i < static_cast<int>(p.size()); i++) {
    scale = std::max(scale, std::abs(p[i]));
}
for (int i = 0; i < static_cast<int>(p.size()); i++) {
    p[i] /= scale;
}
int n = p.size() - 1;
if (n == 1) {
    roots.push_back(-p[0] / p[1]);
    return roots;
}
cpoly z(n);
LD radius = root_bound(p), offset = PI / (2 * n);
LD max_radius = 2 * radius;
for (int i = 0; i < n; i++) {
    LD angle = offset + 2 * PI * i / n;
    z[i] = radius * cdouble(cosl(angle), sinl(angle));
}
for (int it = 0; it < ITERATIONS; it++) {
    bool done = true;
    cpoly next = z;
    for (int i = 0; i < n; i++) {
        auto [fx, dfx] = eval_with_derivative(p, z[i]);
        if (std::abs(fx) <= EPS) {
            continue;
        }
        cdouble repulsion = 0;
        for (int j = 0; j < n; j++) {
            if (i != j) {
                cdouble diff = z[i] - z[j];
                if (std::abs(diff) <= EPS * EPS) {
                    LD angle = 2 * PI * (i + 1) / (n + 1);
                    diff = EPS * (1 + std::abs(z[i])) * cdouble(cosl(angle), sinl(angle));
                }
                repulsion += cdouble(1) / diff;
            }
        }
        cdouble denom = dfx - fx * repulsion;
        if (is_zero(denom)) {
            continue;
        }
        cdouble step = fx / denom;
        if (!is_finite(step) && !is_zero(dfx)) {
            step = fx / dfx;
        }
        if (!is_finite(step)) {
            continue;
        }
        LD limit = 2 * max_radius;
        if (std::abs(step) > limit) {
            step *= limit / std::abs(step);
        }
        next[i] = z[i] - step;
        if (!is_finite(next[i])) {
            next[i] = z[i];
            continue;
        }
        if (std::abs(next[i]) > max_radius) {
            next[i] *= max_radius / std::abs(next[i]);
        }
        if (std::abs(step) > EPS * (1 + std::abs(next[i]))) {
            done = false;
        }
    }
    z = next;
    if (done) {
        break;
    }
}

```

```

    }
}
for (int i = 0; i < n; i++) {
    roots.emplace_back(z[i]);
}
std::sort(roots.begin(), roots.end(), root_less);
return roots;
}

std::vector<cdouble> find_all_roots(
    const std::vector<LD> &p, const LD EPS = ROOT_EPS, const int ITERATIONS = 2000
) {
    cpoly q(p.begin(), p.end());
    return find_all_roots(q, EPS, ITERATIONS);
}

```

Example Usage

```

#include <cassert>
#include <cstdio>
using namespace std;

cdouble eval(const cpoly &p, const cdouble &x) {
    return eval_with_derivative(p, x).first;
}

void print_roots(vector<cdouble> x) {
    sort(x.begin(), x.end(), [](const cdouble &a, const cdouble &b) {
        return abs(a.real() - b.real()) > 0.5e-5 ? a.real() < b.real() : a.imag() < b.imag();
    });
    for (const auto &z : x) {
        LD real = abs(z.real()) < 0.5e-5 ? 0 : z.real();
        LD imag = abs(z.imag()) < 0.5e-5 ? 0 : z.imag();
        printf("%.5Lf, %.5Lf\n", real, imag);
    }
}

int main() {
    { // -1 + 2x - 6x^2 + 2x^3
        printf("Roots of -1 + 2x - 6x^2 + 2x^3:\n");
        vector<LD> p{-1.0, 2.0, -6.0, 2.0};
        vector<cdouble> roots = find_all_roots(p);
        assert(roots.size() == 3);
        for (const auto &root : roots) {
            assert(abs(eval(cpoly(p.begin(), p.end()), root)) < CHECK_EPS);
        }
        print_roots(roots);
    }
    { // (-24+36i) + (-26+12i)x + (-30+40i)x^2 + (-26+12i)x^3 + (-6+4i)x^4
      // = ((2 + 3i)x + 6)(x + i)(2x + (6 + 4i))(xi + 1):
        printf("Roots of ((2 + 3i)x + 6)(x + i)(2x + (6 + 4i))(xi + 1):\n");
        cpoly p;
        p.emplace_back(-24, 36);
        p.emplace_back(-26, 12);
        p.emplace_back(-30, 40);
        p.emplace_back(-26, 12);
        p.emplace_back(-6, 4);
        vector<cdouble> roots = find_all_roots(p);
        assert(roots.size() == 4);
        for (const auto &root : roots) {
            assert(abs(eval(p, root)) < CHECK_EPS);
        }
        print_roots(roots);
    }
    return 0;
}

```

Example Output

```

Roots of -1 + 2x - 6x^2 + 2x^3:
(0.15098, -0.40314)
(0.15098, 0.40314)
(2.69805, 0.00000)
Roots of ((2 + 3i)x + 6)(x + i)(2x + (6 + 4i))(xi + 1):
(-3.00000, -2.00000)
(-0.92308, 1.38462)
(0.00000, -1.00000)
(0.00000, 1.00000)

```

5.4.6 Integration (Simpson)

Computes the definite integral from a to b for a continuous function f using the composite Simpson's rule: the interval is split into n equal subintervals and each consecutive pair is approximated by a parabola, giving integral $\sim h/3 \cdot [f(x_0) + 4f(x_1) + 2f(x_2) + \dots + 4f(x_{n-1}) + f(x_n)]$ with step $h = (b - a)/n$.

- `integrate(f, a, b, n)` returns the definite integral of a function f from a to b using n subintervals, where n must be even. Larger n trades runtime for accuracy: the error is $O((b - a) \cdot h^4)$, so for a smooth f a value of n on the order of 10^6 reaches near double precision.

Unlike adaptive quadrature, the step size here is uniform, so a feature narrower than h may be under-resolved. The benefit is a simple, non-recursive routine with predictable cost that is not fooled by the premature-termination heuristic that adaptive Simpson's rule can suffer on functions whose coarse sample points happen to fit a parabola.

Time Complexity: $O(n)$ per call to `integrate()`, requiring $n + 1$ evaluations of f .

Space Complexity: $O(1)$ auxiliary.

```

template<class Fn>
double integrate(Fn f, double a, double b, int n = 1000000) {
    double h = (b - a) / n, s = f(a) + f(b);
    for (int i = 1; i < n; i++) {
        s += f(a + h * i) * (i & 1 ? 4 : 2);
    }
    return s * h / 3;
}

```

Example Usage

```

#include <cassert>
#include <cmath>
#include <cstdio>
using namespace std;

double f(double x) {
    return sin(x);
}

int main() {
    const double PI = acos(-1.0);
    assert(fabs(integrate(f, 0.0, PI / 2) - 1) < 1e-10);
    return 0;
}

```

5.4.7 Linear Programming (Simplex)

Solves a linear programming problem using Dantzig's simplex algorithm. The canonical form of a linear programming problem is to maximize (or minimize) the dot product cx , subject to $ax \leq b$ and $x \geq 0$, where x is a

vector of unknowns to be solved, c is a vector of coefficients, a is a matrix of linear equation coefficients, and b is a vector of boundary coefficients.

- `simplex_solve(a, b, c, &x)` solves the linear programming problem for an m by n matrix `a` of real values, a length m vector `b`, and a length n vector `c`, returning 0 if an optimum was found, -1 if there are no feasible solutions, or 1 if the objective is unbounded. If an optimum is found, then the vector pointed to by `x` is populated with a dense solution vector.

Time Complexity: Polynomial (average) on the number of equations and unknowns, but exponential in the worst case.

Space Complexity: $O(m \cdot n)$ auxiliary heap space.

```
#include <cfloat>
#include <cmath>
#include <vector>

template<class Matrix>
int simplex_solve(
    const Matrix &a, const std::vector<double> &b, const std::vector<double> &c,
    std::vector<double> *x, const bool MAXIMIZE = true, const double EPS = 1e-10
) {
    int m = a.size(), n = c.size();
    Matrix t(m + 2, std::vector<double>(n + 2));
    t[1][1] = 0;
    for (int j = 1; j <= n; j++) {
        t[1][j + 1] = MAXIMIZE ? c[j - 1] : -c[j - 1];
    }
    for (int i = 1; i <= m; i++) {
        for (int j = 1; j <= n; j++) {
            t[i + 1][j + 1] = -a[i - 1][j - 1];
        }
        t[i + 1][1] = b[i - 1];
    }
    for (int j = 1; j <= n; j++) {
        t[0][j + 1] = j;
    }
    for (int i = n + 1; i <= m + n; i++) {
        t[i - n + 1][0] = i;
    }
    for (;;) {
        int p1 = 0, p2 = 0;
        double mn = DBL_MAX, xmax = EPS;
        for (int j = 2; j <= n + 1; j++) {
            if (t[1][j] > xmax) {
                p2 = j;
                xmax = t[1][j];
            }
        }
        if (p2 == 0) {
            break;
        }
        for (int i = 2; i <= m + 1; i++) {
            if (t[i][p2] < -EPS) {
                double v = fabs(t[i][1] / t[i][p2]);
                if (mn <= v) {
                    continue;
                }
                mn = v;
                p1 = i;
            }
        }
        if (p1 == 0) {
            return 1;
        }
        std::swap(t[p1][0], t[0][p2]);
        for (int i = 1; i <= m + 1; i++) {
```

```

    if (i != p1) {
        for (int j = 1; j <= n + 1; j++) {
            if (j != p2) {
                t[i][j] -= t[p1][j] * t[i][p2] / t[p1][p2];
            }
        }
    }
}
t[p1][p2] = 1.0 / t[p1][p2];
for (int j = 1; j <= n + 1; j++) {
    if (j != p2) {
        t[p1][j] *= fabs(t[p1][p2]);
    }
}
for (int i = 1; i <= m + 1; i++) {
    if (i != p1) {
        t[i][p2] *= t[p1][p2];
    }
}
for (int i = 2; i <= m + 1; i++) {
    if (t[i][1] < 0) {
        return -1;
    }
}
}
x->assign(n, 0);
for (int j = 1; j <= n; j++) {
    for (int i = 2; i <= m + 1; i++) {
        if (fabs(t[i][0] - j) < EPS) {
            (*x)[j - 1] = t[i][1];
        }
    }
}
return 0;
}

```

Example Usage

```

#include <cassert>
#include <iostream>
using namespace std;

int main() {
    // Solve [x, y] that maximizes 3x + 4y, subject to x, y >= 0 and:
    // -2x + 1y <= 0
    // 1x + 0.85y <= 9
    // 1x + 2y <= 14
    vector<vector<double>> va{{-2, 1}, {1, 0.85}, {1, 2}};
    vector<double> vb{0, 9, 14};
    vector<double> vc{3, 4};
    vector<double> x;
    assert(simplex_solve(va, vb, vc, &x) == 0);
    double maxval = 0;
    for (int i = 0; i < static_cast<int>(x.size()); i++) {
        maxval += vc[i] * x[i];
    }
    cout << "Solution = " << maxval << " at (" << x[0];
    for (int i = 1; i < static_cast<int>(x.size()); i++) {
        cout << ", " << x[i];
    }
    cout << ")." << endl;
    return 0;
}

```


Example Output

`Solution = 33.3043 at (5.30435, 4.34783).`

Chapter 6

Mathematics

6.1 Math Utilities

Common mathematic constants and functions, most of which already have standard STL equivalents:

`std::signbit()`, `std::copysign()`, `std::trunc()`, `std::round()`, `std::erf()`, `std::tgamma()`, `std::lgamma()`. These are written for educational purposes, without heavy optimizations.

Time Complexity: $O(1)$ for all operations.

Space Complexity: $O(1)$ auxiliary for all operations.

```
#include <algorithm>
#include <cfloat>
#include <climits>
#include <cmath>
#include <limits>
#include <random>
#include <string>
#include <vector>

#ifdef M_PI
const double M_PI = acos(-1.0);
#endif
#ifdef M_E
const double M_E = exp(1.0);
#endif
const double M_PHI = (1.0 + sqrt(5.0)) / 2.0;
const double M_INF = std::numeric_limits<double>::infinity();
const double M_NAN = std::numeric_limits<double>::quiet_NaN();
```

Epsilon Comparisons:

- `EQ()`, `NE()`, `LT()`, `GT()`, `LE()`, and `GE()` relationally compare two values x and y accounting for absolute error. For any x , the range of values considered equal barring absolute error is $[x - \text{EPS}, x + \text{EPS}]$. Values outside of this range are considered not equal (strictly less or strictly greater).
- `rEQ()` returns whether x and y are equal barring relative error. For any x , the range of values considered equal is $[x(1 - \text{EPS}), x(1 + \text{EPS})]$.

```
const double EPS = 1e-9;

#define EQ(x, y) (fabs((x) - (y)) <= EPS)
#define NE(x, y) (fabs((x) - (y)) > EPS)
#define LT(x, y) ((x) < (y) - EPS)
```

```

#define GT(x, y) ((x) > (y) + EPS)
#define LE(x, y) ((x) <= (y) + EPS)
#define GE(x, y) ((x) >= (y) - EPS)
#define rEQ(x, y) (fabs((x) - (y)) <= EPS * fabs(x))

```

Sign Functions:

- `sgn(x)` returns -1 (if $x < 0$), 0 (if $x = 0$), or 1 (if $x > 0$). Unlike `signbit()` or `copysign()`, this does not handle the sign of `NaN`.
- `signbit_(x)` is analogous to `std::signbit()`, returning whether the sign bit of the floating point number is set to true. If so, then `x` is considered “negative.” Note that this works as expected on `+0.0`, `-0.0`, `Inf`, `-Inf`, `NaN`, as well as `-NaN`. Warning: This assumes that the sign bit is the leading (most significant) bit in the internal representation of the IEEE floating point value.
- `copysign_(x, y)` is analogous to `std::copysign()`, returning a number with the magnitude of `x` but the sign of `y`.

```

template<class T>
int sgn(const T &x) {
    return (T(0) < x) - (x < T(0));
}

template<class Double>
bool signbit_(Double x) {
    return (((unsigned char *)&x)[sizeof(x) - 1] >> (CHAR_BIT - 1)) & 1;
}

template<class Double>
Double copysign_(Double x, Double y) {
    return signbit_(y) ? -fabs(x) : fabs(x);
}

```

Rounding Functions:

- `floor0(x)` returns `x` rounded down, symmetrically towards zero. This function is analogous to `std::trunc()`.
- `ceil0(x)` returns `x` rounded up, symmetrically away from zero. This function is analogous to `std::round()`.
- `round_half_up(x)` returns `x` rounded half up, towards positive infinity.
- `round_half_down(x)` returns `x` rounded half down, towards negative infinity.
- `round_half_to0(x)` returns `x` rounded half down, symmetrically towards zero.
- `round_half_from0(x)` returns `x` rounded half up, symmetrically away from zero.
- `round_half_even(x)` returns `x` rounded half to even, using banker’s rounding.
- `round_half_alterate(x)` returns `x` rounded, where ties are broken by alternating rounds towards positive and negative infinity.
- `round_half_alterate0(x)` returns `x` rounded, where ties are broken by alternating symmetric rounds towards and away from zero.
- `round_half_random(x)` returns `x` rounded, where ties are broken randomly.
- `round_n_places(x, n, f)` returns `x` rounded to `n` digits after the decimal, using the specified rounding function `f(x)`.

```

template<class Double>
Double floor0(const Double &x) {
    Double res = floor(fabs(x));
    return (x < 0.0) ? -res : res;
}

template<class Double>
Double ceil0(const Double &x) {
    Double res = ceil(fabs(x));
    return (x < 0.0) ? -res : res;
}

```

```

template<class Double>
Double round_half_up(const Double &x) {
    return floor(x + 0.5);
}

template<class Double>
Double round_half_down(const Double &x) {
    return ceil(x - 0.5);
}

template<class Double>
Double round_half_to0(const Double &x) {
    Double res = round_half_down(fabs(x));
    return (x < 0.0) ? -res : res;
}

template<class Double>
Double round_half_from0(const Double &x) {
    Double res = round_half_up(fabs(x));
    return (x < 0.0) ? -res : res;
}

template<class Double>
Double round_half_even(const Double &x, const Double &eps = 1e-9) {
    if (x < 0.0) {
        return -round_half_even(-x, eps);
    }
    Double ipart;
    modf(x, &ipart);
    if (fabs(x - (ipart + 0.5)) < eps) { // exactly halfway: break the tie towards even
        return (fmod(ipart, 2.0) < eps) ? ipart : ceil0(ipart + 0.5);
    }
    return round_half_from0(x);
}

template<class Double>
Double round_half_alterate(const Double &x) {
    static bool up = true;
    return (up = !up) ? round_half_up(x) : round_half_down(x);
}

template<class Double>
Double round_half_alterate0(const Double &x) {
    static bool up = true;
    return (up = !up) ? round_half_from0(x) : round_half_to0(x);
}

template<class Double>
Double round_half_random(const Double &x) {
    static std::mt19937 rng(std::random_device{}());
    return (rng() % 2 == 0) ? round_half_from0(x) : round_half_to0(x);
}

template<class Double, class RoundingFunction>
Double round_n_places(const Double &x, unsigned int n, RoundingFunction f) {
    return f(x * pow(10, n)) / pow(10, n);
}

```

Error Function:

- `erf(x)` returns the error encountered in integrating the normal distribution. Its value is $2/\sqrt{\pi} \int_0^x e^{-t^2} dt$. This function is analogous to `std::erf(x)`.
- `erfc(x)` returns the error function complement, that is, $1 - \text{erf}(x)$. This function is analogous to `std::erfc(x)`.

```
#define ERF_EPS 1e-14
```

```
double erfc_(double x);

double erf_(double x) {
    if (signbit_(x)) {
        return -erf_(-x);
    }
    if (fabs(x) > 2.2) {
        return 1.0 - erfc_(x);
    }
    double sum = x, term = x, xx = x * x;
    int j = 1;
    do {
        term *= xx / j;
        sum -= term / (2 * (j++) + 1);
        term *= xx / j;
        sum += term / (2 * (j++) + 1);
    } while (fabs(term) > sum * ERF_EPS);
    return 2 / sqrt(M_PI) * sum;
}

double erfc_(double x) {
    if (fabs(x) < 2.2) {
        return 1.0 - erf_(x);
    }
    if (signbit_(x)) {
        return 2.0 - erfc_(-x);
    }
    double a = 1, b = x, c = x, d = x * x + 0.5, q1, q2 = 0, n = 1.0, t;
    do {
        t = a * n + b * x;
        a = b;
        b = t;
        t = c * n + d * x;
        c = d;
        d = t;
        n += 0.5;
        q1 = q2;
        q2 = b / d;
    } while (fabs(q1 - q2) > q2 * ERF_EPS);
    return 1 / sqrt(M_PI) * exp(-x * x) * q2;
}

#undef ERF_EPS
```

Gamma Functions:

- `tgamma_(x)` returns the gamma function of `x`. Unlike `std::tgamma()`, this version only supports positive `x`, returning `NaN` if `x` ≤ 0 .
- `lgamma_(x)` returns the natural logarithm of the absolute value of the gamma function of `x`. Unlike `std::lgamma()`, this version only supports positive `x`, returning `NaN` if `x` ≤ 0 .

```
double lgamma_(double x);

double tgamma_(double x) {
    if (x <= 0) {
        return M_NAN;
    }
    if (x < 1e-3) {
        return 1.0 / (x * (1.0 + 0.57721566490153286060651209 * x));
    }
    if (x < 12) {
        double y = x;
        int n = 0;
        bool arg_was_less_than_one = (y < 1);
        if (arg_was_less_than_one) {
```

```

    y += 1;
} else {
    n = static_cast<int>(floor(y)) - 1;
    y -= n;
}
static const double p[] = {-1.71618513886549492533811e+0, 2.47656508055759199108314e+1,
                             -3.79804256470945635097577e+2, 6.29331155312818442661052e+2,
                             8.66966202790413211295064e+2, -3.14512729688483675254357e+4,
                             -3.61444134186911729807069e+4, 6.64561438202405440627855e+4};
static const double q[] = {-3.08402300119738975254353e+1, 3.15350626979604161529144e+2,
                             -1.01515636749021914166146e+3, -3.10777167157231109440444e+3,
                             2.25381184209801510330112e+4, 4.75584627752788110767815e+3,
                             -1.34659959864969306392456e+5, -1.15132259675553483497211e+5};

double num = 0, den = 1, z = y - 1;
for (int i = 0; i < 8; i++) {
    num = (num + p[i]) * z;
    den = den * z + q[i];
}
double result = num / den + 1;
if (arg_was_less_than_one) {
    result /= (y - 1);
} else {
    for (int i = 0; i < n; i++) {
        result *= y++;
    }
}
return result;
}
return (x > 171.624) ? 2 * DBL_MAX : exp(lgamma(x));
}

double lgamma_(double x) {
    if (x <= 0) {
        return M_NAN;
    }
    if (x < 12) {
        return log(fabs(tgamma_(x)));
    }
    static const double c[] = {1.0 / 12, -1.0 / 360, 1.0 / 1260, -1.0 / 1680,
                                1.0 / 1188, -691.0 / 360360, 1.0 / 156, -3617.0 / 122400};
    double z = 1.0 / (x * x), sum = c[7];
    for (int i = 6; i >= 0; i--) {
        sum = sum * z + c[i];
    }
    return (x - 0.5) * log(x) - x + 0.91893853320467274178032973640562 + sum / x;
}

```

Base Conversion:

- Given an integer in base *a* as a vector *d* of digits (where *d*[0] is the least significant digit), `convert_base(d, a, b)` returns a vector of the integer's digits when converted base *b* (again with index 0 storing the least significant digit). The actual value of the entire integer to be converted must be able to fit within an unsigned 64-bit integer for intermediate storage.
- `to_base(x, b)` returns the digits of the unsigned integer *x* in base *b*, where index 0 of the result stores the least significant digit.
- `to_roman(x)` returns the Roman numeral representation of the unsigned integer *x* as an `std::string`.

```

std::vector<int> convert_base(const std::vector<int> &d, int a, int b) {
    unsigned long long x = 0, power = 1;
    for (int di : d) {
        x += di * power;
        power *= a;
    }
    std::vector<int> res;
    do { // do-while so that a value of 0 yields the single digit {0}

```

```

    res.push_back(x % b);
    x /= b;
} while (x != 0);
return res;
}

std::vector<int> to_base(unsigned int x, int b = 10) {
    std::vector<int> res;
    do { // do-while so that a value of 0 yields the single digit {0}
        res.push_back(x % b);
        x /= b;
    } while (x != 0);
    return res;
}

std::string to_roman(unsigned int x) {
    static const std::string h[] = {"", "C", "CC", "CCC", "CD", "D", "DC", "DCC", "DCCC", "CM"};
    static const std::string t[] = {"", "X", "XX", "XXX", "XL", "L", "LX", "LXX", "LXXX", "XC"};
    static const std::string o[] = {"", "I", "II", "III", "IV", "V", "VI", "VII", "VIII", "IX"};
    std::string prefix(x / 1000, 'M');
    x %= 1000;
    return prefix + h[x / 100] + t[x / 10 % 10] + o[x % 10];
}

```

Example Usage

```

#include <cassert>
#include <iostream>
using namespace std;

int main() {
    assert(EQ(M_PI, 3.14159265359));
    assert(EQ(M_E, 2.718281828459));
    assert(EQ(M_PHI, 1.61803398875));

    double x = -12345.6789;
    assert((-M_INF < x) && (x < M_INF));
    assert((M_INF + x == M_INF) && (M_INF - x == M_INF));
    assert((M_INF + M_INF == M_INF) && (-M_INF - M_INF == -M_INF));
    assert((M_NAN != x) && (M_NAN != M_INF) && (M_NAN != M_NAN));
    assert(!(M_NAN < x) && !(M_NAN > x) && !(M_NAN <= x) && !(M_NAN >= x));
    assert(isnan(0.0 * M_INF) && isnan(0.0 * -M_INF) && isnan(M_INF / -M_INF));
    assert(isnan(M_NAN) && isnan(-M_NAN) && isnan(M_INF - M_INF));

    assert(sgn(x) == -1 && sgn(0.0) == 0 && sgn(5678) == 1);
    assert(signbit(x) && !signbit(0.0) && signbit(-0.0));
    assert(!signbit(M_INF) && signbit(-M_INF));
    assert(!signbit(M_NAN) && signbit(-M_NAN));
    assert(copysign(1.0, +2.0) == +1.0 && copysign(M_INF, -2.0) == -M_INF);
    assert(copysign(1.0, -2.0) == -1.0 && std::signbit(copysign(M_NAN, -2.0)));

    assert(EQ(floor0(1.5), 1.0) && EQ(ceil0(1.5), 2.0));
    assert(EQ(floor0(-1.5), -1.0) && EQ(ceil0(-1.5), -2.0));
    assert(EQ(round_half_up(+1.5), +2) && EQ(round_half_down(+1.5), +1));
    assert(EQ(round_half_up(-1.5), -1) && EQ(round_half_down(-1.5), -2));
    assert(EQ(round_half_to0(+1.5), +1) && EQ(round_half_from0(+1.5), +2));
    assert(EQ(round_half_to0(-1.5), -1) && EQ(round_half_from0(-1.5), -2));
    assert(EQ(round_half_even(+1.5), +2) && EQ(round_half_even(-1.5), -2));
    assert(EQ(round_half_even(3.1), 3) && EQ(round_half_even(3.4), 3)); // non-ties round normally
    assert(NE(round_half_alternate(+1.5), round_half_alternate(+1.5)));
    assert(NE(round_half_alternate0(-1.5), round_half_alternate0(-1.5)));
    assert(EQ(round_n_places(-1.23456, 3, round_half_to0<double>), -1.235));

    assert(EQ(erf(1.0), 0.8427007929) && EQ(erf(-1.0), -0.8427007929));
    assert(EQ(tgamma(0.5), 1.7724538509) && EQ(tgamma(1.0), 1.0));
    assert(EQ(lgamma(0.5), 0.5723649429) && EQ(lgamma(1.0), 0.0));
}

```

```

std::vector<int> digits{6, 5, 4, 3, 2, 1};
std::vector<int> base20 = to_base(123456, 20);
assert(convert_base(base20, 20, 10) == digits);
assert(to_roman(1234) == "MCCXXXIV");
assert(to_roman(5678) == "MMMMDCLXXVIII");
return 0;
}

```

6.2 Combinatorics

6.2.1 Combinatorial Calculations

The following functions implement common operations in combinatorics. All inputs must be non-negative. All return values and table entries are computed as 64-bit integers modulo an input argument m or p .

- `factorial(n, m)` returns $n! \bmod m$.
- `factorialp(n, p)` returns $n! \bmod p$, where p is prime.
- `binomial_table(n, m)` returns rows 0 to n of Pascal's triangle as a table t such that $t[i][j] = \binom{i}{j} \bmod m$.
- `permute(n, k, m)` returns $(n \text{ permute } k) \bmod m$.
- `choose(n, k, p)` returns $\binom{n}{k} \bmod p$, where p is prime.
- `multichoose(n, k, p)` returns $(n \text{ multichoose } k) \bmod p$, where p is prime.
- `catalan(n, p)` returns the n th Catalan number mod p , where p is prime.
- `partitions(n, m)` returns the number of partitions of n , mod m .
- `partitions(n, k, m)` returns the number of partitions of n into k parts, mod m .
- `stirling1(n, k, m)` returns the (n, k) unsigned Stirling number of the 1st kind mod m .
- `stirling2(n, k, m)` returns the (n, k) Stirling number of the 2nd kind mod m .
- `eulerian1(n, k, m)` returns the (n, k) Eulerian number of the 1st kind mod m , where $n > k$.
- `eulerian2(n, k, m)` returns the (n, k) Eulerian number of the 2nd kind mod m , where $n > k$.

Time Complexity:

- $O(n)$ for `factorial(n, m)`.
- $O(p \log n)$ for `factorialp(n, p)`.
- $O(n^2)$ for `binomial_table(n, m)`.
- $O(k)$ for `permute(n, k, p)`.
- $O(\min(k, n - k))$ for `choose(n, k, p)`.
- $O(k)$ for `multichoose(n, k, p)`.
- $O(n)$ for `catalan(n, p)`.
- $O(n^2)$ for `partitions(n, m)`.
- $O(n \cdot k)$ for `partitions(n, k, m)`, `stirling1(n, k, m)`, `stirling2(n, k, m)`, `eulerian1(n, k, m)`, and `eulerian2(n, k, m)`.

Space Complexity:

- $O(n^2)$ auxiliary heap space for `binomial_table(n, m)`.
- $O(n \cdot k)$ auxiliary heap space for `partitions(n, k, m)`, `stirling1(n, k, m)`, `stirling2(n, k, m)`, `eulerian1(n, k, m)`, and `eulerian2(n, k, m)`.
- $O(1)$ auxiliary for all other operations.

```

#include <vector>

using int64 = long long;
using table = std::vector<std::vector<int64>>>;

int64 factorial(int n, int m = 1000000007) {
    int64 res = 1;

```



```

    for (int i = 2; i <= n; i++) {
        res = (res * i) % m;
    }
    return res % m;
}

int64 factorialp(int64 n, int64 p = 1000000007) {
    int64 res = 1;
    while (n > 1) {
        if (n / p % 2 == 1) {
            res = res * (p - 1) % p;
        }
        int max = n % p;
        for (int i = 2; i <= max; i++) {
            res = (res * i) % p;
        }
        n /= p;
    }
    return res % p;
}

table binomial_table(int n, int64 m = 1000000007) {
    table t(n + 1);
    for (int i = 0; i <= n; i++) {
        for (int j = 0; j <= i; j++) {
            if (i < 2 || j == 0 || i == j) {
                t[i].push_back(1);
            } else {
                t[i].push_back((t[i - 1][j - 1] + t[i - 1][j]) % m);
            }
        }
    }
    return t;
}

int64 permute(int n, int k, int64 m = 1000000007) {
    if (n < k) {
        return 0;
    }
    int64 res = 1;
    for (int i = 0; i < k; i++) {
        res = res * (n - i) % m;
    }
    return res % m;
}

int64 mulmod(int64 x, int64 n, int64 m) {
    int64 a = 0, b = x % m;
    for (; n > 0; n >>= 1) {
        if (n & 1) {
            a = (a + b) % m;
        }
        b = (b << 1) % m;
    }
    return a % m;
}

int64 powmod(int64 x, int64 n, int64 m) {
    int64 a = 1, b = x;
    for (; n > 0; n >>= 1) {
        if (n & 1) {
            a = mulmod(a, b, m);
        }
        b = mulmod(b, b, m);
    }
    return a % m;
}

```

```

int64 choose(int n, int k, int64 p = 1000000007) {
    if (n < k) {
        return 0;
    }
    if (k > n - k) {
        k = n - k;
    }
    int64 num = 1, den = 1;
    for (int i = 0; i < k; i++) {
        num = num * (n - i) % p;
    }
    for (int i = 1; i <= k; i++) {
        den = den * i % p;
    }
    return num * powmod(den, p - 2, p) % p;
}

int64 multichoose(int n, int k, int64 p = 1000000007) {
    return choose(n + k - 1, k, p);
}

int64 catalan(int n, int64 p = 1000000007) {
    return choose(2 * n, n, p) * powmod(n + 1, p - 2, p) % p;
}

int64 partitions(int n, int64 m = 1000000007) {
    std::vector<int64> t(n + 1, 0);
    t[0] = 1;
    for (int i = 1; i <= n; i++) {
        for (int j = i; j <= n; j++) {
            t[j] = (t[j] + t[j - i]) % m;
        }
    }
    return t[n] % m;
}

int64 partitions(int n, int k, int64 m = 1000000007) {
    table t(n + 1, std::vector<int64>(k + 1, 0));
    t[0][1] = 1;
    for (int i = 1; i <= n; i++) {
        for (int j = 1, h = k < i ? k : i; j <= h; j++) {
            t[i][j] = (t[i - 1][j - 1] + t[i - j][j]) % m;
        }
    }
    return t[n][k] % m;
}

int64 stirling1(int n, int k, int64 m = 1000000007) {
    table t(n + 1, std::vector<int64>(k + 1, 0));
    t[0][0] = 1;
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= k; j++) {
            t[i][j] = (i - 1) * t[i - 1][j] % m;
            t[i][j] = (t[i][j] + t[i - 1][j - 1]) % m;
        }
    }
    return t[n][k] % m;
}

int64 stirling2(int n, int k, int64 m = 1000000007) {
    table t(n + 1, std::vector<int64>(k + 1, 0));
    t[0][0] = 1;
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= k; j++) {
            t[i][j] = j * t[i - 1][j] % m;
            t[i][j] = (t[i][j] + t[i - 1][j - 1]) % m;
        }
    }
}

```

```

    return t[n][k] % m;
}

int64 eulerian1(int n, int k, int64 m = 1000000007) {
    if (k > n - 1 - k) {
        k = n - 1 - k;
    }
    table t(n + 1, std::vector<int64>(k + 1, 1));
    for (int j = 1; j <= k; j++) {
        t[0][j] = 0;
    }
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= k; j++) {
            t[i][j] = (i - j) * t[i - 1][j - 1] % m;
            t[i][j] = (t[i][j] + ((j + 1) * t[i - 1][j] % m)) % m;
        }
    }
    return t[n][k] % m;
}

int64 eulerian2(int n, int k, int64 m = 1000000007) {
    table t(n + 1, std::vector<int64>(k + 1, 1));
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= k; j++) {
            if (i == j) {
                t[i][j] = 0;
            } else {
                t[i][j] = (j + 1) * t[i - 1][j] % m;
                t[i][j] = ((2 * i - 1 - j) * t[i - 1][j - 1] % m + t[i][j]) % m;
            }
        }
    }
    return t[n][k] % m;
}

```

Example Usage

```

#include <cassert>

int main() {
    table t = binomial_table(10);
    for (int i = 0; i < static_cast<int>(t.size()); i++) {
        for (int j = 0; j < static_cast<int>(t[i].size()); j++) {
            assert(t[i][j] == choose(i, j));
        }
    }
    assert(factorial(10) == 3628800);
    assert(factorialp(123456) == 639390503);
    assert(permute(10, 4) == 5040);
    assert(choose(20, 7) == 77520);
    assert(multichoose(20, 7) == 657800);
    assert(catalan(10) == 16796);
    assert(partitions(4) == 5);
    assert(partitions(100, 5) == 38225);
    assert(stirling1(4, 2) == 11);
    assert(stirling2(4, 3) == 6);
    assert(eulerian1(9, 5) == 88234);
    assert(eulerian2(8, 3) == 195800);
    return 0;
}

```

6.2.2 Enumerating Arrangements

For the purposes of this section, we define a “size k arrangement of n ” to be a permutation of a size k subset of the integers from 0 to $n - 1$, for $0 \leq k \leq n$. There are n permute k possible arrangements, but n^k possible arrangements if repeated values are allowed.

- `next_arrangement(n, a)` tries to rearrange `a` to the next lexicographically greater arrangement, returning true if such an arrangement exists or false if the array is already in descending order (in which case `a` is unchanged). The input `a` must consist of distinct integers in the range $[0, n)$.
- `arrangement_by_rank(n, k, r)` returns the size k arrangement of n which is lexicographically ranked r out of all size k arrangements of n , where r is a zero-based rank in the range $[0, n \text{ permute } k)$.
- `rank_by_arrangement(n, a)` returns an integer representing the zero-based rank of arrangement `a`, which must consist of distinct integers in the range $[0, n)$.
- `next_arrangement_with_repeats(n, a)` tries to rearrange `a` to the next lexicographically greater arrangement with repeats, returning true if such an arrangement exists or false if the array is already in descending order (in which case `a` is unchanged). The input `a` must consist of integers in the range $[0, n)$. If `a` were interpreted as a k digit integer in base n , this function could be thought of as incrementing the integer.

Time Complexity:

- $O(n \cdot k)$ for `next_arrangement(n, a)`, `arrangement_by_rank(n, k, r)`, and `rank_by_arrangement(n, a)`, where k is the size of `a`.
- $O(k)$ for `next_arrangement_with_repeats(n, a)`.

Space Complexity:

- $O(n)$ auxiliary heap space for `next_arrangement()`, `arrangement_by_rank()`, and `rank_by_arrangement()`.
- $O(1)$ auxiliary for `next_arrangement_with_repeats()`.

```
#include <algorithm>
#include <numeric>
#include <vector>

bool next_arrangement(int n, std::vector<int> &a) {
    int k = a.size();
    std::vector<bool> used(n);
    for (int i = 0; i < k; i++) {
        used[a[i]] = true;
    }
    for (int i = k - 1; i >= 0; i--) {
        used[a[i]] = false;
        for (int j = a[i] + 1; j < n; j++) {
            if (!used[j]) {
                a[i++] = j;
                used[j] = true;
                for (int x = 0; x < k; x++) {
                    if (!used[x]) {
                        a[i++] = x;
                    }
                }
            }
        }
        return true;
    }
}

return false;
}

long long n_permute_k(int n, int k) {
    long long res = 1;
    for (int i = 0; i < k; i++) {
        res *= n - i;
    }
    return res;
}
```

```

std::vector<int> arrangement_by_rank(int n, int k, long long r) {
    std::vector<int> values(n), res(k);
    std::iota(values.begin(), values.end(), 0);
    for (int i = 0; i < k; i++) {
        long long count = n_permute_k(n - 1 - i, k - 1 - i);
        int pos = r / count;
        res[i] = values[pos];
        std::copy(values.begin() + pos + 1, values.end(), values.begin() + pos);
        r %= count;
    }
    return res;
}

long long rank_by_arrangement(int n, const std::vector<int> &a) {
    int k = a.size();
    long long res = 0;
    std::vector<bool> used(n);
    for (int i = 0; i < k; i++) {
        int count = 0;
        for (int j = 0; j < a[i]; j++) {
            if (!used[j]) {
                count++;
            }
        }
        res += count * n_permute_k(n - i - 1, k - i - 1);
        used[a[i]] = true;
    }
    return res;
}

bool next_arrangement_with_repeats(int n, std::vector<int> &a) {
    int k = a.size();
    for (int i = k - 1; i >= 0; i--) {
        if (a[i] < n - 1) {
            a[i]++;
            std::fill(a.begin() + i + 1, a.end(), 0);
            return true;
        }
    }
    return false;
}

```

Example Usage

```

#include <cassert>
#include <iostream>
using namespace std;

template<class It>
void print_range(It lo, It hi) {
    cout << "{";
    for (; lo != hi; ++lo) {
        cout << *lo << (lo == hi - 1 ? "" : "," );
    }
    cout << "} ";
}

int main() {
    {
        int n = 4, k = 3;
        vector<int> a{0, 1, 2};
        cout << n << " permute " << k << " arrangements:" << endl;
        int count = 0;
        do {
            print_range(a.begin(), a.end());

```

```

    auto b = arrangement_by_rank(n, k, count);
    assert(a == b);
    assert(rank_by_arrangement(n, a) == count);
    count++;
    if (count == 10 || count == 20) {
        cout << endl;
    }
} while (next_arrangement(n, a));
cout << endl;
}
{
    int n = 4, k = 2;
    vector<int> a{0, 0};
    cout << endl << n << "^" << k << " arrangements with repeats:" << endl;
    int count = 0;
    do {
        print_range(a.begin(), a.end());
        count++;
        if (count == 13) {
            cout << endl;
        }
    } while (next_arrangement_with_repeats(n, a));
    cout << endl;
}
return 0;
}

```

Example Output

```

4 permute 3 arrangements:
{0,1,2} {0,1,3} {0,2,1} {0,2,3} {0,3,1} {0,3,2} {1,0,2} {1,0,3} {1,2,0} {1,2,3}
{1,3,0} {1,3,2} {2,0,1} {2,0,3} {2,1,0} {2,1,3} {2,3,0} {2,3,1} {3,0,1} {3,0,2}
{3,1,0} {3,1,2} {3,2,0} {3,2,1}

4^2 arrangements with repeats:
{0,0} {0,1} {0,2} {0,3} {1,0} {1,1} {1,2} {1,3} {2,0} {2,1} {2,2} {2,3} {3,0}
{3,1} {3,2} {3,3}

```

6.2.3 Enumerating Permutations

A permutation is an ordered list consisting of n (not necessarily distinct) elements.

- `next_permutation(lo, hi)` is analogous to `std::next_permutation(lo, hi)`, taking two BidirectionalIterators `lo` and `hi` as a range `[lo, hi)` for which the function tries to rearrange to the next lexicographically greater permutation. The function returns true if such a permutation exists, or false if the range is already in descending order (in which case the values are unchanged). This implementation requires an ordering on the set of possible elements defined by the `<` operator on the iterator's value type.
- `next_permutation(a)` is analogous to `next_permutation()`, except that it takes a vector instead of a range.
- `next_permutation(x)` returns the next lexicographically greater permutation of the binary digits of the integer `x`, that is, the lowest integer greater than `x` with the same number of 1-bits. This can be used to generate combinations of a set of n items by treating each 1 bit as whether to “take” the item at the corresponding position.
- `permutation_by_rank(n, r)` returns the permutation of the integers in the range $[0, n)$ which is lexicographically ranked r , where r is a zero-based rank in the range $[0, n!)$.
- `rank_by_permutation(a)` returns an integer representing the zero-based rank of permutation `a`, which must be a permutation of the integers $[0, n)$.
- `permutation_cycles(a)` returns the decomposition of the permutation `a` into cycles. A permutation cycle is a subset of a permutation whose elements are consecutively swapped, relative to a sorted set. For example, `{3, 1, 0, 2}` decomposes to `{0, 3, 2}` and `{1}`, meaning that starting from the sorted order `{0, 1, 2, 3}`, the 0th value is replaced by the 3rd, the 3rd by the 2nd, and the 2nd by the 0th ($0 \rightarrow 3 \rightarrow 2 \rightarrow 0$).

Time Complexity:

- $O(n^2)$ per call to `next_permutation(lo, hi)`, where n is the distance between `lo` and `hi`.
- $O(n^2)$ per call to `next_permutation(a)`, `permutation_by_rank(n, r)`, and `rank_by_permutation(a)`.
- $O(1)$ per call to `next_permutation(x)`.
- $O(n)$ per call to `permutation_cycles()`.

Space Complexity:

- $O(1)$ auxiliary for `next_permutation()` and `next_permutation()`.
- $O(n)$ auxiliary heap space for `permutation_by_rank()`, `rank_by_permutation()`, and `permutation_cycles()`.

```
#include <algorithm>
#include <numeric>
#include <vector>

template<class It>
bool next_permutation_(It lo, It hi) {
    if (lo == hi) {
        return false;
    }
    It i = lo;
    if (++i == hi) {
        return false;
    }
    i = hi;
    --i;
    for (;;) {
        It j = i;
        if (*--i < *j) {
            It k = hi;
            while (!(*i < *--k)) {
            }
            std::iter_swap(i, k);
            std::reverse(j, hi);
            return true;
        }
        if (i == lo) {
            std::reverse(lo, hi);
            return false;
        }
    }
}

template<class T>
bool next_permutation(std::vector<T> &a) {
    int n = a.size();
    for (int i = n - 2; i >= 0; i--) {
        if (a[i] < a[i + 1]) {
            for (int j = n - 1; j--;) {
                if (a[i] < a[j]) {
                    std::swap(a[i++], a[j]);
                    for (j = n - 1; i < j; i++, j--) {
                        std::swap(a[i], a[j]);
                    }
                    return true;
                }
            }
        }
    }
    return false;
}

long long next_permutation(long long x) {
    long long s = x & -x, r = x + s;
    return r | (((x ^ r) >> 2) / s);
}
```

```

std::vector<int> permutation_by_rank(int n, long long x) {
    std::vector<long long> factorial(n);
    std::vector<int> values(n), res(n);
    factorial[0] = 1;
    for (int i = 1; i < n; i++) {
        factorial[i] = i * factorial[i - 1];
    }
    std::iota(values.begin(), values.end(), 0);
    for (int i = 0; i < n; i++) {
        int pos = x / factorial[n - 1 - i];
        res[i] = values[pos];
        std::copy(values.begin() + pos + 1, values.end(), values.begin() + pos);
        x %= factorial[n - 1 - i];
    }
    return res;
}

long long rank_by_permutation(const std::vector<int> &a) {
    int n = a.size();
    std::vector<long long> factorial(n);
    factorial[0] = 1;
    for (int i = 1; i < n; i++) {
        factorial[i] = i * factorial[i - 1];
    }
    long long res = 0;
    for (int i = 0; i < n; i++) {
        int v = a[i];
        for (int j = 0; j < i; j++) {
            if (a[j] < a[i]) {
                v--;
            }
        }
        res += v * factorial[n - 1 - i];
    }
    return res;
}

using cycles = std::vector<std::vector<int>>>;

cycles permutation_cycles(const std::vector<int> &a) {
    int n = a.size();
    std::vector<bool> visit(n);
    cycles res;
    for (int i = 0; i < n; i++) {
        if (!visit[i]) {
            int j = i;
            std::vector<int> curr;
            do {
                curr.push_back(j);
                visit[j] = true;
                j = a[j];
            } while (j != i);
            res.push_back(std::move(curr));
        }
    }
    return res;
}

```

Example Usage

```

#include <bitset>
#include <cassert>
#include <iostream>
using namespace std;

```



```

template<class It>
void print_range(It lo, It hi) {
    cout << "{";
    for (; lo != hi; ++lo) {
        cout << *lo << (lo == hi - 1 ? "" : ",");
    }
    cout << "} ";
}

int main() {
    {
        const int n = 4;
        vector<int> a{0, 1, 2, 3}, b = a, c = a;
        cout << "Permutations of [0, " << n << "):" << endl;
        int count = 0;
        do {
            print_range(a.begin(), a.end());
            assert(b == a);
            assert(c == a);
            vector<int> d = permutation_by_rank(n, count);
            assert(d == a);
            assert(rank_by_permutation(a) == count);
            count++;
            if (count == 8 || count == 16) {
                cout << endl;
            }
            std::next_permutation(b.begin(), b.end());
            next_permutation(c);
        } while (next_permutation(a));
        cout << endl;
    }
    { // Permutations of binary digits.
        const int n = 5;
        cout << "\nPermutations of 2 zeros and 3 ones:" << endl;
        int lo = bitset<5>(string("00111")).to_ulong();
        int hi = bitset<6>(string("100011")).to_ulong();
        do {
            cout << bitset<n>(lo).to_string() << " ";
        } while ((lo = next_permutation(lo)) != hi);
        cout << endl;
    }
    { // Decomposition into cycles.
        const int n = 4;
        vector<int> a{3, 1, 0, 2};
        cout << "\nDecomposition of {3,1,0,2} into cycles:" << endl;
        cycles c = permutation_cycles(a);
        for (const auto &cycle : c) {
            print_range(cycle.begin(), cycle.end());
        }
        cout << endl;
    }
    return 0;
}

```

Example Output

```

Permutations of [0, 4):
{0,1,2,3} {0,1,3,2} {0,2,1,3} {0,2,3,1} {0,3,1,2} {0,3,2,1} {1,0,2,3} {1,0,3,2}
{1,2,0,3} {1,2,3,0} {1,3,0,2} {1,3,2,0} {2,0,1,3} {2,0,3,1} {2,1,0,3} {2,1,3,0}
{2,3,0,1} {2,3,1,0} {3,0,1,2} {3,0,2,1} {3,1,0,2} {3,1,2,0} {3,2,0,1} {3,2,1,0}

Permutations of 2 zeros and 3 ones:
00111 01011 01101 01110 10011 10101 10110 11001 11010 11100

Decomposition of {3,1,0,2} into cycles:
{0,3,2} {1}

```

6.2.4 Enumerating Combinations

A combination is a subset of size k chosen from a total of n (not necessarily distinct) elements, where order does not matter.

- `next_combination(lo, mid, hi)` takes random-access iterators `lo`, `mid`, and `hi` as a range `[lo, hi)` of n elements for which the function will rearrange such that the k elements in `[lo, mid)` become the next lexicographically greater combination. The function returns true if such a combination exists, or false if `[lo, mid)` already consists of the lexicographically greatest combination of the elements in `[lo, hi)` (in which case the values are unchanged). This implementation requires an ordering on the set of possible elements defined by `operator<` on the iterator's value type.
- `next_combination(n, a)` rearranges `a` to become the next lexicographically greater combination of distinct integers in the range $[0, n)$. The vector `a` must be sorted and contain distinct integers in the range $[0, n)$.
- `next_combination_mask(x)` interprets the bits of an integer `x` as a mask with 1-bits specifying the chosen items for a combination and returns the mask of the next lexicographically greater combination (that is, the lowest integer greater than `x` with the same number of 1 bits). Note that this does not generate combinations in the same order as `next_combination()`, nor does it work if the corresponding n items are not distinct (in that case, duplicate combinations will be generated).
- `combination_by_rank(n, k, r)` returns the combination of k distinct integers in the range $[0, n)$ that is lexicographically ranked r , where r is a zero-based rank in the range $[0, \binom{n}{k})$.
- `rank_by_combination(n, a)` returns an integer representing the zero-based rank of combination `a`, which must contain sorted distinct integers in $[0, n)$.
- `next_combination_with_repeats(n, a)` rearranges `a` to become the next lexicographically greater combination of not necessarily distinct integers in the range $[0, n)$. The vector `a` must be sorted. Note that there is a total of n multichoose k combinations if repetition is allowed, where n multichoose $k = \binom{n+k-1}{k}$.

Time Complexity:

- $O(n)$ per call to `next_combination(lo, hi)`, where n is the distance between `lo` and `hi`.
- $O(k)$ per call to `next_combination(n, a)` and `next_combination_with_repeats(n, a)`, where k is the size of `a`.
- $O(1)$ per call to `next_combination_mask(x)`.
- $O(n \cdot k)$ per call to `combination_by_rank()` and `rank_by_combination()`.

Space Complexity:

- $O(k)$ auxiliary heap space for `combination_by_rank(n, k, r)`.
- $O(1)$ auxiliary for all other operations.

```
#include <algorithm>
#include <iterator>
#include <vector>

template<class It>
bool next_combination(It lo, It mid, It hi) {
    if (lo == mid || mid == hi) {
        return false;
    }
    It l = mid - 1, h = hi - 1;
    int len1 = 1, len2 = 1;
    while (l != lo && !(l < *h)) {
        --l;
        ++len1;
    }
    if (l == lo && !(l < *h)) {
        std::rotate(lo, mid, hi);
        return false;
    }
    for (; mid < h; ++len2) {
        if (!(l < *--h)) {
            ++h;
            break;
        }
    }
}
```

```

}
if (len1 == 1 || len2 == 1) {
    std::iter_swap(l, h);
} else if (len1 == len2) {
    std::swap_ranges(l, mid, h);
} else {
    std::iter_swap(l++, h++);
    int total = (--len1) + (--len2), gcd = total;
    for (int i = len1; i != 0; i--) {
        std::swap(gcd %= i, i);
    }
    int skip = total / gcd - 1;
    for (int i = 0; i < gcd; i++) {
        It curr = (i < len1) ? (l + i) : (h + (i - len1));
        int k = i;
        auto prev = *curr;
        for (int j = 0; j < skip; j++) {
            k = (k + len1) % total;
            It next = (k < len1) ? (l + k) : (h + (k - len1));
            *curr = *next;
            curr = next;
        }
        *curr = prev;
    }
}
return true;
}

bool next_combination(int n, std::vector<int> &a) {
    int k = a.size();
    for (int i = k - 1; i >= 0; i--) {
        if (a[i] < n - k + i) {
            a[i]++;
            while (++i < k) {
                a[i] = a[i - 1] + 1;
            }
            return true;
        }
    }
    return false;
}

long long next_combination_mask(long long x) {
    if (x == 0) {
        return 0;
    }
    long long s = x & -x, r = x + s;
    return r | (((x ^ r) >> 2) / s);
}

long long n_choose_k(long long n, long long k) {
    if (k > n - k) {
        k = n - k;
    }
    long long res = 1;
    for (int i = 0; i < k; i++) {
        res = res * (n - i) / (i + 1);
    }
    return res;
}

std::vector<int> combination_by_rank(int n, int k, long long r) {
    std::vector<int> res(k);
    int count = n;
    for (int i = 0; i < k; i++) {
        int j = 1;
        for (;;) {
            long long am = n_choose_k(count - j, k - 1 - i);

```

```

        if (r < am) {
            break;
        }
        r -= am;
    }
    res[i] = (i > 0) ? (res[i - 1] + j) : (j - 1);
    count -= j;
}
return res;
}

long long rank_by_combination(int n, const std::vector<int> &a) {
    int k = a.size();
    long long res = 0;
    int prev = -1;
    for (int i = 0; i < k; i++) {
        for (int j = prev + 1; j < a[i]; j++) {
            res += n_choose_k(n - 1 - j, k - 1 - i);
        }
        prev = a[i];
    }
    return res;
}

bool next_combination_with_repeats(int n, std::vector<int> &a) {
    int k = a.size();
    for (int i = k - 1; i >= 0; i--) {
        if (a[i] < n - 1) {
            for (++a[i]; ++i < k;) {
                a[i] = a[i - 1];
            }
            return true;
        }
    }
    return false;
}

```

Example Usage

```

#include <cassert>
#include <iostream>
using namespace std;

template<class It>
void print_range(It lo, It hi) {
    cout << "{";
    for (; lo != hi; ++lo) {
        cout << *lo << (lo == hi - 1 ? "" : ",");
    }
    cout << "} ";
}

int main() {
    {
        int k = 3;
        string s = "11234";
        cout << "\n" << s << "\n choose " << k << ":" << endl;
        do {
            cout << s.substr(0, k) << " ";
        } while (next_combination(s.begin(), s.begin() + k, s.end()));
        cout << endl;
    }
    { // Unordered combinations using masks.
        int n = 5, k = 3;
        string char_set = "abcde"; // Must be distinct.
        cout << "\n\n" << char_set << "\n choose " << k << " with masks:" << endl;
    }
}

```

```

long long mask = 0, dest = 0;
for (int i = 0; i < k; i++) {
    mask |= (1 << i);
}
for (int i = k - 1; i < n; i++) {
    dest |= (1 << i);
}
do {
    for (int i = 0; i < n; i++) {
        if ((mask >> i) & 1) {
            cout << char_set[i];
        }
    }
    cout << " ";
    mask = next_combination_mask(mask);
} while (mask != dest);
cout << endl;
}
{ // Combinations of distinct integers from 0 to n - 1.
    int n = 5, k = 3;
    vector<int> a{0, 1, 2};
    cout << "\n" << n << " choose " << k << ":" << endl;
    int count = 0;
    do {
        print_range(a.begin(), a.end());
        vector<int> b = combination_by_rank(n, k, count);
        assert(a == b);
        assert(rank_by_combination(n, a) == count);
        count++;
    } while (next_combination(n, a));
    cout << endl;
}
{ // Combinations with repeats.
    int n = 3, k = 2;
    vector<int> a{0, 0};
    cout << "\n" << n << " multichoose " << k << ":" << endl;
    do {
        print_range(a.begin(), a.end());
    } while (next_combination_with_repeats(n, a));
    cout << endl;
}
return 0;
}

```

Example Output

```

"11234" choose 3:
112 113 114 123 124 134 234

"abcde" choose 3 with masks:
abc abd acd bcd abe ace bce ade bde

5 choose 3:
{0,1,2} {0,1,3} {0,1,4} {0,2,3} {0,2,4} {0,3,4} {1,2,3} {1,2,4} {1,3,4} {2,3,4}

3 multichoose 2:
{0,0} {0,1} {0,2} {1,1} {1,2} {2,2}

```

6.2.5 Enumerating Partitions

A partition of a natural number n is a way to write n as a sum of positive integers where the order of the addends does not matter.

- `next_partition(p)` takes a reference to a vector `p` of positive integers as a partition of n for which the function will re-assign to become the next lexicographically greater partition. The function returns true if

such a partition exists, or false if `p` already consists of the lexicographically greatest partition (i.e. the single integer n).

- `partition_by_rank(n, r)` returns the partition of n that is lexicographically ranked r if addends in each partition were sorted in non-increasing order, where r is a zero-based rank in the range $[0, \text{partitions}(n))$.
- `rank_by_partition(p)` returns an integer representing the zero-based rank of the partition specified by vector `p`, which must consist of positive integers sorted in non-increasing order.
- `generate_increasing_partitions(n, f)` calls the function `f(lo, hi)` on strictly increasing partitions of n in lexicographically increasing order of partition, where `lo` and `hi` are random-access iterators to a range `[lo, hi)` of integers. Note that non-strictly increasing partitions like $\{1, 1, 1, 1\}$ are skipped.

Time Complexity:

- $O(n)$ per call to `next_partition()`.
- $O(n^2)$ per call to `partition_by_rank(n, r)` and `rank_by_partition(p)`.
- $O(p(n))$ per call to `generate_increasing_partitions(n, f)`, where $p(n)$ is the number of partitions of n .

Space Complexity:

- $O(1)$ auxiliary for `next_partition()`.
- $O(n^2)$ auxiliary heap space for `partition_function()`, `partition_by_rank()`, and `rank_by_partition()`.
- $O(n)$ auxiliary stack space for `generate_increasing_partitions()`.

```
#include <vector>

bool next_partition(std::vector<int> &p) {
    int n = p.size();
    if (n <= 1) {
        return false;
    }
    int s = p[n - 1] - 1, i = n - 2;
    p.pop_back();
    for (; i > 0 && p[i] == p[i - 1]; i--) {
        s += p[i];
        p.pop_back();
    }
    for (p[i]++; s > 0; s--) {
        p.push_back(1);
    }
    return true;
}

long long partition_function(int a, int b) {
    static std::vector<std::vector<long long>> p(1, std::vector<long long>(1, 1));
    if (a >= static_cast<int>(p.size())) {
        int old = p.size();
        p.resize(a + 1);
        p[0].resize(a + 1);
        for (int i = 1; i <= a; i++) {
            p[i].resize(a + 1);
            for (int j = old; j <= i; j++) {
                p[i][j] = p[i - 1][j - 1] + p[i - j][j];
            }
        }
    }
    return p[a][b];
}

std::vector<int> partition_by_rank(int n, long long r) {
    std::vector<int> res;
    for (int i = n, j; i > 0; i -= j) {
        for (j = 1;; j++) {
            long long count = partition_function(i, j);
            if (r < count) {
                break;
            }
        }
    }
}
```

```

        r -= count;
    }
    res.push_back(j);
}
return res;
}

long long rank_by_partition(const std::vector<int> &p) {
    long long res = 0;
    int sum = 0;
    for (int x : p) {
        sum += x;
    }
    for (int x : p) {
        for (int j = 0; j < x; j++) {
            res += partition_function(sum, j);
        }
        sum -= x;
    }
    return res;
}

using Fn = void (*)(std::vector<int>::iterator, std::vector<int>::iterator);

void generate_increasing_partitions(int left, int prev, int i, std::vector<int> &p, Fn f) {
    if (left == 0) {
        f(p.begin(), p.begin() + i);
        return;
    }
    for (p[i] = prev + 1; p[i] <= left; p[i]++) {
        generate_increasing_partitions(left - p[i], p[i], i + 1, p, f);
    }
}

void generate_increasing_partitions(int n, Fn f) {
    std::vector<int> p(n, 0);
    generate_increasing_partitions(n, 0, 0, p, f);
}

```

Example Usage

```

#include <cassert>
#include <iostream>
using namespace std;

template<class It>
void print_range(It lo, It hi) {
    cout << "{";
    for (; lo != hi; ++lo) {
        cout << *lo << (lo == hi - 1 ? "" : ",");
    }
    cout << "} ";
}

int main() {
    {
        int n = 4;
        vector<int> a(n, 1);
        cout << "Partitions of " << n << ":" << endl;
        int count = 0;
        do {
            print_range(a.begin(), a.end());
            vector<int> b = partition_by_rank(n, count);
            assert(equal(a.begin(), a.end(), b.begin()));
            assert(rank_by_partition(a) == count);
            count++;
        }
    }
}

```

```

    } while (next_partition(a));
    cout << endl;
}
{
    int n = 8;
    cout << "\nIncreasing partitions of " << n << ":" << endl;
    generate_increasing_partitions(n, print_range);
    cout << endl;
}
return 0;
}

```

Example Output

```

Partitions of 4:
{1,1,1,1} {2,1,1} {2,2} {3,1} {4}

Increasing partitions of 8:
{1,2,5} {1,3,4} {1,7} {2,6} {3,5} {8}

```

6.2.6 Enumerating Generic Combinatorial Sequences

Enumerate combinatorial sequences by inheriting an abstract class. Child classes of `AbstractEnumerator` must implement the `count()` function which should return the number of combinatorial sequences starting with the given prefix.

- `to_rank(a)` returns an integer representing the zero-based rank of the combinatorial sequence `a`.
- `from_rank(r)` returns a combinatorial sequence of integers that is lexicographically ranked `r`, where `r` is a zero-based rank in the range `[0, total_count())`.
- `enumerate(f)` calls the function `f(lo, hi)` on every specified combinatorial sequence in lexicographically increasing order, where `lo` and `hi` are two random-access iterators to a range `[lo, hi)` of integers.

Time Complexity: $O(n^2)$ calls will be made to `count()` per call to all operations, where n is the length of the combinatorial sequence.

Space Complexity: $O(n)$ auxiliary heap space per call to all operations.

```

#include <vector>

class AbstractEnumerator {
protected:
    int range, length;

    AbstractEnumerator(int r, int l) : range(r), length(l) {}
    virtual ~AbstractEnumerator() = default;
    virtual long long count(const std::vector<int> &prefix) { return 0; }
    std::vector<int> next(std::vector<int> &a) { return from_rank(to_rank(a) + 1); }
    long long total_count() { return count(std::vector<int>(0)); }

public:
    long long to_rank(const std::vector<int> &a) {
        long long res = 0;
        for (int i = 0; i < static_cast<int>(a.size()); i++) {
            std::vector<int> prefix(a.begin(), a.end());
            prefix.resize(i + 1);
            for (prefix[i] = 0; prefix[i] < a[i]; prefix[i]++) {
                res += count(prefix);
            }
        }
        return res;
    }

    std::vector<int> from_rank(long long r) {
        std::vector<int> a(length);
    }
}

```



```

    for (int i = 0; i < static_cast<int>(a.size()); i++) {
        std::vector<int> prefix(a.begin(), a.end());
        prefix.resize(i + 1);
        for (prefix[i] = 0; prefix[i] < range; ++prefix[i]) {
            long long curr = count(prefix);
            if (r < curr) {
                break;
            }
            r -= curr;
        }
        a[i] = prefix[i];
    }
    return a;
}

// Accepts any callable f(lo, hi), including capturing lambdas and functors.
template<class Fn>
void enumerate(Fn f) {
    long long total = total_count();
    for (long long i = 0; i < total; i++) {
        std::vector<int> curr = from_rank(i);
        f(curr.begin(), curr.end());
    }
}

};

class ArrangementEnumerator : public AbstractEnumerator {
public:
    ArrangementEnumerator(int n, int k) : AbstractEnumerator(n, k) {}

    long long count(const std::vector<int> &prefix) {
        int n = prefix.size();
        for (int i = 0; i < n - 1; i++) {
            if (prefix[i] == prefix[n - 1]) {
                return 0;
            }
        }
        long long res = 1;
        for (int i = 0; i < length - n; i++) {
            res *= range - n - i;
        }
        return res;
    }
};

class PermutationEnumerator : public ArrangementEnumerator {
public:
    PermutationEnumerator(int n) : ArrangementEnumerator(n, n) {}
};

class CombinationEnumerator : public AbstractEnumerator {
    std::vector<std::vector<long long>> table;

public:
    CombinationEnumerator(int n, int k)
        : AbstractEnumerator(n, k), table(n + 1, std::vector<long long>(n + 1)) {
        for (int i = 0; i <= n; i++) {
            for (int j = 0; j <= i; j++) {
                table[i][j] = (j == 0) ? 1 : table[i - 1][j - 1] + table[i - 1][j];
            }
        }
    }

    long long count(const std::vector<int> &prefix) {
        int n = prefix.size();
        if (n >= 2 && prefix[n - 1] <= prefix[n - 2]) {
            return 0;
        }
    }
};

```

```

    if (n == 0) {
        return table[range][length - n];
    }
    return table[range - prefix[n - 1] - 1][length - n];
}
};

class PartitionEnumerator : public AbstractEnumerator {
    std::vector<std::vector<long long>> table;

public:
    PartitionEnumerator(int n)
        : AbstractEnumerator(n + 1, n), table(n + 1, std::vector<long long>(n + 1)) {
        std::vector<std::vector<long long>> tmp(table);
        tmp[0][0] = 1;
        for (int i = 1; i <= n; i++) {
            for (int j = 1; j <= i; j++) {
                tmp[i][j] = tmp[i - 1][j - 1] + tmp[i - j][j];
            }
        }
        for (int i = 1; i <= n; i++) {
            for (int j = 1; j <= n; j++) {
                table[i][j] = tmp[i][j] + table[i][j - 1];
            }
        }
    }

    long long count(const std::vector<int> &prefix) {
        int n = static_cast<int>(prefix.size()), sum = 0;
        for (int x : prefix) {
            sum += x;
        }
        if (sum == range - 1) {
            return 1;
        }
        if (sum > range - 1 || (n > 0 && prefix[n - 1] == 0) ||
            (n >= 2 && prefix[n - 1] > prefix[n - 2])) {
            return 0;
        }
        if (n == 0) {
            return table[range - sum - 1][range - 1];
        }
        return table[range - sum - 1][prefix[n - 1]];
    }
};

```

Example Usage

```

#include <iostream>
using namespace std;

template<class It>
void print_range(It lo, It hi) {
    cout << "{";
    for (; lo != hi; ++lo) {
        cout << *lo << (lo == hi - 1 ? " " : ",");
    }
    cout << "} ";
}

int main() {
    {
        cout << "3 permute 2 arrangements:" << endl;
        ArrangementEnumerator arr(3, 2);
        arr.enumerate([&auto lo, &auto hi] { print_range(lo, hi); });
        cout << endl;
    }
}

```

```

}
{
    cout << "\nPermutations of [0, 3):" << endl;
    PermutationEnumerator perm(3);
    perm.enumerate([](auto lo, auto hi) { print_range(lo, hi); });
    cout << endl;
}
{
    cout << "\n4 choose 3 combinations:" << endl;
    CombinationEnumerator comb(4, 3);
    comb.enumerate([](auto lo, auto hi) { print_range(lo, hi); });
    cout << endl;
}
{
    cout << "\nPartition of 4:" << endl;
    PartitionEnumerator part(4);
    part.enumerate([](auto lo, auto hi) { print_range(lo, hi); });
    cout << endl;
}
return 0;
}

```

Example Output

```

3 permute 2 arrangements:
{0,1} {0,2} {1,0} {1,2} {2,0} {2,1}

Permutations of [0, 3):
{0,1,2} {0,2,1} {1,0,2} {1,2,0} {2,0,1} {2,1,0}

4 choose 3 combinations:
{0,1,2} {0,1,3} {0,2,3} {1,2,3}

Partition of 4:
{1,1,1,1} {2,1,1,0} {2,2,0,0} {3,1,0,0} {4,0,0,0}

```

6.3 Number Theory

6.3.1 GCD, LCM, Mod Inverse, Chinese Remainder

Common number theory operations relating to modular arithmetic.

- `gcd(a, b)` and `gcd2(a, b)` both return the greatest common divisor of `a` and `b` using the Euclidean algorithm.
- `lcm(a, b)` returns the lowest common multiple of `a` and `b`.
- `extended_euclid(a, b)` and `extended_euclid2(a, b)` both return a pair (x, y) of integers such that $\text{gcd}(a, b) = ax + by$.
- `mod(a, b)` returns the value of $a \bmod b$ under the true Euclidean definition of modulo, that is, the smallest nonnegative integer m satisfying $a + bn = m$ for some integer n . Note that this is identical to the remainder operator `%` in C++ for nonnegative operands `a` and `b`, but the result will differ when an operand is negative.
- `mod_inverse(a, m)` and `mod_inverse2(a, m)` both return an integer x such that $ax \equiv 1 \pmod{m}$, where the arguments must satisfy $m > 0$ and $\text{gcd}(a, m) = 1$.
- `generate_inverse(p)` returns a vector v of integers where for each index i in the vector, $i \cdot v[i] \equiv 1 \pmod{p}$, where the argument p is prime.
- `simple_restore(a, p)` and `Garner_restore(a, p)` both return the solution x for the system of simultaneous congruences $x \equiv a[i] \pmod{p[i]}$ for all indices i in $[0, n)$, where `p` consists of pairwise coprime integers. The solution x is guaranteed to be unique by the Chinese remainder theorem.

Time Complexity:

- $O(\log(a + b))$ per call to `gcd(a, b)`, `gcd2(a, b)`, `lcm(a, b)`, `extended_euclid(a, b)`, `extended_euclid2(a, b)`, `mod_inverse(a, b)`, and `mod_inverse2(a, b)`.
- $O(1)$ for `mod(a, b)`.
- $O(p)$ for `generate_inverse(p)`.
- Exponential for `simple_restore(a, p)`.
- $O(n^2)$ for `garner_restore(a, p)`.

Space Complexity:

- $O(\log(a + b))$ auxiliary stack space for `gcd2(a, b)`, `extended_euclid2(a, b)`, and `mod_inverse2(a, b)`.
- $O(p)$ auxiliary heap space for `generate_inverse(p)`.
- $O(n)$ auxiliary heap space for `garner_restore(a, p)`.
- $O(1)$ auxiliary space for all other operations.

```
#include <cstdlib>
#include <utility>
#include <vector>

template<class Int>
Int gcd(Int a, Int b) {
    while (b != 0) {
        Int t = b;
        b = a % b;
        a = t;
    }
    return (a < 0 ? -a : a);
}

template<class Int>
Int gcd2(Int a, Int b) {
    return (b == 0) ? (a < 0 ? -a : a) : gcd(b, a % b);
}

template<class Int>
Int lcm(Int a, Int b) {
    if (a == 0 || b == 0) {
        return 0;
    }
    Int res = a / gcd(a, b) * b;
    return (res < 0 ? -res : res);
}

template<class Int>
std::pair<Int, Int> extended_euclid(Int a, Int b) {
    Int x = 1, y = 0, x1 = 0, y1 = 1;
    while (b != 0) {
        Int q = a / b, prev_x1 = x1, prev_y1 = y1, prev_b = b;
        x1 = x - q * x1;
        y1 = y - q * y1;
        b = a - q * b;
        x = prev_x1;
        y = prev_y1;
        a = prev_b;
    }
    return (a > 0) ? std::pair<Int, Int>{x, y} : std::pair<Int, Int>{-x, -y};
}

template<class Int>
std::pair<Int, Int> extended_euclid2(Int a, Int b) {
    if (b == 0) {
        return (a > 0) ? std::pair<Int, Int>{1, 0} : std::pair<Int, Int>{-1, 0};
    }
    std::pair<Int, Int> r = extended_euclid2(b, a % b);
    return {r.second, r.first - a / b * r.second};
}
```

```

}

template<class Int>
Int mod(Int a, Int m) {
    Int r = a % m;
    return (r >= 0) ? r : (r + m);
}

template<class Int>
Int mod_inverse(Int a, Int m) {
    a = mod(a, m);
    return (a == 0) ? 0 : mod((1 - m * mod_inverse(m % a, a)) / a, m);
}

template<class Int>
Int mod_inverse2(Int a, Int m) {
    return mod(extended_euclid(a, m).first, m);
}

std::vector<int> generate_inverse(int p) {
    std::vector<int> res(p);
    res[1] = 1;
    for (int i = 2; i < p; i++) {
        res[i] = (p - (p / i) * res[p % i] % p) % p;
    }
    return res;
}

long long simple_restore(const std::vector<int> &a, const std::vector<int> &p) {
    int n = a.size();
    long long res = 0, m = 1;
    for (int i = 0; i < n; i++) {
        while (res % p[i] != a[i]) {
            res += m;
        }
        m *= p[i];
    }
    return res;
}

long long garner_restore(const std::vector<int> &a, const std::vector<int> &p) {
    int n = a.size();
    if (n == 0) {
        return 0;
    }
    std::vector<long long> x(a.begin(), a.end());
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < i; j++) {
            x[i] =
                mod_inverse(static_cast<long long>(p[j]), static_cast<long long>(p[i])) * (x[i] - x[j]);
        }
        x[i] = (x[i] % p[i] + p[i]) % p[i];
    }
    long long res = x[0], m = 1;
    for (int i = 1; i < n; i++) {
        m *= p[i - 1];
        res += x[i] * m;
    }
    return res;
}

```

Example Usage

```

#include <cassert>
using namespace std;

```

```

int main() {
{
    for (int steps = 0; steps < 10000; steps++) {
        int a = rand() % 200 - 10, b = rand() % 200 - 10, g = gcd(a, b);
        assert(g == gcd2(a, b));
        if (g == 1 && b > 1) {
            int inv = mod_inverse(a, b);
            assert(inv == mod_inverse2(a, b) && mod(a * inv, b) == 1);
        }
        auto res = extended_euclid(a, b);
        assert(res == extended_euclid2(a, b));
        auto [rx, ry] = res;
        assert(g == a * rx + b * ry);
    }
}
{
    int p = 17;
    auto res = generate_inverse(p);
    for (int i = 0; i < p; i++) {
        if (i > 0) {
            assert(mod(i * res[i], p) == 1);
        }
    }
}
{
    vector<int> a{2, 3, 1}, m{3, 4, 5};
    int n = a.size();
    int x = simple_restore(a, m);
    assert(x == garner_restore(a, m));
    for (int i = 0; i < n; i++) {
        assert(mod(x, m[i]) == a[i]);
    }
    assert(x == 11);
}
return 0;
}

```

6.3.2 Modular Arithmetic

Wraps arithmetic modulo a compile-time constant in a small value type. This is a common contest helper for dynamic programming, combinatorics, polynomial operations, and any calculation where all answers are taken modulo a number such as $10^9 + 7$.

The implementation is intentionally close to common contest “Mint” templates: normalization happens at construction, arithmetic operators are overloaded, mixed integer operations are supported through implicit construction, and combinations can be computed from lazy factorial tables.

Division uses the extended Euclidean algorithm, so the divisor only needs to be coprime to the modulus. For the factorial-table combination helper, the usual contest assumption is that `MOD` is prime and the requested factorials are invertible modulo `MOD`.

- `Modular<T>(x)` constructs the residue class of integer `x` modulo `T::value`.
- `value()` and `operator()()` return the stored representative in `[0, MOD)`.
- `pow(e)` returns this value raised to nonnegative exponent `e`.
- `inv()` returns the multiplicative inverse, asserting it exists.
- Operators `+`, `-`, `*`, `/`, comparison, increment, decrement, and stream I/O are overloaded.
- `ModCombinatorics<Mint>::choose(n, k)` returns $\binom{n}{k}$ using lazy factorial and inverse-factorial tables.

Time Complexity:

- $O(1)$ per addition, subtraction, multiplication, comparison, and stream output.
- $O(\log e)$ per call to `pow(e)`.

- $O(\log \text{MOD})$ per call to `inv()` and division.
- $O(n)$ total table growth to answer combinations up to size `n`.

Space Complexity:

- $O(1)$ auxiliary for Modular arithmetic.
- $O(n)$ for factorial and inverse-factorial tables grown through `choose(n, k)`.

```
#include <cassert>
#include <iostream>
#include <vector>

template<class T>
T inverse(T a, T m) {
    T original_m = m;
    T u = 0, v = 1;
    while (a != 0) {
        T q = m / a;
        m -= q * a;
        T tmp = a;
        a = m;
        m = tmp;
        u -= q * v;
        tmp = v;
        v = u;
        u = tmp;
    }
    assert(m == 1);
    u %= original_m;
    return u < 0 ? u + original_m : u;
}

template<class T>
class Modular {
public:
    using value_type = typename T::value_type;

private:
    value_type v;

public:
    Modular(long long x = 0) { v = normalize(x); }

    static value_type mod() { return T::value; }

    static value_type normalize(long long x) {
        if (-static_cast<long long>(mod()) <= x && x < static_cast<long long>(mod())) {
            value_type y = (value_type)x;
            return y < 0 ? y + mod() : y;
        }
        x %= mod();
        if (x < 0) {
            x += mod();
        }
        return (value_type)x;
    }

    value_type value() const { return v; }
    value_type operator()() const { return v; }

    Modular pow(long long e) const {
        assert(e >= 0);
        Modular base = *this, res = 1;
        while (e > 0) {
            if (e & 1) {
                res *= base;
            }
            base *= base;
            e /= 2;
        }
        return res;
    }
};
```

```

    e >>= 1;
}
return res;
}

Modular inv() const {
    assert(v != 0);
    return Modular(inverse(static_cast<long long>(v), static_cast<long long>(mod())));
}

Modular &operator+=(const Modular &other) {
    v += other.v;
    if (v >= mod()) {
        v -= mod();
    }
    return *this;
}

Modular &operator-=(const Modular &other) {
    v -= other.v;
    if (v < 0) {
        v += mod();
    }
    return *this;
}

Modular &operator*=(const Modular &other) {
    v = normalize(static_cast<long long>(v) * other.v);
    return *this;
}

Modular operator-(const Modular &other) const { return Modular(-v); }
Modular &operator++() { return *this += 1; }
Modular &operator--() { return *this -= 1; }
Modular &operator/=(const Modular &other) { return *this *= other.inv(); }

Modular operator++(int) {
    Modular res = *this;
    ++*this;
    return res;
}

Modular operator--(int) {
    Modular res = *this;
    --*this;
    return res;
}

friend Modular operator+(Modular a, const Modular &b) { return a += b; }
friend Modular operator-(Modular a, const Modular &b) { return a -= b; }
friend Modular operator*(Modular a, const Modular &b) { return a *= b; }
friend Modular operator/(Modular a, const Modular &b) { return a /= b; }
friend bool operator==(const Modular &a, const Modular &b) { return a.v == b.v; }
friend bool operator!=(const Modular &a, const Modular &b) { return !(a == b); }
friend bool operator<(const Modular &a, const Modular &b) { return a.v < b.v; }
friend std::ostream &operator<<(std::ostream &os, const Modular &x) { return os << x.v; }

friend std::istream &operator>>(std::istream &is, Modular &x) {
    long long value;
    is >> value;
    x = Modular(value);
    return is;
}
};

template<class Mint>
class ModCombinatorics {
    std::vector<Mint> fact, inv_fact;

```



```

void ensure(int n) {
    auto old = static_cast<int>(fact.size());
    if (old > n) {
        return;
    }
    fact.resize(n + 1);
    inv_fact.resize(n + 1);
    for (int i = old; i <= n; i++) {
        fact[i] = fact[i - 1] * i;
    }
    inv_fact[n] = fact[n].inv();
    for (int i = n; i > old; i--) {
        inv_fact[i - 1] = inv_fact[i] * i;
    }
}

public:
ModCombinatorics() : fact(1, Mint(1)), inv_fact(1, Mint(1)) {}

Mint factorial(int n) {
    ensure(n);
    return fact[n];
}

Mint choose(int n, int k) {
    if (k < 0 || k > n) {
        return 0;
    }
    ensure(n);
    return fact[n] * inv_fact[k] * inv_fact[n - k];
}

Mint permute(int n, int k) {
    if (k < 0 || k > n) {
        return 0;
    }
    ensure(n);
    return fact[n] * inv_fact[n - k];
}

Mint multichoose(int n, int k) { return choose(n + k - 1, k); }
};

struct Mod1000000007 {
    using value_type = int;
    static const int value = 1000000007;
};

using Mint = Modular<Mod1000000007>;

```

Example Usage

```

#include <cassert>
#include <sstream>
using namespace std;

int main() {
    Mint a = 1000000008LL;
    Mint b = -2;
    assert(a.value() == 1);
    assert(b() == Mint::mod() - 2);
    assert(a + b == -1);
    assert(2 + Mint(3) == 5);
    assert(Mint(2) * 3 == 6);
    assert(Mint(2).pow(10) == 1024);
}

```

```

assert(Mint(5) / 5 == 1);

Mint x = 0;
x++;
++x;
assert(x == 2);
stringstream ss("1000000008");
ss >> x;
assert(x == 1);

ModCombinatorics<Mint> comb;
assert(comb.factorial(5) == 120);
assert(comb.choose(5, 2) == 10);
assert(comb.permute(5, 2) == 20);
assert(comb.multichoose(3, 2) == 6);
return 0;
}

```

6.3.3 Prime Generation

Generate prime numbers using the Sieve of Eratosthenes.

- `sieve(n)` returns a vector of all the primes less than or equal to `n`.
- `sieve(lo, hi)` returns a vector of all the primes in the range `[lo, hi]`.

Time Complexity:

- $O(n \log(\log(n)))$ per call to `sieve(n)`.
- $O(\sqrt{hi} * \log(\log(hi - lo)))$ per call to `sieve(lo, hi)`.

Space Complexity:

- $O(n)$ auxiliary heap space per call to `sieve(n)`.
- $O(hi - lo + \sqrt{hi})$ auxiliary heap space per call to `sieve(lo, hi)`.

```

#include <algorithm>
#include <cmath>
#include <vector>

std::vector<int> sieve(int n) {
    if (n < 2) {
        return {};
    }
    std::vector<bool> prime(n + 1, true);
    for (int i = 2; i <= n / i; i++) {
        if (prime[i]) {
            for (long long j = 1LL * i * i; j <= n; j += i) {
                prime[j] = false;
            }
        }
    }
    std::vector<int> res;
    for (int i = 2; i <= n; i++) {
        if (prime[i]) {
            res.push_back(i);
        }
    }
    return res;
}

std::vector<int> sieve(int lo, int hi) {
    if (hi < 2 || lo > hi) {
        return {};
    }
    int sqrt_hi = sqrt(hi);

```

```

while (1LL * (sqrt_hi + 1) * (sqrt_hi + 1) <= hi) {
    sqrt_hi++;
}
int fourth_root_hi = sqrt(sqrt_hi);
while (1LL * (fourth_root_hi + 1) * (fourth_root_hi + 1) <= sqrt_hi) {
    fourth_root_hi++;
}
std::vector<bool> prime1(sqrt_hi + 1, true), prime2(hi - lo + 1, true);
for (int i = 2; i <= fourth_root_hi; i++) {
    if (prime1[i]) {
        for (long long j = 1LL * i * i; j <= sqrt_hi; j += i) {
            prime1[j] = false;
        }
    }
}
for (int i = 2, len = hi - lo; i <= sqrt_hi; i++) {
    if (prime1[i]) {
        long long start = std::max(1LL * i * i, ((static_cast<long long>(lo) + i - 1) / i) * i);
        for (long long j = start; j <= hi; j += i) {
            prime2[j - lo] = false;
        }
    }
}
std::vector<int> res;
for (int i = (lo > 1) ? lo : 2; i <= hi; i++) {
    if (prime2[i - lo]) {
        res.push_back(i);
    }
}
return res;
}

```

Example Usage

```

#include <ctime>
#include <iostream>
using namespace std;

int main() {
    int pmax = 100000000;
    time_t start;
    double delta;

    start = clock();
    auto p = sieve(pmax);
    delta = static_cast<double>((clock() - start)) / CLOCKS_PER_SEC;
    cout << "sieve(n=" << pmax << "): " << delta << "s" << endl;

    int l = 1000000000, h = 1005000000;
    start = clock();
    p = sieve(l, h);
    delta = static_cast<double>((clock() - start)) / CLOCKS_PER_SEC;
    cout << "sieve([" << l << ", " << h << "]): " << delta << "s" << endl;
    return 0;
}

```

Example Output

```

sieve(n=100000000): 0.042641s
sieve([1000000000, 1005000000]): 0.01969s

```

6.3.4 Primality Testing

Determine whether an integer n is prime. This can be done deterministically by testing all numbers under $\sqrt{n}$ using trial division, probabilistically using the Miller-Rabin test, or deterministically using the Miller-Rabin test if the maximum input is known ($2^{63} - 1$ for the purposes here).

- `is_prime(n)` returns whether the integer `n` is prime using an optimized trial division technique based on the fact that all primes greater than 6 must take the form $6n + 1$ or $6n - 1$.
- `is_probable_prime(n, k)` returns true if the integer `n` is prime, or false with an error probability of $(1/4)^k$ if `n` is composite. In other words, the result is guaranteed to be correct if `n` is prime, but could be wrong with probability $(1/4)^k$ if `n` is composite. This implementation uses exponentiation by squaring to support all signed 64-bit integers (up to and including $2^{63} - 1$).
- `is_prime_fast(n)` returns whether the signed 64-bit integer `n` is prime using a fully deterministic version of the Miller-Rabin test.

Time Complexity:

- $O(\sqrt{n})$ per call to `is_prime(n)`.
- $O(k \log^3(n))$ per call to `is_probable_prime(n, k)`.
- $O(\log^3(n))$ per call to `is_prime_fast(n)`.

Space Complexity: $O(1)$ auxiliary space for all operations.

```
#include <random>

template<class Int>
bool is_prime(Int n) {
    if (n == 2 || n == 3) {
        return true;
    }
    if (n < 2 || n % 2 == 0 || n % 3 == 0) {
        return false;
    }
    for (Int i = 5, w = 4; i <= n / i; i += w) {
        if (n % i == 0) {
            return false;
        }
        w = 6 - w;
    }
    return true;
}

using uint64 = unsigned long long;

// Hot path: Can simply `return (unsigned __int128)x * n % m;` for a major constant-factor speedup
// if the GCC/Clang __int128 extension is available. Otherwise, use the portable double-and-add
// version below, which still won't overflow a signed 64-bit `long long`.
uint64 mulmod(uint64 x, uint64 n, uint64 m) {
    uint64 a = 0, b = x % m;
    for (; n > 0; n >>= 1) {
        if (n & 1) {
            a = (a + b) % m;
        }
        b = (b << 1) % m;
    }
    return a % m;
}

uint64 powmod(uint64 x, uint64 n, uint64 m) {
    uint64 a = 1, b = x;
    for (; n > 0; n >>= 1) {
        if (n & 1) {
            a = mulmod(a, b, m);
        }
        b = mulmod(b, b, m);
    }
}
```

```

    }
    return a % m;
}

uint64 rand64u() {
    static std::mt19937_64 rng(std::random_device{}());
    return rng();
}

bool is_probable_prime(long long n, int k = 20) {
    if (n == 2 || n == 3) {
        return true;
    }
    if (n < 2 || n % 2 == 0 || n % 3 == 0) {
        return false;
    }
    uint64 s = n - 1, p = n - 1;
    while (!(s & 1)) {
        s >>= 1;
    }
    for (int i = 0; i < k; i++) {
        uint64 x, r = powmod(rand64u() % (n - 3) + 2, s, n); // random witness base in [2, n - 2]
        for (x = s; x != p && r != 1 && r != p; x <= 1) {
            r = mulmod(r, r, n);
        }
        if (r != p && !(x & 1)) {
            return false;
        }
    }
    return true;
}

bool is_prime_fast(long long n) {
    static const int np = 12, p[] = {2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37};
    for (int i = 0; i < np; i++) {
        if (n % p[i] == 0) {
            return n == p[i];
        }
    }
    if (n < p[np - 1]) {
        return false;
    }
    uint64 t;
    int s = 0;
    for (t = n - 1; !(t & 1); t >>= 1) {
        s++;
    }
    for (int i = 0; i < np; i++) {
        uint64 r = powmod(p[i], t, n);
        if (r == 1) {
            continue;
        }
        bool ok = false;
        for (int j = 0; j < s && !ok; j++) {
            ok |= (r == (uint64)n - 1);
            r = mulmod(r, r, n);
        }
        if (!ok) {
            return false;
        }
    }
    return true;
}

```

Example Usage

```
#include <cassert>
#include <vector>

int main() {
    std::vector<long long> tests{
        -1,
        0,
        1,
        2,
        3,
        4,
        5,
        1000000LL,
        772023803LL,
        792904103LL,
        813815117LL,
        834753187LL,
        855718739LL,
        876717799LL,
        897746119LL,
        2147483647LL,
        5705234089LL,
        5914686649LL,
        6114145249LL,
        6339503641LL,
        6548531929LL
    };
    for (long long test : tests) {
        bool p = is_prime(test);
        assert(p == is_prime_fast(test));
        assert(p == is_probable_prime(test));
    }
    return 0;
}
```

6.3.5 Integer Factorization

Compute the prime factorization of an integer. In the following implementations, the prime factorization of n is represented as a sorted vector of prime integers which together multiply to n . Note that factors are duplicated in the vector in accordance to their multiplicity in the prime factorization of n . For 0, 1, and prime numbers, the prime factorization is considered to be a vector consisting of a single element - the input itself.

- `prime_factorize(n)` returns the prime factorization of `n` using trial division.
- `get_divisors(n)` returns a sorted vector of all (not merely prime) divisors of `n` using trial division.
- `fermat(n)` returns a factor of `n` (possibly 1 or itself) that is not necessarily prime. This algorithm is efficient for integers with two factors near $\sqrt{n}$, but is roughly as slow as trial division otherwise.
- `pollards_rho_brent(n)` returns a factor of `n` that is not necessarily prime using Pollard's rho algorithm with Brent's optimization. If `n` is prime, then `n` itself is returned. While this algorithm is non-deterministic and may fail to detect factors on certain runs of the same input, it can be placed in a loop to deterministically factor large integers, as done in `prime_factorize_big()`.
- `prime_factorize_big(n, trial_division_cutoff)` returns the prime factorization of a 64-bit integer `n` using a combination of trial division, the Miller-Rabin primality test, and Pollard's rho algorithm. `trial_division_cutoff` specifies the largest factor to test with trial division before falling back to the rho algorithm. This supports 64-bit integers up to and including $2^{63} - 1$.

Time Complexity:

- $O(\sqrt{n})$ per call to `prime_factorize(n)`, `get_divisors(n)`, and `fermat(n)`.
- Unknown, but approximately $O(n^{1/4})$ per call to `pollards_rho_brent(n)` and `prime_factorize_big(n)`.

Space Complexity: $O(f)$ auxiliary heap space for all operations, where f is the number of factors returned.

```

#include <algorithm>
#include <cmath>
#include <random>
#include <vector>

template<class Int>
std::vector<Int> prime_factorize(Int n) {
    if (n <= 3) {
        return std::vector<Int>(1, n);
    }
    std::vector<Int> res;
    for (Int i = 2;; i++) {
        int p = 0;
        Int q = n / i, r = n - q * i;
        if (i > q || (i == q && r > 0)) {
            break;
        }
        while (r == 0) {
            p++;
            n = q;
            q = n / i;
            r = n - q * i;
        }
        for (int j = 0; j < p; j++) {
            res.push_back(i);
        }
    }
    if (n > 1) {
        res.push_back(n);
    }
    return res;
}

template<class Int>
std::vector<Int> get_divisors(Int n) {
    if (n <= 1) {
        return (n < 1) ? std::vector<Int>() : std::vector<Int>(1, 1);
    }
    std::vector<Int> res;
    for (Int i = 1; i <= n / i; i++) {
        if (n % i == 0) {
            res.push_back(i);
            if (i != n / i) {
                res.push_back(n / i);
            }
        }
    }
    std::sort(res.begin(), res.end());
    return res;
}

long long fermat(long long n) {
    if (n % 2 == 0) {
        return 2;
    }
    long long x = sqrt(n), y = 0, r = x * x - y * y - n;
    while (r != 0) {
        if (r < 0) {
            r += x + x + 1;
            x++;
        } else {
            r -= y + y + 1;
            y++;
        }
    }
    return (x == y) ? (x + y) : (x - y);
}

```

```

}

using uint64 = unsigned long long;

// Hot path: Can simply `return (unsigned __int128)x * n % m;` for a major constant-factor speedup
// if the GCC/Clang __int128 extension is available. Otherwise, use the portable double-and-add
// version below, which still won't overflow a signed 64-bit `long long`.
uint64 mulmod(uint64 x, uint64 n, uint64 m) {
    uint64 a = 0, b = x % m;
    for (; n > 0; n >>= 1) {
        if (n & 1) {
            a = (a + b) % m;
        }
        b = (b << 1) % m;
    }
    return a % m;
}

uint64 powmod(uint64 x, uint64 n, uint64 m) {
    uint64 a = 1, b = x;
    for (; n > 0; n >>= 1) {
        if (n & 1) {
            a = mulmod(a, b, m);
        }
        b = mulmod(b, b, m);
    }
    return a % m;
}

uint64 rand64u() {
    static std::mt19937_64 rng(std::random_device{}());
    return rng();
}

uint64 gcd(uint64 a, uint64 b) {
    while (b != 0) {
        uint64 t = b;
        b = a % b;
        a = t;
    }
    return a;
}

long long pollards_rho_brent(long long n) {
    if (n % 2 == 0) {
        return 2;
    }
    uint64 y = rand64u() % (n - 1) + 1;
    uint64 c = rand64u() % (n - 1) + 1;
    uint64 m = rand64u() % (n - 1) + 1;
    uint64 g = 1, r = 1, q = 1, ys = 0, x = 0;
    for (r = 1; g == 1; r <= 1) {
        x = y;
        for (int i = 0; i < r; i++) {
            y = (mulmod(y, y, n) + c) % n;
        }
        for (long long k = 0; k < r && g == 1; k += m) {
            ys = y;
            long long lim = std::min(m, r - k);
            for (int j = 0; j < lim; j++) {
                y = (mulmod(y, y, n) + c) % n;
                q = mulmod(q, (x > y) ? (x - y) : (y - x), n);
            }
            g = gcd(q, n);
        }
    }
    if (g == n) {
        do {

```



```

        ys = (mulmod(ys, ys, n) + c) % n;
        g = gcd((x > ys) ? (x - ys) : (ys - x), n);
    } while (g <= 1);
}
return g;
}

bool is_prime_fast(long long n) {
    static const int np = 12, p[] = {2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37};
    for (int i = 0; i < np; i++) {
        if (n % p[i] == 0) {
            return n == p[i];
        }
    }
    if (n < p[np - 1]) {
        return false;
    }
    uint64 t;
    int s = 0;
    for (t = n - 1; !(t & 1); t >>= 1) {
        s++;
    }
    for (int i = 0; i < np; i++) {
        uint64 r = powmod(p[i], t, n);
        if (r == 1) {
            continue;
        }
        bool ok = false;
        for (int j = 0; j < s && !ok; j++) {
            ok |= (r == (uint64)n - 1);
            r = mulmod(r, r, n);
        }
        if (!ok) {
            return false;
        }
    }
    return true;
}

std::vector<long long> prime_factorize_big(
    long long n, long long trial_division_cutoff = 1000000LL
) {
    if (n <= 3) {
        return std::vector<long long>(1, n);
    }
    std::vector<long long> res;
    for (; n % 2 == 0; n /= 2) {
        res.push_back(2);
    }
    for (; n % 3 == 0; n /= 3) {
        res.push_back(3);
    }
    for (long long i = 5, w = 4; i <= trial_division_cutoff && i <= n / i; i += w) {
        for (; n % i == 0; n /= i) {
            res.push_back(i);
        }
        w = 6 - w;
    }
    for (long long p; n > trial_division_cutoff && !is_prime_fast(n); n /= p) {
        do {
            p = pollards_rho_brent(n);
        } while (p == n);
        res.push_back(p);
    }
    if (n != 1) {
        res.push_back(n);
    }
    sort(res.begin(), res.end());
}

```

```

    return res;
}

```

Example Usage

```

#include <cassert>
#include <set>
using namespace std;

void validate(long long n, const vector<long long> &factors) {
    if (n == 1 || is_prime_fast(n)) {
        assert(factors == vector<long long>(1, n));
        return;
    }
    long long prod = 1;
    for (long long f : factors) {
        assert(is_prime_fast(f));
        prod *= f;
    }
    assert(prod == n);
}

int main() {
    { // Small tests.
        for (int i = 1; i <= 10000; i++) {
            auto v1 = prime_factorize(static_cast<long long>(i));
            auto v2 = prime_factorize_big(i);
            validate(i, v1);
            assert(v1 == v2);
            auto d = get_divisors(i);
            set<int> s(d.begin(), d.end());
            assert(d.size() == s.size());
            for (int j = 1; j <= i; j++) {
                if (i % j == 0) {
                    assert(s.count(j));
                }
            }
        }
    }
    { // Fermat works best for numbers with two factors close to each other.
        long long n = 1000003LL * 100000037;
        assert(fermat(n) == 1000003);
    }
    { // Large tests.
        const vector<long long> tests{
            3LL * 3 * 5 * 7 * 9949 * 9967 * 1000003,
            2LL * 1000003 * 1000000007,
            999961LL * 1000033,
            357267896789127671LL,
            2LL * 2 * 2 * 2 * 2 * 2 * 2 * 2 * 3 * 3 * 3 * 3 * 3 * 5 * 5 * 7 * 7 * 11 * 13 * 17 * 19 * 23 * 29 *
            31 * 37,
            2LL * 2 * 2 * 2 * 2 * 2 * 2 * 2 * 3 * 3 * 3 * 3 * 3 * 5 * 5 * 7 * 7 * 35336848213,
            2LL * 2 * 2 * 2 * 2 * 2 * 2 * 2 * 3 * 3 * 3 * 3 * 3 * 5 * 5 * 7 * 7 * 186917 * 186947,
        };
        for (const auto t : tests) {
            validate(t, prime_factorize_big(t));
        }
    }
    return 0;
}

```

6.3.6 Euler's Totient Function

Euler's totient function `phi(n)` returns the number of positive integers less than or equal to n that are relatively prime to n . That is, `phi(n)` is the number of integers k in the range $[1, n]$ for which $\gcd(n, k) = 1$. The computation of `phi(1..n)` can be performed simultaneously, as done so by `phi_table(n)` which returns a vector `v` such that `v[i]` stores `phi(i)` for i in the range $[0, n]$.

Time Complexity: $O(n \log(\log(n)))$ per call to `phi(n)` and `phi_table(n)`.

Space Complexity:

- $O(1)$ auxiliary space for `phi(n)`.
- $O(n)$ auxiliary heap space for `phi_table(n)`.

```
#include <numeric>
#include <vector>

int phi(int n) {
    int res = n;
    for (int i = 2; i <= n / i; i++) {
        if (n % i == 0) {
            while (n % i == 0) {
                n /= i;
            }
            res -= res / i;
        }
    }
    if (n > 1) {
        res -= res / n;
    }
    return res;
}

std::vector<int> phi_table(int n) {
    std::vector<int> res(n + 1);
    std::iota(res.begin(), res.end(), 0);
    for (int i = 1; i <= n; i++) {
        for (int j = 2 * i; j <= n; j += i) {
            res[j] -= res[i];
        }
    }
    return res;
}
```

Example Usage

```
#include <cassert>
using namespace std;

int main() {
    assert(phi(1) == 1);
    assert(phi(9) == 6);
    assert(phi(1234567) == 1224720);
    const int n = 1000;
    vector<int> v = phi_table(n);
    for (int i = 0; i <= n; i++) {
        assert(v[i] == phi(i));
    }
    return 0;
}
```

6.3.7 Binary Exponentiation

Given three unsigned 64-bit integers x , n , and m , `powmod()` returns x raised to the power of n (modulo m). `mulmod()` returns x multiplied by n (modulo m). Despite the fact that both functions use unsigned 64-bit integers for arguments and intermediate calculations, arguments x and n must not exceed $2^{63} - 1$ (the maximum value of a signed 64-bit integer) for the result to be correctly computed without overflow.

Binary exponentiation, also known as exponentiation by squaring, decomposes the exponentiation into a logarithmic number of multiplications while avoiding overflow. To further prevent overflow in the intermediate squaring computations, multiplication is performed using a similar principle of repeated addition.

Time Complexity: $O(\log n)$ per call to `mulmod()` and `powmod()`, where n is the second argument.

Space Complexity: $O(1)$ auxiliary.

```
using uint64 = unsigned long long;

// Hot path: Can simply `return (unsigned __int128)x * n % m;` for a major constant-factor speedup
// if the GCC/Clang __int128 extension is available. Otherwise, use the portable double-and-add
// version below, which still won't overflow a signed 64-bit `long long`.
uint64 mulmod(uint64 x, uint64 n, uint64 m) {
    uint64 a = 0, b = x % m;
    for (; n > 0; n >>= 1) {
        if (n & 1) {
            a = (a + b) % m;
        }
        b = (b << 1) % m;
    }
    return a % m;
}

uint64 powmod(uint64 x, uint64 n, uint64 m) {
    uint64 a = 1, b = x;
    for (; n > 0; n >>= 1) {
        if (n & 1) {
            a = mulmod(a, b, m);
        }
        b = mulmod(b, b, m);
    }
    return a % m;
}
```

Example Usage

```
#include <cassert>

int main() {
    assert(powmod(2, 10, 1000000007) == 1024);
    assert(powmod(2, 62, 1000000) == 387904);
    assert(powmod(10001, 10001, 100000) == 10001);
    return 0;
}
```

6.4 Arbitrary Precision Arithmetic

6.4.1 Big Integers (Simple)

Perform simple arithmetic operations on arbitrary precision big integers whose digits are internally represented as an `std::string` in little-endian order.

- `BigInt(n)` constructs a big integer from a long long (default: 0).
- `BigInt(s)` constructs a big integer from a string `s`, which must strictly consist of a sequence of numeric digits, optionally preceded by a minus sign.
- `str()` returns the string representation of the big integer.
- `comp(a, b)` returns -1 , 0 , or 1 depending on whether the big integers `a` and `b` compare less, equal, or greater, respectively.
- `add(a, b)` returns the sum of big integers `a` and `b`.
- `sub(a, b)` returns the difference of big integers `a` and `b`.
- `mul(a, b)` returns the product of big integers `a` and `b`.
- `div(a, b)` returns the quotient of big integers `a` and `b`.

Time Complexity:

- $O(n)$ per call to the constructor, `str()`, `comp()`, `add()`, and `sub()`, where n is total number of digits in the arguments and result for each operation.
- $O(n \cdot m)$ per call to `mul(a, b)` and `div(a, b)` where n is the number of digits in `a` and m is the number of digits in `b`.

Space Complexity:

- $O(n)$ for storage of the big integer, where n is the number of the digits.
- $O(n)$ auxiliary heap space for `str()`, `add()`, `sub()`, `mul()`, and `div()`, where n is the total number of digits in the arguments and result for each operation.

```
#include <algorithm>
#include <cctype>
#include <stdexcept>
#include <string>

class BigInt {
    std::string digits;
    int sign;

    void normalize() {
        size_t pos = digits.find_last_not_of('0');
        if (pos != std::string::npos) {
            digits.erase(pos + 1);
        } else {
            digits = "0"; // An all-zero (or empty) magnitude collapses to a single "0".
        }
        if (digits.size() == 1 && digits[0] == '0') {
            sign = 1;
        }
    }
}

static int comp(const std::string &a, const std::string &b, int asign, int bsign) {
    if (asign != bsign) {
        return asign < bsign ? -1 : 1;
    }
    if (a.size() != b.size()) {
        return a.size() < b.size() ? -asign : asign;
    }
    for (int i = static_cast<int>(a.size()) - 1; i >= 0; i--) {
        if (a[i] != b[i]) {
            return a[i] < b[i] ? -asign : asign;
        }
    }
    return 0;
}

static BigInt add(const std::string &a, const std::string &b, int asign, int bsign) {
    if (asign != bsign) {
        return (asign == 1) ? sub(a, b, asign, 1) : sub(b, a, bsign, 1);
    }
    BigInt res;
```

```

    res.sign = assign;
    res.digits.resize(std::max(a.size(), b.size()) + 1, '0');
    for (int i = 0, carry = 0; i < static_cast<int>(res.digits.size()); i++) {
        int d = carry;
        if (i < static_cast<int>(a.size())) {
            d += a[i] - '0';
        }
        if (i < static_cast<int>(b.size())) {
            d += b[i] - '0';
        }
        res.digits[i] = '0' + (d % 10);
        carry = d / 10;
    }
    res.normalize();
    return res;
}

static BigInt sub(const std::string &a, const std::string &b, int assign, int bsign) {
    if (assign == -1 || bsign == -1) {
        return add(a, b, assign, -bsign);
    }
    BigInt res;
    if (comp(a, b, assign, bsign) < 0) {
        res = sub(b, a, bsign, assign);
        res.sign = -1;
        return res;
    }
    res.digits.assign(a.size(), '0');
    for (int i = 0, borrow = 0; i < static_cast<int>(res.digits.size()); i++) {
        int d = (i < static_cast<int>(b.size())) ? a[i] - b[i] : a[i] - '0' - borrow;
        if (a[i] > '0') {
            borrow = 0;
        }
        if (d < 0) {
            d += 10;
            borrow = 1;
        }
        res.digits[i] = '0' + (d % 10);
    }
    res.normalize();
    return res;
}

public:
    BigInt(long long n = 0) {
        sign = (n < 0) ? -1 : 1;
        if (n == 0) {
            digits = "0";
            return;
        }
        for (n = (n > 0) ? n : -n; n > 0; n /= 10) {
            digits += '0' + (n % 10);
        }
        normalize();
    }

    BigInt(const std::string &s) {
        if (s.empty() || (s[0] == '-' && s.size() == 1)) {
            throw std::runtime_error("Invalid string format to construct BigInt.");
        }
        digits.assign(s.rbegin(), s.rend());
        if (s[0] == '-') {
            sign = -1;
            digits.erase(digits.size() - 1);
        } else {
            sign = 1;
        }
        if (digits.find_first_not_of("0123456789") != std::string::npos) {

```

```

        throw std::runtime_error("Invalid string format to construct BigInt.");
    }
    normalize();
}

std::string to_string() const {
    return (sign < 0 ? "-" : "") + std::string(digits.rbegin(), digits.rend());
}

friend int comp(const BigInt &a, const BigInt &b) {
    return comp(a.digits, b.digits, a.sign, b.sign);
}

friend BigInt add(const BigInt &a, const BigInt &b) {
    return add(a.digits, b.digits, a.sign, b.sign);
}

friend BigInt sub(const BigInt &a, const BigInt &b) {
    return sub(a.digits, b.digits, a.sign, b.sign);
}

friend BigInt mul(const BigInt &a, const BigInt &b) {
    BigInt res, row(a);
    for (char dc : b.digits) {
        for (int j = 0; j < (dc - '0'); j++) {
            res = add(res.digits, row.digits, res.sign, row.sign);
        }
        if (row.digits.size() > 1 || row.digits[0] != '0') {
            row.digits.insert(0, "0");
        }
    }
    res.sign = a.sign * b.sign;
    res.normalize();
    return res;
}

friend BigInt div(const BigInt &a, const BigInt &b) {
    if (b.digits.size() == 1 && b.digits[0] == '0') {
        throw std::runtime_error("Division by zero in BigInt.");
    }
    BigInt res, row;
    res.digits.assign(a.digits.size(), '0');
    row.digits.clear(); // Running remainder; kept normalized and nonnegative below.
    for (int i = static_cast<int>(a.digits.size()) - 1; i >= 0; i--) {
        row.digits.insert(row.digits.begin(), a.digits[i]);
        row.normalize(); // Strip the spurious high-order zero left by the previous remainder.
        // Subtract while the remainder is >= b. A strict ">" undercounts the quotient digit when the
        // remainder equals b, and leaves the remainder too large, overflowing the next digit past '9'.
        while (comp(row.digits, b.digits, 1, 1) >= 0) {
            res.digits[i]++;
            row = sub(row.digits, b.digits, 1, 1);
        }
    }
    res.sign = a.sign * b.sign;
    res.normalize();
    return res;
}
};

```

Example Usage

```

#include <cassert>

int main() {
    BigInt a("-9899819294989142124"), b("12398124981294214");
    assert(add(a, b).to_string() == "-9887421170007847910");
}

```

```

assert(sub(a, b).to_string() == "-9912217419970436338");
assert(mul(a, b).to_string() == "-122739196911503356525379735104870536");
assert(div(a, b).to_string() == "-798");
assert(comp(a, b) == -1 && comp(a, a) == 0 && comp(b, a) == 1);
return 0;
}

```

6.4.2 Big Integers

Perform operations on arbitrary precision big integers internally represented as a vector of base-10⁹ digits in little-endian order. Typical arithmetic operations involving mixed numeric primitives and strings are supported using templates and operator overloading, as long as at least one operand is a `BigInt` at any given level of evaluation.

- `BigInt(n)` constructs a big integer from a long long (default: 0).
- `BigInt(s)` constructs a big integer from a C string or an `std::string` `s`.
- `operator=` is defined to copy from another big integer or to assign from an 64-bit integer primitive.
- `size()` returns the number of digits in the base-10 representation.
- Operators `>>` and `<<` are defined to support stream-based input and output.
- `v.to_string()`, `v.to_llong()`, `v.to_double()`, and `v.to_ldouble()` return the big integer `v` converted to an `std::string`, `long long`, `double`, and `long double` respectively. For the latter three data types, overflow behavior is based on that of inputting from `std::istream`.
- `v.abs()` returns the absolute value of big integer `v`.
- `a.comp(b)` returns `-1`, `0`, or `1` depending on whether the big integers `a` and `b` compare less, equal, or greater, respectively.
- Operators `<`, `>`, `<=`, `>=`, `==`, `!=`, `+`, `-`, `*`, `/`, `%`, `++`, `-`, `+=`, `-=`, `*=`, `/=`, and `%=` are defined analogous to those on integer primitives. Addition, subtraction, and comparisons are performed using the standard linear algorithms. Multiplication is performed using a combination of the grade school algorithm (for smaller inputs) and either the Karatsuba algorithm (if the `USE_FFT_MULT` flag is set to false) or the Schonhage-Strassen algorithm (if `USE_FFT_MULT` is set to true). Division and modulo are computed simultaneously using the grade school method.
- `a.div(b)` returns a pair consisting of the quotient and remainder.
- `v.pow(n)` returns `v` raised to the power of `n`.
- `v.sqrt()` returns the integral part of the square root of big integer `v`.
- `v.nth_root(n)` returns the integral part of the n th root of big integer `v`.
- `rand(n)` returns a random, positive big integer with n digits.

Time Complexity:

- $O(n)$ per call to the constructors, `size()`, `to_string()`, `to_llong()`, `to_double()`, `to_ldouble()`, `abs()`, `comp()`, `rand()`, and all comparison and arithmetic operators except multiplication, division, and modulo, where n is total number of digits in the arguments and result for each operation.
- $O(n \cdot \log(n) \cdot \log(\log(n)))$ or $O(n^{1.585})$ per call to multiplication operations, depending on whether `USE_FFT_MULT` is set to true or false.
- $O(n \cdot m)$ per call to division and modulo operations, where n and m are the number of digits in the dividend and divisor, respectively.
- $O(M(m) \log n)$ per call to `pow(n)`, where $M(m)$ is cost to multiply two big integers of length m , depending on the multiplication method used.

Space Complexity:

- $O(n)$ for storage of the big integer.
- $O(n)$ auxiliary heap space for negation, addition, subtraction, multiplication, division, `abs()`, `sqrt()`, `pow()`, and `nth_root()`.
- $O(1)$ auxiliary space for all other operations.


```

#include <algorithm>
#include <cmath>
#include <complex>
#include <cstdlib>
#include <cstring>
#include <iomanip>
#include <istream>
#include <ostream>
#include <sstream>
#include <stdexcept>
#include <string>
#include <utility>
#include <vector>

class BigInt {
    static const int BASE = 1000000000, BASE_DIGITS = 9;
    static const bool USE_FFT_MULT = true;

    using vint = std::vector<int>;
    using vll = std::vector<long long>;
    using vcd = std::vector<std::complex<double>>;

    vint digits;
    int sign;

    void normalize() {
        while (!digits.empty() && digits.back() == 0) {
            digits.pop_back();
        }
        if (digits.empty()) {
            sign = 1;
        }
    }

    void read(int n, const char *s) {
        sign = 1;
        digits.clear();
        int pos = 0;
        while (pos < n && (s[pos] == '-' || s[pos] == '+')) {
            if (s[pos] == '-') {
                sign = -sign;
            }
            pos++;
        }
        for (int i = n - 1; i >= pos; i -= BASE_DIGITS) {
            int x = 0;
            for (int j = std::max(pos, i - BASE_DIGITS + 1); j <= i; j++) {
                x = x * 10 + s[j] - '0';
            }
            digits.push_back(x);
        }
        normalize();
    }

    static int comp(const vint &a, const vint &b, int asign, int bsign) {
        if (asign != bsign) {
            return asign < bsign ? -1 : 1;
        }
        if (a.size() != b.size()) {
            return a.size() < b.size() ? -asign : asign;
        }
        for (int i = static_cast<int>(a.size()) - 1; i >= 0; i--) {
            if (a[i] != b[i]) {
                return a[i] < b[i] ? -asign : asign;
            }
        }
        return 0;
    }
}

```

```

static BigInt add(const vint &a, const vint &b, int assign, int bsign) {
    if (assign != bsign) {
        return (assign == 1) ? sub(a, b, assign, 1) : sub(b, a, bsign, 1);
    }
    BigInt res;
    res.digits = a;
    res.sign = assign;
    int carry = 0, size = static_cast<int>(std::max(a.size(), b.size()));
    for (int i = 0; i < size || carry; i++) {
        if (i == static_cast<int>(res.digits.size())) {
            res.digits.push_back(0);
        }
        res.digits[i] += carry + (i < static_cast<int>(b.size()) ? b[i] : 0);
        carry = (res.digits[i] >= BASE) ? 1 : 0;
        if (carry) {
            res.digits[i] -= BASE;
        }
    }
    return res;
}

static BigInt sub(const vint &a, const vint &b, int assign, int bsign) {
    if (assign == -1 || bsign == -1) {
        return add(a, b, assign, -bsign);
    }
    BigInt res;
    if (comp(a, b, assign, bsign) < 0) {
        res = sub(b, a, bsign, assign);
        res.sign = -1;
        return res;
    }
    res.digits = a;
    res.sign = assign;
    for (int i = 0, borrow = 0; i < static_cast<int>(a.size()) || borrow; i++) {
        res.digits[i] -= borrow + (i < static_cast<int>(b.size()) ? b[i] : 0);
        borrow = res.digits[i] < 0;
        if (borrow) {
            res.digits[i] += BASE;
        }
    }
    res.normalize();
    return res;
}

static vint convert_base(const vint &digits, int l1, int l2) {
    vll p(std::max(l1, l2) + 1);
    p[0] = 1;
    for (int i = 1; i < static_cast<int>(p.size()); i++) {
        p[i] = p[i - 1] * 10;
    }
    vint res;
    long long curr = 0;
    for (int i = 0, curr_digits = 0; i < static_cast<int>(digits.size()); i++) {
        curr += digits[i] * p[curr_digits];
        curr_digits += l1;
        while (curr_digits >= l2) {
            res.push_back(static_cast<int>((curr % p[l2])));
            curr /= p[l2];
            curr_digits -= l2;
        }
    }
    res.push_back(static_cast<int>(curr));
    while (!res.empty() && res.back() == 0) {
        res.pop_back();
    }
    return res;
}

```

```

template<class It>
static vll karatsuba(It alo, It ahi, It blo, It bhi) {
    int n = std::distance(alo, ahi), k = n / 2;
    vll res(n * 2);
    if (n <= 32) {
        for (int i = 0; i < n; i++) {
            for (int j = 0; j < n; j++) {
                res[i + j] += alo[i] * blo[j];
            }
        }
        return res;
    }
    auto a1b1 = karatsuba(alo, alo + k, blo, blo + k);
    auto a2b2 = karatsuba(alo + k, ahi, blo + k, bhi);
    vll a2(alo + k, ahi), b2(blo + k, bhi);
    for (int i = 0; i < k; i++) {
        a2[i] += alo[i];
        b2[i] += blo[i];
    }
    auto r = karatsuba(a2.begin(), a2.end(), b2.begin(), b2.end());
    for (int i = 0; i < static_cast<int>(a1b1.size()); i++) {
        r[i] -= a1b1[i];
        res[i] += a1b1[i];
    }
    for (int i = 0; i < static_cast<int>(a2b2.size()); i++) {
        r[i] -= a2b2[i];
        res[i + n] += a2b2[i];
    }
    for (int i = 0; i < static_cast<int>(r.size()); i++) {
        res[i + k] += r[i];
    }
    return res;
}

template<class It>
static vcd fft(It lo, It hi, bool invert = false) {
    int n = std::distance(lo, hi), k = 0, high1 = -1;
    while ((1 << k) < n) {
        k++;
    }
    std::vector<int> rev(n, 0);
    for (int i = 1; i < n; i++) {
        if (!(i & (i - 1))) {
            high1++;
        }
        rev[i] = rev[i ^ (1 << high1)];
        rev[i] |= (1 << (k - high1 - 1));
    }
    vcd roots(n), res(n);
    for (int i = 0; i < n; i++) {
        double alpha = 2 * 3.14159265358979323846 * i / n;
        roots[i] = std::complex<double>(cos(alpha), sin(alpha));
        res[i] = *(lo + rev[i]);
    }
    for (int len = 1; len < n; len <= 1) {
        vcd tmp(n);
        int rstep = roots.size() / (len <= 1);
        for (int pdest = 0; pdest < n; pdest += len) {
            int p = pdest;
            for (int i = 0; i < len; i++) {
                std::complex<double> c = roots[i * rstep] * res[p + len];
                tmp[pdest] = res[p] + c;
                tmp[pdest + len] = res[p] - c;
                pdest++;
                p++;
            }
        }
    }
}

```

```

        res.swap(tmp);
    }
    if (invert) {
        for (int i = 0; i < static_cast<int>(res.size()); i++) {
            res[i] /= n;
        }
        std::reverse(res.begin() + 1, res.end());
    }
    return res;
}

public:
BigInt() : sign(1) {}
BigInt(int v) { *this = static_cast<long long>(v); }
BigInt(long long v) { *this = v; }
BigInt(const char *s) { read(strlen(s), s); }
BigInt(const std::string &s) { read(s.size(), s.c_str()); }

BigInt &operator=(const BigInt &v) {
    sign = v.sign;
    digits = v.digits;
    return *this;
}

BigInt &operator=(long long v) {
    sign = 1;
    if (v < 0) {
        sign = -1;
        v = -v;
    }
    digits.clear();
    for (; v > 0; v /= BASE) {
        digits.push_back(v % BASE);
    }
    return *this;
}

int size() const {
    if (digits.empty()) {
        return 1;
    }
    std::ostringstream oss;
    oss << digits.back();
    return oss.str().length() + BASE_DIGITS * (digits.size() - 1);
}

friend std::istream &operator>>(std::istream &in, BigInt &v) {
    std::string s;
    in >> s;
    v.read(s.size(), s.c_str());
    return in;
}

friend std::ostream &operator<<(std::ostream &out, const BigInt &v) {
    if (v.sign == -1) {
        out << '-';
    }
    out << (v.digits.empty() ? 0 : v.digits.back());
    for (int i = static_cast<int>(v.digits.size()) - 2; i >= 0; i--) {
        out << std::setw(BASE_DIGITS) << std::setfill('0') << v.digits[i];
    }
    return out;
}

std::string to_string() const {
    std::ostringstream oss;
    if (sign == -1) {
        oss << '-';
    }

```

```

    }
    oss << (digits.empty() ? 0 : digits.back());
    for (int i = static_cast<int>(digits.size()) - 2; i >= 0; i--) {
        oss << std::setw(BASE_DIGITS) << std::setfill('0') << digits[i];
    }
    return oss.str();
}

long long to_llong() const {
    long long res = 0;
    for (int i = static_cast<int>(digits.size()) - 1; i >= 0; i--) {
        res = res * BASE + digits[i];
    }
    return res * sign;
}

double to_double() const {
    std::stringstream ss(to_string());
    double res;
    ss >> res;
    return res;
}

long double to_ldouble() const {
    std::stringstream ss(to_string());
    long double res;
    ss >> res;
    return res;
}

int comp(const BigInt &v) const { return comp(digits, v.digits, sign, v.sign); }
bool operator<(const BigInt &v) const { return comp(v) < 0; }
bool operator>(const BigInt &v) const { return comp(v) > 0; }
bool operator<=(const BigInt &v) const { return comp(v) <= 0; }
bool operator>=(const BigInt &v) const { return comp(v) >= 0; }
bool operator==(const BigInt &v) const { return comp(v) == 0; }
bool operator!=(const BigInt &v) const { return comp(v) != 0; }

template<class T> friend bool operator<(const T &a, const BigInt &b) { return BigInt(a) < b; }
template<class T> friend bool operator>(const T &a, const BigInt &b) { return BigInt(a) > b; }
template<class T> friend bool operator<=(const T &a, const BigInt &b) { return BigInt(a) <= b; }
template<class T> friend bool operator>=(const T &a, const BigInt &b) { return BigInt(a) >= b; }
template<class T> friend bool operator==(const T &a, const BigInt &b) { return BigInt(a) == b; }
template<class T> friend bool operator!=(const T &a, const BigInt &b) { return BigInt(a) != b; }

BigInt abs() const { BigInt res(*this); res.sign = 1; return res; }
BigInt operator-() const { BigInt res(*this); res.sign = -sign; return res; }
BigInt operator+(const BigInt &v) const { return add(digits, v.digits, sign, v.sign); }
BigInt operator-(const BigInt &v) const { return sub(digits, v.digits, sign, v.sign); }

void operator*=(int v) {
    if (v < 0) {
        sign = -sign;
        v = -v;
    }
    for (int i = 0, carry = 0; i < static_cast<int>(digits.size()) || carry; i++) {
        if (i == static_cast<int>(digits.size())) {
            digits.push_back(0);
        }
        long long curr = digits[i] * static_cast<long long>(v) + carry;
        carry = static_cast<int>((curr / BASE));
        digits[i] = static_cast<int>((curr % BASE));
    }
    normalize();
}

BigInt operator*(int v) const {
    BigInt res(*this);

```

```

    res *= v;
    return res;
}

BigInt operator*(const BigInt &v) const {
    static const int TEMP_BASE = 10000, TEMP_BASE_DIGITS = 4;
    vint a = convert_base(digits, BASE_DIGITS, TEMP_BASE_DIGITS);
    vint b = convert_base(v.digits, BASE_DIGITS, TEMP_BASE_DIGITS);
    int n = 1;
    while (n < 2 * static_cast<int>(std::max(a.size(), b.size()))) {
        n <= 1;
    }
    a.resize(n, 0);
    b.resize(n, 0);
    vll c;
    if (USE_FFT_MULT) {
        auto at = fft(a.begin(), a.end()), bt = fft(b.begin(), b.end());
        for (int i = 0; i < n; i++) {
            at[i] *= bt[i];
        }
        at = fft(at.begin(), at.end(), true);
        c.resize(n);
        for (int i = 0; i < n; i++) {
            c[i] = at[i].real() + 0.5;
        }
    } else {
        c = karatsuba(a.begin(), a.end(), b.begin(), b.end());
    }
    BigInt res;
    res.sign = sign * v.sign;
    for (int i = 0, carry = 0; i < static_cast<int>(c.size()); i++) {
        long long d = c[i] + carry;
        res.digits.push_back(d % TEMP_BASE);
        carry = d / TEMP_BASE;
    }
    res.digits = convert_base(res.digits, TEMP_BASE_DIGITS, BASE_DIGITS);
    res.normalize();
    return res;
}

BigInt &operator/=(int v) {
    if (v == 0) {
        throw std::runtime_error("Division by zero in BigInt.");
    }
    if (v < 0) {
        sign = -sign;
        v = -v;
    }
    for (int i = static_cast<int>(digits.size()) - 1, rem = 0; i >= 0; i--) {
        long long curr = digits[i] + rem * static_cast<long long>(BASE);
        digits[i] = static_cast<int>((curr / v));
        rem = static_cast<int>((curr % v));
    }
    normalize();
    return *this;
}

BigInt operator/(int v) const {
    BigInt res(*this);
    res /= v;
    return res;
}

int operator%(int v) const {
    if (v == 0) {
        throw std::runtime_error("Division by zero in BigInt.");
    }
    if (v < 0) {

```

```

    v = -v;
}
int m = 0;
for (int i = static_cast<int>(digits.size()) - 1; i >= 0; i--) {
    m = (digits[i] + m * static_cast<long long>(BASE)) % v;
}
return m * sign;
}

std::pair<BigInt, BigInt> div(const BigInt &v) const {
    if (v == 0) {
        throw std::runtime_error("Division by zero in BigInt.");
    }
    if (comp(digits, v.digits, 1, 1) < 0) {
        return {BigInt(0), *this};
    }
    int norm = BASE / (v.digits.back() + 1);
    BigInt an = abs() * norm, bn = v.abs() * norm, q, r;
    q.digits.resize(an.digits.size());
    for (int i = static_cast<int>(an.digits.size()) - 1; i >= 0; i--) {
        r *= BASE;
        r += an.digits[i];
        int s1 = (r.digits.size() <= bn.digits.size()) ? 0 : r.digits[bn.digits.size()];
        int s2 = (r.digits.size() <= bn.digits.size() - 1) ? 0 : r.digits[bn.digits.size() - 1];
        int d = (static_cast<long long>(s1) * BASE + s2) / bn.digits.back();
        for (r -= bn * d; r < 0; r += bn) {
            d--;
        }
        q.digits[i] = d;
    }
    q.sign = sign * v.sign;
    r.sign = sign;
    q.normalize();
    r.normalize();
    return {q, r / norm};
}

BigInt operator/(const BigInt &v) const { return div(v).first; }
BigInt operator%(const BigInt &v) const { return div(v).second; }
BigInt operator++(int){ BigInt t(*this); operator++; return t; }
BigInt operator--(int){ BigInt t(*this); operator--; return t; }
BigInt &operator++() { *this = *this + BigInt(1); return *this; }
BigInt &operator--() { *this = *this - BigInt(1); return *this; }
BigInt &operator+=(const BigInt &v) { *this = *this + v; return *this; }
BigInt &operator-=(const BigInt &v) { *this = *this - v; return *this; }
BigInt &operator*=(const BigInt &v) { *this = *this * v; return *this; }
BigInt &operator/=(const BigInt &v) { *this = *this / v; return *this; }
BigInt &operator%=(const BigInt &v) { *this = *this % v; return *this; }

template<class T>
friend BigInt operator+(const T &a, const BigInt &b) { return BigInt(a) + b; }

template<class T>
friend BigInt operator-(const T &a, const BigInt &b) { return BigInt(a) - b; }

BigInt pow(int n) const {
    if (n == 0) {
        return BigInt(1);
    }
    if (*this == 0 || n < 0) {
        return BigInt(0);
    }
    BigInt x(*this), res(1);
    for (; n != 0; n >>= 1) {
        if (n & 1) {
            res *= x;
        }
        x *= x;
    }
}

```

```

    }
    return res;
}

BigInt sqrt() const {
    if (sign == -1) {
        throw std::runtime_error("Cannot take square root of a negative number.");
    }
    BigInt v(*this);
    while (v.digits.empty() || v.digits.size() % 2 == 1) {
        v.digits.push_back(0);
    }
    int n = v.digits.size();
    int ldig = (int)::sqrt(static_cast<double>(v.digits[n - 1]) * BASE + v.digits[n - 2]);
    int norm = BASE / (ldig + 1);
    v *= norm;
    v *= norm;
    while (v.digits.empty() || v.digits.size() % 2 == 1) {
        v.digits.push_back(0);
    }
    BigInt r(static_cast<long long>(v.digits[n - 1]) * BASE + v.digits[n - 2]);
    int q = ldig = (int)::sqrt(static_cast<double>(v.digits[n - 1]) * BASE + v.digits[n - 2]);
    BigInt res;
    for (int j = n / 2 - 1; j >= 0; j--) {
        for (; q-- > 0) {
            BigInt r1 =
                (r - (res * 2 * BASE + q) * q) * BASE * BASE +
                (j > 0 ? static_cast<long long>(v.digits[2 * j - 1]) * BASE + v.digits[2 * j - 2] : 0);
            if (r1 >= 0) {
                r = r1;
                break;
            }
        }
        res = res * BASE + q;
        if (j > 0) {
            int sz1 = res.digits.size(), sz2 = r.digits.size();
            int d1 = (sz1 + 2 < sz2) ? r.digits[sz1 + 2] : 0;
            int d2 = (sz1 + 1 < sz2) ? r.digits[sz1 + 1] : 0;
            int d3 = (sz1 < sz2) ? r.digits[sz1] : 0;
            q = (static_cast<long long>(d1) * BASE * BASE + static_cast<long long>(d2) * BASE + d3) /
                (ldig * 2);
        }
    }
    res.normalize();
    return res / norm;
}

BigInt nth_root(int n) const {
    if (sign == -1 && n % 2 == 0) {
        throw std::runtime_error("Cannot take even root of a negative number.");
    }
    if (*this == 0 || n < 0) {
        return BigInt(0);
    }
    if (n >= size()) {
        int p = 1;
        while (comp(BigInt(p).pow(n)) > 0) {
            p++;
        }
        return comp(BigInt(p).pow(n)) < 0 ? p - 1 : p;
    }
    BigInt lo(BigInt(10).pow(static_cast<int>(ceil(static_cast<double>(size()) / n) - 1)),
        hi(lo * 10), mid;
    while (lo < hi) {
        mid = (lo + hi) / 2;
        int cmp = comp(digits, mid.pow(n).digits, 1, 1);
        if (lo < mid && cmp > 0) {
            lo = mid;
        }
    }
}

```



```

    } else if (mid < hi && cmp < 0) {
        hi = mid;
    } else {
        return (sign == -1) ? -mid : mid;
    }
}
return (sign == -1) ? -(mid + 1) : (mid + 1);
}

static BigInt rand(int n) {
    if (n == 0) {
        return BigInt(0);
    }
    std::string s(1, '1' + (::rand() % 9));
    for (int i = 1; i < n; i++) {
        s += '0' + (::rand() % 10);
    }
    return BigInt(s);
}

friend int comp(const BigInt &a, const BigInt &b) { return a.comp(b); }
friend BigInt abs(const BigInt &v) { return v.abs(); }
friend BigInt pow(const BigInt &v, int n) { return v.pow(n); }
friend BigInt sqrt(const BigInt &v) { return v.sqrt(); }
friend BigInt nth_root(const BigInt &v, int n) { return v.nth_root(n); }
};

```

Example Usage

```

#include <cassert>

int main() {
    BigInt a("-9899819294989142124"), b("12398124981294214");
    assert(a + b == "-9887421170007847910");
    assert(a - b == "-9912217419970436338");
    assert(a * b == "-122739196911503356525379735104870536");
    assert(a / b == "-798");
    assert(BigInt(20).pow(12345).size() == 16062);
    assert(BigInt("9812985918924981892491829").nth_root(4) == 1769906);
    for (int i = -100; i <= 100; i++) {
        if (i >= 0) {
            assert(BigInt(i).sqrt() == static_cast<int>(sqrt(i)));
        }
        for (int j = -100; j <= 100; j++) {
            assert(BigInt(i) + BigInt(j) == i + j);
            assert(BigInt(i) - BigInt(j) == i - j);
            assert(BigInt(i) * BigInt(j) == i * j);
            if (j != 0) {
                assert(BigInt(i) / BigInt(j) == i / j);
            }
            if (0 < i && i <= 10 && 0 < j && j <= 10) {
                assert(BigInt(i).nth_root(j) == static_cast<long long>((pow(i, 1.0 / j) + 1E-5)));
                long long p = 1;
                for (int k = 0; k < j; k++) {
                    p *= i;
                }
                assert(BigInt(i).pow(j) == p);
            }
        }
    }
    for (int i = 0; i < 20; i++) {
        int n = rand() % 100 + 1;
        BigInt a(BigInt::rand(n)), s(a.sqrt()), xx(s * s), yy(s + 1);
        yy *= yy;
        assert(xx <= a && a < yy);
        BigInt b(BigInt::rand(rand() % n + 1) + 1), q(a / b);
    }
}

```

```

    xx = q * b;
    yy = b * (q + 1);
    assert(a >= xx && a < yy);
}
BigInt x(-6);
assert(x.to_string() == "-6");
assert(x.to_llong() == -6LL);
assert(x.to_double() == -6.0);
assert(x.to_ldouble() == -6.0);
return 0;
}

```

6.4.3 Rational Numbers

Perform operations on Rational numbers internally represented as two integers, a numerator and a denominator. The template integer type must support streamed input/output, comparisons, and arithmetic operations. Overflow is not checked for in internal operations: comparisons and arithmetic cross-multiply numerators and denominators, so instantiate with a wider integer type (such as `__int128`) if the values may grow large.

- `Rational(n)` constructs a Rational with numerator `n` and denominator 1.
- `Rational(n, d)` constructs a Rational with numerator `n` and denominator `d`.
- `operator>>` inputs a Rational using the next integer from the stream as the numerator and 1 as the denominator.
- `operator<<` outputs a Rational as a string consisting of possibly a minus sign followed by the numerator, followed by a slash, followed by the denominator.
- `v.to_string()`, `v.to_llong()`, `v.to_double()`, and `v.to_ldouble()` return the rational `v` converted to an `std::string`, `long long`, `double`, and `long double` respectively.
- Operators `<`, `>`, `<=`, `>=`, `==`, `!=`, `+`, `-`, `*`, `/`, `%`, `++`, `-`, `+=`, `-=`, `*=`, `/=`, and `%=` are defined analogous to those on numerical primitives.

Time Complexity:

- $O(\log(n + d))$ per call to constructor `Rational(n, d)`.
- $O(1)$ per call to all other operations, assuming that corresponding operations on the template integer type are $O(1)$ as well.

Space Complexity:

- $O(1)$ for storage of the Rational.
- $O(1)$ auxiliary space for all operations.

```

#include <istream>
#include <ostream>
#include <sstream>
#include <string>

template<class Int = long long>
class Rational {
    Int num, den;

public:
    Rational() : num(0), den(1) {}
    Rational(const Int &n) : num(n), den(1) {}

    template<class T1, class T2>
    Rational(const T1 &n, const T2 &d) : num(n), den(d) {
        if (den == 0) {
            throw std::runtime_error("Division by zero in Rational.");
        }
        if (den < 0) {
            num = -num;
        }
    }
};

```

```

    den = -den;
}
Int a(num < 0 ? -num : num), b(den), tmp;
while (a != 0 && b != 0) {
    tmp = a % b;
    a = b;
    b = tmp;
}
Int gcd = (b == 0) ? a : b;
num /= gcd;
den /= gcd;
}

friend std::istream &operator>>(std::istream &in, Rational &r) {
    in >> r.num;
    r.den = 1;
    return in;
}

friend std::ostream &operator<<(std::ostream &out, const Rational &r) {
    out << r.num << "/" << r.den;
    return out;
}

std::string to_string() const {
    std::stringstream ss;
    ss << num << "/" << den;
    return ss.str();
}

// to_llong/to_double/to_ldouble round-trip through a stringstream so that any Int supporting
// streamed I/O (e.g. a big-integer type) converts, even without a direct cast to the target type.
long long to_llong() const {
    std::stringstream ss;
    ss << num << " " << den;
    long long n, d;
    ss >> n >> d;
    return n / d;
}

double to_double() const {
    std::stringstream ss;
    ss << num << " " << den;
    double n, d;
    ss >> n >> d;
    return n / d;
}

long double to_ldouble() const {
    long double n, d;
    std::stringstream ss;
    ss << num << " " << den;
    ss >> n >> d;
    return n / d;
}

bool operator<(const Rational &r) const { return num * r.den < r.num * den; }
bool operator>(const Rational &r) const { return r.num * den < num * r.den; }
bool operator<=(const Rational &r) const { return !(r < *this); }
bool operator>=(const Rational &r) const { return !(*this < r); }
bool operator==(const Rational &r) const { return num == r.num && den == r.den; }
bool operator!=(const Rational &r) const { return num != r.num || den != r.den; }

template<class T>
friend bool operator<(const T &a, const Rational &b) { return Rational(a) < b; }

template<class T>
friend bool operator>(const T &a, const Rational &b) { return Rational(a) > b; }

```

```

template<class T>
friend bool operator<=(const T &a, const Rational &b) { return Rational(a) <= b; }

template<class T>
friend bool operator>=(const T &a, const Rational &b) { return Rational(a) >= b; }

template<class T>
friend bool operator==(const T &a, const Rational &b) { return Rational(a) == b; }

template<class T>
friend bool operator!=(const T &a, const Rational &b) { return Rational(a) != b; }

Rational abs() const { return Rational(num < 0 ? -num : num, den); }
friend Rational abs(const Rational &r) { return r.abs(); }
Int floor() const { return num < 0 ? -((-num + den - 1) / den) : num / den; }
Int ceil() const { return num < 0 ? -(-num / den) : (num + den - 1) / den; }

Rational operator+(const Rational &r) const {
    return Rational(num * r.den + r.num * den, den * r.den);
}

Rational operator-(const Rational &r) const {
    return Rational(num * r.den - r.num * den, r.den * den);
}

Rational operator*(const Rational &r) const { return Rational(num * r.num, r.den * den); }
Rational operator/(const Rational &r) const { return Rational(num * r.den, den * r.num); }

Rational operator%(const Rational &r) const {
    return *this - r * Rational(num * r.den / (r.num * den), 1);
}

Rational operator-() const { return Rational(-num, den); }
Rational operator++(int) { Rational t(*this); operator++(); return t; }
Rational operator--(int) { Rational t(*this); operator--(); return t; }
Rational &operator++() { *this = *this + 1; return *this; }
Rational &operator--() { *this = *this - 1; return *this; }
Rational &operator+=(const Rational &r) { *this = *this + r; return *this; }
Rational &operator-=(const Rational &r) { *this = *this - r; return *this; }
Rational &operator*=(const Rational &r) { *this = *this * r; return *this; }
Rational &operator/=(const Rational &r) { *this = *this / r; return *this; }
Rational &operator%=(const Rational &r) { *this = *this % r; return *this; }

template<class T>
friend Rational operator+(const T &a, const Rational &b) { return Rational(a) + b; }

template<class T>
friend Rational operator-(const T &a, const Rational &b) { return Rational(a) - b; }

template<class T>
friend Rational operator*(const T &a, const Rational &b) { return Rational(a) * b; }

template<class T>
friend Rational operator/(const T &a, const Rational &b) { return Rational(a) / b; }

template<class T>
friend Rational operator%(const T &a, const Rational &b) { return Rational(a) % b; }
};

```

Example Usage

```

#include <cassert>
#include <cmath>

int main() {

```

```

#define EQ(a, b) (fabs((a) - (b)) <= 1E-9)
using Rational = Rational<long long>;

assert(Rational(-21, 1) % 2 == -1);
Rational r(Rational(-53, 10) % Rational(-17, 10));
assert(EQ(r.to_ldouble(), fmod(-5.3, -1.7)));
assert(r.to_string() == "-1/5");
return 0;
}

```

6.5 Linear Algebra

6.5.1 Matrix Utilities

Basic matrix operations defined on a two-dimensional vector of numeric values.

- `make_matrix(r, c, v)` constructs and returns a matrix with `r` rows and `c` columns where the value at every index is initialized to `v`.
- `make_matrix(a)` returns a matrix constructed from the two-dimensional vector `a`.
- `identity_matrix(n)` returns the n by n identity matrix, that is, a matrix where `a[i][j]` equals 1 (if $i = j$), or 0 otherwise, for every i and j in $[0, n)$.
- `rows(a)` returns the number of rows r in an r by c matrix `a`.
- `columns(a)` returns the number of columns c in an r by c matrix `a`.
- `a[i][j]` may be used to access or modify the entry at row i , column j of an r by c matrix `a`, for every `i` in $[0, r)$ and `j` in $[0, c)$.
- Operators `<`, `>`, `<=`, `>=`, `==`, and `!=` define lexicographical comparison based on that of `std::vector`.
- Operators `+`, `-`, `*`, `/`, `+=`, `-=`, `*=`, and `/=` define scalar addition, subtraction, multiplication, and division involving a matrix and a numeric scalar value.
- Operators `*` and `*=` define vector and matrix multiplication.
- Operators `^` and `^=` define matrix exponentiation of a square matrix a by an integer power p .
- `power_sum(a, p)` returns the power sum of a square matrix a up to an integer power p , that is, $a + a^2 + \dots + a^p$.
- `transpose(a)` returns the transpose of an r by c matrix `a`, that is, a new c by r matrix b such that `a[i][j] = b[j][i]` for every `i` in $[0, r)$ and `j` in $[0, c)$.
- `transpose_in_place(a)` assigns the square matrix `a` to its transpose, returning a reference to the modified argument itself.
- `rotate(a, d)` returns the matrix `a` rotated `d` degrees clockwise. A negative `d` specifies a counter-clockwise rotation, and `d` must be a multiple of 90.
- `rotate_in_place(a, d)` assigns the square matrix `a` to its rotation by `d` degrees clockwise, returning a reference to the modified argument itself. A negative `d` specifies a counter-clockwise rotation, and `d` must be a multiple of 90.

Time Complexity:

- $O(n \cdot m)$ for construction, output, comparison, and scalar arithmetic of n by m matrices.
- $O(1)$ for `rows(a)` and `columns(a)`.
- $O(n \cdot m)$ for matrix-matrix addition and subtraction of n by m matrices.
- $O(n \cdot m \cdot \log(p))$ for exponentiation of an n by m matrix to power p .
- $O(n \cdot m \cdot \log^2(p))$ for power sum of an n by m matrix to power p .
- $O(n \cdot m \cdot k)$ for multiplication of an n by m matrix by an m by k matrix.
- $O(n \cdot m)$ for `transpose()`, `transpose_in_place()`, `rotate()`, and `rotate_in_place()` of n by m matrices.

Space Complexity:

- $O(1)$ auxiliary space for `rows()`, `columns()`, `a[i][j]` access, comparison operators, and in-place operations.

- $O(n \cdot m \cdot \log(p))$ auxiliary stack and heap space for exponentiation of an n by m matrix to power p , as well as the power sum of an n by m matrix up to power p .
- $O(n \cdot m)$ auxiliary heap space for all non-in-place operations returning an n by m matrix, `transpose()`, and `rotate()`.

```

#include <algorithm>
#include <cstdint>
#include <iomanip>
#include <ostream>
#include <stdexcept>
#include <vector>

using matrix = std::vector<std::vector<int>>>;

matrix make_matrix(int r, int c) {
    return matrix(r, matrix::value_type(c));
}

template<class T>
matrix make_matrix(int r, int c, const T &v) {
    return matrix(r, matrix::value_type(c, v));
}

template<class T>
matrix make_matrix(const std::vector<std::vector<T>>> &a) {
    int r = a.size(), c = a.empty() ? 0 : a[0].size();
    matrix res(r, matrix::value_type(c));
    for (int i = 0; i < r; i++) {
        for (int j = 0; j < c; j++) {
            res[i][j] = a[i][j];
        }
    }
    return res;
}

matrix identity_matrix(int n) {
    matrix res(n, matrix::value_type(n, 0));
    for (int i = 0; i < n; i++) {
        res[i][i] = 1;
    }
    return res;
}

int rows(const matrix &a) {
    return a.size();
}

int columns(const matrix &a) {
    return a.empty() ? 0 : a[0].size();
}

std::ostream &operator<<(std::ostream &out, const matrix &a) {
    static const int W = 10, P = 5;
    for (int i = 0; i < rows(a); i++) {
        for (int j = 0; j < columns(a); j++) {
            out << std::setw(W) << std::fixed << std::setprecision(P) << a[i][j];
        }
        out << std::endl;
    }
    return out;
}

template<class T>
matrix &operator+=(matrix &a, const T &v) {
    for (int i = 0; i < rows(a); i++) {
        for (int j = 0; j < columns(a); j++) {
            a[i][j] += v;
        }
    }
}

```

```

    }
}
return a;
}

template<class T>
matrix &operator-=(matrix &a, const T &v) {
    for (int i = 0; i < rows(a); i++) {
        for (int j = 0; j < columns(a); j++) {
            a[i][j] -= v;
        }
    }
    return a;
}

template<class T>
matrix &operator*=(matrix &a, const T &v) {
    for (int i = 0; i < rows(a); i++) {
        for (int j = 0; j < columns(a); j++) {
            a[i][j] *= v;
        }
    }
    return a;
}

template<class T>
matrix &operator/=(matrix &a, const T &v) {
    for (int i = 0; i < rows(a); i++) {
        for (int j = 0; j < columns(a); j++) {
            a[i][j] /= v;
        }
    }
    return a;
}

matrix &operator+=(matrix &a, const matrix &b) {
    if (rows(a) != rows(b) || columns(a) != columns(b)) {
        throw std::runtime_error("Invalid dimensions for matrix addition.");
    }
    for (int i = 0; i < rows(a); i++) {
        for (int j = 0; j < columns(a); j++) {
            a[i][j] += b[i][j];
        }
    }
    return a;
}

matrix &operator-=(matrix &a, const matrix &b) {
    if (rows(a) != rows(b) || columns(a) != columns(b)) {
        throw std::runtime_error("Invalid dimensions for matrix addition.");
    }
    for (int i = 0; i < rows(a); i++) {
        for (int j = 0; j < columns(a); j++) {
            a[i][j] -= b[i][j];
        }
    }
    return a;
}

matrix operator+(const matrix &a, const matrix &b) {
    matrix c(a);
    return c += b;
}

matrix operator-(const matrix &a, const matrix &b) {
    matrix c(a);
    return c -= b;
}

```

```

template<class T>
matrix &operator+=(matrix &a, const std::vector<T> &v) {
    if (columns(a) != static_cast<int>(v.size()) || v.empty()) {
        throw std::runtime_error("Invalid dimensions for matrix multiplication.");
    }
    for (int i = 0; i < rows(a); i++) {
        a[i][0] *= v[0];
        for (int j = 1; j < columns(a); j++) {
            a[i][j] += a[i][0] * v[j];
        }
    }
    for (int i = 0; i < rows(a); i++) {
        a[i].resize(1);
    }
    return a;
}

matrix operator*(const matrix &a, const matrix &b) {
    if (columns(a) != rows(b)) {
        throw std::runtime_error("Invalid dimensions for matrix multiplication.");
    }
    matrix res = make_matrix(rows(a), columns(b), 0);
    for (int i = 0; i < rows(a); i++) {
        for (int j = 0; j < columns(b); j++) {
            for (int k = 0; k < rows(b); k++) {
                res[i][j] += a[i][k] * b[k][j];
            }
        }
    }
    return res;
}

matrix &operator*(matrix &a, const matrix &b) { return a = a * b; }
template<class T> matrix operator+(const matrix &a, const T &v) { matrix m(a); return m += v; }
template<class T> matrix operator-(const matrix &a, const T &v) { matrix m(a); return m -= v; }
template<class T> matrix operator*(const matrix &a, const T &v) { matrix m(a); return m *= v; }
template<class T> matrix operator/(const matrix &a, const T &v) { matrix m(a); return m /= v; }
template<class T> matrix operator+(const T &v, const matrix &a) { return a + v; }
template<class T> matrix operator-(const T &v, const matrix &a) { return a - v; }
template<class T> matrix operator*(const T &v, const matrix &a) { return a * v; }
template<class T> matrix operator/(const T &v, const matrix &a) { return a / v; }

matrix operator^(const matrix &a, unsigned int p) {
    if (rows(a) != columns(a)) {
        throw std::runtime_error("Matrix must be square for exponentiation.");
    }
    if (p == 0) {
        return identity_matrix(rows(a));
    }
    return (p % 2 == 0) ? (a * a) ^ (p / 2) : a * (a ^ (p - 1));
}

matrix operator^(matrix &a, unsigned int p) {
    return a = a ^ p;
}

matrix power_sum(const matrix &a, unsigned int p) {
    if (rows(a) != columns(a)) {
        throw std::runtime_error("Matrix must be square for power_sum.");
    }
    if (p == 0) {
        return make_matrix(rows(a), rows(a));
    }
    return (p % 2 == 0) ? power_sum(a, p / 2) * (identity_matrix(rows(a)) + (a ^ (p / 2)))
        : (a + a * power_sum(a, p - 1));
}

```



```

matrix transpose(const matrix &a) {
    matrix res = make_matrix(columns(a), rows(a));
    for (int i = 0; i < rows(res); i++) {
        for (int j = 0; j < columns(res); j++) {
            res[i][j] = a[j][i];
        }
    }
    return res;
}

matrix &transpose_in_place(matrix &a) {
    if (rows(a) != columns(a)) {
        throw std::runtime_error("Matrix must be square for transpose_in_place.");
    }
    for (int i = 0; i < rows(a); i++) {
        for (int j = i + 1; j < columns(a); j++) {
            std::swap(a[i][j], a[j][i]);
        }
    }
    return a;
}

matrix rotate(const matrix &a, int degrees = 90) {
    if (degrees % 90 != 0) {
        throw std::runtime_error("Rotation must be by a multiple of 90 degrees.");
    }
    if (degrees < 0) {
        degrees = 360 - ((-degrees) % 360);
    }
    matrix res;
    switch (degrees % 360) {
        case 90: {
            res = make_matrix(columns(a), rows(a));
            for (int i = 0; i < columns(a); i++) {
                for (int j = 0; j < rows(a); j++) {
                    res[i][j] = a[rows(a) - j - 1][i];
                }
            }
            break;
        }
        case 180: {
            res = make_matrix(rows(a), columns(a));
            for (int i = 0; i < rows(a); i++) {
                for (int j = 0; j < columns(a); j++) {
                    res[i][j] = a[rows(a) - i - 1][columns(a) - j - 1];
                }
            }
            break;
        }
        case 270: {
            res = make_matrix(columns(a), rows(a));
            for (int i = 0; i < columns(a); i++) {
                for (int j = 0; j < rows(a); j++) {
                    res[i][j] = a[j][columns(a) - i - 1];
                }
            }
            break;
        }
        default: {
            res = a;
        }
    }
    return res;
}

matrix &rotate_in_place(matrix &a, int degrees = 90) {
    if (degrees % 90 != 0) {
        throw std::runtime_error("Rotation must be by a multiple of 90 degrees.");
    }

```

```

}
if (degrees % 180 != 0 && rows(a) != columns(a)) {
    throw std::runtime_error("Matrix must be square for rotate_in_place.");
}
if (degrees < 0) {
    degrees = 360 - ((-degrees) % 360);
}
int n = rows(a);
switch (degrees % 360) {
    case 90: {
        transpose_in_place(a);
        for (auto &row : a) {
            std::reverse(row.begin(), row.end());
        }
        break;
    }
    case 180: {
        // A 180-degree rotation is a horizontal flip (reverse each row) followed by a vertical flip
        // (reverse the row order). This works for non-square matrices too, unlike a row/column index
        // scheme keyed on a single dimension.
        for (auto &row : a) {
            std::reverse(row.begin(), row.end());
        }
        std::reverse(a.begin(), a.end());
        break;
    }
    case 270: {
        transpose_in_place(a);
        for (int j = 0; j < n; j++) {
            for (int i = 0, k = columns(a) - 1; i < k; i++, k--) {
                std::swap(a[i][j], a[k][j]);
            }
        }
        break;
    }
}
return a;
}

```

Example Usage

```

#include <cassert>
#include <iostream>
using namespace std;

int main() {
    matrix a{{1, 2, 3}, {4, 5, 6}};
    matrix a90{{4, 1}, {5, 2}, {6, 3}};
    matrix a180{{6, 5, 4}, {3, 2, 1}};
    matrix a270{{3, 6}, {2, 5}, {1, 4}};
    cout << a << endl;
    assert(rotate(a, -270) == a90);
    assert(rotate(a, -180) == a180);
    assert(rotate(a, -90) == a270);
    assert(rotate(a, 0) == a);
    assert(rotate(a, 90) == a90);
    assert(rotate(a, 180) == a180);
    assert(rotate(a, 270) == a270);
    assert(rotate(a, 360) == a);

    matrix b{{1, 2, 3}, {4, 5, 6}, {7, 8, 9}};
    for (int d = -360; d <= 360; d += 90) {
        matrix m = b;
        assert(rotate_in_place(m, d) == rotate(b, d));
    }
}

```

```

matrix m = make_matrix(5, 5, 10) + 10;
vector<int> v{1, 2, 3, 4, 5};
matrix mv{{300}, {300}, {300}, {300}, {300}};
assert(m * v == mv);

m[0][0] += 5;
assert(m[0][0] == 25 && m[1][1] == 20);
assert(power_sum(m, 3) == m + m * m + (m ^ 3));
return 0;
}

```

6.5.2 Row Reduction

Converts a matrix to reduced row echelon form using Gaussian elimination to solve a system of linear equations as well as compute the determinant. In practice, this method is prone to rounding error on certain matrices. For a more accurate algorithm for solving systems of linear equations, LU decomposition with row partial pivoting should be used.

- `row_reduce(a)` assigns the matrix `a` to its reduced row echelon form, returning a reference to the modified argument itself.
- `solve_system(a, b, &x)` solves the system of linear equations $ax = b$ given an r by c matrix `a` of real values, and a length r vector `b`, returning 0 if there is one solution, -1 if there are zero solutions, or -2 if there are infinite solutions. If there is exactly one solution, then the vector pointed to by `x` is populated with the solution vector of length c .

Time Complexity: $O(r^2 \cdot c)$ per call to `row_reduce(a)` and `solve_system(a)`, where r and c are the number of rows and columns of `a` respectively.

Space Complexity:

- $O(1)$ auxiliary for `row_reduce(a)`.
- $O(r \cdot c)$ auxiliary heap space for `solve_system(a)`.

```

#include <cmath>
#include <cstdint>
#include <stdexcept>
#include <vector>

const double EPS = 1e-9;

template<class Matrix>
Matrix &row_reduce(Matrix &a) {
    if (a.empty()) {
        return a;
    }
    int r = a.size(), c = a[0].size(), lead = 0;
    for (int row = 0; row < r && lead < c; row++) {
        int i = row;
        while (fabs(a[i][lead]) < EPS) {
            if (++i == r) {
                i = row;
                if (++lead == c) {
                    return a;
                }
            }
        }
        std::swap(a[i], a[row]);
        auto lv = a[row][lead];
        for (int j = 0; j < c; j++) {
            a[row][j] /= lv;
        }
        for (int i = 0; i < r; i++) {
            if (i != row) {

```

```

        lv = a[i][lead];
        for (int j = 0; j < c; j++) {
            a[i][j] -= lv * a[row][j];
        }
    }
    for (int j = 0; j < lead; j++) {
        a[row][j] = 0;
    }
    a[row][lead++] = 1;
}
return a;
}

template<class Matrix, class T>
int solve_system(const Matrix &a, const std::vector<T> &b, std::vector<T> *x) {
    if (x == nullptr || a.empty() || a.size() != b.size()) {
        return -1;
    }
    int r = a.size(), c = a[0].size();
    if (r < c) {
        return -2;
    }
    Matrix m(a);
    for (int i = 0; i < r; i++) {
        m[i].push_back(b[i]);
    }
    row_reduce(m);
    int rank = 0;
    for (int i = 0; i < r; i++) {
        int lead = -1;
        for (int j = 0; j < c && lead < 0; j++) {
            if (fabs(m[i][j]) > EPS) {
                lead = j;
            }
        }
        if (lead < 0) {
            // A zero coefficient row with a nonzero constant means 0 = nonzero (no solution).
            if (fabs(m[i][c]) > EPS) {
                return -1;
            }
        } else {
            rank++;
        }
    }
    // A unique solution requires a pivot in every column; fewer means free variables remain. This
    // also catches trailing free columns, which a per-row `lead > i` test alone would miss.
    if (rank < c) {
        return -2;
    }
    x->resize(c);
    for (int i = 0; i < c; i++) {
        (*x)[i] = m[i][c];
    }
    return 0;
}

```

Example Usage

```

#include <cassert>
using namespace std;

int main() {
    vector<vector<double>> a{{-1, 2, 5}, {1, 0, -6}, {-4, 2, 2}};
    vector<double> b{3, 1, -2};
    vector<double> x;
}

```

```

assert(solve_system(a, b, &x) == 0);
for (int i = 0; i < static_cast<int>(a.size()); i++) {
    double sum = 0;
    for (int j = 0; j < static_cast<int>(a[i].size()); j++) {
        sum += a[i][j] * x[j];
    }
    assert(fabs(sum - b[i]) < EPS);
}
return 0;
}

```

6.5.3 Determinant and Inverse

Computes the determinant and inverse of a square matrix using Gaussian elimination. The inverse of a matrix a is another matrix b such that ab equals the identity matrix. The inverse of a exists if and only if the determinant of a is nonzero. In this case, a is called invertible or non-singular. In practice, simple Gaussian elimination is prone to rounding error on certain matrices. For a more accurate algorithm for solving systems of linear equations, use LU decomposition with row partial pivoting.

- `det_naive(a)` returns the determinant of an n by n matrix `a`, using the classic divide-and-conquer algorithm by Laplace expansions.
- `det(a)` returns the determinant of an n by n matrix `a` using Gaussian elimination.
- `invert(a)` assigns the n by n matrix `a` to its inverse (if it exists), returning a reference to the modified argument itself. If `a` is not invertible, then its assigned values after the function call will be undefined (`+/-Inf` or `+/-NaN`).

Time Complexity:

- $O(n!)$ per call to `det_naive()`, where n is the dimension of the matrix.
- $O(n^3)$ per call to `det()` and `invert()` where n is the dimension of the matrix.

Space Complexity:

- $O(n)$ auxiliary stack space and $O(n! \cdot n)$ auxiliary heap space for `det_naive()`, where n is the dimension of the matrix.
- $O(n^2)$ auxiliary heap space for `det()` and `invert()`.

```

#include <cmath>
#include <vector>

template<class SquareMatrix>
double det_naive(const SquareMatrix &a) {
    int n = a.size();
    if (n == 1) {
        return a[0][0];
    }
    if (n == 2) {
        return a[0][0] * a[1][1] - a[0][1] * a[1][0];
    }
    double res = 0;
    SquareMatrix temp(n - 1, typename SquareMatrix::value_type(n - 1));
    for (int p = 0; p < n; p++) {
        int h = 0, k = 0;
        for (int i = 1; i < n; i++) {
            for (int j = 0; j < n; j++) {
                if (j == p) {
                    continue;
                }
                temp[h][k++] = a[i][j];
            }
            if (k == n - 1) {
                h++;
                k = 0;
            }
        }
        res += a[0][p] * det_naive(temp);
    }
    return res;
}

```

```

    }
    }
    }
    res += (p % 2 == 0 ? 1 : -1) * a[0][p] * det_naive(temp);
}
return res;
}

template<class SquareMatrix>
double det(const SquareMatrix &a, double EPS = 1e-10) {
    SquareMatrix b(a);
    int n = a.size();
    double res = 1.0;
    for (int i = 0; i < n; i++) {
        // Partial pivoting: bring the largest-magnitude entry in column i to the diagonal. Each row
        // swap negates the determinant, so the swap sign must be tracked. Selecting pivot rows out of
        // order without this sign correction can return |det| with the wrong sign.
        int p = i;
        for (int r = i + 1; r < n; r++) {
            if (fabs(b[r][i]) > fabs(b[p][i])) {
                p = r;
            }
        }
        if (fabs(b[p][i]) < EPS) {
            return 0;
        }
        if (p != i) {
            std::swap(b[p], b[i]);
            res = -res;
        }
        res *= b[i][i];
        for (int j = i + 1; j < n; j++) {
            double z = b[j][i] / b[i][i];
            for (int k = i; k < n; k++) {
                b[j][k] -= z * b[i][k];
            }
        }
    }
    return res;
}

template<class SquareMatrix>
SquareMatrix &invert(SquareMatrix &a) {
    int n = a.size();
    for (int i = 0; i < n; i++) {
        a[i].resize(2 * n);
        for (int j = n; j < n * 2; j++) {
            a[i][j] = (i == j - n ? 1 : 0);
        }
    }
    for (int i = 0; i < n; i++) {
        // Partial pivoting: without it a zero on the diagonal divides by zero and fails even for an
        // invertible matrix (e.g. a permutation matrix). Picking the largest-magnitude pivot also
        // improves numerical stability.
        int p = i;
        for (int r = i + 1; r < n; r++) {
            if (fabs(a[r][i]) > fabs(a[p][i])) {
                p = r;
            }
        }
        std::swap(a[p], a[i]);
        double z = a[i][i];
        for (int j = i; j < n * 2; j++) {
            a[i][j] /= z;
        }
        for (int j = 0; j < n; j++) {
            if (i != j) {
                double z = a[j][i];

```

```

        for (int k = 0; k < n * 2; k++) {
            a[j][k] -= z * a[i][k];
        }
    }
}
for (int i = 0; i < n; i++) {
    a[i].erase(a[i].begin(), a[i].begin() + n);
}
return a;
}

```

Example Usage

```

#include <cassert>
using namespace std;

int main() {
    vector<vector<double>> a{{6, 1, 1}, {4, -2, 5}, {2, 8, 7}};
    int n = static_cast<int>(a.size());
    vector<vector<double>> res(n, vector<double>(n, 0));
    double d = det(a);
    assert(fabs(d - det_naive(a)) < 1e-10);
    vector<vector<double>> inv = a;
    invert(inv);
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < n; j++) {
            for (int k = 0; k < n; k++) {
                res[i][j] += a[i][k] * inv[k][j];
            }
        }
    }
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < n; j++) {
            assert(fabs(res[i][j] - (i == j ? 1 : 0)) < 1e-10);
        }
    }
    return 0;
}

```

6.5.4 LU Decomposition

The LU decomposition of a matrix a with row-partial pivoting is a factorization of a (after some rows are possibly permuted by a permutation matrix p) as a product of a lower triangular matrix l and an upper triangular matrix u . This factorization can be used to tackle many common problems in linear algebra such as solving systems of linear equations and computing determinants. An improvement on basic row reduction, LU decomposition by row-partial pivoting keeps the relative magnitude of matrix values small, thus reducing the relative error due to rounding in computed solutions.

- `lu_decompose(a, &p1col)` assigns the r by c matrix `a` to merged LU decomposition matrix `lu`, returning either 0 or 1 denoting the “sign” of the permutation parity (0 if the number of overall row swaps performed is even, or 1 if it is odd), or -1 denoting a degenerate matrix (i.e. singular for square matrices). The merged matrix `lu` has `lu[i][j] = l[i][j]` for $i > j$ and `lu[i][j] = u[i][j]` for $i \leq j$. Note that the algorithm always yields an atomic lower triangular matrix for which the diagonal entries `l[i][i]` are always equal to 1, so this is not explicitly stored in the resulting merged matrix. For general i and j , the values of the lower and upper triangular matrices should be accessed via the `getl(lu, i, j)` and `getu(lu, i, j)` functions. Optionally, a `vector<int>` pointer `p1col` may be passed to return the permutation vector `p1col` where `p1col[i]` stores the only column that is equal to 1 in row i of the permutation matrix P (all other columns in row i of P are implicitly 0). The resulting P corresponds to `p1col` will satisfy $P * a = lu$.

- `solve_system(a, b, &x)` solves the system of linear equations $ax = b$ given an r by c matrix `a` of real values, and a length r vector `b`, returning 0 if there is one solution or -1 if there are zero or infinite solutions. If there is exactly one solution, then the vector pointed to by `x` is populated with the solution vector of length c .
- `det(a)` returns the determinant of an n by n matrix `a` using LU decomposition.
- `invert(a)` assigns the n by n matrix `a` to its inverse (if it exists), returning 0 if the inversion was successful or -1 if `a` has no inverse.

Time Complexity:

- $O(r^2 \cdot c)$ per call to `lu_decompose(a)` and `solve_system(a, b)`, where r and c are the number of rows and columns respectively, in accordance to the functions' descriptions above.
- $O(n^3)$ per call to `det(a)` and `inverse(a)`, where n is the dimension of `a`.

Space Complexity:

- $O(1)$ auxiliary for `lu_decompose()`.
- $O(n^2)$ for `det(a)` and `inverse(a)`.
- $O(r \cdot c)$ auxiliary heap space for `solve_system(a, b)`.

```
#include <algorithm>
#include <cmath>
#include <cstdint>
#include <limits>
#include <numeric>
#include <vector>

template<class Matrix>
int lu_decompose(Matrix &a, std::vector<int> *picol = nullptr, const double EPS = 1e-10) {
    int r = a.size(), c = a[0].size(), parity = 0;
    if (picol != nullptr) {
        picol->resize(r);
        std::iota(picol->begin(), picol->end(), 0);
    }
    for (int i = 0; i < r && i < c; i++) {
        int pi = i;
        for (int k = i + 1; k < r; k++) {
            if (fabs(a[k][i]) > fabs(a[pi][i])) {
                pi = k;
            }
        }
        if (fabs(a[pi][i]) < EPS) {
            return -1;
        }
        if (pi != i) {
            if (picol != nullptr) {
                std::iter_swap(picol->begin() + i, picol->begin() + pi);
            }
            std::iter_swap(a.begin() + i, a.begin() + pi);
            parity = 1 - parity;
        }
        for (int j = i + 1; j < r; j++) {
            a[j][i] /= a[i][i];
            for (int k = i + 1; k < c; k++) {
                a[j][k] -= a[j][i] * a[i][k];
            }
        }
    }
    return parity;
}

template<class Matrix>
double getl(const Matrix &lu, int i, int j) {
    return i > j ? lu[i][j] : (i < j ? 0 : 1);
}
```



```

template<class Matrix>
double getu(const Matrix &lu, int i, int j) {
    return i <= j ? lu[i][j] : 0;
}

template<class Matrix, class T>
int solve_system(
    const Matrix &a, const std::vector<T> &b, std::vector<T> *x, const double EPS = 1e-10
) {
    int r = a.size(), c = a[0].size();
    if (x == nullptr || a.empty() || a.size() != b.size() || r < c) {
        return -1;
    }
    x->resize(c);
    std::vector<int> pcol;
    Matrix lu;
    int status = lu_decompose(lu = a, &pcol, EPS);
    if (status < 0) {
        return status;
    }
    for (int i = 0; i < c; i++) {
        (*x)[i] = b[pcol[i]];
        for (int k = 0; k < i; k++) {
            (*x)[i] -= getl(lu, i, k) * (*x)[k];
        }
    }
    for (int i = c - 1; i >= 0; i--) {
        for (int k = i + 1; k < c; k++) {
            (*x)[i] -= getu(lu, i, k) * (*x)[k];
        }
        (*x)[i] /= getu(lu, i, i);
    }
    for (int i = 0; i < r; i++) {
        double val = 0;
        for (int j = 0; j < c; j++) {
            val += a[i][j] * (*x)[j];
        }
        // Mixed absolute/relative tolerance: dividing by b[i] alone would skip the check for negative
        // b[i] (ratio goes negative) and divide by zero when b[i] == 0.
        if (fabs(val - b[i]) > EPS * (1.0 + fabs(b[i]))) {
            return -1;
        }
    }
    return 0;
}

template<class T>
double det(const T &a) {
    int n = a.size();
    T lu;
    int status = lu_decompose(lu = a);
    if (status < 0) {
        return 0;
    }
    double res = 1;
    for (int i = 0; i < n; i++) {
        res *= lu[i][i];
    }
    return status == 0 ? res : -res;
}

template<class T>
int invert(T &a) {
    int n = a.size();
    std::vector<int> pcol;
    int status = lu_decompose(a, &pcol);
    if (status < 0) {
        return status;
    }

```

```

}
T ia(n, typename T::value_type(n, 0));
for (int j = 0; j < n; j++) {
    for (int i = 0; i < n; i++) {
        if (p1col[i] == j) {
            ia[i][j] = 1.0;
        } else {
            ia[i][j] = 0.0;
        }
        for (int k = 0; k < i; k++) {
            ia[i][j] -= get1(a, i, k) * ia[k][j];
        }
    }
    for (int i = n - 1; i >= 0; i--) {
        for (int k = i + 1; k < n; k++) {
            ia[i][j] -= getu(a, i, k) * ia[k][j];
        }
        ia[i][j] /= getu(a, i, i);
    }
}
a.swap(ia);
return 0;
}

```

Example Usage

```

#include <cassert>
using namespace std;

int main() {
    { // Solve a system.
        vector<vector<double>> a{{-1, 2, 5}, {1, 0, -6}, {-4, 2, 2}};
        vector<double> b{3, 1, -2};
        vector<double> x;
        assert(solve_system(a, b, &x) == 0);
        for (int i = 0; i < static_cast<int>(a.size()); i++) {
            double sum = 0;
            for (int j = 0; j < static_cast<int>(a[i].size()); j++) {
                sum += a[i][j] * x[j];
            }
            assert(fabs(sum - b[i]) < 1e-10);
        }
    }
    { // Find the determinant.
        vector<vector<double>> a{{1, 3, 5}, {2, 4, 7}, {1, 1, 0}};
        assert(fabs(det(a) - 4) < 1e-10);
    }
    { // Find the inverse.
        vector<vector<double>> a{{6, 1, 1}, {4, -2, 5}, {2, 8, 7}};
        vector<vector<double>> inv = a;
        int n = static_cast<int>(a.size());
        vector<vector<double>> res(n, vector<double>(n, 0));
        assert(invert(inv) == 0);
        for (int i = 0; i < n; i++) {
            for (int j = 0; j < n; j++) {
                for (int k = 0; k < n; k++) {
                    res[i][j] += a[i][k] * inv[k][j];
                }
            }
        }
        for (int i = 0; i < n; i++) {
            for (int j = 0; j < n; j++) {
                assert(fabs(res[i][j] - (i == j ? 1 : 0)) < 1e-10);
            }
        }
    }
}

```

```
    return 0;
}
```

6.6 Game Theory

6.6.1 Nim

Solves normal-play Nim and related impartial games where each position is the disjoint sum of independent piles. In normal play, the player who makes the last move wins. A Nim position is losing exactly when the xor of all pile sizes is 0.

This is the base case of the Sprague-Grundy theorem: every finite impartial game position is equivalent to a Nim pile of size equal to its Grundy number, and sums of games combine by xor.

- `nim_sum(piles)` returns the xor of all pile sizes.
- `first_player_wins(piles)` returns whether the player to move has a winning strategy.
- `winning_move(piles)` returns a move `(pile, new_size)` that moves to xor 0, or `(-1, -1)` if the position is losing.

Time Complexity: $O(n)$ per call, where n is the number of piles.

Space Complexity: $O(1)$ auxiliary.

```
#include <utility>
#include <vector>

int nim_sum(const std::vector<int> &piles) {
    int x = 0;
    for (int p : piles) {
        x ^= p;
    }
    return x;
}

bool first_player_wins(const std::vector<int> &piles) {
    return nim_sum(piles) != 0;
}

std::pair<int, int> winning_move(const std::vector<int> &piles) {
    int x = nim_sum(piles);
    if (x == 0) {
        return {-1, -1};
    }
    for (int i = 0; i < static_cast<int>(piles.size()); i++) {
        if (int target = piles[i] ^ x; target < piles[i]) {
            return {i, target};
        }
    }
    return {-1, -1};
}
```

Example Usage

```
#include <cassert>
using namespace std;

int main() {
    vector<int> piles{3, 4, 5};
    assert(nim_sum(piles) == 2);
    assert(first_player_wins(piles));
}
```

```

auto [pile_idx, new_size] = winning_move(piles);
piles[pile_idx] = new_size;
assert(nim_sum(piles) == 0);

vector<int> losing{1, 2, 3};
assert(!first_player_wins(losing));
assert(winning_move(losing).first == -1);
return 0;
}

```

6.6.2 Sprague-Grundy

Computes Grundy numbers for finite impartial games. The Grundy number $g[p]$ of a position p is the minimum excluded nonnegative integer (MEX) of the Grundy numbers reachable in one move. A position is losing exactly when its Grundy number is 0.

For a sum of independent impartial games, the combined Grundy number is the xor of the component Grundy numbers. This is the Sprague-Grundy theorem.

- `mex(values)` returns the minimum excluded nonnegative integer.
- `subtraction_game_grundy(max_stones, moves)` computes Grundy numbers for a game where a move subtracts one value in `moves` from the pile.
- `sum_grundy(grundies)` returns the xor of component Grundy numbers.

Time Complexity: $O(n \cdot m)$ for `subtraction_game_grundy(max_stones, moves)`, where n is `max_stones` and m is the number of allowed moves.

Space Complexity: $O(n + m)$ auxiliary heap space.

```

#include <vector>

int mex(const std::vector<int> &values) {
    std::vector<bool> seen(values.size() + 1, false);
    for (int v : values) {
        if (0 <= v && v < static_cast<int>(seen.size())) {
            seen[v] = true;
        }
    }
    for (int i = 0; i < static_cast<int>(seen.size()); i++) {
        if (!seen[i]) {
            return i;
        }
    }
    return seen.size();
}

std::vector<int> subtraction_game_grundy(int max_stones, const std::vector<int> &moves) {
    std::vector<int> grundy(max_stones + 1, 0);
    for (int stones = 1; stones <= max_stones; stones++) {
        std::vector<int> reachable;
        for (int m : moves) {
            if (m <= stones) {
                reachable.push_back(grundy[stones - m]);
            }
        }
        grundy[stones] = mex(reachable);
    }
    return grundy;
}

int sum_grundy(const std::vector<int> &grundies) {
    int x = 0;
    for (int g : grundies) {

```

```

    x ^= g;
}
return x;
}

```

Example Usage

```

#include <cassert>
using namespace std;

int main() {
    vector<int> moves{1, 3, 4};
    auto g = subtraction_game_grundy(10, moves);
    assert(g[0] == 0);
    assert(g[1] == 1);
    assert(g[2] == 0);
    assert(g[4] == 2);
    assert(g[7] == 0);

    vector<int> heaps{g[5], g[7], g[10]};
    assert(sum_grundy(heaps) == (g[5] ^ g[7] ^ g[10]));
    return 0;
}

```

6.6.3 Grundy Numbers on DAG

Computes Grundy numbers for impartial games whose positions form a directed acyclic graph. Each vertex is a game position, and each outgoing edge is a legal move. Terminal vertices have Grundy number 0; all other vertices take the MEX of their successors' Grundy numbers.

The implementation uses memoized DFS. It assumes the graph is acyclic; if cycles are present, the usual finite impartial-game Grundy recurrence is not directly valid without additional analysis.

- `grundy_on_dag(g)` returns the Grundy number of every vertex in graph `g`.
- `g[u]` contains all positions reachable from position `u` in one move.

Time Complexity: $O(n + m + s)$, where n is the number of vertices, m is the number of edges, and s is the total cost of MEX sets over outgoing edges.

Space Complexity: $O(n + d)$ auxiliary heap and stack space, where d is DFS depth.

```

#include <vector>

int grundy_dfs(int u, const std::vector<std::vector<int>> &g, std::vector<int> *memo) {
    if ((*memo)[u] != -1) {
        return (*memo)[u];
    }
    std::vector<bool> seen(g[u].size() + 1, false);
    for (int v : g[u]) {
        int x = grundy_dfs(v, g, memo);
        if (x < static_cast<int>(seen.size())) {
            seen[x] = true;
        }
    }
    int res = 0;
    while (res < static_cast<int>(seen.size()) && seen[res]) {
        res++;
    }
    (*memo)[u] = res;
    return res;
}

```

```
std::vector<int> grundy_on_dag(const std::vector<std::vector<int>>> &g) {
    std::vector<int> memo(g.size(), -1);
    for (int u = 0; u < static_cast<int>(g.size()); u++) {
        grundy_dfs(u, g, &memo);
    }
    return memo;
}
```

Example Usage

```
#include <cassert>
using namespace std;

int main() {
    vector<vector<int>> g{{1, 2}, {3}, {3, 4}, {}, {}};
    auto grundy = grundy_on_dag(g);
    assert(grundy[3] == 0);
    assert(grundy[4] == 0);
    assert(grundy[1] == 1);
    assert(grundy[2] == 1);
    assert(grundy[0] == 0);
    return 0;
}
```

6.6.4 Minimax and Alpha-Beta Pruning

Searches a finite two-player, zero-sum, perfect-information game tree using minimax and alpha-beta pruning. Minimax assumes both players play optimally: the maximizing player chooses the child with largest value, and the minimizing player chooses the child with smallest value.

Alpha-beta pruning computes the same value as minimax, but avoids exploring branches that cannot affect the final answer. It is most effective when good moves are searched first.

The example game is “take 1 or 2 stones”: starting with `stones` stones, players alternate taking 1 or 2 stones, and the player who takes the last stone wins. The evaluation returns 1 for a win for the player to move at the root and -1 for a loss.

- `minimax(stones, maximizing)` returns the exact game-tree value.
- `alpha_beta(stones, maximizing, alpha, beta)` returns the same value, with pruning.
- `best_take(stones)` returns an optimal first move, either 1 or 2.

Time Complexity: $O(b^d)$ in the worst case, where b is branching factor and d is search depth. Alpha-beta can reduce this substantially with good move ordering.

Space Complexity: $O(d)$ auxiliary recursion stack space.

```
#include <algorithm>

int minimax(int stones, bool maximizing) {
    if (stones == 0) {
        return maximizing ? -1 : 1;
    }
    if (maximizing) {
        int best = -2;
        for (int take = 1; take <= 2 && take <= stones; take++) {
            best = std::max(best, minimax(stones - take, false));
        }
        return best;
    } else {
        int best = 2;
        for (int take = 1; take <= 2 && take <= stones; take++) {
```

```

        best = std::min(best, minimax(stones - take, true));
    }
    return best;
}

int alpha_beta(int stones, bool maximizing, int alpha = -2, int beta = 2) {
    if (stones == 0) {
        return maximizing ? -1 : 1;
    }
    if (maximizing) {
        int value = -2;
        for (int take = 1; take <= 2 && take <= stones; take++) {
            value = std::max(value, alpha_beta(stones - take, false, alpha, beta));
            alpha = std::max(alpha, value);
            if (alpha >= beta) {
                break;
            }
        }
        return value;
    } else {
        int value = 2;
        for (int take = 1; take <= 2 && take <= stones; take++) {
            value = std::min(value, alpha_beta(stones - take, true, alpha, beta));
            beta = std::min(beta, value);
            if (alpha >= beta) {
                break;
            }
        }
        return value;
    }
}

int best_take(int stones) {
    int best_move = 1, best_value = -2;
    for (int take = 1; take <= 2 && take <= stones; take++) {
        int value = alpha_beta(stones - take, false);
        if (value > best_value) {
            best_value = value;
            best_move = take;
        }
    }
    return best_move;
}

```

Example Usage

```

#include <cassert>

int main() {
    assert(minimax(1, true) == 1);
    assert(minimax(3, true) == -1);
    for (int stones = 1; stones <= 10; stones++) {
        assert(minimax(stones, true) == alpha_beta(stones, true));
    }
    assert(best_take(4) == 1); // Move to losing state with 3 stones.
    assert(best_take(5) == 2); // Move to losing state with 3 stones.
    return 0;
}

```

Chapter 7

Geometry

7.1 Geometry Primitives

7.1.1 Point

A 2D point class template supporting epsilon comparisons. The coordinate type `T` defaults to `double`. Integer types (e.g. `int`) fully support all exact operations. Floating-point-only operations (`norm`, `arg`, `normalize`, `rotateCW/CCW`, `reflect` across a line) return coordinates of type `fp_t`, which is `double` when `T` is integral and `T` itself otherwise.

Exact operations (return `point<T>` or `T`, no precision lost for integers):

- element-wise arithmetic, `dot()`, `cross()`, `sqnorm()`, cardinal rotations, `reflect(point)`, comparisons.

Overflow warning: the exact products `dot()`, `cross()`, and `sqnorm()` compute values on the order of the squared coordinate magnitude. With `point<int>` and coordinates around 10^4 to 10^5 they already overflow a 32-bit `int` (and `sqdist`-style differences overflow even sooner). For integer geometry with coordinates beyond a few thousand, use `PointL` (`point<long long>`) so these products stay exact.

Floating-point-only operations (return `point<fp_t>` or `fp_t`):

- `norm()`, `arg()`, `proj()`, `normalize()`, `rotateCW()`, `rotateCCW()`, `reflect(line)`.
- `operator/` also promotes to `fp_t`.

Implicit conversion from `point<U>` to `point<T>` is provided when `U` is integral and `T` is floating-point, so `PointI` can be passed wherever `PointD` is expected.

Non-floating-point coordinate types such as `Modular` or `Rational` compose for the exact operations (arithmetic, `dot`, `cross`, `sqnorm`, comparisons): these route to exact `==/<` rather than epsilon comparisons. The floating-point-only operations are simply not instantiated unless called.

Type aliases:

- `PointI = Point<int>`: exact integer geometry (small coordinates only; see overflow warning)
- `PointL = Point<long long>`: exact integer geometry for large coordinates
- `PointD = Point<double>`: standard floating-point
- `PointLD = Point<long double>`: extra precision
- `Point = PointD`: default point type is double

Time Complexity: $O(1)$ per call to the constructor and all other operations.

Space Complexity:

- $O(1)$ for storage of the point.
- $O(1)$ auxiliary for all operations.


```

#include <cmath>
#include <ostream>
#include <type_traits>
#include <utility>

const double EPS = 1e-9;

#define EQ(a, b) (fabs((a) - (b)) <= EPS)
#define LT(a, b) ((a) < (b) - EPS)

template<typename T>
struct point {
    // fp_t is T for floating-point types, double for integral types.
    using fp_t = std::conditional_t<std::is_integral<T>::value, double, T>;

    T x, y;

    point() : x(0), y(0) {}
    point(T x, T y) : x(x), y(y) {}
    explicit point(const std::pair<T, T> &p) : x(p.first), y(p.second) {}

    // Implicit conversion from integral to floating-point point (optional, can just use to_double()).
    template<
        typename U,
        typename = std::enable_if_t<std::is_integral<U>::value && std::is_floating_point<T>::value>>
    point(const point<U> &p) : x(static_cast<T>(p.x)), y(static_cast<T>(p.y)) {}

    bool operator==(const point &p) const {
        if constexpr (std::is_floating_point<T>::value) {
            return EQ(x, p.x) && EQ(y, p.y);
        }
        return x == p.x && y == p.y; // exact for ints and types like Modular/Rational
    }

    bool operator!=(const point &p) const { return !(*this == p); }

    bool operator<(const point &p) const {
        if constexpr (std::is_floating_point<T>::value) {
            return EQ(x, p.x) ? LT(y, p.y) : LT(x, p.x);
        }
        return x == p.x ? y < p.y : x < p.x; // exact for ints and types like Modular/Rational
    }

    bool operator>(const point &p) const { return p < *this; }
    bool operator<=(const point &p) const { return !(*this > p); }
    bool operator>=(const point &p) const { return !(*this < p); }
    point operator+(const point &p) const { return {x + p.x, y + p.y}; }
    point operator-(const point &p) const { return {x - p.x, y - p.y}; }
    point operator+(T v) const { return {x + v, y + v}; }
    point operator-(T v) const { return {x - v, y - v}; }
    point operator*(T v) const { return {x * v, y * v}; }

    // Division always promotes to fp_t to avoid integer truncation.
    point<fp_t> operator/(fp_t v) const { return {(fp_t)x / v, (fp_t)y / v}; }

    point &operator+=(const point &p) { x += p.x; y += p.y; return *this; }
    point &operator-=(const point &p) { x -= p.x; y -= p.y; return *this; }
    point &operator+=(T v) { x += v; y += v; return *this; }
    point &operator-=(T v) { x -= v; y -= v; return *this; }
    point &operator*=(T v) { x *= v; y *= v; return *this; }
    friend point operator+(T v, const point &p) { return p + v; }
    friend point operator*(T v, const point &p) { return p * v; }

    // --- Exact operations: return T or point<T>, work for any coordinate type ---

    // Overflow warning: these products grow like the squared coordinate magnitude, so for integer T
    // with large coordinates (beyond a few thousand for 32-bit int) use a wider type such as PointL.
    T sqnorm() const { return x * x + y * y; }

```

```

T dot(const point &p) const { return x * p.x + y * p.y; }
T cross(const point &p) const { return x * p.y - y * p.x; }

// --- Floating-point operations: return fp_t or point<fp_t> ---

fp_t norm() const { return sqrt(static_cast<fp_t>(sqnorm())); }
fp_t arg() const { return atan2(static_cast<fp_t>(y), static_cast<fp_t>(x)); }
fp_t proj(const point &p) const { return static_cast<fp_t>(dot(p)) / p.norm(); }

point<fp_t> normalize() const {
    fp_t n = norm();
    if (n < static_cast<fp_t>(1e-30)) { // guard against dividing by a near-zero norm, not a
        return {fp_t(0), fp_t(0)}; // geometric tolerance (EPS is too coarse here)
    }
    return {static_cast<fp_t>(x) / n, static_cast<fp_t>(y) / n};
}

// --- Cardinal rotations: exact for all coordinate types including int ---

// Returns (x, y) rotated 90/180/270 degrees counter-clockwise about the origin.
point rotate90() const { return {-y, x}; }
point rotate180() const { return {-x, -y}; }
point rotate270() const { return {y, -x}; }

// Returns (x, y) rotated 90/180/270 degrees counter-clockwise about point p.
point rotate90(const point &p) const { return (*this - p).rotate90() + p; }
point rotate180(const point &p) const { return (*this - p).rotate180() + p; }
point rotate270(const point &p) const { return (*this - p).rotate270() + p; }

// --- Arbitrary-angle rotations: always return floating-point ---

// Returns (x, y) rotated t radians clockwise about the origin.
point<fp_t> rotateCW(fp_t t) const {
    fp_t fx = (fp_t)x, fy = (fp_t)y;
    return {fx * cos(t) + fy * sin(t), fy * cos(t) - fx * sin(t)};
}

// Returns (x, y) rotated t radians counter-clockwise about the origin.
point<fp_t> rotateCCW(fp_t t) const {
    fp_t fx = (fp_t)x, fy = (fp_t)y;
    return {fx * cos(t) - fy * sin(t), fx * sin(t) + fy * cos(t)};
}

// Returns (x, y) rotated t radians clockwise about point p.
point<fp_t> rotateCW(const point &p, fp_t t) const {
    return point<fp_t>{(fp_t)(x - p.x), (fp_t)(y - p.y)}.rotateCW(t) +
        point<fp_t>{(fp_t)p.x, (fp_t)p.y};
}

// Returns (x, y) rotated t radians counter-clockwise about point p.
point<fp_t> rotateCCW(const point &p, fp_t t) const {
    return point<fp_t>{(fp_t)(x - p.x), (fp_t)(y - p.y)}.rotateCCW(t) +
        point<fp_t>{(fp_t)p.x, (fp_t)p.y};
}

// --- Reflections ---

// Returns (x, y) reflected across point p. Exact for any coordinate type.
point reflect(const point &p) const { return {2 * p.x - x, 2 * p.y - y}; }

// Returns (x, y) reflected across the line containing points p and q.
// Always returns floating-point coordinates.
point<fp_t> reflect(const point &p, const point &q) const {
    point<fp_t> fp{(fp_t)p.x, (fp_t)p.y};
    if (p == q) {
        return point<fp_t>{(fp_t)x, (fp_t)y}.reflect(fp);
    }
    point<fp_t> r{(fp_t)(x - p.x), (fp_t)(y - p.y)};

```

```

    point<fp_t> s{(fp_t)(q.x - p.x), (fp_t)(q.y - p.y)};
    fp_t ssq = s.sqnorm();
    r = point<fp_t>{(r.x * s.x + r.y * s.y) / ssq, (r.x * s.y - r.y * s.x) / ssq};
    return point<fp_t>{r.x * s.x - r.y * s.y + fp.x, r.x * s.y + r.y * s.x + fp.y};
}

// --- Explicit type conversions ---

point<double> to_double() const { return {(double)x, (double)y}; }
point<long double> to_ldouble() const { return {(long double)x, (long double)y}; }

// --- Friend free-function versions ---

friend T sqnorm(const point &p) { return p.sqnorm(); }
friend fp_t norm(const point &p) { return p.norm(); }
friend fp_t arg(const point &p) { return p.arg(); }
friend T dot(const point &p, const point &q) { return p.dot(q); }
friend T cross(const point &p, const point &q) { return p.cross(q); }
friend fp_t proj(const point &p, const point &q) { return p.proj(q); }
friend point<fp_t> normalize(const point &p) { return p.normalize(); }
friend point rotate90(const point &p) { return p.rotate90(); }
friend point rotate180(const point &p) { return p.rotate180(); }
friend point rotate270(const point &p) { return p.rotate270(); }
friend point rotate90(const point &p, const point &q) { return p.rotate90(q); }
friend point rotate180(const point &p, const point &q) { return p.rotate180(q); }
friend point rotate270(const point &p, const point &q) { return p.rotate270(q); }
friend point<fp_t> rotateCW(const point &p, fp_t t) { return p.rotateCW(t); }
friend point<fp_t> rotateCCW(const point &p, fp_t t) { return p.rotateCCW(t); }
friend point<fp_t> rotateCW(const point &p, const point &q, fp_t t) { return p.rotateCW(q, t); }
friend point<fp_t> rotateCCW(const point &p, const point &q, fp_t t) { return p.rotateCCW(q, t); }
friend point reflect(const point &p, const point &q) { return p.reflect(q); }

friend point<fp_t> reflect(const point &p, const point &a, const point &b) {
    return p.reflect(a, b);
}

friend std::ostream &operator<<(std::ostream &out, const point &p) {
    if constexpr (std::is_floating_point<T>::value) {
        return out << "(" << (fabs(p.x) < EPS ? 0 : p.x) << "," << (fabs(p.y) < EPS ? 0 : p.y) << ")";
    } else {
        return out << "(" << p.x << "," << p.y << ")";
    }
}
};

using PointI = point<int>;
using PointL = point<long long>;
using PointD = point<double>;
using PointLD = point<long double>;
using Point = PointD; // Default point type is double.

```

Example Usage

```

#include <cassert>

const double PI = acos(-1.0);

int main() {
    // Float point - same API as before.
    Point p(-10, 3);
    assert(Point(-18, 29) == p + Point(-3, 9) * 6.0 / 2.0 - Point(-1, 1));
    assert(EQ(109, p.sqnorm()));
    assert(EQ(10.44030650891, p.norm()));
    assert(EQ(2.850135859112, p.arg()));
    assert(EQ(0, p.dot(Point(3, 10))));
    assert(EQ(0, p.cross(Point(10, -3))));
}

```

```

assert(EQ(10, p.proj(Point(-10, 0))));
assert(EQ(1, p.normalize().norm()));
assert(Point(-3, -10) == p.rotate90());
assert(Point(10, -3) == p.rotate180());
assert(Point(3, 10) == p.rotate270());
assert(Point(3, 12) == p.rotateCW(Point(1, 1), PI / 2));
assert(Point(1, -10) == p.rotateCCW(Point(2, 2), PI / 2));
assert(Point(10, -3) == p.reflect(Point(0, 0)));
assert(Point(-10, -3) == p.reflect(Point(-2, 0), Point(5, 0)));

// Integer point - exact arithmetic, float-only ops return PointD.
PointI a(3, 4), b(1, 0);
assert(a + b == PointI(4, 4));
assert(a - b == PointI(2, 4));
assert(a * 2 == PointI(6, 8));
assert(a.sqnorm() == 25);
assert(a.dot(b) == 3);
assert(a.cross(b) == -4);
assert(EQ(a.norm(), 5.0));
assert(a.rotate90() == PointI(-4, 3));
assert(a.rotate180() == PointI(-3, -4));
assert(a.rotate270() == PointI(4, -3));
PointD anorm = a.normalize(); // returns PointD
assert(EQ(anorm.norm(), 1.0));
PointD adiv = a / 2.0; // returns PointD (division always promotes)
assert(EQ(adiv.x, 1.5) && EQ(adiv.y, 2.0));
auto rotated = PointI(1, 0).rotateCCW(PI / 4); // returns PointD (arbitrary rotation promotes)
double root2over2 = sqrt(2.0) / 2.0;
assert(EQ(rotated.x, root2over2) && EQ(rotated.y, root2over2));

// Implicit conversion PointI -> PointD.
PointD promoted = PointI(1, 2);
assert(promoted == PointD(1, 2));

return 0;
}

```

7.1.2 Line

A straight line in two dimensions, templated on the coordinate type `T` (default `double`). The line $ax + by + c = 0$ is stored UN-normalized, so a line through integer points keeps the integer coefficients: `a = q.y - p.y`, `b = p.x - q.x`, `c = -(a*p.x + b*p.y)`. Storage and the exact predicates `contains()`, `is_parallel()`, and `is_perpendicular()` therefore stay exact for integer `T`. The inherently-fractional operations `slope()`, `x()`, and `y()` return `fp_t`, which is `double` when `T` is integral and `T` itself otherwise.

Because coefficients are not normalized to a canonical form, `operator==` compares lines by proportionality (whether they are the same geometric line) rather than by coefficient equality.

Overflow warning: the exact predicates form products of coefficients (e.g. cross terms in `is_parallel()`/`is_perpendicular()` and `a*p.x + b*p.y` in `contains()`), which grow like the squared coordinate magnitude. For integer `T`, use `LineL` (`line<long long>`) once coordinates exceed a few tens of thousands, or the 32-bit products overflow.

Operations include `horizontal()`, `vertical()`, `slope()`, evaluating y at some x with `y()` (and vice versa with `x()`), checking if a point falls on the line with `contains()`, checking if another line is parallel or perpendicular with `is_parallel()` or `is_perpendicular()`, and finding the `parallel()` or `perpendicular()` line through a point.

Type aliases:

- `LineI = line<int>`: exact integer-coefficient lines (small values only; see overflow warning)
- `LineL = line<long long>`: exact integer-coefficient lines for large coordinates
- `LineD = line<double>`: standard floating-point

- `LineLD = line<long double>`: extra precision
- `Line = LineD`: default line type is double

Time Complexity: $O(1)$ per call to the constructor and all other operations.

Space Complexity:

- $O(1)$ for storage of the line.
- $O(1)$ auxiliary for all operations.

```
#include <cmath>
#include <limits>
#include <ostream>
#include <type_traits>
#include <utility>

const double EPS = 1e-9;
const double M_NAN = std::numeric_limits<double>::quiet_NaN();

// SFINAE guard: valid only when Pt exposes numeric .x/.y members, so templated point constructors
// don't hijack scalar-argument calls.
template<class Pt>
using if_point = decltype(std::declval<const Pt &>().x, std::declval<const Pt &>().y, void());

template<class T>
struct line {
    // fp_t is T for floating-point types, double for integral types.
    using fp_t = std::conditional_t<std::is_integral<T>::value, double, T>;

    T a, b, c;

    line() : a(0), b(0), c(0) {} // Invalid or uninitialized line.
    line(T a, T b, T c) : a(a), b(b), c(c) {}

    // Line through two points. Coefficients are exact (type T) for integer points.
    template<class Pt, class = if_point<Pt>>
    line(const Pt &p, const Pt &q) : a(q.y - p.y), b(p.x - q.x), c(-(a * p.x + b * p.y)) {}

    // Line of a given slope through a point. Intended for floating-point T (a generic slope
    // produces non-integer coefficients), so this constructor is disabled for integer T.
    template<
        class Pt, class = if_point<Pt>, class U = T,
        class = std::enable_if_t<std::is_floating_point<U>::value>>
    line(fp_t slope, const Pt &p) : a(-slope), b(1), c(slope * p.x - p.y) {}

    // Epsilon-aware for floating-point T; exact for int and for coordinate types like Modular or
    // Rational, which therefore compose for all of the predicates below.
    template<class A, class B>
    static bool eq(const A &p, const B &q) {
        if constexpr (std::is_floating_point<T>::value) {
            return fabs(p - q) <= EPS;
        }
        return p == q;
    }
};

// Two lines are equal iff their coefficient vectors are proportional (the same line).
bool operator==(const line &l) const {
    return eq(a * l.b, l.a * b) && eq(b * l.c, l.b * c) && eq(a * l.c, l.a * c);
}

bool operator!=(const line &l) const { return !(*this == l); }

// Returns whether the line is a valid line (not both a and b zero).
bool valid() const { return !(eq(a, 0) && eq(b, 0)); }
bool horizontal() const { return valid() && eq(a, 0); }
bool vertical() const { return valid() && eq(b, 0); }

// Slope -a/b, or NaN if the line is vertical or invalid.
```

```

fp_t slope() const { return (!valid() || eq(b, 0)) ? M_NAN : -(fp_t)a / b; }

// Solve for x at a given y (NaN if horizontal or invalid).
fp_t x(fp_t y) const { return (!valid() || eq(a, 0)) ? M_NAN : -((fp_t)b * y + c) / a; }

// Solve for y at a given x (NaN if vertical or invalid).
fp_t y(fp_t x) const { return (!valid() || eq(b, 0)) ? M_NAN : -((fp_t)a * x + c) / b; }

// Whether point p lies on the line. Exact for integer T and integer p.
template<class Pt>
bool contains(const Pt &p) const {
    return eq(a * p.x + b * p.y + c, 0);
}

// Parallel iff the normals are parallel (cross == 0); perpendicular iff normals are
// perpendicular (dot == 0). Both exact for integer T.
bool is_parallel(const line &l) const { return eq(a * l.b, l.a * b); }
bool is_perpendicular(const line &l) const { return eq(a * l.a, -(b * l.b)); }

// Parallel/perpendicular line through point p. Exact for integer T and integer p.
template<class Pt, class = if_point<Pt>>
line parallel(const Pt &p) const {
    return line(a, b, -(a * p.x + b * p.y));
}

template<class Pt, class = if_point<Pt>>
line perpendicular(const Pt &p) const {
    return line(-b, a, b * p.x - a * p.y);
}

friend std::ostream &operator<<(std::ostream &out, const line &l) {
    auto clean = [](T v) -> double { return fabs((double)v) < EPS ? 0 : (double)v; };
    return out << clean(l.a) << "x" << std::showpos << clean(l.b) << "y" << clean(l.c) << "=0"
        << std::noshowpos;
}

};

using LineI = line<int>;
using LineL = line<long long>;
using LineD = line<double>;
using LineLD = line<long double>;
using Line = LineD; // Default line type is double.

```

Example Usage

```

#include <cassert>

struct Point {
    double x, y;
    Point(double x = 0, double y = 0) : x(x), y(y) {}
};

struct PointI {
    int x, y;
    PointI(int x = 0, int y = 0) : x(x), y(y) {}
};

int main() {
    // Floating-point line.
    Line l(2, -5, -8);
    Line para = Line(2, -5, -8).parallel(Point(-6, -2));
    Line perp = Line(2, -5, -8).perpendicular(Point(-6, -2));
    assert(l.is_parallel(para) && l.is_perpendicular(perp));
    assert(l.slope() == 0.4);
    // operator== is proportionality-based: para is the same line as -0.4x + y - 0.4 = 0.
    assert(para == Line(-0.4, 1, -0.4));
}

```

```

assert(perp == Line(2.5, 1, 17));
assert(l.contains(Point(1.5, -1))); // 2(1.5) - 5(-1) - 8 = 0.

// Integer line through integer points - exact coefficients and predicates.
LineI il(PointI(0, 0), PointI(2, 1)); // a=1, b=-2, c=0 -> x - 2y = 0
assert(il.a == 1 && il.b == -2 && il.c == 0);
assert(il.contains(PointI(4, 2))); // exact integer check
assert(!il.contains(PointI(4, 3)));
assert(il.is_parallel(LineI(PointI(1, 1), PointI(3, 2)))); // both slope 1/2
assert(il.is_perpendicular(LineI(PointI(0, 0), PointI(1, -2)))); // dot of normals = 0
assert(il.slope() == 0.5); // fp_t result

return 0;
}

```

7.1.3 Circle

A circle in two dimensions supporting epsilon comparisons. The circle centered at (h, k) is represented by the relation $(x - h)^2 + (y - k)^2 = r^2$, where the radius r is normalized to a non-negative number. Operations include constructing a circle from a line segment, constructing a circumcircle, checking if a point falls inside the circle with `contains()` or on its edge with `on_edge()`, and constructing an incircle with `incircle()`.

Circle is floating-point by nature (the radius requires a square root), so it always stores and computes in `double`. The center of a circumcircle is rational and the radius irrational even for integer inputs, so there is no exact integer `Circle` to store; instead, every point-accepting constructor and predicate is templated on the point type `Pt` and only reads `.x/ .y`, accepting `Point/ PointD/ PointI` from 7.1.1 or any struct with numeric `.x/ .y` fields.

When exact integer reasoning about a circumcircle is needed, use `in_circumcircle(a, b, c, d)`, which tests whether `a` lies inside the circle through `a`, `b`, `c` using a determinant (no square root) and is therefore exact for integer coordinates without ever building a `Circle`.

Time Complexity: $O(1)$ per call to the constructors and all other operations.

Space Complexity:

- $O(1)$ for storage of the circle.
- $O(1)$ auxiliary for all operations.

```

#include <cmath>
#include <ostream>
#include <stdexcept>
#include <type_traits>
#include <utility>

const double EPS = 1e-9;

#define EQ(a, b) (fabs((a) - (b)) <= EPS)
#define LT(a, b) ((a) < (b) - EPS)
#define LE(a, b) ((a) <= (b) + EPS)

// SFINAE guard: valid only when Pt exposes numeric .x/.y members. This keeps the templated point
// constructors from hijacking calls like Circle(double, double, double).
template<class Pt>
using if_point = decltype(std::declval<const Pt &>().x, std::declval<const Pt &>().y, void());

struct Circle {
    double h, k, r;

    Circle() : h(0), k(0), r(0) {}
    explicit Circle(double r) : h(0), k(0), r(fabs(r)) {}
    Circle(double h, double k, double r) : h(h), k(k), r(fabs(r)) {}

    template<class Pt, class = if_point<Pt>>

```

```

Circle(const Pt &o, double r) : h(o.x), k(o.y), r(fabs(r)) {}

// Circle with the line segment ab as a diameter.
template<class Pt, class = if_point<Pt>>
Circle(const Pt &a, const Pt &b) {
    h = (a.x + b.x) / 2.0;
    k = (a.y + b.y) / 2.0;
    r = hypot(a.x - h, a.y - k);
}

// Circumcircle of three points.
template<class Pt, class = if_point<Pt>>
Circle(const Pt &a, const Pt &b, const Pt &c) {
    double an = (double)(b.x - c.x) * (b.x - c.x) + (double)(b.y - c.y) * (b.y - c.y);
    double bn = (double)(a.x - c.x) * (a.x - c.x) + (double)(a.y - c.y) * (a.y - c.y);
    double cn = (double)(a.x - b.x) * (a.x - b.x) + (double)(a.y - b.y) * (a.y - b.y);
    double wa = an * (bn + cn - an);
    double wb = bn * (an + cn - bn);
    double wc = cn * (an + bn - cn);
    double w = wa + wb + wc;
    if (EQ(w, 0)) {
        throw std::runtime_error("No circumcircle from collinear points.");
    }
    h = (wa * a.x + wb * b.x + wc * c.x) / w;
    k = (wa * a.y + wb * b.y + wc * c.y) / w;
    r = hypot(a.x - h, a.y - k);
}

// Circle of radius r that contains points a and b. In the general case, there will be two
// possible circles and only one is chosen arbitrarily. However if the diameter is equal to
// dist(a, b) = 2*r, then there is only one possible center. If points a and b are identical, then
// there are infinite circles. If the points are too far away relative to the radius, then there
// is no possible Circle. In the latter two cases, an exception is thrown.
template<class Pt, class = if_point<Pt>>
Circle(const Pt &a, const Pt &b, double r) : r(fabs(r)) {
    if (LE(r, 0) && EQ(a.x, b.x) && EQ(a.y, b.y)) { // Circle with zero area.
        h = a.x;
        k = a.y;
        return;
    }
    double d = hypot(b.x - a.x, b.y - a.y);
    if (EQ(d, 0)) {
        throw std::runtime_error("Identical points, infinite circles.");
    }
    if (LT(r * 2.0, d)) {
        throw std::runtime_error("Points too far away to make Circle.");
    }
    double v = sqrt(r * r - d * d / 4.0) / d;
    double mx = (a.x + b.x) / 2.0, my = (a.y + b.y) / 2.0;
    h = mx + v * (a.y - b.y);
    k = my + v * (b.x - a.x);
    // The other answer is (h, k) = (mx - v*(a.y - b.y), my - v*(b.x - a.x)).
}

bool operator==(const Circle &c) const { return EQ(h, c.h) && EQ(k, c.k) && EQ(r, c.r); }
bool operator!=(const Circle &c) const { return !(*this == c); }

template<class Pt>
bool contains(const Pt &p) const {
    return LE((double)(p.x - h) * (p.x - h) + (double)(p.y - k) * (p.y - k), r * r);
}

template<class Pt>
bool on_edge(const Pt &p) const {
    return EQ((double)(p.x - h) * (p.x - h) + (double)(p.y - k) * (p.y - k), r * r);
}

friend std::ostream &operator<<(std::ostream &out, const Circle &c) {

```



```

        return out << std::showpos << "(x" << -(fabs(c.h) < EPS ? 0 : c.h) << ")^2+"
            << "(y" << -(fabs(c.k) < EPS ? 0 : c.k) << ")^2" << std::noshowpos << "="
            << (fabs(c.r) < EPS ? 0 : c.r * c.r);
    }
};

// Returns the Circle inscribed inside the triangle abc.
template<class Pt>
Circle incircle(const Pt &a, const Pt &b, const Pt &c) {
    double al = hypot(b.x - c.x, b.y - c.y);
    double bl = hypot(a.x - c.x, a.y - c.y);
    double cl = hypot(a.x - b.x, a.y - b.y);
    double l = al + bl + cl;
    double px = a.x - c.x, py = a.y - c.y, qx = b.x - c.x, qy = b.y - c.y;
    return EQ(l, 0) ? Circle(a.x, a.y, 0)
        : Circle(
            (al * a.x + bl * b.x + cl * c.x) / l, (al * a.y + bl * b.y + cl * c.y) / l,
            fabs(px * qy - py * qx) / l
        );
}

// Returns +1 if point d lies strictly inside the circle through a, b, c, 0 if d lies exactly on
// it, or -1 if d is outside. This uses a determinant (no square root), so it is exact for integer
// coordinates: the result is computed in `long long` for integral inputs (watch overflow for large
// coordinates, where the determinant is degree four) and `long double` otherwise. The points a, b,
// c must not be collinear; the result does not depend on their orientation.
template<class Pt>
int in_circumcircle(const Pt &a, const Pt &b, const Pt &c, const Pt &d) {
    using CoordT = decltype(a.x + a.x);
    using W = std::conditional_t<std::is_integral<CoordT>::value, long long, long double>;
    W adx = (W)a.x - d.x, ady = (W)a.y - d.y;
    W bdx = (W)b.x - d.x, bdy = (W)b.y - d.y;
    W cdx = (W)c.x - d.x, cdy = (W)c.y - d.y;
    W det = (adx * adx + ady * ady) * (bdx * cdy - cdx * bdy) +
        (bdx * bdx + bdy * bdy) * (cdx * ady - adx * cdy) +
        (cdx * cdx + cdy * cdy) * (adx * bdy - bdx * ady);
    W orient = ((W)b.x - a.x) * ((W)c.y - a.y) - ((W)b.y - a.y) * ((W)c.x - a.x);
    W val = orient > 0 ? det : -det;
    return val > 0 ? 1 : (val < 0 ? -1 : 0);
}

```

Example Usage

```

#include <cassert>

struct Point {
    double x, y;
    Point(double x = 0, double y = 0) : x(x), y(y) {}
};

struct PointI {
    int x, y;
    PointI(int x = 0, int y = 0) : x(x), y(y) {}
};

int main() {
    Circle c(-2, 5, sqrt(10));
    assert(c == Circle(Point(-2, 5), sqrt(10)));
    assert(c == Circle(Point(1, 6), Point(-5, 4)));
    assert(c == Circle(Point(-3, 2), Point(-3, 8), Point(-1, 8)));
    assert(c == incircle(Point(-12, 5), Point(3, 0), Point(0, 9)));
    assert(c.contains(Point(-2, 8)) && !c.contains(Point(-2, 9)));
    assert(c.on_edge(Point(-1, 2)) && !c.on_edge(Point(-1.01, 2)));

    // Integer-coordinate points are accepted; the Circle is computed in double.
    assert(c == Circle(PointI(1, 6), PointI(-5, 4)));
}

```

```

assert(c == Circle(PointI(-3, 2), PointI(-3, 8), PointI(-1, 8)));
assert(c == incircle(PointI(-12, 5), PointI(3, 0), PointI(0, 9)));
assert(c.contains(PointI(-2, 8)) && !c.contains(PointI(-2, 9)));
assert(c.on_edge(PointI(-1, 2)));

// Exact integer in-circle predicate: unit circle through (1,0), (0,1), (-1,0).
assert(in_circumcircle(PointI(1, 0), PointI(0, 1), PointI(-1, 0), PointI(0, 0)) == 1); // inside
assert(
    in_circumcircle(PointI(1, 0), PointI(0, 1), PointI(-1, 0), PointI(2, 0)) == -1
); // outside
assert(
    in_circumcircle(PointI(1, 0), PointI(0, 1), PointI(-1, 0), PointI(0, -1)) == 0
); // on edge
// Orientation-independent: reversing a, b, c gives the same answer.
assert(in_circumcircle(PointI(-1, 0), PointI(0, 1), PointI(1, 0), PointI(0, 0)) == 1);

return 0;
}

```

7.1.4 Triangle

Common triangle calculations in two dimensions. The functions are templated on the point type `Pt`, which should work with `Point`/`PointD`/`PointI` from 7.1.1 or any struct with numeric `.x` and `.y` fields. `triangle_area()` returns `double` regardless of input type (the area is fractional via the `/2`). `same_side()` and `point_in_triangle()` do all arithmetic in the point's own coordinate type and are exact for integer points: they reduce each cross product to a sign before combining, so no precision is lost and cross products are never multiplied together.

Overflow warning: each individual cross product is still on the order of the squared coordinate magnitude, so for integer point types use a 64-bit coordinate type (e.g. `PointL` from 7.1.1) once coordinates exceed a few tens of thousands.

Note: with an integer point type, coordinates must be integers - fractional literals like `-2.44` cannot be represented. Use a floating-point point type for fractional coordinates.

- `triangle_area(a, b, c)` returns the area of the triangle `abc`.
- `triangle_area_sides(s1, s2, s3)` returns the area of a triangle with side lengths `s1`, `s2`, and `s3`. The given lengths must be non-negative and form a valid triangle.
- `triangle_area_medians(m1, m2, m3)` returns the area of a triangle with medians of lengths `m1`, `m2`, and `m3`. The median of a triangle is a line segment joining a vertex to the midpoint of the opposing edge.
- `triangle_area_altitudes(h1, h2, h3)` returns the area of a triangle with altitudes `h1`, `h2`, and `h3`. An altitude of a triangle is the shortest line between a vertex and the infinite line that is extended from its opposite edge.
- `same_side(p1, p2, a, b)` returns whether points `p1` and `p2` lie on the same side of the line containing points `a` and `b`. If one or both points lie exactly on the line, then the result will depend on the setting of `EDGE_IS_SAME_SIDE`.
- `point_in_triangle(p, a, b, c)` returns whether point `p` lies within the triangle `abc`. If the point lies on or close to an edge (by roughly `EPS`), then the result will depend on the setting of `EDGE_IS_SAME_SIDE` in the function above.

Time Complexity: $O(1)$ for all operations.

Space Complexity: $O(1)$ auxiliary for all operations.

```

#include <algorithm>
#include <cmath>
#include <utility>

const double EPS = 1e-9;

#define EQ(a, b) (fabs((a) - (b)) <= EPS)
#define LT(a, b) ((a) < (b) - EPS)

```

```

#define LE(a, b) ((a) <= (b) + EPS)

template<class Pt>
double triangle_area(const Pt &a, const Pt &b, const Pt &c) {
    double acx = a.x - c.x, acy = a.y - c.y, bcx = b.x - c.x, bcy = b.y - c.y;
    return fabs(acx * bcy - acy * bcx) / 2.0;
}

double triangle_area_sides(double s1, double s2, double s3) {
    double s = (s1 + s2 + s3) / 2.0;
    return sqrt(s * (s - s1) * (s - s2) * (s - s3));
}

double triangle_area_medians(double m1, double m2, double m3) {
    return 4.0 * triangle_area_sides(m1, m2, m3) / 3.0;
}

double triangle_area_altitudes(double h1, double h2, double h3) {
    if (EQ(h1, 0) || EQ(h2, 0) || EQ(h3, 0)) {
        return 0;
    }
    double x = h1 * h1, y = h2 * h2, z = h3 * h3;
    double v = 2.0 / (x * y) + 2.0 / (x * z) + 2.0 / (y * z);
    return 1.0 / sqrt(v - 1.0 / (x * x) - 1.0 / (y * y) - 1.0 / (z * z));
}

template<class Pt>
bool same_side(const Pt &p1, const Pt &p2, const Pt &a, const Pt &b) {
    static const bool EDGE_IS_SAME_SIDE = true;
    // Cross products in the point's native coordinate type (exact for integer points).
    auto c1 = (b.x - a.x) * (p1.y - a.y) - (b.y - a.y) * (p1.x - a.x);
    auto c2 = (b.x - a.x) * (p2.y - a.y) - (b.y - a.y) * (p2.x - a.x);
    // Compare the signs, not the product c1 * c2, to avoid integer overflow.
    int s1 = LT(0, c1) ? 1 : (LT(c1, 0) ? -1 : 0);
    int s2 = LT(0, c2) ? 1 : (LT(c2, 0) ? -1 : 0);
    return EDGE_IS_SAME_SIDE ? (s1 * s2 >= 0) : (s1 * s2 > 0);
}

template<class Pt>
bool point_in_triangle(const Pt &p, const Pt &a, const Pt &b, const Pt &c) {
    return same_side(p, a, b, c) && same_side(p, b, a, c) && same_side(p, c, a, b);
}

```

Example Usage

```

#include <cassert>

struct Point {
    double x, y;
    Point(double x = 0, double y = 0) : x(x), y(y) {}
};

struct PointI {
    int x, y;
    PointI(int x = 0, int y = 0) : x(x), y(y) {}
};

int main() {
    assert(EQ(6, triangle_area(Point(0, -1), Point(4, -1), Point(0, -4))));
    assert(EQ(6, triangle_area_sides(3, 4, 5)));
    assert(EQ(6, triangle_area_medians(3.605551275, 2.5, 4.272001873)));
    assert(EQ(6, triangle_area_altitudes(3, 4, 2.4)));

    // Fractional coordinates require a floating-point point type.
    assert(point_in_triangle(Point(0, 0), Point(-1, 0), Point(0, -2), Point(4, 0)));
    assert(!point_in_triangle(Point(0, 1), Point(-1, 0), Point(0, -2), Point(4, 0)));
}

```

```

assert(point_in_triangle(Point(-2.44, 0.82), Point(-1, 0), Point(-3, 1), Point(4, 0)));
assert(!point_in_triangle(Point(-2.44, 0.7), Point(-1, 0), Point(-3, 1), Point(4, 0)));

// Integer coordinates: same_side / point_in_triangle are computed exactly in int.
assert(EQ(8, triangle_area(PointI(0, 0), PointI(4, 0), PointI(0, 4))));
assert(point_in_triangle(PointI(1, 1), PointI(0, 0), PointI(4, 0), PointI(0, 4)));
assert(!point_in_triangle(PointI(5, 5), PointI(0, 0), PointI(4, 0), PointI(0, 4)));
assert(point_in_triangle(PointI(2, 2), PointI(0, 0), PointI(4, 0), PointI(0, 4))); // on edge

return 0;
}

```

7.1.5 Rectangle

Common rectangle calculations in two dimensions. The functions are templated on the point type `Pt`, which should work with `Point`/`PointD`/`PointI` from 7.1.1, or any struct with numeric `.x` and `.y` fields. All operations use comparisons only and are exact for integer-coordinate points.

- `point_in_rectangle(p, v, w, h)` returns whether point `p` lies within the rectangle defined by a vertex `v`, a width of `w`, and a height of `h`. Note that negative widths and heights are supported. If the point lies on or close to an edge (by roughly `EPS`), then the result will depend on the setting of `EDGE_IS_INSIDE`.
- `point_in_rectangle(p, a, b)` returns whether point `p` lies within the rectangle with opposing vertices `a` and `b`. If the point lies on or close to an edge (by roughly `EPS`), then the result will depend on the setting of `EDGE_IS_INSIDE`.
- `rectangle_intersection(a1, b1, a2, b2, &p, &q)` determines the intersection region of the rectangle with opposing vertices `a1` and `b1` and the rectangle with opposing vertices `a2` and `b2`. Returns `-1` if the rectangles are completely disjoint, `0` if the rectangles partially intersect, `1` if the first rectangle is completely inside the second, and `2` if the second rectangle is completely inside the first. If there is an intersection, the lower-left and upper-right vertices of the normalized intersection rectangle will be stored into pointers `p` and `q` if they are not `nullptr`. If the intersection is a single point or line segment, then the result will depend on the setting of `EDGE_IS_INSIDE` within `point_in_rectangle()`.

Time Complexity: $O(1)$ for all operations.

Space Complexity: $O(1)$ auxiliary for all operations.

```

#include <algorithm>
#include <cmath>
#include <cstdint>
#include <utility>

const double EPS = 1e-9;

#define EQ(a, b) (fabs((a) - (b)) <= EPS)
#define LT(a, b) ((a) < (b) - EPS)
#define GT(a, b) ((a) > (b) + EPS)
#define LE(a, b) ((a) <= (b) + EPS)
#define GE(a, b) ((a) >= (b) - EPS)

template<class Pt, class T>
bool point_in_rectangle(const Pt &p, const Pt &v, const T &w, const T &h) {
    static const bool EDGE_IS_INSIDE = true;
    if (w < 0) {
        return point_in_rectangle(p, Pt(v.x + w, v.y), -w, h);
    }
    if (h < 0) {
        return point_in_rectangle(p, Pt(v.x, v.y + h), w, -h);
    }
    return EDGE_IS_INSIDE ? (GE(p.x, v.x) && LE(p.x, v.x + w) && GE(p.y, v.y) && LE(p.y, v.y + h))
        : (GT(p.x, v.x) && LT(p.x, v.x + w) && GT(p.y, v.y) && LT(p.y, v.y + h));
}

```

```

template<class Pt>
bool point_in_rectangle(const Pt &p, const Pt &a, const Pt &b) {
    auto xl = std::min(a.x, b.x), yl = std::min(a.y, b.y);
    auto xh = std::max(a.x, b.x), yh = std::max(a.y, b.y);
    return point_in_rectangle(Pt(p.x, p.y), Pt(xl, yl), xh - xl, yh - yl);
}

template<class Pt>
int rectangle_intersection(
    const Pt &a1, const Pt &b1, const Pt &a2, const Pt &b2, Pt *p = nullptr, Pt *q = nullptr
) {
    static const bool EDGE_IS_INSIDE = true;
    // Normalize each rectangle to [xl, xh] x [yl, yh] with coordinate-wise min/max only.
    // (Using auto keeps the native coordinate type, so integer rectangles stay exact.)
    auto xl1 = std::min(a1.x, b1.x), xh1 = std::max(a1.x, b1.x);
    auto yl1 = std::min(a1.y, b1.y), yh1 = std::max(a1.y, b1.y);
    auto xl2 = std::min(a2.x, b2.x), xh2 = std::max(a2.x, b2.x);
    auto yl2 = std::min(a2.y, b2.y), yh2 = std::max(a2.y, b2.y);
    // The intersection is the overlap of the two coordinate intervals.
    auto ixl = std::max(xl1, xl2), ixh = std::min(xh1, xh2);
    auto iyl = std::max(yl1, yl2), iyh = std::min(yh1, yh2);
    bool disjoint = EDGE_IS_INSIDE ? (GT(ixl, ixh) || GT(iyl, iyh)) : (GE(ixl, ixh) || GE(iyl, iyh));
    if (disjoint) {
        return -1; // Completely disjoint.
    }
    // Opposing corners of the overlap: bottom-left in p, top-right in q.
    if (p != nullptr) {
        *p = Pt(ixl, iyl);
    }
    if (q != nullptr) {
        *q = Pt(ixh, iyh);
    }
    // The overlap equals a rectangle exactly when that rectangle is contained in the other.
    if (EQ(ixl, xl1) && EQ(ixh, xh1) && EQ(iyl, yl1) && EQ(iyh, yh1)) {
        return 1; // Rectangle 1 completely inside 2.
    }
    if (EQ(ixl, xl2) && EQ(ixh, xh2) && EQ(iyl, yl2) && EQ(iyh, yh2)) {
        return 2; // Rectangle 2 completely inside 1.
    }
    return 0; // Partial intersection.
}

```

Example Usage

```

#include <cassert>

struct Point {
    double x, y;
    Point(double x = 0, double y = 0) : x(x), y(y) {}
    bool operator==(const Point &p) const { return EQ(x, p.x) && EQ(y, p.y); }
};

int main() {
    assert(point_in_rectangle(Point(0, -1), Point(0, -3), 3, 2));
    assert(point_in_rectangle(Point(2, -2), Point(3, -3), -3, 2));
    assert(!point_in_rectangle(Point(0, 0), Point(3, -1), -3, -2));
    assert(point_in_rectangle(Point(2, -2), Point(3, -3), Point(0, -1)));
    assert(!point_in_rectangle(Point(-1, -2), Point(3, -3), Point(0, -1)));

    Point p, q;
    assert(-1 == rectangle_intersection(Point(0, 0), Point(1, 1), Point(2, 2), Point(3, 3)));
    assert(0 == rectangle_intersection(Point(1, 1), Point(7, 7), Point(5, 5), Point(0, 0), &p, &q));
    assert(p == Point(1, 1) && q == Point(5, 5));
    assert(1 == rectangle_intersection(Point(1, 1), Point(0, 0), Point(0, 0), Point(1, 10), &p, &q));
    assert(p == Point(0, 0) && q == Point(1, 1));
}

```

```

assert(2 == rectangle_intersection(Point(0, 5), Point(5, 7), Point(1, 6), Point(2, 5), &p, &q));
assert(p == Point(1, 5) && q == Point(2, 6));

return 0;
}

```

7.2 Angles, Distances, and Intersections

7.2.1 Angles

Angle calculations in two dimensions. All operations are inherently floating-point (`atan2`, `acos`, trigonometry), so these functions use the local double-coordinate `PointD` type. Convert integer points to `PointD` when asking for angles. The constants `DEG` and `RAD` may be used as multipliers to convert between degrees and radians. For example, if t is a value in degrees, then $t \cdot \text{DEG}$ is the equivalent angle in radians; if t is in radians, then $t \cdot \text{RAD}$ is the equivalent angle in degrees.

- `reduce_deg(t)` takes an angle t degrees and returns an equivalent angle in the range $[0, 360)$ degrees. E.g. -630 becomes 90 .
- `reduce_rad(t)` takes an angle t radians and returns an equivalent angle in the range $[0, 2\pi)$ radians. E.g. 720.5 becomes 0.5 .
- `polar_point(r, t)` returns a two-dimensional Cartesian point given radius r and angle t radians in polar coordinates (see `std::polar()`).
- `polar_angle(p)` returns the angle in radians of the line segment from $(0, 0)$ to point p , relative counterclockwise to the positive x -axis.
- `angle(a, o, b)` returns the smallest angle in radians formed by the points a , o , b with vertex at point o .
- `angle_between(a, b)` returns the angle in radians of segment from point a to point b , relative counterclockwise to the positive x -axis.
- `angle_between(a1, b1, a2, b2)` returns the smaller angle in radians between two lines $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$, limited to $[0, \pi/2]$.
- `cross(a, b, o)` returns the magnitude (Euclidean norm) of the three-dimensional cross product between points a and b where the z -component is implicitly zero and the origin is implicitly shifted to point o . This operation is also equal to double the signed area of the triangle from these three points.
- `turn(a, o, b)` returns -1 if the path $a \rightarrow o \rightarrow b$ forms a left turn on the plane, 0 if the path forms a straight line segment, or 1 if it forms a right turn.

Time Complexity: $O(1)$ for all operations.

Space Complexity: $O(1)$ auxiliary for all operations.

```

#include <algorithm>
#include <cmath>

const double EPS = 1e-9;
const double PI = acos(-1.0);
const double DEG = PI / 180;
const double RAD = 180 / PI;

#define EQ(a, b) (fabs((a) - (b)) <= EPS)
#define LT(a, b) ((a) < (b) - EPS)

double reduce_deg(double t) {
    if (t < -360) {
        return reduce_deg(fmod(t, 360));
    }
    if (t < 0) {
        return t + 360;
    }
}

```

```

    return (t >= 360) ? fmod(t, 360) : t;
}

double reduce_rad(double t) {
    if (t < -2 * PI) {
        return reduce_rad(fmod(t, 2 * PI));
    }
    if (t < 0) {
        return t + 2 * PI;
    }
    return (t >= 2 * PI) ? fmod(t, 2 * PI) : t;
}

template<class OutPt>
OutPt polar_point(double r, double t) {
    return OutPt(r * cos(t), r * sin(t));
}

template<class Pt>
double polar_angle(const Pt &p) {
    double t = atan2((double)p.y, (double)p.x);
    return (t < 0) ? (t + 2 * PI) : t;
}

template<class Pt>
double angle(const Pt &a, const Pt &o, const Pt &b) {
    double ux = o.x - a.x, uy = o.y - a.y;
    double vx = o.x - b.x, vy = o.y - b.y;
    double cosine = (ux * vx + uy * vy) / (std::hypot(ux, uy) * std::hypot(vx, vy));
    return acos(std::max(-1.0, std::min(1.0, cosine)));
}

template<class Pt>
double angle_between(const Pt &a, const Pt &b) {
    double t = atan2(a.x * b.y - a.y * b.x, a.x * b.x + a.y * b.y);
    return (t < 0) ? (t + 2 * PI) : t;
}

double angle_between(const double &a1, const double &b1, const double &a2, const double &b2) {
    double t = atan2(a1 * b2 - a2 * b1, a1 * a2 + b1 * b2);
    if (t < 0) {
        t += PI;
    }
    return LT(PI / 2, t) ? (PI - t) : t;
}

template<class Pt>
auto cross(const Pt &a, const Pt &b, const Pt &o = Pt(0, 0)) {
    return (a.x - o.x) * (b.y - o.y) - (a.y - o.y) * (b.x - o.x);
}

// Built on cross(a, b, o), whose sign is the opposite of the path-direction cross product, so a
// positive cross here is a right turn: this returns +1 for right/clockwise (NOT the more common
// +1 = counter-clockwise convention), -1 for left/counter-clockwise, and 0 if collinear.
template<class Pt>
int turn(const Pt &a, const Pt &o, const Pt &b) {
    auto c = cross(a, b, o);
    return LT(c, 0) ? -1 : (LT(0, c) ? 1 : 0);
}

```

Example Usage

```

struct Point {
    double x, y;
    Point(double x = 0, double y = 0) : x(x), y(y) {}
    bool operator==(const Point &p) const { return EQ(x, p.x) && EQ(y, p.y); }
}

```

```
};

#include <cassert>

int main() {
    assert(EQ(123, reduce_deg(-8 * 360 + 123)));
    assert(EQ(1.2345, reduce_rad(2 * PI * 8 + 1.2345)));
    assert(polar_point<Point>(4, PI) == Point(-4, 0));
    assert(polar_point<Point>(4, -PI / 2) == Point(0, -4));
    assert(EQ(45, polar_angle(Point(5, 5)) * RAD));
    assert(EQ(135 * DEG, polar_angle(Point(-4, 4))));
    assert(EQ(90 * DEG, angle(Point(5, 0), Point(0, 5), Point(-5, 0))));
    assert(EQ(225 * DEG, angle_between(Point(0, 5), Point(5, -5))));
    assert(-1 == cross(Point(0, 1), Point(1, 0), Point(0, 0)));
    assert(1 == turn(Point(0, 1), Point(0, 0), Point(-5, -5)));
    return 0;
}
```

7.2.2 Distances

Distance calculations in two dimensions for points, lines, and line segments. The functions are templated on the point type `Pt`: pass any struct with numeric `.x` and `.y` fields (for example `PointD`/`PointI` from 7.1.1, or the local example struct). `sqnorm`, `dot`, `cross`, and `sqdist` preserve the coordinate type and are exact for integer points. Functions that return an actual distance use `double` for the final division/square root. `closest_point` returns a point of the same type as its inputs, so use a floating-point `Pt` there for a meaningful (non-truncated) result.

- `dist(a, b)` and `sqdist(a, b)` respectively return the distance and squared distance between points `a` and `b`.
- `line_dist(p, a, b, c)` returns the distance from point `p` to the line $ax + by + c = 0$.
- `line_dist(p, a, b)` returns the distance from point `p` to the infinite line containing points `a` and `b`. If `a` = `b`, returns the distance from `p` to `a`.
- `line_dist(a1, b1, c1, a2, b2, c2)` returns the distance between two lines.
- `seg_dist(p, a, b)` returns the distance from point `p` to the line segment `ab`.
- `seg_dist(a, b, c, d)` returns the minimum distance between line segments `ab` and `cd`. Returns 0 if the segments touch or intersect.
- `closest_point(a, b, p)` returns the point on segment `ab` closest to point `p`.

Time Complexity: $O(1)$ for all operations.

Space Complexity: $O(1)$ auxiliary for all operations.

```
#include <algorithm>
#include <cmath>
#include <utility>

const double EPS = 1e-9;

#define EQ(a, b) (fabs((a) - (b)) <= EPS)
#define LT(a, b) ((a) < (b) - EPS)
#define LE(a, b) ((a) <= (b) + EPS)
#define GE(a, b) ((a) >= (b) - EPS)

template<class Pt> auto sqnorm(const Pt &a) { return a.x*a.x + a.y*a.y; }
template<class Pt> double norm(const Pt &a) { return sqrt((double)sqnorm(a)); }
template<class Pt> auto dot(const Pt &a, const Pt &b) { return a.x*b.x + a.y*b.y; }
template<class Pt> auto cross(const Pt &a, const Pt &b) { return a.x*b.y - a.y*b.x; }

// sqdist returns the coordinate type (exact for integer points).
template<class Pt>
auto sqdist(const Pt &a, const Pt &b) {
    auto dx = b.x - a.x, dy = b.y - a.y;
    return dx * dx + dy * dy;
}
```



```

template<class Pt>
double dist(const Pt &a, const Pt &b) {
    return sqrt((double)sqdist(a, b));
}

template<class Pt, class T>
double line_dist(const Pt &p, const T &a, const T &b, const T &c) {
    return fabs((double)a * p.x + (double)b * p.y + c) / sqrt((double)a * a + (double)b * b);
}

template<class Pt>
double line_dist(const Pt &p, const Pt &a, const Pt &b) {
    if (EQ(a.x, b.x) && EQ(a.y, b.y)) {
        return dist(p, a);
    }
    auto n = sqdist(a, b);
    auto d = (p.x - a.x) * (b.x - a.x) + (p.y - a.y) * (b.y - a.y);
    double u = (double)d / n;
    double dx = a.x + u * (b.x - a.x) - p.x, dy = a.y + u * (b.y - a.y) - p.y;
    return sqrt(dx * dx + dy * dy);
}

template<class T>
double line_dist(const T &a1, const T &b1, const T &c1, const T &a2, const T &b2, const T &c2) {
    if (EQ(a1 * b2, a2 * b1)) {
        double factor = EQ(b1, 0) ? ((double)a1 / a2) : ((double)b1 / b2);
        return EQ(c1, c2 * factor) ? 0
            : fabs(c2 * factor - c1) / sqrt((double)a1 * a1 + (double)b1 * b1);
    }
    return 0;
}

template<class Pt>
double seg_dist(const Pt &p, const Pt &a, const Pt &b) {
    if (EQ(a.x, b.x) && EQ(a.y, b.y)) {
        return dist(p, a);
    }
    auto n = sqdist(a, b);
    auto d = (p.x - a.x) * (b.x - a.x) + (p.y - a.y) * (b.y - a.y);
    if (LE(d, 0) || EQ(n, 0)) {
        return dist(p, a);
    }
    if (GE(d, n)) {
        return dist(p, b);
    }
    double t = (double)d / n;
    double dx = a.x + t * (b.x - a.x) - p.x, dy = a.y + t * (b.y - a.y) - p.y;
    return sqrt(dx * dx + dy * dy);
}

template<class Pt>
double seg_dist(const Pt &a, const Pt &b, const Pt &c, const Pt &d) {
    if (EQ(a.x, b.x) && EQ(a.y, b.y)) {
        return seg_dist(a, c, d);
    }
    if (EQ(c.x, d.x) && EQ(c.y, d.y)) {
        return seg_dist(c, a, b);
    }
    auto ab_x = b.x - a.x, ab_y = b.y - a.y;
    auto ac_x = c.x - a.x, ac_y = c.y - a.y;
    auto cd_x = d.x - c.x, cd_y = d.y - c.y;
    auto c1 = ab_x * cd_y - ab_y * cd_x;
    auto c2 = ac_x * ab_y - ac_y * ab_x;
    if (EQ(c1, 0) && EQ(c2, 0)) {
        auto less = [](const Pt &u, const Pt &v) { return u.x != v.x ? u.x < v.x : u.y < v.y; };
        auto minp = [](const Pt &u, const Pt &v) -> const Pt & { return less(v, u) ? v : u; };
        auto maxp = [](const Pt &u, const Pt &v) -> const Pt & { return less(u, v) ? v : u; };
    }
}

```

```

    const Pt &res1 = maxp(minp(a, b), minp(c, d));
    const Pt &res2 = minp(maxp(a, b), maxp(c, d));
    if (!less(res2, res1)) {
        return 0;
    }
} else if (!EQ(c1, 0)) {
    auto t_num = ac_x * cd_y - ac_y * cd_x;
    bool c1_pos = c1 > 0;
    bool t_between_01 = c1_pos ? (LE(0, t_num) && LE(t_num, c1)) : (LE(t_num, 0) && LE(c1, t_num));
    bool u_between_01 = c1_pos ? (LE(0, c2) && LE(c2, c1)) : (LE(c2, 0) && LE(c1, c2));
    if (t_between_01 && u_between_01) {
        return 0;
    }
}
return std::min(
    std::min(seg_dist(a, c, d), seg_dist(b, c, d)), std::min(seg_dist(c, a, b), seg_dist(d, a, b))
);
}

template<class Pt>
Pt closest_point(const Pt &a, const Pt &b, const Pt &p) {
    Pt res{};
    if (EQ(a.x, b.x) && EQ(a.y, b.y)) {
        res.x = a.x;
        res.y = a.y;
        return res;
    }
    auto n = sqdist(a, b);
    auto d = (p.x - a.x) * (b.x - a.x) + (p.y - a.y) * (b.y - a.y);
    double t = (double)d / n;
    if (t <= 0) {
        res.x = a.x;
        res.y = a.y;
    } else if (t >= 1) {
        res.x = b.x;
        res.y = b.y;
    } else {
        res.x = a.x + t * (b.x - a.x);
        res.y = a.y + t * (b.y - a.y);
    }
    return res;
}

```

Example Usage

```

#include <cassert>

struct Point {
    double x, y;
    Point(double x = 0, double y = 0) : x(x), y(y) {}
};

struct PointI {
    int x, y;
    PointI(int x = 0, int y = 0) : x(x), y(y) {}
};

int main() {
    assert(EQ(5, dist(Point(-1, -1), Point(2, 3))));
    assert(EQ(25, sqdist(Point(-1, -1), Point(2, 3))));
    assert(EQ(1.2, line_dist(Point(2, 1), -4, 3, -1)));
    assert(EQ(0.8, line_dist(Point(3, 3), Point(-1, -1), Point(2, 3))));
    assert(EQ(1.2, line_dist(Point(2, 1), Point(-1, -1), Point(2, 3))));
    assert(EQ(0, line_dist(-4, 3, -1, 8, 6, 2)));
    assert(EQ(0.8, line_dist(-4, 3, -1, -8, 6, -10)));
    assert(EQ(1.0, seg_dist(Point(3, 3), Point(-1, -1), Point(2, 3))));
}

```

```

assert(EQ(1.2, seg_dist(Point(2, 1), Point(-1, -1), Point(2, 3))));
assert(EQ(0, seg_dist(Point(0, 2), Point(3, 3), Point(-1, -1), Point(2, 3))));
assert(EQ(0.6, seg_dist(Point(-1, 0), Point(-2, 2), Point(-1, -1), Point(2, 3))));

// Integer points: sqdist is exact, dist returns double.
PointI ia{-1, -1}, ib{2, 3};
assert(sqdist(ia, ib) == 25); // int result
assert(dot(ia, ib) == -5);    // int result
assert(EQ(dist(ia, ib), 5.0)); // double result

return 0;
}

```

7.2.3 Line Intersection

Intersection calculations in two dimensions for straight lines and line segments. The functions are templated on the point type `Pt`: pass any struct with numeric `.x` and `.y` fields (for example `PointD`/`PointI` from 7.1.1, or the local example struct). Point outputs are written through a caller-supplied pointer whose type is deduced, so the output point type may differ from the input – e.g. integer endpoints with a floating-point output point.

`seg_intersection()` has a detection-only overload (called without the output pointers) whose calculations are exact when `Pt` has integer coordinates.

Overflow warning: the exact integer paths multiply coordinate differences (cross products and squared lengths), which grow like the squared coordinate magnitude. With 32-bit `int` coordinates these overflow once coordinates exceed a few tens of thousands, so for larger integer inputs use a point type with 64-bit (`long long`) coordinates.

- `line_intersection(a1, b1, c1, a2, b2, c2, &p)` intersects lines $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$, returning -1 (parallel), 0 (one point, stored in `p`), or 1 (identical).
- `line_intersection(p1, p2, p3, p4, &p)` intersects the infinite lines through `p1`-`p2` and `p3`-`p4`.
- `seg_intersection(a, b, c, d, &p, &q)` intersects segments `a`-`b` and `c`-`d`, returning -1 (none), 0 (one point), or 1 (overlapping segment). Touching behavior is controlled by `TOUCH_IS_INTERSECT`.
- `closest_point(a, b, c, p)` returns the closest point on line $ax + by + c = 0$ to `p`.

Time Complexity: $O(1)$ for all operations.

Space Complexity: $O(1)$ auxiliary for all operations.

```

#include <algorithm>
#include <cmath>
#include <cstdint>
#include <utility>

const double EPS = 1e-9;

#define EQ(a, b) (fabs((a) - (b)) <= EPS)
#define LT(a, b) ((a) < (b) - EPS)
#define LE(a, b) ((a) <= (b) + EPS)

// Overflow risk for integer Pt: these products are ~O(max_coord^2); use long long if necessary
template<class Pt> auto sqnorm(const Pt &a) { return a.x*a.x + a.y*a.y; }
template<class Pt> auto dot(const Pt &a, const Pt &b) { return a.x*b.x + a.y*b.y; }
template<class Pt> auto cross(const Pt &a, const Pt &b) { return a.x*b.y - a.y*b.x; }

template<class OutPt>
void set_point(OutPt *p, double x, double y) {
    p->x = x;
    p->y = y;
}

template<class Pt>
bool point_on_segment(const Pt &p, const Pt &a, const Pt &b) {

```

```

// Overflow risk for integer Pt: these products are ~0(max_coord^2); use long long if necessary.
return EQ((p.x - a.x) * (b.y - a.y) - (p.y - a.y) * (b.x - a.x), 0) &&
    LE(std::min(a.x, b.x), p.x) && LE(p.x, std::max(a.x, b.x)) &&
    LE(std::min(a.y, b.y), p.y) && LE(p.y, std::max(a.y, b.y));
}

template<class OutPt>
int line_intersection(double a1, double b1, double c1, double a2, double b2, double c2, OutPt *p) {
    // Cramer's rule for a1*x + b1*y = -c1, a2*x + b2*y = -c2.
    double det = a1 * b2 - a2 * b1;
    if (EQ(det, 0)) { // Parallel (lines need not be given in normalized form).
        // Identical iff the other two 2x2 determinants also vanish.
        return (EQ(a1 * c2 - a2 * c1, 0) && EQ(b1 * c2 - b2 * c1, 0)) ? 1 : -1;
    }
    p->x = (b1 * c2 - b2 * c1) / det;
    if (!EQ(b1, 0)) {
        p->y = -(a1 * p->x + c1) / b1;
    } else {
        p->y = -(a2 * p->x + c2) / b2;
    }
    return 0;
}

template<class Pt, class OutPt>
int line_intersection(const Pt &p1, const Pt &p2, const Pt &p3, const Pt &p4, OutPt *p) {
    auto a1 = p2.y - p1.y, b1 = p1.x - p2.x;
    auto c1 = -(p1.x * p2.y - p2.x * p1.y);
    auto a2 = p4.y - p3.y, b2 = p3.x - p4.x;
    auto c2 = -(p3.x * p4.y - p4.x * p3.y);
    auto x = -(c1 * b2 - c2 * b1), y = -(a1 * c2 - a2 * c1);
    auto det = a1 * b2 - a2 * b1;
    if (EQ(det, 0)) {
        return (EQ(x, 0) && EQ(y, 0)) ? 1 : -1;
    }
    p->x = (double)x / det;
    p->y = (double)y / det;
    return 0;
}

// Detection is exact for integer Pt. Intersection point output may use a separate float OutPt.
template<class Pt, class OutPt>
int seg_intersection(
    const Pt &a, const Pt &b, const Pt &c, const Pt &d, OutPt *p = nullptr, OutPt *q = nullptr
) {
    static const bool TOUCH_IS_INTERSECT = true;
    auto ab_x = b.x - a.x, ab_y = b.y - a.y;
    auto ac_x = c.x - a.x, ac_y = c.y - a.y;
    auto cd_x = d.x - c.x, cd_y = d.y - c.y;
    auto ab2 = ab_x * ab_x + ab_y * ab_y;
    auto cd2 = cd_x * cd_x + cd_y * cd_y;
    if (EQ(ab2, 0)) {
        if (TOUCH_IS_INTERSECT && point_on_segment(a, c, d)) {
            if (p != nullptr) {
                set_point(p, (double)a.x, (double)a.y);
            }
            return 0;
        }
        return -1;
    }
    if (EQ(cd2, 0)) {
        if (TOUCH_IS_INTERSECT && point_on_segment(c, a, b)) {
            if (p != nullptr) {
                set_point(p, (double)c.x, (double)c.y);
            }
            return 0;
        }
        return -1;
    }
}

```

```

auto c1 = ab_x * cd_y - ab_y * cd_x;
auto c2 = ac_x * ab_y - ac_y * ab_x;
if (EQ(c1, 0) && EQ(c2, 0)) { // Collinear.
    Pt res1 = std::max(std::min(a, b), std::min(c, d));
    Pt res2 = std::min(std::max(a, b), std::max(c, d));
    bool overlap = TOUCH_IS_INTERSECT ? !(res2 < res1) : (res1 < res2);
    if (overlap) {
        if (res1 == res2) {
            if (p != nullptr) {
                set_point(p, (double)res1.x, (double)res1.y);
            }
            return 0;
        }
        if (p != nullptr && q != nullptr) {
            set_point(p, (double)res1.x, (double)res1.y);
            set_point(q, (double)res2.x, (double)res2.y);
        }
        return 1;
    }
    return -1;
}
if (EQ(c1, 0)) {
    return -1;
}
auto t_num = ac_x * cd_y - ac_y * cd_x;
bool c1_pos = c1 > 0;
bool t_ok = c1_pos ? (TOUCH_IS_INTERSECT ? (LE(0, t_num) && LE(t_num, c1))
                                           : (LT(0, t_num) && LT(t_num, c1)))
                  : (TOUCH_IS_INTERSECT ? (LE(t_num, 0) && LE(c1, t_num))
                                           : (LT(t_num, 0) && LT(c1, t_num)));
bool u_ok = c1_pos ? (TOUCH_IS_INTERSECT ? (LE(0, c2) && LE(c2, c1)) : (LT(0, c2) && LT(c2, c1)))
                  : (TOUCH_IS_INTERSECT ? (LE(c2, 0) && LE(c1, c2)) : (LT(c2, 0) && LT(c1, c2)));
if (t_ok && u_ok) {
    if (p != nullptr) {
        double t = (double)t_num / c1;
        set_point(p, (double)(a.x + t * ab_x), (double)(a.y + t * ab_y));
    }
    return 0;
}
return -1;
}

// Detection-only overload: no output point needed, exact for integer Pt.
template<class Pt>
int seg_intersection(const Pt &a, const Pt &b, const Pt &c, const Pt &d) {
    return seg_intersection(a, b, c, d, static_cast<Pt *>(nullptr), static_cast<Pt *>(nullptr));
}

template<class Pt>
Pt closest_point(double a, double b, double c, const Pt &p) {
    Pt res{};
    if (EQ(a, 0)) {
        res.x = p.x;
        res.y = -c / b;
    } else if (EQ(b, 0)) {
        res.x = -c / a;
        res.y = p.y;
    } else {
        line_intersection(a, b, c, -b, a, b * p.x - a * p.y, &res);
    }
    return res;
}

```

Example Usage

```
#include <cassert>

struct Point {
    double x, y;
    Point(double x = 0, double y = 0) : x(x), y(y) {}
    bool operator==(const Point &p) const { return EQ(x, p.x) && EQ(y, p.y); }
    bool operator!=(const Point &p) const { return !(*this == p); }
    bool operator<(const Point &p) const { return LT(x, p.x) || (EQ(x, p.x) && LT(y, p.y)); }
    bool operator>(const Point &p) const { return p < *this; }
};

struct PointI {
    int x, y;
    PointI(int x = 0, int y = 0) : x(x), y(y) {}
    bool operator==(const PointI &p) const { return x == p.x && y == p.y; }
    bool operator!=(const PointI &p) const { return !(*this == p); }
    bool operator<(const PointI &p) const { return x != p.x ? x < p.x : y < p.y; }
    bool operator>(const PointI &p) const { return p < *this; }
};

int main() {
    Point p, q;

    assert(line_intersection(-1.0, 1.0, 0.0, 1.0, 1.0, -3.0, &p) == 0);
    assert(p == Point(1.5, 1.5));
    assert(line_intersection(Point(0, 0), Point(1, 1), Point(0, 4), Point(4, 0), &p) == 0);
    assert(p == Point(2, 2));

    {
#define test(a, b, c, d, e, f, g, h) \
        seg_intersection(Point(a, b), Point(c, d), Point(e, f), Point(g, h), &p, &q)

        assert(0 == test(-4, 0, 4, 0, 0, -4, 0, 4) && p == Point(0, 0));
        assert(0 == test(0, 0, 10, 10, 2, 2, 16, 4) && p == Point(2, 2));
        assert(0 == test(-2, 2, -2, -2, -2, 0, 0, 0) && p == Point(-2, 0));
        assert(0 == test(0, 4, 4, 4, 4, 0, 4, 8) && p == Point(4, 4));
        assert(1 == test(10, 10, 0, 0, 2, 2, 6, 6));
        assert(p == Point(2, 2) && q == Point(6, 6));
        assert(-1 == test(6, 8, 8, 10, 12, 12, 4, 4));
        assert(-1 == test(-4, 2, -8, 8, 0, 0, -4, 6));
    }

    assert(Point(2.5, 2.5) == closest_point(-1.0, -1.0, 5.0, Point(0, 0)));
    assert(Point(3, 0) == closest_point(1.0, 0.0, -3.0, Point(0, 0)));

    // Detection-only with integer coordinates (exact, no float needed).
    assert(seg_intersection(PointI(0, 0), PointI(4, 4), PointI(0, 4), PointI(4, 0)) == 0);
    assert(seg_intersection(PointI(0, 0), PointI(1, 0), PointI(2, 0), PointI(3, 0)) == -1);
    assert(seg_intersection(PointI(0, 0), PointI(4, 4), PointI(0, 4), PointI(4, 0), &p) == 0);
    assert(p == Point(2, 2));

    return 0;
}
```

7.2.4 Circle Intersection

Circle tangent and intersection calculations in two dimensions. Such calculations are floating-point by nature, so this section mostly computes in double. The input point `p` to `tangent()` can be any type with numeric `.x` and `.y` fields, and intersection point outputs are written through a caller-supplied point type.

- `tangent(c, p, &l1, &l2)` determines the line(s) tangent to circle `c` that pass through point `p`, returning `-1` if there is no tangent line because `p` is strictly inside `c`, `0` if there is exactly one tangent line because `p` is

on the boundary of `c` (in which case the line will be stored into pointer `l1` if it's not `nullptr`), or 1 if there are two tangent lines because `p` is strictly outside of `c` (in which case the lines will be stored into pointers `l1` and `l2` if they are not `nullptr`).

- `intersection(c, l, &p, &q)` determines the intersection between the circle `c` and line `l`, returning `-1` if there is no intersection, `0` if the line has one intersection point because the line is tangent (in which case it will be stored into pointer `p` if it's not `nullptr`), or `1` if there are two intersection points because the line crosses through the circle (in which case they will be stored into pointers `p` and `q` if they are not `nullptr`).
- `intersection(c1, c2, &p, &q)` determines the intersection points between two circles `c1` and `c2`, returning `-2` if circle `c2` completely encloses circle `c1`, `-1` if circle `c1` completely encloses circle `c2`, `0` if the circles are completely disjoint, `1` if the circles are tangent with one intersection (stored in `p`), `2` if the circles intersect at two points (stored in `p` and `q`), `3` if the circles are equal and intersect at infinite points.
- `intersection_area(c1, c2)` returns the intersection area of circles `c1` and `c2`.

Time Complexity: $O(1)$ for all operations.

Space Complexity: $O(1)$ auxiliary for all operations.

```
#include <algorithm>
#include <cmath>
#include <cstdint>
#include <utility>

const double EPS = 1e-9, PI = acos(-1.0);

#define EQ(a, b) (fabs((a) - (b)) <= EPS)
#define LT(a, b) ((a) < (b) - EPS)
#define LE(a, b) ((a) <= (b) + EPS)

struct Circle {
    double h, k, r;

    Circle(double h, double k, double r) {
        this->h = h;
        this->k = k;
        this->r = r;
    }
};

struct Line {
    double a, b, c;

    Line() : a(0), b(0), c(0) {}

    Line(double a, double b, double c) {
        if (!EQ(b, 0)) {
            this->a = a / b;
            this->c = c / b;
            this->b = 1;
        } else {
            this->c = c / a;
            this->a = 1;
            this->b = 0;
        }
    }

    Line(double px, double py, double qx, double qy) : a(0), b(0), c(0) {
        if (EQ(px, qx)) {
            if (!EQ(py, qy)) { // Vertical line.
                a = 1;
                b = 0;
                c = -px;
            } // Else, invalid line.
        } else {
            a = -(double)(py - qy) / (px - qx);
            b = 1;
        }
    }
};
```

```

    c = -(a * px) - (b * py);
}
}

bool operator==(const Line &l) const { return EQ(a, l.a) && EQ(b, l.b) && EQ(c, l.c); }
};

template<class Pt>
int tangent(const Circle &c, const Pt &p, Line *l1 = nullptr, Line *l2 = nullptr) {
    auto sqnorm = [](double x, double y) { return x * x + y * y; };
    double vopx = p.x - c.h, vopy = p.y - c.k;
    if (EQ(sqnorm(vopx, vopy), c.r * c.r)) { // Point on an edge.
        if (l1 != nullptr) { // Tangent through p is perpendicular to the radius cp.
            *l1 = Line(vopx, vopy, -(vopx * p.x + vopy * p.y));
        }
        return 0;
    }
    if (LE(sqnorm(vopx, vopy), c.r * c.r)) {
        return -1; // Point inside Circle, no intersection.
    }
    double qx = vopx / c.r, qy = vopy / c.r;
    double n = sqnorm(qx, qy), d = qy * sqrt(sqnorm(qx, qy) - 1.0);
    double t1x = (qx - d) / n, t1y = c.k;
    double t2x = (qx + d) / n, t2y = c.k;
    if (!EQ(qy, 0)) { // Common case.
        t1y += c.r * (1.0 - t1x * qx) / qy;
        t2y += c.r * (1.0 - t2x * qx) / qy;
    } else { // Point at center horizontal, y = 0.
        d = c.r * sqrt(1.0 - t1x * t1x);
        t1y += d;
        t2y -= d;
    }
    t1x = t1x * c.r + c.h;
    t2x = t2x * c.r + c.h;
    // Note: here, t1 and t2 are the two points of tangencies
    if (l1 != nullptr) {
        *l1 = Line(p.x, p.y, t1x, t1y);
    }
    if (l2 != nullptr) {
        *l2 = Line(p.x, p.y, t2x, t2y);
    }
    return 1;
}

template<class OutPt>
int intersection(const Circle &c, const Line &l, OutPt *p = nullptr, OutPt *q = nullptr) {
    double v = c.h * l.a + c.k * l.b + l.c;
    double aabb = l.a * l.a + l.b * l.b;
    double disc = v * v / aabb - c.r * c.r;
    if (disc > EPS) {
        return -1;
    }
    double x0 = -l.a * v / aabb, y0 = -l.b * v / aabb;
    if (disc > -EPS) {
        if (p != nullptr) {
            p->x = x0 + c.h;
            p->y = y0 + c.k;
        }
        return 0;
    }
    double k = sqrt(std::max(0.0, disc / -aabb));
    if (p != nullptr && q != nullptr) {
        p->x = x0 + k * l.b + c.h;
        p->y = y0 - k * l.a + c.k;
        q->x = x0 - k * l.b + c.h;
        q->y = y0 + k * l.a + c.k;
    }
    return 1;
}

```



```

}

template<class OutPt>
int intersection(const Circle &c1, const Circle &c2, OutPt *p = nullptr, OutPt *q = nullptr) {
    if (EQ(c1.h, c2.h) && EQ(c1.k, c2.k)) {
        return EQ(c1.r, c2.r) ? 3 : (c1.r > c2.r ? -1 : -2);
    }
    double d12x = c2.h - c1.h, d12y = c2.k - c1.k;
    double d = hypot(d12x, d12y);
    if (LT(c1.r + c2.r, d)) {
        return 0;
    }
    if (LT(d, fabs(c1.r - c2.r))) {
        return c1.r > c2.r ? -1 : -2;
    }
    double a = (c1.r * c1.r - c2.r * c2.r + d * d) / (2 * d);
    double x0 = c1.h + (d12x * a / d), y0 = c1.k + (d12y * a / d);
    double s = sqrt(c1.r * c1.r - a * a), rx = -d12y * s / d, ry = d12x * s / d;
    if (EQ(rx, 0) && EQ(ry, 0)) {
        if (p != nullptr) {
            p->x = x0;
            p->y = y0;
        }
        return 1;
    }
    if (p != nullptr && q != nullptr) {
        p->x = x0 - rx;
        p->y = y0 - ry;
        q->x = x0 + rx;
        q->y = y0 + ry;
    }
    return 2;
}

double intersection_area(const Circle &c1, const Circle &c2) {
    double r = std::min(c1.r, c2.r), R = std::max(c1.r, c2.r);
    double d = std::hypot(c2.h - c1.h, c2.k - c1.k);
    if (LE(d, R - r)) {
        return PI * r * r;
    }
    if (LE(R + r, d)) {
        return 0;
    }
    double alpha = std::max(-1.0, std::min(1.0, (d * d + r * r - R * R) / 2 / d / r));
    double beta = std::max(-1.0, std::min(1.0, (d * d + R * R - r * r) / 2 / d / R));
    return r * r * acos(alpha) + R * R * acos(beta) -
        0.5 * sqrt((-d + r + R) * (d + r - R) * (d - r + R) * (d + r + R));
}

```

Example Usage

```

#include <cassert>

struct Point {
    double x, y;
    Point(double x = 0, double y = 0) : x(x), y(y) {}
    bool operator==(const Point &p) const { return EQ(x, p.x) && EQ(y, p.y); }
};

int main() {
    Line l1, l2;
    assert(-1 == tangent(Circle(0, 0, 4), Point(1, 1), &l1, &l2));
    assert(0 == tangent(Circle(0, 0, sqrt(2)), Point(1, 1), &l1, &l2));
    assert(l1 == Line(-1, -1, 2));
    assert(1 == tangent(Circle(0, 0, 2), Point(2, 2), &l1, &l2));
    assert(l1 == Line(0, -2, 4));
}

```

```

assert(l2 == Line(2, 0, -4));

Point p, q;
assert(-1 == intersection(Circle(1, 1, 3), Line(5, 3, -30), &p, &q));
assert(0 == intersection(Circle(1, 1, 3), Line(0, 1, -4), &p, &q));
assert(p == Point(1, 4));
assert(1 == intersection(Circle(1, 1, 3), Line(0, 1, -1), &p, &q));
assert(p == Point(4, 1));
assert(q == Point(-2, 1));

assert(-2 == intersection(Circle(1, 1, 1), Circle(0, 0, 3), &p, &q));
assert(-1 == intersection(Circle(0, 0, 3), Circle(1, 1, 1), &p, &q));
assert(0 == intersection(Circle(5, 0, 4), Circle(-5, 0, 4), &p, &q));
assert(1 == intersection(Circle(-5, 0, 5), Circle(5, 0, 5), &p, &q));
assert(p == Point(0, 0));
assert(2 == intersection(Circle(-0.5, 0, 1), Circle(0.5, 0, 1), &p, &q));
assert(p == Point(0, -sqrt(3) / 2));
assert(q == Point(0, sqrt(3) / 2));

// Each circle passes through the other's center.
double r = 3, a = intersection_area(Circle(-r / 2, 0, r), Circle(r / 2, 0, r));
assert(EQ(a, r * r * (2 * PI / 3 - sqrt(3) / 2)));

return 0;
}

```

7.3 Polygons and Planar Searches

7.3.1 Polygon Sorting and Area

Given a list of distinct points in two-dimensions, order them into a valid polygon and determine the area. The functions are templated on the point type. The local `Point` struct (`double` coordinates) is the default; replace it with `Point`/`PointD`/`PointI` from 7.1.1 or any struct with numeric `.x` and `.y` fields and `<` / `==` operators.

- `mean_center(lo, hi)` returns the arithmetic mean of a range of points (as the local `Point` type with `double` coordinates).
- `cw_comp(a, b, c)` returns whether point `a` compares clockwise before `b` about `c`.
- `CWComparator(c)` / `CCWComparator(c)` return comparators for `std::sort`.
- `polygon_area_2x(lo, hi)` returns exactly double the area of the polygon with vertices specified by the input range `[lo, hi)`. The return value is integral or floating-point, depending on the input point type). For integer vertices, divide by 2 in the caller if the exact area is needed.
- `polygon_area(lo, hi)` returns the area as `double`.

Overflow warning: `cw_comp` and `polygon_area_2x` form cross products that grow like the squared coordinate magnitude (and the shoelace sum accumulates over all vertices). For integer point types use a 64-bit coordinate type (e.g. `PointL` from 7.1.1) for large or numerous coordinates.

Time Complexity:

- $O(n)$ per call to `mean_center` and `polygon_area`, where n is the number of points.
- $O(1)$ per call to `cw_comp` and the comparators.

Space Complexity: $O(1)$ auxiliary for all operations.

```

#include <algorithm>
#include <cmath>
#include <random>
#include <stdexcept>
#include <utility>

```

```

const double EPS = 1e-9;

#define EQ(a, b) (fabs((a) - (b)) <= EPS)

// Comparator (clockwise angular order about c). Exact: comparisons are pure sign tests on
// coordinate differences and the cross product, so it is a valid strict weak ordering and is
// exact for integer-coordinate points.
template<class Pt>
bool cw_comp(const Pt &a, const Pt &b, const Pt &c) {
    if (a.x - c.x >= 0 && b.x - c.x < 0) {
        return true;
    }
    if (a.x - c.x < 0 && b.x - c.x >= 0) {
        return false;
    }
    if (a.x - c.x == 0 && b.x - c.x == 0) {
        if (a.y - c.y >= 0 || b.y - c.y >= 0) {
            return a.y > b.y;
        }
        return b.y > a.y;
    }
    auto acx = a.x - c.x, acy = a.y - c.y;
    auto bcx = b.x - c.x, bcy = b.y - c.y;
    auto det = acx * bcy - acy * bcx;
    if (det == 0) {
        auto acnorm = acx * acx + acy * acy;
        auto bcnorm = bcx * bcx + bcy * bcy;
        return acnorm > bcnorm;
    }
    return det < 0;
}

// Returns 2 * area. Result is exact (integer) for integer-coordinate points.
template<class It>
auto polygon_area_2x(It lo, It hi) {
    using T = decltype(lo->x * lo->y);
    if (lo == hi) {
        return T(0);
    }
    T area = 0;
    for (It i = lo, j = hi - 1; i != hi; j = i++) {
        area += (T)(j->x - i->x) * (T)(j->y + i->y);
    }
    return area < T(0) ? -area : area;
}

template<class It>
double polygon_area(It lo, It hi) {
    return (double)polygon_area_2x(lo, hi) / 2.0;
}

```

Example Usage

```

#include <cassert>
#include <iostream>
#include <vector>
using namespace std;

struct Point {
    double x, y;
    Point(double x = 0, double y = 0) : x(x), y(y) {}
    bool operator==(const Point &p) const { return EQ(x, p.x) && EQ(y, p.y); }
    bool operator!=(const Point &p) const { return !(*this == p); }
    bool operator<(const Point &p) const { return x != p.x ? x < p.x : y < p.y; }
    bool operator>(const Point &p) const { return p < *this; }
};

```

```

struct PointI {
    int x, y;
    PointI(int x = 0, int y = 0) : x(x), y(y) {}
    bool operator==(const PointI &p) const { return x == p.x && y == p.y; }
    bool operator!=(const PointI &p) const { return !(*this == p); }
    bool operator<(const PointI &p) const { return x != p.x ? x < p.x : y < p.y; }
    bool operator>(const PointI &p) const { return p < *this; }
};

template<class It>
Point mean_center(It lo, It hi) {
    if (lo == hi) {
        throw std::runtime_error("Cannot get center of an empty range.");
    }
    double x_sum = 0, y_sum = 0, n = hi - lo;
    for (It it = lo; it != hi; ++it) {
        x_sum += it->x;
        y_sum += it->y;
    }
    return Point(x_sum / n, y_sum / n);
}

int main() {
    vector<Point> points{Point(1, 3), Point(1, 2), Point(2, 1), Point(0, 0), Point(-1, 3)};
    vector<Point> v(points);
    std::mt19937 rng(1234567);
    std::shuffle(v.begin(), v.end(), rng);
    Point c = mean_center(v.begin(), v.end());
    assert(EQ(c.x, 0.6) && EQ(c.y, 1.8));
    sort(v.begin(), v.end(), [c](const Point &a, const Point &b) { return cw_comp(a, b, c); });
    cout << "Clockwise vertices: ";
    for (const auto &p : v) {
        cout << "(" << p.x << ", " << p.y << ") ";
    }
    cout << endl;
    assert(EQ(polygon_area(v.begin(), v.end()), 5));

    sort(v.begin(), v.end(), [c](const Point &a, const Point &b) { return cw_comp(b, a, c); });
    cout << "Counterclockwise vertices: ";
    for (const auto &p : v) {
        cout << "(" << p.x << ", " << p.y << ") ";
    }
    cout << endl;
    assert(EQ(polygon_area(v.begin(), v.end()), 5));

    // Integer points: polygon_area_2x is exact (no float arithmetic).
    vector<PointI> iv{{0, 0}, {4, 0}, {0, 3}}; // right triangle, area = 6
    assert(polygon_area_2x(iv.begin(), iv.end()) == 12); // exact int
    assert(EQ(polygon_area(iv.begin(), iv.end()), 6.0));

    return 0;
}

```

Example Output

```

Clockwise vertices: (1, 3) (1, 2) (2, 1) (0, 0) (-1, 3)
Counterclockwise vertices: (-1, 3) (0, 0) (2, 1) (1, 2) (1, 3)

```

7.3.2 Point-in-Polygon (Ray Casting)

Given a point `p` and a polygon, determines whether `p` lies inside using ray casting. The function is templated on the point type `Pt`. The local `Point` struct (double coordinates) is the default; replace it with `Point`/`PointD`/`PointI` from 7.1.1 or any struct with numeric `.x` and `.y` fields. All comparisons are exact (no epsilon): the

ray-crossing parity needs a consistent comparison to be correct, and the result is exact for integer-coordinate points.

Overflow warning: the edge orientation test forms a cross product that grows like the squared coordinate magnitude. For integer point types use a 64-bit coordinate type (e.g. `PointL` from 7.1.1) once coordinates exceed a few tens of thousands.

- `point_in_polygon(p, lo, hi)` returns whether `p` lies within the polygon given by range `[lo, hi)` in clockwise or counter-clockwise order. If `p` lies exactly on an edge or vertex, the result depends on `EDGE_IS_INSIDE`.

Time Complexity: $O(n)$ per call, where n is the distance between `lo` and `hi`.

Space Complexity: $O(1)$ auxiliary.

```
template<class Pt>
auto cross(const Pt &a, const Pt &b, const Pt &o) {
    return (a.x - o.x) * (b.y - o.y) - (a.y - o.y) * (b.x - o.x);
}

// Detection is exact for integer-coordinate Pt.
template<class Pt, class It>
bool point_in_polygon(const Pt &p, It lo, It hi) {
    static const bool EDGE_IS_INSIDE = true;
    if (lo == hi) {
        return false;
    }
    bool res = false;
    for (It i = lo, j = hi - 1; i != hi; j = i++) {
        if (i->y == p.y && (i->x == p.x || (j->y == p.y && (i->x <= p.x || j->x <= p.x)))) {
            return EDGE_IS_INSIDE;
        }
        if ((p.y < i->y) != (p.y < j->y)) {
            auto det = cross(*i, *j, p);
            if (det == 0) {
                return EDGE_IS_INSIDE;
            }
            if ((0 < det) != (i->y < j->y)) {
                res = !res;
            }
        }
    }
    return res;
}
```

Example Usage

```
#include <cassert>
#include <vector>
using namespace std;

struct Point {
    double x, y;
    Point(double x = 0, double y = 0) : x(x), y(y) {}
};

struct PointI {
    int x, y;
    PointI(int x = 0, int y = 0) : x(x), y(y) {}
};

int main() {
    vector<Point> poly{Point(-1, 3), Point(1, 3), Point(2, 1), Point(0, 0)};
    assert(point_in_polygon(Point(1, 2), poly.begin(), poly.end()));
    assert(point_in_polygon(Point(0, 3), poly.begin(), poly.end()));
    assert(!point_in_polygon(Point(0, 3.01), poly.begin(), poly.end()));
    assert(!point_in_polygon(Point(2, 2), poly.begin(), poly.end()));
}
```

```
// Integer-coordinate points: exact detection.
vector<PointI> ipoly{{0, 0}, {4, 0}, {4, 4}, {0, 4}};
assert(point_in_polygon(PointI(2, 2), ipoly.begin(), ipoly.end()));
assert(!point_in_polygon(PointI(5, 2), ipoly.begin(), ipoly.end()));

return 0;
}
```

7.3.3 Convex Hull and Diametral Pair

Given a list of points in two dimensions, computes the convex hull using the monotone chain algorithm and the diametral pair using rotating calipers. The functions are templated on the iterator type; the value type of the iterator is used as the point type. Both functions accept either floating-point or integral coordinates, but since they use only cross product comparisons, they happen to be exact for integral points.

Overflow warning: `cross()` and the squared distance in `diametral_pair()` grow like the squared coordinate magnitude. With 32-bit `int` coordinates they overflow once coordinates exceed a few tens of thousands; use a 64-bit (`long long`) coordinate type for larger integer inputs.

- `convex_hull(lo, hi)` returns the convex hull in clockwise order. The input range is sorted lexicographically after the call. To instead return the hull points counter-clockwise order, replace every `>= 0` with `<= 0`.
- `diametral_pair(lo, hi)` returns the maximum-distance pair of points.

Time Complexity: $O(n \log n)$ per call, where n is the distance between `lo` and `hi`.

Space Complexity: $O(n)$ auxiliary for storage of the convex hull.

```
#include <algorithm>
#include <cmath>
#include <random>
#include <utility>
#include <vector>

template<class Pt>
auto cross(const Pt &a, const Pt &b, const Pt &o) {
    // Overflow risk for integer Pt: these products are ~0(max_coord^2); use long long if necessary.
    return (a.x - o.x) * (b.y - o.y) - (a.y - o.y) * (b.x - o.x);
}

// Convex hull: exact for integer-coordinate points.
template<class It>
auto convex_hull(It lo, It hi) {
    using Pt = typename std::iterator_traits<It>::value_type;
    int k = 0;
    if (hi - lo <= 1) {
        return std::vector<Pt>(lo, hi);
    }
    std::vector<Pt> res(2 * (int)(hi - lo));
    std::sort(lo, hi);
    for (It it = lo; it != hi; ++it) {
        while (k >= 2 && cross(res[k - 1], *it, res[k - 2]) >= 0) {
            k--;
        }
        res[k++] = *it;
    }
    int t = k + 1;
    for (It it = hi - 1; it != lo; --it) {
        while (k >= t && cross(res[k - 1], *it, res[k - 2]) >= 0) {
            k--;
        }
        res[k++] = *it;
    }
}
```

```

    }
    res.resize(k - 1);
    return res;
}

// Diametral pair: squared-distance comparisons are exact for integer-coordinate points.
template<class It>
auto diametral_pair(It lo, It hi) {
    using Pt = typename std::iterator_traits<It>::value_type;
    auto h = convex_hull(lo, hi);
    int m = h.size();
    if (m == 1) {
        return std::pair<Pt, Pt>{h[0], h[0]};
    }
    if (m == 2) {
        return std::pair<Pt, Pt>{h[0], h[1]};
    }
    int k = 1;
    while (std::abs(cross(h[0], h[(k + 1) % m], h[m - 1])) > std::abs(cross(h[0], h[k], h[m - 1]))) {
        k++;
    }
    auto sqdist = [](const Pt &a, const Pt &b) {
        // Overflow risk for integer Pt: ~0(max_coord^2); use long long if necessary.
        auto dx = a.x - b.x, dy = a.y - b.y;
        return dx * dx + dy * dy;
    };
    auto maxsq = sqdist(h[0], h[0]);
    std::pair<Pt, Pt> res{h[0], h[0]};
    for (int i = 0, j = k; i <= k && j < m; i++) {
        auto d = sqdist(h[i], h[j]);
        if (d > maxsq) {
            maxsq = d;
            res = {h[i], h[j]};
        }
        while (j < m && std::abs(cross(h[(i + 1) % m], h[(j + 1) % m], h[i])) >
                std::abs(cross(h[(i + 1) % m], h[j], h[i]))) {
            d = sqdist(h[i], h[(j + 1) % m]);
            if (d > maxsq) {
                maxsq = d;
                res = {h[i], h[(j + 1) % m]};
            }
            j++;
        }
    }
    return res;
}

```

Example Usage

```

#include <cassert>
#include <vector>
using namespace std;

struct PointI {
    int x, y;
    PointI(int x = 0, int y = 0) : x(x), y(y) {}
    bool operator==(const PointI &o) const { return x == o.x && y == o.y; }
    bool operator!=(const PointI &o) const { return !(*this == o); }
    bool operator<(const PointI &o) const { return x != o.x ? x < o.x : y < o.y; }
    bool operator>(const PointI &o) const { return o < *this; }
};

int main() {
    {
        vector<PointI> v{{1, 3}, {1, 2}, {2, 1}, {0, 0}, {-1, 3}};
        std::mt19937 rng(12345);
    }
}

```

```

    std::shuffle(v.begin(), v.end(), rng);
    vector<PointI> h{{-1, 3}, {1, 3}, {2, 1}, {0, 0}};
    assert(convex_hull(v.begin(), v.end()) == h);
}
{
    vector<PointI> v{{0, 0}, {3, 0}, {0, 3}, {1, 1}, {4, 4}};
    auto [p1, p2] = diametral_pair(v.begin(), v.end());
    assert(p1 == PointI(0, 0));
    assert(p2 == PointI(4, 4));
}
{
    vector<PointI> v{{0, 0}, {4, 0}, {4, 4}, {0, 4}, {2, 2}};
    auto h = convex_hull(v.begin(), v.end());
    assert(h.size() == 4); // interior point (2,2) excluded
    auto [p1, p2] = diametral_pair(v.begin(), v.end());
    // diametral pair is exact
    assert(
        (p1 == PointI(0, 0) && p2 == PointI(4, 4)) || (p1 == PointI(4, 4) && p2 == PointI(0, 0)) ||
        (p1 == PointI(4, 0) && p2 == PointI(0, 4)) || (p1 == PointI(0, 4) && p2 == PointI(4, 0))
    );
}
return 0;
}

```

7.3.4 Minimum Enclosing Circle

Given a list of points in two dimensions, finds the circle with smallest area containing all points using a randomized algorithm.

- `minimum_enclosing_circle(lo, hi)` returns the minimum enclosing circle given a range `[lo, hi]` of points. The input range is shuffled after the call. The function is templated on the iterator type and works with any point type `Pt` having numeric `.x` and `.y` fields. The output is always a `Circle` with `double` coordinates, so integral inputs are accepted and converted internally.

Time Complexity: $O(n)$ on average, where n is the distance between `lo` and `hi`.

Space Complexity: $O(1)$ auxiliary.

```

#include <algorithm>
#include <cmath>
#include <random>
#include <stdexcept>
#include <utility>

const double EPS = 1e-9;

#define EQ(a, b) (fabs((a) - (b)) <= EPS)
#define LE(a, b) ((a) <= (b) + EPS)
#define LT(a, b) ((a) < (b) - EPS)

double sqnorm(double x, double y) {
    return x * x + y * y;
}

// SFINAE guard: valid only when Pt exposes numeric .x/.y members. This keeps the templated point
// constructors from hijacking calls like Circle(double, double, double).
template<class Pt>
using if_point = decltype(std::declval<const Pt &>().x, std::declval<const Pt &>().y, void());

struct Circle {
    double h, k, r;

    Circle() : h(0), k(0), r(0) {}
    Circle(double h, double k, double r) : h(h), k(k), r(fabs(r)) {}
}

```



```

template<class Pt, class = if_point<Pt>>
Circle(const Pt &a, const Pt &b) {
    h = (a.x + b.x) / 2.0;
    k = (a.y + b.y) / 2.0;
    r = std::hypot(a.x - h, a.y - k);
}

template<class Pt, class = if_point<Pt>>
Circle(const Pt &a, const Pt &b, const Pt &c) {
    double an = sqnorm(b.x - c.x, b.y - c.y);
    double bn = sqnorm(a.x - c.x, a.y - c.y);
    double cn = sqnorm(a.x - b.x, a.y - b.y);
    double wa = an * (bn + cn - an), wb = bn * (an + cn - bn), wc = cn * (an + bn - cn);
    double w = wa + wb + wc;
    if (EQ(w, 0)) {
        throw std::runtime_error("No circumcircle from collinear points.");
    }
    h = (wa * a.x + wb * b.x + wc * c.x) / w;
    k = (wa * a.y + wb * b.y + wc * c.y) / w;
    r = std::hypot(a.x - h, a.y - k);
}

template<class Pt, class = if_point<Pt>>
bool contains(const Pt &p) const {
    return LE(sqnorm(p.x - h, p.y - k), r * r);
}
};

// Input points can be any type with .x and .y fields; Circle output is always double.
template<class It>
Circle minimum_enclosing_circle(It lo, It hi) {
    if (lo == hi) {
        return Circle(0, 0, 0);
    }
    if (lo + 1 == hi) {
        return Circle(lo->x, lo->y, 0);
    }
    static std::mt19937 rng(std::random_device{}());
    std::shuffle(lo, hi, rng);
    Circle res(*lo, *(lo + 1));
    for (It i = lo + 2; i != hi; ++i) {
        if (res.contains(*i)) {
            continue;
        }
        res = Circle(*i, *lo);
        for (It j = lo + 1; j != i; ++j) {
            if (res.contains(*j)) {
                continue;
            }
            res = Circle(*i, *j);
            for (It k = lo; k != j; ++k) {
                if (!res.contains(*k)) {
                    res = Circle(*i, *j, *k);
                }
            }
        }
    }
    return res;
}

```

Example Usage

```

#include <cassert>
#include <vector>
using namespace std;

```

```

struct Point {
    double x, y;
    Point(double x = 0, double y = 0) : x(x), y(y) {}
    bool operator==(const Point &p) const { return EQ(x, p.x) && EQ(y, p.y); }
};

struct PointI {
    int x, y;
    PointI(int x = 0, int y = 0) : x(x), y(y) {}
};

int main() {
    vector<Point> v{{0, 0}, {0, 1}, {1, 0}, {1, 1}};
    Circle res = minimum_enclosing_circle(v.begin(), v.end());
    assert(EQ(res.h, 0.5) && EQ(res.k, 0.5) && EQ(res.r, 1 / sqrt(2)));

    // Integer-coordinate input: Circle output is always double.
    vector<PointI> iv{{0, 0}, {0, 2}, {2, 0}, {2, 2}};
    Circle ic = minimum_enclosing_circle(iv.begin(), iv.end());
    assert(EQ(ic.h, 1.0) && EQ(ic.k, 1.0));

    return 0;
}

```

7.3.5 Closest Pair

Given a list of points, finds the closest pair using divide and conquer.

- `closest_pair(lo, hi, &res)` returns the minimum squared distance between any two points in `[lo, hi)`. The input range is sorted after the call. The closest pair is stored into `res` if non-null. The function is templated on the iterator type. The returned minimum squared distance preserves the coordinate arithmetic type, so integer-coordinate inputs stay exact. The result pair contains the original point type.

Overflow warning: the squared distance grows like the squared coordinate magnitude. For integer inputs use a 64-bit coordinate type (e.g. `PointL` from 7.1.1) once coordinates exceed a few tens of thousands, or the 32-bit products overflow.

Time Complexity: $O(n \log^2 n)$ per call, where n is the distance between `lo` and `hi`.

Space Complexity: $O(n \log^2 n)$ auxiliary stack space.

```

#include <algorithm>
#include <cmath>
#include <iterator>
#include <limits>
#include <utility>
#include <vector>

const double EPS = 1e-9;

#define EQ(a, b) (fabs((a) - (b)) <= EPS)

template<class It, class T>
T closest_pair(
    It lo, It hi,
    std::pair<
        typename std::iterator_traits<It>::value_type,
        typename std::iterator_traits<It>::value_type> *res = nullptr,
    T mindist = std::numeric_limits<T>::max(), bool sort_x = true
) {
    using Pt = typename std::iterator_traits<It>::value_type;
    if (lo == hi) {
        return mindist;
    }
}

```

```

}
if (sort_x) {
    std::sort(lo, hi, [](const Pt &a, const Pt &b) { return a.x != b.x ? a.x < b.x : a.y < b.y; });
}
It mid = lo + (hi - lo) / 2;
auto midx = mid->x;
T d1 = closest_pair(lo, mid, res, mindist, false);
mindist = std::min(mindist, d1);
T d2 = closest_pair(mid + 1, hi, res, mindist, false);
mindist = std::min(mindist, d2);
std::sort(lo, hi, [](const Pt &a, const Pt &b) { return a.y != b.y ? a.y < b.y : a.x < b.x; });
std::vector<It> t;
for (It it = lo; it != hi; ++it) {
    auto dx = it->x - midx;
    if (dx * dx < mindist) {
        t.push_back(it);
    }
}
for (int i = 0; i < static_cast<int>(t.size()); i++) {
    for (int j = i + 1; j < static_cast<int>(t.size()); j++) {
        const auto &a = *t[i], &b = *t[j];
        auto dy = b.y - a.y;
        if (dy * dy >= mindist) {
            break;
        }
        auto dx = a.x - b.x;
        auto dist = dx * dx + dy * dy;
        if (mindist > dist) {
            mindist = dist;
            if (res) {
                *res = {a, b};
            }
        }
    }
}
return mindist;
}

template<class It>
auto closest_pair(
    It lo, It hi,
    std::pair<
        typename std::iterator_traits<It>::value_type,
        typename std::iterator_traits<It>::value_type> *res = nullptr
) {
    using T = decltype((lo->x - lo->x) * (lo->x - lo->x) + (lo->y - lo->y) * (lo->y - lo->y));
    return closest_pair(lo, hi, res, std::numeric_limits<T>::max(), true);
}

```

Example Usage

```

#include <cassert>
#include <vector>
using namespace std;

struct Point {
    double x, y;
    Point(double x = 0, double y = 0) : x(x), y(y) {}
    bool operator==(const Point &p) const { return EQ(x, p.x) && EQ(y, p.y); }
};

struct PointI {
    int x, y;
    PointI(int x = 0, int y = 0) : x(x), y(y) {}
    bool operator==(const PointI &p) const { return x == p.x && y == p.y; }
};

```

```
int main() {
    vector<Point> v{{2, 3}, {12, 30}, {40, 50}, {5, 1}, {12, 10}, {3, 4}};
    pair<Point, Point> res;
    assert(EQ(closest_pair(v.begin(), v.end(), &res), 2));
    auto [p1, p2] = res;
    assert(p1 == Point(2, 3) && p2 == Point(3, 4));

    // Integer-coordinate input: exact pair selection, int squared distance returned.
    vector<PointI> iv{{0, 0}, {10, 10}, {3, 4}};
    pair<PointI, PointI> ires;
    assert(closest_pair(iv.begin(), iv.end(), &ires) == 25); // (0,0)-(3,4) or (3,4)-(0,0)
    auto [i1, i2] = ires;
    assert((i1 == PointI(0, 0) && i2 == PointI(3, 4)) || (i1 == PointI(3, 4) && i2 == PointI(0, 0)));

    return 0;
}
```

7.3.6 Multiple Segment Intersection (Sweep Line)

Given a list of line segments in two dimensions, determine whether any pair of segments intersect using a sweep line algorithm. The input type `Segment<Pt>` is templated on the point type, so endpoints may be integer (`PointI`) or floating-point (`Point`/`PointD`). The cross-product sign tests in `seg_intersection` are exact for integer endpoints, so intersection detection is exact.

- `find_intersection(lo, hi, &res1, &res2)` returns whether any pair of segments intersect given a range `[lo, hi]` of segments, where `lo` and `hi` are random-access iterators. If an intersection is found, then one such pair of segments will be stored into pointers `res1` and `res2`. Touching behavior is controlled by `TOUCH_IS_INTERSECT`.

Overflow warning: `seg_intersection` forms the usual quadratic cross products, but the y -ordering cross-multiplication (`ay * bdx`) is cubic in the coordinate magnitude. With 32-bit `int` endpoints this overflows once coordinates reach the low thousands, so use a 64-bit (`long long`) coordinate type for any non-trivial integer inputs.

Time Complexity: $O(n \log n)$ per call to `find_intersection(lo, hi, &res1, &res2)`, where n is the distance between `lo` and `hi`.

Space Complexity: $O(n)$ auxiliary heap space for `find_intersection(lo, hi, &res1, &res2)`, where n is the distance between `lo` and `hi`.

```
#include <algorithm>
#include <cassert>
#include <set>
#include <type_traits>
#include <utility>
#include <vector>

template<class Pt>
bool point_on_segment(const Pt &p, const Pt &a, const Pt &b) {
    return (p.x - a.x) * (b.y - a.y) == (p.y - a.y) * (b.x - a.x) && std::min(a.x, b.x) <= p.x &&
        p.x <= std::max(a.x, b.x) && std::min(a.y, b.y) <= p.y && p.y <= std::max(a.y, b.y);
}

// Simplified detection-only version of `seg_intersection` from 7.2.3, with exact arithmetic for
// integer-coordinate Pt.
template<class Pt>
int seg_intersection(const Pt &a, const Pt &b, const Pt &c, const Pt &d) {
    static const bool TOUCH_IS_INTERSECT = true;
    auto ab_x = b.x - a.x, ab_y = b.y - a.y;
    auto ac_x = c.x - a.x, ac_y = c.y - a.y;
    auto cd_x = d.x - c.x, cd_y = d.y - c.y;
```

```

auto ab2 = ab_x * ab_x + ab_y * ab_y;
auto cd2 = cd_x * cd_x + cd_y * cd_y;
if (ab2 == 0) {
    return (TOUCH_IS_INTERSECT && point_on_segment(a, c, d)) ? 0 : -1;
}
if (cd2 == 0) {
    return (TOUCH_IS_INTERSECT && point_on_segment(c, a, b)) ? 0 : -1;
}
auto c1 = ab_x * cd_y - ab_y * cd_x;
auto c2 = ac_x * ab_y - ac_y * ab_x;
if (c1 == 0 && c2 == 0) {
    Pt res1 = std::max(std::min(a, b), std::min(c, d));
    Pt res2 = std::min(std::max(a, b), std::max(c, d));
    bool overlap = TOUCH_IS_INTERSECT ? !(res2 < res1) : (res1 < res2);
    if (overlap) {
        return (res1 == res2) ? 0 : 1;
    }
    return -1;
}
if (c1 == 0) {
    return -1;
}
auto t_num = ac_x * cd_y - ac_y * cd_x;
bool c1_pos = c1 > 0;
bool t_between_01 =
    c1_pos ? (TOUCH_IS_INTERSECT ? (0 <= t_num && t_num <= c1) : (0 < t_num && t_num < c1))
           : (TOUCH_IS_INTERSECT ? (t_num <= 0 && c1 <= t_num) : (t_num < 0 && c1 < t_num));
bool u_between_01 = c1_pos ? (TOUCH_IS_INTERSECT ? (0 <= c2 && c2 <= c1) : (0 < c2 && c2 < c1))
                           : (TOUCH_IS_INTERSECT ? (c2 <= 0 && c1 <= c2) : (c2 < 0 && c1 < c2));
return (t_between_01 && u_between_01) ? 0 : -1;
}

// The point type Pt must support operator< (used to canonicalize endpoint order).
template<class Pt>
struct Segment {
    Pt p, q;

    Segment() {}
    Segment(const Pt &p, const Pt &q) : p(std::min(p, q)), q(std::max(p, q)) {}
};

template<class Pt>
bool intersect(const Segment<Pt> &s1, const Segment<Pt> &s2) {
    return seg_intersection(s1.p, s1.q, s2.p, s2.q) >= 0;
}

// Seg is deduced from the output pointers (e.g. Segment<PointI>* or Segment<PointD>*).
template<class It, class Seg>
bool find_intersection(It lo, It hi, Seg *res1, Seg *res2) {
    using Pt = std::decay_t<decltype(lo->p)>;
    struct Event {
        Pt p;
        int type;
        It seg;
    };
    std::vector<Event> e;
    for (It it = lo; it != hi; ++it) {
        if (it->p > it->q) {
            std::swap(it->p, it->q);
        }
        e.push_back(Event{it->p, 1, it});
        e.push_back(Event{it->q, -1, it});
    }
    std::sort(e.begin(), e.end(), [](const Event &a, const Event &b) {
        if (a.p.x != b.p.x) {
            return a.p.x < b.p.x;
        }
        if (a.type != b.type) {

```

```

    return b.type < a.type;
}
return a.p.y < b.p.y;
});
// Compare y-values at sweep coordinate x without division: y = (p.y*dx + dy*(x - p.x)) / dx.
// Vertical segments use their lower endpoint
auto ycmp = [](const auto &a, const auto &b, auto x) {
    // Overflow risk for integer Pt: the final cross-multiply is ~0(max_coord^3); use 64-bit
    // coordinates for non-trivial integer inputs.
    auto adx = a.q.x - a.p.x, bdx = b.q.x - b.p.x;
    auto ay = a.p.y * adx + (a.q.y - a.p.y) * (x - a.p.x);
    auto by = b.p.y * bdx + (b.q.y - b.p.y) * (x - b.p.x);
    if (adx == 0) {
        ay = a.p.y;
        adx = 1;
    }
    if (bdx == 0) {
        by = b.p.y;
        bdx = 1;
    }
    auto lhs = ay * bdx, rhs = by * adx;
    return (lhs < rhs) ? -1 : ((rhs < lhs) ? 1 : 0);
};
auto cmp = [lo, ycmp](It a, It b) {
    if (a == b) {
        return false;
    }
    auto x = std::max(a->p.x, b->p.x);
    int order = ycmp(*a, *b, x);
    return (order == 0) ? (a - lo < b - lo) : (order < 0);
};
using active_set = std::set<It, decltype(cmp)>;
active_set s(cmp);
std::vector<typename active_set::iterator> position(hi - lo);
for (const auto &ev : e) {
    It seg = ev.seg;
    if (ev.type == 1) {
        auto it = s.insert(seg).first;
        position[seg - lo] = it;
        auto next = it;
        if (++next != s.end() && intersect(**next, *seg)) {
            *res1 = **next;
            *res2 = *seg;
            return true;
        }
        if (it != s.begin() && intersect(**--it, *seg)) {
            *res1 = **it;
            *res2 = *seg;
            return true;
        }
    } else {
        auto it = position[seg - lo];
        auto next = it;
        if (++next != s.end() && it != s.begin()) {
            auto prev = it;
            if (intersect(**next, **--prev)) {
                *res1 = **next;
                *res2 = **prev;
                return true;
            }
        }
        s.erase(it);
    }
}
return false;
}

```

Example Usage

```
#include <vector>
using namespace std;

struct Point {
    double x, y;
    Point(double x = 0, double y = 0) : x(x), y(y) {}
    bool operator==(const Point &p) const { return x == p.x && y == p.y; }
    bool operator!=(const Point &p) const { return !(*this == p); }
    bool operator<(const Point &p) const { return x != p.x ? x < p.x : y < p.y; }
    bool operator>(const Point &p) const { return p < *this; }
};

struct PointI {
    int x, y;
    PointI(int x = 0, int y = 0) : x(x), y(y) {}
    bool operator==(const PointI &p) const { return x == p.x && y == p.y; }
    bool operator!=(const PointI &p) const { return !(*this == p); }
    bool operator<(const PointI &p) const { return x != p.x ? x < p.x : y < p.y; }
    bool operator>(const PointI &p) const { return p < *this; }
};

int main() {
    { // Double-coordinate segments.
        vector<Segment<Point>> v{
            Segment<Point>(Point(0, 0), Point(2, 2)), Segment<Point>(Point(3, 0), Point(0, -1)),
            Segment<Point>(Point(0, 2), Point(2, -2)), Segment<Point>(Point(0, 3), Point(9, 0))
        };
        Segment<Point> res1, res2;
        assert(find_intersection(v.begin(), v.end(), &res1, &res2));
        assert(res1.p == Point(0, 0) && res1.q == Point(2, 2));
        assert(res2.p == Point(0, 2) && res2.q == Point(2, -2));
    }
    { // Integer-coordinate segments: detection is exact.
        vector<Segment<PointI>> v{
            Segment<PointI>({0, 0}, {2, 2}), Segment<PointI>({3, 0}, {0, -1}),
            Segment<PointI>({0, 2}, {2, -2}), Segment<PointI>({0, 3}, {9, 0})
        };
        Segment<PointI> res1, res2;
        assert(find_intersection(v.begin(), v.end(), &res1, &res2));

        vector<Segment<PointI>> disjoint{
            Segment<PointI>({0, 0}, {1, 0}), Segment<PointI>({0, 5}, {1, 5})
        };
        assert(!find_intersection(disjoint.begin(), disjoint.end(), &res1, &res2));
    }
    return 0;
}
```

7.4 Advanced Planar Geometry

7.4.1 Convex Polygon Cut

Given a convex polygon and two points specifying an infinite line, cuts off the right part and returns the resulting left part.

- `convex_cut(lo, hi, p, q)` returns the left-side polygon in clockwise order after cutting with the line through `p` and `q`. The function is templated on the input point type. The `turn` classification uses cross products and is exact for integer-coordinate points; the edge-cut intersection point uses floating-point arithmetic, and the output `Point` always has `double` coordinates. Overflow warning: those cross products grow like the

squared coordinate magnitude, so for integer point types use a 64-bit coordinate type (e.g. `PointL` from 7.1.1) for large coordinates.

Time Complexity: $O(n)$ per call to `convex_cut(lo, hi, p, q)`, where n is the distance between `lo` and `hi`.

Space Complexity: $O(n)$ auxiliary for storage of the resulting convex cut.

```
#include <cmath>
#include <cstdint>
#include <stdexcept>
#include <utility>
#include <vector>

const double EPS = 1e-9;

#define EQ(a, b) (fabs((a) - (b)) <= EPS)
#define LT(a, b) ((a) < (b) - EPS)

struct Point {
    double x, y;
    Point(double x = 0, double y = 0) : x(x), y(y) {}
    bool operator==(const Point &p) const { return x == p.x && y == p.y; }
};

template<class PtA, class PtB, class PtO>
double cross(const PtA &a, const PtB &b, const PtO &o) {
    return (a.x - o.x) * (b.y - o.y) - (a.y - o.y) * (b.x - o.x);
}

template<class PtA, class PtO, class PtB>
int turn(const PtA &a, const PtO &o, const PtB &b) {
    double c = cross(a, b, o);
    return LT(c, 0) ? -1 : (LT(0, c) ? 1 : 0);
}

template<class PtA, class PtB>
int line_intersection(
    const PtA &p1, const PtA &p2, const PtB &p3, const PtB &p4, Point *p = nullptr
) {
    double a1 = p2.y - p1.y, b1 = p1.x - p2.x;
    double c1 = -(p1.x * p2.y - p2.x * p1.y);
    double a2 = p4.y - p3.y, b2 = p3.x - p4.x;
    double c2 = -(p3.x * p4.y - p4.x * p3.y);
    double x = -(c1 * b2 - c2 * b1), y = -(a1 * c2 - a2 * c1);
    double det = a1 * b2 - a2 * b1;
    if (EQ(det, 0)) {
        return (EQ(x, 0) && EQ(y, 0)) ? 1 : -1;
    }
    if (p != nullptr) {
        *p = Point(x / det, y / det);
    }
    return 0;
}

template<class It, class Pt>
std::vector<Point> convex_cut(It lo, It hi, const Pt &p, const Pt &q) {
    if (EQ(p.x, q.x) && EQ(p.y, q.y)) {
        throw std::runtime_error("Cannot cut using line from identical points.");
    }
    if (lo == hi) {
        return {};
    }
    std::vector<Point> res;
    for (It i = lo, j = hi - 1; i != hi; j = i++) {
        Point pj((double)j->x, (double)j->y), pi((double)i->x, (double)i->y);
        int d1 = turn(q, p, pj), d2 = turn(q, p, pi);
        if (d1 >= 0) {
            res.push_back(pj);
        }
    }
}
```



```

    }
    if (d1 * d2 < 0) {
        Point r;
        line_intersection(p, q, pj, pi, &r);
        res.push_back(r);
    }
}
return res;
}

```

Example Usage

```

#include <cassert>
using namespace std;

struct PointI {
    int x, y;
    PointI(int x = 0, int y = 0) : x(x), y(y) {}
};

int main() {
    {
        vector<Point> v{{1, 3}, {2, 2}, {2, 1}, {0, 0}, {-1, 3}};
        // Cut using the vertical line through (0, 0).
        vector<Point> c{{-1, 3}, {0, 3}, {0, 0}};
        assert(convex_cut(v.begin(), v.end(), Point(0, 0), Point(0, 1)) == c);
    }
    { // On a non-convex input, the result may be multiple disjoint polygons!
        vector<Point> v{{0, 0}, {2, 2}, {0, 4}, {3, 4}, {3, 0}};
        vector<Point> c{{1, 0}, {0, 0}, {1, 1}, {1, 3}, {0, 4}, {1, 4}};
        assert(convex_cut(v.begin(), v.end(), Point(1, 0), Point(1, 4)) == c);
    }
    {
        vector<PointI> v{{0, 0}, {4, 0}, {4, 4}, {0, 4}};
        vector<Point> c{{0, 4}, {0, 0}, {2, 0}, {2, 4}};
        assert(convex_cut(v.begin(), v.end(), PointI(2, 0), PointI(2, 4)) == c);
    }
    return 0;
}

```

7.4.2 Polygon Intersection and Union

Given two polygons, determine the areas of their intersection and union using a sweep line algorithm and the inclusion-exclusion principle.

- `intersection_area(lo1, hi1, lo2, hi2)` returns the intersection area of two polygons respectively specified by two ranges `[lo1, hi1)` and `[lo2, hi2)` of vertices in clockwise order, where `lo1`, `hi1`, `lo2`, and `hi2` must be random-access iterators.
- `union_area(lo1, hi1, lo2, hi2)` returns the union area of two polygons respectively specified by two ranges `[lo1, hi1)` and `[lo2, hi2)` of vertices in clockwise order, where `lo1`, `hi1`, `lo2`, and `hi2` must be random-access iterators.

Overflow warning: the segment-intersection tests form cross products in the input point's coordinate type, which grow like the squared coordinate magnitude. For integer point types use long long (e.g. `PointL` from 7.1.1) if necessary.

Time Complexity: $O(n^2 \log n)$ per call to `intersection_area(lo1, hi1, lo2, hi2)` and `union_area(lo1, hi1, lo2, hi2)` where n is the sum of distances between `lo1` and `hi1` and `lo2` and `hi2` respectively.

Space Complexity: $O(n)$ auxiliary heap space for `intersection_area(lo1, hi1, lo2, hi2)` and `union_area(lo1, hi1, lo2, hi2)`, where n is the sum of distances between `lo1` and `hi1` and `lo2` and `hi2` respectively.

```
#include <algorithm>
#include <cmath>
#include <set>
#include <utility>
#include <vector>

const double EPS = 1e-9;

#define EQ(a, b) (fabs((a) - (b)) <= EPS)
#define LT(a, b) ((a) < (b) - EPS)
#define LE(a, b) ((a) <= (b) + EPS)

template<class Pt>
int seg_intersection1(
    const Pt &A, const Pt &B, const Pt &C, const Pt &D, double *outx, double *outy
) {
    auto less = [](const Pt &u, const Pt &v) { return u.x != v.x ? u.x < v.x : u.y < v.y; };
    auto minp = [&](const Pt &u, const Pt &v) -> const Pt & { return less(v, u) ? v : u; };
    auto maxp = [&](const Pt &u, const Pt &v) -> const Pt & { return less(u, v) ? v : u; };
    static const bool TOUCH_IS_INTERSECT = true;
    auto ab_x = B.x - A.x, ab_y = B.y - A.y;
    auto ac_x = C.x - A.x, ac_y = C.y - A.y;
    auto cd_x = D.x - C.x, cd_y = D.y - C.y;
    auto c1 = ab_x * cd_y - ab_y * cd_x;
    auto c2 = ac_x * ab_y - ac_y * ab_x;
    if (EQ(c1, 0) && EQ(c2, 0)) { // Collinear.
        const Pt &res1 = maxp(minp(A, B), minp(C, D));
        const Pt &res2 = minp(maxp(A, B), maxp(C, D));
        bool overlap = TOUCH_IS_INTERSECT ? !less(res2, res1) : less(res1, res2);
        if (overlap) {
            if (res1 == res2) {
                *outx = (double)res1.x;
                *outy = (double)res1.y;
                return 0; // Collinear and meeting at an endpoint.
            }
            return 1; // Collinear and overlapping.
        }
        return -1; // Collinear and disjoint.
    }
    if (EQ(c1, 0)) {
        return -1; // Parallel and disjoint.
    }
    auto t_num = ac_x * cd_y - ac_y * cd_x;
    bool c1_pos = c1 > 0;
    bool t_between_01 = c1_pos ? (TOUCH_IS_INTERSECT ? (LE(0, t_num) && LE(t_num, c1))
                                                                    : (LT(0, t_num) && LT(t_num, c1)))
                                : (TOUCH_IS_INTERSECT ? (LE(t_num, 0) && LE(c1, t_num))
                                                                    : (LT(t_num, 0) && LT(c1, t_num)));

    bool u_between_01 =
        c1_pos ? (TOUCH_IS_INTERSECT ? (LE(0, c2) && LE(c2, c1)) : (LT(0, c2) && LT(c2, c1)))
              : (TOUCH_IS_INTERSECT ? (LE(c2, 0) && LE(c1, c2)) : (LT(c2, 0) && LT(c1, c2)));
    if (t_between_01 && u_between_01) {
        double t = (double)t_num / c1;
        *outx = A.x + t * ab_x;
        *outy = A.y + t * ab_y;
        return 0; // Non-parallel with one intersection.
    }
    return -1; // Non-parallel with no intersections.
}

// The two segments may use different point types (e.g. polygon edge vs. sweep line).
template<class PtA, class PtB>
int line_intersection1(
```

```

    const PtA &p1, const PtA &p2, const PtB &p3, const PtB &p4, double *outx, double *outy
) {
    auto a1 = p2.y - p1.y, b1 = p1.x - p2.x;
    auto c1 = -(p1.x * p2.y - p2.x * p1.y);
    auto a2 = p4.y - p3.y, b2 = p3.x - p4.x;
    auto c2 = -(p3.x * p4.y - p4.x * p3.y);
    auto x = -(c1 * b2 - c2 * b1), y = -(a1 * c2 - a2 * c1);
    auto det = a1 * b2 - a2 * b1;
    if (EQ(det, 0)) {
        return (EQ(x, 0) && EQ(y, 0)) ? 1 : -1;
    }
    if (outx != nullptr && outy != nullptr) {
        *outx = (double)x / det;
        *outy = (double)y / det;
    }
    return 0;
}

struct Event {
    double y;
    int mask_delta;
    Event(double y = 0, int mask_delta = 0) : y(y), mask_delta(mask_delta) {}
};

template<class It>
double intersection_area(It lo1, It hi1, It lo2, It hi2) {
    if (lo1 == hi1 || lo2 == hi2) {
        return 0;
    }
    struct _PointD {
        double x, y;
    }; // For line intersection with the sweep line.
    std::vector<It> plo{lo1, lo2}, phi{hi1, hi2};
    std::set<double> xs;
    for (It it = lo1; it != hi1; ++it) {
        xs.insert(it->x);
    }
    for (It it = lo2; it != hi2; ++it) {
        xs.insert(it->x);
    }
    for (It i1 = lo1, j1 = hi1 - 1; i1 != hi1; j1 = i1++) {
        for (It i2 = lo2, j2 = hi2 - 1; i2 != hi2; j2 = i2++) {
            double outx, outy;
            if (seg_intersection1(*i1, *j1, *i2, *j2, &outx, &outy) == 0) {
                xs.insert(outx);
            }
        }
    }
    std::vector<double> xsa(xs.begin(), xs.end());
    double res = 0;
    for (size_t k = 0; k + 1 < xsa.size(); k++) {
        double x = (xsa[k] + xsa[k + 1]) / 2;
        _PointD sweep0{x, 0}, sweep1{x, 1};
        std::vector<Event> events;
        for (int poly = 0; poly < 2; poly++) {
            It lo = plo[poly], hi = phi[poly];
            double area = 0;
            for (It i = lo, j = hi - 1; i != hi; j = i++) {
                area += (j->x - i->x) * (j->y + i->y);
            }
            for (It j = lo, i = hi - 1; j != hi; i = j++) {
                double px, py;
                if (line_intersection1(*j, *i, sweep0, sweep1, &px, &py) == 0) {
                    double y = py, x0 = i->x, x1 = j->x;
                    int sgn_area = (area < 0 ? -1 : (area > 0 ? 1 : 0));
                    if (x0 < x && x1 > x) {
                        events.emplace_back(y, sgn_area * (1 << poly));
                    } else if (x0 > x && x1 < x) {

```

```

        events.emplace_back(y, -sgn_area * (1 << poly));
    }
}
}
std::sort(events.begin(), events.end(), [](const Event &e1, const Event &e2) {
    if (e1.y != e2.y) {
        return e1.y < e2.y;
    }
    return e1.mask_delta < e2.mask_delta;
});
double a = 0;
int mask = 0;
for (size_t j = 0; j < events.size(); j++) {
    if (mask == 3) {
        a += events[j].y - events[j - 1].y;
    }
    mask += events[j].mask_delta;
}
res += a * (xsa[k + 1] - xsa[k]);
}
return res;
}

template<class It>
double polygon_area(It lo, It hi) {
    if (lo == hi) {
        return 0;
    }
    double area = 0;
    for (It i = lo, j = hi - 1; i != hi; j = i++) {
        area += (j->x - i->x) * (j->y + i->y);
    }
    return fabs(area / 2.0);
}

template<class It>
double union_area(It lo1, It hi1, It lo2, It hi2) {
    return polygon_area(lo1, hi1) + polygon_area(lo2, hi2) - intersection_area(lo1, hi1, lo2, hi2);
}

```

Example Usage

```

#include <cassert>
using namespace std;

struct Point {
    double x, y;
    Point(double x = 0, double y = 0) : x(x), y(y) {}
    bool operator==(const Point &p) const { return EQ(x, p.x) && EQ(y, p.y); }
    bool operator!=(const Point &p) const { return !(*this == p); }
    bool operator<(const Point &p) const { return x != p.x ? x < p.x : y < p.y; }
};

struct PointI {
    int x, y;
    PointI(int x = 0, int y = 0) : x(x), y(y) {}
    bool operator==(const PointI &o) const { return x == o.x && y == o.y; }
    bool operator!=(const PointI &p) const { return !(*this == p); }
    bool operator<(const PointI &p) const { return x != p.x ? x < p.x : y < p.y; }
};

int main() {
    vector<Point> p, s;
    // Irregular pentagon a triangle of area 1.5 overlapping quadrant 2.
    p.emplace_back(1, 3);
}

```

```

p.emplace_back(1, 2);
p.emplace_back(2, 1);
p.emplace_back(0, 0);
p.emplace_back(-1, 3);
// Square of area 12.5 in quadrant 2.
s.emplace_back(0, 0);
s.emplace_back(0, 3);
s.emplace_back(-3, 3);
s.emplace_back(-3, 0);
assert(EQ(1.5, intersection_area(p.begin(), p.end(), s.begin(), s.end())));
assert(EQ(12.5, union_area(p.begin(), p.end(), s.begin(), s.end())));

// Integer-coordinate polygons are accepted (computation proceeds in double).
vector<PointI> ip{{1, 3}, {1, 2}, {2, 1}, {0, 0}, {-1, 3}};
vector<PointI> is{{0, 0}, {0, 3}, {-3, 3}, {-3, 0}};
assert(EQ(1.5, intersection_area(ip.begin(), ip.end(), is.begin(), is.end())));
assert(EQ(12.5, union_area(ip.begin(), ip.end(), is.begin(), is.end())));

return 0;
}

```

7.4.3 Delaunay Triangulation (Simple)

Given a set P of two dimensional points, the Delaunay triangulation of P is a set of non-overlapping triangles that covers the entire convex hull of P such that no point in P lies within the circumcircle of any of the resulting triangles. For any point p in the convex hull of P (but not necessarily in P), the nearest point is guaranteed to be a vertex of the enclosing triangle from the triangulation.

The triangulation may not exist (e.g. for a set of collinear points), or may not be unique if it does exist. The following program assumes its existence and produces one such valid result using a simple algorithm which encases each triangle in a circle and rejects the triangle if another point in the tessellation is within the generalized circle.

- `delaunay_triangulation(lo, hi)` returns a Delaunay triangulation for the input range `[lo, hi)` of points, where `lo` and `hi` must be random-access iterators, or an empty vector if a triangulation does not exist.

Time Complexity: $O(n^4)$ per call to `delaunay_triangulation(lo, hi)`, where n is the distance between `lo` and `hi`.

Space Complexity: $O(n)$ auxiliary heap space for storage of the Delaunay triangulation.

```

#include <algorithm>
#include <array>
#include <cmath>
#include <utility>
#include <vector>

const double EPS = 1e-9;

#define EQ(a, b) (fabs((a) - (b)) <= EPS)
#define LT(a, b) ((a) < (b) - EPS)
#define LE(a, b) ((a) <= (b) + EPS)

template<class Pt>
int seg_intersection(const Pt &a, const Pt &b, const Pt &c, const Pt &d) {
    static const bool TOUCH_IS_INTERSECT = false; // false is important!
    double ab_x = b.x - a.x, ab_y = b.y - a.y;
    double ac_x = c.x - a.x, ac_y = c.y - a.y;
    double cd_x = d.x - c.x, cd_y = d.y - c.y;
    double c1 = ab_x * cd_y - ab_y * cd_x;
    double c2 = ac_x * ab_y - ac_y * ab_x;
    if (EQ(c1, 0) && EQ(c2, 0)) { // Collinear.
        Pt res1 = std::max(std::min(a, b), std::min(c, d));
        Pt res2 = std::min(std::max(a, b), std::max(c, d));
        if (TOUCH_IS_INTERSECT ? !(res2 < res1) : (res1 < res2)) {
            return 1; // Collinear and overlapping.
        }
    }
    return 0;
}

```

```

    }
    return -1; // Collinear and disjoint.
}
if (EQ(c1, 0)) {
    return -1; // Parallel and disjoint.
}
double t_num = ac_x * cd_y - ac_y * cd_x;
bool c1_pos = c1 > 0;
bool t_between_01 = (TOUCH_IS_INTERSECT ? (LE(0, t_num) && LE(t_num, c1))
                                           : (LT(0, t_num) && LT(t_num, c1)))
                   : (TOUCH_IS_INTERSECT ? (LE(t_num, 0) && LE(c1, t_num))
                                           : (LT(t_num, 0) && LT(c1, t_num)));

bool u_between_01 =
    c1_pos ? (TOUCH_IS_INTERSECT ? (LE(0, c2) && LE(c2, c1)) : (LT(0, c2) && LT(c2, c1)))
           : (TOUCH_IS_INTERSECT ? (LE(c2, 0) && LE(c1, c2)) : (LT(c2, 0) && LT(c1, c2)));
if (t_between_01 && u_between_01) {
    return 0; // Non-parallel with one intersection.
}
return -1; // Non-parallel with no intersections.
}

struct Triangle {
    double ax, ay, bx, by, cx, cy;

    Triangle(double ax, double ay, double bx, double by, double cx, double cy)
        : ax(ax), ay(ay), bx(bx), by(by), cx(cx), cy(cy) {}

    bool operator==(const Triangle &t) const {
        return EQ(ax, t.ax) && EQ(ay, t.ay) && EQ(bx, t.bx) && EQ(by, t.by) && EQ(cx, t.cx) &&
            EQ(cy, t.cy);
    }
};

// Accepts any point type with numeric .x/.y; coordinates are copied to double arrays.
template<class It>
std::vector<Triangle> delaunay_triangulation(It lo, It hi) {
    using Pt = typename std::iterator_traits<It>::value_type;
    int n = hi - lo;
    std::vector<Pt> pts;
    std::vector<double> x, y, z;
    for (It it = lo; it != hi; ++it) {
        pts.emplace_back(it->x, it->y);
        x.emplace_back(it->x);
        y.emplace_back(it->y);
        z.emplace_back((double)it->x * it->x + (double)it->y * it->y);
    }
    std::vector<Triangle> res;
    std::vector<std::array<int, 3>> res_idx;
    for (int i = 0; i < n - 2; i++) {
        for (int j = i + 1; j < n; j++) {
            for (int k = i + 1; k < n; k++) {
                if (j == k) {
                    continue;
                }
                double nx = (y[j] - y[i]) * (z[k] - z[i]) - (y[k] - y[i]) * (z[j] - z[i]);
                double ny = (x[k] - x[i]) * (z[j] - z[i]) - (x[j] - x[i]) * (z[k] - z[i]);
                double nz = (x[j] - x[i]) * (y[k] - y[i]) - (x[k] - x[i]) * (y[j] - y[i]);
                if (LE(0, nz)) {
                    continue;
                }
                std::vector<Pt> s1{pts[i], pts[j], pts[k], pts[i]};
                for (int m = 0; m < n; m++) {
                    if (nx * (x[m] - x[i]) + ny * (y[m] - y[i]) + nz * (z[m] - z[i]) > 0) {
                        goto skip;
                    }
                }
                // Handle four points on a circle.
                for (const auto &tri : res_idx) {

```

```

        std::vector<Pt> s2{pts[tri[0]], pts[tri[1]], pts[tri[2]], pts[tri[0]]};
        for (int u = 0; u < 3; u++) {
            for (int v = 0; v < 3; v++) {
                if (seg_intersection(s1[u], s1[u + 1], s2[v], s2[v + 1]) == 0) {
                    goto skip;
                }
            }
        }
        res.emplace_back(pts[i].x, pts[i].y, pts[j].x, pts[j].y, pts[k].x, pts[k].y);
        res_idx.push_back({i, j, k});
        skip:;
    }
}
}
return res;
}

```

Example Usage

```

#include <cassert>
using namespace std;

struct Point {
    double x, y;
    Point(double x = 0, double y = 0) : x(x), y(y) {}
    bool operator==(const Point &p) const { return EQ(x, p.x) && EQ(y, p.y); }
    bool operator<(const Point &p) const { return x != p.x ? x < p.x : y < p.y; }
};

struct PointI {
    int x, y;
    PointI(int x = 0, int y = 0) : x(x), y(y) {}
    bool operator==(const PointI &p) const { return x == p.x && y == p.y; }
    bool operator<(const PointI &p) const { return x != p.x ? x < p.x : y < p.y; }
};

int main() {
    vector<Point> v{{1, 3}, {1, 2}, {2, 1}, {0, 0}, {-1, 3}};
    vector<Triangle> t{
        Triangle(1, 3, 1, 2, -1, 3), Triangle(1, 3, 2, 1, 1, 2), Triangle(1, 2, 2, 1, 0, 0),
        Triangle(1, 2, 0, 0, -1, 3)
    };
    assert(delaunay_triangulation(v.begin(), v.end()) == t);

    // Integer-coordinate inputs are accepted (triangulation computed in double).
    vector<PointI> iv{{1, 3}, {1, 2}, {2, 1}, {0, 0}, {-1, 3}};
    assert(delaunay_triangulation(iv.begin(), iv.end()) == t);

    return 0;
}

```

7.4.4 Delaunay Triangulation (Guibas-Stolfi)

Given a set P of two dimensional points, the Delaunay triangulation of P is a set of non-overlapping triangles that covers the entire convex hull of P such that no point in P lies within the circumcircle of any of the resulting triangles. For any point p in the convex hull of P (but not necessarily in P), the nearest point is guaranteed to be a vertex of the enclosing triangle from the triangulation.

The triangulation may not exist (e.g. for a set of collinear points), or may not be unique if it does exist. The following program produces one such valid result using the Guibas-Stolfi divide-and-conquer algorithm with a

quad-edge data structure. A full description is available in the original paper:

<https://people.eecs.berkeley.edu/~jrs/meshpapers/GuibasStolfi.pdf>

- `delaunay_triangulation(lo, hi)` returns a Delaunay triangulation for the input range `[lo, hi)` of points, or an empty vector if a triangulation does not exist. Duplicate points are ignored.

Time Complexity: $O(n \log n)$ per call to `delaunay_triangulation(lo, hi)`, where n is the distance between `lo` and `hi`.

Space Complexity: $O(n)$ auxiliary heap space for storage of the Delaunay triangulation.

```
#include <algorithm>
#include <cmath>
#include <utility>
#include <vector>

const double EPS = 1e-9;

#define EQ(a, b) (fabs((a) - (b)) <= EPS)
#define LT(a, b) ((a) < (b) - EPS)
#define GT(a, b) ((a) > (b) + EPS)

struct Point {
    double x, y;
    Point(double x = 0, double y = 0) : x(x), y(y) {}
    bool operator==(const Point &p) const { return x == p.x && y == p.y; }
    bool operator!=(const Point &p) const { return !(*this == p); }
    bool operator<(const Point &p) const { return x != p.x ? x < p.x : y < p.y; }
    bool operator>(const Point &p) const { return p < *this; }
};

double cross(const Point &a, const Point &b, const Point &o = Point(0, 0)) {
    return (a.x - o.x) * (b.y - o.y) - (a.y - o.y) * (b.x - o.x);
}

long double incircle(const Point &a, const Point &b, const Point &c, const Point &d) {
    long double adx = a.x - d.x, ady = a.y - d.y;
    long double bdx = b.x - d.x, bdy = b.y - d.y;
    long double cdx = c.x - d.x, cdy = c.y - d.y;
    long double abdet = adx * bdy - bdx * ady;
    long double bcdet = bdx * cdy - cdx * bdy;
    long double cadet = cdx * ady - adx * cdy;
    long double alift = adx * adx + ady * ady;
    long double blift = bdx * bdx + bdy * bdy;
    long double clift = cdx * cdx + cdy * cdy;
    return alift * bcdet + blift * cadet + clift * abdet;
}

struct Edge {
    Point origin;
    Edge *rot, *onext;
    bool primal, deleted, used;

    Edge *rev() const { return rot->rot; }
    Edge *lnext() const { return rot->rev()->onext->rot; }
    Edge *oprev() const { return rot->onext->rot; }
    Point dest() const { return rev()->origin; }
};

struct QuadEdgePool {
    std::vector<Edge *> edges;

    QuadEdgePool() = default;

    ~QuadEdgePool() {
        for (Edge *e : edges) {
            delete e;
        }
    }
};
```



```

    }
}

QuadEdgePool(const QuadEdgePool &) = delete;
QuadEdgePool &operator=(const QuadEdgePool &) = delete;

Edge *make_edge(const Point &from, const Point &to) {
    Edge *e1 = new Edge, *e2 = new Edge, *e3 = new Edge, *e4 = new Edge;
    edges.push_back(e1);
    edges.push_back(e2);
    edges.push_back(e3);
    edges.push_back(e4);
    e1->origin = from;
    e2->origin = to;
    e1->rot = e3;
    e2->rot = e4;
    e3->rot = e2;
    e4->rot = e1;
    e1->onext = e1;
    e2->onext = e2;
    e3->onext = e4;
    e4->onext = e3;
    e1->primal = e2->primal = true;
    e3->primal = e4->primal = false;
    e1->deleted = e2->deleted = e3->deleted = e4->deleted = false;
    e1->used = e2->used = e3->used = e4->used = false;
    return e1;
}

void delete_edge(Edge *e) {
    splice(e, e->oprev());
    splice(e->rev(), e->rev()->oprev());
    e->deleted = e->rot->deleted = e->rev()->deleted = e->rev()->rot->deleted = true;
}

Edge *connect(Edge *a, Edge *b) {
    Edge *e = make_edge(a->dest(), b->origin);
    splice(e, a->lnext());
    splice(e->rev(), b);
    return e;
}

static void splice(Edge *a, Edge *b) {
    std::swap(a->onext->rot->onext, b->onext->rot->onext);
    std::swap(a->onext, b->onext);
}

};

bool left_of(const Point &p, Edge *e) {
    return GT(cross(e->origin, e->dest(), p), 0);
}

bool right_of(const Point &p, Edge *e) {
    return LT(cross(e->origin, e->dest(), p), 0);
}

bool in_circle(const Point &a, const Point &b, const Point &c, const Point &d) {
    return incircle(a, b, c, d) > EPS;
}

std::pair<Edge *, Edge *> build_triangulation(
    std::vector<Point> &p, int lo, int hi, QuadEdgePool &pool
) {
    if (hi - lo == 1) {
        Edge *a = pool.make_edge(p[lo], p[hi]);
        return {a, a->rev()};
    }
    if (hi - lo == 2) {

```

```

Edge *a = pool.make_edge(p[lo], p[lo + 1]);
Edge *b = pool.make_edge(p[lo + 1], p[hi]);
pool.splice(a->rev(), b);
int side = GT(cross(p[lo], p[lo + 1], p[hi]), 0)
           ? 1
           : (LT(cross(p[lo], p[lo + 1], p[hi]), 0) ? -1 : 0);
if (side == 0) {
    return {a, b->rev()};
}
Edge *c = pool.connect(b, a);
if (side == 1) {
    return {a, b->rev()};
}
return {c->rev(), c};
}
int mid = lo + (hi - lo) / 2;
auto [ldo, ldi] = build_triangulation(p, lo, mid, pool);
auto [rdi, rdo] = build_triangulation(p, mid + 1, hi, pool);
for (;;) {
    if (left_of(rdi->origin, ldi)) {
        ldi = ldi->lnext();
    } else if (right_of(ldi->origin, rdi)) {
        rdi = rdi->rev()->onext;
    } else {
        break;
    }
}
Edge *base = pool.connect(rdi->rev(), ldi);
if (ldi->origin == ldo->origin) {
    ldo = base->rev();
}
if (rdi->origin == rdo->origin) {
    rdo = base;
}
for (;;) {
    Edge *lcand = base->rev()->onext;
    if (right_of(lcand->dest(), base)) {
        while (in_circle(base->dest(), base->origin, lcand->dest(), lcand->onext->dest())) {
            Edge *next = lcand->onext;
            pool.delete_edge(lcand);
            lcand = next;
        }
    }
    Edge *rcand = base->oprev();
    if (right_of(rcand->dest(), base)) {
        while (in_circle(base->dest(), base->origin, rcand->dest(), rcand->oprev()->dest())) {
            Edge *prev = rcand->oprev();
            pool.delete_edge(rcand);
            rcand = prev;
        }
    }
    bool lvalid = right_of(lcand->dest(), base);
    bool rvalid = right_of(rcand->dest(), base);
    if (!lvalid && !rvalid) {
        break;
    }
    if (!lvalid ||
        (rvalid && in_circle(lcand->dest(), lcand->origin, rcand->origin, rcand->dest()))) {
        base = pool.connect(rcand, base->rev());
    } else {
        base = pool.connect(base->rev(), lcand->rev());
    }
}
return {ldo, rdo};
}

struct Triangle {
    Point a, b, c;

```

```

Triangle(const Point &a, const Point &b, const Point &c) : a(a), b(b), c(c) {
    if (b < a && b < c) {
        this->a = b;
        this->b = c;
        this->c = a;
    } else if (c < a && c < b) {
        this->a = c;
        this->b = a;
        this->c = b;
    }
}

bool operator==(const Triangle &t) const { return a == t.a && b == t.b && c == t.c; }

bool operator<(const Triangle &t) const {
    return a != t.a ? a < t.a : (b != t.b ? b < t.b : c < t.c);
}
};

// Accepts any point type with numeric .x/.y; inputs are copied to double Points.
template<class It>
std::vector<Triangle> delaunay_triangulation(It lo, It hi) {
    std::vector<Point> p;
    for (It it = lo; it != hi; ++it) {
        p.emplace_back(it->x, it->y);
    }
    std::sort(p.begin(), p.end());
    p.erase(std::unique(p.begin(), p.end()), p.end());
    if (p.size() < 3) {
        return std::vector<Triangle>();
    }
    QuadEdgePool pool;
    build_triangulation(p, 0, p.size() - 1, pool);
    std::vector<Triangle> res;
    for (Edge *start : pool.edges) {
        if (!start->primal || start->deleted || start->used) {
            continue;
        }
        std::vector<Point> face;
        Edge *curr = start;
        do {
            curr->used = true;
            face.push_back(curr->origin);
            curr = curr->lnext();
        } while (curr != start);
        if (face.size() == 3 && GT(cross(face[0], face[1], face[2]), 0)) {
            res.emplace_back(face[0], face[1], face[2]);
        }
    }
    std::sort(res.begin(), res.end());
    return res;
}

```

Example Usage

```

#include <cassert>
using namespace std;

struct PointI {
    int x, y;
    PointI(int x = 0, int y = 0) : x(x), y(y) {}
};

int main() {
    vector<Point> v{{1, 3}, {1, 2}, {2, 1}, {0, 0}, {-1, 3}};

```

```
vector<Triangle> t{
    Triangle(Point(-1, 3), Point(0, 0), Point(1, 2)),
    Triangle(Point(-1, 3), Point(1, 2), Point(1, 3)),
    Triangle(Point(0, 0), Point(2, 1), Point(1, 2)),
    Triangle(Point(1, 2), Point(2, 1), Point(1, 3))
};
assert(delaunay_triangulation(v.begin(), v.end()) == t);

// Integer-coordinate inputs are accepted (triangulation computed in double).
vector<PointI> iv{{1, 3}, {1, 2}, {2, 1}, {0, 0}, {-1, 3}};
assert(delaunay_triangulation(iv.begin(), iv.end()) == t);

return 0;
}
```